

THE COMPLETE GUITAR COMPENDIUM

The Fretboard Atlas

Theory, technique, rhythm and style: from the first chord to jazz, flamenco, fingerstyle and beyond

CLAUDINE FRETWELL

13

PARTS

59

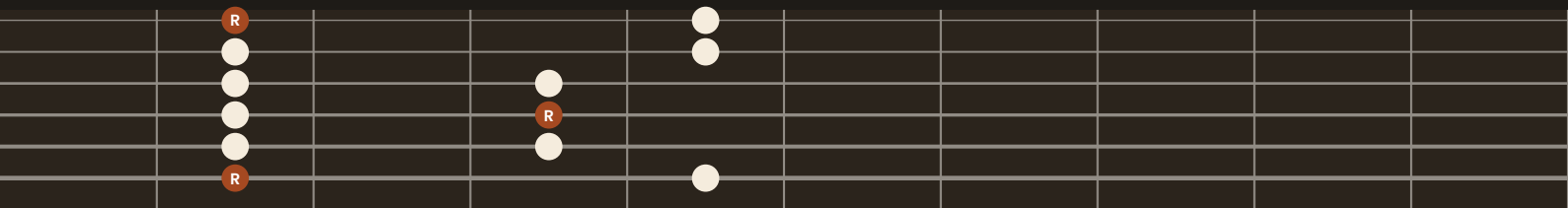
CHAPTERS

2500+

DIAGRAMS

551

AUDIO TRACKS



For nylon-string, steel-string and electric guitar

Second edition · v2

COMPANION APP AND AUDIO



THE COMPLETE GUITAR COMPENDIUM

The Fretboard Atlas

Theory, technique, rhythm and style: from the first chord to jazz, flamenco, fingerstyle and beyond

Claudine Fretwell



Web companion

<https://amalics.github.io/fretboard-atlas-audio/>

Every audio track and practice loop, plus an interactive fretboard, chord finder, metronome, tuner and ear trainer. Scan the code or click the link. Each chapter opener has its own code for that chapter's tracks.

Every diagram, score and recording in this book is generated from the theory it illustrates, so the shapes, the notation and the rules always agree. Written and produced with an AI system (Claude by Anthropic) under human direction and editorial review. Second edition, v2, 7 October 2026. Licensed under CC BY-NC-SA 4.0.

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EDITION NOTES

About This Book

How this book was made, how you may use it, and the people and projects it owes thanks to.

How it was made

The Fretboard Atlas was written and produced with an AI system, Claude by Anthropic, under human direction and editorial review. The text, the music theory engine that computes every diagram, the study pieces, the audio and the companion website were all generated with AI assistance; a human editor chose the scope, reviewed the results and decided what went into the book. Claudine Fretwell is a pen name. This is the second edition (v2) of 7 October 2026; the companion website is built together with it, so track numbers always match.

Every chord shape, scale map, score and recording is computed from the same theory code, and the build checks each chord voicing note by note and each bar of notation beat by beat. Facts about history, songs and artists were checked against published sources where the text makes specific claims; if you find a mistake, treat the book as you would any reference and check the original source.

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Songs, artists, recordings and brands are named only to show where a technique or idea can be heard. No lyrics or melodies of copyrighted songs are reproduced, and the book is not affiliated with or endorsed by any artist, publisher or manufacturer named in it.

Health note

Playing should not hurt. The advice on posture, warming up and tension in this book is general guidance; if you have pain that persists, see a doctor or physiotherapist.

INTRODUCTION

How to Use This Book

This book is a reference you return to, not a novel to read once. Each chapter adds one layer, and each layer makes the earlier ones easier to remember.

Guitar theory looks endless from the outside. In practice it is a small set of ideas that keep coming back in new forms. A major scale is seven notes. A chord is three or more of those notes played together. A progression is a series of chords. An arpeggio is a chord played one note at a time. Each topic is built from the one before it, and the chapters follow that order.

The book has thirteen parts. Parts One to Four cover the basics of the instrument, the fretboard, scales and harmony. Part Five is about the picking hand: fingerstyle in depth, strumming and pick technique. Part Six covers open and alternate tunings. Part Seven teaches reading notation and rhythm, ear training and transcription. Part Eight goes further into theory. Parts Nine to Eleven cover expressive technique, rhythm, practice methods and gear. Part Twelve is a set of fourteen style workshops, and Part Thirteen is a collection of graded study pieces in notation and TAB.

How each chapter works

Every chapter ends with a review in three parts. **Using it in real music** shows where the idea appears in actual songs and pieces. **How to train it** gives short, timed drills with a clear goal. **Check your understanding** is a short quiz (answers in Appendix J), followed by ruled space for your own notes.

The boxes

TIP

Practical advice that saves time or frustration.

SHORTCUT

A pattern or memory aid that replaces a lot of counting.

TRY THIS

A short exercise. Do these with the guitar in your hands: this is where most of the learning happens.

NYLON · STEEL

Advice that differs between nylon-string and steel-string guitars (technique, sound or comfort).

HOW IT CONNECTS

A link back to an earlier chapter, showing how the current topic grows out of it.

CAREFUL

A warning about something that can hurt your hands, your instrument or your progress.

UNDER THE HOOD

An optional look under the hood: why a rule works. You can skip it on a first reading.

LISTEN FOR

What to listen for in a recording of the style or piece.

Audio tracks

Every example marked with a ► **TRACK** number has a recording. Chapter 2 explains how the tracks work and how to practise with them; Appendix K lists them all.



Web companion

<https://amalics.github.io/fretboard-atlas-audio/>

The companion site plays every track and practice loop (with a slow version, speed control and looping) and adds tools a book cannot: an interactive fretboard for any scale, chord or tuning, a chord finder that names any shape you tap, a metronome, a tuner and an ear trainer that scores you. Your progress through the 52-week plan (Appendix A) can be ticked off there too, and every chapter's flashcards are there with spaced repetition. Every chapter opener with audio has its own small code that opens just that chapter's tracks.

Learning tracks

The chapters are grouped by topic, but you should not learn them strictly in order. Follow one of these tracks instead. Each mixes theory, technique and rhythm so that what you learn is used straight away. Whatever track you choose, start the daily rhythm routine from Chapter 37 in your first week.

Track	Chapters, in this order	Typical time
1 • Foundations (new to guitar)	Chapter 1, Chapter 2, Chapter 3, Chapter 37 routine, Chapter 11 (open chords), Chapter 22, Chapter 17, Chapter 4, Chapter 5, Chapter 38	3–4 months
2 • Building (open chords secure)	Chapter 7, Chapter 18, Chapter 10, Chapter 13, Chapter 14, Chapter 43, Chapter 6, Chapter 12, Chapter 26, Chapter 19, Chapter 57	4–6 months
3 • Expanding (colour and style)	Chapter 9, Chapter 25, Chapter 20, Chapter 28, Chapter 32, Chapter 48, any style workshops, Chapter 58	ongoing
4 • Advanced (fluent player)	Chapter 31, Chapter 34, Chapter 35, Chapter 33, Chapter 30, Chapter 59	ongoing
If you mainly want to...	Go to	
Strum and sing songs	Chapter 11, Chapter 22, Chapter 13, Chapter 14, Chapter 47	
Play fingerstyle guitar	Chapter 17 to Chapter 21, Chapter 25, Chapter 54, Chapter 57	
Play classical guitar	Chapter 1, Chapter 17, Chapter 26, Chapter 20, Chapter 57 to Chapter 59	
Improvise and solo	Chapter 5, Chapter 7, Chapter 36, Chapter 16, Chapter 9, Chapter 35	
Play by ear	Chapter 3, Chapter 28, Chapter 30	
Buy a guitar or set up gear	Chapter 41, Chapter 42	

TIP • The golden rule

Learn a little and use it at once. Ten minutes of theory followed by twenty minutes of applying it to music you like is worth more than an hour of reading.

Your Skill Map

Every chapter is a box on this map. Work roughly top to bottom and tick a box once you have worked through the chapter's drills and answered its quiz without looking.

PART 1 FOUNDATIONS

1 Your instrument ☐

2 Diagrams ☐

3 Building blocks ☐

PART 2 FRETBOARD

4 Fretboard map ☐

PART 3 SCALES

5 Major scale ☐

6 Minor scales ☐

7 Pentatonic ☐

8 Horizontal ☐

9 Modes ☐

PART 4 HARMONY

10 Triads ☐

11 Chords ☐

12 CAGED ☐

13 Chords in a key ☐

14 Progressions ☐

15 Spices ☐

16 Arpeggios ☐

PART 5 PICKING HAND

17 Fingerstyle basics ☐

18 Picking patterns ☐

19 Travis picking ☐

20 Melody & bass ☐

21 Colours ☐

22 Strumming ☐

23 Songbook ☐

24 Pick technique ☐

PART 6 TUNINGS

25 Open tunings ☐

PART 7 READING & EAR

26 Notation ☐

27 Rhythm reading ☐

28 Ear training ☐

29 Ear course ☐

30 Transcription ☐

PART 8 DEEPER THEORY

31 Minor harmony ☐

32 Secondary dom. ☐

33 Modulation ☐

34 Voice leading ☐

35 Over changes ☐

PART 9 TECHNIQUE & TIME

36 Expression ☐

37 Rhythm ☐

PART 10 PUTTING IT TO WORK

38 Practice ☐

39 What to avoid ☐

40 Fitting together ☐

PART 11 GEAR & SOUND

41 Choosing a guitar ☐

42 Sound and loopers ☐

PART 12 STYLES

43 Blues ☐

44 Rock & riffs ☐

45 Funk rhythm ☐

46 Country ☐

47 Folk & songs ☐

48 Jazz harmony ☐

49 Gypsy jazz ☐

50 Bossa & Latin ☐

51 Choro & Brazil ☐

52 Reggae ☐

53 Flamenco ☐

54 Celtic ☐

55 Neo-soul ☐

56 African guitar ☐

PART 13 STUDY PIECES

57 Pieces 1–2 ☐

58 Pieces 3–5 ☐

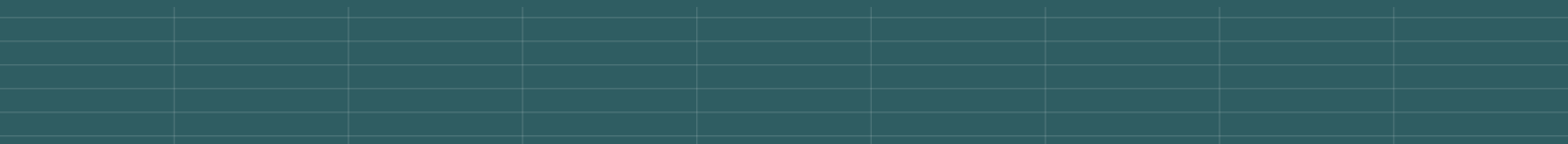
59 Pieces 6+ ☐

PART ONE

Foundations

The vocabulary every other chapter depends on: the instrument, the symbols, and the twelve notes all music is made from.

- 1 Know Your Instrument
- 2 Reading the Diagrams
- 3 The Building Blocks of Music



CHAPTER 1

Know Your Instrument



TRACKS

1

Nylon-string, steel-string and electric guitars share the same tuning and the same fretboard, so every shape in this book works on all of them. What differs is the feel, the sound and the technique.

All three guitars have six strings tuned, from lowest to highest, E – A – D – G – B – E, and frets spaced one half step apart. A chord, scale or arpeggio shape played on one of them gives the same notes on the others. The differences are in string material, neck shape and string tension, and those decide which techniques and styles feel natural.

Nylon, steel and electric compared

	Nylon-string (classical)	Steel-string acoustic	Electric
Strings	Nylon trebles; bass strings wound over a nylon core	Steel, usually bronze-wound	Steel, nickel-wound; lighter gauges
Neck width at nut	Wide: about 50–52 mm	About 42–45 mm	About 41–43 mm
Fingerboard	Usually flat	Slightly curved (radiused)	Curved; low frets, low action
Neck joins body at	12th fret	14th fret	16th–22nd fret (cutaway)
String tension	Low	High	Low to medium
Usual picking hand	Fingers and nails: p, i, m, a	Pick or fingers	Pick, fingers or hybrid
Typical repertoire	Classical, flamenco, bossa nova, solo arrangements	Folk, pop, country, blues, singer-songwriter	Rock, blues, jazz, funk, pop
Sound	Warm, round, short sustain	Bright, ringing, loud	Depends on amplifier and effects
String bends	Small; vibrato is used instead	Up to a half step	Up to two whole steps

What it means for learning

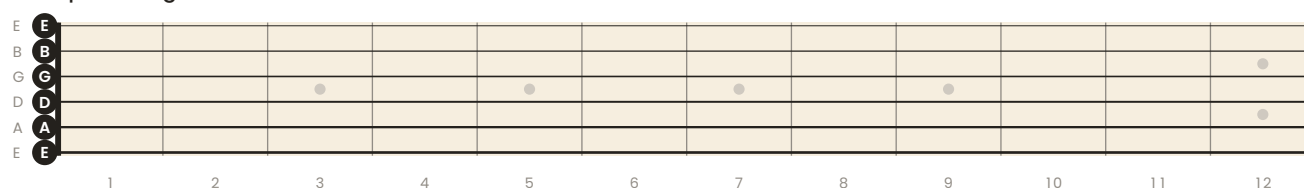
The theory is identical on all three guitars. The differences show up in which shapes are comfortable and which sounds suit the instrument.

- **Stretches.** The wide nylon neck makes stretches across the strings longer, but low tension makes notes easier to press. The narrower steel-string neck is easier to reach across, but each note needs more pressure. Barre chords are hard on both, for different reasons.
- **Upper frets.** On a nylon-string guitar the neck meets the body at the 12th fret, so frets 13–19 are harder to reach. Steel-string players reach the 14th fret comfortably, electric players usually the 20th.
- **Sustain and texture.** Nylon strings fade quickly and blend well, which suits arpeggios, counterpoint and melody over bass. Steel strings ring long and loud, which suits strumming, drones and open tunings.
- **Notation.** Classical music is mostly written in standard notation with fingering marks. Steel-string and electric music is more often written in TAB and chord charts. This book uses both, and Chapter 26 teaches you to read notation fluently.

The parts of the guitar

Part	What it does
Headstock and tuners	Hold the strings and change their pitch when turned
Nut	Slotted strip at the end of the fingerboard; sets the string height and spacing
Frets	Metal strips; each one raises the pitch by a half step
Fret markers	Dots at frets 3, 5, 7, 9, 12 (double), 15, 17 and 19 for orientation
Body and soundhole	Amplify the string vibration (acoustic)
Pickups	Turn string vibration into an electric signal (electric and electro-acoustic)
Bridge and saddle	Anchor the strings and transfer their vibration to the top

The open strings and the fret markers



Strings are numbered from the thinnest (1st, high E) to the thickest (6th, low E). Diagrams in this book show the 1st string at the top.

Posture

NYLON · STEEL

Nylon: sit with the guitar on the left leg (for a right-handed player), raised with a footstool or a guitar support so that the headstock is about shoulder height. The fretting-hand thumb rests behind the neck, roughly opposite the 2nd finger.

Steel and electric: usually played on the right leg or standing with a strap. The thumb may sit lower for barres and wide stretches, or wrap over the top of the neck for some chords.

Whichever guitar you play, check these four points before every session:

1. Shoulders level and relaxed; the guitar comes to you, not the other way round.
2. Fretting wrist fairly straight, never sharply bent.
3. Fingertips press just behind the fret, not on top of it or in the middle of the gap.
4. Picking forearm rests on the edge of the body, so the hand hangs freely over the strings.

Finger names

The picking-hand fingers use the Spanish letters from the classical tradition. Fretting-hand fingers are numbered.

Picking hand	Spanish name	Finger	Fretting hand
p	pulgar	thumb	T (thumb, occasionally for the 6th string)
i	índice	index	1
m	medio	middle	2
a	anular	ring	3
c	chiquito	little finger (rarely used)	4

Two strokes for the fingers

The picking fingers use two strokes. Chapter 17 develops both in detail.

- **Free stroke** (*tirando*): the finger plucks the string and moves up into the palm, clear of the next string. Used for arpeggios and chords. It is the default stroke in steel-string fingerstyle.
- **Rest stroke** (*apoyando*): the finger plucks through the string and comes to rest on the next lower-pitched string. It is fuller and louder, and brings out a melody.

TIP · Nails on nylon strings

Shape the nails of i, m and a into a smooth curve that follows the fingertip and extends one to two millimetres. Polish them with very fine sandpaper (1000–2000 grit). A rough edge makes a scratchy tone.

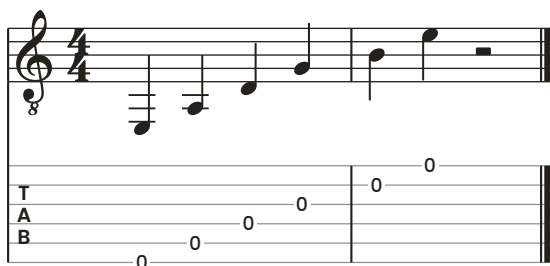
TIP · Picks and strings on steel

Start with a medium pick (about 0.6–0.8 mm). Thin picks are forgiving for strumming; thicker ones give more control for single notes. Light-gauge strings and a low action make barre chords much easier.

Tuning

Tune with a clip-on or app tuner at first. Pluck each open string, let it ring, and turn the tuning peg until the tuner reads the target note. Always tune **up** to a note: if you are sharp, go below the note and come back up. This keeps the string from slipping. Chapter 4 shows how to tune the strings to each other without a tuner.

▶ TRACK 1 Open strings in notation and TAB



Guitar music is written an octave higher than it sounds, which is what the small 8 under the clef means. Chapter 26 explains this.

Chapter 1 review · use it, train it, test it

Using it in real music

Choosing an instrument for a style. Arpeggiated ballads, bossa nova and solo pieces sit naturally on nylon strings; strummed folk and pop on steel; rock, funk and lead playing on electric. Knowing why helps you pick the right guitar for a song, or adapt a song to the guitar you own.

Moving between guitars. Every chord and scale works on all three. Expect to press harder on steel, stretch further on nylon and use a lighter touch on electric. Give your hands a week to adjust.

How to train it

Drill	How	Time	Goal
Posture check	Sit in front of a mirror; check guitar angle, shoulders and fretting wrist	1 min	Same posture every session without thinking
Free and rest strokes	Open 1st string: alternate i and m, 8 free strokes then 8 rest strokes. Repeat on strings 2 and 3	3 min	Even volume and tone from both fingers
Alternate picking	Open strings, down-up eighth notes with a metronome at 60–80 bpm	3 min	Even upstrokes, relaxed wrist

Check your understanding

Write your answers, then check them in Appendix J.

1. List the open strings from the 6th to the 1st.

2. At which fret does the neck usually join the body on a nylon-string guitar?

3. What do the letters p, i, m and a stand for?

4. What is the difference between a rest stroke and a free stroke?

5. Does a C major chord shape learned on steel strings work unchanged on nylon strings?

6. Which fretting-hand finger is number 3?

7. Why should you tune up to a note rather than down?

MY NOTES

CHAPTER 2

Reading the Diagrams



TRACKS

2

Guitar music is written down in several ways. This chapter shows each kind of diagram used in the book, what every symbol means, and how the recorded examples work.

Guitarists use four main kinds of written music: chord boxes, fretboard diagrams, TAB and standard notation. This book adds three smaller ones for the picking hand and for rhythm: picking maps, strum grids and rhythm grids. All of them describe the same thing, the notes and where to play them, from a different angle. Learn to read each one here and the rest of the book becomes much faster to use.

Diagram	Shows	Best for
Chord box	One chord shape, neck upright	Learning chords
Fretboard diagram	Many notes, neck lying flat	Scales, arpeggios, note maps
TAB	Strings and frets in time order	Riffs, melodies, licks
Standard notation	Pitch and rhythm exactly	Classical music, reading in ensembles
Picking map	Which finger plucks which string	Fingerstyle patterns
Strum grid	Down and up strums on a beat grid	Strumming patterns
Rhythm grid	Onsets on a grid, without pitch	Grooves and rhythmic figures

Chord boxes

A chord box shows the neck standing upright, as if the guitar hangs on a wall facing you. The vertical lines are the strings: low E (6th string) on the left, high E (1st string) on the right. The horizontal lines are the frets. A thick line at the top is the nut.

- A **dot** is a fretted note. The number inside it is the fretting-hand finger: 1 index, 2 middle, 3 ring, 4 little finger.
- A **rust dot** is the root, the note the chord is named after.
- A **small circle** above a string: play the string open.
- An **x** above a string: do not play it (mute it or avoid it).
- A **long bar** across several strings is a barre: one finger, usually the 1st, flattened across them.
- A **fret number** beside the box (for example 5fr) means the top line is that fret, not the nut.
- The letters under the strings are the notes that sound, from low to high.

Three open chords and a barre chord

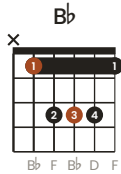


Finger numbers inside the dots, sounding notes underneath. The F chord shows a barre with the 1st finger across all six strings.

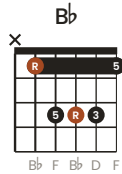
The same box can show other information inside the dots. With **degrees** inside, you see the job of each note in the chord (R, 3, 5, and so on) instead of the finger. This is how the book shows movable shapes: the shape stays the same in every key, only the fret changes.

Movable shapes away from the nut

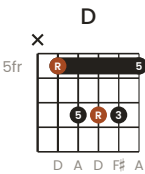
Fingers



Degrees



Same shape at 5fr



An A-shape major barre chord. Moved up four frets, Bb becomes D; the degrees do not change.

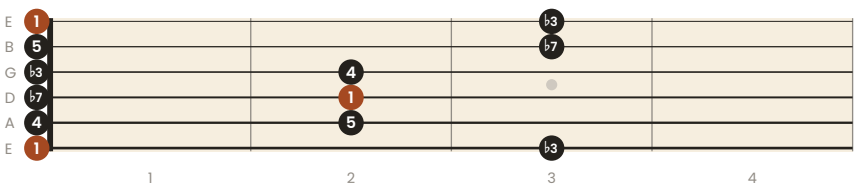
TIP · Which way up

If chord boxes confuse you, hold the book up next to the guitar's neck with the headstock pointing to the ceiling. The box now matches what you see.

Fretboard diagrams

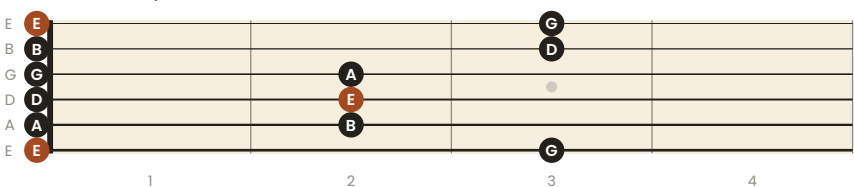
For scales, arpeggios and anything that spans more than four frets, the book draws the neck lying flat, the way you see it when you look down at the guitar on your lap. The high E string is at the **top** and the low E string at the **bottom**, exactly as in TAB. Fret numbers run along the bottom; open strings are at the left edge. Fret markers are shaded as on a real neck.

E minor pentatonic in open position, with degrees



Degrees inside the dots. The rust dots are the root (E); the other numbers say how far each note is from it. The dots at the left edge are open strings.

The same shape with note names



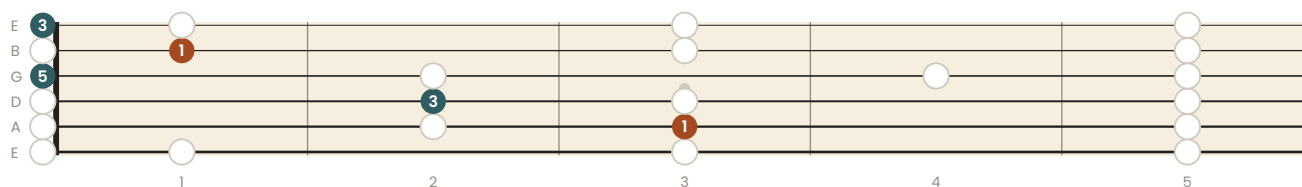
The same five notes, E G A B D, labelled by name.

Inside the dots you see either note names (E, G, A) or degrees (b3, 4, 5, b7, with the root marked R or 1). Degrees are usually more useful: they tell you what each note does, and they stay the same when you move the shape to a new key. Chapter 3 explains degrees and intervals.

The dots use five colours:

Colour	Meaning
Rust	The root
Dark (ink)	Other notes of the scale or pattern
Teal	Notes picked out for attention, usually chord tones
Gold	A characteristic or added note, such as a blue note or a mode's characteristic note
Light grey	Background notes, shown for context

A C chord inside its scale



Grey dots are the C major scale in open position; the C chord is drawn on top, root in rust, 3rd and 5th in teal.

TAB (tablature)

TAB uses six horizontal lines for the six strings, high E on top. A number on a line is the fret to play on that string; 0 is the open string. Read left to right. Numbers stacked vertically are played together.

TAB on its own shows where to play but not how long each note lasts. In this book every TAB example is printed under a staff of notation, so the rhythm is always there. If you do not read notation yet, use the stems in the TAB and the recording.

▶ **TRACK 2** A C major chord played one note at a time, then together

The first bar is an arpeggio (one note after another); in the second bar the five notes are stacked, so they are struck together.

TAB uses letters and symbols between the numbers for left-hand techniques.

Symbol	Technique	Covered in
h	Hammer-on: sound the next note by hammering a finger down	Chapter 36
p	Pull-off: sound the next note by pulling the finger off	Chapter 36
/ or \	Slide up or down to the next note	Chapter 36
b, full, ½	Bend the string up by a whole step (full) or half step (½)	Chapter 36
~	Vibrato	Chapter 36
Diamond note head, or <12>	Natural harmonic	Chapter 36
x	Muted, percussive note	Chapter 22

▶ **TRACK 3** Slurs, a slide, a bend and vibrato

Hammer-on, pull-off, slide, full bend and vibrato, in that order. The curved lines and labels above the TAB show each technique.

Standard notation at a glance

Standard notation shows pitch by the position of a note on the five-line staff and rhythm by the shape of the note. It is the main language of classical guitar and of any music shared with other instruments. Guitar music is written on the treble clef, one octave higher than it sounds: that is what the small 8 under the clef means.

You do not need notation to use most of this book: every example also has TAB. Chapter 26 teaches note reading step by step, and Chapter 27 teaches rhythm reading. For now, notice two things in every example: notes higher on the staff sound higher, and the TAB under each note tells you where to play it.

TRACK 4 C major up and down in first position

As the notes climb the staff, the TAB moves from the 5th string to the 2nd; as they fall, it moves back. Chapter 26 adds every natural note in first position.

Classical scores add fingering marks to the notes. Steel-string and electric music in notation sometimes uses them too.

Mark	Meaning
1 2 3 4 beside a note head	Fretting-hand finger; 0 means open string
p i m a above or below a note	Picking-hand finger: thumb, index, middle, ring
A number in a circle, such as a circled 5	The string to play the note on
C V, CV or BV	Full barre at the 5th fret (C for <i>cejilla</i> , B for barre); ½C V is a half barre
Roman numeral, such as VII	Position: the 1st finger is at that fret
A straight line between two finger numbers	Guide finger: keep the finger on the string while it slides

Picking maps

A picking map shows a fingerstyle pattern on a grid. Each column is one rhythmic slot (an eighth note in most patterns), counted along the bottom, and each line is a string, high E on top as in TAB. A dot means: pluck this string now. The colour of the dot is the picking-hand finger:

Colour	Finger
Rust	p (thumb)
Teal	i (index)
Gold	m (middle)
Dark (ink)	a (ring)

The chord name is printed above each bar; hold that chord while the pattern plays.

TRACK 5 A simple p-i-m-a pattern on A minor

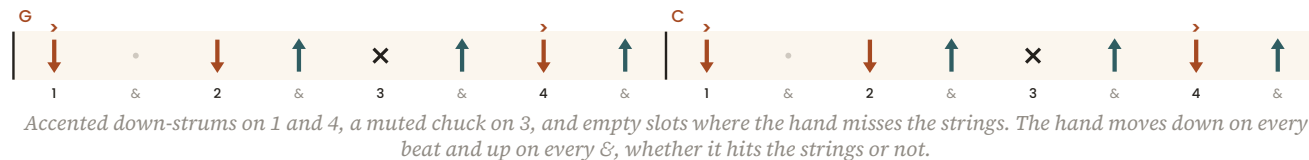
The thumb alternates between the root and another bass string; i, m and a stay on strings 3, 2 and 1.

Chapter 17 and Chapter 18 use picking maps throughout.

Strum grids

A strum grid shows a strumming pattern on the beats of a bar. The count is written along the bottom (1 & 2 & 3 & 4 &) and the chord name above the bar. An arrow down is a down-strum (D), an arrow up is an up-strum (U). An empty slot means the hand keeps moving but misses the strings. An accent mark (>) means strum harder; an × is a muted chuck.

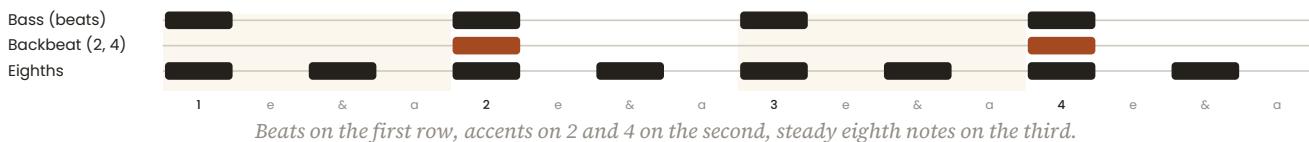
▶ TRACK 6 Reading a strum grid: accents and a chuck



Rhythm grids

A rhythm grid shows *when* notes happen, without pitch. The grid is counted along the bottom; in 4/4 there are usually 16 slots, one per sixteenth note (1 e & a, 2 e & a...), with every other beat shaded. A dark bar is a note, a rust bar an accented note, and a longer bar a note held for more than one slot. Several rows can be stacked to show how parts fit together.

▶ TRACK 7 Three layers of a basic groove



Chord charts

A chord chart shows the chords of a song bar by bar. Each box is one bar; two chords in a box share the bar. Roman numerals under the chords give each chord's number in the key, so you can move the progression to any key. Chapter 13 explains the numerals.

A four-bar progression in G



Other diagrams

- **Keyboard.** A piano keyboard is the clearest picture of half and whole steps, so it appears where the book explains scale construction. Chapter 3 uses it first.
- **Circle of fifths.** A circle of the twelve keys, used for key signatures and related keys. See Chapter 5.
- **Flow charts.** Boxes and arrows that show how one idea leads to another.

The audio tracks

Every example marked ▶ **TRACK** followed by a number has a recording. The tracks are generated from the same data as the printed example, so the sound always matches the page. Each track is available at full tempo and at a slow practice tempo with a one-bar count-in. Track numbers run through the whole book; Appendix K lists them all with their chapter and title.

To use a track:

1. Listen once without the guitar while following the notation or TAB.
2. Play along with the slow version until it is clean three times in a row.
3. Move to the full-tempo version.

TRY THIS · Read one example in every format

Take the A minor picking pattern above. Find the chord in a chord box (Am, x02210), then read the pattern from the picking map, then play along with the track. If all three agree with what your hands do, you can read the diagrams in this book.

Chapter 2 review · use it, train it, test it**Using it in real music**

Songbooks and online TAB. Most pop, rock and folk songs are written as chord boxes, a chord chart and TAB. Reading them quickly means you can learn a song from the page in minutes.

Classical scores. Classical pieces use standard notation with fingering marks such as circled string numbers, C V and p i m a. These tell you not only which notes to play but how the composer or editor wants them played.

Working with other musicians. A chord chart with Roman numerals is how many bands and session players communicate a song. You can transpose it on sight.

How to train it

Drill	How	Time	Goal
TAB sight-reading	Take a short riff in TAB you have not heard and play it slowly without listening first	5 min	Play an unfamiliar riff correctly first time
Box to fretboard	Pick a chord box from Chapter 11; find the same notes on a fretboard diagram drawing	3 min	Translate between upright and flat views without hesitation
Draw your songs	Draw the chord boxes for a song you know from memory, then check them	5 min	Diagrams match what your hands do
Staff note of the day	Pick one note from the C major example; find it on the staff and on the guitar	2 min	Name first-position notes in under 2 seconds

Check your understanding

Write your answers, then check them in Appendix J.

1. In a chord box, which string is on the left?

2. In TAB, which line is the high E (1st) string?

3. What does *5fr* beside a chord box mean?

4. What does × above a string in a chord box mean?

5. In a fretboard diagram, which string is drawn at the top?

6. Guitar music is written on the treble clef. How does it sound compared with the written pitch?

7. In a classical score, what does a circled 3 next to a note mean?

8. What does C VII mean?

CHAPTER 3

The Building Blocks of Music



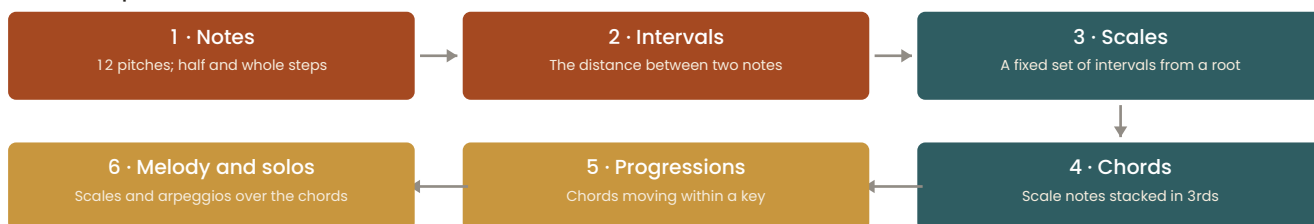
TRACKS

3

Twelve notes, two kinds of step, a set of intervals, and a steady beat. Every scale, chord and rhythm in this book is built from the material in this chapter.

Music theory describes two things: pitch (how high or low a note is) and time (when it happens and how long it lasts). On the pitch side there are only twelve notes, and the distances between them, called intervals, are what give scales and chords their sound. On the time side there is a steady beat, grouped into bars and divided into smaller values. This chapter covers both.

How the topics in this book build on each other



The musical alphabet

Western music uses seven letter names, A B C D E F G, and then starts again. These are the **natural notes**. Between most pairs of natural notes there is one more note, named with a sharp (#, raised) or a flat (b, lowered). Two pairs have no note between them: B–C and E–F.

That gives twelve different notes in total, the **chromatic scale**:

1	2	3	4	5	6	7	8	9	10	11	12
C	C# / Db	D	D# / Eb	E	F	F# / Gb	G	G# / Ab	A	A# / Bb	B

After B comes C again, one **octave** higher. An octave higher means the same note name at double the frequency; the two notes sound so alike that they share a name.

C# and Db are the same pitch with two names. Notes like these are **enharmonic** equivalents. Which name is correct depends on the key and the scale: a scale uses each letter name once, so in D major the note between F and G is F#, not Gb. Chapter 5 explains the rule.

CAREFUL · E# and Cb exist

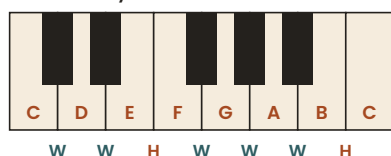
Because B–C and E–F are half steps, E# is the same pitch as F, and Cb is the same pitch as B. They look odd but appear in keys such as F# major (which contains E#) and Gb major (which contains Cb). Use them when the spelling of a scale requires them.

Half steps and whole steps

The smallest distance in Western music is the **half step**, also called a semitone. On the guitar a half step is **one fret**. A **whole step** (a tone) is two half steps: **two frets**.

A piano keyboard shows this clearly. Every pair of neighbouring keys, white or black, is a half step apart. The two places where white keys touch with no black key between them are B–C and E–F.

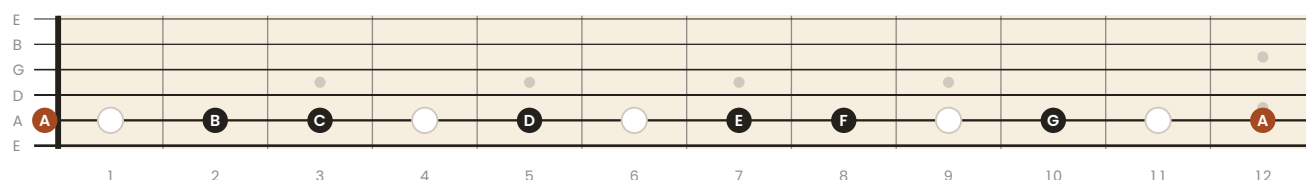
One octave of C major on the keyboard



W = whole step, H = half step. The white keys from C to C follow the pattern W W H W W W W H.

On one string the guitar works like a ruler of half steps. Each fret is one half step; twelve frets is an octave. That is why the 12th fret has the same notes as the open strings, and why every pattern on the neck repeats from the 12th fret.

The chromatic scale on the 5th string



Twelve frets, twelve half steps, back to A. Natural notes are named; the grey dots are the sharps and flats between them. B-C (frets 2-3) and E-F (frets 7-8) have no fret between them.

TRY THIS · Find the half steps by ear

Play the 5th string from the open A up to the 12th fret, one fret at a time, saying each note name aloud. Then play only the natural notes: A B C D E F G A. Listen for the two short steps, B-C and E-F.

Intervals

An **interval** is the distance between two notes. Intervals are the real building blocks of music: a scale is a series of intervals from a root, a chord is intervals stacked together, and recognising intervals by ear is the basis of playing by ear (Chapter 28).

Every interval has a name, a short symbol used as a scale or chord degree, and a size in half steps. On a single string the size in half steps is also the number of frets.

Half steps	Interval	Degree	From C	Sound	Heard at the start of
0	Unison	1 (R)	C	The same note	
1	Minor 2nd	b2	D $\flat$	Tense, close	The <i>Jaws</i> theme
2	Major 2nd	2	D	A plain step	"Frère Jacques"
3	Minor 3rd	b3	E $\flat$	Dark, soft	"Greensleeves"
4	Major 3rd	3	E	Bright	"When the Saints Go Marching In"
5	Perfect 4th	4	F	Open, strong	"Here Comes the Bride", "Amazing Grace"
6	Tritone	b5 (#4)	G $\flat$ / F $\sharp$	Unstable	"Maria" (<i>West Side Story</i>)
7	Perfect 5th	5	G	Hollow, stable	"Twinkle, Twinkle, Little Star"
8	Minor 6th	b6	A $\flat$	Bittersweet	"The Entertainer" (the first leap up)
9	Major 6th	6	A	Warm	"My Bonnie Lies over the Ocean"
10	Minor 7th	b7	B $\flat$	Bluesy, unresolved	"Somewhere" (<i>West Side Story</i>)
11	Major 7th	7	B	Sharp, wants to rise	An octave that falls a half step short
12	Octave	8 (R)	C	The same note, higher	"Somewhere Over the Rainbow"

TRACK 8 Every interval above C

The image shows two systems of musical notation and fretboard diagrams. The first system is in 4/4 time, starting on C4 (8th fret, 5th string). It shows intervals from C to D♭ (minor 2nd) to C to C5 (octave). The second system starts on C5 (3rd fret, 5th string) and shows intervals from C to D♭ (minor 2nd) to C to C6 (octave). Fretboard diagrams below each staff show fingerings: 3, 4, 3, 0 for the first system and 3, 5, 3, 1, 2, 3, 4, 5 for the second.

C followed by each note above it, from the minor 2nd (D♭) to the octave. The one-flat key signature makes the notation spell the notes as flats, matching the interval names. Sing the second note of each pair before you play it.

UNDER THE HOOD · Why C to E♭ is a 3rd but C to D♯ is not

The number of an interval counts letter names, including both ends. C to E covers C D E, three letters, so it is a 3rd. C to E♭ is also a 3rd, a smaller one: a minor 3rd. C to D♯ covers only C D, so it is a 2nd (an augmented 2nd), even though D♯ and E♭ are the same fret. The sound is the same; the name follows the spelling. This matters when you spell scales and chords, and Chapter 5 and Chapter 10 rely on it.

Intervals larger than an octave are called **compound intervals**. The ones you will meet in chord names are the 9th (an octave plus a 2nd, 14 half steps), the 11th (an octave plus a 4th, 17 half steps) and the 13th (an octave plus a 6th, 21 half steps). Chapter 11 uses them.

Intervals as shapes on the neck

Because most adjacent strings are tuned a perfect 4th apart, every interval has a fixed shape across the strings. The map below takes C on the 5th string (3rd fret) as the root and labels every note in reach by its interval from C. The same shape works from any root on the 5th string: move the whole map up or down the neck. The eight shapes after the map are the ones you will use most, and they work the same from a root on the 6th string.

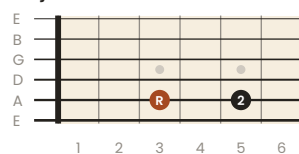
The interval map around a root on the 5th string

The diagram shows a fretboard with strings E, B, G, D, A, E. It displays six positions (1-6) for a root on the 5th string. Each position shows fret numbers for intervals: 1 (root), 2 (major 2nd), 3 (major 3rd), 4 (perfect 4th), 5 (perfect 5th), 6 (major 6th), 7 (major 7th), and 8 (octave). The root is always on the 5th string. The diagram uses color coding: root is black, 3rds and 5ths are teal, and 2nds, 4ths, 6ths, and 7ths are orange.

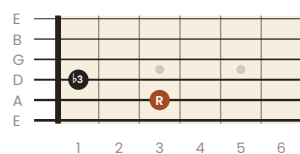
Root in rust; 3rds and the 5th, the notes that make chords, in teal. Every interval up to the octave fits under one hand.

Eight interval shapes from a root on the 5th string

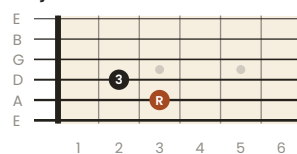
Major 2nd



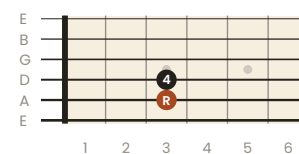
Minor 3rd



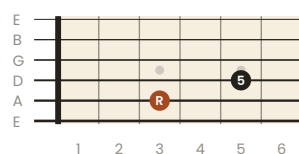
Major 3rd



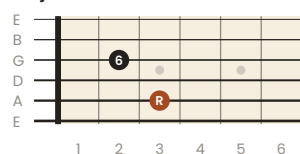
Perfect 4th



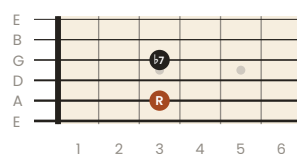
Perfect 5th



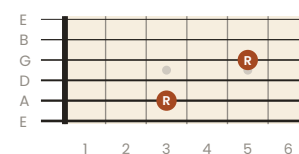
Major 6th



Minor 7th



Octave

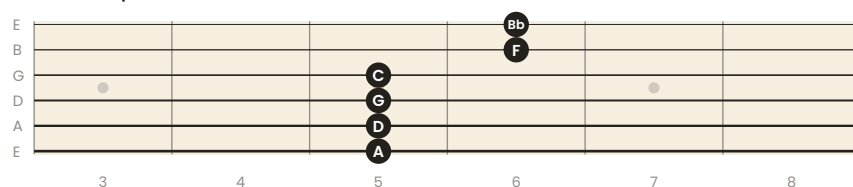


Same string: major 2nd 2 frets up. Next string: minor 3rd 2 frets back, major 3rd 1 back, perfect 4th same fret, perfect 5th 2 up. Two strings over: major 6th 1 back, minor 7th same fret, octave 2 up.

The B-string shift

There is one exception in the tuning. The G and B strings are a major 3rd (4 frets) apart, not a perfect 4th. Any interval shape that crosses from the 3rd string to the 2nd string must move **one fret higher** on the 2nd string to give the same interval.

A stack of perfect 4ths



Each note is a perfect 4th above the one below it: A, D, G, C, F, B $\flat$. On the lower strings the 4th is at the same fret; from the 3rd string to the 2nd it is one fret higher, and the shape stays shifted from there on.

SHORTCUT · Octaves on the neck

From the 6th or 5th string: skip a string and go up 2 frets. From the 4th or 3rd string: skip a string and go up 3 frets (the B-string shift). Chapter 4 uses these shapes to find every note on the neck.

Rhythm and time

Pitch is half of music. The other half is time. Music sits on a steady **beat** (the pulse you tap your foot to). Beats are grouped into **bars** (also called measures), and the **time signature** at the start of a piece says how.

- The **top number** is the number of beats in a bar.
- The **bottom number** is the note value that gets one beat: 4 means a quarter note, 8 an eighth note.

The speed of the beat is the **tempo**, in beats per minute (bpm); 60 bpm is one beat a second.

Note values

Note value	Beats in 4/4	Count it as
Whole note	4	1 – 2 – 3 – 4 (one note held)
Half note	2	1 – 2, 3 – 4
Quarter note	1	1, 2, 3, 4
Eighth note	$\frac{1}{2}$	1 & 2 & 3 & 4 &
Sixteenth note	$\frac{1}{4}$	1 e & a 2 e & a ...
Eighth-note triplet	$\frac{1}{3}$	1 trip let 2 trip let ...
Dotted note	value $\times 1\frac{1}{2}$	A dotted half note lasts 3 beats; a dotted quarter $1\frac{1}{2}$

Each value has a matching **rest** of the same length: a silence that still counts.

TRACK 9 One note in every common value

4

On the open A string, a whole note, half notes, quarters, eighths, sixteenths and eighth-note triplets. The beat stays the same; only the subdivision changes.

TRACK 10 Subdivisions of a 4/4 bar

Whole
Half
Quarter
Eighth
Sixteenth

1 e & a 2 e & a 3 e & a 4 e & a

Each row divides the bar into twice as many notes as the row above.

Time signatures

Time signature	Feel	Where you hear it
4/4	Four even beats; the default	Most pop, rock, folk, blues
3/4	Three beats; waltz feel	Waltzes, minuets, many folk songs
2/4	Two beats; march feel	Marches, polkas, some Latin styles
6/8	Two main beats, each divided into three eighths	Jigs, ballads, many Irish tunes
12/8	Four main beats, each divided into three	Slow blues, gospel, soul ballads

In 3/4 and 4/4 each beat divides naturally into two. In 6/8 and 12/8 each beat divides into three, which gives a rolling, lilting feel. Count 6/8 as "1 2 3 4 5 6" with two strong pulses per bar.

TRACK 11 Waltz in 3/4: accent on beat 1

Bass
Chord

1 e & a 2 e & a 3 e & a

Three beats per bar. A bass note on beat 1, the chord on beats 2 and 3, the classic waltz accompaniment.

TRY THIS • Count, then strum

Set a metronome to 60. Count aloud "1 & 2 & 3 & 4 &" and tap your foot on the numbers. Strum one chord with down-strums on the numbers only, then down on the numbers and up on the &s. Your strumming hand should move down and up like a pendulum on every beat, whether it touches the strings or not. This one habit keeps your rhythm steady. Chapter 37 builds a full daily routine on it.

Chapter 3 review · use it, train it, test it

Using it in real music

Working out a melody by ear. Hear that a tune jumps up a perfect 4th, as "Amazing Grace" does on its first two notes? That is 5 frets up on one string, or the same fret on the next string. Interval knowledge turns what you hear into finger positions.

Building chords from any root. A major chord is a root, a major 3rd and a perfect 5th. With the interval shapes in this chapter you can find those three notes from any root on the neck, without memorising note names first.

Keeping time with others. Counting "1 & 2 &" and knowing the time signature lets you come in at the right place, hold notes for the right length, and play along with a drummer or a recording.

How to train it

Drill	How	Time	Goal
Chromatic recital	Say the chromatic scale up with sharps and down with flats while playing it on one string	2 min	No hesitation at B–C and E–F
One-string intervals	On the 2nd string, play a root and then each interval up to the octave, naming each one aloud	3 min	Instant link between name, fret distance and sound
Interval shapes	Choose a root on the 5th string; play the 3rd, 5th and octave using the cross-string shapes	3 min	Find each shape without counting frets
Interval songs	Play a random root, then sing an interval above it using the song list; check on the guitar	4 min	Recognise 2nds, 3rds, 4ths, 5ths and octaves by ear
Clap and count	Metronome at 70: clap quarters, eighths, triplets and sixteenths for 4 bars each, counting aloud	4 min	Switch subdivisions without losing the beat

Check your understanding

Write your answers, then check them in Appendix J.

1. How many half steps make a whole step, and how many frets is that?

2. Between which two pairs of natural notes is there no sharp or flat?

3. What is the enharmonic name for G#?

4. Name the interval from C up to G.

5. Name the interval from A up to C.

6. In 3/4 time, how many quarter-note beats are in each bar?

7. How many beats does a dotted half note last in 4/4?

8. Which two adjacent open strings are a major 3rd apart rather than a perfect 4th?

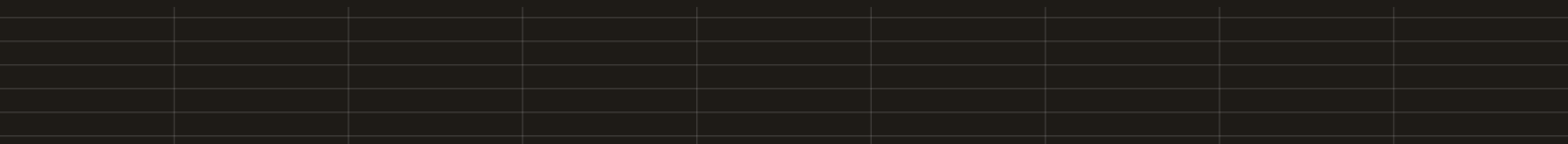
MY NOTES

PART TWO

The Fretboard

Find any note anywhere on the neck, using landmarks, octave shapes and the logic of the tuning.

4 Mapping the Fretboard



CHAPTER 4

Mapping the Fretboard



TRACKS

4

Knowing the notes on the neck turns shapes into music. You do not need to memorise every fret one by one; you need a few landmarks and the patterns that connect them.

The neck of a guitar with 19 or more frets holds well over a hundred note positions, but only twelve different notes. The same few patterns repeat everywhere, so you learn the neck through landmarks and shapes rather than by memorising positions one at a time. This chapter builds the map in five steps, then gives a four-week plan to make it automatic.

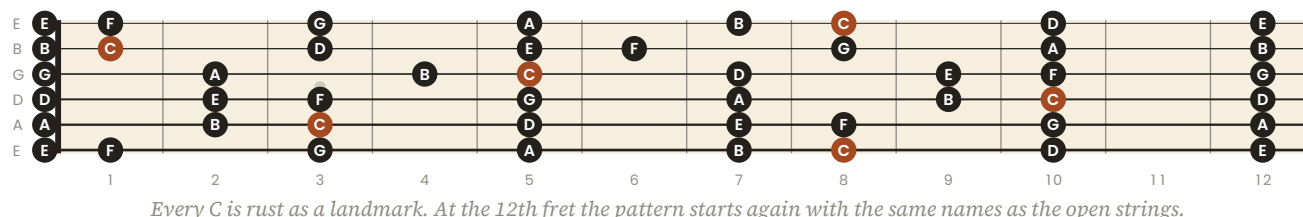
Step 1: the open strings

From the thickest (6th) string to the thinnest (1st): **E A D G B E**. A sentence such as "Eddie Ate Dynamite, Good Bye Eddie" helps at first. The 6th and 1st strings are both E, two octaves apart, so everything you learn on the 6th string also works on the 1st.

Step 2: the natural notes

Every fret raises the pitch by a half step. Count up from each open string, remembering that there is no fret between B and C or between E and F, and you can work out any note. Here are all the natural notes (no sharps or flats) up to the 12th fret.

Natural notes on the fretboard, frets 0–12



From the 12th fret upwards the neck repeats: the 13th fret has the same notes as the 1st, the 15th the same as the 3rd, and so on. Anything you learn in frets 0–12 you also know in frets 12–24.

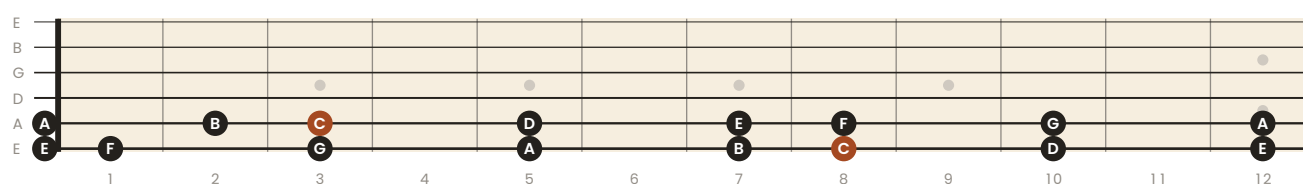
Step 3: the 6th and 5th strings first

The 6th and 5th strings matter most. Almost every movable chord shape, power chord and scale box has its root on one of them, so these two strings let you place anything on the neck.

Note	E	F	G	A	B	C	D	E
6th string fret	0	1	3	5	7	8	10	12

Note	A	B	C	D	E	F	G	A
5th string fret	0	2	3	5	7	8	10	12

The natural notes on the 6th and 5th strings



Learn these sixteen positions first. Say each note name aloud as you play it.

SHORTCUT · Use the fret markers

The fret markers at 3, 5, 7, 9 and 12 are landmarks. On the 6th string they are G, A, B, C $\sharp$ and E. On the 5th string they are C, D, E, F $\sharp$ and A. Learn the notes at the markers and every other fret is one step away.

String	Fret 3	Fret 5	Fret 7	Fret 9	Fret 12
6th (E)	G	A	B	C $\sharp$	E
5th (A)	C	D	E	F $\sharp$	A
4th (D)	F	G	A	B	D
3rd (G)	B $\flat$	C	D	E	G
2nd (B)	D	E	F $\sharp$	G $\sharp$	B
1st (E)	G	A	B	C $\sharp$	E

TRACK 12 Natural notes on the 6th string

0 1 3 5 7 8 10 12 12 10 8 7 5 3 1 0

E F G A B C D E, up the string and back. Say the names aloud as you play.

Step 4: octave shapes

Every note appears several times on the neck. An **octave shape** takes you from a note you know to the same note on another string. Once you know a note on the 6th or 5th string, octave shapes find it everywhere else.

Every C up to the 17th fret

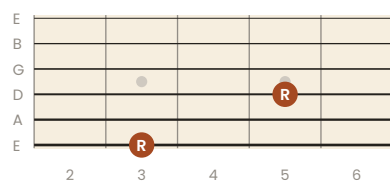
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Nine Cs between the nut and the 17th fret. Each one connects to its neighbours with one of the octave shapes below.

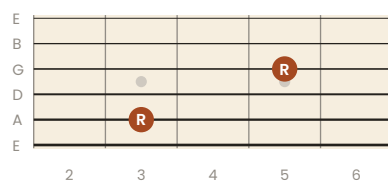
From	To	Shape	Example
6th string	4th string	Skip a string, up 2 frets	G at 6/3 → G at 4/5
5th string	3rd string	Skip a string, up 2 frets	C at 5/3 → C at 3/5
4th string	2nd string	Skip a string, up 3 frets (B-string shift)	F at 4/3 → F at 2/6
3rd string	1st string	Skip a string, up 3 frets (B-string shift)	A at 3/2 → A at 1/5
6th string	3rd string	Skip two strings, down 3 frets	C at 6/8 → C at 3/5
5th string	2nd string	Skip two strings, down 2 frets	C at 5/3 → C at 2/1
4th string	1st string	Skip two strings, down 2 frets	C at 4/10 → C at 1/8
6th string	1st string	Same fret, two octaves higher	G at 6/3 → G at 1/3
Any string	Same string	12 frets higher	A at 6/5 → A at 6/17

The octave shapes

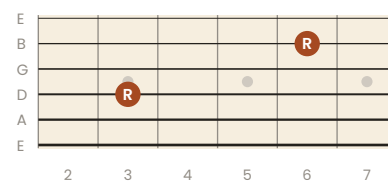
6th to 4th: skip a string, up 2



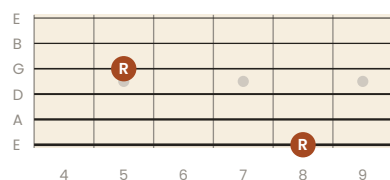
5th to 3rd: skip a string, up 2



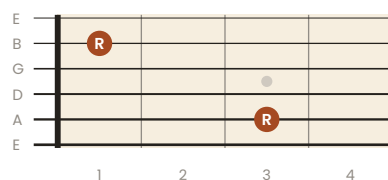
4th to 2nd: skip a string, up 3



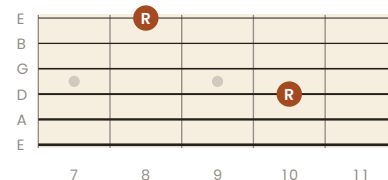
6th to 3rd: skip two strings, down 3



5th to 2nd: skip two strings, down 2



4th to 1st: skip two strings, down 2



The shapes that cross from the 3rd string to the 2nd string are one fret longer than the others. This is the B-string shift from Chapter 3: the G and B strings are a major 3rd apart instead of a perfect 4th.

TRACK 13 One note everywhere: every C from low to high

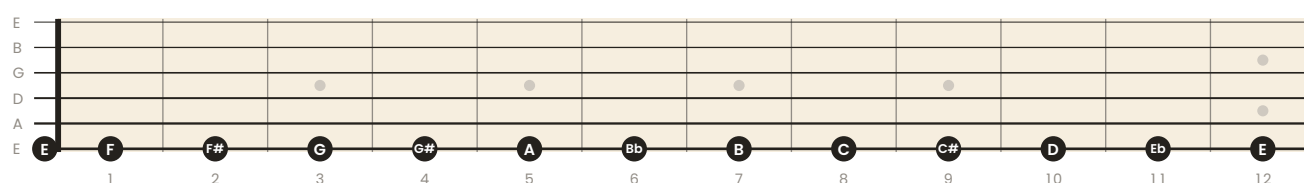


Three octaves of C. The low C on two strings, middle C on three, high C on three. Notes of the same pitch should sound identical.

Step 5: sharps and flats

Once you know the natural notes, sharps and flats come for free. A sharp is one fret higher than the natural note, a flat is one fret lower. F# is the fret above F; Bb is the fret below B. A# and Bb are the same fret: which name you use depends on the key (Chapter 5).

Sharps and flats around the naturals on the 6th string



Gold dots are the five notes between the naturals. Each has two names (F# = Gb, G# = Ab, A# = Bb, C# = Db, D# = Eb).

TRY THIS · The C major scale on one string

Play the C major scale on the 2nd string only. Start at the 1st fret (C) and move up in the pattern whole, whole, half, whole, whole, whole, half. You finish on C at the 13th fret. Then do the same on the 5th string, starting at the 3rd fret.

TRACK 14 C major on the 2nd string

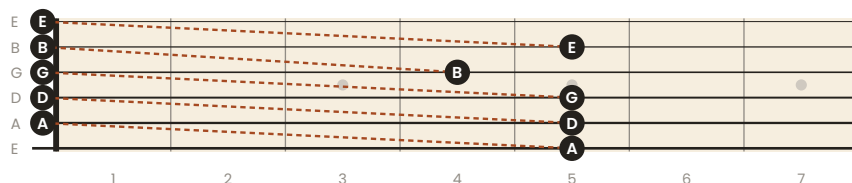
C D E F G A B C. Two-fret jumps for whole steps, one-fret moves for the half steps E-F and B-C.

Tuning by ear

Because adjacent strings are a perfect 4th apart (a major 3rd between the G and B strings), the 5th fret of each string gives the same note as the next open string. The exception is the G string, where the 4th fret matches the open B. This is the standard way to tune by ear, and it is also a good daily reminder of the B-string shift.

Press	Should match
6th string, 5th fret (A)	Open 5th string (A)
5th string, 5th fret (D)	Open 4th string (D)
4th string, 5th fret (G)	Open 3rd string (G)
3rd string, 4th fret (B)	Open 2nd string (B)
2nd string, 5th fret (E)	Open 1st string (E)

The 5th-fret tuning positions



Each fretted note matches the next open string. Only the 3rd string uses the 4th fret.

How to do it:

1. Tune the 6th string to E with a tuner or a piano.
2. Play the 6th string at the 5th fret and let it ring, then play the open 5th string.
3. If the two notes "beat" (a slow wobble in the sound), they are not quite in tune. Turn the 5th string's peg until the wobble slows and disappears.
4. Repeat up the strings, using the 4th fret on the 3rd string.
5. Check the result with chords: an open E major and an open C major should both sound clean.

TRACK 15 Tuning reference: fretted note, then the next open string

Five pairs; in each pair the two notes should sound identical.

CAREFUL · Harmonics and the B string

Some players tune with harmonics (5th-fret harmonic of one string against the 7th-fret harmonic of the next). Each pair comes out as a pure 4th, about 2 cents narrower than an equal-tempered 4th, so E–A, A–D and D–G are each within a cent or two, although the error adds up across the strings. The method does not work between the G and B strings: the two harmonics are different notes. Matching the 4th-fret harmonic of the G string to the open B gives a pure major 3rd, which leaves the B about 14 cents flat. Use the fretted 4th-fret method for that pair, or a tuner.

NYLON · STEEL

Nylon: new nylon strings stretch for days, and you will retune many times in the first week. Tune slightly above pitch and let them settle, then tune up to the note again.

Steel and electric: new strings settle within an hour or two of playing. Stretch each new string gently along its length after fitting it to speed this up.

A four-week plan

Ten minutes a day is enough. Do each week's work before moving on.

When	Work	Test at the end of the week
Week 1	Natural notes on the 6th and 5th strings. Say each note aloud, up and down. Then call out random notes and find them.	Any natural note on either string in under 2 seconds
Week 2	Add the 4th and 3rd strings, using octave shapes from the 6th and 5th.	Find any natural note on strings 6–3
Week 3	Add the 2nd and 1st strings. The 1st string is the same as the 6th.	Name any natural note on any string, frets 0–12
Week 4	One note everywhere: choose a note each day and find it on all six strings. Add sharps and flats.	One note on every string in under 10 seconds
Afterwards	Whenever you play a chord or scale, name its root aloud.	Knowing the note under each finger becomes automatic

Chapter 4 review · use it, train it, test it

Using it in real music

Placing barre chords. Asked for B $\flat$ or F $\sharp$ m? Find the root on the 6th or 5th string and put the shape there. Chapter 11 and Chapter 12 rely on this.

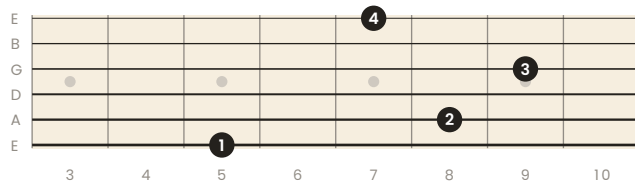
Talking to other musicians. When a bandmate says "start on the D at the 10th fret", you know where to go. When a teacher says "play the G on the 4th string", you know it is at the 5th fret.

Choosing where to play a melody. Every note exists in several places. Knowing them all lets you choose the position that keeps a melody smooth, which matters in classical fingerings and fingerstyle arrangements (Chapter 20).

How to train it

Drill	How	Time	Goal
Note finder	Name a random natural note; find it on the 6th string, then the 5th	2 min	Any note within 2 seconds
One note everywhere	Choose today's note; play it on all six strings up to the 12th fret, low to high	2 min	All positions in under 15 seconds
Octave jumps	Play a note on the 6th string, then its octave on the 4th; repeat for 10 random notes	2 min	Octave shape without looking
Name as you play	Play any scale you know slowly, saying each note name aloud	3 min	No hesitation on any string

Quiz question 1



Check your understanding

Write your answers, then check them in Appendix J.

1. Name the four numbered notes in the diagram above.

2. What note is at the 8th fret of the 6th string?

3. What note is at the 7th fret of the 5th string?

4. At which fret is G on the 5th string?

5. Where is F# on the 6th string?

6. You have C at the 3rd fret of the 5th string. Where is its octave on the 3rd string?

7. Which fret of the G string matches the open B string when tuning?

8. What note is at the 10th fret of the 1st string?

MY NOTES

PART THREE

Scales

How the major scale is built, and how minor, pentatonic, blues and modal sounds all come from it.

- 5 The Major Scale and Keys
- 6 Minor Scales
- 7 Pentatonic and Blues Scales
- 8 Playing Horizontally
- 9 Modes



CHAPTER 5

The Major Scale and Keys



TRACKS

5

The major scale is the reference point for nearly all Western music theory. Chords, keys, modes and chord numbers are all defined against it.

The major scale is seven notes built from a fixed pattern of whole and half steps. Every major key, every chord number (I, IV, V) and every mode in Chapter 9 is described by comparing it with this scale. Learn how it is built, how keys and key signatures follow from it, and how it lies on the neck.

The formula

Start on any note and follow this pattern of whole steps (W, two frets) and half steps (H, one fret):

W – W – H – W – W – W – H

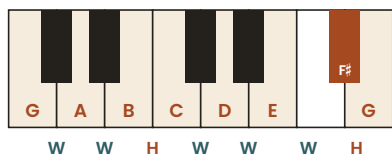
The half steps fall between the 3rd and 4th degrees and between the 7th degree and the octave. From C the pattern lands only on natural notes, the white keys of the keyboard in Chapter 3, which is why C major has no sharps or flats.

From any other root, the pattern needs sharps or flats to keep the half steps in the right places. Each letter name is used exactly once, in order; that rule decides whether a note is written as a sharp or a flat.

Degree	1	2	3	4	5	6	7	8 (1)
C major	C	D	E	F	G	A	B	C
G major	G	A	B	C	D	E	F#	G
D major	D	E	F#	G	A	B	C#	D
F major	F	G	A	Bb	C	D	E	F
Eb major	Eb	F	G	Ab	Bb	C	D	Eb
Step to next	W	W	H	W	W	W	H	

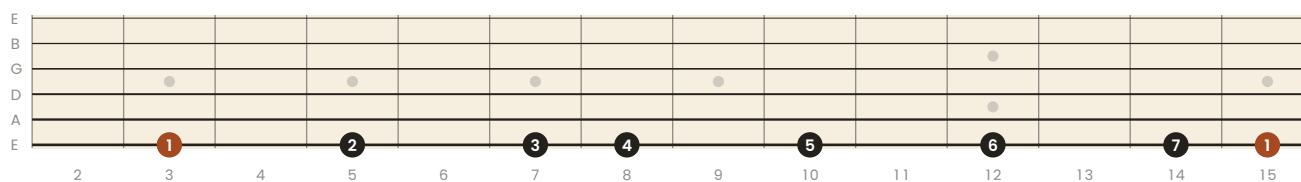
G major needs F# because the step from E to the 7th degree must be a whole step and from the 7th to G a half step: F would be wrong on both counts. F major needs Bb because the step from A to the 4th degree must be a half step. In D major the 7th degree is C#, not Db, because the letter C has not yet been used.

G major: one black key



The same W W H W W W H from G. The 7th degree falls on a black key, F#.

The major scale on one string



G major on the 6th string. Two frets for each whole step, one fret for the half steps 3–4 and 7–8.

Names of the degrees

Degree	Name	Role
1	Tonic	Home: the note the music resolves to
2	Supertonic	A step above home; often a passing note
3	Mediant	Decides whether the sound is major or minor
4	Subdominant	Leads away from home
5	Dominant	The strongest pull back to the tonic
6	Submediant	The root of the relative minor key
7	Leading note	A half step below the tonic; it leads up to it

The leading note is called the leading tone in American usage. In a minor key the note a whole step below the tonic is called the subtonic.

Keys and key signatures

A key is the set of notes of one scale with one note acting as home. A song "in G" mostly uses the seven notes of G major and sounds finished on G. The sharps or flats that a key needs are written once at the start of each line of music: the **key signature**. Every F in a piece with one sharp is played as F $\sharp$, unless an accidental cancels it.

TRACK 16 G major, one octave, with its key signature

The sharp on the top line of the staff is F $\sharp$. The scale needs no other accidentals.

Key	Sharps	Key	Flats
C	none	F	B $\flat$
G	F $\sharp$	B $\flat$	B $\flat$ E $\flat$
D	F $\sharp$ C $\sharp$	E $\flat$	B $\flat$ E $\flat$ A $\flat$
A	F $\sharp$ C $\sharp$ G $\sharp$	A $\flat$	B $\flat$ E $\flat$ A $\flat$ D $\flat$
E	F $\sharp$ C $\sharp$ G $\sharp$ D $\sharp$	D $\flat$	B $\flat$ E $\flat$ A $\flat$ D $\flat$ G $\flat$
B	F $\sharp$ C $\sharp$ G $\sharp$ D $\sharp$ A $\sharp$	G $\flat$	B $\flat$ E $\flat$ A $\flat$ D $\flat$ G $\flat$ C $\flat$
F $\sharp$	F $\sharp$ C $\sharp$ G $\sharp$ D $\sharp$ A $\sharp$ E $\sharp$	C $\flat$	B $\flat$ E $\flat$ A $\flat$ D $\flat$ G $\flat$ C $\flat$ F $\flat$
C $\sharp$	all seven		

F $\sharp$ major (six sharps) and G $\flat$ major (six flats) sound identical, as do B major and C $\flat$ major, and C $\sharp$ major and D $\flat$ major. In practice you will meet F $\sharp$ /G $\flat$ in both spellings, B far more than C $\flat$, and D $\flat$ far more than C $\sharp$.

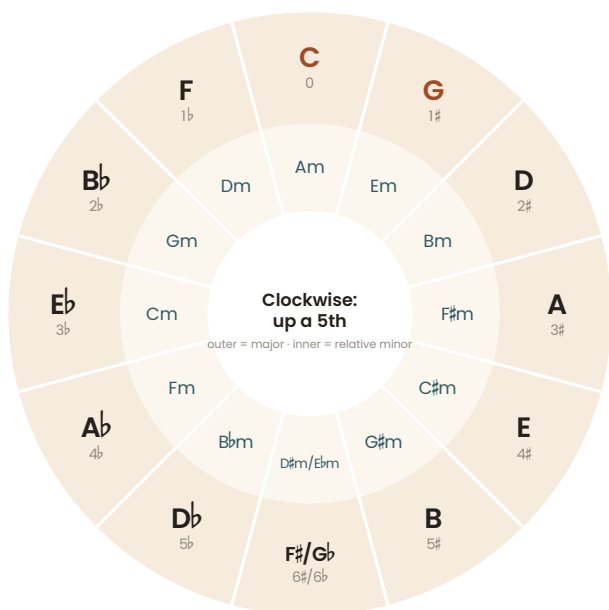
SHORTCUT · Reading a key signature

Order of sharps: F C G D A E B ("Father Charles Goes Down And Ends Battle"). **Order of flats:** the same in reverse, B E A D G C F.

Sharp keys: the key is a half step above the last sharp. Last sharp C $\sharp$ $\rightarrow$ D major. **Flat keys:** the key is the second-to-last flat. B $\flat$ E $\flat$ A $\flat$ $\rightarrow$ E $\flat$ major. The one flat key to memorise is F major (one flat, B $\flat$).

The circle of fifths

Arrange the twelve keys so that each one is a perfect 5th above the last, and you get the circle of fifths. Clockwise from C, each key adds one sharp. Anticlockwise, each key is a perfect 4th higher (a 5th lower) and adds one flat. The inner ring shows each key's relative minor, which shares its key signature (Chapter 6).



Neighbouring keys on the circle share six of their seven notes, so they sound closely related. C major and G major differ only by F versus F#. The three most important chords in a key, I, IV and V, always sit side by side: for C they are F, C and G; for G they are C, G and D. Chapter 13 builds on this, and Chapter 33 uses the circle to move between keys.

▶ TRACK 17 A short melody in D major

Musical notation for a short melody in D major (4/4 time). The melody is written on a treble clef staff with a key signature of two sharps (F# and C#). The melody consists of two phrases, each four bars long.

Phrase 1 (Bars 1-4):

- Bar 1: D4 (quarter), E4 (quarter), F#4 (quarter), G4 (quarter)
- Bar 2: A4 (quarter), B4 (quarter), C#5 (quarter), B4 (quarter)
- Bar 3: A4 (quarter), G4 (quarter), F#4 (quarter), E4 (quarter)
- Bar 4: D4 (half), D4 (half)

Phrase 2 (Bars 5-8):

- Bar 5: D4 (quarter), E4 (quarter), F#4 (quarter), G4 (quarter)
- Bar 6: A4 (quarter), B4 (quarter), C#5 (quarter), B4 (quarter)
- Bar 7: A4 (quarter), G4 (quarter), F#4 (quarter), E4 (quarter)
- Bar 8: D4 (half), D4 (half)

Fingerings (T = Thumb, A = Index, B = Middle) are indicated below the staff:

- Bar 1: T (0), A (0), B (4)
- Bar 2: T (2), A (4), B (2)
- Bar 3: T (0), A (4), B (2)
- Bar 4: T (0), A (4), B (2)
- Bar 5: T (3), A (2), B (0)
- Bar 6: T (0), A (4), B (2)
- Bar 7: T (4), A (0), B (2)
- Bar 8: T (3), A (4), B (2)

An original eight-bar melody. Every F and C is sharp because of the key signature; no accidentals are needed.

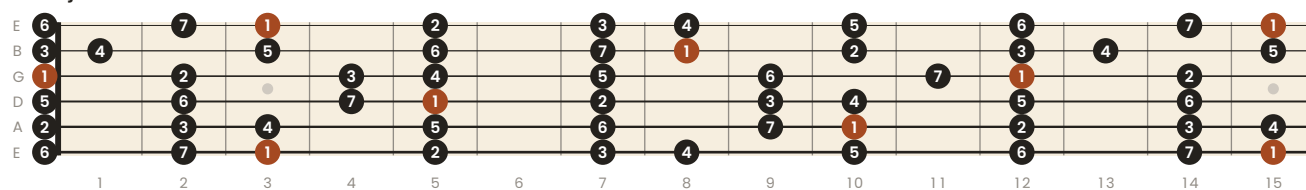
NYLON · STEEL

Guitar-friendly keys are the ones that use the open strings: E, A, D, G and C major and their relative minors E minor, A minor, B minor and F# minor. Much of the nylon-string repertoire is in E minor, A minor, D major and G major for this reason. Steel-string and electric players often use a capo to keep these open-string shapes in other keys.

The major scale across the neck

Here is G major over the whole neck, with degrees. Do not try to memorise it as one picture. The shapes below break it into pieces that fit under the hand.

G major, frets 0–15



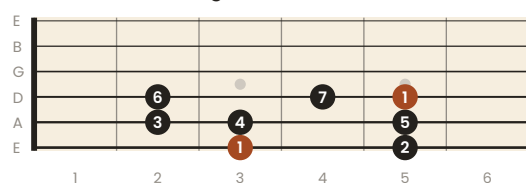
Every G is rust. The same layout repeats from the 12th fret.

One-octave shapes

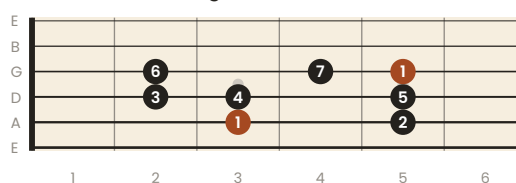
The quickest way to start is a one-octave shape with the root on the 6th or 5th string. Move the shape so the root lands on any note and you have that note's major scale, the same idea as moving a barre chord. Use one finger per fret.

One-octave major scale shapes

Root on the 6th string (G)



Root on the 5th string (C)



Both shapes use the same fingering, 2 4 1 2 4 1 3 4, with the 2nd finger on the root: the same shape, one string apart.

TRY THIS · Move the shapes

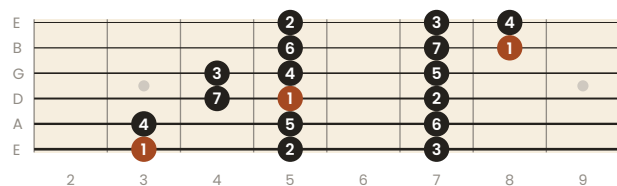
Play the 6th-string shape from G (3rd fret), then from A (5th fret), then from F (1st fret). Play the 5th-string shape from C (3rd fret), then from D (5th fret) and B $\flat$ (1st fret). Name the key each time.

The seven three-notes-per-string patterns

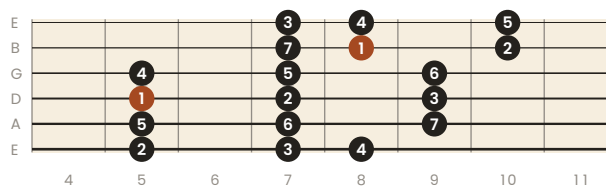
Playing exactly three notes on every string gives seven patterns that cover the neck and link into one another. They are popular for speed and for smooth alternate picking because every string has the same number of notes. Pattern 1 starts on the root on the 6th string, pattern 2 on the 2nd degree, and so on up the neck. They are shown in G major with degrees, so they work in every key.

The seven three-notes-per-string patterns of G major

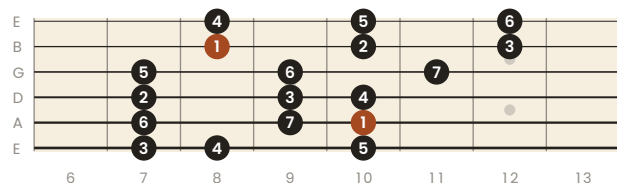
Pattern 1: from the root



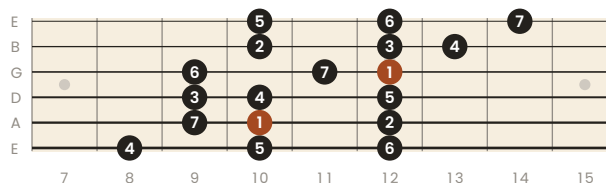
Pattern 2: from the 2nd



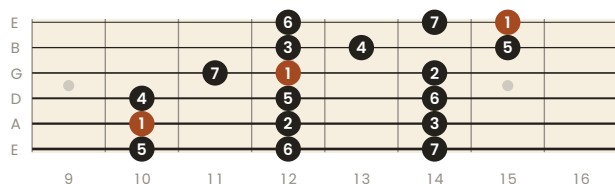
Pattern 3: from the 3rd



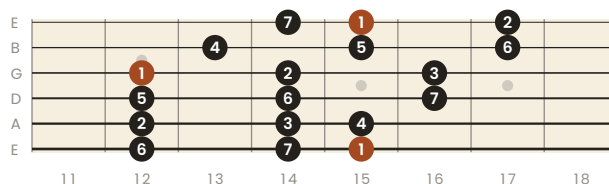
Pattern 4: from the 4th



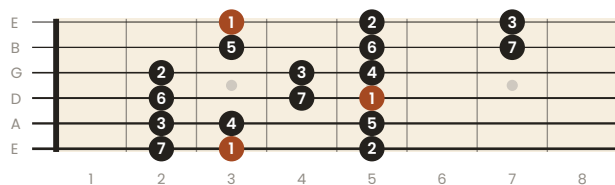
Pattern 5: from the 5th



Pattern 6: from the 6th



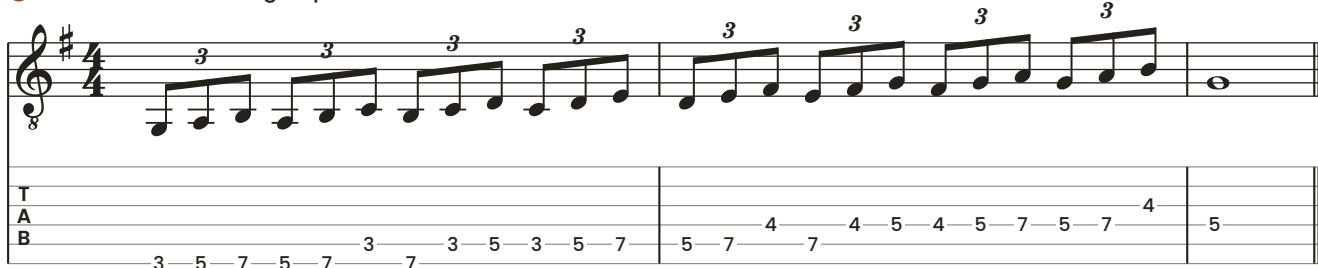
Pattern 7: from the 7th



Patterns 1–6 climb the neck from the 3rd fret. Pattern 7 is shown at the 2nd fret, just below pattern 1; it also lies at the 14th fret, above pattern 6.

Each pattern begins where the previous one has its second note on each string, so neighbouring patterns overlap by two notes per string. After pattern 7 comes pattern 1 again, twelve frets higher than before. On each string use 1-3-4 when the steps are whole then half, 1-2-4 when they are half then whole, and 1-2-4 with a stretch where both are whole steps (a five-fret span).

TRACK 18 Pattern 1 in groups of three



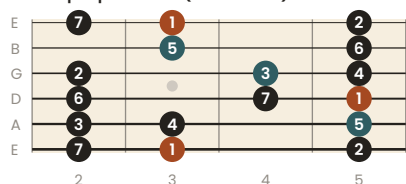
1-2-3, 2-3-4, 3-4-5 and so on, through the first octave of pattern 1. Sequences like this turn a shape into musical lines; try the same with every pattern.

CAGED position views

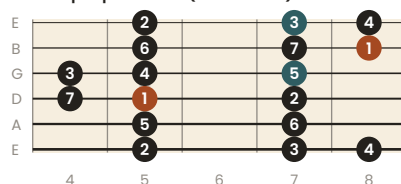
A second way to divide the neck uses five positions, each built around one of the five open chord shapes C, A, G, E and D moved up the neck (Chapter 12 covers the system in full). Each position spans four or five frets with the hand in one place, which suits melodic playing and links scales directly to chords.

G major in five positions, with the G chord shape inside each

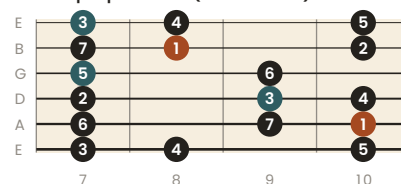
E-shape position (frets 2–5)



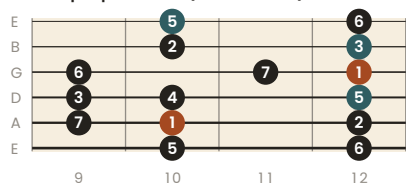
D-shape position (frets 4–8)



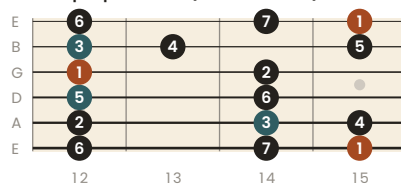
C-shape position (frets 7–10)



A-shape position (frets 9–12)



G-shape position (frets 12–15)



Teal and rust dots are the notes of the G chord shape (root, 3rd, 5th); dark dots are the rest of the scale. The E-shape position returns at frets 14–17.

TIP · Which system to learn first

Learn the one-octave shapes first, then the CAGED positions, which tie scales to chords you already know. Add the three-notes-per-string patterns when you want fast, even runs. All three describe the same notes; they are different ways of cutting up the same map.

Practice loops

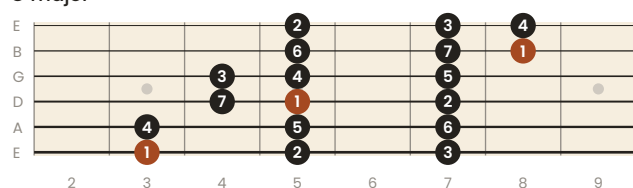
Two loops for putting the shapes to work. The first is a chord progression in G; the second holds one chord so that you can hear what each degree sounds like against it.

TRACK 19 G major pop loop: I – vi – IV – V · LOOP · pop · 92 bpm

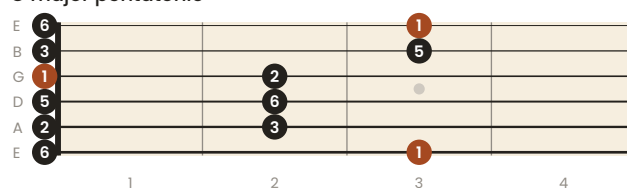
G I	Em vi	C IV	D V
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IMPROVISE WITH

G major



G major pentatonic



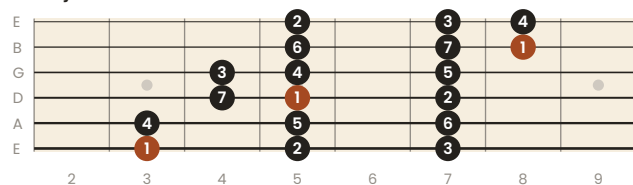
Use any G major pattern from this chapter. Land on the 3rd of each chord on beat 1 (B over G, G over Em, E over C, F# over D), then end a phrase on G over the G chord and hear it close.

TRACK 20 G drone: hear the seven degrees · LOOP · modal drone · 72 bpm

G I	G I
--------	--------

IMPROVISE WITH

G major



Hold each degree of G major for a whole bar, from 1 up to 7. The 1, 3 and 5 sound at rest; the 2 and 6 add colour; the 4 wants to fall to 3 and the 7 wants to rise to 1.

Chapter 5 review · use it, train it, test it

Using it in real music

Finding the key of a song. Look at the key signature, or at the first and last chords. A song built on G, C, D and Em is almost certainly in G major, so the G major scale fits a melody or solo over all of it.

Writing and improvising melodies. Any melody built from the notes of the key sounds settled over the key's chords. Ending on the tonic makes a phrase sound finished; ending on the 2nd or the 7th leaves it open.

Using the circle. Neighbouring keys share six notes, which is why many songs move one step round the circle for a bridge or a key change, and why I, IV and V sit side by side.

Drill	How	Time	Goal
Spell a key	Pick a random root; spell its major scale aloud using W H W W W H, one letter per degree	2 min	Any major scale spelled in under 15 seconds
Pattern of the week	Play one three-notes-per-string pattern up and down with a metronome from 60 bpm, adding 5 bpm per clean run	5 min	Clean eighth notes at 100 bpm
Sequences	In one pattern, play groups of three (1-2-3, 2-3-4) and then 3rds (1-3, 2-4)	4 min	Smooth, musical lines from a shape
Circle walk	Go round the circle clockwise, playing each root on the 6th or 5th string and naming its key signature	2 min	Know the sharps or flats of every key

Write your answers, then check them in Appendix J.

CHAPTER 6

Minor Scales



TRACKS

6

Every major key has a relative minor that uses the same notes from a different starting point. From that one scale come the two variations, harmonic and melodic minor, that give minor keys their pull.

A minor scale has a minor 3rd ($\flat 3$) above its root, where a major scale has a major 3rd. That one note gives minor keys their darker colour. There are three common forms of the minor scale. All three share the first five notes, 1 2 $\flat 3$ 4 5, and differ only in the 6th and 7th degrees.

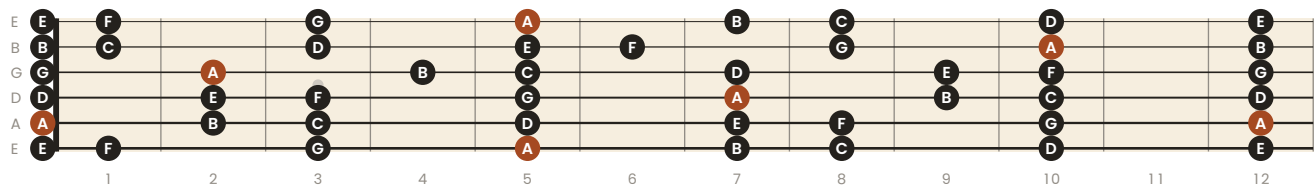
The relative minor

Play the C major scale but start and finish on A: A B C D E F G A. These are the same seven notes as C major, but with A as home the sound is quite different. This is A **natural minor**, the **relative minor** of C major. Every major key has a relative minor, which starts on the major scale's 6th degree and shares its key signature.

To find it on the guitar, go down three frets (a minor 3rd) from the major root. To go the other way, from a minor key to its **relative major**, go up three frets.

Major key	C	G	D	A	E	F	B $\flat$	E $\flat$
Relative minor	Am	Em	Bm	F $\sharp$ m	C $\sharp$ m	Dm	Gm	Cm
Key signature	none	1 $\sharp$	2 $\sharp$	3 $\sharp$	4 $\sharp$	1 $\flat$	2 $\flat$	3 $\flat$

C major and A minor share every note



The notes of C major, with A as the root. Play from any A to the next A and the scale is A natural minor; play from C to C and it is C major.

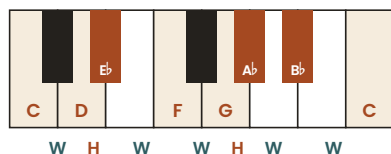
Relative and parallel

Two different relationships between major and minor are both useful:

- **Relative** keys share their notes and key signature but have different roots: C major and A minor.
- **Parallel** keys share their root but not their notes: C major and C minor.

C minor compared with C major has a lowered 3rd, 6th and 7th: C D E $\flat$ F G A $\flat$ B $\flat$. That is why C minor has three flats, the key signature of its own relative major, E $\flat$ major.

C natural minor

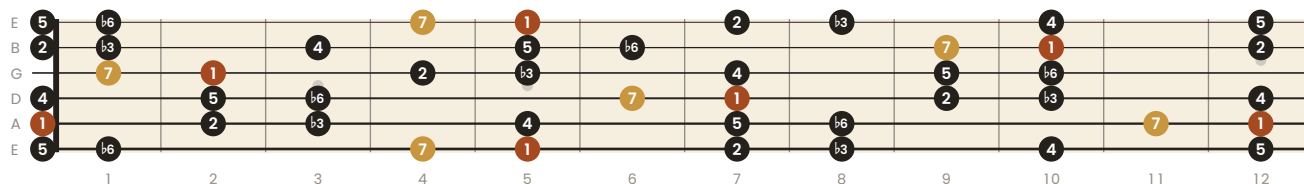


W H W W H W W. Compared with C major, E, A and B are each lowered a half step.

TIP · Which view to use

Use the **relative** view to find shapes: every A minor shape is a C major shape with a new home note. Use the **parallel** view to hear the sound: play C major, then C minor, over a C bass. Chapter 9 uses both views.

A harmonic minor, frets 0–12



The raised 7th (G#) in gold. Note the three-fret gap between the b6 and the 7th on each string.

▶ TRACK 23 A harmonic minor, one octave



The sharp in bar 2 is the raised 7th. Listen for the wide step from F to G#.

Melodic minor

Melodic minor raises both the 6th and the 7th, which removes the awkward augmented 2nd and gives a smooth line up to the tonic. Its upper half is identical to the major scale; only the b3 makes it minor.

In classical music the melodic minor is used ascending only. Descending, melodies return to the natural minor, because the raised notes are needed only when the line is moving up to the tonic. In jazz the ascending form is used in both directions as a scale in its own right, often called **jazz minor**.

▶ TRACK 24 A melodic minor, up in the melodic form, down in the natural form



F# and G# on the way up, F and G on the way down, as in classical usage.

CAREFUL · Spelling the raised notes

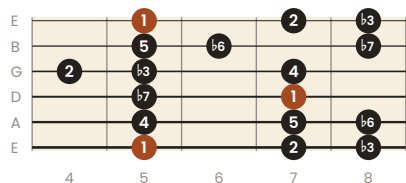
Raised degrees in minor keys are spelled with the same letter, sharpened or naturalised: in A minor the raised 7th is G#, never Ab, and in C minor the raised 6th and 7th are A natural and B natural, never B double-flat or Cb. Each letter appears once, as in every seven-note scale.

Shapes on the neck

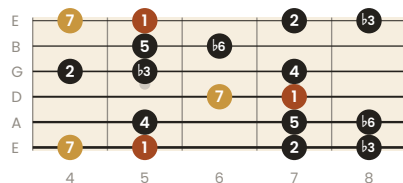
The quickest minor shapes put the root under the 1st finger on the 6th string. Here is A minor in 5th position (frets 4–8) in all three forms, with the notes that differ in gold.

A minor in 5th position: natural, harmonic, melodic

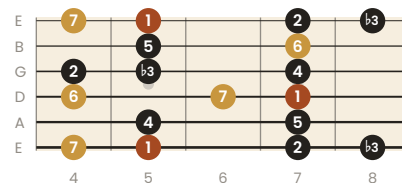
Natural minor



Harmonic minor



Melodic minor

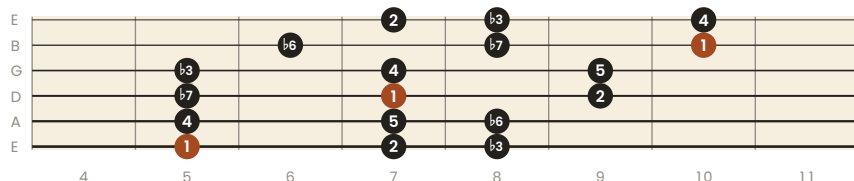


Same root, same hand position. Only the 6th and 7th degrees move.

HOW IT CONNECTS · Minor shapes are major shapes

A natural minor uses exactly the same notes as C major, so every C major shape from Chapter 5 is also an A minor shape. The natural-minor diagram above is the C major scale in 5th position, with A as home. Learn a shape once and use it twice.

The whole natural minor scale across the neck works the same way as the major scale: break it into one-octave shapes, three-notes-per-string patterns or CAGED positions. The three-notes-per-string pattern that starts on A on the 6th string is pattern 6 of C major.

A natural minor as a three-notes-per-string pattern

Root on the 6th string at the 5th fret. The same dots form pattern 6 of C major.

Minor in practice: the major V chord

In a minor key, songs and pieces mix the three forms. Melodies mostly use the natural minor, but the chord on the 5th degree is very often made major (E or E7 in A minor), borrowing the raised 7th from harmonic minor. When that chord appears, play G# over it, and return to G when the music moves away.

TRACK 25 A minor and E major: the raised 7th in the harmony

Am **E**

Am **E7** **Am**

The G# in the E and E7 bars comes from harmonic minor. Chapter 31 explains the chords of minor keys in full.

TRACK 26 Harmonic minor loop: i – iv – V7 · LOOP · waltz · 108 bpm

Am
i

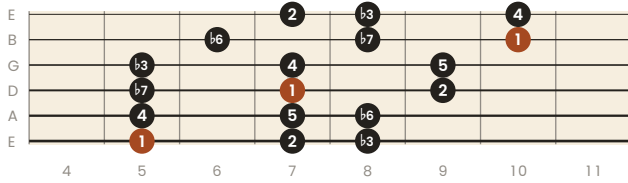
Dm
iv

E7
V7

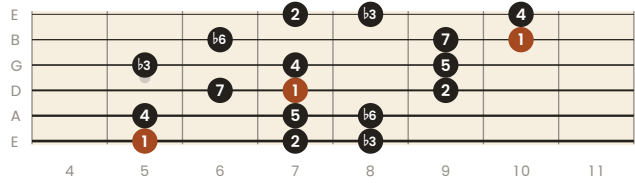
E7
V7

IMPROVISE WITH

A minor



A harmonic minor



Natural minor over Am and Dm, harmonic minor over E7. Put G# on beat 1 of each E7 bar and let it rise to A when Am returns.

NYLON · STEEL

Harmonic minor is the sound of much classical and flamenco guitar music. On nylon strings, play A harmonic minor slowly with rest strokes, then play an E major chord followed by A minor: the G# in the E chord is the harmonic minor at work. On steel and electric, the same scale over E and Am drives many metal and neoclassical riffs.

The harmonic and melodic minor scales also have seven modes each, which jazz players and flamenco guitarists use over particular chords. Those are covered in Chapter 31.

Chapter 6 review · use it, train it, test it

Using it in real music

Minor-key songs. Most minor songs use the natural minor for their melody. If a song in A minor contains an E or E7 chord, switch to A harmonic minor over that chord so that G# matches the chord.

Classical and Spanish music. Harmonic minor gives the dramatic colour of many classical and flamenco pieces. Melodic minor appears in melodies that climb to the tonic, and is common in Baroque music, including Bach's.

Reusing shapes. Every major shape is also a shape for its relative minor. If you know the G major patterns, you know E minor; if you know C major, you know A minor.

How to train it

Drill	How	Time	Goal
Natural against harmonic	In 5th position play A natural minor, then A harmonic minor; listen for the G#	3 min	Hear the difference with eyes closed
Drone practice	Play each minor scale slowly over a sustained open A string	4 min	Feel the root as home in all three forms
Relative pairs	Play C major then A minor in the same position; repeat for G/Em and F/Dm	3 min	Name any relative minor in under 3 seconds
Parallel switch	Play C major, then C natural minor, over a C bass note	3 min	Hear and name the three changed notes

Check your understanding

Write your answers, then check them in Appendix J.

1. What is the relative minor of G major?

2. What is the relative major of D minor?

3. Which note differs between A natural minor and A harmonic minor?

4. Write the formula of the ascending melodic minor.

5. What is the relative minor of E $\flat$ major?

6. What is the 7th note of E harmonic minor?

7. What is the parallel minor of C major, and how many flats does it

8. Which interval occurs between the 6th and 7th degrees of harmonic

MY NOTES

CHAPTER 7

Pentatonic and Blues Scales



TRACKS

7

Five notes, five shapes. The pentatonic scales are the most used scales for improvising on guitar, because they leave out the notes most likely to clash with the chords.

"Penta" means five. A pentatonic scale has five notes instead of seven. The two notes it leaves out are the ones that form half steps, so no two notes in the scale are a half step apart. That makes the scale hard to play badly: almost any note fits over the key's main chords. Rock, blues, country, folk and much of the world's traditional music are built on it.

Building the pentatonic scales

Take the major scale and remove the 4th and the 7th, the two notes that sit a half step above or below another scale note. What remains is the **major pentatonic**: 1 2 3 5 6. Do the same to the natural minor, removing the 2nd and the $b6$, and you get the **minor pentatonic**: 1 $b3$ 4 5 $b7$.

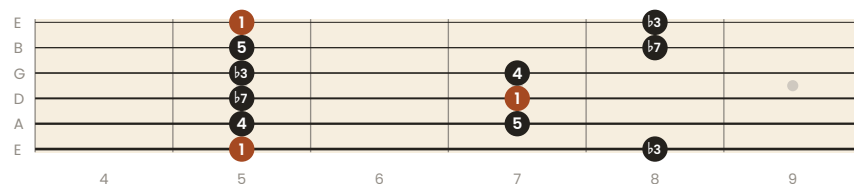
Scale	Formula	In A	In C
Minor pentatonic	1 $b3$ 4 5 $b7$	A C D E G	C E b F G B b
Major pentatonic	1 2 3 5 6	A B C $\sharp$ E F $\sharp$	C D E G A
Blues scale	1 $b3$ 4 $b5$ 5 $b7$	A C D E b E G	C E b F G b G B b

The minor pentatonic is a minor scale without its 2nd and 6th. The major pentatonic is a major scale without its 4th and 7th. Because the relative minor shares its notes with the relative major, A minor pentatonic and C major pentatonic contain the same five notes: A C D E G and C D E G A.

The five boxes

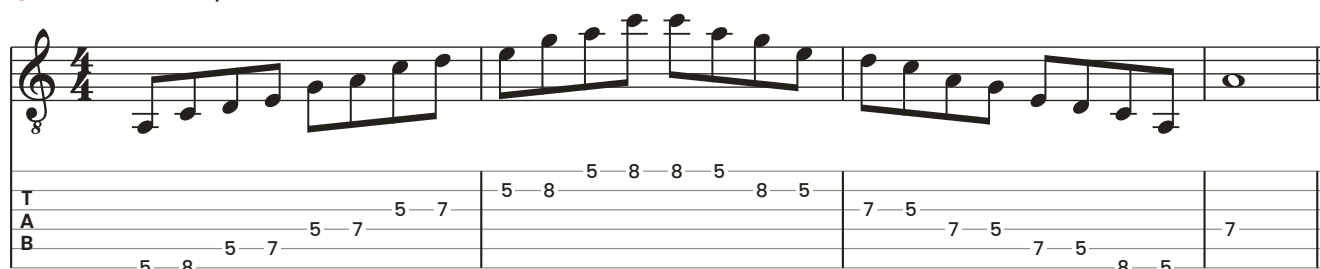
Played two notes per string, the minor pentatonic forms five shapes across the neck, called **boxes** or positions. The top edge of each box is the bottom edge of the next. After box 5 the cycle starts again with box 1, twelve frets higher. Here they are in A minor.

Box 1: root on the 6th string at the 5th fret



The most used shape on the guitar. Play it with the 1st finger at the 5th fret and the 3rd or 4th finger at the 7th and 8th.

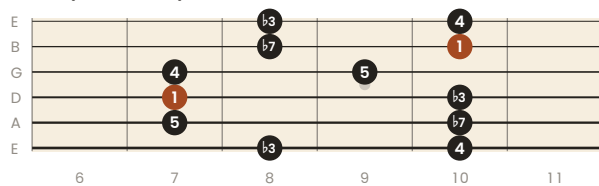
▶ TRACK 27 Box 1 up and down



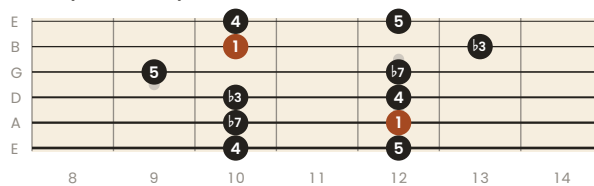
Alternate picking, or alternate i and m on nylon strings. End on the A on the 4th string.

Boxes 2 to 5

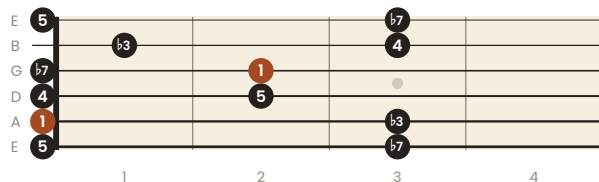
Box 2 (frets 7–10)



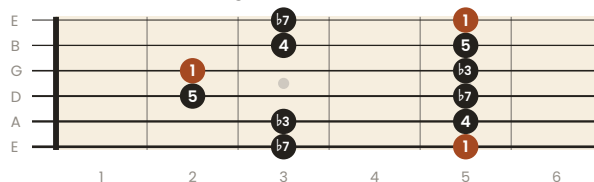
Box 3 (frets 9–13)



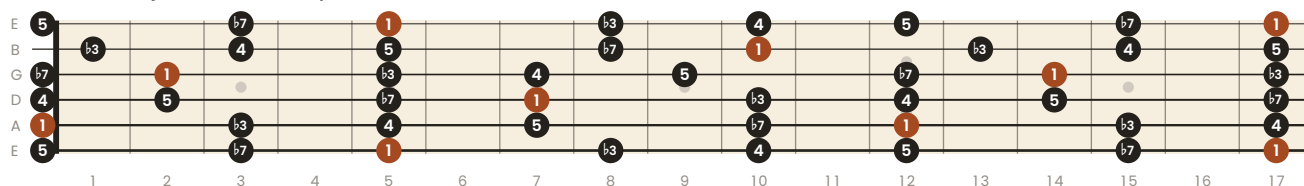
Box 4 (frets 12–15)



Box 5 (frets 2–5, and again at 14–17)



All five boxes joined: A minor pentatonic across the neck



The whole map. Box 5 at frets 2–5 leads into box 1 at the 5th fret; box 4 at 12–15 leads into box 5 again at 14–17.

Joining the boxes

Learn box 1 first, then box 2, and practise moving between them before adding the others. The easiest way to change box is to slide: play a note in one box, then slide the same finger up to the next scale note on that string, which puts your hand in the next box.

 TRACK 28 Sliding from box 1 into box 2

[illegible]

The slide on the 3rd string from D (7th fret) to E (9th fret) moves the hand from box 1 to box 2. The phrase ends on A with vibrato.

TRY THIS • Three notes, then five

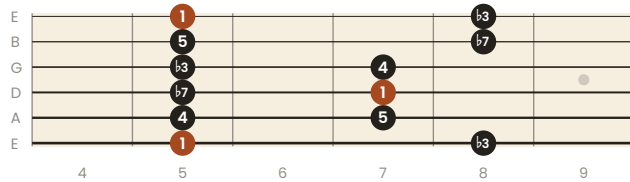
Put on the A minor blues loop below. Improvise for one minute using only three notes of box 1: the A, C and D on the 3rd and 2nd strings. Leave space between phrases. Then use the whole box, and only after that slide into box 2. End every phrase on A or E.

TRACK 29 A minor blues loop · LOOP · ballad · 80 bpm

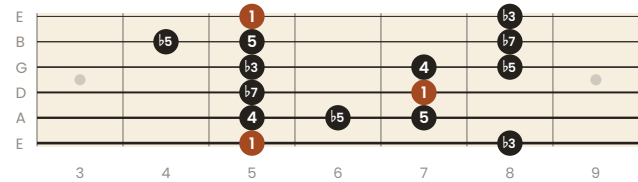
Am7 <i>i7</i>	Dm7 <i>iv7</i>	Am7 <i>i7</i>	Am7 <i>i7</i>
Dm7 <i>iv7</i>	Dm7 <i>iv7</i>	Am7 <i>i7</i>	Am7 <i>i7</i>
F7 <i>bVI7</i>	E7 <i>V7</i>	Am7 <i>i7</i>	E7 <i>V7</i>

IMPROVISE WITH

A minor pentatonic



A blues



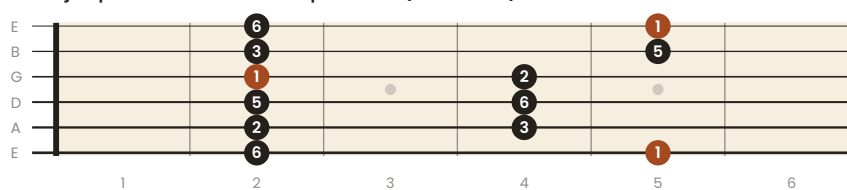
Box 1 covers the whole form. Over E7 land on B or D, chord tones that are in the box, and resolve to A or C when Am7 returns in bar 11.

Chapter 8 takes the next step: one-string and two-string routes, diagonal shapes and the shifts that join the boxes along the whole neck.

Major pentatonic

The major pentatonic uses the same five shapes as the minor pentatonic; only the root moves. A major pentatonic (A B C# E F#) has the same notes as F# minor pentatonic, its relative minor. So to play A major pentatonic, take the A minor pentatonic boxes and move them **down three frets**: box 1 now starts at the 2nd fret.

A major pentatonic in box 1 position (frets 2–5)



The box 1 shape at the 2nd fret, with A as the root. The roots now fall on the 3rd string at the 2nd fret and on the 6th and 1st strings at the 5th fret.

SHORTCUT · One shape, two scales

Every pentatonic box is two scales: a minor pentatonic with its root on the 6th string, and the major pentatonic three frets higher. Box 1 at the 5th fret is A minor pentatonic or C major pentatonic, depending on whether the music treats A or C as home.

TRACK 30 A major pentatonic phrase

A D A

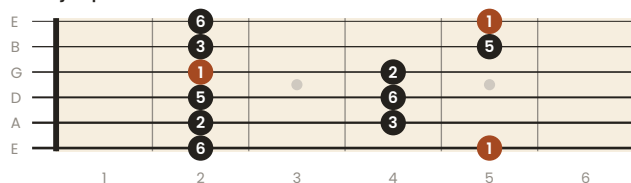
Bright and open, the sound of country, southern rock and many folk melodies. Ends on the root, A.

▶ **TRACK 31** Country-pop loop in A · LOOP · pop · 104 bpm

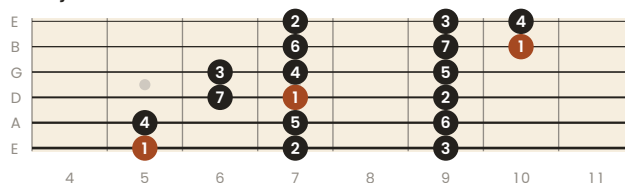
A <i>I</i>	D <i>IV</i>	A <i>I</i>	E <i>V</i>
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IMPROVISE WITH

A major pentatonic



A major



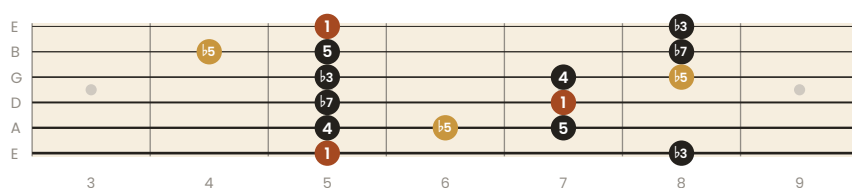
Use A major pentatonic box 1 at the 2nd fret. Land on C# (the 3rd) over A, F# over D and B or E over E; play the same phrase over A in bars 1 and 3 and change only its last note.

The major pentatonic sounds bright and open; it suits country, folk, pop and major-key blues. Over a major-key song in G (G, C, D, Em), use G major pentatonic, which uses the same shapes as E minor pentatonic.

The blues scale

Add one note to the minor pentatonic, the $\flat 5$, and you have the **blues scale**: 1 $\flat 3$ 4 $\flat 5$ 5 $\flat 7$. The $\flat 5$ is the **blue note** of the blues scale. (The $\flat 3$ and $\flat 7$ are also called blue notes when they are played against a major chord.) Use it as a passing note between the 4th and the 5th, or bend up to it; it is a note to move through, not one to stop on.

A blues scale in box 1



The $\flat 5$ (Eb) in gold, on the 5th string at the 6th fret and the 3rd string at the 8th fret. A third one lies a stretch below the box, on the 2nd string at the 4th fret.

▶ **TRACK 32** A blues phrase through the blue note

A7

The $\flat 5$ appears twice, each time passing from the 4th to the 5th. The phrase ends on the root.

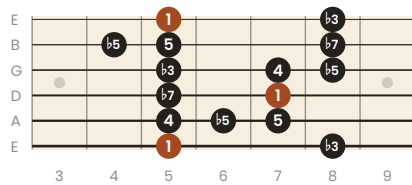
A related scale, the **major blues scale** (1 2 $\flat 3$ 3 5 6), adds the $\flat 3$ to the major pentatonic as a passing note into the major 3rd. It is the sound of much country and swing guitar, and Chapter 43 uses both scales.

TRACK 33 12-bar shuffle in A · LOOP · blues shuffle · 96 bpm

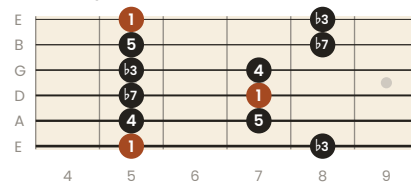
A7 I7	D7 IV7	A7 I7	A7 I7
D7 IV7	D7 IV7	A7 I7	A7 I7
E7 V7	D7 IV7	A7 I7	E7 V7

IMPROVISE WITH

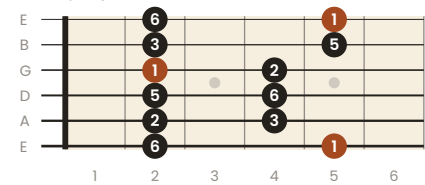
A blues



A minor pentatonic



A major pentatonic

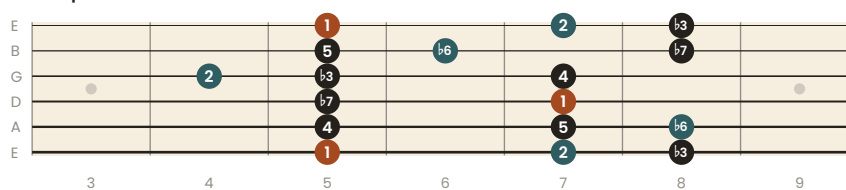


Play the blues scale first, using the $b5$ only as a passing note. Then aim for the 3rd of each chord: bend or slide C up to $C\sharp$ over A7, land on $F\sharp$ (from A major pentatonic) over D7, and on $G\sharp$, a fret below A, over E7.

From pentatonic to the full scale

Each pentatonic box is missing two notes of the full seven-note scale. Add them back and the box becomes a complete scale shape in the same position. Add the 2 and the $b6$ to A minor pentatonic box 1 and you get A natural minor in 5th position, the same notes as C major.

Box 1 plus two notes: A natural minor



The two added notes, B (2) and F ($b6$), in teal. Everything else is box 1.

The same idea works with other notes. Add the 2 and the natural 6 instead and you get the Dorian mode. On the major side, add the missing 4 and 7 to the major pentatonic and you have the major scale. Chapter 9 builds on this.

NYLON · STEEL

Steel and electric: bend the 4th up a whole step to the 5th, and the $b7$ up a whole step to the root. Bend the $b3$ slightly (a quarter step) for a bluesy colour. Hammer-ons and pull-offs between the two notes on each string make the boxes flow.

Nylon: wide bends barely work on nylon strings. Use slides, slurs (hammer-ons and pull-offs) and side-to-side vibrato instead, and play melodies with rest strokes for a full, singing tone. Chapter 36 covers all of these.

Chapter 7 review · use it, train it, test it

Using it in real music

Blues jam. Over a 12-bar blues in A (Chapter 14 explains the form), A minor pentatonic or the A blues scale fits the whole form. Aim for the root over A7 and for the 4th (D) over D7. The 12-bar shuffle loop in this chapter is the place to try it.

Rock and pop solos. Many rock solos and pop fills use the minor pentatonic. Over a minor-key progression such as Am – F – G, A minor pentatonic is the safe first choice.

Country and bright major songs. Over a major-key song such as G – C – D, use G major pentatonic, which uses the E minor pentatonic shapes. Mixing major and minor pentatonic over a dominant-7th blues is a hallmark of the blues sound.

Write your answers, then check them in Appendix J.

2. Spell E minor pentatonic.

[illegible]

CHAPTER 8

Playing Horizontally



TRACKS

8

The boxes teach the neck from top to bottom. This chapter teaches it from end to end, with one-string scales, two-string boxes, diagonal shapes and the shifts that join them.

The five boxes of Chapter 7 are vertical: the hand stays in one position and the notes come from crossing the strings. Playing horizontally means moving along the neck instead, on one string, on a pair of strings or on a diagonal line. The notes are the same; the route is different. This chapter uses A minor pentatonic (A C D E G) throughout and the box numbers from Chapter 7.

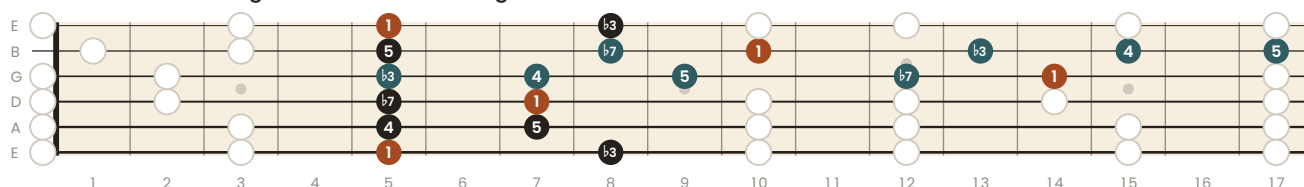
Why horizontal

Boxes are good for three things: they keep every note within reach of the four fingers, they make the roots easy to find, and they let you play fast without moving the hand. What they hide is the rest of the neck. A player who knows only boxes tends to stay in one of them for a whole solo, and the phrases stay inside the range of that box.

Moving along the neck gives you three things a box cannot:

- **One map.** The five boxes become one connected neck rather than five separate pictures.
- **Phrases that travel.** A line can rise for an octave and more and keep going, the way a singer's melody does.
- **A singing tone.** Fewer string changes mean one string's colour for several notes, and slides between them connect the notes like a voice.

Box 1 and a route along the 3rd and 2nd strings



Box 1 in black and rust, the route of the phrase below in teal (its two As in rust), every other A minor pentatonic note as an open circle. The route starts inside box 1 and leaves it on the third note.

▶ TRACK 34 A phrase that travels along two strings

Tablature for Track 34:

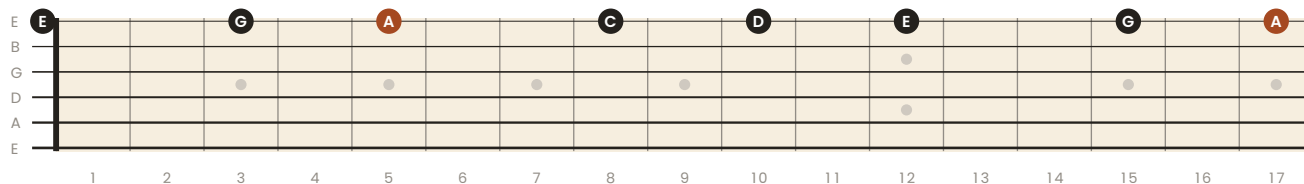
String	Fret	String	Fret	String	Fret
T	5	A	7	B	9
T	8	A	10	B	9
T	12	A	10	B	13
T	15	A	17	B	15
T	13	A	15	B	17
T	15	A	13	B	14

Two strings, three positions, one rising line. Slide the 3rd finger on the 3rd string and on the 2nd string; end on A with vibrato.

One-string scales

The most horizontal way to play a scale is on one string. Every interval becomes a distance you can see: one fret is a half step, two frets a whole step, three frets a minor 3rd. Nothing changes between strings, so this is the clearest picture of a scale's structure that the guitar gives you.

A minor pentatonic on the 1st string



Eight notes from the open E to the A at the 17th fret. From A the distances are 3, 2, 2, 3 and 2 frets.

From	To	Frets	Interval
A	C	3	Minor 3rd
C	D	2	Whole step
D	E	2	Whole step
E	G	3	Minor 3rd
G	A	2	Whole step

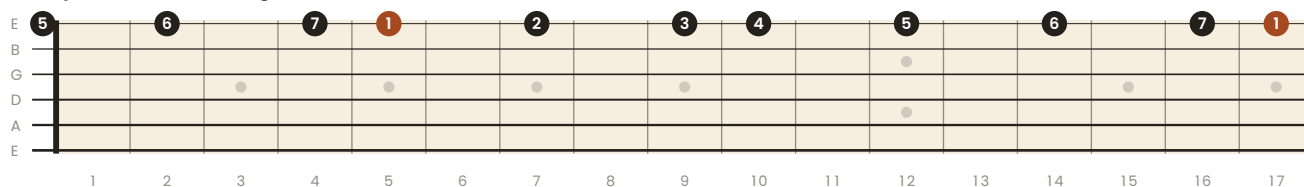
The pattern 3 2 2 3 2 is the minor pentatonic itself. Learn it as a rhythm of distances, and you can play the scale on any string from any root.

▶ TRACK 35 A minor pentatonic up the 1st string

Two notes per hand position, with the 1st and 3rd fingers, then shift. Keep the thumb loose so the hand can move; going down, reverse the fingering.

The major scale on one string shows its formula in the same way: whole, whole, half, whole, whole, whole, half, or 2 2 1 2 2 2 1 frets. Chapter 5 explains the formula; here you see it as a ruler.

A major on the 1st string



From the A at the 5th fret to the A at the 17th. The two half steps (3 to 4 and 7 to 1) are the only notes one fret apart.

▶ **TRACK 36** A major scale on one string

5 — 7 — 9 — 10 — 12 — 14 — 16 — 17 — 16 — 14 — 12 — 10 — 9 — 7 — 5

T
A
B

Sing the degree numbers as you play. The half steps feel and sound different from the whole steps.

A one-string line is not only an exercise. Slides make it sound like a voice. The lick below stays on the 2nd string from box 1 up to box 5 and back.

▶ **TRACK 37** One-string lick on the 2nd string
Am

5 — 8 — 10 — 10 — 13 — 15 — 17 — 15 — 13 — 10 — 8 — 10 — 10

T
A
B

Pick the first note of each slide and let the slide sound the second. The 3rd finger does most of the work.

TRY THIS · One string, any root

Play A minor pentatonic on the 3rd string from the A at the 2nd fret, then on the 5th string from the open A. Say the distances aloud (3 2 2 3 2) as you go. Then play the A major scale on the 4th string from the A at the 7th fret.

Two-string boxes

Between one string and a full box lies the **two-string box**: two notes on each of two neighbouring strings, four notes under the hand. Restrict A minor pentatonic to the 1st and 2nd strings and the whole neck becomes a strip that you can read at a glance.

A minor pentatonic on the 1st and 2nd strings

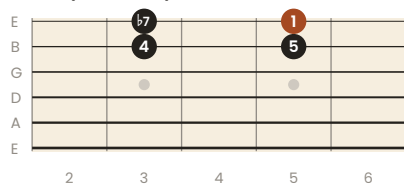
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Two strings, the whole neck. Each square or slanted group of four neighbouring notes is one two-string box.

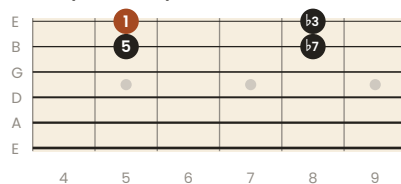
Each two-string box is one of the five boxes cut down to two strings, so the units come in the same order as the boxes: 4, 5, 1, 2, 3, 4, 5 going up from the nut. Together they form the box cycle seen through a two-string window. On the 1st and 2nd strings, a perfect 4th apart, three of the units are squares (the same two frets on both strings) and two are slanted by one fret.

The two-string boxes on the 1st and 2nd strings

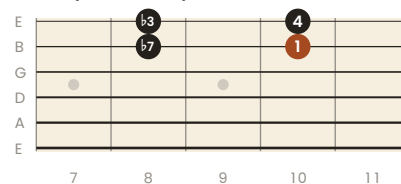
Box 5 (frets 3–5)



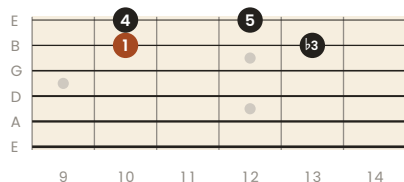
Box 1 (frets 5–8)



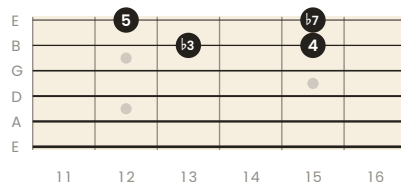
Box 2 (frets 8–10)



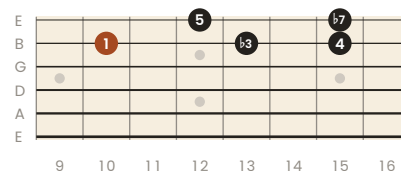
Box 3 (frets 10–13)



Box 4 (frets 12–15)



A 3+2 unit from A

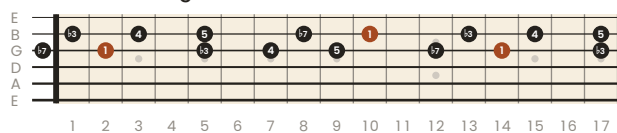


Play each unit with the 1st finger on the lower fret and the 3rd or 4th finger on the higher one. The last diagram is a 3+2 unit: three notes on the 2nd string and two on the 1st cover one octave.

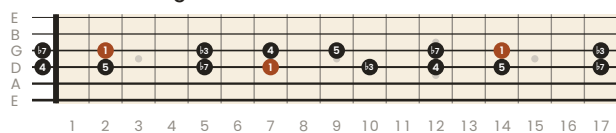
The same idea works on any pair of strings. The 2nd and 3rd strings are a major 3rd apart instead of a perfect 4th (the B-string shift from Chapter 3), so every 2nd-string note sits one fret further up than it would on a pair tuned in 4ths, and the squares become slanted shapes. The 3rd and 4th strings are a perfect 4th apart again, like the 1st and 2nd.

The next two pairs of strings

2nd and 3rd strings



3rd and 4th strings



Compare the 2nd string with the diagram above; only the 3rd string is new. The 3rd and 4th strings have the same tuning interval as the 1st and 2nd, so their units have the same shapes, three frets lower.

The run below climbs the 1st and 2nd strings one two-string box at a time. Each group of four starts one scale note higher than the last, and each new group is the next box.

▶ TRACK 38 Two-string run through six boxes

Boxes 5, 1, 2, 3, 4 and 5 again, two per bar. Play the 1st finger on the lower note of each string and shift the hand on the first note of every group.

TRACK 39 Two-string lick on the 4th and 3rd strings

Tablature: 5 7 5 7 9 10 12 9 12 14 14 12 10 12 14

From box 1 to box 4 on two strings, using slides to change position. The lower register gives the same line a darker colour.

Diagonal pentatonic shapes

A **diagonal pattern** plays three notes on every string instead of two. Each string then reaches one note into the next box, so the hand climbs as it crosses the strings, about one box per string. Each diagonal is named after the note it starts on, on the 6th string. They are the pentatonic cousins of the three-notes-per-string patterns in Chapter 5.

Diagonal from the open E

Starts on the open 6th string and ends on the A at the 17th fret of the 1st string. It passes through box 5, then boxes 1, 2, 3 and 4, and ends in box 5 an octave higher.

Diagonal from the root (6th to 3rd string)

From the A at the 5th fret to the C at the 17th, two octaves and a minor 3rd. It runs from box 1 through boxes 2, 3 and 4 to box 5. On the 2nd and 1st strings it would need frets 15–22, so it stops here.

Diagonal from G (6th to 2nd string)

From the G at the 3rd fret to the E at the 17th fret of the 2nd string. It covers boxes 5, 1, 2, 3 and 4 and ends in box 5.

Diagonal	Strings 6 and 5	Strings 4 and 3	Strings 2 and 1
From the open E	Boxes 5 and 1 (open E from box 4)	Boxes 1, 2 and 3	Boxes 3, 4 and 5
From the root	Boxes 1, 2 and 3	Boxes 3, 4 and 5	(past the 20th fret)
From G	Boxes 5, 1 and 2	Boxes 2, 3 and 4	Boxes 4 and 5 (2nd string only)

Three pentatonic notes on one string span four or five frets, more than one finger per fret allows. Above the 9th fret most hands can stretch 1-2-4 or 1-3-4 across them. Lower down, use the slide fingering: 1st finger on the first note, 3rd finger on the second, then slide the 3rd finger up to the third. That slide is the position shift, and the 1st finger lands ready for the next string.

Up: 1st finger, 3rd finger, slide the 3rd finger. Down: 3rd finger, slide it back, 1st finger. Twelve notes each way, one string per beat.

Do not force a five-fret stretch at the 5th fret. Use the slide fingering there, and save the stretch for higher on the neck where the frets are closer together.

Every horizontal line needs **position shifts**: moments where the whole hand moves to a new place on the neck. A clean shift is silent (unless you want to hear the slide), on time and lands with the fingers already in shape. These are the five ways to do it.

Pick a note, slide the same finger to the next scale note and let the hand follow. The slide is heard. Chapter 36 covers the slide itself.

T 12 12
 A 5 7 *sl.* 9 12 10 13 13 10
 B

Stretch shift

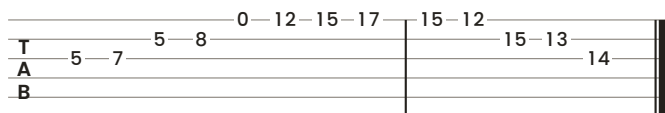
Reach a finger one or two frets outside the position for the next note, then let the rest of the hand settle around it. Nothing is heard but the notes.

[illegible]

Open-string shift

58

TRACK 43 Open-string shift to the 12th fret

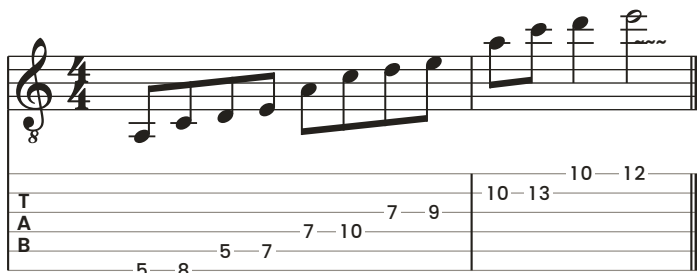


The open 1st string gives the hand an eighth note to jump from the 5th position to the 12th. The open E and the E at the 12th fret are an octave apart.

Octave-shape jump

Play a short cell, then play it again an octave higher with an octave shape from Chapter 4: skip a string and go up two frets from the 6th or 5th string, up three frets from the 4th or 3rd. The line travels diagonally in large steps.

TRACK 44 Octave-shape jump with one cell

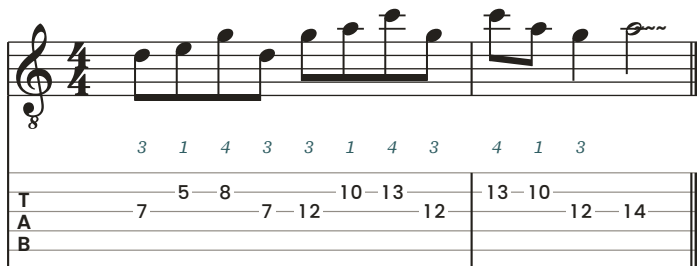


The cell A C D E three times: from the 6th string at the 5th fret, from the 4th at the 7th, and from the 2nd at the 10th.

Guide finger

The finger that plays the last note before a shift stays on its string and glides, with light pressure, to the first note after it. The guide keeps the hand on track, and the slide is barely heard. In classical notation a dash before a finger number marks it.

TRACK 45 Guide-finger shift on the 3rd string



The same four-note shape (fingers 3 1 4 3) five frets apart. The 3rd finger glides from D at the 7th fret to G at the 12th without leaving the 3rd string.

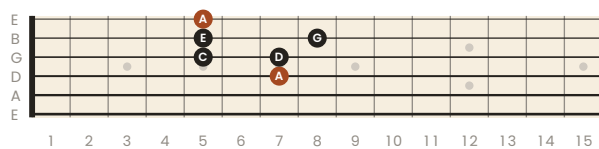
TIP · Look where you are going

During a shift, look at the fret you are moving to, not at the finger that is leaving. Move the hand a moment early, on the last note before the shift, so the arrival is on time.

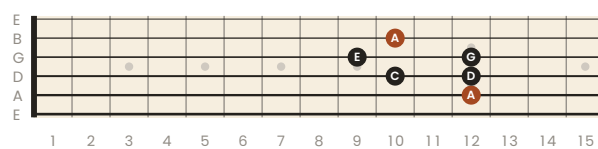
One lick, five places

The same short phrase, A C D E G A G E D C A, fits in many places on the neck. The grid shows five routes: two boxes, one string, two strings and a diagonal. The pitches are identical in all five; only the hand's route and the tone change.

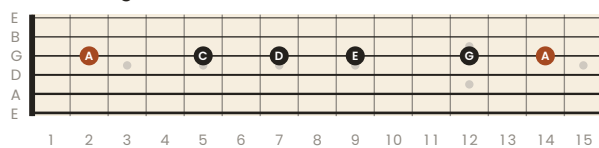
1. Box 1



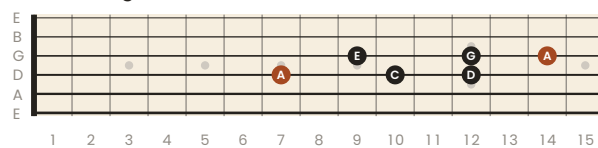
2. Box 3



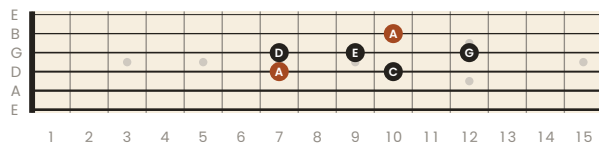
3. One string (3rd)



4. Two strings (4th and 3rd)



5. Diagonal



A3 C4 D4 E4 G4 A4 in each diagram. The diagonal route follows the diagonal from the open E.

▶ **TRACK 46** The lick in five places

First example of the lick in five places. The lick is shown on a staff with a treble clef and a 4/4 time signature. The notes are A3, C4, D4, E4, G4, A4. The fretboard diagram below shows the fret numbers for each note on the strings T, A, and B.

String	Fret	String	Fret	String	Fret
T	7	A	5	B	7
T	5	A	7	B	5
T	8	A	5	B	7
T	5	A	7	B	5
T	8	A	5	B	7

Second example of the lick in five places. The lick is shown on a staff with a treble clef and a 4/4 time signature. The notes are A3, C4, D4, E4, G4, A4. The fretboard diagram below shows the fret numbers for each note on the strings T, A, and B.

String	Fret	String	Fret	String	Fret
T	12	A	10	B	12
T	10	A	12	B	10
T	9	A	12	B	10
T	12	A	10	B	12
T	10	A	12	B	10

Third example of the lick in five places. The lick is shown on a staff with a treble clef and a 4/4 time signature. The notes are A3, C4, D4, E4, G4, A4. The fretboard diagram below shows the fret numbers for each note on the strings T, A, and B.

String	Fret	String	Fret	String	Fret
T	2	A	5	B	7
T	9	A	12	B	14
T	12	A	14	B	12
T	9	A	7	B	5
T	12	A	5	B	2

Fourth example of the lick in five places. The lick is shown on a staff with a treble clef and a 4/4 time signature. The notes are A3, C4, D4, E4, G4, A4. The fretboard diagram below shows the fret numbers for each note on the strings T, A, and B.

String	Fret	String	Fret	String	Fret
T	7	A	10	B	12
T	9	A	12	B	14
T	12	A	14	B	12
T	9	A	12	B	10
T	12	A	10	B	7

9

Chapter 8 review · use it, train it, test it

Using it in real music

Rock and blues solos. Long solos often start low in a box and climb the neck over several phrases, each in a higher position. Two-string routes and slide shifts are the usual way to get there.

Melodies. Playing a song melody on one or two strings gives it an even tone and lets you slide between notes like a singer. Many slow instrumental guitar melodies are played this way, high on the 2nd and 3rd strings.

Classical and fingerstyle. On nylon strings, guide-finger and open-string shifts are part of standard technique. Classical scale fingerings often change position several times in one scale for exactly the reasons in this chapter.

How to train it

Drill	How	Time	Goal
Distance ladder	Play A minor pentatonic on each string in turn, from the lowest A to the highest, saying 3 2 2 3 2	3 min	No wrong notes on any string
Box slices	On one pair of strings, play the two-string boxes from box 5 to box 5 an octave higher and back	3 min	No pause at the shifts
Diagonal run	Play the two-octave diagonal run with the slide fingering at 60 bpm, then 80	3 min	Even triplets through every slide
Five places	Play the lick in all five places without stopping between them	2 min	Same rhythm and volume on every route

Check your understanding

Write your answers, then check them in Appendix J.

1. On one string, how many frets lie between E and G in A minor pentatonic, and what interval is that?

2. Why are the two-string boxes on the 2nd and 3rd strings shaped differently from those on the 1st and 2nd?

3. Which three notes does the diagonal from the root play on the 6th string, and at which frets?

4. In the slide fingering for a diagonal, which finger slides?

5. What gives the hand time to move in an open-string shift?

6. A cell starts on the A at the 7th fret of the 4th string. Where does the same cell start an octave higher?

MY NOTES

CHAPTER 9

Modes



TRACKS

9

A mode is a scale that uses the notes of a parent scale but treats a different note as home. The seven modes of the major scale give seven distinct sounds from shapes you already know.

Play the C major scale from D to D and you get **D Dorian**. Play it from E to E and you get **E Phrygian**. Each degree of the major scale can serve as a home note, and each gives a different arrangement of whole and half steps above that note. Those seven arrangements are the **modes** of the major scale. You already know two of them: the major scale itself is the Ionian mode, and the natural minor is the Aeolian mode.

The seven modes

Starts on	Mode	Formula	In C (white keys)	Sound
1st degree	Ionian	1 2 3 4 5 6 7	C D E F G A B	The major scale; bright, settled
2nd degree	Dorian	1 2 $\flat$ 3 4 5 6 $\flat$ 7	D E F G A B C	Minor with a bright 6th
3rd degree	Phrygian	1 $\flat$ 2 $\flat$ 3 4 5 $\flat$ 6 $\flat$ 7	E F G A B C D	Dark, Spanish
4th degree	Lydian	1 2 3 $\sharp$ 4 5 6 7	F G A B C D E	Major with a raised 4th; floating
5th degree	Mixolydian	1 2 3 4 5 6 $\flat$ 7	G A B C D E F	Major with a bluesy $\flat$ 7
6th degree	Aeolian	1 2 $\flat$ 3 4 5 $\flat$ 6 $\flat$ 7	A B C D E F G	The natural minor; sad, serious
7th degree	Locrian	1 $\flat$ 2 $\flat$ 3 4 $\flat$ 5 $\flat$ 6 $\flat$ 7	B C D E F G A	Unstable; rarely used as a home

SHORTCUT · Order and brightness

Order: "I Don't Particularly Like Modes A Lot": Ionian, Dorian, Phrygian, Lydian, Mixolydian, Aeolian, Locrian.

Brightness: from brightest to darkest, Lydian, Ionian, Mixolydian, Dorian, Aeolian, Phrygian, Locrian. Each step towards the dark end lowers exactly one note by a half step: $\sharp 4 \rightarrow 4$, then $7 \rightarrow \flat 7$, $3 \rightarrow \flat 3$, $6 \rightarrow \flat 6$, $2 \rightarrow \flat 2$, and finally $5 \rightarrow \flat 5$.

Dorian is common in folk, funk, jazz and rock; Mixolydian in rock, folk, Celtic music and blues; Lydian in film and TV music. Songs where you can hear the modes: "Scarborough Fair" (traditional) and Santana's "Oye Como Va" are Dorian; the Beatles' "Norwegian Wood" is Mixolydian; the theme of *The Simpsons* is Lydian; "All Along the Watchtower" moves through Aeolian chords. Phrygian is the basis of flamenco and of many heavy metal riffs.

Two ways to think about modes

Relative (parent-scale) view. D Dorian has the same notes as C major, so every C major shape is also a D Dorian shape. This is the fastest way to find a mode on the neck: name the parent major scale and play its patterns, keeping the new root in mind.

Parallel (character) view. D Dorian is D natural minor with a raised 6th. Compare each mode with the major or natural minor scale on the **same root** and listen for the note that differs. That note gives the mode its character. This is the way to hear and use a mode.

Both views describe the same notes. Use the relative view to find the fingerings and the parallel view to choose a mode and make it sound like itself.

Finding the parent scale

Count back to the degree the mode starts on. Dorian starts on the 2nd degree, so its parent is a whole step below: D Dorian → C major, A Dorian → G major. Mixolydian starts on the 5th degree, so its parent is a perfect 4th above (or a perfect 5th below): A Mixolydian → D major.

Mode	Parent major scale is	Example
Dorian	A whole step below the root	E Dorian → D major
Phrygian	A major 3rd below the root	E Phrygian → C major
Lydian	A perfect 4th below (a 5th above) the root	F Lydian → C major
Mixolydian	A perfect 5th below (a 4th above) the root	G Mixolydian → C major
Aeolian	A minor 3rd above the root	A Aeolian → C major
Locrian	A half step above the root	B Locrian → C major

Characteristic notes

In the parallel view, each mode differs from major or natural minor by one note (Locrian by two). Learn these and you know what to emphasise when you play the mode.

Mode	Compare with	Characteristic note	On D
Lydian	Major	#4	G#
Mixolydian	Major	b7	C
Dorian	Natural minor	6 (natural)	B
Phrygian	Natural minor	b2	Eb
Locrian	Natural minor	b2 and b5	Eb and Ab

All seven modes on one root show the differences most clearly. Here they are on D:

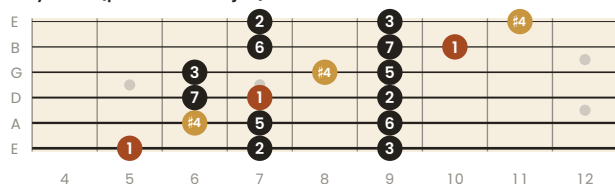
Mode	Notes on D	Parent major
D Lydian	D E F# G# A B C#	A
D Ionian	D E F# G A B C#	D
D Mixolydian	D E F# G A B C	G
D Dorian	D E F G A B C	C
D Aeolian	D E F G A Bb C	F
D Phrygian	D Eb F G A Bb C	Bb
D Locrian	D Eb F G Ab Bb C	Eb

The modes on the neck

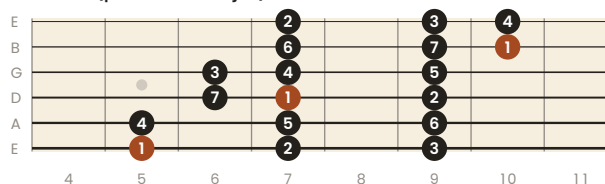
The diagrams show all seven modes on the same root, A, as three-notes-per-string patterns starting on the 6th string at the 5th fret. The characteristic note is in gold. Each one is also a pattern of its parent major scale from Chapter 5.

Seven modes on A, three notes per string

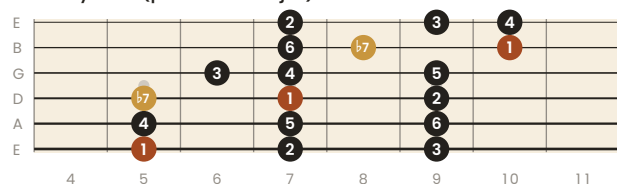
A Lydian (parent: E major)



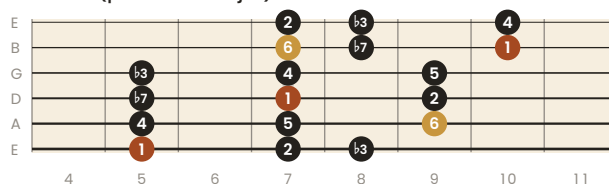
A Ionian (parent: A major)



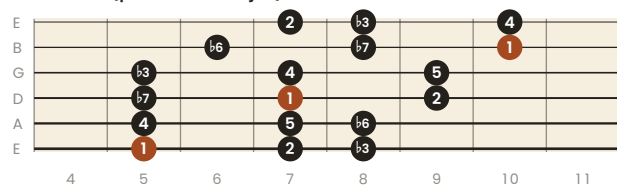
A Mixolydian (parent: D major)



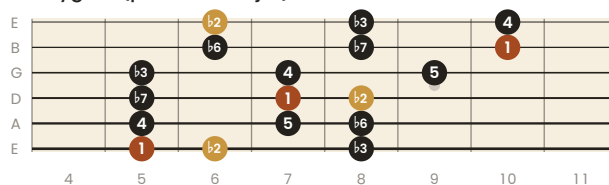
A Dorian (parent: G major)



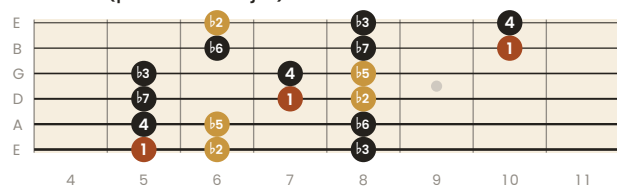
A Aeolian (parent: C major)



A Phrygian (parent: F major)



A Locrian (parent: B♭ major)



Ordered from brightest to darkest. Compare neighbouring diagrams to see the one note that changes each time.

▶ TRACK 48 Ionian, Mixolydian and Dorian on A

The same fingering three times. Bar 2 lowers the 7th (G♯ → G) for Mixolydian; bar 3 also lowers the 3rd (C♯ → C) for Dorian.

Hearing a mode

A mode only sounds like itself when the music around it makes its root feel like home. Play a mode over a **drone** (one sustained bass note, such as an open string) or a **vamp** (one or two chords repeated). If you play D Dorian over a C major progression, it simply sounds like C major.

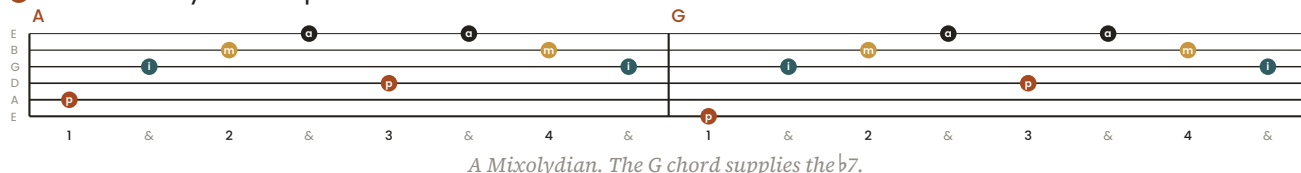
The best vamps contain the characteristic note in one of the chords:

Mode	Vamp	Why it works
Dorian	Am – D, or Dm7 – G	The major IV chord contains the natural 6 (F♯ in A Dorian)
Mixolydian	A – G, or D – C	The ♭VII chord contains the ♭7
Lydian	C – D/C, or F – G/F	The major II chord over the root contains the ♯4
Phrygian	Em – F, or Am – B♭	The ♭II chord a half step above contains the ♭2
Aeolian	Am – F – G	The ♭VI and ♭VII chords

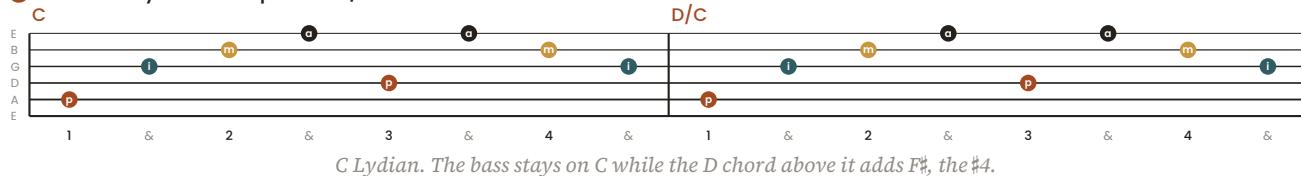
▶ TRACK 49 Dorian vamp: Am to D

A Dorian. The F♯ in the D chord is the natural 6 that separates Dorian from natural minor.

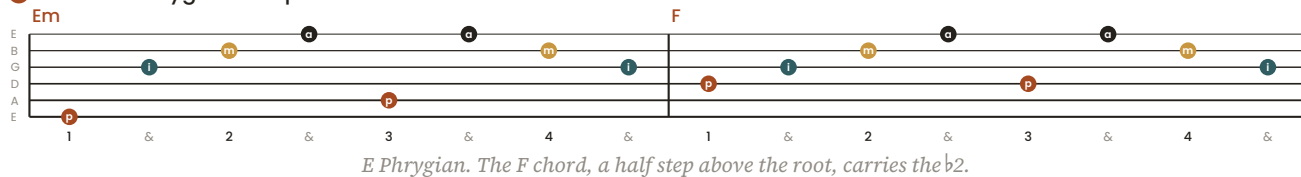
TRACK 50 Mixolydian vamp: A to G



TRACK 51 Lydian vamp: C to D/C



TRACK 52 Phrygian vamp: Em to F



TRY THIS · The drone switch

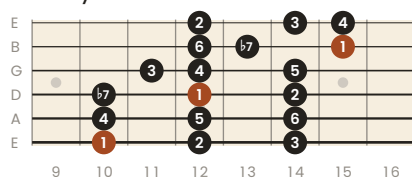
Let the open 4th string (D) ring, or play over the D drone loop below. Play D Ionian, then D Mixolydian, then D Dorian, then D Aeolian, then D Phrygian, using the table of notes on D. One note changes each time. Listen for the moment the mood changes from major to minor (Mixolydian to Dorian) and for the darkening when the 2nd drops (Aeolian to Phrygian).

TRACK 53 D drone for the drone switch · LOOP · modal drone · 72 bpm

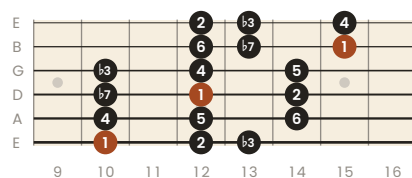


IMPROVISE WITH

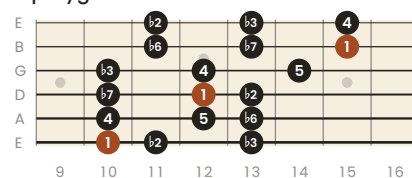
D mixolydian



D dorian



D phrygian



D5 has no 3rd, so every mode on D fits over it. Hold the characteristic note of each mode (C in Mixolydian, B in Dorian, E_b in Phrygian) for a full bar before moving on.

NYLON · STEEL

Steel-string: open and drone tunings make modes easy to hear. In DADGAD (Chapter 25) the open D strings form a drone under D Mixolydian and D Dorian melodies, the basis of much Celtic fingerstyle playing (Chapter 54).

Nylon: the Phrygian mode over an E bass is the harmonic heart of flamenco. Play Am – G – F – E with a strong E at the end and the E chord's $G\sharp$ turns the mode into Phrygian dominant, covered in Chapter 53.

The harmonic minor and melodic minor scales also have seven modes each, including Phrygian dominant, Lydian dominant and the altered scale. They are covered in Chapter 31.

Practice loops

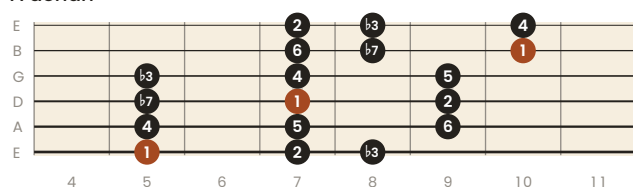
One loop for each of the five modes in everyday use. Every loop contains the chord that carries the mode's characteristic note, so the mode is heard rather than just played. Start each one by landing on the characteristic note as its chord arrives.

▶ TRACK 54 Dorian vamp: Am7 – D9 · LOOP · funk · 96 bpm

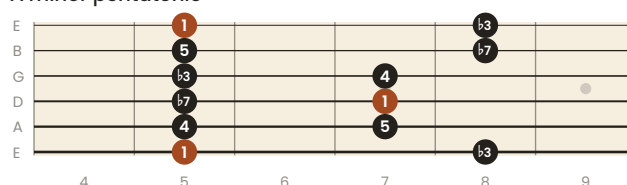
Am7 <i>i7</i>	Am7 <i>i7</i>	D9 <i>IV9</i>	D9 <i>IV9</i>
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IMPROVISE WITH

A dorian



A minor pentatonic



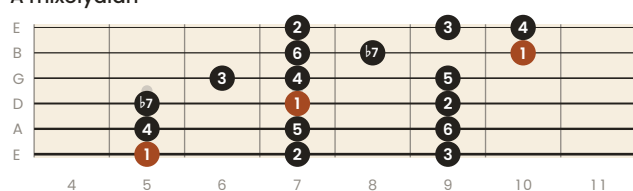
Think of A minor pentatonic plus F# and B. Land on F# (the 6) at the start of the D9 bars, where it is the chord's 3rd; then play four bars with F instead and hear it clash.

▶ TRACK 55 Mixolydian vamp: A – G · LOOP · rock · 112 bpm

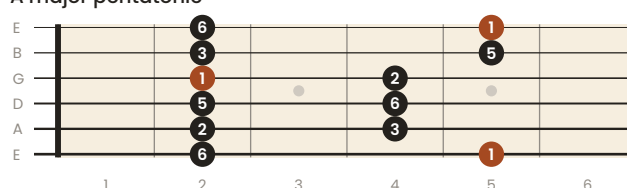
A <i>I</i>	G <i>bVII</i>	A <i>I</i>	G <i>bVII</i>
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IMPROVISE WITH

A mixolydian



A major pentatonic



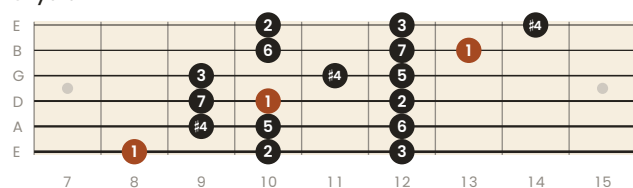
Start with A major pentatonic, then add D and G. Land on G (the b7) as the G chord arrives and on C# when A returns; G# never appears.

▶ TRACK 56 Lydian vamp: Cmaj7 – D/C · LOOP · ballad · 72 bpm

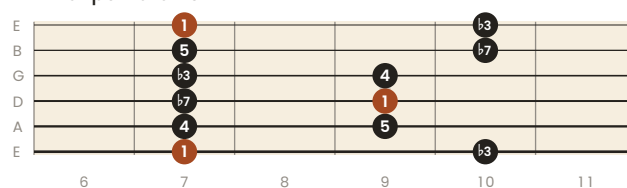
Cmaj7 <i>I maj7</i>	D/C <i>II/1</i>	Cmaj7 <i>I maj7</i>	D/C <i>II/1</i>
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IMPROVISE WITH

C lydian



B minor pentatonic



Land on F# (the #4) as the D chord arrives, where it is the chord's 3rd, and hold it into Cmaj7. B minor pentatonic (B D E F# A) over C gives 7, 9, 3, #4 and 6: a quick Lydian sound from a familiar box.

TRACK 57 Phrygian vamp: Em – F · LOOP · rock · 100 bpm

Em <i>i</i>	F <i>bII</i>	Em <i>i</i>	F <i>bII</i>
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IMPROVISE WITH

E phrygian

E minor pentatonic

Play F over the F chord and let it fall a half step to E as Em returns: that b2 to 1 move is the Phrygian sound. Exercise: phrases in E minor pentatonic that add F only as the last note of each F bar.

TRACK 58 Aeolian loop: Em – C – D · LOOP · pop · 92 bpm

Em <i>i</i>	C <i>bVI</i>	D <i>bVII</i>	Em <i>i</i>
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IMPROVISE WITH

E natural minor

E minor pentatonic

The b6 (C) separates Aeolian from Dorian. Hold C over the C chord, then try C# in the same place and hear it clash.

Chapter 9 review · use it, train it, test it

Using it in real music

Funk and minor jams. Over a minor vamp such as Dm7 – G or Am – D, Dorian is the sound: minor, but with a bright 6th. Land on the 6 to make it obvious.

Rock, folk and Celtic tunes. Progressions built on I – bVII (A – G, D – C, G – F) are Mixolydian: play the major scale with a b7, or the parent major scale a 4th above the root.

Film and atmospheric music. A major chord moving to the major chord a whole step above over the same bass (C – D/C) gives the floating Lydian sound.

Flamenco and metal. Phrygian over an E bass, with the F chord a half step above, gives the dark Spanish colour on nylon strings and the heavy sound of many metal riffs on electric.

How to train it

Drill	How	Time	Goal
Drone switch	Over a D drone, play D Ionian, Mixolydian, Dorian, Aeolian and Phrygian in turn	5 min	Hear each mode's character change
Characteristic-note phrases	Improvise over a vamp and land deliberately on the characteristic note (6 in Dorian, b7 in Mixolydian)	4 min	Make the mode audible, not just correct
Parent lookup	Name a mode and root (E Phrygian, F Lydian); name the parent major scale and play one of its patterns	2 min	Find any mode's shapes in under 5 seconds
Brightness ladder	On one root, play the seven modes from Lydian to Locrian, lowering one note each time	4 min	Play the ladder from memory

Check your understanding

Write your answers, then check them in Appendix J.

1. Which mode starts on the 5th degree of the major scale?

2. What is the characteristic note of Lydian?

3. Spell E Phrygian.

4. D Dorian has the same notes as which major scale?

5. A Mixolydian has the same notes as which major scale?

6. Which mode is the natural minor scale?

7. Which mode is the brightest?

8. G Dorian has the same notes as which major scale?

MY NOTES

PART FOUR

Harmony

From three-note triads to full chord vocabularies, the CAGED map, keys, progressions and arpeggios.

- 10 Triads
- 11 Chords: Open, Barre and Beyond
- 12 The CAGED System
- 13 Chords in a Key
- 14 Chord Progressions
- 15 Spices: Making Progressions Your Own
- 16 Arpeggios



CHAPTER 10

Triads



TRACKS

10

A triad is three notes stacked in 3rds. Every chord in this book is a triad, a triad with doubled notes, or a triad with notes added, so learning triads on small string sets teaches you the whole neck.

A triad is the smallest complete chord: a root, a 3rd above it and a 5th above the root. Take any note of a scale, skip one, take the next, skip one, take the next. From C in C major that gives C (skip D) E (skip F) G: the C major triad.

On the guitar you rarely play just three notes in a fixed order. Open chords and barre chords double some of the notes across five or six strings. Underneath, though, they are still triads, and the clearest way to see them is on three adjacent strings. This chapter builds the four triad qualities, shows their inversions on every adjacent string set, and uses them to connect chords smoothly.

The four triad qualities

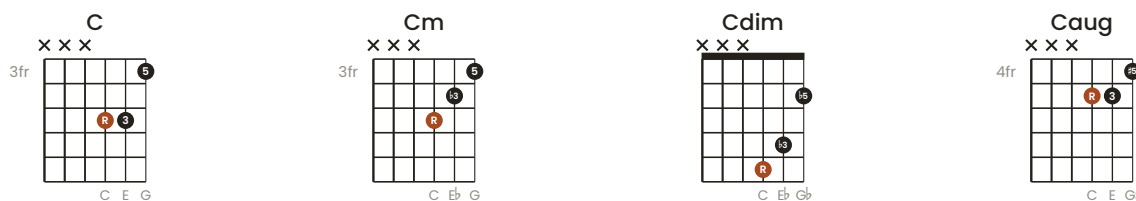
Two kinds of 3rd make all triads: the **major 3rd** (4 half steps, 4 frets) and the **minor 3rd** (3 half steps, 3 frets). Stack two of them and you get one of four qualities.

Triad	Formula	Stacked 3rds	In C	Symbol	Sound
Major	1 3 5	major + minor	C E G	C	Bright, stable
Minor	1 b3 5	minor + major	C E $\flat$ G	Cm	Darker, stable
Diminished	1 b3 b5	minor + minor	C E $\flat$ G $\flat$	Cdim	Tense, unstable
Augmented	1 3 #5	major + major	C E G $\sharp$	Caug	Restless, unresolved

Other books and charts also write Cmin or C $-$ for minor, C $^{\circ}$ for diminished and C $^{+}$ for augmented; this book uses the symbols in the table.

The outer interval tells the qualities apart too. Major and minor triads span a perfect 5th (7 half steps). The diminished triad spans a diminished 5th (6 half steps); the augmented triad spans an augmented 5th (8 half steps).

One fret changes the quality: C triads on strings 3–2–1



Lower the 3rd for minor; lower the 3rd and 5th for diminished; raise the 5th for augmented. The root stays put.

SHORTCUT • Changing quality

Start from a major triad. Lower the 3rd one fret for minor. Lower the 5th as well for diminished. Raise the 5th one fret from major for augmented. On an open E chord, lifting the finger from the 3rd string turns E into Em: that finger was holding the 3rd.

TRACK 59 The four qualities on C

C Cm Cdim Caug C

3 3 2 4 3
5 5 4 5 5
5 5 5 5 5

T
A
B

Listen to how the major and minor triads sound finished, while the diminished and augmented triads want to move on.

Spelling triads

Triads are spelled on every other letter: C–E–G, D–F–A, E–G–B and so on. The letters never change with the quality; only the accidentals do. C minor is C–E $\flat$ –G, never C–D $\sharp$ –G, because the 3rd of C must be some kind of E.

Root	Major	Minor	Root	Major	Minor
C	C E G	C E $\flat$ G	F $\sharp$	F $\sharp$ A $\sharp$ C $\sharp$	F $\sharp$ A C $\sharp$
C $\sharp$ /D $\flat$	D $\flat$ F A $\flat$	C $\sharp$ E G $\sharp$	G	G B D	G B $\flat$ D
D	D F $\sharp$ A	D F A	G $\sharp$ /A $\flat$	A $\flat$ C E $\flat$	G $\sharp$ B D $\sharp$
E $\flat$	E $\flat$ G B $\flat$	E $\flat$ G $\flat$ B $\flat$	A	A C $\sharp$ E	A C E
E	E G $\sharp$ B	E G B	B $\flat$	B $\flat$ D F	B $\flat$ D $\flat$ F
F	F A C	F A $\flat$ C	B	B D $\sharp$ F $\sharp$	B D F $\sharp$

Where two spellings exist, the table gives the one used in practice: D $\flat$ major (five flats) rather than C $\sharp$ major (seven sharps), but C $\sharp$ minor (four sharps) rather than D $\flat$ minor.

HOW IT CONNECTS

The intervals behind these formulas are in Chapter 3. The triads that occur naturally in each key, and why they follow the pattern major, minor, minor, major, major, minor, diminished, are the subject of Chapter 13.

Inversions

The lowest note decides the inversion. The order of the upper notes does not matter.

Position	Lowest note	C major	Slash symbol	Sound
Root position	Root	C–E–G	C	Solid, settled
1st inversion	3rd	E–G–C	C/E	Lighter, wants to move
2nd inversion	5th	G–C–E	C/G	Floating; often a passing chord

In **close position** the three notes fit inside one octave. On three adjacent strings each inversion gives one close-position shape, and the three shapes follow each other up the neck in a fixed cycle: root position, 1st inversion, 2nd inversion, root position an octave higher, and so on. Each shape's top note becomes the next shape's middle note, so learning one shape tells you where the next one is.

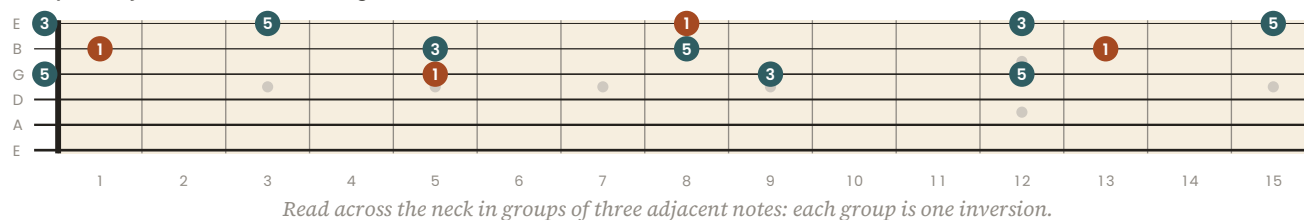
UNDER THE HOOD · Why the bass note matters

The 3rd in the bass weakens the sense of arrival, and the 5th in the bass, with the root a 4th above it, sounds unstable. That is why a 2nd-inversion chord in classical harmony usually passes on to something else, for example C/G to G at a cadence. On guitar the inversion matters most when you play with a bass player: the bass note they play decides the inversion the listener hears, whatever your three strings contain.

Triads on strings 3–2–1

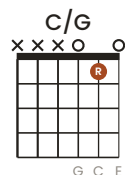
The top three strings (G, B, E) are where triads cut through a band and where most melody-on-top arranging happens. Here are all the C major triad notes on these strings, and the three shapes they form.

Every C major triad tone on strings 3–2–1

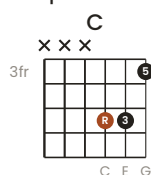


C major on strings 3–2–1

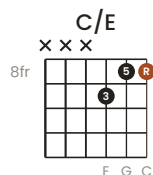
2nd inversion



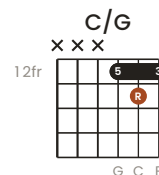
Root position



1st inversion



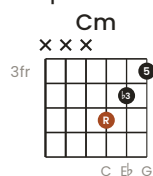
2nd inversion



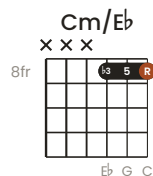
The cycle repeats at the 12th fret. The bass of each shape is the note on the 3rd string.

C minor on strings 3–2–1

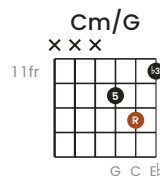
Root position



1st inversion



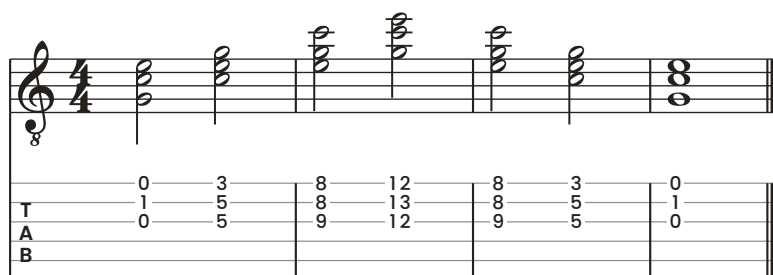
2nd inversion



Compare each shape with its major partner: only the 3rd has moved, one fret down.

▶ **TRACK 60** Inversion ladder: C major up strings 3–2–1

C/G C C/E C/G C/E C C/G

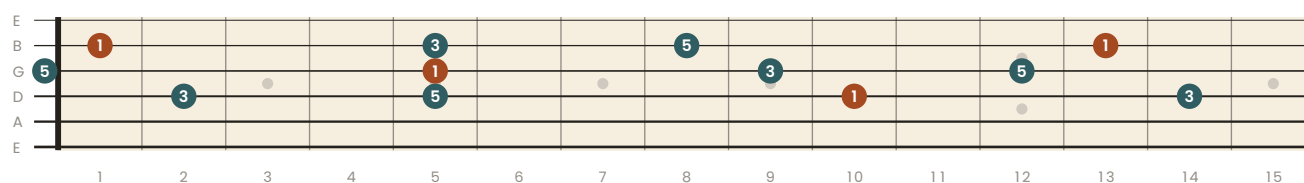


The same chord in four shapes, up the neck and back. On the way up the top note climbs E, G, C, E.

Triads on strings 4–3–2

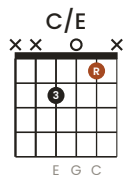
The middle strings (D, G, B) give a warmer sound and sit well under a singer. The shapes differ from those on strings 3–2–1 because the interval between the G and B strings is a major 3rd, not a 4th: any shape that crosses that pair shifts one fret.

Every C major triad tone on strings 4–3–2

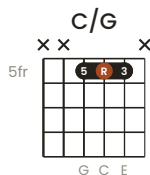


C major on strings 4–3–2

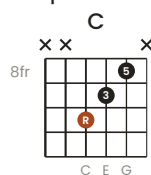
1st inversion



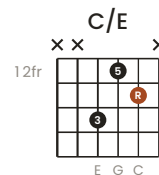
2nd inversion



Root position

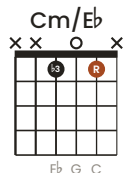


1st inversion

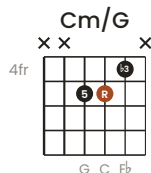


C minor on strings 4–3–2

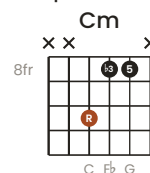
1st inversion



2nd inversion



Root position

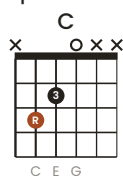


Triads on strings 5–4–3 and 6–5–4

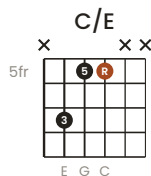
The lower string sets have no G–B pair inside them: all three strings are a 4th apart. The shapes on strings 5–4–3 are therefore exactly the shapes on strings 6–5–4, moved across one string. These low triads sound thick and are used less for strumming, but they matter for bass lines, for classical writing and for seeing where chord tones lie under your barre chords.

C major on strings 5–4–3

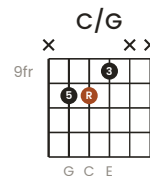
Root position



1st inversion

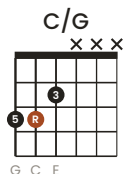


2nd inversion

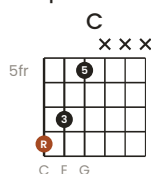


C major on strings 6–5–4

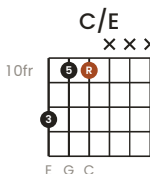
2nd inversion



Root position

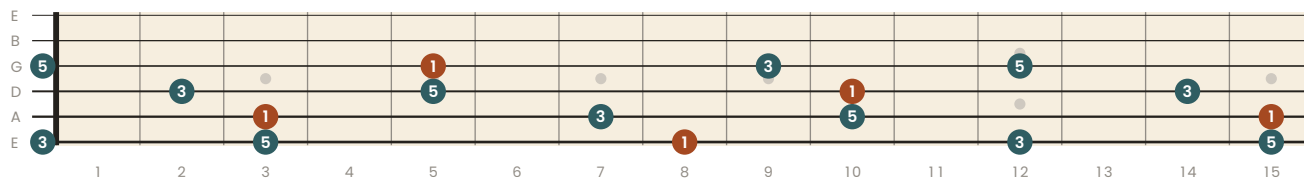


1st inversion



Same three shapes as on strings 5–4–3, one string lower. On 6–5–4 they spell C starting from a different inversion.

All C major triad tones on the lower strings



Pick out three adjacent strings to see the inversions on 6–5–4 or 5–4–3.

CAREFUL · Low triads and muddy sound

Close triads below about the 5th fret on strings 6–5–4 crowd three notes into a low register and can sound muddy, particularly on a steel-string acoustic. Use them for single bass notes, short stabs or with a palm mute, and move up the neck for sustained chords.

Diminished and augmented triads on the neck

The **augmented triad** divides the octave into three equal major 3rds. Its inversions are the same shape moved four frets, and Caug, Eaug and G[♯]aug contain the same three notes. Learn one shape and slide it.

The **diminished triad** is usually heard as part of a larger chord: a dominant 7th without its root (B–D–F inside G7), or the top of a m7b5 or dim7 chord. On strings 3–2–1, B diminished sits at x x x 4 3 1 (root position), x x x 7 6 7 (1st inversion) and x x x 10 12 10 (2nd inversion).

Augmented and diminished on strings 3–2–1



The three augmented shapes are identical and four frets apart. Bdim in root position and 1st inversion.

TRY THIS · Find every inversion of G

Using the fretboards above as a model, find G major (G–B–D) in all three inversions on each of the four string sets. Write the frets down, then check them against your eyes on the neck: every shape must contain one G, one B and one D, on three adjacent strings, within four frets.

Voice leading with triads

Each triad has three notes, or **voices**. Moving from one chord to the next, each voice travels to a note of the new chord. Good voice leading follows two rules:

1. **Keep common tones.** If both chords share a note, keep it on the same string.
2. **Move the other voices by the smallest step.** A half step or whole step is best; avoid leaps.

Played this way, a progression on three strings needs almost no hand movement, and the top voice forms a small melody. Take C – Am – F – G on strings 3–2–1:

C – Am – F – G with the nearest inversions



C to Am keeps C and E and moves G up to A. Am to F keeps C and A and moves E up to F. F to G moves all three voices down: C to B, F to D and A to G.

▶ TRACK 61 Voice-led C – Am – F – G

Diagram illustrating the voice leading for the progression C – Am – F – G on strings 3–2–1. The top voice (T) sings G, A, A, G, G; the middle voice (A) sings E, E, F, D, E. The hand stays between frets 3 and 6.

	C	Am	F	G	C
T	3	5	5	3	3
A	5	5	5	4	5
B	5	5	5	5	5

The top voice sings G, A, A, G, G; the middle voice E, E, F, D, E. The hand stays between frets 3 and 6.

Compare that with the open chords C, Am, F and G, where the hand jumps across the neck and the top voice leaps. Neither is wrong; they are different sounds. Close voice leading sounds smooth and connected, and it leaves room for a bass player and a singer. Chapter 34 adds a third rule (only the bass may leap) and applies all three to 7th chords.

Triads over an open bass

An open string under a moving triad holds the music together. With the open 4th string (D) as a drone, triads on strings 3–2–1 can move freely while the bass stays put. The chords become slash chords, such as G/D, but you only need the triad shapes you already know.

TRACK 62 Triads over an open D drone

Diagram showing triads over an open D drone (4th string) on strings 3–2–1. The chords are D, Em/D, G/D, and D. The fretboard diagram shows the fingerings for the triads on strings 3, 2, and 1 for each chord.

	D	Em/D	G/D	D
T	5	7	10	5
A	7	8	12	7
B	7	9	12	7
B	0	0	0	0

The open D rings under every chord. G/D sounds open and suspended because the bass is the 5th of G.

NYLON · STEEL

Nylon: a triad on the top strings over an open or fretted bass is the basic texture of 19th-century guitar studies by Sor, Giuliani and Carcassi. Practise the triads with p on the bass and i, m, a plucking the three upper strings together.

Steel: high triads are the usual second-guitar part next to someone strumming open chords. At capo 5 or capo 7 they give the bright, separated sound used in many folk and pop recordings.

Chapter 10 review · use it, train it, test it

Using it in real music

Second guitar parts. When a piano or another guitar plays full chords, a triad on strings 3–2–1 or 4–3–2 fills the upper register without crowding the sound.

Funk, soul and reggae rhythm. Short triad stabs on the top three or four strings are the standard rhythm-guitar vocabulary in these styles. Chapter 45 and Chapter 52 build on the shapes in this chapter.

Arranging a melody. Put the melody note on the 1st string and choose the inversion that has it on top. This is the starting point of chord-melody arranging and of Chapter 20.

Reading the bass line. Slash chords such as C/E and G/B in a chart are inversions. Knowing the inversion shapes lets you play them exactly as written.

How to train it

Drill	How	Time	Goal
Inversion ladder	Play C major on one string set in all inversions from the open position to the 15th fret and back; name each inversion aloud	3 min	All four string sets without stopping
Quality switch	Hold any major triad shape; change it to minor, diminished and augmented by moving one note at a time	2 min	Change quality without looking
Key of the day	Pick a major triad; find it on all four string sets in all three inversions	4 min	12 shapes in under 2 minutes
Voice-leading loop	C – Am – F – G on strings 3–2–1, then 4–3–2, keeping each voice within two frets	4 min	No voice moves more than two frets

Check your understanding

Write your answers, then check them in Appendix J.

1. Spell a G major and a D minor triad.

2. Spell a B diminished triad and an A^b augmented triad.

3. Which two intervals, stacked, make a minor triad?

4. A C major triad with E as its lowest note is in which inversion, and how is it written as a slash chord?

5. On strings 3–2–1, the frets x x x 9 8 8 give which C triad shape?

6. Why do triad shapes on strings 4–3–2 differ from those on strings 3–2–1?

7. Why are C^{aug}, E^{aug} and G^{#aug} the same shape moved four frets?

8. State the two rules of voice leading used in this chapter.

MY NOTES

CHAPTER 11

Chords: Open, Barre and Beyond



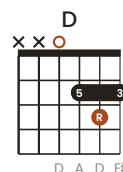
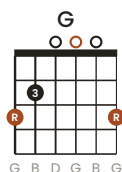
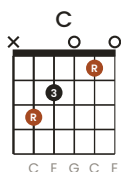
TRACKS

The open chords every guitarist learns first, the movable barre shapes that work in every key, and the 7th, sus, add, 6, 9 and slash chords that give music its colour.

A chord on the guitar is a triad, or a 7th chord, with some notes doubled to fill five or six strings. The open C chord, x32010, contains C, E, G, C and E: a C triad with the root and 3rd doubled. Knowing which degree each finger plays is what lets you change a chord on purpose rather than by accident.

This chapter collects the chord vocabulary used in the rest of the book: open chords, power chords, barre chords, 7th chords, colour chords and slash chords, and ends with a reference library of movable shapes.

What the fingers play: degrees inside the open C, G and D chords

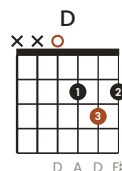
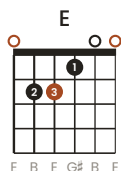
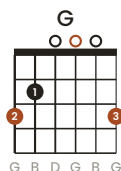
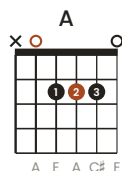
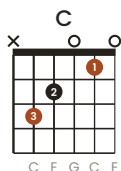


Each chord contains only 1, 3 and 5. The doubled notes add fullness, not new harmony.

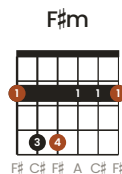
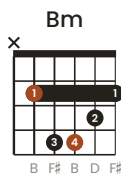
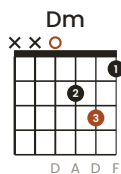
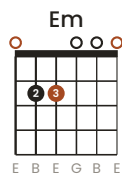
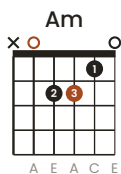
The essential open chords

Open chords use open strings, so they ring longer and sound fuller than any other shape. Learn them in families. The five major chords C, A, G, E and D are also the basis of the CAGED system in Chapter 12.

Major

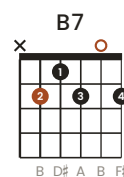
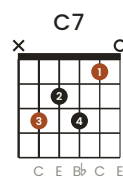
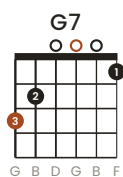
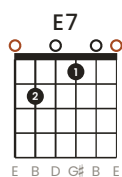
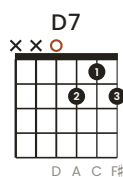
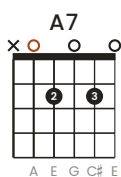


Minor

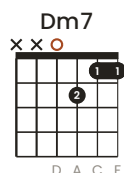
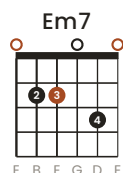
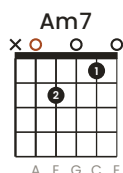
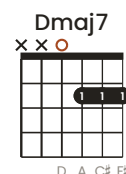
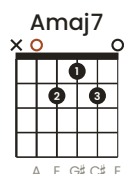
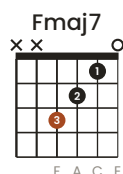
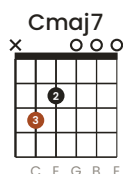


Bm and F#m have no open strings; they are barre shapes, but they appear in so many open-position songs that they belong here.

Dominant 7th



Major 7th and minor 7th

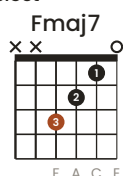


The F chord

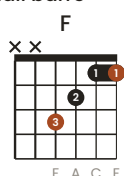
F is the first chord that cannot use the open strings in the usual way, because the open low E and the open B are not in an F chord. Start with the four-string version, then add the full barre when your hand is ready.

F in three stages

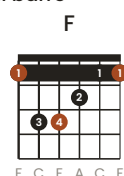
Easiest



Small barre



Full barre



Fmaj7 is a common substitute while you build strength. The small F barres only strings 1 and 2.

TIP · Changing chords cleanly

Look for **anchor fingers** that stay put: C to Am keeps fingers 1 and 2 in place; G to Em7 keeps finger 3 if you play G as 320033. Many players finger the open G with 3, 2 and 4 (fingers 320004) instead of 2, 1 and 3: it makes the change to C quicker. Both fingerings appear in this book. Move all fingers as one unit, not one after another. Practise a single change back and forth for one minute and count the clean changes (a **one-minute change**). Press just behind the fret with the fingertips upright, so they do not touch the neighbouring strings.

▶ TRACK 63 Open-chord changes: G – Em – C – D



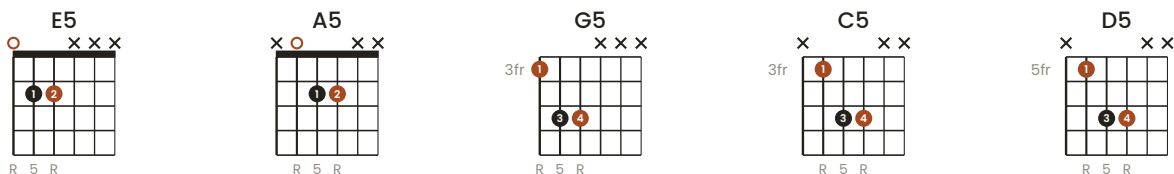
Down-strums on the beats and one up-strum on the last &, where the fretting hand leaves for the next chord. Keep the strumming hand moving in eighth notes throughout, even if the open strings sound for a moment. Chapter 22 builds full patterns on this motion.

Power chords

A **power chord** is the root and the 5th, usually with the root doubled an octave higher. It has no 3rd, so it is neither major nor minor and fits over either. The symbol is the root with a 5: E5, A5, C5. Power chords are the rhythm vocabulary of rock and metal, especially with distortion, which makes full major and minor chords sound muddy.

The same two- or three-finger shape moves anywhere on the neck, with the root on the 6th or the 5th string.

Power chords: open and movable



The movable shape is root, 5th, octave. Mute the unused strings with the underside of the 1st finger.

TRACK 64 Power-chord riff in E

Musical notation for a power-chord riff in E major. The riff consists of two measures of four eighth notes each, with power chords E5, G5, A5, E5, D5, and C5. The notation includes a treble clef, a key signature of one sharp (F#), and a 4/4 time signature. The fretboard diagrams below the staff show the fingerings for each chord.

Downstrokes throughout. On electric guitar, palm-mute the eighth notes: rest the picking hand lightly on the strings at the bridge.

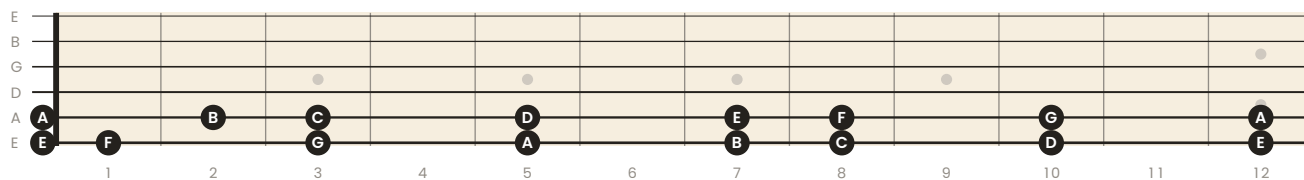
Barre chords

A barre chord is an open chord shape moved up the neck, with the 1st finger lying across the strings in place of the nut. Two shapes do most of the work:

- The **E shape**: the open E chord fingered with 2, 3 and 4, barred with the 1st finger. Root on the 6th string.
- The **A shape**: the open A chord moved up, with the root on the 5th string.

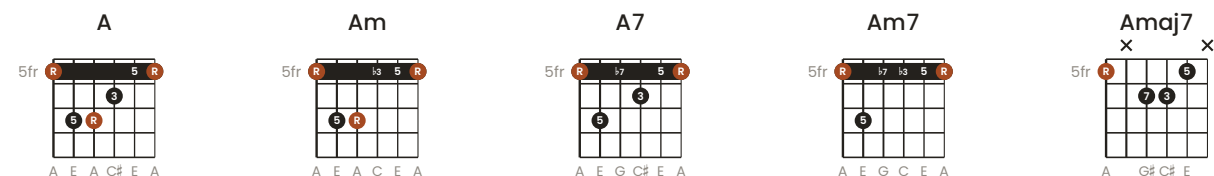
To play any major or minor chord, find its root on the 6th or 5th string and put the shape there.

Roots for barre chords: natural notes on strings 6 and 5



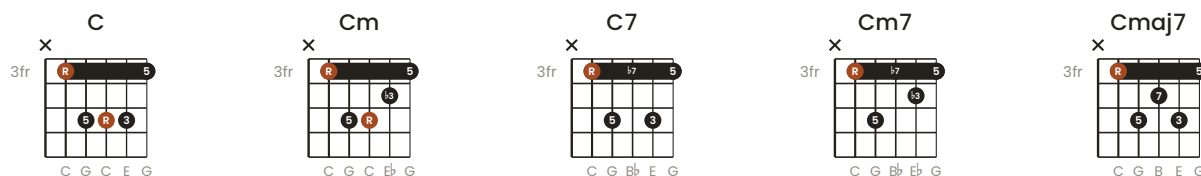
Learn these and you can place any E-shape or A-shape barre chord. Sharps and flats are one fret either side. Chapter 4 covers the whole neck.

E-shape family at the 5th fret (A chords)



Major to minor lifts the 2nd finger (the 3rd). Major to 7 lifts the 4th finger (the octave on the 4th string drops to the b7 under the barre).

A-shape family at the 3rd fret (C chords)



The A-shape major is often played with the 3rd finger barring strings 4–2, leaving the 1st string muted or lightly touched.

TIP · Making a barre sound

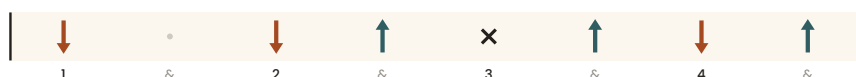
Roll the 1st finger slightly onto its bony outer edge and keep it close to the fret. Let the weight of the arm pull the neck back gently instead of squeezing with the thumb. Then check each string one at a time and adjust. Start around the 5th to 7th fret, where the strings are closer to the frets, and move down towards F at the 1st fret over several days.

NYLON · STEEL

Nylon: the wide neck spreads the shape out, but the low tension needs less pressure. Classical music uses the **half barre** (written $\frac{1}{2}C$ or CII) across three to five strings far more than the full barre.

Steel: strings need much more pressure. Light-gauge strings and a well set-up nut and action make the biggest single difference to barre chords on a steel-string acoustic.

▶ TRACK 65 Barre progression in F



One bar each of F (E shape, 1st fret), Dm (A shape, 5th fret), B $\flat$ (A shape, 1st fret) and C (A shape, 3rd fret). The x is a muted chuck: release the pressure of the fretting hand for that strum.

Seventh chords

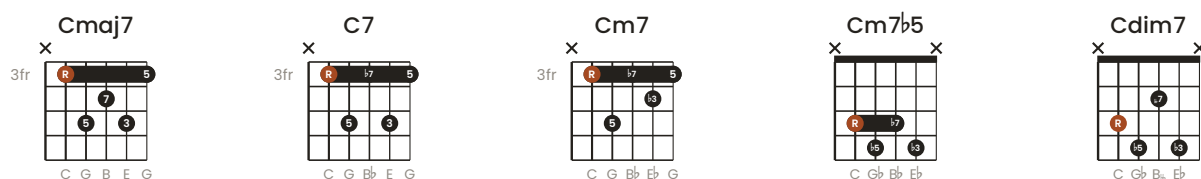
Add another 3rd on top of a triad and you get a **7th chord**: four different notes. 7th chords are the basic harmony of jazz, soul, bossa nova and the blues.

Chord	Formula	In C	Symbol	Use
Major 7th	1 3 5 7	C E G B	Cmaj7	Soft; tonic chord in jazz
Dominant 7th	1 3 5 b7	C E G B $\flat$	C7	Tense; resolves a 5th down
Minor 7th	1 b3 5 b7	C E $\flat$ G B $\flat$	Cm7	Mellow; ii in major keys
Half-diminished	1 b3 b5 b7	C E $\flat$ G $\flat$ B $\flat$	Cm7b5	Dark; ii in minor keys
Diminished 7th	1 b3 b5 $\flat\flat$ 7	C E $\flat$ G $\flat$ B $\flat\flat$ (= A)	Cdim7	Very tense; repeats every 3 frets
Minor-major 7th	1 b3 5 7	C E $\flat$ G B	CmMaj7	Dark, unsettled; film, jazz

The words in the symbol are precise. **maj** refers only to the 7th: Cmaj7 has a major 3rd and a major 7th. A plain 7 always means the b7. **m** always means the minor 3rd. So Cm7 has a minor 3rd and a b7, and CmMaj7 has a minor 3rd and a major 7th.

Jazz charts often use other symbols for the same chords: C Δ 7 for Cmaj7, C-7 for Cm7, C $\circ$ 7 for Cm7b5, C $\circ$ 7 for Cdim7 and Cm(maj7) for CmMaj7. This book uses the symbols in the table.

Five 7th chords on C, root on the 5th string



Same root, one note different each time. In Cdim7 the $\flat\flat$ 7 is the note A: the chord is built from minor 3rds only.

UNDER THE HOOD · The diminished 7th repeats every three frets

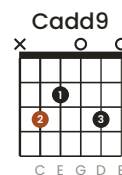
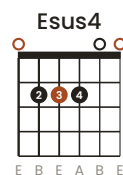
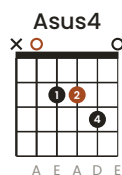
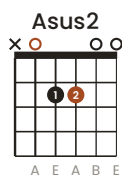
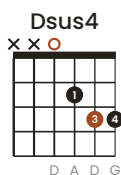
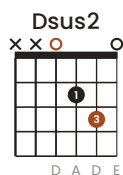
A dim7 chord stacks three minor 3rds, and a fourth minor 3rd brings you back to the root an octave up. All four notes are therefore equally spaced, and moving the shape three frets gives the same four notes in a new order. Cdim7, E♭dim7, F♯dim7 and Adim7 contain the same notes. There are only three different dim7 chords.

Colour chords: sus, add, 6 and 9

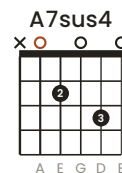
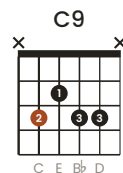
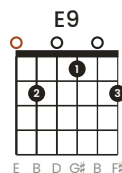
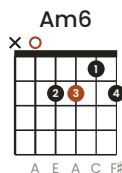
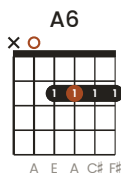
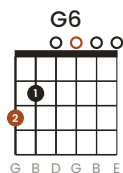
These chords add a note to a triad, or replace its 3rd, to change the colour without changing the basic harmony.

Chord	Formula	In C	Use
sus2 (suspended 2nd)	1 2 5	C D G	No 3rd; open, neither major nor minor
sus4 (suspended 4th)	1 4 5	C F G	No 3rd; the 4th usually falls to the 3rd
add9 (added 9th)	1 3 5 9	C E G D	Bright; no 7th; common in pop and folk
6 (sixth)	1 3 5 6	C E G A	Sweet, settled; replaces a major tonic
m6 (minor sixth)	1 ♭3 5 6	C E♭ G A	Minor with a bright edge; bossa, gypsy jazz
6/9 (six-nine)	1 3 5 6 9	C E G A D	Rich major; jazz and bossa nova endings
9 (dominant 9th)	1 3 5 ♭7 9	C E G B♭ D	Dominant 7th plus 9th; funk, soul, blues
7sus4 (7th, suspended 4th)	1 4 5 ♭7	C F G B♭	Dominant without a 3rd; folk, pop, modal jazz

Open-position colour chords



Sixth and ninth chords



C9 is shown without its 5th, the usual way to play it. A barre of the 3rd finger across strings 4–2 is also common.

▶ TRACK 66 Suspensions around D: D – Dsus4 – D – Dsus2 – D

D Dsus4 D Dsus2 D

	2	2	3	3	2	2	0	0	2
T	3	3	3	3	3	3	3	3	3
A	2	2	2	2	2	2	2	2	2
B	0	0	0	0	0	0	0	0	0

Only the 1st string moves: the 4th falls to the 3rd (G to F♯), the 2nd rises to it (E to F♯). A staple of folk and rock rhythm guitar.

Reading any chord symbol

Chord symbols follow a fixed order: **root + quality + extension + alterations + /bass**. A few rules cover nearly everything.

- A letter alone means a major triad: G = G B D.
- **m** means a minor 3rd: Gm = G B $\flat$ D.
- **maj** refers only to the 7th: Gmaj7 = G B D F $\sharp$.
- A plain 7, 9, 11 or 13 means a dominant chord, so the $\flat 7$ is included: G9 = G B D F A.
- **add** adds a note without the 7th: Gadd9 = G B D A.
- **sus** replaces the 3rd: Gsus4 = G C D, Gsus2 = G A D.
- **b** and **#** with a number alter that note: G7 $\flat 9$ = G B D F A $\flat$.
- A **slash** names the bass note: G/B is a G chord with B as the lowest note.

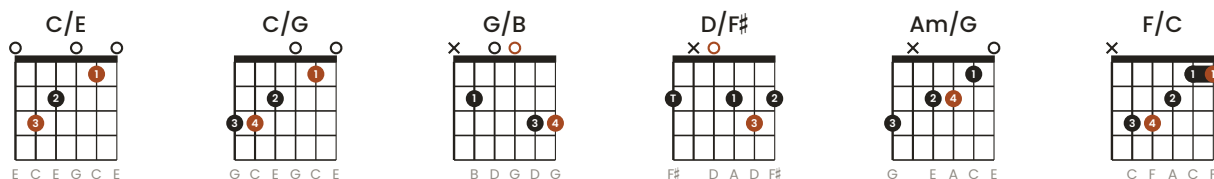
Symbol	Notes	Read as
G	G B D	G major
Gm7	G B $\flat$ D F	G minor 7th
Gmaj7	G B D F $\sharp$	G major 7th
G7	G B D F	G dominant 7th
G9	G B D F A	G dominant 9th
Gadd9	G B D A	G major with an added 9th
G6	G B D E	G sixth
Gm7 $\flat 5$	G B $\flat$ D $\flat$ F	G half-diminished
G7 $\sharp 9$	G B D F A $\sharp$	G dominant 7th with a sharp 9th
G/B	B, then G B D	G major in 1st inversion

Upper extensions (9, 11, 13) and altered dominants are developed in Chapter 48.

Slash chords

A slash chord names a chord and, after the slash, its lowest note. Sometimes the bass note is in the chord, which makes it an inversion: C/E, G/B. Sometimes it is not: Am/G is an A minor chord over a G bass, which sounds as Am7 with the 7th in the bass. Either way, play the named chord and make sure the bass note is the lowest note you sound.

Common slash chords



D/F $\sharp$: fret the F $\sharp$ with the thumb over the neck, or with the 1st finger (fingers 1x0243). Mute the 5th string in D/F $\sharp$ and Am/G.

Slash chords make bass lines walk. Under a series of ordinary chords, the bass can move down step by step:

TRACK 67 Descending bass with slash chords

C G/B Am Am/G Fmaj7 C/E Dm7 G C

	0	3	0	0	0	0	1	3	0
T	1	3	1	1	1	1	1	0	1
A	0	0	2	2	2	0	2	0	0
B	2	0	2	2	3	2	0	0	2
	3	2	0	3	1	0	3	2	3

The bass falls C, B, A, G, F, E, D, then G to C. This descending line under familiar chords is common in pop ballads and classical guitar alike.

HOW IT CONNECTS

Slash chords are inversions or bass pedals. Chapter 10 shows the inversion shapes on three strings; Chapter 14 uses descending and ascending bass lines as complete progressions.

Reference library: movable shapes

Every shape below is movable: slide it so the root (rust) is on the note you want. Root-on-6th-string shapes are shown with G at the 3rd fret, root-on-5th-string shapes with C at the 3rd fret.

Root on the 6th string (G)

G G D G B D G	Gm G D G Bb D G	G7 G D F B D G	Gm7 G D Fb Bb D G	Gmaj7 G F# B D	G5 G D G
G7 G F B	Gm7 G F Bb	Gmaj7 G F# B	Gm7b5 G F Bb Db	G13 G F B E	G6 G E B E

Root on the 5th string (C)

C C G C E G	Cm C G C Eb G	C7 C G Bb E G	Cm7 C G Bb Eb G	Cmaj7 C G B E G	C5 C G C
Cm7b5 C Gb Bb Eb	Cdim7 C Gb Bb Eb	C9 C E Bb D G	Cm9 C Eb Bb D	C7#9 C E Bb D#	C6/9 C E A D G

Shell voicings (root, 3rd and 7th only) and the 9th, 13th and 7#9 shapes are the basis of jazz, funk and blues comping.

Chapter 11 review · use it, train it, test it

Using it in real music

Playing in any key. A song in B $\flat$ or F $\sharp$ has few comfortable open chords. With E-shape and A-shape barre chords you can play it anyway, or use a capo (see Chapter 13).

Choosing a voicing. Open chords ring and suit strumming and fingerpicking. Barre chords sound tighter and mute when you release pressure, which suits funk and rock rhythm. Power chords suit distortion.

Adding colour. Replace plain chords with maj7, m7, sus and add9 versions to make a simple progression sound richer. Keep the root and the function; change only the colour.

Following the bass. When a chart shows slash chords, the bass line usually matters to the song. Play the written bass note even if you simplify the chord above it.

How to train it

Drill	How	Time	Goal
One-minute changes	Two chords back and forth for 60 seconds	3 min	60 clean changes a minute
Barre build-up	Hold an E-shape barre at the 7th fret, check each string, release. Move one fret lower each day	3 min	Clean F at the 1st fret
Family switch	On one barre, cycle major, minor, 7, m7, maj7	3 min	Five qualities in 10 seconds
Chord of the day	Learn one new chord; use it in a song you know	3 min	A growing, usable vocabulary

Check your understanding

Write your answers, then check them in Appendix J.

1. Spell G7 and Gmaj7.

2. What is the difference between Cmaj7 and C7?

3. An E-shape major barre chord at the 3rd fret is which chord?

4. An A-shape minor barre chord at the 7th fret is which chord?

5. Which notes does C5 contain, and why does it fit over both C major and C minor?

6. What does D/F $\sharp$ mean?

7. Spell Csus4 and Cadd9.

8. Why do Cdim7 and E $\flat$ dim7 contain the same notes?

MY NOTES

CHAPTER 12

The CAGED System



TRACKS

12

The five open chord shapes C, A, G, E and D can each be moved up the neck. In any key they follow each other in the same order and together cover the whole fretboard, linking chords, arpeggios and scales into one map.

Every major chord can be played with any of the five open shapes C, A, G, E and D, moved up the neck. Where the open chord used the nut, a barre or a finger takes its place. The idea is not new: the E-shape and A-shape barre chords in Chapter 11 are two of the five. CAGED adds the other three and shows how all five fit together.

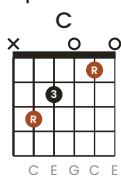
The five shapes always follow each other in the same order going up the neck: C, A, G, E, D, then C again an octave higher. Neighbouring shapes share notes, so the end of one shape is the start of the next.

The five shapes in C

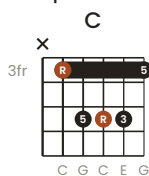
Here is C major played with all five shapes, from the open position up to the 13th fret.

C major in all five CAGED shapes

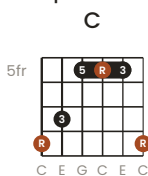
C shape



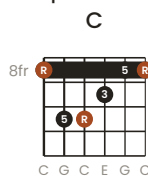
A shape



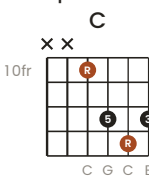
G shape



E shape



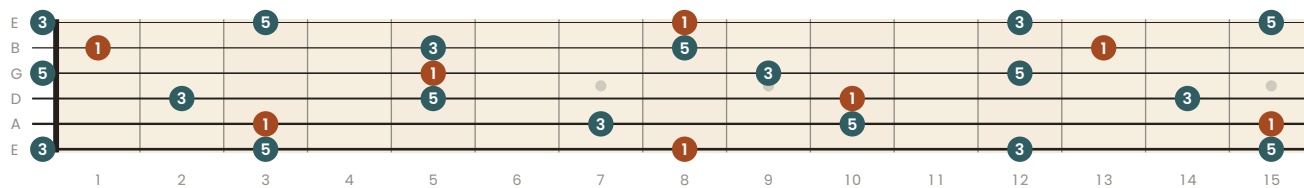
D shape



Open position, 3rd fret, 5th fret, 8th fret, 10th fret. After the D shape comes the C shape again at the 12th fret.

Put the five shapes together and you have every C, E and G on the neck. Each shape is one window onto this map.

Every C major chord tone from the nut to the 15th fret



Shaded frets mark where each shape's root sits: C shape and A shape at fret 3 (5th string), G shape and E shape at fret 8 (6th string), D shape at fret 10 (4th string).

Shape	Root strings	Root under	Usual fingering
C shape	5th (and 2nd)	4th finger	Partial barre; often strings 5–2 only
A shape	5th (and 3rd)	1st-finger barre	Barre; 3rd finger across strings 4–2
G shape	6th (and 3rd, 1st)	4th finger	Awkward in full; usually played in part
E shape	6th (and 4th, 1st)	1st-finger barre	Full barre
D shape	4th (and 2nd)	1st finger	Strings 4–1 only

SHORTCUT · The order never changes

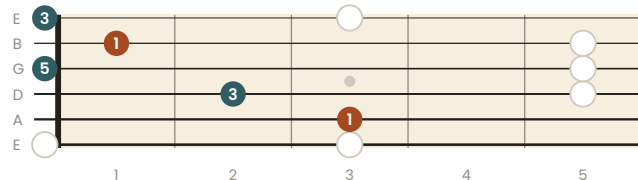
The word CAGED gives the order going up the neck. Start from any shape and continue the word, wrapping round from D to C. An E-shape G chord at the 3rd fret is followed by a D-shape G at the 5th fret, a C-shape G at the 7th, an A-shape G at the 10th and a G-shape G at the 12th (the open G chord an octave up).

How the shapes connect

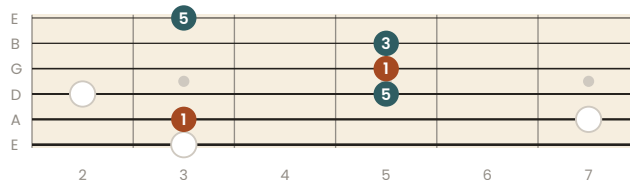
The highest notes of one shape are the lowest notes of the next. The C shape ends on the root at the 3rd fret of the 5th string, and the A shape begins with its barre on that same root. The A shape's notes at the 5th fret (5th, root and 3rd on strings 4-2) are the bottom of the G shape. Learn each join and the neck stops being five separate shapes.

The five C shapes inside the full map of C chord tones

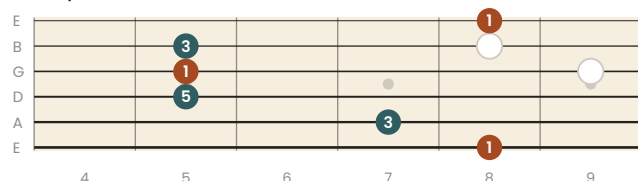
C shape (frets 0–3)



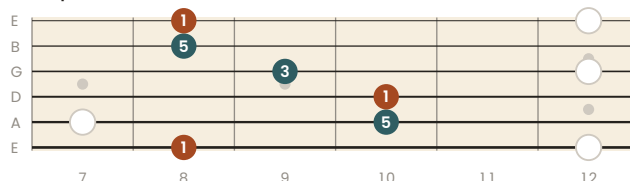
A shape (frets 3–5)



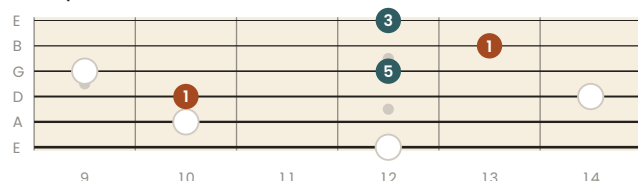
G shape (frets 5–8)



E shape (frets 8–10)



D shape (frets 10–13)



Grey dots are chord tones belonging to the neighbouring shapes. Each window overlaps the next by one fret or so.

TRACK 68 C major up the neck through the five shapes

C

	0	3	8	8	12	12	0
T	1	5	5	8	13	13	1
A	0	5	5	9	12	12	0
B	2	5	5	10	10	14	2
	3	3	7	10	15	3	

The same chord six times. Listen for how the register and the top note change while the harmony stays the same.

Finding the shapes in any key

Each shape is fixed to its root. Find the root on the right string and the shape follows.

Key	C shape	A shape	G shape	E shape	D shape
C	open	3	5	8	10
D	2	5	7	10	open
E	4	7	9	open	2
F	5	8	10	1	3
G	7	10	open	3	5
A	9	open	2	5	7
B♭	10	1	3	6	8

The numbers give the lowest fret the shape uses (the barre or the nut). Read any row from the smallest number upwards and you get the CAGED order starting from a different letter: in G, the order from the nut is G, E, D, C, A.

TRY THIS · One key, five shapes

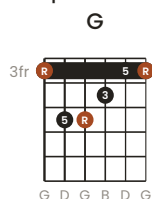
Choose a key from the table, such as A. Play the chord in all five shapes from the nut upwards, saying the shape name aloud. Then play them in reverse. Within a week you will see the neck as five linked zones instead of fifteen separate frets.

Three positions for one progression

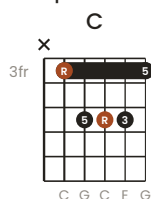
CAGED lets you play a whole progression without leaving one area of the neck. Here is I – IV – V in G in three positions: each chord uses whichever shape lies nearest.

I – IV – V in G, frets 2–5

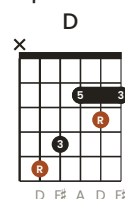
E shape



A shape

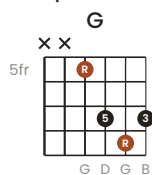


C shape

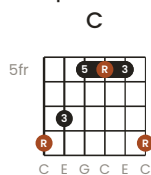


I – IV – V in G, frets 5–8

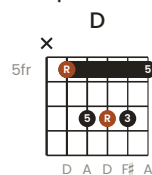
D shape



G shape

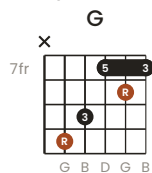


A shape

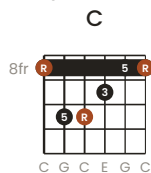


I – IV – V in G, frets 7–10

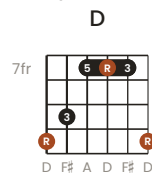
C shape



E shape



G shape



▶ TRACK 69 I – IV – V in G, low and high positions

G C D G G C D G

	3	3	2	3	7	8	10	7
T	3	5	3	3	8	8	7	8
A	4	5	2	4	7	9	7	7
B	5	5	4	5	9	10	7	9
	5	3	5	5	10	10	9	10
	3			3		8	10	

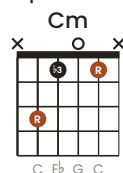
Bars 1–2 stay between frets 2 and 5; bars 3–4 between frets 7 and 10. The second position is brighter and suits a second guitar part.

CAGED for minor chords

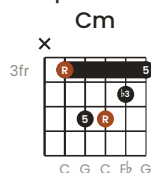
Lower every 3rd in a major shape by one fret and you get the minor version. Three of the minor shapes are familiar open chords (Am, Em, Dm). The C-shape and G-shape minors are awkward as full chords, so players use parts of them.

C minor in the five shapes

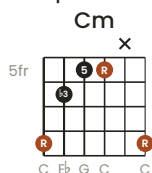
C shape



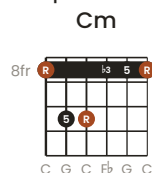
A shape



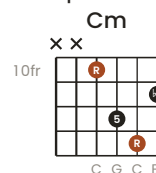
G shape



E shape



D shape



Compare with the major row above. Only the dots labelled b3 have moved.

Scales around the shapes

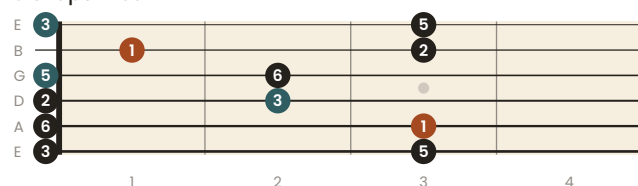
Each chord shape sits inside one pentatonic box and one position of the major scale. Find the chord and you have found the scale that surrounds it: the chord tones are the strongest notes to land on, and the scale notes around them are the ones to move through.

Major pentatonic

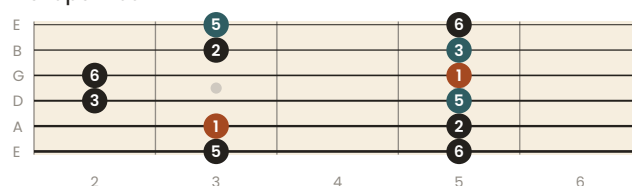
The C major pentatonic scale (C D E G A) adds only the 2nd and 6th to the chord tones. The five boxes of Chapter 7 line up with the five CAGED shapes. The box numbers are those of the relative minor (A minor pentatonic), as in that chapter.

C major pentatonic around each CAGED shape

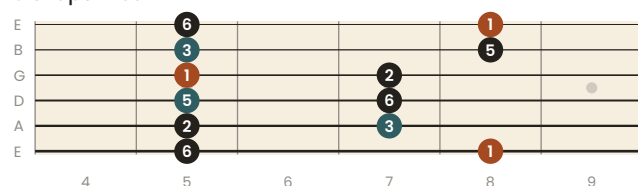
C shape + box 4



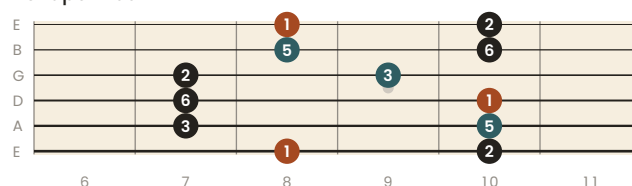
A shape + box 5



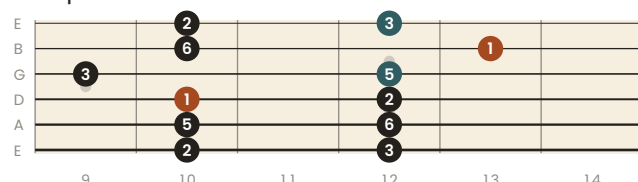
G shape + box 1



E shape + box 2



D shape + box 3



Rust = root, teal = 3rd and 5th, ink = the added 2nd and 6th.

TRACK 70 Pentatonic phrase around the E-shape C chord

C

Every note comes from box 2. The phrase starts on the 3rd (E) and lands on the 3rd at the bottom of the shape before the final chord.

Major scale positions

Fill in the 4th and 7th and you have the full major scale in the same position. The fret window around each shape is the "position" used in Chapter 5.

C major scale around the E shape and the A shape

E shape (frets 7–10)

A shape (frets 2–5)

The chord shape is the skeleton of the scale position. Play the scale, then the chord, then the scale again, until you see both at once.

Arpeggio shapes

Remove everything except the chord tones in the same window and you have an arpeggio: the chord played one note at a time. Each CAGED shape gives one arpeggio fingering. Chapter 16 develops them, including the 7th-chord versions.

C major arpeggio in each CAGED position

C shape

A shape

G shape

E shape

D shape

Every C, E and G within each four-fret window. Play from the lowest note to the highest and back.

NYLON · STEEL

Nylon: the neck joins the body at the 12th fret, so the shapes above about the 10th fret are hard to reach. In practice CAGED on a classical guitar covers the open position to the 10th fret, and the fingerboard positions of classical method books (I, II, V, VII) correspond closely to the windows in this chapter.

Steel and electric: with a 14-fret neck or a cutaway you can use all five shapes in most keys, and the higher shapes become the usual places to solo.

Chapter 12 review · use it, train it, test it

Using it in real music

Same chord, new register. If a second guitar or a chorus needs a higher G, CAGED shows you four more Gs up the neck. Choose the one whose top note fits the melody.

Staying in one position. In a song that moves between I, IV and V, play every chord in the nearest shape instead of jumping between open chords. Your hand stays still and the voices connect, as in Chapter 10.

Soloing over changes. Each CAGED shape sits inside a scale position. When the chord changes, find its shape near your hand and aim for its chord tones. Chapter 35 builds on this.

Playing without a capo. You can play a song in its original key higher up the neck using the higher shapes, instead of moving a capo.

How to train it

Drill	How	Time	Goal
CAGED ladder	Play one major chord in all five shapes from the nut to the 12th fret, naming each shape	3 min	All five shapes without pausing
Random key	Pick a random major chord; find it in all five shapes; check against the table	3 min	Under one minute per key
Chord and box	Strum a shape, then play the pentatonic box around it, then the chord again	4 min	See chord and scale as one picture
One position	Play I – IV – V in a key in each of the three positions shown in this chapter	3 min	No chord change takes longer than one beat

Check your understanding

Write your answers, then check them in Appendix J.

1. In the CAGED order going up the neck, which shape follows the E shape?

2. A C major chord with its root at the 8th fret of the 6th string, barred, uses which shape?

3. You play G with the E shape at the 3rd fret. Which shape gives the next G up the neck, and at which fret does it start?

4. On which string is the root of the D shape?

5. How do you turn a CAGED major shape into its minor version?

6. Which pentatonic box surrounds the E-shape chord, using the box numbers of the relative minor?

7. At which fret does the A-shape A chord start, and why?

CHAPTER 13

Chords in a Key



TRACKS

13

Build a chord on every note of a scale, using only the notes of that scale, and you get the chords that belong to the key. Roman numerals name them in every key at once, and the same numbers make transposing and using a capo simple.

A key is a scale used as home base. The chords that belong to the key are built from that scale alone: start on each note in turn and stack two 3rds above it, using only scale notes. This is called **harmonising** the scale, and the chords it produces are the **diatonic** chords of the key.

Seven notes give seven chords. Their qualities follow the same pattern in every major key, which is why a song's key tells you most of its chords before you hear it.

Harmonising the major scale

Stack 3rds on each note of C major (C D E F G A B):

Degree	Notes	Chord	Quality	Numeral
1	C E G	C	major	I
2	D F A	Dm	minor	ii
3	E G B	Em	minor	iii
4	F A C	F	major	IV
5	G B D	G	major	V
6	A C E	Am	minor	vi
7	B D F	Bdim	diminished	vii°

The pattern **major – minor – minor – major – major – minor – diminished** never changes for a major key. It comes from the order of whole and half steps in the major scale: each chord's 3rds are major or minor depending on where the half steps (E–F and B–C in C major) fall.

Roman numerals name each chord by its degree: upper case for major, lower case for minor, ° for diminished. They let you talk about chords independently of the key. "ii – V – I" means Dm – G – C in C major and Am – D – G in G major.

▶ TRACK 71 The diatonic triads of C major, root position on strings 5–4–3

C Dm Em F G Am Bdim C C

	0	2	4	5	7	9	10	12	0
T									
A	2	3	5	7	9	10	12	14	2
B	3	5	7	8	10	12	14	15	3

The same three-string shape family climbs the scale. Listen for the change of colour on Dm, Em and Am, and the tension of Bdim.

The seven chords of C major with their numerals

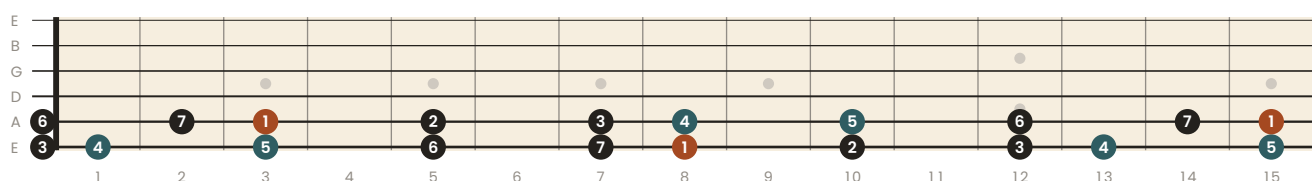
C I	Dm ii	Em iii	F IV	G V	Am vi	Bdim vii°	C I
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The chords in the common keys

Key	I	ii	iii	IV	V	vi	vii°
C	C	Dm	Em	F	G	Am	Bdim
G	G	Am	Bm	C	D	Em	F#dim
D	D	Em	F#m	G	A	Bm	C#dim
A	A	Bm	C#m	D	E	F#m	G#dim
E	E	F#m	G#m	A	B	C#m	D#dim
F	F	Gm	Am	Bb	C	Dm	Edim
Bb	Bb	Cm	Dm	Eb	F	Gm	Adim
Eb	Eb	Fm	Gm	Ab	Bb	Cm	Ddim

The guitar's open strings favour the sharp keys G, D, A and E: their I, IV and V chords are all open chords. In flat keys, barre chords or a capo are the usual answer.

Diatonic chord roots in C major on strings 6 and 5



Each dot is the root of a diatonic chord: an E-shape chord on the 6th string or an A-shape chord on the 5th. Rust and teal (1, 4, 5) are major chords; ink (2, 3, 6) minor; 7 diminished.

SHORTCUT · Four chords for most songs

The three major chords I, IV and V, plus the minor vi, carry a very large share of folk, pop, rock and country songs. In G that is G, C, D and Em; in D it is D, G, A and Bm. If you know a song's key, try these four first when learning it by ear.

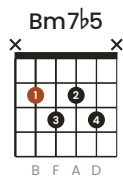
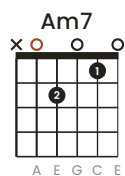
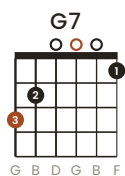
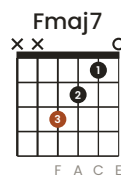
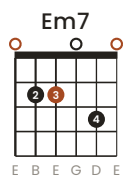
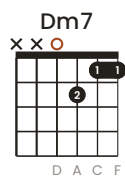
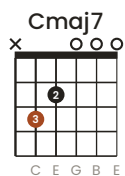
Seventh chords in a key

Stack one more 3rd on each triad and you get the diatonic 7th chords. The pattern in every major key is:

Imaj7 – ii7 – iii7 – IVmaj7 – V7 – vi7 – vii°7

In C: Cmaj7, Dm7, Em7, Fmaj7, G7, Am7, Bm7b5. (A lower-case numeral with 7, such as ii7, means a minor 7th chord.) Only one chord in the key is a dominant 7th: the V7. That is why a dominant 7th chord points so strongly to a key: G7 occurs only in C major (and as V7 in C minor).

The diatonic 7th chords of C major in open position



TRACK 72 Diatonic 7th chords in C

Chords: Cmaj7, Dm7, Em7, Fmaj7, G7, Am7, Bm7^b5, Cmaj7

	Cmaj7	Dm7	Em7	Fmaj7	G7	Am7	Bm7 ^b 5	Cmaj7
T	0	1	0	0	1	0	3	0
A	0	2	0	1	0	0	2	0
B	2	0	2	2	0	2	3	2
	3		2	3	2	0	2	3

Compare with the triads. Every chord sounds softer and fuller, apart from G7 and Bm7^b5, which sound unresolved.

UNDER THE HOOD · Why V7 pulls to I

G7 contains B and F, a tritone (three whole steps) apart. That interval is unstable, and in the key of C each note has a half step to move to: B rises to C, F falls to E. Those two notes are the root and 3rd of C. The pull is so strong that V7 – I is the standard way to end a phrase. Bdim (vii°) contains the same B and F, which is why it has the same dominant function.

Chord function: home, away and tension

Each diatonic chord has a **function**, a role in the movement of the music.

Function	Chords	Contains	Feeling
Tonic (T)	I, also vi and iii	The tonic note, no tension	Rest, arrival, home
Subdominant (S)	IV, also ii	The 4th degree	Moving away, opening up
Dominant (D)	V, V7, vii°	The 7th degree (leading note) and usually the 4th	Tension that wants to return to I

A great deal of music follows the cycle **tonic** → **subdominant** → **dominant** → **tonic**. Chords of the same function can replace each other: vi for I, ii for IV. The progression keeps its sense while the colour changes.

The basic harmonic cycle



TRACK 73 Same functions, different chords

Chords: C, F, G7, C, Am, Dm, G7, C

	C	F	G7	C	Am	Dm	G7	C
T	0	1	1	0	0	1	1	0
A	1	1	0	1	1	3	0	1
B	2	3	0	2	2	0	0	2
	3		2	3	0		2	3

Bars 1–4: I – IV – V7 – I. Bars 5–8: vi – ii – V7 – I. The second version replaces the tonic with vi and the subdominant with ii: the shape of the phrase is the same, the colour is darker.

Chords in minor keys

Harmonising the natural minor scale gives a different pattern:

i – ii° – $\flat$ III – iv – v – $\flat$ VI – $\flat$ VII

In A minor: Am, Bdim, C, Dm, Em, F, G. These are the same seven chords as C major, because A minor and C major share the same notes (they are **relative** keys). What changes is the home chord.

In practice the v chord is usually made major (V, often V7), borrowed from the harmonic minor scale. Raising the 7th degree (G to G# in A minor) creates a leading note a half step below the tonic, and E or E7 pulls back to Am far more strongly than Em does.

Minor key	i	ii°	$\flat$ III	iv	v / V	$\flat$ VI	$\flat$ VII
Am	Am	Bdim	C	Dm	Em / E	F	G
Em	Em	F#dim	G	Am	Bm / B	C	D
Dm	Dm	Edim	F	Gm	Am / A	B $\flat$	C
Bm	Bm	C#dim	D	Em	F#m / F#	G	A
Gm	Gm	Adim	B $\flat$	Cm	Dm / D	E $\flat$	F

A minor with the major V

Am <i>i</i>	Dm <i>iv</i>	E7 V7	Am <i>i</i>	F $\flat$ VI	G $\flat$ VII	E7 V7	Am <i>i</i>
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The numerals are measured from the tonic A, so F and G show as $\flat$ VI and $\flat$ VII. E7 is the V7 borrowed from A harmonic minor.

Chapter 31 treats the three minor scales and their chords in detail.

Numbers instead of letters

Because the pattern is the same in every key, a progression written in numbers works in any key. Session musicians in Nashville write charts as plain numbers: 1 – 5 – 6m – 4 instead of G – D – Em – C. Roman numerals do the same job in theory books, and this book uses them throughout.

A progression in G

G <i>I</i>	D <i>V</i>	Em <i>vi</i>	C <i>IV</i>
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The same progression in D: the numerals do not change

D <i>I</i>	A <i>V</i>	Bm <i>vi</i>	G <i>IV</i>
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The circle of fifths (Chapter 5) shows how far apart two keys are. Neighbouring keys share six of their seven notes and four of their seven diatonic triads (C and G major share C, Em, G and Am), so moving to a neighbour is the smoothest change.

Transposing

To move a song into another key:

1. Find the song's key and write each chord as a number.
2. Write out the chords of the new key from the table above.
3. Replace each number with the chord of the new key. Keep the quality and any additions (7, sus4, /bass).

Numbers	In G	In A	In C	In E
I – V – vi – IV	G D Em C	A E F#m D	C G Am F	E B C#m A
ii – V – I	Am D G	Bm E A	Dm G C	F#m B E
vi – IV – I – V7	Em C G D7	F#m D A E7	Am F C G7	C#m A E B7

Using a capo

A capo clamps all six strings at one fret and acts as a movable nut. You play the same chord shapes, and everything sounds higher by the number of frets. Each fret raises the key by a half step.

Capo fret	0	1	2	3	4	5	7
G shapes sound in	G	A $\flat$	A	B $\flat$	B	C	D
C shapes sound in	C	D $\flat$	D	E $\flat$	E	F	G
D shapes sound in	D	E $\flat$	E	F	F $\sharp$	G	A
E shapes sound in	E	F	F $\sharp$	G	A $\flat$	A	B
A shapes sound in	A	B $\flat$	B	C	D $\flat$	D	E

Read the table the other way round to choose a capo for a song. A song in B $\flat$ can be played with A shapes at capo 1, or with G shapes at capo 3. A song in E $\flat$ works with D shapes at capo 1 or C shapes at capo 3. Pick the shapes that suit the song: G and C shapes give a bright, ringing sound with bass runs; D shapes put the melody notes high; E and A shapes suit blues and rock.

TRACK 74 G shapes at capo 2, sounding in A
Capo 2 (TAB numbers count from the capo)

Chord symbols: A, D, E, F $\sharp$ m, D, E, A

TAB numbers (count from capo):

	Chord 1 (A)	Chord 2 (D)	Chord 3 (E)	Chord 4 (F $\sharp$ m)	Chord 5 (D)	Chord 6 (E)	Chord 7 (A)
T	3	0	0	0	0	2	3
A	0	0	2	2	0	2	0
B	2	3	0	2	3	0	2

You finger G, C, D and Em; the chord symbols show what sounds. The TAB numbers count from the capo.

A capo also lets two guitarists play the same key with different shapes: one plays open G shapes at capo 2, the other open A shapes without a capo. Both are in A, but the voicings differ, so the two parts do not blur together.

NYLON · STEEL

Steel: capos are standard equipment for singers who need a song in a particular key, and for keeping the ringing open-string sound in keys such as A $\flat$ or B $\flat$.

Nylon: in flamenco the capo (*cejilla*) is used constantly to suit the singer. In classical concert music it is rare. Use a capo made for a flat fingerboard on a classical guitar; a capo curved for a steel-string neck leaves the middle strings buzzing.

Chapter 13 review · use it, train it, test it

Using it in real music

Learning songs by ear. Find the key from the last chord or the bass note the song keeps returning to. Then try the seven diatonic chords, starting with I, IV, V and vi. Guessing from seven chords is far quicker than from all twelve roots.

Transposing for a singer. Write the song in numbers, then rebuild it in the singer's key, or keep your shapes and use the capo table.

Substituting chords. Keep the function, change the colour: vi instead of I, ii instead of IV, or a 7th chord in place of a triad. Chapter 14 shows these substitutions in full progressions.

How to train it

Drill	How	Time	Goal
Key charts	Write the seven chords of a new key each day, then play them up the neck as E-shape and A-shape barre chords	4 min	Name the chords of any common key without looking
Numbers aloud	Play a song you know and say the number of each chord as you play it	3 min	Think in numbers, not letters
Transpose a song	Play the same song in three different keys, with and without a capo	5 min	Each key at the same tempo
I – IV – V round the circle	Play I – IV – V – I in C, G, D, A, E, then F, B $\flat$, E $\flat$	3 min	Fluent primary chords in eight keys

Check your understanding

Write your answers, then check them in Appendix J.

1. List the seven diatonic triads of D major.

2. What is the V chord in A major, and the vi chord in F major?

3. Write a ii – V – I in G major using 7th chords.

4. Which chord in C major is diminished, and which is the only dominant 7th?

5. Name the function of IV and ii, and of V and vii $^\circ$.

6. Why is the v chord in a minor key usually made major?

7. At capo 3, which key do G shapes sound in?

8. A song in E $\flat$: which shapes and capo fret could you use?

MY NOTES

CHAPTER 14

Chord Progressions



TRACKS

14

A progression is a series of chords that leaves home, builds tension and returns. A handful of common progressions carry much of the music you know; a few less obvious ones show how voice leading, borrowed chords and bass lines create their effect.

A progression is a sequence of chords. Most progressions move away from the tonic, build tension and come back, following the functions in Chapter 13: tonic, subdominant, dominant, tonic. Written in Roman numerals, a progression works in every key.

This chapter covers the progressions you will meet most often, the 12-bar blues, the cadences that end phrases, and a set of less obvious progressions chosen because each one shows a technique you can reuse: a chromatic line, a borrowed chord, a pedal bass or a chain of dominants.

The common progressions

Name	Numerals	In C	In G	Typical use
Three-chord	I – IV – V	C F G	G C D	Folk, early rock and roll, country
Pop progression	I – V – vi – IV	C G Am F	G D Em C	Pop and rock since the 1980s
Minor-start pop	vi – IV – I – V	Am F C G	Em C G D	The same four chords, darker start
1950s progression	I – vi – IV – V	C Am F G	G Em C D	Doo-wop ballads, "Stand by Me"
Jazz ii – V – I	ii7 – V7 – Imaj7	Dm7 G7 Cmaj7	Am7 D7 Gmaj7	Jazz standards, bossa nova
Circle	vi – ii – V – I	Am Dm G C	Em Am D G	Jazz turnarounds, Baroque sequences
Canon	I V vi iii IV I IV V	C G Am Em F C F G	G D Em Bm C G C D	Pachelbel's Canon and many pop songs
Mixolydian	I – bVII – IV – I	C B $\flat$ F C	G F C G	Rock anthems, Celtic and folk tunes

I – IV – V in G

G <i>I</i>	C <i>IV</i>	D <i>V</i>	G <i>I</i>
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I – V – vi – IV in C

C <i>I</i>	G <i>V</i>	Am <i>vi</i>	F <i>IV</i>
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I – vi – IV – V in G

G <i>I</i>	Em <i>vi</i>	C <i>IV</i>	D <i>V</i>
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ii – V – I in C

Dm7 <i>ii7</i>	G7 <i>V7</i>	Cmaj7 <i>Imaj7</i>	Cmaj7 <i>Imaj7</i>
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TRACK 75 I – V – vi – IV in C with open chords

C G Am F

0 0 0 0 3 3 3 3 0 0 0 0 1 1 1 1
 T 1 1 1 1 0 0 0 0 1 1 1 1 1 1 1 1
 A 0 0 0 0 0 0 0 0 2 2 2 2 2 2 2 2
 B 2 2 2 2 0 0 0 0 2 2 2 2 3 3 3 3
 B 3 3 3 3 3 3 3 3 0 0 0 0 3 3 3 3

Four strums per bar. Any of the strum patterns in Chapter 22 fits this progression.

TIP · One loop, four songs

I – V – vi – IV, vi – IV – I – V, IV – I – V – vi and V – vi – IV – I are the same four chords started at different points. Which chord comes first changes which chord the ear hears as home, and with it the mood.

Minor-key progressions

Name	Numerals	In Am	In Em	Typical use
Andalusian cadence	i – bVII – bVI – V	Am G F E	Em D C B	Flamenco, classical, surf, pop
Minor rock	i – bVI – bVII	Am F G	Em C D	Rock, film, metal
Minor folk	i – iv – v	Am Dm Em	Em Am Bm	Folk ballads, modal tunes
Classical minor	i – iv – V – i	Am Dm E Am	Em Am B Em	Classical studies, Baroque
Minor ii – V – i	iiø7 – V7 – i	Bm7b5 E7 Am	F#m7b5 B7 Em	Jazz, bossa nova, gypsy jazz

The **Andalusian cadence** is a descending bass line, A – G – F – E, harmonised with four chords. The last chord is a major V borrowed from harmonic minor, so the progression can loop back to i. Del Shannon's "Runaway" and Ray Charles's "Hit the Road Jack" are built on it.

Andalusian cadence in A minor

Am G F E
 i bVII bVI V

TRACK 76 Andalusian cadence

Am G F E

0 0 3 3 1 1 0
 T 1 1 0 0 1 1 0
 A 2 2 0 0 2 2 1
 B 2 2 0 0 3 3 2
 B 0 0 2 3 3 1 0

The bass falls A, G, F, E. The G# in the E chord is the leading note that pulls back to A.

NYLON · STEEL

Nylon: the Andalusian cadence is the harmonic basis of much flamenco, where it is played in the Phrygian sense (heard from E rather than A; see Chapter 53). Play it with a rasgueado, a fan of fingers flicked outwards across the strings.

Steel and electric: the same four chords make a strong strummed or palm-muted riff. Power chords work, but the G# in the full E chord is what makes the return to Am convincing.

The 12-bar blues

The 12-bar blues uses the I, IV and V chords over twelve bars of 4/4, usually all as dominant 7th chords. It is the form of most blues and of much early rock and roll and jazz.

12-bar blues in A

A7 <i>I7</i>	A7 <i>I7</i>	A7 <i>I7</i>	A7 <i>I7</i>
D7 <i>IV7</i>	D7 <i>IV7</i>	A7 <i>I7</i>	A7 <i>I7</i>
E7 <i>V7</i>	D7 <i>IV7</i>	A7 <i>I7</i>	E7 <i>V7</i>

Common variations:

- **Quick change:** IV in bar 2 (A7 | D7 | A7 | A7 | ...), which gives the first line more movement.
- **Turnaround:** bar 12 on V leads back to the start. On the last time through, bar 12 stays on I.
- **Minor blues:** i, iv and v as minor chords (Am7, Dm7, Em7 or E7).

Play the blues with a shuffle feel: each beat divided long-short, as in a triplet with the first two notes tied. The pattern below is written in straight eighths; swing it.

TRACK 77 12-bar boogie pattern in A

A7

D7 **A7**

E7 **D7** **A7** **E7**

Root and 5th alternating with root and 6th on the open bass strings. Palm-mute lightly. Chapter 43 adds licks, turnarounds and the full shuffle feel.

Cadences

A **cadence** is the chord movement that ends a phrase. Four types cover almost all music.

Cadence	Chords	Effect
Authentic (perfect)	V → I, or V7 → I	Full stop; the strongest ending
Plagal	IV → I	Soft ending; the "Amen" of hymns
Half	any chord → V	A comma; the phrase stays open
Deceptive (interrupted)	V → vi	The ear expects I and gets the relative minor instead

▶ TRACK 78 The four cadences in C

G7 C F C C G G7 Am

	1	0	1	0	0	3	1	0
T	0	1	1	1	1	0	0	1
A	0	0	2	0	0	0	0	2
B	0	2	3	2	2	0	0	2
	2	3	3	3	2	2	2	0
	3				3	3		

Authentic, plagal, half, deceptive. Listen to how settled each bar sounds at its end.

Progressions worth studying

Each progression below is chosen for one technique. Read why it works, learn the chart, play the audio example, then try the variation. Every one transposes to other keys through the numerals.

Descending chromatic line in minor

Why it works. The harmony is essentially one chord, A minor, held for four bars. What moves is a single voice, here the bass: A, G#, G, F#, one half step at a time. Each new bass note recolours the same chord: major 7th, minor 7th, then the 6th (F#), which turns Am into a D/F# sound. The F# is the raised 6th of A Dorian, so the line also brightens the minor key on its way down. This device is called a **line cliché**; it opens Led Zeppelin's "Stairway to Heaven" and runs through the jazz standard "My Funny Valentine".

Line cliché in A minor

Am	AmMaj7/G#	Am7/G	D/F#
i	i42	i42	IV6

▶ TRACK 79 Line cliché in A minor

Am AmMaj7/G# Am7/G D/F#

	0	0	0	0	0	0	2
T	1	1	1	1	1	1	3
A	2	2	2	2	2	2	2
B	2	2	2	2	2	2	0
	0	0	2	2	2	2	
			4	4	3	3	2

The upper strings barely change; mute the 5th string in bars 2–4 with the fingers fretting the bass.

TRY THIS · Rising line in major

Move a single voice up instead of down: D – Daug – D6 – D7, a line of A, A#, B, C (the 5th, #5, 6th and b7 of D). The D7 then leads to G. The Beatles' "Something" uses the descending major version C – Cmaj7 – C7.

Borrowed iv: I – iii – IV – iv

Why it works. C to Em keeps two common tones (E and G) and moves only C down to B, so it is a gentle step away from home. F to Fm changes one note: A falls to Ab. Fm is **borrowed** from C minor, and the Ab then falls again to G in the final C chord. The result is a slow chromatic line, A – Ab – G, that gives the ending a bittersweet colour. Fm to C is a **minor plagal cadence** (iv – I), a softer and darker close than V – I. Radiohead's "Creep" uses a related I – III – IV – iv.

I – iii – IV – iv – I in C

C <i>I</i>	Em <i>iii</i>	F <i>IV</i>	Fm <i>iv</i>	C <i>I</i>
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TRACK 80 I – iii – IV – iv – I in C

C Em F Fm C

	0	0	1	1	0
T	1	0	1	1	1
A	0	0	2	1	0
B	2	2	3	3	2
	3	2			3
		0			

Listen to the 3rd string in bars 3–4. It plays A, then Ab, then the open G.

TRY THIS · Variations

Replace Em with E (III major, the V of Am). The G# rises to A in F, a second chromatic line. Or end a song with F – Fm – C instead of G – C.

Backdoor ii – V: iv7 – bVII7 – I

Why it works. The usual way into C is ii7 – V7 (Dm7 – G7), the "front door". The **backdoor** route uses Fm7 and Bb7, both borrowed from C minor. Bb7 (Bb D F Ab) has no leading note B, but its F falls to E, its Ab falls to G and its D moves to C or E, while the root rises a whole step from Bb to C. The resolution is softer and warmer than G7 – C. Jazz standards use it often; Tadd Dameron's "Lady Bird" reaches its tonic this way.

Front door and back door to C

Dm7 <i>ii7</i>	G7 <i>V7</i>	Cmaj7 <i>Imaj7</i>	Cmaj7 <i>Imaj7</i>
Fm7 <i>iv7</i>	Bb7 <i>bVII7</i>	Cmaj7 <i>Imaj7</i>	Cmaj7 <i>Imaj7</i>

TRACK 81 Front door, then back door

Dm7 G7 Cmaj7 Fm7 Bb7 Cmaj7

	1	1	0	1	0
T	1	0	0	1	3
A	2	0	0	1	1
B	0	0	2	1	3
		2	3	3	1
	3				

Fm7 is an E-shape barre at the 1st fret, Bb7 an A-shape barre at the 1st fret.

TRY THIS · Use only the back door

Play Cmaj7 – B♭7 – Cmaj7 as a two-chord vamp. Then try Cmaj7 – Fm7 – Cmaj7, the minor plagal version, which needs no B♭ at all.

Mixolydian rock: I – ♭VII – IV – I

Why it works. The C chord does not belong to D major; its C natural is the ♭7 of D, the note that defines the Mixolydian mode (Chapter 9). From C the roots fall by 4ths, C to G to D, so the progression ends with two plagal steps in a row instead of a V – I. Without a leading note the ending sounds open and strong rather than final, which suits rock. The long coda of the Beatles' "Hey Jude" is this progression.

I – ♭VII – IV – I in D

D I	C ♭VII	G IV	D I
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TRACK 82 Mixolydian rock in D

D C G D

	2	2	0	0	3	3	2
T	3	3	1	1	0	0	3
A	2	2	0	0	0	0	2
B	0	0	2	2	0	0	0
			3	3	2	2	
					3	3	

All open chords. Accent beats 2 and 4 with the strumming hand for a rock feel.

TRY THIS · Other keys and orders

Move it to A (A – G – D – A) and E (E – D – A – E): both lie under open chords. Then reverse the last two chords, I – ♭VII – I, as a two-chord rock vamp.

TRACK 83 Mixolydian rock loop in D · LOOP · rock · 104 bpm

D I	C ♭VII	G IV	D I
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IMPROVISE WITH**D major pentatonic**
D mixolydian

Start with D major pentatonic, then add the C natural and land on it as the C chord arrives. Over G aim for B (its 3rd) and fall back to A or F# when D returns.

Aeolian loop: i – ♭VI – ♭III – ♭VII

Why it works. These are the same four chords as vi – IV – I – V in C. Starting on Am and avoiding the E chord removes the leading note G#, so nothing pulls strongly back to Am. The loop circles instead of resolving, which is the sound of the natural minor (Aeolian) mode. It suits songs that should feel driven and unresolved; the Cranberries' "Zombie" uses the same pattern in E minor.

i – ♭VI – ♭III – ♭VII in A minor

Am i	F ♭VI	C ♭III	G ♭VII
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▶ **TRACK 84** Aeolian loop with open chords

Am F C G

	0	0	1	1	0	0	3	3
T	1	1	1	1	1	1	0	0
A	2	2	2	2	0	0	0	0
B	2	2	3	3	2	2	0	0
	0	0	3	3	3	3	2	2
			1	1			3	3

Loop it. The top voice moves only by small steps (E, F, E, G), which keeps the texture smooth.

TRY THIS · Make it resolve

Replace the last bar with E7. The loop now ends with a dominant and lands firmly on Am: the mode has become ordinary A minor.

▶ **TRACK 85** Aeolian loop in A minor · LOOP · pop · 92 bpm

Am *i* F *bVI* C *bIII* G *bVII*

IMPROVISE WITH

A minor pentatonic

A natural minor

Land on the 3rd of each chord on beat 1: C over Am, A over F, E over C and B over G. A minor pentatonic box 1 has the first three; B comes from the full natural minor.

Rising bass: I – I/3 – IV – #iv°7 – V

Why it works. The bass climbs C, E, F, G. C/E is the tonic in 1st inversion, so the harmony stands still while the bass moves. F#dim7 is a **passing chord**: its F# is a half step below G and pulls up to it, and its other notes (A, C, Eb) form most of a D7, the V7 of G. The diminished chord therefore works as a dominant aimed at V. The progression is a staple of gospel, doo-wop and stride piano.

Rising bass in C

C *I* C/E *I6* F *IV* F#dim7 *#iv°7* G *V* C *I*

▶ **TRACK 86** Rising bass in C

C C/E F F#dim7 G C

	0	0	1	1	3	0
T	1	1	1	1	0	1
A	0	0	2	2	0	0
B	2	2	3	1	0	2
	3	3	1	2	2	3
	0	0	1	2	3	3

Bass notes C (5th string), then E, F, F# and G on the 6th string. F#dim7 is 2x121x.

TRY THIS · Replace the diminished chord

Play D7/F# (2x0212) instead of F#dim7. It is the full secondary dominant of G; compare it with the darker dim7.

Stepwise bass under the Canon pattern

Why it works. Pachelbel's Canon (about 1700) repeats the chords I – V – vi – iii – IV – I – IV – V. Choosing inversions so that the bass moves by step turns it into a falling scale: D, C#, B, A, G, F#, E, then A to begin again. Here the second IV is replaced by ii (Em) to keep the bass stepping. Every chord is diatonic; the effect comes entirely from the bass line, which gives the progression its sense of gradual descent. Procol Harum's "A Whiter Shade of Pale" is built on a similar falling bass.

Descending bass in D

D	A/C#	Bm	F#m/A	G	D/F#	Em	A
I	V6	vi	iii6	IV	I6	ii	V

TRACK 87 Descending bass in D

D A/C# Bm F#m/A G D/F# Em A

	D	A/C#	Bm	F#m/A	G	D/F#	Em	A
T	2	2	2	2	3	2	0	0
A	3	2	3	2	0	3	0	2
B	2	2	4	2	0	2	0	2
B	0	2	4	4	0	0	2	2
B		4	2	0	2	2	2	0
B					3	2	0	

Follow the lowest note of each chord. It falls one scale step at a time from D to E, then turns up to A.

TRY THIS · The short version

Play only the first five chords with one change: D – A/C# – Bm – D/A – G (I – V/7 – vi – I/5 – IV). The bass falls D, C#, B, A, G. D/A is xx0232 with the open 5th string added.

Deceptive cadences: V – vi and V – bVI

Why it works. After G7 the ear expects C. In V – vi the upper voices resolve as they would to C (B rises to C, F falls to E), but the bass rises a step to A, and Am shares two notes with C. It sounds like an ending that has been postponed. V – bVI goes further: Ab major is borrowed from C minor and its bass also rises a step, G to Ab, but it brings in two notes foreign to the key (Ab and Eb). The surprise is darker and more dramatic, and film music uses it often. Both cadences extend a phrase so that the real V – I can come later.

Two ways to avoid the expected ending

C	F	G7	Am	F	G7	Ab	G7	C
I	IV	V7	vi	IV	V7	bVI	V7	I

TRACK 88 Two deceptive cadences, then the real ending

C F G7 Am F G7 Ab G7 C

	C	F	G7	Am	F	G7	Ab	G7	C
T	0	1	1	0	1	1	4	1	0
A	1	1	0	1	1	0	4	0	1
B	0	2	0	2	2	0	5	0	0
B	2	3	0	2	3	0	6	0	2
B	3		2	0	2	2	6	2	3
B			3		3	3	4	3	

Ab is an E-shape barre at the 4th fret. The last bar finally gives the authentic cadence.

TRY THIS · Deceptive in minor

In A minor the same move is E7 – F (V – $\flat$ VI), which is diatonic there. Play Am – Dm – E7 – F – Dm – E7 – Am.

Pedal point: chords over an open string

Why it works. A **pedal** is a bass note that stays put while the harmony above it changes. Here the open 4th string (D) sounds under every chord. G/D, Em/D and A/D all contain notes that clash mildly with D (A/D has C# and E against it), but the ear hears the pedal as a separate layer and accepts the friction as colour. Pedals are common in folk, Celtic and open-tuning guitar music, where an open string can ring for a whole passage (Chapter 25).

Chords over a D pedal

D <i>I</i>	G/D <i>IV64</i>	Em/D <i>ii/1</i>	A/D <i>V/1</i>	D <i>I</i>
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▶ TRACK 89 Chords over an open D

D G/D Em/D A/D D

	D	G/D	Em/D	A/D	D
T	2	3	0	0	2
A	3	3	0	2	3
B	2	4	0	2	2
B	0	0	0	0	0

The open 4th string is in every chord. Em/D is the four open top strings.

TRY THIS · An E pedal

Use the open 6th string: E (022100) – D/E (0x0232) – A/E (002220) – E. Strum from the 6th string every time so the pedal is always heard.

A chain of dominants round the circle

Why it works. Each dominant 7th resolves to a chord a 5th lower, and that chord is itself made into a dominant 7th, so the tension never lets go until the final C. The roots E, A, D, G, C move round the circle of fifths. E7, A7 and D7 are **secondary dominants**: the V7 of Am, of Dm and of G (written V7/vi, V7/ii, V7/V). Each one borrows a leading note from outside the key (G#, C#, F#). This chain, III7 – VI7 – II7 – V7 – I, is often called the ragtime progression.

Secondary dominants falling by 5ths

C <i>I</i>	E7 <i>III7</i>	A7 <i>VI7</i>	D7 <i>II7</i>	G7 <i>V7</i>	C <i>I</i>
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▶ TRACK 90 Dominants round the circle

C E7 A7 D7 G7 C

	C	E7	A7	D7	G7	C
T	0	0	0	2	1	0
A	1	0	2	1	0	1
B	0	1	0	2	0	0
B	2	0	2	0	0	2
B	3	2	0	0	2	3
B		0			3	

All open 7th chords. Listen for the chromatic notes G#, C# and F#, each resolving up a half step into the next chord.

Why it works. In A natural minor the iv chord is Dm. Making it major, D, brings in F#, the raised 6th that defines the Dorian mode (Chapter 9). Neither chord is a dominant, so the vamp can repeat for minutes without wanting to resolve, which makes it ideal for improvising. Santana's "Oye Como Va" is built on the same two-chord Dorian move (Am7 – D9).

Am7 <i>i7</i>	D <i>IV</i>	Am7 <i>i7</i>	D <i>IV</i>
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The image displays a musical score for the song "The Sound of Silence" by Simon & Garfunkel. The score is written for guitar and bass, with a key signature of one sharp (F#) and a 4/4 time signature. The guitar part is in the treble clef, and the bass part is in the bass clef. The score is divided into four measures, each with a chord diagram above it. The chords are Am7, D, Am7, and D. The guitar part features a melodic line with eighth and quarter notes, while the bass part provides a steady accompaniment with eighth and quarter notes. The chord diagrams show the fingerings for each chord: Am7 (0, 2, 0, 0) and D (2, 3, 2, 0).

Chord Diagrams:

- Am7:** 0 2 0 0
- D:** 2 3 2 0

Score Structure:

- Measure 1:** Am7 (0 2 0 0) | D (2 3 2 0) | Am7 (0 2 0 0) | D (2 3 2 0)
- Measure 2:** Am7 (0 2 0 0) | D (2 3 2 0) | Am7 (0 2 0 0) | D (2 3 2 0)
- Measure 3:** Am7 (0 2 0 0) | D (2 3 2 0) | Am7 (0 2 0 0) | D (2 3 2 0)
- Measure 4:** Am7 (0 2 0 0) | D (2 3 2 0) | Am7 (0 2 0 0) | D (2 3 2 0)

TRY THIS · The Lydian vamp
Play D – E – D – E (I – II in D). The E chord contains G#, the raised 4th of D, which is the note of the Lydian mode. The sound is bright and floating; film and television music use it for a sense of wonder. Chapter 9 gives the scales to play over both vamps.

Several progressions above use chords that are not diatonic. They come from two sources, both developed in full in Chapter 32:

- **Borrowed chords** come from the parallel minor key. In this chapter: Fm (iv) in I – iii – IV – iv, Fm7 and Bb7 in the backdoor ii – V, Ab (bVI) in the deceptive V – bVI, and C (bVII) in the Mixolydian rock progression in D. bIII (Eb in C, as in C – Eb – F – C) is the other common choice.
- **Secondary dominants** are V7 chords aimed at a diatonic chord other than I. In this chapter: E7, A7 and D7 in the chain of dominants, and D7/F# in the rising-bass variation.

- **Substitute by function:** vi for I, ii for IV (Chapter 13).
- **Add 7ths:** C – Am – F – G becomes Cmaj7 – Am7 – Fmaj7 – G7.
- **Suspend and resolve:** G – Gsus4 – G, or D – Dsus4 – D – Dsus2 – D (Chapter 11).
- **Move the bass:** use inversions and slash chords so the bass walks by step.
- **Borrow one chord:** replace IV with iv, or V with $\flat$ VII, at one point in the song.

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Chapter 14 review · use it, train it, test it

Using it in real music

Recognising songs. Once you know I – V – vi – IV, the Andalusian cadence and the 12-bar blues by ear, you can name the chords of many songs after one hearing. Chapter 28 trains this directly.

Songwriting. Start with a common progression, then make one change from this chapter: a chromatic line, a borrowed iv, a pedal or a deceptive cadence. One unexpected chord is usually enough.

Jamming. "12-bar in E" tells you everything: E7, A7 and B7 in a known order. A Dorian or Mixolydian vamp tells you which scale to use.

How to train it

Drill	How	Time	Goal
Four keys	Play I – V – vi – IV in C, G, D and A with open chords, four strums per chord	4 min	No gaps at chord changes or key changes
Progression of the week	Take one example from "Progressions worth studying"; play it, then its variation, then transpose both to a second key	5 min	Play both keys from memory
Cadence ear test	A partner plays a phrase ending with one of the four cadences; name it	3 min	8 out of 10 correct
Write daily	Write a four-bar progression using one borrowed chord or one secondary dominant	3 min	A notebook of your own ideas

Check your understanding

Write your answers, then check them in Appendix J.

1. Write I – V – vi – IV in D.

2. Which three chords make a 12-bar blues in E, and in which bars does the V chord appear in the standard form?

3. Write the Andalusian cadence in E minor.

4. Name the cadences V → vi and IV → I.

5. In C major, which secondary dominant leads to Am, and which note does it borrow?

6. Which chord is the borrowed iv in G major?

7. Why does a descending line such as Am – AmMaj7/G# – Am7/G – D/F# sound connected?

8. What makes Am7 – D a Dorian vamp rather than an A minor progression?

MY NOTES

CHAPTER 15

Spices: Making Progressions Your Own



TRACKS

15

Small changes turn a plain chord progression into an arrangement. The examples are played plain and then spiced, so you can hear exactly what each change does.

A **spice** is a small change to a progression that keeps it recognisable but adds colour, motion or tension. The chords and their order stay the same: you change one note in a chord, add a chord between two others, move the bass, or play a chord a little early. Chapter 14 gives you the progressions themselves; this chapter shows how to dress them.

Every audio example uses the same format. The plain version comes first, then a double barline, then the spiced version. Work through each one in three steps:

1. **Hear it.** Listen to the track and find the moment where the two versions differ. The caption names the change.
2. **Play it plain.** Learn the plain version until the changes are automatic.
3. **Play it spiced.** Add the change, then switch between the two versions until you can choose the spice at will.

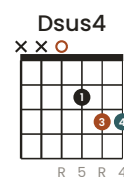
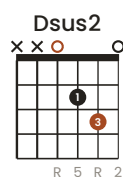
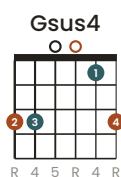
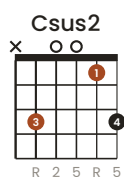
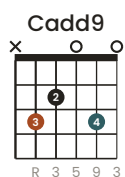
One spice is often enough. A song in which every chord is spiced loses the contrast that makes each spice audible.

One-note spices on open chords

Most open chords have a neighbour that differs by one note, often by one finger. Replacing the chord with its neighbour for a beat or two, or alternating the two, gives a progression movement without changing its harmony. The formulas for sus, add and 7th chords are in Chapter 11; here the point is which finger moves and what you hear.

Plain	Spiced	What moves	What you hear
C	Cadd9	The C on the 2nd string rises to D (4th finger, 3rd fret)	Bright, open, modern pop and folk
C	Csus2	Both Es (4th and 1st strings) give way to D and G	No 3rd: neutral, airy
G	Gsus4	Both Bs (5th and 2nd strings) rise to C	Tension that wants to fall back to G
D	Dsus4, Dsus2	The 1st string: F# up to G, or down to open E	Movement inside one chord
Am	Am7	Lift the 3rd finger: A on the 3rd string becomes open G	Softer, less final
Am	Am(add9)	The 3rd finger's A becomes B (4th finger, 4th fret)	Bittersweet, film-music minor
E	E7	Lift the 3rd finger: E on the 4th string becomes open D	Bluesy, pulls towards A
E	Esus4	The G# on the 3rd string rises to A	Tension before the full E or E7
F	Fmaj7	Lift the 1st finger from the 1st string: F becomes open E	Gentle, dreamy, and easier to play

C, G and D with one note changed



Compare each shape with the plain C, G and D in Chapter 11. Fretted changed notes are teal; the degrees under the strings show every note, including changes on open strings.

▶ **TRACK 92** Plain, then spiced: G – C – D – G

Diagram showing the fretboard for Track 92, illustrating the progression of chords G, C, D, and G. The diagram includes a treble clef staff with a key signature of one sharp (F#) and a 4/4 time signature. The fretboard is shown with fingerings (0-3) for the Treble (T), Alto (A), and Bass (B) staves.

Chord Progression: G, C, D, G

Fingerings:

- G:** T (3, 0, 0, 2), A (0, 0, 2, 3), B (2, 3, 0, 3)
- C:** T (0, 1, 0, 2), A (0, 2, 2, 3), B (3, 3, 0, 3)
- D:** T (2, 3, 2, 0), A (3, 3, 0, 0), B (0, 0, 0, 0)
- G:** T (3, 0, 0, 2), A (0, 0, 2, 3), B (2, 3, 0, 3)

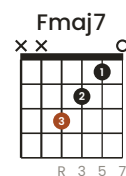
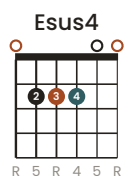
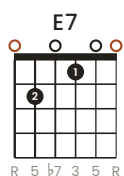
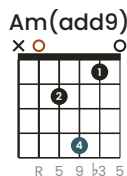
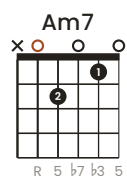
Spiced Chords (Bars 5-8): G, Gsus4, G, Cadd9, Csus2, C, Dsus4, D, Dsus2, D, G

Fingerings:

- G:** T (3, 0, 0, 2), A (0, 0, 2, 3), B (2, 3, 0, 3)
- Gsus4:** T (3, 0, 1, 0), A (0, 0, 0, 0), B (3, 3, 0, 3)
- G:** T (3, 0, 0, 2), A (0, 0, 2, 3), B (2, 3, 0, 3)
- Cadd9:** T (0, 3, 0, 2), A (0, 0, 0, 0), B (3, 3, 0, 3)
- Csus2:** T (3, 0, 1, 0), A (0, 0, 0, 0), B (3, 3, 0, 3)
- C:** T (0, 1, 0, 2), A (0, 2, 2, 3), B (3, 3, 0, 3)
- Dsus4:** T (3, 0, 3, 0), A (3, 3, 0, 0), B (0, 0, 0, 0)
- D:** T (2, 3, 2, 0), A (3, 3, 0, 0), B (0, 0, 0, 0)
- Dsus2:** T (2, 3, 2, 0), A (3, 3, 0, 0), B (0, 0, 0, 0)
- D:** T (2, 3, 2, 0), A (3, 3, 0, 0), B (0, 0, 0, 0)
- G:** T (3, 0, 0, 2), A (0, 0, 2, 3), B (2, 3, 0, 3)

Bars 5–8 keep every root but change the notes above it: G to Gsus4 and back, Cadd9 to Csus2 and back to C, and D with both suspensions.

Am, E and F with one note changed



Am(add9) needs a stretch from the 1st to the 4th fret; keep the 2nd finger on the 4th string. For Esus4, keep the E shape and add the 4th finger on the 3rd string at the 2nd fret.

▶ **TRACK 93** Plain, then spiced: Am – F – E – Am

Diagram showing the fretboard for Track 93, illustrating the progression of chords Am, F, E, and Am. The diagram includes a treble clef staff with a key signature of one sharp (F#) and a 4/4 time signature. The fretboard is shown with fingerings (0-3) for the Treble (T), Alto (A), and Bass (B) staves.

Chord Progression: Am, F, E, Am

Fingerings:

- Am:** T (0, 1, 2, 0), A (2, 2, 2, 0), B (0, 0, 0, 0)
- F:** T (1, 1, 2, 3), A (2, 2, 2, 3), B (3, 3, 0, 3)
- E:** T (0, 0, 1, 2), A (1, 2, 2, 0), B (2, 2, 2, 0)
- Am:** T (0, 1, 2, 0), A (2, 2, 2, 0), B (0, 0, 0, 0)

Spiced Chords (Bars 5-8): Am, Am7, Fmaj7, Esus4, E, E7, Am(add9)

Fingerings:

- Am:** T (0, 1, 2, 0), A (2, 2, 2, 0), B (0, 0, 0, 0)
- Am7:** T (0, 1, 2, 0), A (2, 2, 2, 0), B (0, 0, 0, 0)
- Fmaj7:** T (0, 1, 2, 0), A (2, 2, 2, 0), B (3, 3, 0, 3)
- Esus4:** T (0, 0, 1, 2), A (1, 2, 2, 0), B (2, 2, 2, 0)
- E:** T (0, 0, 1, 2), A (1, 2, 2, 0), B (2, 2, 2, 0)
- E7:** T (0, 0, 1, 2), A (1, 2, 2, 0), B (2, 2, 2, 0)
- Am(add9):** T (0, 1, 2, 0), A (2, 2, 2, 0), B (0, 0, 0, 0)

Spiced: Am falls to Am7, F becomes Fmaj7, the E chord starts suspended and adds its 7th, and the last chord is Am(add9).

TIP · Where one-note spices fit

Use a sus chord where the chord lasts a whole bar or more: the suspension needs time to resolve. Use add9, maj7 and m7 as straight replacements when a song feels too square. The sus4 moves in the Who's "Pinball Wizard" and the Dsus4 – D figure in Tom Petty's "Free Fallin'" are well-known examples of the first kind.

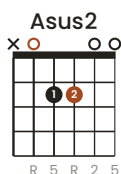
Moving inner voices inside a held chord

A one-note spice can also be played as a hammer-on or pull-off while the rest of the chord rings. You strum the chord, lift one finger, and put it back with a hammer-on, or pull it off to an open string. The chord symbol does not change: the movement is an ornament, not a new chord. Jimi Hendrix's "Little Wing" is built on this kind of embellished chord playing.

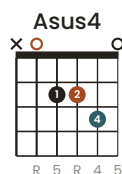
The useful moves are the same notes as in the previous section: 2 – 3 and 4 – 3 on top of D and A, and 9 – 3 in the bass of G and C.

Where the hammer-ons start

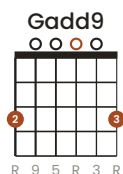
A: hammer to fret 2



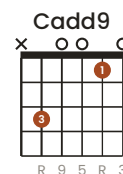
A: pull off to fret 2



G: hammer the 5th string



C: hammer the 4th string



Each shape is the plain chord with one finger lifted or added. The hammer-on or pull-off restores the plain chord.

TRACK 94 Plain, then spiced: D – G – A – D

Diagram showing two musical staves for the progression D – G – A – D. The first staff shows the plain progression, and the second staff shows the progression with one-note spices (hammer-ons or pull-offs) indicated by 'H' and 'P' above the notes.

Staff 1: Plain progression

	D	G	A	D
T	2	3	0	2
A	3	0	2	3
B	2	0	2	0

Staff 2: Spiced progression

	D	G	A	Dsus4	D
T	2	3	0	3	2
A	3	0	2	2	3
B	2	0	2	0	0

Only one string moves in each spiced bar: the 1st string over D, the 5th string under G, the 2nd string over A. Let the other strings ring.

TRY THIS · Embellish your own progression

Take any progression of open chords. In each bar, strum on beat 1, then on beat 2 play one hammer-on or pull-off from the table above. Keep the strumming hand moving even when it does not strike the strings.

A related device moves one voice by half steps over a held chord, as in Am – AmMaj7/G# – Am7/G – D/F#. That is the **line cliché**, covered in Chapter 14 and Chapter 34.

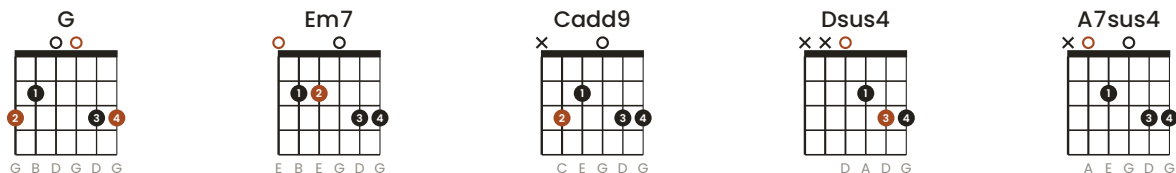
Pedal and drone spices

A **pedal** is a note that stays the same while the chords change around it. Chapter 14 uses a low open string as a pedal under the harmony. On the guitar a pedal on top is at least as useful: a fretted or open note on the 1st and 2nd strings that sounds in every chord.

Anchored top strings

In G major, the notes D and G at the 3rd fret of the 2nd and 1st strings belong to G, Em7, Cadd9, Dsus4 and A7sus4. Keep the 3rd and 4th fingers there, and change only the lower strings. The top of every chord stays the same, so the progression sounds connected and the hand barely moves. Oasis's "Wonderwall" is played with shapes of this family.

Anchored on the 3rd fret



The 3rd and 4th fingers never leave the 2nd and 1st strings. Only the 1st and 2nd fingers move.

TRACK 95 Plain, then spiced: Em – C – G – D

Em C G D

5

Em7 Cadd9 G Dsus4 A7sus4

Spiced: D and G sound on top of every chord. Em becomes Em7, C becomes Cadd9, D becomes Dsus4, and A7sus4 leads back to the start.

Open-string drone

In E major the open 2nd and 1st strings (B and E) can ring through every chord. Move the shapes on the lower strings and leave the top two strings open. Some chords gain a colour note: over C $\sharp$ m the open B is the minor 7th, over A it replaces the 3rd, over B the open E is a 4th.

Shapes with the top two strings open



C $\sharp$ m7 and Bsus4 use the same fingering, two frets apart. Mute the 6th string with the tip of the 1st finger.

TRACK 96 Plain, then spiced: E – C[♯]m – A – B

Diagram showing two musical staves and their corresponding fretboard diagrams for the progression E – C[♯]m – A – B.

Staff 1 (Plain): Shows the progression E, C[♯]m, A, B. The fretboard diagrams below show the fingerings for each chord.

	E		C [♯] m		A		B	
T	0	0	4	4	0	0	2	2
A	0	0	5	5	2	2	4	4
B	1	1	6	6	2	2	4	4
	2	2	6	6	2	2	4	4
	2	2	4	4	0	0	2	2
	0	0						

Staff 2 (Spiced): Shows the progression E, C[♯]m7, Asus2, Bsus4. The fretboard diagrams below show the fingerings for each chord.

	E		C [♯] m7		Asus2		Bsus4	
T	0	0	0	0	0	0	0	0
A	0	0	6	6	2	2	4	4
B	1	1	6	6	2	2	4	4
	2	2	4	4	0	0	2	2
	2	2						
	0	0						

The plain version uses two barre chords. The spiced version keeps B and E ringing on top of every chord, which is both easier and fuller.

Passing and approach chords

A passing chord fills the space between two chords of the progression; it lasts a beat or two and is never the goal. Three kinds cover most uses.

- **Chromatic approach.** Play the target chord's shape a half step above or below, just before the target. A[♭]7 slides down into G7; B slides up into C. Every voice moves by a half step, so almost any chord can be approached this way (Chapter 34 explains why).
- **Diminished passing chord.** Between I and ii, a dim7 chord on the raised root lets the bass climb by half steps: C – C[♯]dim7 – Dm. Chapter 32 shows why it works as a dominant of Dm.
- **Walk-up to IV.** Fill the gap between I and IV with chords on the scale steps between them: G – A[♭]m7 – G/B – C. The bass climbs G, A, B, C.

CAREFUL · Keep passing chords short

A passing chord that lasts as long as the main chords stops sounding like a passing chord and starts to change the progression. One or two beats is right; place it just before the target, not just after the previous chord.

The passing chords used in the examples

Diagram showing five fretboard diagrams for passing chords: C[♯]dim7, A[♭]7, G7, B, and G/B.

C[♯]dim7: Fretboard diagram showing the shape of the diminished chord. Fingering: 1 (C[♯]), 2 (E), 3 (B[♭]), 4 (C[♯]).

A[♭]7: Fretboard diagram showing the shape of the dominant chord. Fingering: 1 (A[♭]), 2 (E[♭]), 3 (G[♭]), 4 (B[♭]).

G7: Fretboard diagram showing the shape of the dominant chord. Fingering: 1 (G), 2 (D), 3 (F), 4 (B).

B: Fretboard diagram showing the shape of the dominant chord. Fingering: 1 (B), 2 (F[♯]), 3 (D[♯]), 4 (F[♯]).

G/B: Fretboard diagram showing the shape of the dominant chord. Fingering: 1 (B), 2 (D), 3 (G), 4 (B).

C[♯]dim7 keeps C[♯] in the bass, between C and D. A[♭]7 and G7 are the same E-shape barre a fret apart; B slides up a fret into the A-shape barre of C.

▶ **TRACK 97** Plain, then spiced: C – Dm – G7 – C

C Dm G7 C

	0	0	1	1	1	1	0
T	1	1	3	3	0	0	1
A	0	0	2	2	0	0	0
B	2	2	0	0	2	2	2
	3	3			3	3	3

C C#dim7 Dm Ab7 G7 B C

	0	3	1	4	3	2	3
T	1	2	3	4	3	4	5
A	0	3	2	5	4	4	5
B	2	2	0	4	3	4	5
	3	4		6	5	2	3
				4	3		

Spiced: C#dim7 between C and Dm, Ab7 a half step above G7, and B a half step below the final C.

▶ **TRACK 98** Plain, then spiced: G – C with a walk-up

G C

	3	3	3	3	0	0	0
T	0	0	0	0	1	1	1
A	0	0	0	0	0	0	0
B	2	2	2	2	2	2	2
	3	3	3	3	3	3	3

G Am7 G/B C

	3	3	3	0	3	0	0
T	0	0	0	1	0	1	1
A	0	0	0	0	0	0	0
B	2	2	2	2	0	2	2
	3	3	3	0	2	3	3

Spiced: Am7 and G/B take one beat each at the end of the second bar, so the bass climbs G, A, B into C.

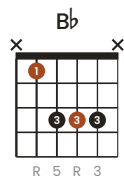
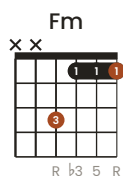
Borrowed colour

A **borrowed chord** comes from the parallel minor key (Chapter 32 gives the theory). For spicing, two swaps do most of the work:

- **iv instead of IV.** In C, Fm instead of F. The Ab (the b6 of C) falls a half step to G, the 5th of the next C chord. Play it in place of IV, or right after IV as in Chapter 14.
- **bVII before I.** In C, Bb instead of G. The tonic arrives from a whole step below, with no leading note, which sounds strong and open rather than final.

Ending a song with iv – I (Fm – C) instead of V – I is the **minor plagal cadence**: a soft, slightly wistful close.

Borrowed chords in C, small shapes



Both shapes avoid the full barre. For B $\flat$, flatten the 3rd finger across strings 4–2 and let the tip mute the 1st string.

TRACK 99 Plain, then spiced: C – F – G – C

Diagram showing two musical staves for the progression C – F – G – C. The first staff shows the 'Plain' version with natural notes. The second staff shows the 'Spiced' version with borrowed chords Fm and B $\flat$.

Plain version (C – F – G – C):

	C	F	G	C
T	0	1	3	0
A	1	1	0	1
B	0	2	0	0
	3	3	2	3

Spiced version (C – Fm – B $\flat$ – C):

	C	Fm	B $\flat$	C
T	0	1	3	0
A	1	1	3	1
B	0	1	3	0
	3	3	1	3

Spiced: Fm replaces F (iv for IV) and B $\flat$ replaces G (bVII for V). Both come from C minor; C itself stays major.

TRY THIS • One borrowed chord per song

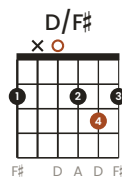
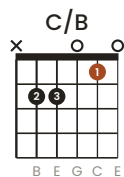
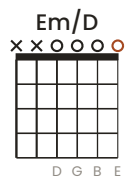
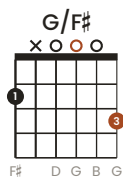
Choose a song you play in G and replace one C with Cm, or one D before G with F. Use the swap once, at the end of a verse or in the last line, so it stands out.

Bass-line spices

Changing only the lowest note of a chord turns jumps in the bass into steps. Two methods do this.

- **Slash chords.** Keep the chord and put a passing note in the bass: G – G/F $\sharp$ – Em, C – C/B – Am. Each slash chord lasts two beats or less.
- **Walking notes.** Between two chords, play single bass notes on the last beats of the bar, moving by step towards the next root. In country and bluegrass rhythm guitar these runs are a standard part of the style (Chapter 46).

Slash chords for a falling bass



Mute the 5th string in G/F $\sharp$ and D/F $\sharp$ with the finger on the 6th string. Em/D is just the four open top strings.

▶ **TRACK 100** Plain, then spiced: G – Em – C – Am – D – G

Chord progression: G, Em, C, Am, D, G

Chord progression: G, G/F#, Em, Em/D, C, C/B, Am, D, D/F#, G

Spiced: the bass falls G, F#, E, D, C, B, A, walks up B, C# to D, and steps from F# to G.

Rhythmic spices

A rhythmic spice changes when a chord arrives, not which chord it is. The most common is the **push** or anticipation: the new chord is played an eighth note early, on the "&" of 4, and tied over the barline, so nothing is struck on the next beat 1. The music seems to lean forward into the bar. Chapter 37 trains the timing; here it is applied to a full strummed progression.

▶ **TRACK 101** Plain, then spiced: Am – F – C – G

Chord progression: Am, F, C, G

Chord progression: Am, F, C, G

Spiced: F, C and G each arrive on the "&" of 4 and are held over the barline. The chords are the same; only their timing changes.

Two further rhythmic spices need no new chords at all. A **stop** is one chord struck on beat 1 and then left to ring, or muted, for the rest of the bar; use it once, just before a chorus. A **half-bar change** shortens a chord that normally lasts two bars: change to the next chord halfway through the second bar, so the harmony moves faster just where the song builds.

Spicing a whole song

The three tracks below are one 8-bar progression in G, I - vi - IV - V - I - vi - IV V - I, played three ways. The plain version uses four strums per bar. The lightly spiced version adds three one-note spices and one slash chord. The fully spiced version combines most of the chapter. The practice loop plays the plain progression so that you can try your own spices over it.

The progression

G <i>I</i>	Em <i>vi</i>	C <i>IV</i>	D <i>V</i>	G <i>I</i>	Em <i>vi</i>	C <i>IV</i>	D <i>V</i>	G <i>I</i>
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▶ **TRACK 102** Spice practice loop in G · LOOP · folk · 88 bpm

G <i>I</i>	Em <i>vi</i>	C <i>IV</i>	D <i>V</i>
G <i>I</i>	Em <i>vi</i>	C <i>IV</i>	D <i>V</i>

STRUM THIS PATTERN IN EVERY BAR



The loop plays the plain chords. Strum along with one spice at a time: first Cadd9 and Dsus4, then the anchored top strings, then the pushes. Keep a spice only if it still sounds right after several times round.

▶ **TRACK 103** Whole song, plain

Chord progression: G, Em, C, D, G

5

3 3 3 3 0 0 0 0 0 0 0 0 2 2 2 2

0 0 0 0 0 0 0 0 1 1 1 1 3 3 3 3

0 0 0 0 0 0 0 0 0 0 0 0 2 2 2 2

2 2 2 2 2 2 2 2 3 3 3 3 0 0 0 0

3 3 3 3 0 0 0 0 0 0 0 0 3 3 3 3

Open chords, four even strums per bar.

TRACK 104 Whole song, lightly spiced

Chords: G, G/F#, Em, Em7, Cadd9, Dsus4, D

5

G/F# walks the bass from G to E, Em gains its 7th, C becomes Cadd9, and every D starts as Dsus4.

TRACK 105 Whole song, fully spiced

Chords: G, Em7, Cadd9, Dsus4, D, G

5

Anchored top strings throughout, a hammer-on under G, an F# bass note leading to Em7, pushes into bars 2 to 7, a Dsus4 pull-off, and Cm(add9), the borrowed iv, in bar 7.

The fully spiced version is a ceiling, not a target. In a real arrangement you would use the light version for the verse and save two or three of the stronger spices for the chorus or the last line.

The spice rack

The table sums up the chapter. The last column names the "Plain, then spiced" track that demonstrates each spice.

Spice	What changes	Effect	Where it fits	Example
sus2, sus4	The 3rd becomes a 2nd or 4th, then returns	Tension and release inside one chord	Folk, rock, pop	G – C – D – G
add9, maj7, m7	One note added or lowered	Colour without new harmony	Pop, folk, ballads	Am – F – E – Am
Hammer-on embellishment	One string moves while the chord rings	Melody inside the chord	Soul, folk, rock ballads	D – G – A – D
Top-string pedal	The same notes on top of every chord	Connected, ringing sound	Acoustic pop, folk rock	Em – C – G – D; E – C#m – A – B
Chromatic approach	Target shape a half step above or below	Slide, tension just before the target	Soul, jazz, funk, rock	C – Dm – G7 – C
Diminished passing chord	dim7 on the raised root between I and ii	Bass climbs by half steps	Gospel, swing, choro	C – Dm – G7 – C
Walk-up	Chords on the steps between I and IV	Forward motion into IV	Country, folk, gospel	G – C with a walk-up
Borrowed iv, bVII	Chords from the parallel minor	Bittersweet or strong, open colour	Pop, rock, film	C – F – G – C
Slash chords and walking bass	Only the bass note	Stepwise bass line	Ballads, country, singer-songwriter	G – Em – C – Am – D – G
Push	The chord arrives an eighth early	Drive, lean into the bar	Rock, pop, funk	Am – F – C – G

Chapter 15 review · use it, train it, test it

Using it in real music

Accompanying a singer. Spices belong where the voice is resting: at the end of a line, in a gap, in a bar where the melody holds a long note. Under a busy melody, play the plain chords.

Learning songs. When a recording sounds richer than the chord chart, listen for the spices in this chapter. Many charts write only "D" where the guitarist plays Dsus4 – D, or "C" where it is Cadd9 with the anchored top strings.

Arranging. Decide which spices go in which section before you play the song through. A common plan: plain verse, light spices in the second verse, the strongest spices (pushes, borrowed chords) in the chorus and the ending.

How to train it

Drill	How	Time	Goal
One-finger switches	Alternate each plain chord with its one-note spice (C – Cadd9, Am – Am7, E – E7, F – Fmaj7), four strums each	3 min	Every switch clean at 80 bpm
Embellish in time	Play D – G – A – D with one hammer-on or pull-off per bar, as in the inner-voice example	4 min	Hammer-ons as loud as the strums
Anchor practice	Loop Em7 – Cadd9 – G – Dsus4 without lifting the 3rd and 4th fingers	3 min	No break in the top notes
Spice a song	Take a song you know and add exactly one spice from each of three rows of the spice rack	5 min	A version you can play twice the same way

Check your understanding

Write your answers, then check them in Appendix J.

1. Starting from an open C chord (x32010), which finger moves to make Cadd9 (x32030), and which note does it change?
2. Why does Fmaj7 (xx3210) count as a spice on F rather than a new chord?
3. Name the chord that fills the gap in C – ? – Dm so that the bass rises by half steps.
4. Which two notes stay at the 3rd fret on the 2nd and 1st strings in the anchored G-major shapes, and which chords share them?
5. In the key of C, which borrowed chords replace F and G in C – F – G – C, and from which key do they come?
6. What does a push do to the next chord, and what do you play on the following beat 1?

MY NOTES

CHAPTER 16

Arpeggios



TRACKS

16

An arpeggio is a chord played one note at a time. It connects harmony and melody, shows where every chord tone lies on the neck, and is the basis of both melodic soloing and fingerstyle accompaniment.

An arpeggio (from the Italian *arpa*, harp) plays the notes of a chord one after another instead of together. The notes are the same as the chord's; only the order and timing change. That makes arpeggios useful in three ways:

- **Soloing.** Landing on chord tones as the chords change makes a line sound connected to the harmony.
- **Accompaniment.** Held chords picked one string at a time are the main texture of classical and fingerstyle guitar.
- **Fretboard knowledge.** Arpeggio shapes show where every root, 3rd, 5th and 7th lies in each position.

Arpeggio formulas

An arpeggio has exactly the formula of its chord: the triads are built in Chapter 10, the 7th chords in Chapter 11. Here they are side by side, in C and in A.

Arpeggio	Formula	In C	In A
Major	1 3 5	C E G	A C# E
Minor	1 b3 5	C Eb G	A C E
Diminished	1 b3 b5	C Eb Gb	A C Eb
Augmented	1 3 #5	C E G#	A C# E#
Dominant 7th	1 3 5 b7	C E G Bb	A C# E G
Major 7th	1 3 5 7	C E G B	A C# E G#
Minor 7th	1 b3 5 b7	C Eb G Bb	A C E G
Half-diminished (m7b5)	1 b3 b5 b7	C Eb Gb Bb	A C Eb G
Diminished 7th	1 b3 b5 #7	C Eb Gb Bbb (= A)	A C Eb Gb

The **3rd** and **7th** decide a chord's quality: major or minor, dominant or major 7th. They are the strongest notes to aim for in a solo. The root and 5th are stable but say less about which chord is playing.

Movable arpeggio shapes

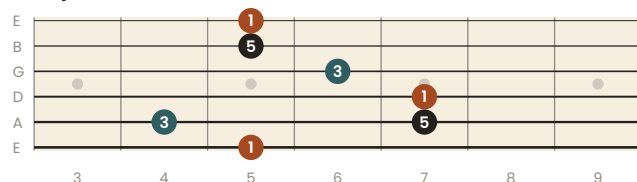
Each shape below covers one position of about four frets and contains every chord tone in that position. Practise each from the lowest note to the highest and back, one note per beat at first, then in eighth notes and triplets.

Root on the 6th string

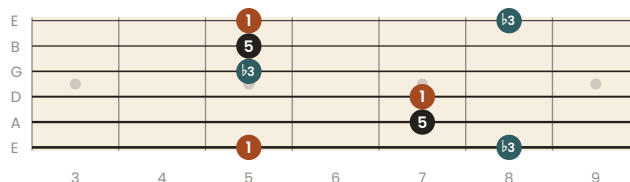
These shapes surround the E-shape barre chords of Chapter 11. Shown in A, with the root at the 5th fret.

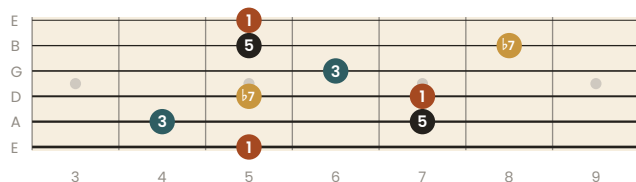
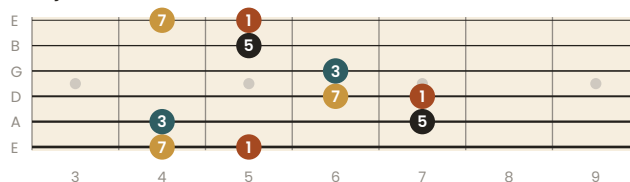
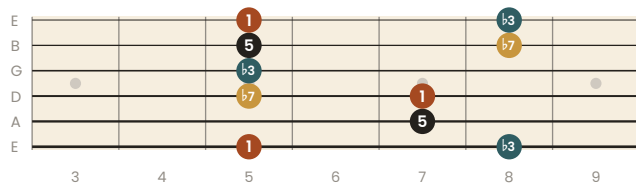
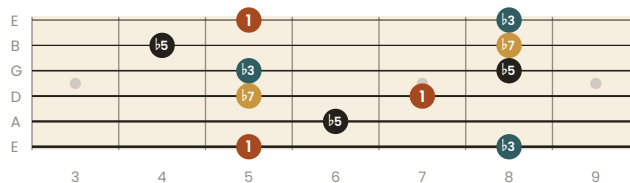
Arpeggios with the root on the 6th string (A)

A major



A minor

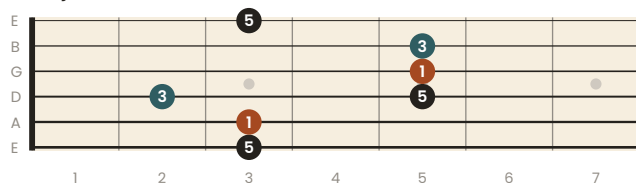
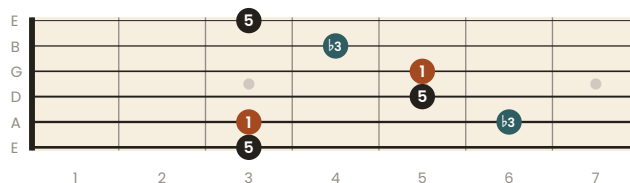
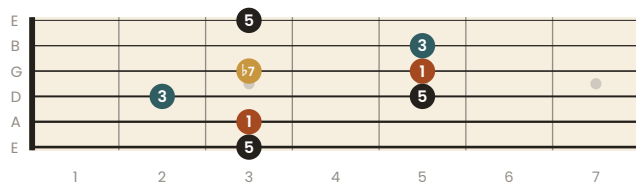
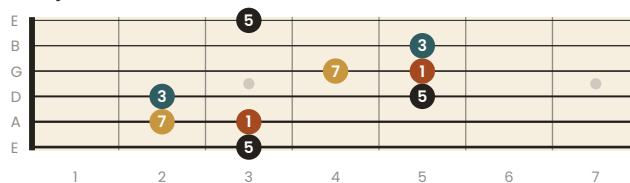
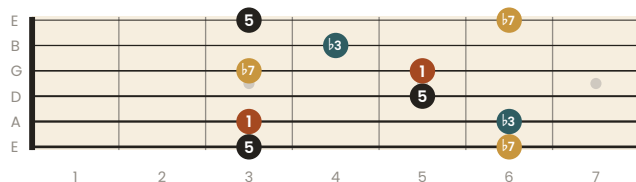
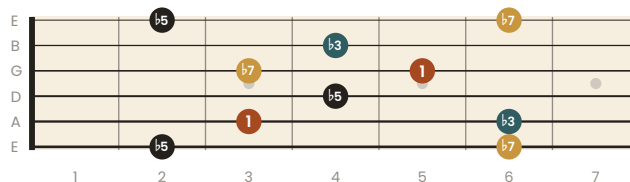


A7**Amaj7****Am7****Am7b5**

Rust = root, teal = 3rd, gold = 7th, ink = 5th.

Root on the 5th string

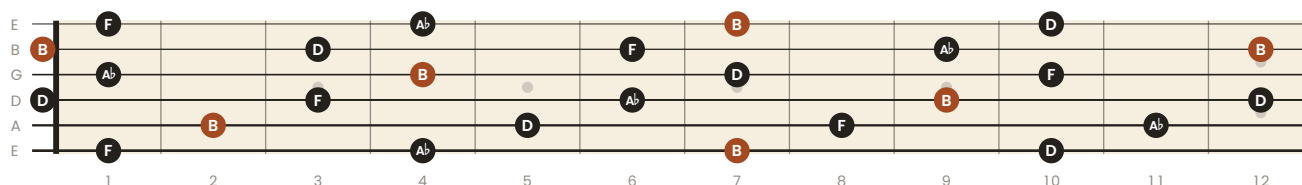
These surround the A-shape barre chords. Shown in C, with the root at the 3rd fret.

Arpeggios with the root on the 5th string (C)**C major****C minor****C7****Cmaj7****Cm7****Cm7b5**

Compare each shape with the one above it in the grid. Only one or two notes move.

Symmetrical arpeggios

The diminished 7th arpeggio is built from minor 3rds only, so, like the dim7 chord in Chapter 11, its shape repeats every three frets along a string, and any of its four notes can be treated as the root. The augmented arpeggio is built from major 3rds and repeats every four frets. Learn one fingering of each and slide it.

B diminished 7th (B D F Ab) on the neck

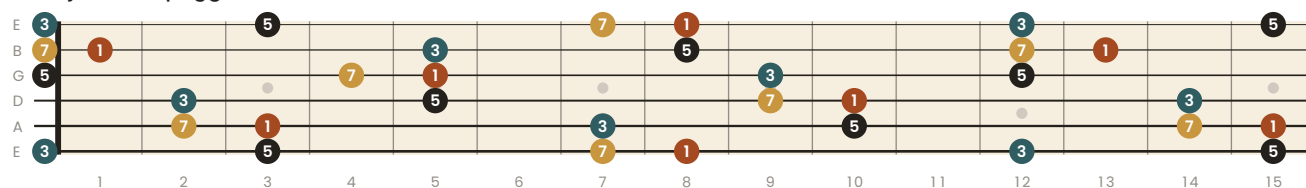
Along each string the notes are three frets apart. Bdim7, Ddim7, Fdim7 and Abdim7 are the same four notes.

A practical use: over a G7 chord, a Bdim7 arpeggio gives the four upper notes of G7b9 (B, D, F, Ab), a tense sound that resolves strongly to C. Gypsy jazz and bebop players use this substitution constantly.

SHORTCUT · An arpeggio is a scale with gaps

Every arpeggio shape is a scale position with the in-between notes removed. Play a major scale position, then play only degrees 1, 3 and 5 from it: that is the major arpeggio. Add 7 for maj7. The five CAGED positions in Chapter 12 each give one arpeggio fingering, so five shapes cover a chord across the whole neck.

C major 7th arpeggio across the neck



Every C, E, G and B on the neck. Pick any four-fret window and you have one Cmaj7 arpeggio shape.

Arpeggios through a progression

To outline a progression, stay in one position and change arpeggio with each chord. Move to the nearest note of the next chord rather than jumping back to its root. Common tones and half steps between the chords then become the smooth joins of your line.

▶ TRACK 106 Outlining C – Am – F – G in open position

C **Am**

F **G** **C**

Each bar is one arpeggio, rising and falling. Most fingers stay where the open chord would put them.

Targeting the 3rd and 7th

In a ii – V – I, the 7th of each chord falls by a half step to the 3rd of the next: the C of Dm7 falls to the B of G7, and the F of G7 falls to the E of Cmaj7. Arpeggios that pass through these notes at the change make the harmony audible even with no chords playing.

TRACK 107 7th-chord arpeggios through ii – V – I in C

Dm7 **G7**

Cmaj7 **Cmaj7**

3

0 1 0 2 3 2 4 1

0 3 0 2 3

0 0 0 2 3

The last C of the Dm7 bar moves to B on the downbeat of G7; the F at the end of the G7 bar resolves to E on the downbeat of Cmaj7.

HOW IT CONNECTS

Chord-tone targeting is the starting point of improvising over changes. Chapter 35 builds whole solos from it, and Chapter 48 applies it to longer progressions.

Sweep picking

When an arpeggio has one note per string, the pick can travel across the strings in one continuous motion: down through the strings going up in pitch, up through them coming back. This is **sweep picking** (or economy picking applied to arpeggios). The pick does not stop and restart on each string; it pushes through and comes to rest briefly against the next string, ready to play it. Chapter 24 covers the picking motion, and its link with economy picking, in detail.

The key skill is muting. Each note must stop as the next one sounds, otherwise the arpeggio blurs into a strummed chord. Lift the fretting finger slightly as soon as the pick has left its string, and let the side of the picking hand rest on the lower strings.

Start with three-string triads from Chapter 10. The voice-led shapes on strings 3–2–1 make an ideal first sweeping exercise because the hand barely moves between chords.

▶ **TRACK 108** Three-string sweeps through C – Am – F – G

C **Am**

F **G** **C**

Use downstrokes while the line rises and upstrokes while it falls. Start slowly enough that every note is separate.

CAREFUL · Speed comes last

Sweeping fast before the notes are separated produces a blur. Practise at a tempo where you can hear each note stop as the next begins, with a metronome, and raise the tempo only when that is consistent.

Fingerstyle arpeggios

In fingerstyle playing, an arpeggio usually means holding a chord with the fretting hand while the picking hand plays its notes in a pattern. The thumb (p) plays the bass strings; i, m and a play the 3rd, 2nd and 1st strings. Mauro Giuliani's 120 right-hand studies (Op. 1) are built entirely on such patterns, and the same patterns underlie folk and pop fingerpicking.

Pattern	Notes per beat	Character
p – i – m – a	4 (sixteenths) or 2 (eighths)	Rising, flowing
p – a – m – i	4 or 2	Falling, harp-like
p – i – m – a – m – i	6 (one bar of 3/4 or 6/8 in eighths)	Rolling; the classic ballad pattern
p – i – m – i	4 or 2	Gentle, rocking

▶ **TRACK 109** p – i – m – a over Am and E

Am **E**

The thumb alternates between the root and another bass string; the fingers rise through strings 3, 2 and 1.

Chapter 17 teaches the strokes and Chapter 18 develops these patterns into full accompaniments, including Travis picking.

NYLON · STEEL

Nylon: practise arpeggio patterns with free strokes and **preparation:** place each finger on its string just before it plays. This improves accuracy and evenness. In slow pieces, let the thumb play rest strokes on bass notes that carry the melody.

Steel: a thumbpick gives the bass a punchier attack. For pick-style arpeggios, crosspicking (rolling a pick across three strings in a repeating pattern) is the bluegrass equivalent of a fingerstyle arpeggio.

Chapter 16 review · use it, train it, test it

Using it in real music

Solos that follow the chords. Over C – Am – F – G, aim for E over C, C over Am, A over F and B over G: the 3rd of each chord. The line will outline the changes even over a bass and drums alone.

Fingerstyle accompaniment. Hold the chords of a song and replace strumming with p – i – m – a or a rolling p – i – m – a – m – i. The song becomes an arrangement at once.

Intros and endings. A slow arpeggiated final chord, or a rising arpeggio into the first verse, is a standard way to frame a song on any guitar.

How to train it

Drill	How	Time	Goal
Shape of the day	One arpeggio shape from this chapter, up and down in eighth notes with a metronome	3 min	Clean at 80 bpm, then raise by 4 bpm per day
Through the changes	One-bar arpeggios over C – Am – F – G in one position; never jump more than three frets	4 min	No gaps between chords
3rds only	Over a practice loop of a four-chord progression, play only the 3rd of each chord, on beat 1	3 min	Find each 3rd without stopping
Three-string sweep	The sweep exercise in this chapter at a slow tempo	3 min	Every note separate at 80 bpm in triplets

Check your understanding

Write your answers, then check them in Appendix J.

1. Spell an Am7 and a D major arpeggio.

2. Spell a Cmaj7 arpeggio and a Bm7b5 arpeggio.

3. Write the formula of a half-diminished (m7b5) arpeggio.

4. Which two chord tones best define a 7th chord's quality?

5. Over a G chord, which three notes are the arpeggio's targets, and which is the most specific to G major?

6. In a ii – V – I in C, which note of Dm7 moves to the 3rd of G7?

7. In sweep picking, in which direction does the pick move when the arpeggio rises in pitch?

MY NOTES

The Picking Hand

Fingerstyle from the first p-i-m-a pattern to independent melody and bass, plus strumming and pick technique.

Fingerstyle Foundations

Fingerpicking Patterns

Travis Picking

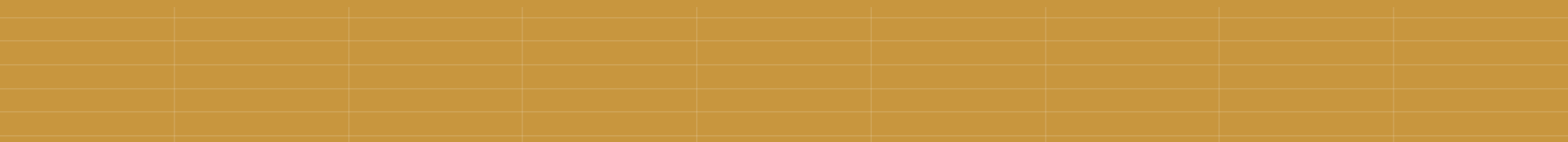
Melody over Bass

Fingerstyle Colours

Strumming Patterns

Strumming Songbook

Pick Technique



CHAPTER 17

Fingerstyle Foundations



TRACKS

17

Before patterns, arrangements and colours, the picking hand needs a stable position, two reliable strokes, an independent thumb and a consistent tone. This chapter builds those foundations and gives you a daily routine to keep them.

Fingerstyle means plucking the strings with the thumb and fingers instead of a pick. It lets you play a bass line, a chord and a melody at the same time, which is why a single guitarist can sound like a small band. The letters p, i, m and a (thumb, index, middle, ring) from Chapter 1 are used throughout this part of the book.

Everything in Chapter 18, Chapter 19, Chapter 20 and Chapter 21 rests on the material here. Work through it slowly; a week spent on the strokes saves months of correcting habits later.

Hand position

The picking hand hangs from a relaxed forearm that rests on the edge of the guitar body. The contact point on the forearm is roughly a hand's width below the elbow. From there, check five points:

1. **Wrist.** Slightly arched, never collapsed onto the top and never bent sharply sideways. A straight line from the forearm through the back of the hand to the knuckles is a good default.
2. **Knuckles.** Roughly parallel to the strings, so that i, m and a reach strings 3, 2 and 1 without stretching.
3. **Fingers.** Curved, with the tips meeting the strings at a slight angle. Each finger moves mainly from the big knuckle (the joint where the finger meets the hand), with the middle and tip joints following.
4. **Thumb.** Extended to the left of the fingers (as you look down), pointing along the strings towards the neck. The thumb should be ahead of i, so the two never collide.
5. **Height.** The hand floats over the soundhole or slightly towards the bridge. The palm does not touch the strings unless you are muting on purpose.

NYLON · STEEL · Where the hand sits

Nylon: the classical position keeps the wrist arched and free, with nothing resting on the top. The hand plays over the end of the soundhole nearest the bridge for a balanced tone.

Steel: many players anchor lightly, with the little finger touching the top below the 1st string or the heel of the hand brushing the bridge pins. A light anchor helps orientation; a heavy one locks the hand and kills the bass. If you anchor, keep it a touch, not a lean.

The default string assignment

For most accompaniment patterns, each finger has a home string. Keeping it fixed means the hand only has to think about *when*, not *where*.

Finger	Home string	Job
p	6, 5 or 4 (moves with the chord)	Bass: root, 5th, alternating bass
i	3	Inner harmony
m	2	Inner harmony
a	1	Top voice, often the melody

When a chord's root is on string 4 (D, Dm, F small shape), the thumb plays string 4 and the fingers stay on 3, 2 and 1. Chapter 18 gives the thumb's home string for every open chord.

Free stroke (tirando)

The free stroke is the default stroke for arpeggios and chords. The finger presses the string slightly towards the top, then releases it as the tip moves up and towards the palm, passing over the next string without touching it.

- **Prepare.** The tip touches the string before it plays. Contact first, then motion.
- **Push, do not hook.** Press into the string a little and let it slip off the tip (or nail). Pulling the string outward, away from the top, makes it slap against the frets.
- **Follow through.** The finger finishes its arc towards the palm and then relaxes back. A short, complete motion is better than a large one.
- **Keep the others still.** When *i* plays, *m* and *a* should not flick in sympathy. Finger independence (below) trains this.

▶ TRACK 110 Free stroke: *i* and *m* alternating on open strings

The first system of notation shows a 4/4 measure. The treble staff has four eighth notes (G4, A4, B4, C5) and the bass staff has four eighth notes (E2, D2, C2, B1). Fingerings alternate: *i* (treble), *m* (bass), *i* (treble), *m* (bass), *i* (treble), *m* (bass), *i* (treble), *m* (bass). The second system shows a 3/4 measure. The treble staff has three eighth notes (G4, A4, B4) and the bass staff has three eighth notes (E2, D2, C2). Fingerings alternate: *i* (treble), *m* (bass), *i* (treble), *m* (bass), *i* (treble), *m* (bass), *i* (treble), *m* (bass). The tablature lines for T, A, and B strings show '0' for open strings.

Alternate *i* and *m* strictly, even when crossing strings. Bar 3 practises the awkward crossing (*i* on the thinner string, *m* on the thicker one). Then repeat starting with *m*.

Rest stroke (apoyando)

In a rest stroke the finger plucks through the string and comes to rest on the next lower-pitched string (the next string towards the 6th). The motion is more downward into the guitar, so the string vibrates more perpendicular to the top. The result is louder, rounder and fuller than a free stroke with the same effort.

- Use the rest stroke for single-line melodies and scale passages, and to bring one note of a texture forward.
- The resting finger also mutes the string it lands on. In a descending scale this is useful; in an arpeggio it stops the chord from ringing, which is why arpeggios use free strokes.
- Alternate *i* and *m*. Never play two consecutive notes with the same finger unless the music requires it.

TRACK 111 Rest stroke: one octave of C major, alternating i and m

Each finger lands on the next lower-pitched string and stays there until the next note is played. Repeat starting with m.

TIP · Crossing strings with i and m

i sits naturally over the thicker string and m over the thinner one. Crossing with i on the lower-pitched string and m on the higher one is comfortable; the reverse (i on the 1st string, m on the 2nd) is awkward, because the fingers have to cross over each other. Bar 3 of the free-stroke example practises the awkward crossing on purpose. When you have a choice of fingering in a melody, prefer the comfortable crossing.

The thumb

The thumb moves from its base joint near the wrist, not from the tip joint. Keep the tip joint firm and let the whole thumb swing like a pendulum.

- **Free stroke thumb.** The thumb plucks down and moves slightly outward, clearing the next string. This is the normal stroke for alternating bass and arpeggio patterns: it lets the next string ring.
- **Rest stroke thumb.** The thumb plucks through and lands on the next higher-pitched string. It gives a strong, singing bass and mutes the string it lands on, which can clean up a bass line.
- **Contact point.** Use the left side of the thumb tip (for a right-handed player), with flesh and nail together if you grow the nail. A thumb that strikes with its pad gives a dull tone.

TRACK 112 Thumb: rest strokes on the bass strings, then a walking line

Bars 1–2 on open strings, bars 3–4 on the notes of E minor pentatonic. The thumb lands on the next string after each note; let it stay there until the next note.

Thumb independence

The thumb must keep time without being pulled along by the fingers. The exercise below holds a steady quarter-note bass while the fingers play off the beat. In bar 3 the melody notes are held across the beat, so the thumb plays while a finger note is still ringing. If the thumb starts to drift onto the off-beats, slow down.

TRACK 113 Thumb independence: steady bass, off-beat fingers

Em Am

3

Em Am

Hold the full Em and Am shapes throughout. Bars 1–2 alternate; bars 3–4 syncopate, so finger notes start on "&" and continue over the next thumb note.

Planting

Planting means placing a finger on its string just before it plays. It gives accuracy (the finger finds the string by touch, not sight) and control (the planted finger stops the string it touches, which you can use to cut a note short).

- **Full planting.** All the fingers needed for the next group are placed together, then play one after another. It suits ascending arpeggios such as p i m a, where each finger moves away from the strings still waiting to be played.
- **Sequential planting.** Each finger is placed on its string as the previous finger plays. Many teachers use it for descending patterns such as p a m i; some prefer not to plant descending arpeggios at all.
- **Planting the thumb.** In fast bass lines, place the thumb on its next string as soon as it has played. This keeps the hand anchored by the thumb rather than by the little finger.

TRACK 114 Planting practice: ascending (full) and descending (sequential)

Am Am E

1 & 2 & 3 & 4 & 1 & 2 & 3 & 4 & 1 & 2 & 3 & 4 & 1 & 2 & 3 & 4 &

Practise each bar slowly, stopping before each group to plant. Listen for the small click of the fingers touching the strings; at performance speed the planting becomes a light touch just before each note and the click disappears.

CAREFUL · Planting is a practice tool

Exaggerated planting cuts notes short and makes arpeggios sound choppy. Plant deliberately while learning; once the pattern is secure, let the fingers arrive just in time.

Finger independence

Each finger must move without dragging its neighbours. The ring finger (a) is the hardest, because it shares a tendon with the middle finger. Two kinds of exercise help: permutations on open strings, and pinches that make two fingers play together while the others stay still.

Permutations

Play each of these on open strings 3, 2 and 1 (i, m, a on their home strings), with the thumb on the open 5th string at the start of each group. Four times each, then move on.

Group	Patterns
Three fingers	p i m a, p a m i, p i a m, p m i a, p m a i, p a i m
Four-note rolls	p i m a m i (ascending and back), p a m i m a
Two fingers on one string	i m i m, m a m a, i a i a (all on string 1)

Pinches

A pinch is the thumb and a finger (or several) plucking at the same moment. In the map below, the thumb and a play together on the beat, then i and m answer. The aim is a perfectly simultaneous pinch with both notes at the same volume.

▶ **TRACK 115** Pinch independence on C and G7

Bars 3 and 4 move the pinch to m and then to i, so a different finger has to play alone each time. Watch the fingers that are not playing: they should hover over their strings, not flick.

Nails, flesh and picks

The surface that touches the string sets the tone more than any other single factor.

Surface	Tone	Suits	Notes
Flesh only	Warm, soft, rounded; less volume	Folk accompaniment, blues, quiet nylon playing	No maintenance. Play with the side of the tip, not the pad
Flesh and nail	Clear, projecting, wide range of colour	Classical, flamenco, solo steel-string	Flesh touches first, the nail releases the string. Needs shaping and polishing
Thumbpick	Loud, bright, defined bass	Travis picking, ragtime, country blues on steel	Fingers can stay bare. Muting with the palm becomes easier
Fingerpicks (metal or plastic)	Very bright, loud	Bluegrass banjo-style rolls, slide, dobro	Pick only forward (towards the floor); no rest strokes

NYLON · STEEL · Nails on the two guitars

Nylon: nails of about 1–2 mm beyond the fingertip, filed to follow the curve of the tip and polished smooth with 1000–2000 grit paper. The contact is flesh first, then nail. Rough or hooked nails catch the string and click.

Steel: steel strings wear nails quickly and the bright tone needs less help. Many players keep short nails, use bare flesh, or strengthen the nails with acrylic or a thin coat of hardener. A thumbpick is common for alternating bass.

Tone production

Three variables change the tone of a single plucked note:

1. **Point of contact.** Near the soundhole and over the fingerboard the tone is warm and soft; near the bridge it becomes bright and thin. The normal position is between the two. Chapter 21 uses this deliberately as *sul tasto* and *sul ponticello*.
2. **Angle of attack.** A finger that meets the string at a slight angle (the tip tilted towards the thumb) slides off smoothly and gives a round tone. A finger that meets the string square on gives a sharper, thinner tone, and with nails it may click.
3. **Direction of release.** Pushing the string towards the top makes it vibrate perpendicular to the soundboard: full and loud. Pulling it sideways along the top, or outward, gives a weaker, buzzier tone.

LISTEN FOR · Matching the fingers

Play the open 1st string four times with *i*, then four times with *m*, then four times with *a*. Most players hear a difference: *a* is often weaker and *m* often louder. The goal is to make all three indistinguishable. Record yourself, because the difference is easier to hear on playback than while playing.

A daily fingerstyle warm-up

Ten minutes, before any repertoire. Use a metronome and start each step at 60 bpm.

1. **Free strokes** (1 min): *i m* alternation on strings 1–3, the first example in this chapter. Focus: even volume from both fingers.
2. **Rest strokes** (1 min): the C major octave. Focus: fullness, landing on the next string.
3. **Thumb** (1 min): bass strings, rest strokes then free strokes. Focus: movement from the base joint.
4. **Permutations** (2 min): the arpeggio permutations on one chord. Focus: independence, especially *a*.
5. **Pinches** (2 min): the pinch map on C and G7. Focus: simultaneous attack.
6. **Study** (3 min): the warm-up study below, which combines all of the above.

TRACK 116 Warm-up study in E minor

The musical notation for the warm-up study in E minor is as follows:

System 1:

- Measure 1: Labeled **Em**. Treble clef, 4/4 time. Notes: G4 (quarter), A4 (quarter), B4 (quarter), C5 (quarter). Bass staff: T (0), A (0), B (0).
- Measure 2: Labeled **Am/E**. Treble clef, 4/4 time. Notes: E4 (quarter), F#4 (quarter), G4 (quarter), A4 (quarter). Bass staff: T (0), A (2), B (0).

System 2:

- Measure 3: Labeled **B7**. Treble clef, 4/4 time. Notes: G4 (quarter), A4 (quarter), B4 (quarter), C5 (quarter). Bass staff: T (2), A (0), B (2).
- Measure 4: Labeled **Em**. Treble clef, 4/4 time. Notes: E4 (quarter), F#4 (quarter), G4 (quarter), A4 (quarter). Bass staff: T (0), A (0), B (0).

*The thumb plays half notes on strings 6 and 4; the fingers play *i m a m i m a*. In bar 3 the fretting hand plays B7 (x21202) and the thumb moves to strings 5 and 4.*

Common faults

Fault	What you hear or see	Fix
Hand bounces with every note	Uneven rhythm, missed strings	Move the fingers from the big knuckle; keep the wrist and forearm still. Plant to build a reference
Fingers pull the string outward	Snapping, buzzing notes	Push slightly into the string before releasing
Thumb and i collide	Missed bass notes, clicks	Keep the thumb ahead of the fingers, pointing towards the neck
The a finger is weak	Top note drops out	Pinch drills (a with p); permutations starting on a
Thumb drags the tempo	Bass on the off-beat, rushing fills	Play the bass line alone with a metronome, then add one finger at a time
Tension in the shoulder or forearm	Fatigue after a few minutes, harsh tone	Check posture; play softer; take breaks every 20 minutes
Little finger pressed hard on the top	Stiff hand, short reach	Anchor lightly or not at all

Chapter 17 review · use it, train it, test it

Using it in real music

Accompaniment. Free strokes and an independent thumb are all that most song accompaniments need. The patterns in Chapter 18 and Chapter 19 are built from the moves in this chapter.

Melody playing. Rest strokes bring a melody forward over an accompaniment. In Chapter 20 the a finger often uses a rest stroke on the melody note while i and m play free strokes underneath.

Tone as an instrument. Changing the point of contact and the angle of attack gives one guitar several voices. Listen for this in classical players and in solo steel-string players alike.

How to train it

Drill	How	Time	Goal
i m alternation	Open strings 1–3, eighth notes, strict alternation, free then rest strokes	2 min	Even volume at 80 bpm
Thumb pendulum	Strings 6, 5, 4 in quarter notes, rest stroke, thumb moving from the base joint	2 min	Steady at 90 bpm, no finger movement
Planting cycle	Am and E bars of the planting map, plant before every group	3 min	Clean, no accidental muting at 70 bpm
Pinch alignment	Pinch map on C and G7	3 min	Pinches sound as one note
Tone matching	One string, i then m then a, four notes each; record	2 min	No audible difference between fingers

Check your understanding

Write your answers, then check them in Appendix J.

1. Where does a rest stroke with m come to rest after plucking the 2nd string?

2. Which stroke is normally used for arpeggios, and why?

3. Where does a thumb rest stroke on the 5th string land?

4. Which planting suits a descending p a m i pattern?

5. Name three things that change the tone of a plucked note.

6. Which finger is usually weakest, and why?

7. Why does the thumb point towards the neck, ahead of the fingers?

8. What is a pinch?

MY NOTES

CHAPTER 18

Fingerpicking Patterns



Hold a chord, let the thumb carry the bass and let the fingers fill in the harmony above it. This chapter is a pattern library, followed by a set of progressions and textures that put the patterns to work, each chosen to train one specific skill.

A fingerpicking pattern is a fixed order of thumb and finger strokes repeated over each chord. Once the pattern runs without thought, the fretting hand changes chords and the picking hand keeps going. Every pattern in this chapter uses the strokes from Chapter 17: free strokes for the fingers, free or rest strokes for the thumb, and planting where it helps.

The two jobs of the picking hand

Every pattern divides the work as set out in Chapter 17: the thumb (p) plays the bass, usually the root on the strong beats, and the fingers keep their home strings (i on the 3rd, m on the 2nd, a on the 1st). The thumb moves between strings as the chords change; the fingers stay where they are, so the hand only decides *when* to play, never *where*.

Reading a picking map

The maps in this part of the book look like TAB, but instead of fret numbers they show which finger plucks which string. Hold the chord named above the bar and pluck the strings marked, left to right. The 1st string is at the top. Counts run underneath.

Colour	Finger
Rust	p (thumb)
Teal	i (index)
Gold	m (middle)
Black	a (ring)

Each dot also carries its finger letter. Dots stacked in one column are played together, as a pinch. Let every note ring until the chord changes: a map shows where the notes start, not how long they last.

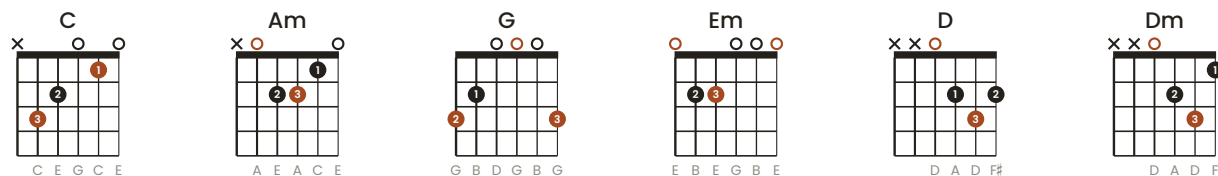
The thumb's home string

The thumb's home string is the one that holds the lowest root of the chord shape. The alternate bass is a second bass note, usually the 5th of the chord (on C and B7 the 3rd), used in patterns that rock between two bass strings. Chapter 19 builds a whole style on it.

Chord (shape)	Home string (B)	Alternate bass (B2)
C (x32010)	5 (C)	4 (E); or 6 at fret 3 (G)
Am (x02210)	5 (A)	4 (E)
A (x02220)	5 (A)	4 (E)
G (320003)	6 (G)	4 (D)
Em, E (022000, 022100)	6 (E)	4 (E) or 5 (B)
D, Dm (xx0232, xx0231)	4 (D)	5 open (A)
F (xx3211)	4 (F)	5 at fret 3 (C), adding the little finger (x33211)
G7 (320001)	6 (G)	4 (D)

Chord (shape)	Home string (B)	Alternate bass (B2)
B7 (x21202)	5 (B)	4 (D#) or 6 at fret 2 (F#)

Open chords and their bass strings



The lowest root of each shape is the thumb's home. On D and Dm the open 5th string (A) is the alternate bass, even though the shapes are usually drawn with it muted.

SHORTCUT · The thumb's home

When you learn any new chord, find the lowest root in the shape. That one habit makes every pattern below work with every chord you know.

The pattern library: 4/4

Each pattern below is shown on one chord. Learn it on that chord until it runs automatically, then move it to other chords. All are written in eighth notes, eight slots per bar.

Pattern	Strokes	Character
1. Rising arpeggio, alternating bass	p i m a p i m a	Flowing, even; the basic ballad pattern
2. p i m i	p i m i p i m i	Classical and folk; keeps a on the shelf for melody
3. Falling arpeggio	p a m i p a m i	Harp-like, cascading
4. Rise and hover	p i m a m i m i	Pop ballads; the top note sounds once per bar
5. Pinch pattern	[p a] i m i [p a] i m i	Marks each half bar; the top note can carry a tune
6. Double stops	p i [m a] i p i [m a] i	Fuller, more rhythmic
7. 3 + 3 + 2 roll	p i a p i a p m	Banjo-style forward roll; accents fall across the beat
8. Block pluck	[p i m a] - p [i m a] - [i m a] p -	Piano-like; the anticipation on "&" of 2 pushes the bar

▶ TRACK 117 Patterns 1–4 on C, Am, G and Em

One bar each. Repeat each bar four times before moving on.

▶ TRACK 118 Patterns 5–8 on D, Dm, E and A

In pattern 7 the thumb plays on 1, the "&" of 2 and 4, so the accents group the bar as 3 + 3 + 2 eighth notes.

NYLON · STEEL · Patterns on the two guitars

Nylon: play from the knuckles with free strokes, and use a rest stroke with a on the melody note when it should sing over the pattern. Plant in slow practice.

Steel: a light palm mute on the bass strings with the edge of the hand makes the thumb punchy and the top strings clear. A thumbpick adds volume to the bass.

Patterns in 3/4 and 6/8

The two time signatures are explained in Chapter 27. In 3/4 there are three beats per bar; the thumb plays beat 1 and the fingers play beats 2 and 3 (the "oom-pah-pah" of a waltz), or a six-note arpeggio fills the bar. In 6/8 there are two main beats per bar, on 1 and 4, each divided into three eighth notes. A six-note pattern in 6/8 sounds rolling rather than waltzing, because the accent falls on 1 and 4 instead of on 1 only.

▶ TRACK 119 A waltz in E minor: i – iv – V7 – i

Bars 1 and 3 pluck block chords on beats 2 and 3; bars 2 and 4 fill the bar with a six-note arpeggio. Accent beat 1 gently and keep 2 and 3 light.

▶ TRACK 120 The Andalusian cadence as a 6/8 ballad

i – bVII – bVI – V in A minor. The open 1st string rings through Am, Fmaj7 and E. Bars 2 and 4 move the thumb to the alternate bass on beat 4.

Classical tremolo: an introduction

In the classical tremolo the thumb plays a bass note and a, m and i then play the same melody note three times in quick succession: p a m i, four notes per beat. At speed the three repeated notes blend into a sustained melody over a moving bass. Francisco Tárrega's *Recuerdos de la Alhambra* is the best-known tremolo piece.

Start slowly, with the thumb on the beat and the three finger notes perfectly even. Unevenness (a "gallop", usually because a is late) is the main problem. Chapter 21 develops the technique further.

▶ TRACK 121 Tremolo study on A minor and E

Each beat is p a m i. The melody moves on the 1st string (E, F, G#), the thumb plays the chord tones on strings 3 to 6. Practise at 50 bpm until all four notes of each beat are equal in volume.

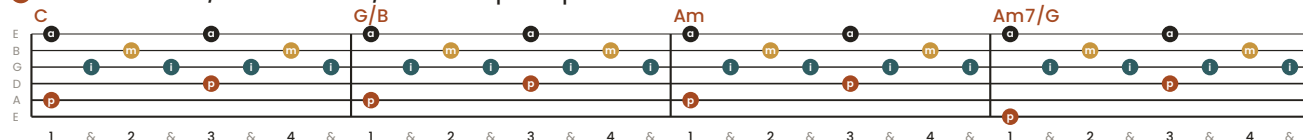
Patterns over progressions

The examples below apply the patterns to progressions and textures that each train one specific skill. Each one starts with what it trains; read that first, then play it slowly.

Descending bass with pinches

Trains: moving the thumb off its home string onto a bass line, while the fingers repeat the same pinch pattern.

TRACK 122 C – G/B – Am – Am7/G with the pinch pattern

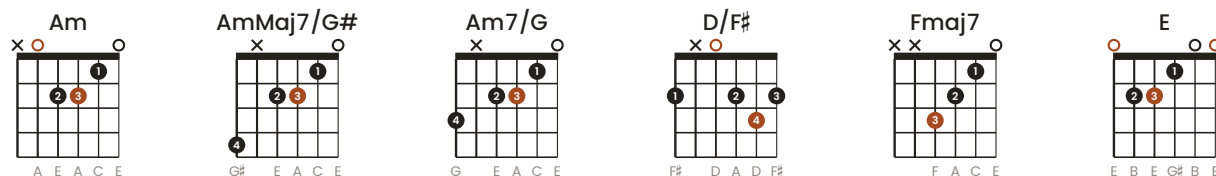


The thumb walks C – B – A – G on the beat. Fret the G in bar 4 with the little finger. Continue with F and G and end on C.

The minor line cliché

Trains: a chromatic bass line under a static finger pattern. The upper strings play the same notes for three bars while the bass descends one fret at a time: A, G#, G, F#, F, E. This kind of line, where one voice moves by half steps and the rest stays still, is called a line cliché.

The line cliché shapes



TRACK 123 Line cliché in A minor

Am AmMaj7/G# Am7/G D/F#

Fmaj7 E Am

The fingers play i m a in every bar; only the thumb and the fretting hand change. Bars 2 and 3 keep the Am shape and add a low G# and G with the little finger. In bar 4 fret the F# with the 1st finger (or the thumb over the neck on steel).

i – bVI – bIII – bVII in A minor

Trains: the falling half of an arpeggio, and keeping a common tone. The open 1st string (E) is in all four chords, so the top of the pattern never changes while the harmony moves.

▶ TRACK 124 Am – Fmaj7 – C – G6 with a rise-and-fall pattern

Am Fmaj7 C G6

Am, Fmaj7, C and G6 all contain E, so the open 1st string is the top note of every bar. Fmaj7 is played x33210, with C in the alternate bass.

Shapes over a pedal

Trains: moving a two-finger shape up the neck while the thumb repeats one open note. A note that stays the same while the harmony above it changes is a pedal. Here the open 4th string (D) is the pedal and the open 1st string (E) rings on top, so only strings 3 and 2 change.

▶ TRACK 125 Sliding shapes over an open D pedal

The shapes on strings 3 and 2 are the 3rds of the D major scale (A–C#, B–D, C#–E, D–F#). Against the D pedal they sound like Dmaj9, D6/9 and Dadd9 colours. Use fingers 1 and 2 for every shape and slide between them without lifting.

Suspensions inside the pattern

Trains: changing one note of a chord while the pattern continues. Lifting or adding one fretting finger turns D into Dsus4 or Dsus2 and A into Asus4 or Asus2. In bars 1–2 the moving note is on top; in bars 3–4 it is the middle voice, under the open E.

TRACK 126 Dsus4 – D – Dsus2 and Asus4 – A – Asus2 colours

D **A**

4

D

The thumb plays on the beat, the fingers on every "&". In bar 2 the Dsus4 arrives early on the "&" of 3 and resolves on the "&" of 4. For the A bars fret 4.2 with finger 1, 3.2 with 2, 2.2 with 3 and 2.3 with 4.

A moving bass under static pinches

Trains: the reverse of a pedal: the fingers repeat the same open-string pinch while the thumb plays a scale. The bass walks up and down the E Dorian scale (E F# G A B C# D) under the open B and E strings.

TRACK 127 Walking bass in E Dorian under open-string pinches

m and a pinch on every "&"; the thumb plays every beat. Fret the bass notes with whichever finger is free and let the open strings ring. The C# in bar 2 is the note that makes the sound Dorian rather than minor.

Three against the beat

Trains: finger independence over a steady thumb. The fingers repeat a three-note cycle (a m i) while the thumb plays on every beat. Because three does not divide eight, the accented a finger lands on a different part of the beat in each bar, and the cycle only returns to its starting point after three bars. Accent every a stroke.

TRACK 128 A three-note finger cycle against a steady thumb

Em Cmaj7 D

1 & 2 & 3 & 4 & 1 & 2 & 3 & 4 & 1 & 2 & 3 & 4 &

Read the fingers alone first, a m i a m i ..., across all three bars; then add the thumb on every beat. The accented a falls on 1, the "&" of 2 and 4 in bar 1, on the "&" of 1, 3 and the "&" of 4 in bar 2, on 2 and the "&" of 3 in bar 3.

A pattern in 5/4

Trains: hearing a long bar as 3 + 2 beats. The thumb marks beats 1 and 4; the fingers fill the first three beats with a six-note arpeggio and the last two with a four-note one.

TRACK 129 Am7 – Fmaj7 – Am7 – E7 in 5/4

Am7 Fmaj7 Am7 E7

1 & 2 & 3 & 4 & 5 & 1 & 2 & 3 & 4 & 5 & 1 & 2 & 3 & 4 & 5 & 1 & 2 & 3 & 4 & 5 &

Count "1 & 2 & 3 & 4 & 5 &" with a slight accent on 1 and 4. Fmaj7 is played x33210 so that the alternate bass is the C on the 5th string.

A pattern in 7/8

Trains: an uneven pulse. The seven eighth notes group as 2 + 2 + 3, and the thumb plays the first note of each group. The long group at the end gives the bar a forward lurch.

TRACK 130 D – C – G – D in 7/8 (2 + 2 + 3)

D C G D

1 2 3 4 5 6 7 1 2 3 4 5 6 7 1 2 3 4 5 6 7 1 2 3 4 5 6 7

Count "1 2, 1 2, 1 2 3" and play the thumb on every "1". D – C – G – D is a Mixolydian progression (I – $\flat$ VII – IV – I in D).

A Dorian vamp with a moving melody

Trains: making the a finger carry a tune while the thumb and the inner fingers keep the pattern. Am7 to D is the characteristic i – IV vamp of A Dorian. The melody moves on the 1st string; the inner voices stay as pinches on the "&".

▶ **TRACK 131** A Dorian vamp, melody on the 1st string

Am7 D Am7

4

D Am7

The melody notes start on beats 2 and 4 together with the thumb, as pinches. The F# over Am7 in bar 3 is the Dorian 6th; fret it with the 3rd finger while the 1st finger keeps the C on the 2nd string.

▶ **TRACK 132** Practice loop: a 6/8 ballad in A minor · LOOP · 6/8 ballad · 44 bpm (dotted quarter)

Am <i>i</i>	G <i>bVII</i>	Fmaj7 <i>bVI maj7</i>	E <i>V</i>
Am <i>i</i>	Am7/G <i>i42</i>	D/F# <i>IV6</i>	E <i>V</i>

IMPROVISE WITH

A minor pentatonic

A natural minor

Play the six-note roll (B i m a m i) over the loop, then the alternate-bass version. In bars 6 and 7 the thumb follows the descending bass G and F#.

Making patterns musical

- **Let it ring.** Hold the full chord shape even when you only pluck some of its strings.
- **Balance.** The bass is steady and slightly softer; the top notes are a little louder, because they carry the melody.
- **Vary the fourth bar.** Repeat a pattern three times, then change the last bar of the phrase (a single pinch and silence, or a switch to a block pluck) to mark the end of the phrase.
- **Add hammer-ons.** Lift a finger of the chord and hammer it back after the bass note: Asus2 to A, Dsus2 to D, or the open G string hammered to A over Am.
- **Find the melody.** Once a pattern is automatic, move the a-finger note to follow a tune. That is the first step towards the arrangements in Chapter 20.

Style	Typical pattern
Singer-songwriter ballad	1, 4 or 5 in 4/4; 6/8 rolls
Folk and country	Alternating bass (Chapter 19)
Classical studies	2 (p i m i), 3 and the 3/4 arpeggio; tremolo
Waltz and folk dance	3/4 bass + block pluck
Bossa nova	Thumb on 1 and 3 with syncopated block plucks (Chapter 50)
Celtic and modal	Pedal and drone textures (Chapter 54)

TRY THIS · Change the time signature

Take a song you already strum in 4/4 and fingerpick it with pattern 1. Then play the same chords in 6/8 with the six-note roll, and in 3/4 with the waltz pattern. The chords stay the same; the mood changes completely.

Chapter 18 review · use it, train it, test it**Using it in real music**

Turning strummed songs into fingerstyle. Any song with a chord chart can be fingerpicked: find each chord's home string, choose a pattern and keep it going.

Accompanying a singer. Arpeggio patterns leave space for the voice. The pinch pattern marks the start of each half bar, which helps a singer keep their place.

Classical repertoire. Many classical studies (Giuliani, Carcassi, Sor) are exactly these patterns over changing chords. Knowing the patterns makes reading those pieces much easier.

Arranging. Pedals, line clichés and suspensions turn a plain chord sequence into an arrangement without changing a single chord symbol.

How to train it

Drill	How	Time	Goal
Thumb alone	Play only the bass part of a map through its progression	2 min	Thumb lands on the right string every time
Pattern on one chord	One pattern on one chord with a metronome, from 60 bpm up	4 min	Even and automatic, eyes closed
Pattern through changes	Apply one pattern to a four-chord progression from this chapter	5 min	No gap at the chord changes
Three-note cycle	The three-against-the-beat map, fingers only, then with the thumb	3 min	Accents stay on a for three bars at 70 bpm
Odd metres	5/4 and 7/8 maps, counting aloud	3 min	Thumb on the group starts without hesitation

Check your understanding

Write your answers, then check them in Appendix J.

1. Which pattern in the 4/4 library keeps a free for a melody, and what are its strokes?

2. Which string is the thumb's home for a C chord? For a G chord? For a D?

3. In Am – Fmaj7 – C – G6, why does the top note of the pattern never change?

4. In the 6/8 Andalusian cadence map, where does the thumb play the alternate bass in bars 2 and 4?

5. In the classical tremolo, which fingers play the repeated melody

6. What is a pedal tone?

7. In the line cliché Am - AmMaj7/G# - Am7/G - D/F#, which notes do

8. How are the seven eighth notes of the 7/8 example grouped, and why?

MY NOTES

CHAPTER 19

Travis Picking



TRACKS

In Travis picking the thumb plays a steady alternating bass on every beat while the fingers play the harmony and melody between and against it. One hand sounds like a bass player and a second guitarist at once.

Travis picking is named after Merle Travis (1917–1983), whose thumb-style playing from Muhlenberg County, Kentucky, made it famous in the 1940s. Chet Atkins built on it, and it became the foundation of folk, country and fingerstyle blues accompaniment. Its defining feature is the thumb: it plays on every beat, alternating between two bass strings, and it never stops. Everything the fingers do sits on top of that pulse.

This chapter builds the pattern in steps, then varies it, adds melody and applies it in several keys. You need the free strokes and the independent thumb from Chapter 17 and the home strings from Chapter 18.

The thumb part

The thumb plays quarter notes: the root of the chord on beats 1 and 3, an alternate bass note on beats 2 and 4. The alternate note is usually on the next string up (towards the 1st string); on D and F it is the string below. On some chords the thumb uses three strings, which makes the bass line more interesting.

TRACK 133 The thumb alone: C, G7 and the three-string bass

8

p

C G7 C G7

5

C G7 C C

Bars 1–2 use two strings per chord. From bar 3 the C bass uses three strings (C, E, G, E) and the G7 bass adds the B on the 5th string. For the low G under C, move the 3rd finger from the 5th string to the 6th string, 3rd fret.

Practise the thumb part on its own until you can talk over it. If the thumb hesitates at a chord change, the whole texture falls apart; a dropped finger note is barely noticed.

Alternate-bass choices per chord

Chord (shape)	Root (beats 1, 3)	Alternate (beats 2, 4)	Three-string option
C (x32010)	5th string, C	4th string, E	5 – 4 – 6 – 4, with G at 6th string fret 3
G (320003)	6th string, G	4th string, D	6 – 4 – 5 – 4, with B on the 5th string
G7 (320001)	6th string, G	4th string, D	6 – 4 – 5 – 4
D (xx0232), D7 (xx0212)	4th string, D	5th string open, A	4 – 3 – 5 – 3 is possible but crowds the fingers
A (x02220), A7 (x02020)	5th string, A	4th string, E	5 – 4 – 6 – 4, with open E on the 6th string
Am (x02210)	5th string, A	4th string, E	5 – 4 – 6 – 4
E, Em, E7	6th string, E	4th string, E (octave) or D on E7	6 – 4 – 5 – 4, with B at 5th string fret 2
F (xx3211)	4th string, F	5th string fret 3, C	Barre F (133211): 6 – 4 – 5 – 4
B7 (x21202)	5th string, B	4th string, D#	5 – 4 – 6 – 4, with F# at 6th string fret 2

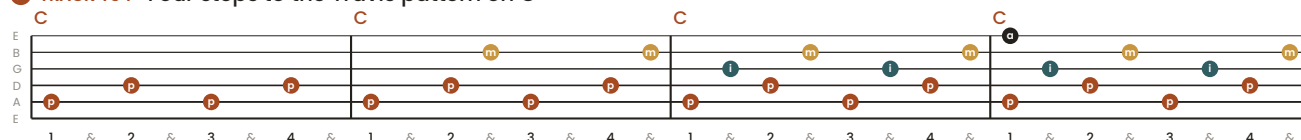
The rule behind the table: the root on the strong beats, another chord tone on a neighbouring bass string on the weak beats. That is the 5th wherever the shape has one there (G, D, A, Am, F), the 3rd on C (E) and B7 (D#), and the octave on E and Em. On E7 the 4th string gives D, the $b7$, which is fine because it is a chord tone.

Building the pattern

Add the fingers one step at a time. Each step is one bar of the map below; repeat each bar until it is comfortable before moving on. The fingers play on the "&" between the thumb notes, so the two parts interlock.

1. **Thumb only.** Root and alternate bass.
2. **One finger.** m on the 2nd string on the "&" of 2 and 4.
3. **Basic Travis pattern.** i on the 3rd string on the "&" of 1 and 3, m on the 2nd string on the "&" of 2 and 4.
4. **Pinch on 1.** As step 3, with a pinching the 1st string together with the thumb on beat 1.

TRACK 134 Four steps to the Travis pattern on C



The thumb never changes. Step 3 is the pattern most players mean by "Travis picking"; step 4 adds a top note on the downbeat.

The same pattern in notation shows how the parts fit: the thumb in quarter notes with stems down, the fingers in eighth notes on the off-beats with stems up.

TRACK 135 The basic pattern through C – Am – F – G7

Step 4 on four chords. F is played x3211, so the thumb alternates between F on the 4th string and C on the 5th. Hold every shape completely so that the notes ring into each other.

Step 4 on four chords. F is played x3211, so the thumb alternates between F on the 4th string and C on the 5th. Hold every shape completely so that the notes ring into each other.

Variations

Once the basic pattern is automatic, the fingers can move freely while the thumb stays put. Three common variations:

- **Syncopated pattern.** Fingers on 1, the "&" of 2 and the "&" of 3. The notes group as 3 + 3 + 2 eighth notes, which gives a rolling, slightly lopsided feel against the even thumb.
- **Backbeat pinch.** The pinch moves to beats 2 and 4, together with the alternate bass, so the top strings accent the backbeat like a snare drum.
- **Double stops.** i and m pluck strings 3 and 2 together on the off-beats for a fuller, more rhythmic sound.

TRACK 136 Syncopated, backbeat and double-stop variations

Bars 1–2 syncopated, bar 3 backbeat pinch, bar 4 double stops ending on a single a.

TRY THIS · Change variation every bar

Play G – Em – C – D twice. The first time use the basic pattern throughout; the second time switch variation every bar. The thumb must not notice the difference.

Palm muting the bass

On a steel-string guitar the bass strings ring for a long time, and an alternating bass soon blurs into a drone. Many Travis players rest the edge of the picking hand lightly on the bass strings, just in front of the saddle. The thumb notes then sound short and percussive, a thump rather than a boom, while the fingers play the unmuted treble strings and ring freely.

- **Touch, do not press:** the bass notes should still have pitch.

- Mute strings 6, 5 and 4 only. If the 3rd string is muted the i notes go dull.
- The mute stays in place as the thumb moves between strings; the hand pivots at the contact point.
- Use it selectively. Mute in verses for a tight, driving sound; lift the hand in a chorus or an ending for a fuller sound.

NYLON · STEEL · Thumbpick, palm mute and nylon

Steel: a thumbpick gives the bass a sharp attack and more volume, and makes palm muting easier because the pick, not the flesh, strikes the string. Many players use a thumbpick with bare fingers.

Nylon: the bass strings already decay quickly, so palm muting is rarely needed. Play the thumb with free strokes and keep the bass lighter than the treble. To shorten a bass note, touch the string with the side of the thumb as it moves to the next one.

Adding melody on top

The step from accompaniment to solo playing is to let the fingers play a tune instead of a fixed pattern. The thumb continues exactly as before; the fingers now play melody notes, either on the off-beats between thumb notes or together with the thumb as pinches. Melody notes are usually on strings 1 and 2 and come from the chord shape you are holding, or from one extra finger added to it.

The study below is an original melody over C – F – C – G7. Learn the thumb part alone first (it is the alternating bass you know), then the melody alone, then put them together a bar at a time.

Steps for adding a melody

1. Play the thumb part alone through the whole progression.
2. Play the melody alone, with the fretting hand already in the chord shapes.
3. Combine one bar at a time, very slowly. Find which melody notes fall together with the thumb (pinches) and which fall between.
4. If a melody note is too awkward to fret inside the shape, change its octave, or replace it with the nearest chord tone and keep the rhythm.

▶ TRACK 137 Thumb-style study in C

The musical notation for Track 137 is presented in two systems. Each system contains four measures of music. The first system covers measures 1-4, and the second system covers measures 5-8. The chords are C, F, C, and G7. The notation includes a treble clef, a key signature of one flat (Bb), and a bass line with fingerings for thumb (T), index (A), and middle (B) fingers. The melody is written in the treble staff, and the bass line is written in the bass staff. The notation includes various note values, rests, and accidentals.

Many melody notes start on the "&" and are held over the next beat, so the thumb plays under a sustained note. That syncopation is the sound of thumb-style picking. F is x33211; on G7 fret the D on the 2nd string with the little finger.

Travis picking in several keys

The pattern depends only on knowing each chord's bass strings. These maps take the basic pattern (step 4) through I – IV – V progressions in three keys.

▶ TRACK 138 G – C – D – G

Key of G. On D the thumb alternates between the 4th string and the open 5th string.

▶ TRACK 139 A – D – E7 – A

Key of A. On E7 the alternate bass is the open D on the 4th string.

Hammer-ons inside the pattern

Hammer-ons and pull-offs add notes without extra picking-hand strokes. The finger plucks one note and the fretting hand sounds the next, so the pattern gains a note while the thumb part stays exactly the same. Sus2 chords are the natural source: Asus2 to A, Dsus2 to D.

▶ TRACK 140 Hammer-ons on A, D and E

Each hammer-on starts from an open string on an off-beat and lands a sixteenth note later. On A, lift the finger on the 2nd string and hammer it back; on D, the 1st string; on E, the 1st finger hammers the G# on the 3rd string.

A minor-key progression

The pattern works the same way in minor keys. This progression descends from the tonic: i – $\flat$ VII – $\flat$ VI – V in E minor.

▶ **TRACK 141** Em – D – C – B7

Em D C B7

1 & 2 & 3 & 4 & 1 & 2 & 3 & 4 & 1 & 2 & 3 & 4 & 1 & 2 & 3 & 4 &

The thumb's home moves 6 – 4 – 5 – 5. On B7 the alternate bass is the D# on the 4th string, the leading tone that pulls back to Em.

Around the circle of fifths

Chords that move by 4ths (E7 to A7 to D7 to G) are the backbone of ragtime and much early country and blues. Each chord is the dominant of the next, so the progression keeps pulling forward until it reaches the tonic. Chapter 32 explains why.

▶ **TRACK 142** E7 – A7 – D7 – G

E7 A7 D7 G

1 & 2 & 3 & 4 & 1 & 2 & 3 & 4 & 1 & 2 & 3 & 4 & 1 & 2 & 3 & 4 &

VI7 – II7 – V7 – I in G (V7/ii – V7/V – V7 – I). The thumb's home moves 6 – 5 – 4 – 6. Listen to the top note too, which is E, E, F# and G.

▶ **TRACK 143** C – A7 – D7 – G7: the same chain, ending in C

C A7 D7 G7

1 & 2 & 3 & 4 & 1 & 2 & 3 & 4 & 1 & 2 & 3 & 4 & 1 & 2 & 3 & 4 &

The ragtime progression I – VI7 – II7 – V7 in C, with the syncopated variation. Repeat it, and finish on C.

▶ **TRACK 144** Practice loop: ragtime changes in C · LOOP · pop · 92 bpm

C I	A7 VI7	D7 II7	G7 V7
C I	A7 VI7	D7 II7	G7 V7
C I			

IMPROVISE WITH

C major pentatonic

4 5 6 7 8 9

C major

7 8 9 10 11 12 13 14

Keep the alternating bass going through every chord. In bar 7 the chord changes on beat 3, so the thumb plays D and A on beats 1–2, then G and D on beats 3–4.

Walking bass between chords

The thumb can also join chords with a short scale line. On beats 3 and 4 of the bar before the change, the thumb walks to the next root by step. The fingers continue unchanged.

TRACK 145 Walk-ups and walk-downs in G

G **C**

4

D7 **G**

Bar 2 walks up G – D – A – B into C; bar 4 walks down C – E – B – A towards D7. In bar 2, lift the 2nd finger to play the open A, then put it back for the B.

Common problems

Problem	Cause	Fix
Thumb skips a beat at chord changes	Fretting hand late; thumb waits for it	Practise changes with the thumb only; change on beat 4 if needed and let the last finger note go
Thumb plays on the off-beat	Fingers drag the thumb	Slow down; say "thumb" on each beat; play the thumb with a metronome alone
Pattern sounds mechanical	Every note equal	Accent the pinch on beat 1, keep the thumb slightly softer, mute the bass a little
Bass notes blur together	Long steel-string sustain	Palm mute strings 6–4 lightly
a misses the 1st string in the pinch	Hand moves towards the bass	Keep the forearm still; the thumb moves from its base joint

Chapter 19 review · use it, train it, test it

Using it in real music

Folk and country accompaniment. Many fingerpicked folk songs from the 1960s onwards use an alternating bass under a singer. Use the basic pattern in verses and the syncopated one to lift a chorus.

Ragtime and country blues. Circle-of-fifths progressions with an alternating bass are the core of the style. Players such as Mississippi John Hurt and Elizabeth Cotten used a steady thumb and syncopated melody, often with only the thumb and one finger.

Solo arrangements. A melody on top of a Travis bass makes a complete performance without a singer. Chapter 20 develops this into full arrangements, and Chapter 46 and Chapter 43 apply it in style.

Check your understanding

1. On which beats does the thumb play the root in a Travis pattern, and on which the alternate bass?

2. In the basic pattern, where do the fingers play?

3. Which bass notes does the three-string bass on C use?

4. Which string gives the alternate bass for a D chord?

5. Why do many steel-string players palm mute the bass strings?

6. What makes a melody note "syncopated" in thumb-style playing?

7. In E7 - A7 - D7 - G, what is the relationship between each chord and the next?

8. What should you drop first if a passage is too hard: a bass note or a finger note?

MY NOTES

CHAPTER 20

Melody over Bass



TRACKS

Solo fingerstyle guitar is two-part music, with a melody on top, played by the fingers, and a bass line underneath, played by the thumb. This chapter teaches you to read the two voices, play them independently, and arrange a tune for solo guitar step by step.

In a pattern, the fingers repeat the same shape whatever the music does. In melody-over-bass playing the fingers play a tune, with its own rhythm, while the thumb plays a bass line with a different rhythm. It is the texture of classical guitar studies, thumb-style country, ragtime and almost all solo fingerstyle arrangements.

The skills come from earlier chapters: the independent thumb from Chapter 17, the alternating bass from Chapter 19 and the chord shapes from Chapter 11. Notation reading is covered in Chapter 26; here we focus on how two voices are written on one staff.

Reading two voices

Guitar music writes both parts on the same staff, distinguished by stem direction:

- **Upper voice (melody):** stems up. Played by i, m and a, usually on strings 1–3.
- **Lower voice (bass):** stems down. Played by p, usually on strings 4–6.
- **Each voice fills every bar on its own.** If the bass is silent, the lower voice shows a rest. Count each voice separately: in 4/4 the stems-up notes add up to four beats, and so do the stems-down notes.
- **Notes that start together** are vertically aligned. They are pinches: thumb and finger together.
- **Held notes.** A bass half note under four melody eighth notes means the bass string keeps ringing while the fingers move. Do not lift the fretting finger until the note's value has run out.

The TAB shows the same notes, but not which voice they belong to or how long they last. Use the notation for rhythm and voice, the TAB for where to play.

TRACK 146 Two voices, two rhythms

The notation shows a melody line (stems up) and a bass line (stems down) in 4/4 time. The melody moves in half notes, and the bass moves in quarter notes. The melody notes are G, C, D, G. The bass notes are G, C, D, G. The TAB below shows the fret numbers for each note.

Measure	Melody (Stems Up)	Bass (Stems Down)
1	G (3)	G (3)
2	C (1)	C (3)
3	D (2)	D (0)
4	G (0)	G (3)

The melody moves in half notes, the bass in quarter notes. On every melody note the thumb plays with it (a pinch); on the other beats the thumb plays alone while the melody note rings.

Bass on the beat, melody in between

The thumb usually keeps a regular pulse, and the melody is placed against it. Three relationships occur constantly:

1. **Together.** Melody note and bass note start at the same time. Play a clean pinch; the melody note slightly louder.

2. **Melody between bass notes.** The melody note starts on an off-beat. The thumb must not move to meet it.
3. **Melody held over a bass note.** The melody starts on an off-beat and continues over the next beat (syncopation). The thumb plays under a ringing note.

Learn every two-voice passage with the method from Chapter 19: the bass alone, the melody alone (with the fretting hand in its real positions), then both, one bar at a time and slowly.

TRACK 147 A moving melody over held bass notes

Am E

3 Am E Am

The bass is played once per bar and must ring for the whole bar. Keep the fretting hand in the Am and E shapes so that nothing stops the bass. The G (fret 3) in bar 1 and the D (2nd string fret 3) in bar 3 are passing notes, not chord tones.

Two independent lines

When the bass becomes a melody of its own, the two lines can move in the same direction (parallel motion) or in opposite directions (contrary motion). Lines a 10th apart (an octave plus a 3rd) sound especially full. The study below begins with parallel 10ths and then moves in contrary motion.

TRACK 148 Parallel tenths and contrary motion in C

C C G7 C

Bars 1–2 are parallel 10ths (C–E, D–F, E–G, F–A and back). In bar 3 the melody rises while the bass falls, arriving on G7 with F on top; the F resolves down to E over C in bar 4.

Melody in sixths

A second line can also run close beneath the melody. Sixths on strings 3 and 1 move up and down the neck in one shape, and the open bass strings keep ringing underneath. This is a common sound in folk and country fingerstyle.

TRACK 149 Sixths on strings 3 and 1 over an open D and A bass

D

	2	3	5	7	5	3	2	7	9	7	5	3	5	2
T	2	4	6	7	6	4	2	7	9	7	6	4	6	2
A	0				0			0				0		0
B			0			0			0					

The pairs are the 6ths of the D major scale (A–F $\sharp$, B–G, C $\sharp$ –A, D–B, E–C $\sharp$). Fret them with fingers 2 and 3 (or 1 and 2) and keep the shape as you slide. The thumb plays the open D and A strings as a drone under every pair.

Arranging a melody for solo guitar

A chord-melody arrangement keeps the tune on top and builds the harmony underneath. The work divides into four steps.

The four steps of an arrangement



Step 1: choose the key

- Write out the melody and find its lowest and highest notes.
- Choose a key where the melody fits on strings 1–3 in the first few frets. On the guitar the written range from E4 (4th string, fret 2) to A5 (1st string, fret 5) is the easiest.
- Prefer keys whose main chord roots are open bass strings: E, A and D (strings 6, 5, 4), with G and C close behind. Minor tunes sit well in E minor, A minor and D minor.
- If a singer needs another key, arrange in an easy key and use a capo.

Step 2: place the melody on the top strings

- Play the melody alone and pick positions that keep it on strings 1–3 as much as possible, leaving strings 4–6 for the bass.
- The melody must be the highest note at every moment. A harmony note above the melody turns the harmony note into the tune.
- Prefer positions that coincide with chord shapes. A melody note in a chord shape costs nothing extra; a note outside it needs a spare finger.

Step 3: harmonise with a bass

- Choose the chords: one per bar in a slow tune, often two per bar at cadences. Most folk tunes need only I, IV, V and their relatives (vi, ii, iii).
- On the first beat of each bar, put the root of the chord in the bass. Check that the melody note on that beat is a chord tone; if not, try another chord, or treat the note as a passing note and harmonise the next one.
- Avoid moving the melody and the bass in parallel octaves or parallel 5ths for more than a note or two; it sounds empty.

Step 4: fill in

- Add one harmony note directly below the melody on the strong beats (a 3rd or 6th below sounds fullest).
- Give the bass a second note in each bar (the 5th or the octave), or a walking line into the next chord.
- In long melody notes, fill the gap with an arpeggio or a moving bass.
- Then simplify: anything that needs a stretch or a jump the hand cannot make in time is removed. Melody first, bass second, inner notes last.

Worked example: Greensleeves

Greensleeves is a traditional English tune, first recorded in the late 16th century. It is in a minor key with a triple-time lilt, and it fits naturally in A minor on the guitar. The arrangement below follows the four steps.

Step 1. The melody runs from E4 to G5 (written). In A minor that is entirely within the first position: strings 1–4, frets 0–4. The chords (Am, C, G, Em, F, E) all have open or first-position bass notes.

Step 2. The melody alone, in its first-position fingering. Learn it until you can sing it while playing.

TRACK 150 Step 2: the melody alone (first eight bars)

The musical notation for Step 2 shows the melody alone for the first eight bars of Greensleeves. The notation is in 3/4 time, A minor key. It consists of two phrases of four bars each. The first phrase starts on an upbeat (the A in the pickup bar) and ends on a long E. The last A of bar 8 is the upbeat to the next phrase. The notation includes a treble clef, a key signature of one flat, and a common time signature. Below the staff are three staves labeled T, A, and B, which are empty.

The phrase begins on an upbeat (the A in the pickup bar) and ends on a long E. The last A of bar 8 is the upbeat to the next phrase.

Step 3. Add the root of each chord in the bass on beat 1, held for the whole bar. Already this sounds like a complete piece.

Each melody note on beat 1 is a chord tone (C over Am, E over C, D over G, G over Em, C over F). In bar 6 the A on beat 1 is a suspension over E that resolves to G#.

Traditional English, arr. for guitar · fingerstyle · 3/4 · 92 bpm

 TRACK 152

7

E E Am C G Em Am

T	0	1	1	2	2	1	3	0	1	0	3	0	0	0	2	0	1	0
A	1	1	2	2	2	2	3	0	0	0	0	0	2	0	2	2	2	2
B	0	2	2	0	0	0	2	3	2	0	3	0	2	0	2	0	0	2

14

E Am C Em G Em

T	1	4	1	2	2	(2)	3	1	3	2	0	3	0	0	0	2	0	0
A	1	4	1	2	2	(2)	3	1	3	2	0	3	0	0	0	2	0	0
B	0	2	2	0	0	0	3	2	0	2	0	3	0	0	0	2	0	2

21

F E E E C Em G

T	1	2	2	2	1	2	0	1	1	2	3	1	3	2	0	3	0	0
A	2	2	2	2	1	2	1	1	2	2	3	1	3	2	0	3	0	0
B	3	3	0	0	2	2	0	2	2	0	3	2	0	2	0	3	0	0

28

Em Am E Am

T	0	2	0	1	0	2	1	4	1	2	(2)
A	0	2	0	2	2	2	1	4	1	2	(2)
B	0	2	0	0	2	2	0	2	0	0	(2)

PLAYING IT · Performance notes

Bring out the melody: play it with a slightly firmer free stroke (or a rest stroke with a or m where the next string is not needed) and keep the bass and harmony notes lighter.

Bar 2: play the F with the 1st finger; the open G under the E leaves the finger free. Bar 5: fret the F on the 4th string with the 3rd finger and the C on the 5th string (beat 3) with the little finger.

PLAYING IT · Performance notes (cont.)

Bars 14 and 30: the F# on the 4th string (fret 4) is a passing note; take it with the little finger while the 1st finger holds the G#.

Bars 17 and 25: the high G on the 1st string (fret 3) is the climax of the tune. Fret it with the little finger over the C shape and let it ring for the full bar.

TIP · Harmonic choices

The chords here are one of several traditional harmonisations. Bars 5 and 21 can use Am instead of F; bars 2 and 10 can use Am instead of C. Try both: F and C give more colour, Am gives a darker, more modal sound. The melody stays the same; only the bass notes change.

Arranging checklist

Check	Question to ask
Melody on top	Is the melody the highest note at every moment?
Melody audible	Is it louder than the accompaniment and played legato?
Bass	Does each chord have its root (or a deliberate inversion) on beat 1?
Playability	Can every bar be fretted without a stretch of more than four frets in the low positions?
Sustain	Do held notes in either voice actually ring for their full value?
Rhythm	Is the melody's rhythm the same as when it is sung, with no notes delayed to fit the bass?

TRY THIS · Arrange another tune

Choose a traditional tune you can sing from memory, such as *Amazing Grace*, *Scarborough Fair*, *Shenandoah* or *The Water Is Wide*. Work through the four steps: write out the melody, choose a key, add bass roots, then fill. Record each stage. If the arrangement feels crowded, remove inner notes until the melody is clear again.

Chapter 20 review · use it, train it, test it

Using it in real music

Solo performance. A melody with a bass line is enough for a complete instrumental performance of a song. Many solo fingerstyle arrangements are no more complicated than the Greensleeves arrangement here.

Classical repertoire. Studies by Carcassi, Carulli, Sor and Giuliani are written in exactly this two-voice notation; the pieces in Chapter 57 and Chapter 58 use it throughout.

Accompanying with melody. Even when someone else sings, a short melodic fill on top of the bass between vocal phrases makes an accompaniment sound arranged.

How to train it

Drill	How	Time	Goal
Voice by voice	Any example in this chapter: bass alone, melody alone, then both	5 min	Both voices together at a steady tempo
Counting two voices	Say the melody rhythm aloud while playing only the bass	2 min	No change in the bass when the melody syncopates
Melody balance	Greensleeves bars 1–8, melody louder than the bass	3 min	A listener can sing the tune from the recording
Sixths	Sixths study in D, then the same pairs over an open A bass	3 min	Shape kept while sliding, clean pitch
Arranging	Steps 1–3 on a new tune, eight bars	10 min	Melody and bass roots written out and playable

CHAPTER 21

Fingerstyle Colours



TRACKS

Beyond plucking notes, the picking hand can make bells, harps, drums and a dozen different tone colours from the same six strings. This chapter covers harmonics, campanella, tremolo, percussive techniques, rasgueado and the control of dynamics and tone.

The previous chapters were about *what* to play: patterns, bass lines and melodies. This one is about *how* it sounds. Each technique here adds a colour that can be used in any pattern or arrangement. Use them sparingly: a single harmonic at the end of a phrase or one percussive hit on a backbeat says more than a whole piece of effects.

Harmonics

Natural harmonics (a string touched lightly above the 12th, 7th, 5th or 4th fret and plucked) are explained in Chapter 36, with a map of the notes they produce and their use for tuning. This section adds what the picking hand contributes: natural harmonics inside fingerstyle textures, and the artificial and harp harmonics that only the picking hand can play.

Natural harmonics in a fingerstyle texture

- Pluck closer to the bridge than usual. The harmonic sounds clearer when the picking point is away from the node.
- A harmonic is quieter than a normal note. Pluck firmly, and use a rest stroke for single harmonics if the next string is not needed.
- The 4th-fret harmonic needs precise placement, slightly towards the nut from the fret wire; keep it for slow passages.

Because natural harmonics need no fretting, the fretting hand is free to play a bass line under them. In the study below, the fingers play 12th-fret harmonics on strings 3, 2 and 1 (G, B and E an octave up), while the thumb plays fretted bass notes that turn the same three harmonics into different chords.

▶ TRACK 153 Harmonics over a moving bass

Em Cmaj7 G6 Am7 Em

The fretting hand touches the 12th fret of strings 3–1 for every harmonic, and frets the bass notes between them. Bass and harmonics share the notes G, B and E, so every chord sounds full. Am7 is incomplete here (A, E, G, B), which gives it an open, modern sound.

Artificial harmonics

An artificial harmonic can be produced from any fretted note. The fretting hand holds the note normally. The picking hand touches the same string lightly with the tip of *i* exactly 12 frets higher, and plucks it with *a* (or with *p*) behind the touching finger. The note sounds one octave above the fretted note.

- Point *i* along the string, so that the fleshy tip lies directly over the fret wire.
- Pluck with *a* for single notes. Pluck with *p* when the fingers need to play other strings at the same time.
- In TAB, artificial harmonics are usually written at the touching fret (fretted fret + 12) and marked as harmonics. That is how they appear in this book.

Harp harmonics

Harp harmonics alternate artificial harmonics with ordinary notes on other strings. The harmonics and the plucked notes lie close together in pitch and blend into a cascading, harp-like sound. Chet Atkins and Lenny Breau made the technique famous on the steel-string and electric guitar.

The usual method: hold a chord shape; *i* touches the harmonic point and *p* plucks the harmonic; *a* plucks an ordinary note two strings higher. Then move both up one string.

▶ TRACK 154 Harp harmonics on C and G

C G C

p a p a p a p a

T 0 1 0 3
A 0 <12> <13> 3
B <15> <14> <12> <12> 0 1

Hold C (x32010, with the little finger adding G on the 1st string) and then G (320003). Every harmonic (diamond) is touched 12 frets above the fretted note and plucked with *p*; each ordinary note is plucked with *a*. Let everything ring.

Campanella

Campanella (Italian for "little bell") means playing consecutive notes of a scale or melody on different strings, so that each note keeps ringing while the next one sounds. Open strings are mixed with fretted notes in a higher position. The technique comes from the Baroque guitar and lute repertoire and is common in harp-like Celtic and modern fingerstyle playing (Chapter 54).

The scale below is G major from D to E, with every note on a different string from the one before. Hold each fretted note as long as the hand allows.

▶ TRACK 155 A campanella scale in G

p i p i p m i m a m i m p i p i

T 0 0 3 0 3 0 0
A 0 4 0 5 3 5 0 0
B 7 7 7 7 7 7 4 7

Fourth position. Use the 4th finger for the notes at fret 7, the 2nd for C (3rd string fret 5) and the 1st for F# (4th string fret 4) and for D (2nd string fret 3). Played normally on one or two strings, the same scale sounds dry; here the notes overlap like bells.

TRY THIS · Campanella over a drone

Play the same scale while the thumb repeats the open 4th string (D) on every beat. Then try the scale fragments B – C – D – E (2.0 – 3.5 – 2.3 – 1.0) and E – D – C – B in different rhythms over the D drone, as a short improvisation.

Tremolo

Chapter 18 introduced the classical tremolo: p plays a bass note, then a, m and i repeat one melody note, four notes per beat. The aim is a melody that seems sustained, over an independent bass.

- **Evenness first.** The three finger notes must be equal in time and volume. Practise a m i on one open string with a metronome, accenting a, then m, then i in turn.
- **Small movements.** Each finger moves only from its big knuckle. The hand stays still.
- **Plant the thumb.** Many players let p land on the next bass string after playing, which steadies the hand.
- **Slow and fast.** Alternate practice at a slow, exact tempo with short bursts at performance tempo; speed comes from relaxation, not effort.

 TRACK 156 Tremolo study on the 2nd string

The image displays a musical score for the song "The Sound of Silence" by Simon & Garfunkel. It includes a guitar part in standard notation and a bass part in tablature. The guitar part is written in G major (one sharp) and 4/4 time. The first system covers measures 1 through 4, with chords Em and Am indicated. The second system covers measures 5 through 8, with chords B7 and Em indicated. The bass part is written in a simplified tablature format, showing fret numbers for the right hand and string numbers for the left hand. The lyrics "p a m i" are written under the first four measures of the guitar part.

System 1 (Measures 1-4):

- Chords:** Em (Measures 1-2), Am (Measures 3-4)
- Lyrics:** p a m i

System 2 (Measures 5-8):

- Chords:** B7 (Measures 5-6), Em (Measures 7-8)

The thumb plays on every beat, including the 3rd string under a melody on the 2nd string. The C over B7 in bar 3 is the b9, a typical tremolo colour in minor keys.

NYLON · STEEL · Tremolo on steel

Nylon: the classical p a m i tremolo, with nails, is the standard. Some players use p i a m i for a five-note tremolo.

Steel: fingers on steel strings sound thinner in tremolo, and the long sustain blurs the bass. Steel-string players often use a pick tremolo instead (Chapter 24), or a slower *am i roll* that is heard as separate notes.

Percussive techniques

Percussive fingerstyle uses the guitar body and the strings as a drum kit while the guitar keeps playing chords and melody. Players such as Michael Hedges, Preston Reed and Andy McKee developed it on the steel-string guitar. Four basic sounds cover most of it:

Sound	How	Imitates
Body hit (low)	The heel of the hand or the side of the thumb strikes the top near the bridge, below the strings	Bass drum
String slap	The fingers (flat, relaxed) slap all strings against the frets near the end of the fingerboard	Snare drum

Sound	How	Imitates
Thumb slap	The side of the thumb strikes a bass string so that it hits the frets: a pitched note with a snap	Slap bass, snare
Fingertip tap	The fingertips or nails tap the top or the upper side	Hi-hat, rim

The first exercise uses only the picking hand. Hold an open Em chord (or let the strings ring open) and play the three rows together.

▶ TRACK 157 A percussive groove

Body hit (heel) String slap Fingers (chord)

The heel of the hand plays beats 1 and 3 (and the "&" of 3); the fingers slap the strings on 2 and 4; between them, i m a pluck the chord. Start at 60 bpm.

The thumb slap is the most useful single percussive sound because it keeps pitch. Slap the bass note on the beat and add a muted chuck with the fingers on the backbeat:

▶ TRACK 158 Thumb slap with a backbeat chuck

Em C

The dot is the slapped bass note; x is a chuck (the fingers strike muted strings); the up arrow is a light upward brush with i.

CAREFUL · Protect the guitar

Body hits on a thin top can crack it, and nails on the finish leave marks. Hit with the soft heel of the hand, close to the bridge where the top is strongest, and never with rings or a watch on. Many players fit a clear protective film over the hitting area.

Rasgueado basics

Rasgueado is the flamenco technique of strumming by flicking the fingers out one after another. For fingerstylists it is a way to add a burst of rhythm and volume to a piece without a pick. Close the hand loosely, with the fingertips resting against the thumb or the palm, then flick them out across the strings, one at a time, with the backs of the nails.

- **Four-stroke rasgueado:** c, a, m, i in turn (little finger first). Each finger is a separate, even stroke; together they make a roll.
- **Continuous rasgueado:** the four-stroke roll repeated, with i on the beat.
- **Single strokes:** i flicks down (towards the floor) and the back of i or the thumb returns upwards.
- Practise on muted strings first, at a slow tempo, until each finger sounds clearly.

▶ TRACK 159 Two rasgueado rhythms

Continuous (i c a m) Roll into 1 and 3

Row 1 plays i on each beat followed by c, a, m. In row 2, i plays beats 2 and 4 and is followed by c a m, so that the roll leads into the accented chord on the next beat.

Chapter 53 uses rasgueado in its proper rhythmic context (compás).

Dynamics and tone colour

Dynamics

Marking	Meaning	How to produce it
pp	Very soft	Minimal stroke; flesh more than nail
p	Soft	Free stroke with a short follow-through
mp, mf	Moderately soft, moderately loud	Normal free stroke
f	Loud	Deeper preparation; push into the string; rest strokes for melody
ff	Very loud	Full rest strokes, heavier thumb; take care not to buzz
crescendo, diminuendo	Gradually louder, softer	Change the depth of the stroke a little on every note

Volume comes from how far the string is pushed towards the top before it is released, not from speed of movement. A tense, fast stroke gives a harsh, buzzing tone without much more volume.

Balance

In a texture with melody and accompaniment, the parts should not all be at the same level. The melody is a step louder than the accompaniment (for example *mf* against *p*); the bass sits between the two. Practise any arrangement three ways: everything equal, melody too loud, then balanced, and listen to the difference.

Tone colour: where you pluck

Term	Where	Sound
Sul tasto (<i>dolce</i>)	Over the end of the fingerboard	Soft, warm, flute-like
Normale (<i>ord.</i>)	Between the soundhole and the bridge	Balanced
Sul ponticello (<i>metallico</i>)	Close to the bridge	Bright, thin, metallic

Contrasting colour is one of the simplest ways to vary a repeat. Play a phrase normally the first time, sul tasto and softer the second time.

▶ TRACK 160 One pattern, three colours

Bar 1 sul tasto and *p*; bar 2 normale and *mf*; bar 3 sul ponticello and *f*. Keep the tempo exactly the same.

NYLON · STEEL · Tone colour on the two guitars

Nylon: the angle of the nail also changes colour. A finger angled towards the thumb gives a rounder tone; a square attack gives a sharper one. Classical scores mark colour with *tasto*, *pont.* and *ord.*

Steel: the change from tasto to ponticello is even greater. The soundhole area booms on many steel-string guitars, so the normal position is usually closer to the bridge than on nylon.

Chapter 21 review · use it, train it, test it

Using it in real music

Endings. A chord of natural harmonics, or one harp-harmonic run, is a classic ending for a fingerstyle arrangement.

Variety in repeats. Change the tone colour, the dynamic or the texture (tremolo instead of single notes) on the second time through a section.

Solo groove. Body hits and thumb slaps let a solo guitarist imply a drum part. Add them to a pattern from Chapter 19: body hit on 1 and 3, string slap on 2 and 4.

How to train it

Drill	How	Time	Goal
Harmonics over a bass	The moving-bass study, harmonics alone, then with the bass	3 min	Every harmonic clear, bass notes in time
Artificial harmonics	C major scale in first position, each note as an artificial harmonic	4 min	Even volume, no fretted note sounding through
Tremolo evenness	a m i on the open 2nd string, sixteenths at 60 bpm	3 min	No gallop; record and check
Percussive groove	Groove diagram, hands only, then with an Em chord	3 min	Steady at 80 bpm
Colour change	One-pattern-three-colours map	2 min	Clear audible difference at the same tempo

Check your understanding

Write your answers, then check them in Appendix J.

1. Why is it easy to play a bass line under 12th-fret natural harmonics?

2. How is an artificial harmonic produced?

3. What are harp harmonics?

4. What is campanella?

5. In which order do the fingers move in a four-stroke rasgueado?

6. What does sul ponticello mean, and how does it sound?

7. Which percussive sound keeps a definite pitch?

8. What produces more volume in a free stroke: a faster finger or a deeper push into the string?

MY NOTES

CHAPTER 22

Strumming Patterns



Strumming is drumming with chords. Keep the strumming hand moving like a pendulum and every rhythm in popular music becomes a choice of when to touch the strings.

A strumming pattern is a rhythm played with chords. Hundreds of patterns exist, but they all come from one motion: a hand that moves down and up in strict time, and touches the strings only on some of its passes. Learn the motion first and the patterns follow.

The pendulum rule

The strumming hand moves down and up continuously, like a pendulum, whether or not it hits the strings.

- **Eighth-note patterns:** down on every beat (1 2 3 4), up on every "&".
- **Sixteenth-note patterns:** the hand moves twice as fast: down on the beat and on "&", up on "e" and "a".
- **A rhythm is a choice of connections.** To leave out a stroke, the hand still makes the motion but passes just above the strings (a miss).

Because the direction of each stroke is fixed by its place in the bar, the hand cannot get lost: a stroke on a beat is always down, a stroke on an "&" always up. This also keeps the time steady, because the hand never stops.

Reading the strum grids

Symbol	Meaning
Rust arrow down	Downstroke (D)
Teal arrow up	Upstroke (U)
Grey arrow	Light stroke: fewer strings, softer (d, u)
> above an arrow	Accent: play it louder
×	Muted chuck: strike strings deadened by the hand
Teal dot	Bass note: pick a single bass string (B)
Small grey dot	Miss: the hand moves but does not touch the strings

Bold counts are the beats; light counts are the subdivisions.

The pattern library

Play each pattern on a single chord (G or Em) for at least a minute, at 70 bpm, before applying it to a song.

▶ TRACK 161 1. All downs

↓

•

↓

•

↓

•

↓

•

1&2&3&4&

The foundation. Punchy rock and first songs. The up motions are misses.

▶ TRACK 162 2. Steady eighths

↓

↑

↓

↑

↓

↑

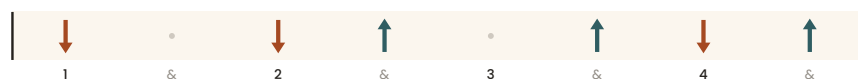
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1&2&3&4&

The pendulum made audible. Keep the upstrokes lighter than the downstrokes.

▶ **TRACK 163** 3. Old faithful



The most-used acoustic pattern. The miss on beat 3 creates the syncopation: the upstroke on the "&" of 2 is held over beat 3.

▶ **TRACK 164** 4. Folk drive



Busier than old faithful, without the gap on beat 3.

▶ **TRACK 165** 5. Accented old faithful



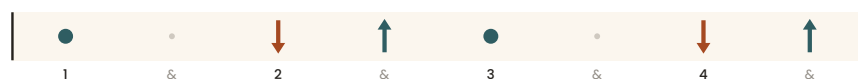
Accents on 1 and 4, a light stroke on 2. Accents give a pattern its shape.

▶ **TRACK 166** 6. Backbeat chuck



Muted chucks on 2 and 4 act like a snare drum.

▶ **TRACK 167** 7. Boom-chick



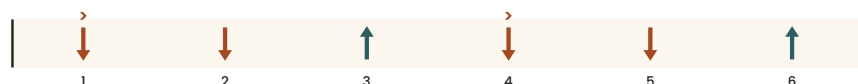
Bass note on 1, strum on 2, alternate bass on 3, strum on 4. Country and folk.

▶ **TRACK 168** 8. Waltz



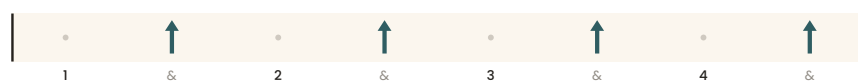
3/4: accent beat 1.

▶ **TRACK 169** 9. 6/8 lilt



Two big beats per bar (1 and 4). In 6/8 the hand moves down-down-up for each group of three, so both main beats are downstrokes.

▶ **TRACK 170** 10. Off-beat skank



Short upstrokes on every "&", with the fretting hand releasing pressure right after each one. Reggae and ska (Chapter 52).

▶ **TRACK 171** 11. Sixteenth funk



Sixteenth-note pendulum: down on the beats and "&", up on "e" and "a". Chucks on 2 and 4. Keep the wrist loose (Chapter 45).

▶ **TRACK 172** 12. Pop push



The last upstroke anticipates the next bar and is held over its downbeat. Change chord on that upstroke.

SHORTCUT • The invisible chord change

In most patterns the last stroke of the bar is an upstroke on the "&" of 4. Let it hit the open strings while the fretting hand is already moving to the next chord. Nobody hears it, and the rhythm never stops for a chord change.

From grid to notation

The grid and standard rhythm notation show the same thing. Old faithful, written out on a G chord, is quarter – eighth – quarter – eighth – eighth – eighth: the upstroke on "&" of 2 lasts through beat 3.

TRACK 173 Old faithful in notation

The fretboard diagram shows the following fret numbers for strings T, A, B across 8 beats:

Beat	T	A	B
1	3	0	2
2	3	0	3
3	3	0	3
4	3	0	3
5	3	0	3
6	3	0	3
7	0	1	2
8	0	1	2

The upstrokes are written with fewer notes, because an upstroke usually catches only the top three or four strings.

Accent patterns and odd metres

Accents can group the eight eighth notes of a 4/4 bar as 3 + 3 + 2 instead of 2 + 2 + 2 + 2. The pendulum still runs normally; only the accents move. The 3 + 3 + 2 grouping is found in Latin, pop and rock rhythms everywhere.

TRACK 174 3 + 3 + 2 accents

Accents on 1, the "&" of 2 and 4. The accented upstroke is the hard part; give it the same weight as a downstroke.

In an odd metre such as 7/8 the bar is grouped into twos and threes. The hand plays down on the first eighth of each group, so it sometimes plays two downstrokes in a row.

TRACK 175 7/8 as 2 + 2 + 3

Count "1 2, 1 2, 1 2 3". The extra downstroke at the end of the three-group resets the hand for the next bar.

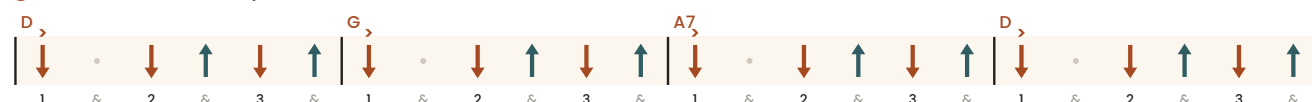
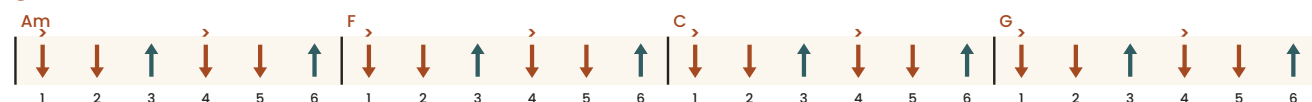
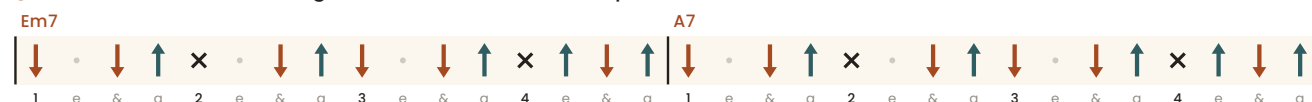
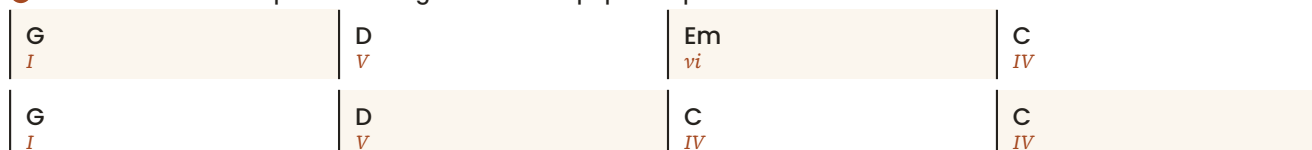
Strumming meets progressions

Patterns become songs when the chords change. Each chart below repeats one pattern per bar.

TRACK 176 Pop: I – V – vi – IV in G with old faithful

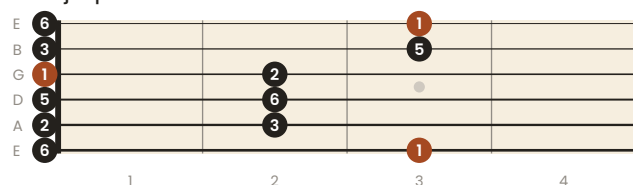
TRACK 177 Country: boom-chick with alternating bass in C

Pick the root on 1 and the alternate bass on 3 (C: strings 5 and 4; F: 4 and 5; G7: 6 and 4), as in Chapter 19.

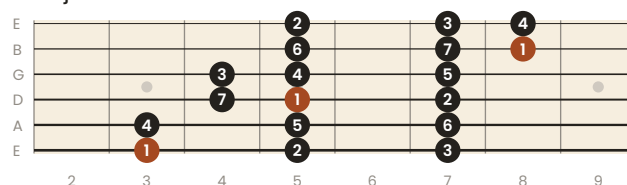
TRACK 178 Waltz in 3/4: I – IV – V7 – I in D**TRACK 179** Ballad in 6/8: vi – IV – I – V in C**TRACK 180** Funk: sixteenth groove on a two-chord vamp**TRACK 181** Practice loop: strum along in G · LOOP · pop · 84 bpm

IMPROVISE WITH

G major pentatonic



G major

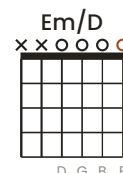
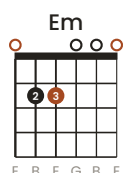
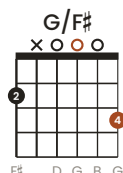
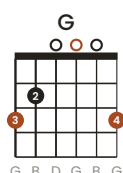


Play old faithful first, then switch pattern every four bars: all downs, steady eighths, folk drive, accented old faithful.

A moving bass inside the strum

Strummed chords can carry a bass line. Change only the lowest note of the chord from bar to bar and accent the first downstroke, so that the bass is heard. G – G/F# – Em – Em/D is a common descending line.

G – G/F# – Em – Em/D

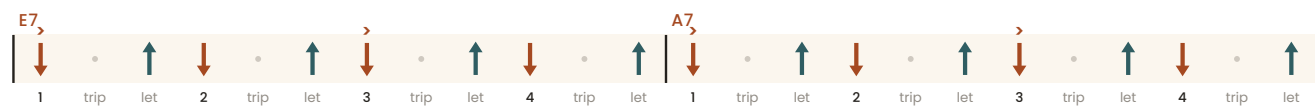


G is fingered 320004 here (or 210003, see Chapter 11). Keep the little finger on the G (1st string, fret 3) for the first two chords; on G/F# the fretting finger on the 6th string also touches and mutes the 5th string. Em/D is just the open D, G, B and E strings: strum strings 4–1 only.

TRACK 182 The descending bass, strummed

Shuffle and swing

In blues, jazz, swing and much country music the eighth notes are swung: long-short pairs that sit on the first and last notes of a triplet. Chapter 27 explains how swing is counted and written. For the strumming hand the change is simple: the pendulum now swings in triplets. The downstroke stays on the beat, the hand passes the strings on the middle triplet, and the upstroke falls on the last third of the beat. Play pattern 2 this way and you have the shuffle of the 12-bar blues in Chapter 43.

TRACK 183 Shuffle strum on E7 and A7

Three slots per beat. Down on each beat, a miss on the middle slot, up on the last one. Accent beats 1 and 3, or 2 and 4 for a heavier backbeat.

Dynamics, accents and muting

- **Accent the backbeat** (2 and 4) for rock and pop; accent beat 1 for folk, country and waltzes.
- **Strum fewer strings on upstrokes**, usually the top three or four. It sounds lighter and more controlled.
- **Aim the downstrokes**. On chords with a root on the 5th or 4th string, start the downstroke on that string, not on the 6th.
- **Palm muting**: rest the edge of the strumming hand on the strings next to the bridge for a short, chunky sound.
- **Fretting-hand muting**: relax the pressure on the chord immediately after a strum (staccato), or for a chuck relax it before the strum so that the strings click without pitch.
- **Build a song**: a lighter pattern in verses (all downs, boom-chick or a palm-muted pattern), the full pattern in choruses. A change of pattern lifts a chorus without any change of chords.

NYLON · STEEL · Strumming on the two guitars

Steel: a thin-to-medium pick (0.5–0.7 mm) is easiest for strumming. Hold it loosely enough that it can flex slightly as it crosses the strings.

Nylon: strum with the thumb (soft and warm) for downstrokes, and with the back of the nail of i for brighter strokes in both directions. The flamenco rasgueado (Chapter 21, Chapter 53) flicks the fingers out one after another for a rolling burst of sound.

TRY THIS · Strum along muted

Put on a song you like, mute the strings with the fretting hand and strum along with only the rhythm. When the pattern locks in with the recording, add the chords.

Chapter 22 review · use it, train it, test it

Using it in real music

Playing with others. A steady strum is the backbone of an acoustic band or a singalong, and more important than any chord voicing.

Arranging a song. Use a lighter pattern in verses and a fuller one in choruses to give the song shape. End with a single accented downstroke and let it ring.

Different styles. Boom-chick says country, off-beat skanks say reggae, sixteenths with chucks say funk, swung eighths say blues: the pattern sets the genre.

How to train it

Drill	How	Time	Goal
Silent pendulum	Mute the strings and strum a pattern along with a song	3 min	The hand never stops; locked to the beat
Pattern ladder	Patterns 1 to 4 for four bars each without stopping	4 min	Smooth switching between patterns
Last-upstroke changes	A four-chord progression, changing on the final upstroke	3 min	No gap in the rhythm at chord changes
Accent shift	Steady eighths with the accent on 1, then on 2, then on "&" of 2	3 min	Accent moves, tempo does not
Odd metre	7/8 pattern on D and C, counting aloud	3 min	Two complete minutes without losing the count

CHAPTER 23

Strumming Songbook



Fifteen well-known songs that guitar teachers use to teach strumming, each with its chord shapes, the commonly taught pattern and a practice loop to play along with.

The patterns in Chapter 22 become music when you play them in songs. This chapter gives fifteen songs that are widely taught as strumming songs, from 1961 to 2017, in straight 4/4, 6/8, 3/4, reggae and shuffle feels. Every song uses open chord shapes, and every one has a practice loop (drums, bass and a guitar playing the same strum) so you can hear the pattern and play along with it.

How to use the songbook

Each song has the same layout:

- **A summary above the chord boxes:** the key you hear on the recording, the capo position, the tempo, the time signature and the level.
- **Chord boxes** of every shape used.
- **A play-along loop** of the main section. The chart shows the chords and numerals; the grid below it is the strum to play in every bar.
- **A second chart** where the song has a clearly different chorus or bridge.
- **Notes** on the difficult change and where the accents fall.

The strums are the common simplified patterns that teachers and songbooks use, not transcriptions of the recordings. Real recordings vary the strum from bar to bar, add fills and drop out. Learn the basic pattern first, then listen to the record and copy its variations. No lyrics or melodies are printed: have the recording or a lyric sheet at hand to sing along.

Capo and shapes

A capo raises every open string by the same number of half steps. With a capo at fret 2, the G shape sounds as A, Em as F#m, C as D and D as E. Chord charts for capo songs name the **shapes**, because those are what you finger; the summary line gives the **sounding key**. The play-along loops sound at the capo pitch, so put the capo on before you play along. The numerals under each chart refer to the shapes, so they are the same as in Chapter 13 without the capo.

Practising a song's strum

1. **Strum the pattern on one chord** with the strings muted until the hand plays it without thought.
2. **Practise the hardest change** on its own: the two chords, back and forth, one strum each, then two, then the full pattern.
3. **Play along at a slow tempo.** Set the player to about 70% speed (or count the pattern at a lower tempo against a metronome) and play the whole loop without stopping. A wrong chord is fine; a stopped hand is not.
4. **Play along at full tempo.** If a change is late, play only the first downstroke of that bar until it lands on time, then fill the pattern back in.

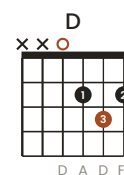
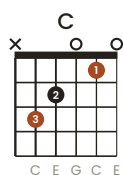
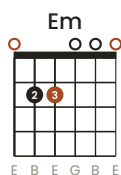
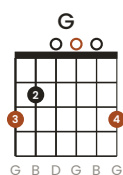
TIP · Keep the hand moving

Every pattern here follows the pendulum rule from Chapter 22: down on the beats, up on the "&" (or down on the eighths and up on the sixteenths in faster patterns). When a change is late, let the last upstroke of the bar hit the open strings while the fretting hand moves.

Straight 4/4: first songs

Stand By Me (Ben E. King, 1961)

Key A · capo 2 (G shapes) · 119 bpm · 4/4 · beginner



TRACK 184 Stand By Me: verse and chorus · LOOP · pop · capo 2 · 119 bpm

G <i>I</i>	G <i>I</i>	Em <i>vi</i>	Em <i>vi</i>
C <i>IV</i>	D <i>V</i>	G <i>I</i>	G <i>I</i>

STRUM THIS PATTERN IN EVERY BAR

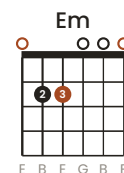
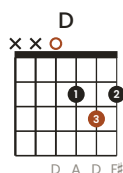
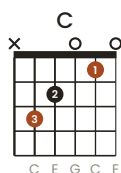
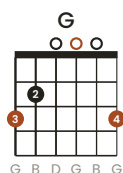


Old faithful over the I – vi – IV – V progression, which runs through the verse and the chorus.

G is fingered 320004 throughout this chapter (or 210003, see Chapter 11); the 320004 fingering leaves the 2nd finger in place for the change to Em. The C to D change happens after only one bar each, so it is the one to practise. Both shapes share the open D string; let it ring while the fingers move. Accent beats 2 and 4 lightly to lock in with the backbeat. Without a capo the song is A, F#m, D, E, which needs a barre for F#m.

Brown Eyed Girl (Van Morrison, 1967)

Key G · no capo · 148 bpm · 4/4 · beginner



TRACK 185 Brown Eyed Girl: verse · LOOP · pop · 148 bpm

G <i>I</i>	C <i>IV</i>	G <i>I</i>	D <i>V</i>
G <i>I</i>	C <i>IV</i>	G <i>I</i>	D <i>V</i>

STRUM THIS PATTERN IN EVERY BAR



Old faithful from Chapter 22. Start at about 110 bpm.

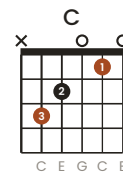
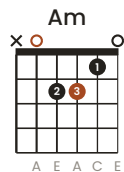
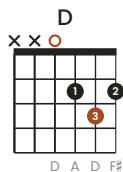
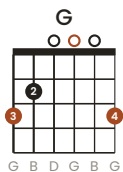
Brown Eyed Girl: second section

C <i>IV</i>	D <i>V</i>	G <i>I</i>	Em <i>vi</i>	C <i>IV</i>	D <i>V</i>	G <i>I</i>	D <i>V</i>
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The tempo is the challenge: at 148 bpm the chords change every 1.6 seconds. Practise G to C with the third finger as a pivot (it moves from the 6th string to the 5th string, fret 3 in both). Accent the downstroke on 1 of each bar; the upstrokes stay light.

Knockin' on Heaven's Door (Bob Dylan, 1973)

Key G · no capo · 69 bpm · 4/4 · beginner



TRACK 186 Knockin' on Heaven's Door: verse and chorus · LOOP · folk · 69 bpm

G <i>I</i>	D <i>V</i>	Am <i>ii</i>	Am <i>ii</i>
G <i>I</i>	D <i>V</i>	C <i>IV</i>	C <i>IV</i>

STRUM THIS PATTERN IN EVERY BAR

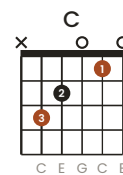
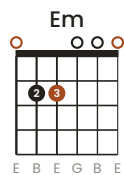
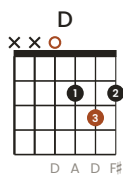
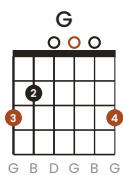


Down on 1, down on 3, down-up on 4. The slow tempo leaves time for every change.

The same eight bars cover verse and chorus. Teachers give several strums for this song; this sparse one is the easiest and fits the slow tempo. Let the downstroke on 1 ring for two full beats. Practise G to D: keep the third finger moving to the 2nd string, fret 3, as one motion. Dylan's recording uses Am7 (x02010) in the first line; either works.

Wagon Wheel (Old Crow Medicine Show, 2004)

Key A · capo 2 (G shapes) · 145 bpm · 4/4 · beginner



TRACK 187 Wagon Wheel: verse and chorus · LOOP · folk · capo 2 · 145 bpm

G <i>I</i>	D <i>V</i>	Em <i>vi</i>	C <i>IV</i>
G <i>I</i>	D <i>V</i>	C <i>IV</i>	C <i>IV</i>

STRUM THIS PATTERN IN EVERY BAR

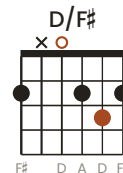
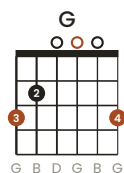
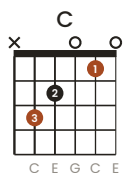
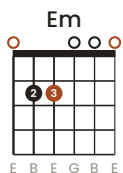


Old faithful over the I – V – vi – IV progression from Chapter 14, one chord per bar, the same in verse and chorus.

The recording has a light swing, and so does the loop. Once the changes are secure, lengthen each downstroke slightly and shorten each upstroke. Accent beats 2 and 4. The only quick move is D to Em: lift the D shape and drop the second and third fingers onto the 5th and 4th strings.

Zombie (The Cranberries, 1994)

Key E minor · no capo · 84 bpm · 4/4 · beginner



TRACK 188 Zombie: the whole song · LOOP · rock · 84 bpm

Em <i>i</i>	C <i>bVI</i>	G <i>bIII</i>	D/F# <i>bVII6</i>
Em <i>i</i>	C <i>bVI</i>	G <i>bIII</i>	D/F# <i>bVII6</i>

STRUM THIS PATTERN IN EVERY BAR

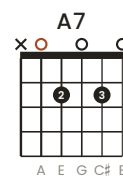
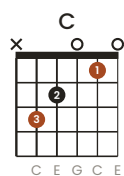
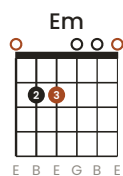
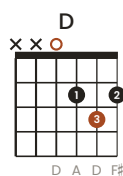
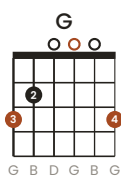


A sixteenth-note grid: downs on every eighth, plus an upstroke on the "a" of 2 and of 4 that leads into the next beat.

The four chords repeat through the whole song; the band builds intensity by playing harder, not by changing chords. For D/F#, wrap the thumb over the neck to fret the 6th string, fret 2, and let it mute the 5th string; plain D is a fine substitute. Practise G to D/F#: both have a bass note on the 6th string. Hit the downstrokes on 2 and 4 harder, like a snare drum.

I'm Yours (Jason Mraz, 2008)

Key B · capo 4 (G shapes) · 76 bpm · 4/4 · beginner



TRACK 189 I'm Yours: verse and chorus · LOOP · pop · capo 4 · 76 bpm

G <i>I</i>	D <i>V</i>	Em <i>vi</i>	C <i>IV</i>
G <i>I</i>	D <i>V</i>	Em <i>vi</i>	C <i>IV</i>

STRUM THIS PATTERN IN EVERY BAR



Old faithful with a relaxed, slightly swung feel; many teachers call it the "island strum".

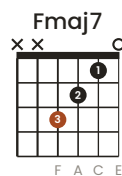
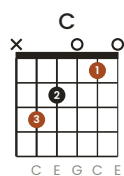
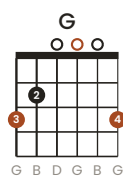
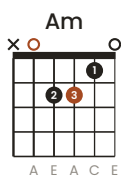
I'm Yours: bridge

G <i>I</i>	D/F# <i>V6</i>	Em <i>vi</i>	D <i>V</i>	C <i>IV</i>	A7 <i>II7</i>
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The verse and chorus use the same four chords. The bridge changes chord every two beats: practise G to D/F# to Em as a falling bass line on the 6th string. On the recording the guitar has a choppy, reggae-like sound; for that, relax the fretting hand right after each upstroke or replace the downstrokes on 2 and 4 with muted chucks (D - x U - U x U).

Riptide (Vance Joy, 2013)

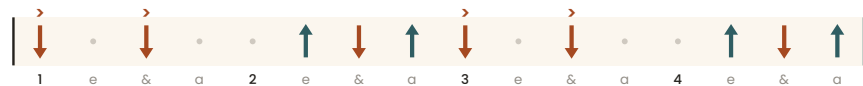
Key Bb (relative minor Bbm) · capo 1 (Am, G, C shapes) · 102 bpm · 4/4 · beginner



TRACK 190 Riptide: verse and chorus · LOOP · folk · capo 1 · 102 bpm

Am <i>vi</i>	G <i>V</i>	C <i>I</i>	Am <i>vi</i>	G <i>V</i>	C <i>I</i>
Am <i>vi</i>	G <i>V</i>	C <i>I</i>	Am <i>vi</i>	G <i>V</i>	C <i>I</i>

STRUM THIS PATTERN IN EVERY BAR



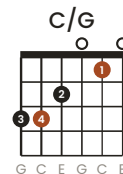
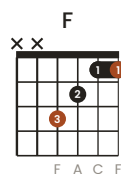
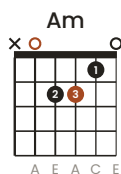
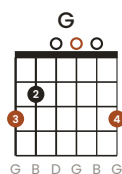
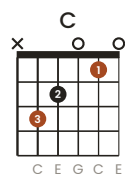
One short figure per two beats: down, down, up, down, up, with the first two downs accented.

Am and G get two beats each, C gets a full bar, so the hand plays the short figure once per chord on Am and G and twice on C. The Am to G change is the hard one, because it comes after only two beats: practise it with the second and third fingers moving together across to the 5th and 6th strings. The bridge adds Fmaj7 to the same chords.

Straight 4/4: intermediate

Let It Be (The Beatles, 1970)

Key C · no capo · 72 bpm · 4/4 · intermediate



TRACK 191 Let It Be: verse · LOOP · ballad · 72 bpm

C <i>I</i>	G <i>V</i>	Am <i>vi</i>	F <i>IV</i>	C <i>I</i>	G <i>V</i>	F <i>IV</i>	C <i>I</i>
C <i>I</i>	G <i>V</i>	Am <i>vi</i>	F <i>IV</i>	C <i>I</i>	G <i>V</i>	F <i>IV</i>	C <i>I</i>

STRUM THIS PATTERN IN EVERY BAR



Two chords per bar, one pattern half per chord. The simplest version is a downstroke on every beat.

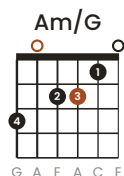
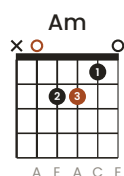
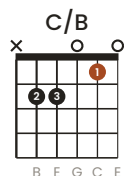
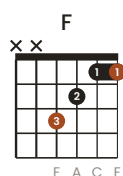
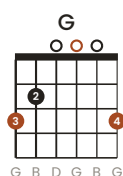
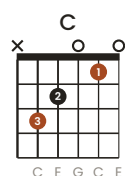
Let It Be: chorus

Am <i>vi</i>	C/G <i>I64</i>	F <i>IV</i>	C <i>I</i>	C <i>I</i>	G <i>V</i>	F <i>IV</i>	C <i>I</i>
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The F chord is the hurdle: the small F shape shown here barres only the top two strings with the first finger. Fmaj7 (xx3210) is an easier substitute until the barre sounds clean. In the chorus, C/G is the C shape with the third finger moved to the 6th string, fret 3, so the bass walks down A – G – F. Accent the first downstroke of each chord.

Have You Ever Seen the Rain (Creedence Clearwater Revival, 1970)

Key C · no capo · 116 bpm · 4/4 · intermediate



TRACK 192 Have You Ever Seen the Rain: verse · LOOP · rock · 116 bpm

C <i>I</i>	C <i>I</i>	C <i>I</i>	C <i>I</i>
G <i>V</i>	G <i>V</i>	C <i>I</i>	C <i>I</i>

STRUM THIS PATTERN IN EVERY BAR



The backbeat chuck pattern from Chapter 22. The muted chucks on 2 and 4 replace the snare drum.

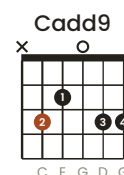
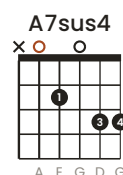
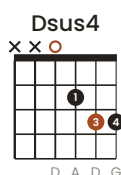
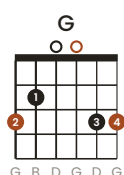
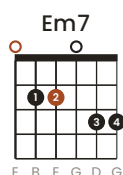
Have You Ever Seen the Rain: chorus

F <i>IV</i>	G <i>V</i>	C <i>I</i>	C/B <i>I/7</i>	Am <i>vi</i>	Am/G <i>vi/5</i>
F <i>IV</i>	G <i>V</i>	C <i>I</i>	C/B <i>I/7</i>	Am <i>vi</i>	Am/G <i>vi/5</i>
F <i>IV</i>	G <i>V</i>	C <i>I</i>	C <i>I</i>		

The chorus bass walks down from C through B, A and G to F, two beats per chord. Practise that bar on its own: C to C/B moves only the bass note, Am to Am/G adds the little finger on the 6th string. For the chucks, relax the fretting hand so the strings stop ringing but keep touching them. Beginners can play old faithful instead.

Wonderwall (Oasis, 1995)

Key F#m · capo 2 (E minor shapes) · 87 bpm · 4/4 · intermediate



TRACK 193 Wonderwall: verse · LOOP · pop · capo 2 · 87 bpm

Em7 <i>i7</i>	G <i>bIII</i>	Dsus4 <i>bVIIIsus4</i>	A7sus4 <i>IV7sus4</i>	Em7 <i>i7</i>	G <i>bIII</i>	Dsus4 <i>bVIIIsus4</i>	A7sus4 <i>IV7sus4</i>
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STRUM THIS PATTERN IN EVERY BAR



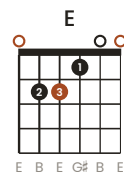
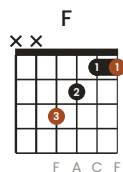
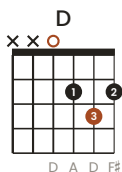
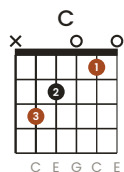
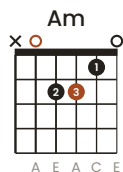
A sixteenth-note pattern with accents on 2 and 4. Two beats per chord.

The third and fourth fingers stay on the 2nd and 1st strings at fret 3 for every shape; only the first two fingers move. That anchor makes the chord changes easy and leaves all the work to the strumming hand, which runs in sixteenths and never stops. The pre-chorus and chorus use the same anchored shapes with Cadd9 added. Practise the pattern muted on one chord at 60 bpm before adding the changes.

Other metres and feels

The House of the Rising Sun (The Animals, 1964)

Key A minor · no capo · dotted quarter = 78 bpm · 6/8 · beginner



▶ TRACK 194 The House of the Rising Sun: verse · LOOP · 6/8 ballad · 78 bpm (dotted quarter)

Am <i>i</i>	C <i>bIII</i>	D <i>IV</i>	F <i>bVI</i>	Am <i>i</i>	C <i>bIII</i>	E <i>V</i>	E <i>V</i>
Am <i>i</i>	C <i>bIII</i>	D <i>IV</i>	F <i>bVI</i>	Am <i>i</i>	E <i>V</i>	Am <i>i</i>	E <i>V</i>

STRUM THIS PATTERN IN EVERY BAR

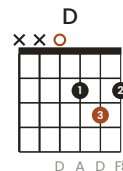
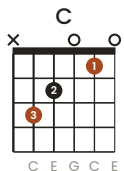
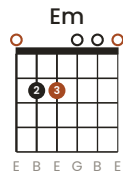
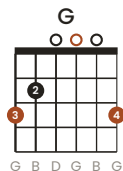


Count 1 2 3 4 5 6 per bar; accent 1, and let 4 sound clearly. The simplest version is six downstrokes per bar.

The recording is played with picked arpeggios, but the song is just as often strummed. There is no separate chorus: the same sixteen bars return for every verse. D to F is the awkward change; keep the first finger moving toward the 2nd string, fret 1. Accent beat 1 and keep the 6/8 lilt: two big beats per bar, not six equal ones.

Perfect (Ed Sheeran, 2017)

Key A $\flat$ · capo 1 (G shapes) · dotted quarter = 63 bpm · 12/8 (counted in 6/8) · beginner

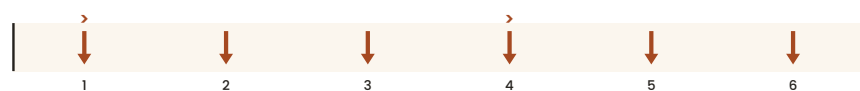


The shapes of Stand By Me, with the note names of each shape under the strings.

▶ TRACK 195 Perfect: verse · LOOP · 6/8 ballad · capo 1 · 63 bpm (dotted quarter)

G <i>I</i>	G <i>I</i>	Em <i>vi</i>	Em <i>vi</i>
C <i>IV</i>	C <i>IV</i>	D <i>V</i>	D <i>V</i>

STRUM THIS PATTERN IN EVERY BAR



All downstrokes, accents on 1 and 4. Each chord lasts one 12/8 bar, which is two bars of 6/8 here.

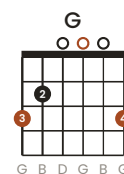
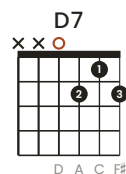
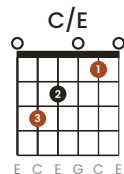
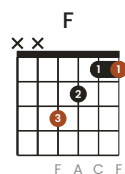
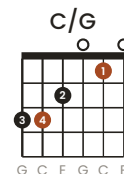
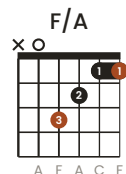
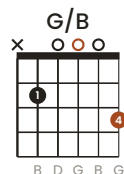
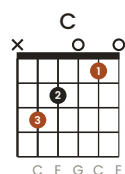
Perfect: chorus (one chord per 6/8 bar)

Em <i>vi</i>	C <i>IV</i>	G <i>I</i>	D <i>V</i>	Em <i>vi</i>	C <i>IV</i>	G <i>I</i>	D <i>V</i>
-----------------	----------------	---------------	---------------	-----------------	----------------	---------------	---------------

In the chorus the chords change twice as fast as in the verse, so practise Em to C to G to D with one 6/8 bar each before playing along. The note names under the boxes show why the slow changes sound smooth: G and Em share G and B, Em and C share E and G, so only one or two notes move. The all-down strum is slow enough to stay relaxed; let the accents on 1 and 4 carry the sway. When it is secure, replace the third and sixth strokes with light upstrokes for a softer sound.

Piano Man (Billy Joel, 1973)

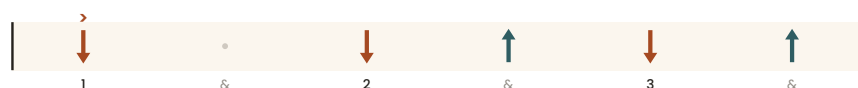
Key C · no capo · 178 bpm · 3/4 · intermediate



▶ TRACK 196 Piano Man: first half of the verse · LOOP · waltz · 178 bpm

C <i>I</i>	G/B <i>V6</i>	F/A <i>IV6</i>	C/G <i>I64</i>
F <i>IV</i>	C/E <i>I6</i>	D7 <i>II7</i>	G <i>V</i>

STRUM THIS PATTERN IN EVERY BAR

*The waltz strum from Chapter 22. Accent beat 1 so that the falling bass line is heard.*

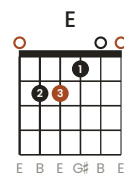
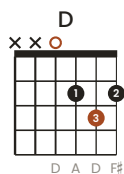
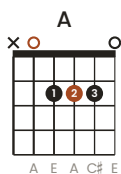
Piano Man: second half of the verse

C <i>I</i>	G/B <i>V6</i>	F/A <i>IV6</i>	C/G <i>I64</i>	F <i>IV</i>	G <i>V</i>	C <i>I</i>	C <i>I</i>
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This is a piano song, so the chords carry a bass line that walks down C - B - A - G - F - E. Aim the accented downstroke at the bass note of each slash chord (5th or 6th string) and let it ring. Many charts write Am instead of F/A; both work. Practise C/G to F, where the whole hand moves, as a separate exercise.

Three Little Birds (Bob Marley and the Wailers, 1977)

Key A · no capo · 74 bpm · 4/4 · beginner



TRACK 197 Three Little Birds: chorus · LOOP · reggae one drop · 74 bpm

A I	A I	D IV	A I
A I	A I	D IV	A I

STRUM THIS PATTERN IN EVERY BAR



The off-beat skank: short upstrokes on every "&", the hand still moving down on the beats without touching the strings.

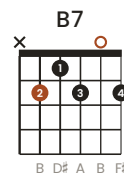
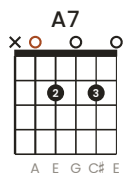
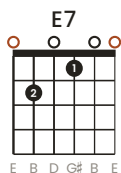
Three Little Birds: verse

A I	E V	A I	D IV	A I	E V	D IV	A I
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The guitar plays only on the off-beats, between the drum and bass hits; this is the feel of Chapter 52. Cut each upstroke short by releasing the fretting-hand pressure right after it. If the off-beats feel strange, first play four downstrokes on the beats, then move every stroke half a beat later.

Before You Accuse Me (Eric Clapton, 1992)

Key E · no capo · 96 bpm · 4/4 (shuffle) · beginner to intermediate



TRACK 198 Before You Accuse Me: 12-bar blues · LOOP · blues shuffle · 96 bpm

E7 I7	A7 IV7	E7 I7	E7 I7
A7 IV7	A7 IV7	E7 I7	E7 I7
B7 V7	A7 IV7	E7 I7	B7 V7

STRUM THIS PATTERN IN EVERY BAR



Each beat is split into three; the down falls on the beat and the up on the last third, which gives the long-short shuffle.

This Bo Diddley song, on Clapton's Unplugged album, is a 12-bar blues with the quick change to A7 in bar 2 (see Chapter 43). Clapton plays it fingerstyle with a thumb bass; the swung strum shown here is the common simplified way to play along with a pick. B7 is the hardest shape: practise B7 to A7 to E7 (bars 9 to 11) on its own. Accent beats 2 and 4.

Finding the strum of any song

Most songs are not printed with a strum. You can work one out by ear in a few steps, using the methods of Chapter 30.

1. **Find the beat and the metre.** Tap your foot and count. If the count falls naturally in fours, the song is in 4/4; in threes, 3/4; if each beat divides into three (1 & a, 2 & a), it is 6/8 or 12/8 or a shuffle.
2. **Find the subdivision.** Strum steady eighths (D U D U) on muted strings with the recording. If it feels too slow and busy-sounding parts are missing, try sixteenths; if it feels rushed, try quarters.
3. **Listen for where the guitar is loud.** Stop strumming where the guitar leaves gaps and keep the hand moving. The gaps turn the steady pattern into the song's pattern, exactly as in *old faithful*.
4. **Find the accents.** Most pop and rock songs accent 2 and 4 with the snare; folk and country songs accent 1; reggae plays only off-beats.
5. **Simplify.** Choose one pattern that fits most bars. Variations come later. A slow-down tool (70% speed, same pitch) and live videos, where the stroke directions are easy to see, both help.

Chapter 23 review · use it, train it, test it

Using it in real music

Building a repertoire. Learn two or three of these songs by heart rather than all fifteen from the page.

Changing pattern within a song. Once a song is secure, play a lighter pattern in the verse (all downs, or fewer strokes) and the full pattern in the chorus.

Moving songs to your voice. Keep the shapes and move the capo until the key suits your singing range.

How to train it

Drill	How	Time	Goal
Hardest change	The song's most difficult change, one strum per chord, then the full pattern	3 min	Change on time, no gap in the strum
Slow play-along	The loop at about 70% speed, from start to end	3 min	Two passes without stopping
Full-tempo play-along	The loop at full tempo	3 min	Every first downstroke on time
Feel switch	Wagon Wheel, then Three Little Birds, then The House of the Rising Sun	4 min	A clear 4/4, reggae and 6/8 feel
Strum by ear	A new song: find metre, subdivision and accents, then a simplified pattern	5 min	A pattern that fits most bars

Check your understanding

Write your answers, then check them in Appendix J.

1. With a capo at fret 2, which chord sounds when you play an Em shape?
2. Why is the strum in this chapter called a simplified pattern?
3. In *Riptide*, why is the Am to G change the hardest one?
4. Where does the guitar play in a reggae strum such as *Three Little Birds*?
5. In the chorus of *Have You Ever Seen the Rain*, which bass notes does the walking line pass through?
6. What is the first step in working out the strum of a song by ear?

CHAPTER 24

Pick Technique



A pick gives attack, volume and speed that fingers cannot match. This chapter covers the grip, alternate and economy picking, sweeps, string skipping, crosspicking, hybrid picking and a method for building speed with a metronome.

Most steel-string and electric players use a pick for strumming and single-note lines; many classical and fingerstyle players use one for particular styles. The pick turns the motion of the whole hand into one point of contact, so small faults in grip and motion are clearly audible. The good news is the same: small corrections make a large difference.

Strumming with a pick is covered in Chapter 22. This chapter is about single notes and arpeggios.

Holding the pick

- Lay the pick on the side of the index finger at the first joint, pointing at right angles to the finger.
- Place the pad of the thumb flat on top, so that the pick is gripped between the thumb pad and the side of the index finger.
- Let 2–3 mm of the tip show. Less gives control; more gives a looser, brushier attack.
- Grip just firmly enough that the pick does not turn in the fingers. A tight grip tenses the forearm and slows you down.
- Curl the remaining fingers loosely, or let them rest lightly on the top or the pickguard as a reference.

Pick shapes and thicknesses

Type	Thickness	Suits
Thin	Under 0.6 mm	Strumming; light and forgiving, little control for single notes
Medium	0.6–0.8 mm	All-round acoustic use, strumming and simple lines
Heavy	0.8–1.2 mm	Single-note lines, bluegrass flatpicking, electric lead
Extra heavy and jazz shapes	1.2 mm and over, small pointed tip	Fast lines, jazz and gypsy jazz, maximum control

The picking motion

- The motion comes mainly from the wrist, with a small contribution from the forearm rotating (as when turning a door handle). The fingers holding the pick stay almost still.
- Keep the motion small. The pick should travel only a few millimetres past the string.
- Angle the pick slightly so that it meets the string with its edge rather than flat (edge picking). It slides through the string more easily and sounds smoother.
- The forearm rests on the body; the hand floats or lightly touches the bridge. Keep the same reference every time.

Alternate picking

Alternate picking is strict down-up alternation: downstrokes on the beats and on "&" in sixteenth-note passages, upstrokes on the off-beats in between. As with the strumming pendulum, the direction of each stroke depends on its rhythmic place, not on the string. It is the most reliable basis for single-note playing.

The exercise below is the classic chromatic warm-up: one finger per fret, four notes per string, strict alternation. Pick directions are shown as D and U above the TAB.

TRACK 199 Chromatic 1-2-3-4 with alternate picking

First system: Treble clef, 4/4 time. Notes: G4, A4, B4, C5, D5, E5, F5, G5. Picking: D, U, D, U, D, U, D, U. Fingering: 5-6-7-8. Second system: Notes: G5, F5, E5, D5, C5, B4, A4, G4. Picking: D, U, D, U, D, U, D, U. Fingering: 8-7-6-5.

Fifth position. Keep each finger down until the next string. Each string starts with a downstroke, because four notes per string is an even number.

Inside and outside picking

When the pick moves from one string to the next it either crosses *between* the two strings (inside picking) or plays the *outer* sides of both (outside picking).

- **Inside:** a downstroke on the higher-pitched string followed by an upstroke on the lower-pitched one, or an upstroke on the lower string followed by a downstroke on the higher one. The pick is trapped between the strings.
- **Outside:** an upstroke on the higher-pitched string followed by a downstroke on the lower one, or the reverse. The pick moves around the strings.

Most players find one of the two easier. Practise both with the exercise below; it alternates between them with each string change.

TRACK 200 String crossing: inside and outside

First system: Treble clef, 4/4 time. Notes: G4, A4, B4, C5, D5, E5, F5, G5. Picking: D, U, D, U, D, U, D, U. Fingering: 3. Second system: Notes: G5, F5, E5, D5, C5, B4, A4, G4. Picking: D, U, D, U, D, U, D, U. Fingering: 2, 3.

Bar 1 is outside picking (down on the 2nd string, up on the 1st string, so the pick goes around both). Bar 2 is inside picking (down on the 1st string, up on the 2nd). Bar 3 mixes the two on strings 3 and 2.

- The pick plays the bass and the lower strings; m and a play the upper strings, often together with the pick (a pinch).
- A sharp, snapping pluck with m (pulling the string slightly outward so it slaps the frets) gives the "chicken picking" sound of country guitar.

TRACK 205 Hybrid picking: alternating bass with pick, m and a above

[illegible]

The pick plays the alternating bass on the beats (all downstrokes); m and a pluck the 2nd and 1st strings on the off-beats. This is the Travis pattern of Chapter 19 with a pick instead of the thumb.

 TRACK 206 Hybrid double stops: pick and m together

Sixths on strings 3 and 1: the pick plays the 3rd string, m plucks the 1st string at the same moment, skipping the 2nd string. Slide between shapes with the fretting hand.

Tremolo picking

Tremolo picking is very fast alternate picking on one note, so that the note seems to be sustained. It is the pick's equivalent of the classical tremolo (Chapter 21), used in surf, mandolin-style and Mediterranean music. Keep the wrist loose and the motion tiny; do not tense up to go faster.

TRACK 207 Tremolo-picked melody in A minor

Four picks per note, strict down-up. At speed the repetitions merge into a sustained line. The D $\sharp$ (2nd string, fret 4) is the leading tone of E.

Building speed with a metronome

Speed is accuracy repeated. A metronome makes progress measurable and stops you from practising mistakes at a tempo you cannot control.

1. **Find the clean tempo.** The fastest tempo at which you can play the passage four times in a row with no mistakes, even tone and relaxed hands. Write it down.
2. **Work below it.** Practise for a few minutes at about 10% below the clean tempo, focusing on economy of motion.
3. **Step up in small increments.** Raise the tempo by 4–8 bpm. When it is clean four times, raise it again. When it breaks down, go back two steps.
4. **Use bursts.** Play a short group (four or six notes) at a tempo well above your clean tempo, then stop. Short bursts train the motion without letting tension build up.
5. **Change the subdivision.** At a fixed metronome setting, play the passage in eighths, then triplets, then sixteenths. This trains the hand to change speed while the beat stays the same; Chapter 37 practises it as the subdivision ladder.
6. **Stop at tension.** If the forearm or shoulder tightens, stop, shake out and drop the tempo.

TRACK 208 Four subdivisions at one tempo

Same click, four speeds. Accent the note on each beat so that you always know where you are.

Week	Clean tempo (sixteenths)	Practice
1	60	Chromatic exercise and economy scale at 54–60
2	66	Add bursts at 80

Week	Clean tempo (sixteenths)	Practice
3	72	Change subdivisions at 72
4	80	Apply to a piece or a solo

The numbers are an example. Keep your own log: progress is rarely steady, and a written record shows it more clearly than memory.

NYLON · STEEL · Pick and nylon

Nylon: a pick on nylon strings sounds dull and wears the strings; classical players get the same evenness with i m alternation and rest strokes (Chapter 17). Gypsy jazz and some Latin styles play steel-string guitars with very heavy picks.

Steel: acoustic flatpicking (bluegrass) uses heavy picks, often 1 mm or more, and plays close to the bridge for volume and clarity.

Chapter 24 review · use it, train it, test it

Using it in real music

Fiddle tunes and bluegrass. Alternate picking and crosspicking are the core of flatpicking (Chapter 46).

Rock and blues solos. Alternate and economy picking for scale runs, string skipping for wide intervals, sweeps for fast arpeggios (Chapter 44, Chapter 43).

Country and jazz. Hybrid picking for chord-melody, double stops and snapping country lines; jazz players often use economy picking for long lines.

How to train it

Drill	How	Time	Goal
Grip check	Play the chromatic exercise and stop every bar to check grip pressure	2 min	Relaxed thumb and index, pick does not turn
Inside and outside	String-crossing exercise	3 min	Both bars equally clean at 80 bpm
Economy scale	Three-notes-per-string G major, up and down	4 min	Even triplets, no accent at string changes
Sweeps	Am and E sweeps at 50 bpm	3 min	Every note separate, no chord sound
Speed log	One exercise, find the clean tempo, write it down	3 min	Daily record

Check your understanding

Write your answers, then check them in Appendix J.

1. Which part of the thumb and which part of the index finger grip the pick?

2. In alternate picking, what decides whether a note is picked up or down?

3. What is the difference between inside and outside picking?

4. In economy picking ascending three notes per string, which strokes play each string?

5. What must the fretting hand do in a sweep?

6. What is hybrid picking?

7. Into what groups does crosspicking usually divide a bar of eight eighth notes?

8. What is a "clean tempo"?

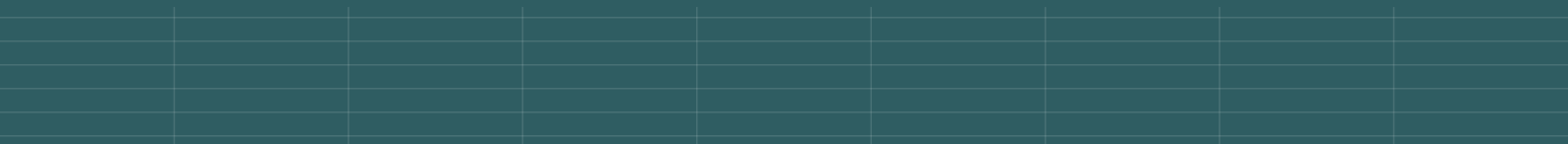
MY NOTES

PART SIX

Beyond Standard Tuning

Open and alternate tunings: new chord shapes, ringing drones and sounds standard tuning cannot reach.

25 Open and Alternate Tunings



CHAPTER 25

Open and Alternate Tunings



TRACKS

25

Retuning one or more strings changes which chords, drones and keys are easy. In an open tuning the open strings already form a chord, so a single finger or a slide across the neck plays full harmony.

Standard tuning is a compromise: it makes scales and chords playable in every key, but no key gets special treatment. An **alternate tuning** changes the pitch of one or more open strings. An **open tuning** is an alternate tuning whose open strings form a chord. Both trade some flexibility for a bigger, more resonant sound in one or two keys.

The theory does not change. A major 3rd is still four frets along one string, an octave is still twelve frets, and the notes on each string still follow the chromatic scale. Only the starting note of each string moves, so the shapes you know from standard tuning move with it.

Why retune

- **Resonance.** More open strings belong to the key, so they ring in sympathy and the guitar sounds larger.
- **Easy chords.** In an open tuning one finger barred across the strings plays a major chord at any fret.
- **Drones.** Low open strings sustain under a melody: the sound of much folk, Celtic and blues guitar.
- **Slide.** A glass or metal slide can only play straight across the strings. An open tuning makes that straight line a chord.
- **New voicings.** Familiar fingerings produce unfamiliar intervals, often with close seconds and unisons that standard tuning cannot reach.

The main tunings

Tuning	Strings (6 1)	Change from standard	Typical use
Standard	E A D G B E	none	Everything
Drop D	D A D G B E	6th down a whole step	Pieces and songs in D, heavy riffs
Double drop D	D A D G B D	6th and 1st down a whole step	Fingerstyle in D
Open D	D A D F# A D	6th, 2nd, 1st down a whole step; 3rd down a half step	Slide, folk, blues
Open G	D G D G B D	6th, 5th, 1st down a whole step	Slide, blues, rock rhythm
DADGAD	D A D G A D	6th, 2nd, 1st down a whole step	Celtic, modal, fingerstyle
Open C	C G C G C E	6th down a major 3rd; 5th and 4th down a whole step; 2nd up a half step	Fingerstyle, low drones

Drop D, double drop D and DADGAD are not open major chords, but they are used in the same way: for drones and ringing open strings in D.

CAREFUL · Tuning up adds tension

Every tuning in the table except Open C only lowers strings. Lowering is always safe. Raising a string (Open E, Open A, the 2nd string in Open C) adds tension to the string and the neck. Raise by a half step at most on normal strings. To get Open E or Open A, tune to Open D or Open G and put a capo at the 2nd fret.

How open tunings work

Because the open strings form a major chord, a straight barre at any fret gives another major chord with the same voicing. The open strings are chord I. The 5th fret is IV, the 7th fret is V, and the 12th fret is I an octave higher. A three-chord song needs three fret positions.

Fret	0	2	3	5	7	9	10	12
Open G	G (I)	A	B $\flat$	C (IV)	D (V)	E	F	G (I)
Open D	D (I)	E	F	G (IV)	A (V)	B	C	D (I)
Open C	C (I)	D	E $\flat$	F (IV)	G (V)	A	B $\flat$	C (I)

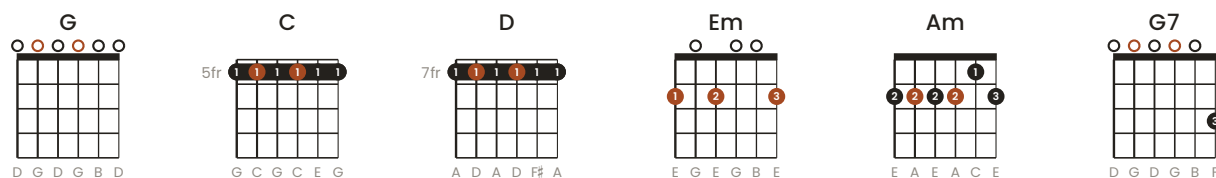
SHORTCUT · Minor chords in an open tuning

Lower the string that holds the 3rd of the chord by one fret. In Open G that is the 2nd (B) string, in Open D the 3rd (F $\sharp$) string, in Open C the 1st (E) string. The diatonic minor chords sit at frets 2 (ii), 4 (iii) and 9 (vi).

Open G

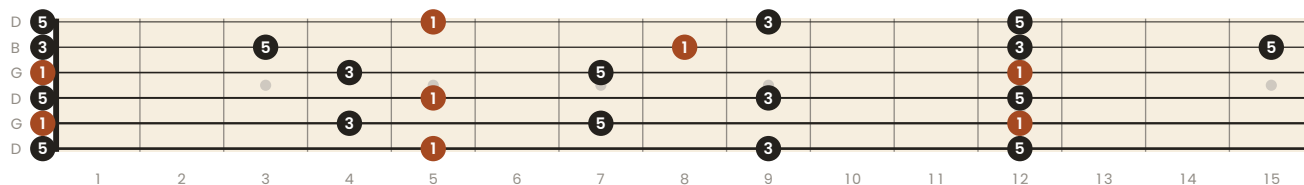
Open G is the most common tuning for slide and for open-chord rock rhythm. Many players leave the 6th string unplayed, so the lowest note is the root G on the 5th string.

The main chords in Open G



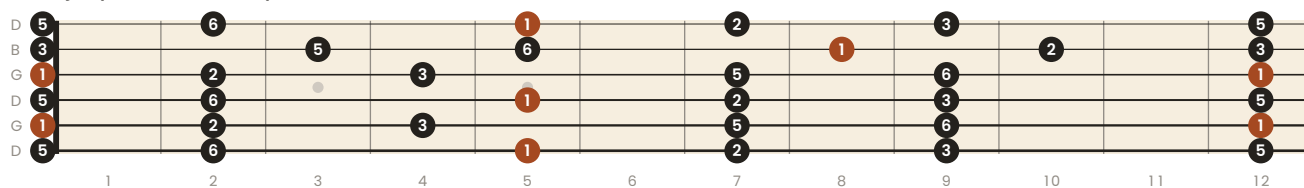
C and D are straight barres. Am is the fret-2 barre with the 2nd string (the 3rd of the chord) lowered by one fret; Em is an open-position shape.

G major chord tones in Open G



At frets 0 and 12 every string is a chord tone. In between, the chord tones form small inversions, for example B, D and G at frets 4, 3 and 5 on the top three strings.

G major pentatonic in Open G



The roots fall on open strings 5 and 3, at fret 5 on strings 6 and 4, at fret 8 on the 2nd string, and at fret 12.

TRACK 209 I – IV – V in Open G

Tuning: D G D G B D

One finger barred at frets 5 and 7. Leave the 6th string silent so the bass note is always the root.

Open D

Open D is a whole step lower in pitch than the E major chord shape and sits well under the voice for many singers. The 3rd of the chord is on the 3rd string, near the middle of the strings, which gives a warm, dark open chord.

The main chords in Open D

G and A are straight barres. Bm and Em lower the 3rd string by one fret.

D major pentatonic in Open D

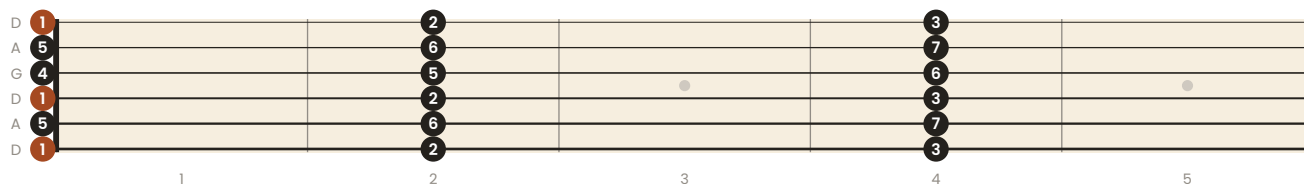
DADGAD

DADGAD is not an open major chord. The open strings are D, A and G: a Dsus4 chord with no 3rd. That neutral sound fits both major and minor melodies, which is why it is so common in Celtic and modal music. The 5th, 4th and 3rd strings are unchanged from standard tuning, so shapes on those strings still work.

Common DADGAD chords

The open 1st and 2nd strings (D and A) ring through almost every chord in D.

D major scale in DADGAD, first position



The open 1st, 2nd and 4th strings give D, A and D. Play scale runs across these open strings so that neighbouring notes overlap and ring together.

 TRACK 210 DADGAD air over a drone

Tuning: D A D G A D



An original eight-beat phrase. Let every note ring until the next note on the same string replaces it.

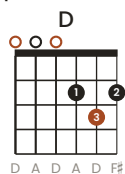
Drop D and double drop D

Drop D lowers only the 6th string, from E to D. Strings 5 to 1 stay the same, so every scale and chord you know on those strings still works, with a deep D bass added. Any shape that uses the 6th string must move up two frets. It is the most common alternate tuning in classical guitar music and in rock. The three lowest strings now form a D power chord, so one finger across them at any fret plays a power chord.

Double drop D also lowers the 1st string to D. Both outer strings are now D, which suits fingerstyle pieces in D and G with a drone on top and bottom.

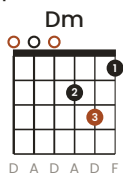
Drop D and double drop D chords

drop D



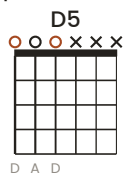
D A D A D F#

drop D



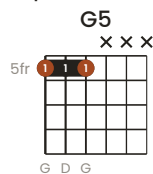
D A D A D F

drop D



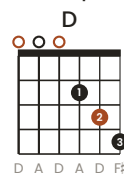
D A D

drop D



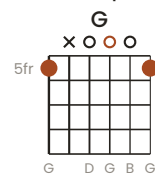
G D G

double drop D



D A D A D F:

double drop D



G D G B G

In drop D a one-finger barre across strings 6, 5 and 4 is a power chord at every fret.

▶ TRACK 211 Drop D riff

Tuning: D A D G B E

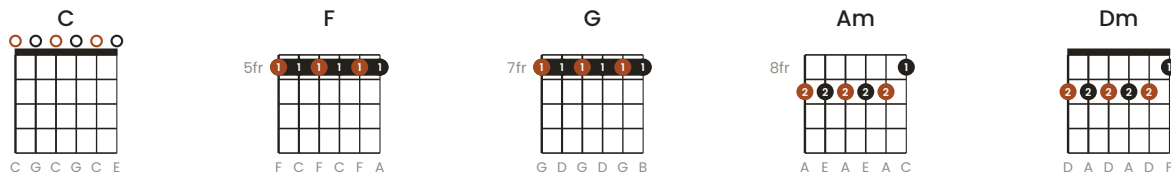


The open D bass and one-finger power chords (G5, F5, E5). Mute the strings with the picking-hand palm between chords.

Open C

Open C is the lowest of the common open tunings. The 6th string drops a major 3rd to C, which needs a heavier string to stay in tune and avoid buzzing. The 3rd of the chord is on the 1st string, so it sits at the top of every barre.

The main chords in Open C



Minor chords lower the 1st string. Am at fret 9 and Dm at fret 2 are the vi and ii chords.

TRACK 212 Open C arpeggios

Tuning: C G C G C E

Two musical staves showing arpeggios for Open C (C) and F (F) chords. The first staff shows the C chord arpeggio (C, G, C, G, C, E) and the F chord arpeggio (F, C, F, C, F, A). The second staff shows the G chord arpeggio (G, D, G, D, G, B) and the C chord arpeggio (C, G, C, G, C, E). The notation includes fret numbers (0, 5, 7) and a picking pattern (T, A, B) for the strings.

The same picking pattern at frets 0, 5 and 7. Keep the barre finger flat so all six strings sound.

NYLON · STEEL

Nylon: drop D is standard repertoire, and many pieces in G also lower the 5th string to G (6 = D, 5 = G). Nylon strings stretch for some time after retuning, so tune slightly sharp, play for a minute and retune. DADGAD works well on nylon for Celtic arrangements; Open C and slide are rare.

Steel: steel-string is the natural home of open tunings, slide and drones because of its long sustain. For Open C or Open D a medium-gauge 6th string, or a set made for low tunings, keeps the low strings from flapping against the frets.

Slide basics

A slide is a glass, metal or ceramic tube worn on one finger of the fretting hand, usually the 3rd or 4th finger. It touches the strings without pressing them to the fretboard.

- Position.** Hold the slide directly over the fret wire, not behind it as with a fingertip. Behind the wire the note is flat.
- Pressure.** Rest the slide on the strings with just enough weight to stop buzz. Pressing too hard pushes the strings onto the frets and causes rattles.

3. **Damping.** Let the fingers behind the slide lie lightly on the strings to silence the part of the string between nut and slide. Mute unwanted strings with the picking hand.
 4. **Vibrato.** Move the slide a few millimetres either side of the fret wire, evenly. Slide vibrato is wider than finger vibrato and sits at the centre of the pitch.
 5. **Angle.** Keep the slide parallel to the frets for chords. Tilt it to play two strings at different frets.
- Slide is usually played fingerstyle or with a thumbpick and fingers, so the picking hand can choose and mute strings precisely. Higher action and heavier strings help on a dedicated slide guitar.

TRACK 213 Slide phrase in Open G
Tuning: D G D G B D

G C G

	10	12	12		12	10	10		3	5	5		3	0	0
T	10	12	12		14	12	10		3	5	5		3	0	0
A	10	12	12				10		3	5	5			0	0
B	10	12	12				10		3	5	5			0	0

Original phrase on strings 4 to 2. Slide into the 12th and 5th frets from two frets below. Fret 10 on the 3rd string is F, the $b7$ of G, which gives the phrase its blues sound.

TRY THIS · Slide on three strings

In Open G, rest the slide across strings 1–3 at the 12th fret and pick the three strings together. Move to fret 5 and fret 7. Listen for buzz and for flat notes: if the chord sounds sour, move the slide closer to the fret wire.

Retuning safely and quickly

- **Tune down to the note when you lower a string, then come back up to it** so the string settles under tension.
- **Retune every string twice.** Changing the tension of one string pulls the neck slightly and detunes the others, especially when several strings change at once.
- **Use the unchanged strings as references.** In drop D, the 6th string open should match the open 4th string an octave lower, and the 7th fret of the 6th string should match the open 5th string.
- **Open G by ear.** Lower the 6th to match the 4th (an octave lower). Lower the 5th until it matches the 5th fret of the 6th string (G). Lower the 1st until it matches the open 4th string an octave higher.
- **Keep a written list** of the new open-string notes on the music stand until the tuning feels familiar.

TIP · Finding your way in a new tuning

Find the roots first: on the open strings and at the 12th fret. Then build the scale by ear on one string, then two. Think in intervals rather than shapes: a major 3rd is four frets on one string in every tuning, and strings tuned a 4th apart (as in standard tuning) keep their usual shapes between them.

Chapter 54 uses DADGAD and drop D for fingerstyle arrangements, Chapter 36 covers slides, vibrato and muting in more detail, and Chapter 58 includes a complete air in DADGAD.

Chapter 25 review · use it, train it, test it

Using it in real music

Slide and blues. Open G and Open D let a slide play full chords. Much Delta blues and later blues-rock slide playing uses these two tunings on steel-string or resonator guitars.

Rock rhythm. Open G with the 6th string removed or left unplayed gives the ringing one-finger chords and suspended-chord riffs heard in many rock songs. Drop D gives heavy one-finger power chords.

Celtic and modal music. DADGAD's droning open strings suit Irish and Scottish dance tunes and modal melodies on either guitar.

Classical repertoire. Many pieces in D major and D minor ask for drop D; some in G ask for 6 = D and 5 = G. Retune before you start the piece, not halfway through.

How to train it

Drill	How	Time	Goal
Tuning of the week	Spend one session a week in a new tuning; find I, IV and V first, then ii and vi	10 min	Comfortable in three tunings
Fast retune	Standard to drop D and back, then standard to Open G and back, with a tuner, then by ear	3 min	Under 30 seconds each way, in tune
Minor barres	In Open G, play G, Am, C, D, Em using barres and the lowered 2nd string	4 min	Clean changes at 70 bpm, one chord per bar
Slide accuracy	Open G: slide to frets 3, 5, 7, 12 on the top three strings and check each with a tuner	5 min	Every target in tune without correction
Write in the tuning	Write a four-bar idea that only works because of the open strings	5 min	A riff you could not play in standard tuning

Check your understanding

Write your answers, then check them in Appendix J.

1. List the strings of Open G from the 6th to the 1st.

2. Which chord do the open strings of DADGAD form?

3. In Open D, a straight barre at the 5th fret gives which chord?

4. In Open G, which string holds the 3rd of the chord, and how do you make a minor barre?

5. What is the safest way to get Open E?

6. In drop D, which note is at the 5th fret of the 6th string?

7. Where should a slide sit relative to the fret wire?

8. Which strings change from standard tuning to DADGAD?

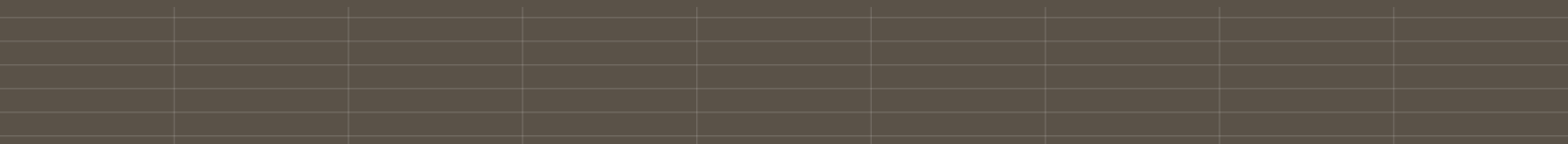
MY NOTES

PART SEVEN

Reading and Listening

Read standard notation and rhythm fluently, train your ear, and learn music directly from recordings.

- 26 Reading Standard Notation
- 27 Reading Rhythm
- 28 Ear Training
- 29 Ear-Training Course
- 30 Transcription



CHAPTER 26

Reading Standard Notation



TRACKS

26

Standard notation shows pitch and rhythm exactly and is the common language of all written music. A few minutes of graded reading a day is enough to read simple melodies within a month.

TAB tells you where to put your fingers. Standard notation tells you what the music is: the pitch, the exact rhythm, the key and the phrasing. Classical guitar music is published almost entirely in notation, and most other instruments read nothing else, so notation is what lets you play from a piano book, a lead sheet or a part written for a violin.

This book prints notation and TAB together, so you can always check one against the other. This chapter teaches pitch: the staff, the notes in first position, ledger lines, key signatures, accidentals and positions. Chapter 27 teaches the rhythm side.

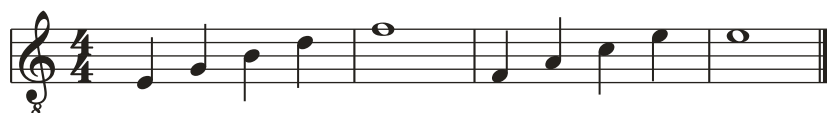
The staff and the clef

Notation is written on a **staff** of five lines. Each line and each space is one letter name; moving from a line to the next space up raises the note by one letter. The **treble clef** (G clef) curls around the second line from the bottom and fixes that line as G.

- Lines, bottom to top: **E G B D F**.
- Spaces, bottom to top: **F A C E**.

Notes above or below the staff use short extra lines called **ledger lines**.

Line notes, then space notes



Bars 1–2: the line notes E G B D F. Bars 3–4: the space notes F A C E.

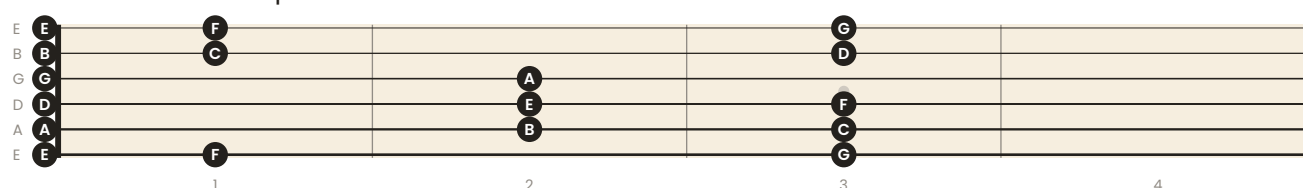
UNDER THE HOOD · Guitar sounds an octave lower than written

Guitar music is written one octave higher than it sounds, so that the guitar's range fits on the treble staff with few ledger lines. The small 8 under the clef shows this. The open 1st string sounds E₄ (the E above middle C) but is written E₅ in the top space of the staff. The open 6th string sounds E₂ and is written E₃, below three ledger lines. When you read from a piano or violin part, the written notes sound an octave higher than the same notes written for guitar, but you play them in the same places on the neck.

Notes in first position

First position means the 1st finger covers the 1st fret, the 2nd finger the 2nd fret, and so on. With the open strings, this position contains every natural note from the low open E to G on the 1st string.

The natural notes in first position



Natural notes only. A sharp or flat is one fret higher or lower.

String	Notes in first position (fret)	Where they sit on the staff
1st (E)	E (0), F (1), G (3)	Top space to just above the staff
2nd (B)	B (0), C (1), D (3)	Middle line to the fourth line
3rd (G)	G (0), A (2)	Second line and second space
4th (D)	D (0), E (2), F (3)	Just below the staff to the first space
5th (A)	A (0), B (2), C (3)	Second ledger line below to the first ledger line below
6th (E)	E (0), F (1), G (3)	Below the third ledger line to below the second

TRACK 214 Every natural note in first position

E F G A B C D E F G A B C D E F G. Say each note name aloud as you play it.

Learn the strings in pairs, top first: the 1st and 2nd strings sit inside the staff and are the easiest to read. Then add the 3rd string, then the three bass strings with their ledger lines.

TIP · Fixed landmarks

Memorise five landmarks and read the rest by counting lines and spaces from them: low E (open 6th, three ledger lines below), C (5th string, 3rd fret: written as middle C, on the first ledger line below the staff), G (open 3rd string, second line), B (open 2nd string, middle line) and high E (open 1st string, top space).

Ledger lines

The three bass strings lie almost entirely below the staff. Count ledger lines downwards like lines and spaces: the first ledger line below the staff is C, the space below it B, the second ledger line A, then G, the third ledger line F, and E just below it.

Above the staff the same counting works upwards: G sits on top of the staff, A on the first ledger line above, C on the second, E on the third. These notes come up as soon as you move up the 1st string beyond the 3rd fret.

TRACK 215 Bass strings: reading below the staff

D E F G, C B A, G F E F, E. Count the ledger lines from middle C downwards.

▶ TRACK 216 The 1st string above the staff

E F G A B C D E on the 1st string alone, up to the 12th fret and back. A, C and E sit on the first, second and third ledger lines above the staff.

Sharps, flats and naturals

An accidental placed before a note raises or lowers it:

Sign	Name	Effect	On the guitar
#	Sharp	Raises the note a half step	One fret higher
b	Flat	Lowers the note a half step	One fret lower
♮	Natural	Cancels a sharp or flat	Back to the natural note

Three rules decide how long an accidental lasts:

1. An accidental applies to every later note **on the same line or space** until the end of the bar.
2. The barline cancels it. In the next bar the note returns to its natural form (or to the key signature).
3. A natural sign cancels an accidental, or a sharp or flat from the key signature, for the rest of the bar.

▶ TRACK 217 Accidentals in C major

Bar 2: the F after F# needs a natural sign, because the sharp would otherwise last to the end of the bar. Bar 3: the same on C# and C.

Key signatures

A **key signature** is the group of sharps or flats placed after the clef on every line. It applies to that note in every octave for the whole piece, unless an accidental overrides it. It saves writing the same accidental again and again, and it tells you the key.

The sharps and flats always appear in a fixed order (sharps F C G D A E B, flats the reverse), and Chapter 5 explains how each key gets its signature. For reading, the table below is the reference.

Key signature	Major key	Relative minor
none	C	A minor
1 sharp (F#)	G	E minor
2 sharps (F# C#)	D	B minor
3 sharps (F# C# G#)	A	F# minor
4 sharps (F# C# G# D#)	E	C# minor
1 flat (Bb)	F	D minor
2 flats (Bb Eb)	Bb	G minor

Key signature	Major key	Relative minor
3 flats (B $\flat$ E $\flat$ A $\flat$)	E $\flat$	C minor

SHORTCUT · Major key or relative minor?

Each signature serves a major key and its relative minor. To decide between them, look at the last note and the first chord: a piece in E minor usually ends on E and often uses D $\sharp$ as a leading note, written as an accidental because the signature does not contain it.

TRACK 218 Key of D major: two sharps from the signature

Fretboard diagram for Track 218 (D major):

String	1st Measure	2nd Measure	3rd Measure	4th Measure
T	2	0	2	3
A			2	
B			4	0

A B C $\sharp$ D, C $\sharp$ B A, F $\sharp$ G A C $\sharp$, D. No sharps are written in the music: the key signature supplies them.

TRACK 219 Key of F major: one flat from the signature

Fretboard diagram for Track 219 (F major):

String	1st Measure	2nd Measure	3rd Measure	4th Measure
T	3	0	2	3
A			1	3
B			2	3

F G A B $\flat$, C B $\flat$, A B $\flat$ C B $\flat$ A G, F. Every B is B $\flat$: 3rd string, 3rd fret.

Positions

A **position** is named after the fret where the 1st finger sits. In position V the 1st finger covers the 5th fret, the 2nd the 6th, the 3rd the 7th and the 4th the 8th. Classical scores mark positions with Roman numerals above the staff. Each position covers four frets on six strings, about two octaves, without moving the hand.

Position	Roman numeral	Fingers cover	Typical use
First	I	frets 1–4 (with open strings)	Most beginner music, open-string keys
Second	II	frets 2–5	Keys of D and A without open strings
Fifth	V	frets 5–8	C, F and A minor away from the open strings
Seventh	VII	frets 7–10	G, D and E minor in the middle of the neck
Ninth	IX	frets 9–12	High melodies up to the 12th fret

Three positions**Position II: D major**
Position V: C major
Position VII: G major

The same written note exists in several places on the neck. The E written in the top space of the staff can be played on the open 1st string, at the 5th fret of the 2nd string, the 9th fret of the 3rd string or the 14th

fret of the 4th string. The position, or a circled string number, tells you which one the editor intends. Chapter 4 shows how to find these unisons quickly.

One written note, four places

All four are the same pitch: the E written in the top space of the staff.

▶ TRACK 220 Reading in position V

1 2 4 1 3 4 3 1 4 2 4 4

5 6 8 5 7 8 7 5 8 6 8 8

T
A
B

Fretting-hand fingers above the TAB. E F G A, B C B, A G F G, C: notes above the staff without leaving the position.

▶ TRACK 221 Reading in position VII

1 3 1 2 4 1 2 4 2 1 4 2 2

7 9 7 8 10 7 8 10 8 7 10 8 8

T
A
B

D E F# G, A B C D, C B A G, G. Every F# comes from the key signature.

Fingering marks

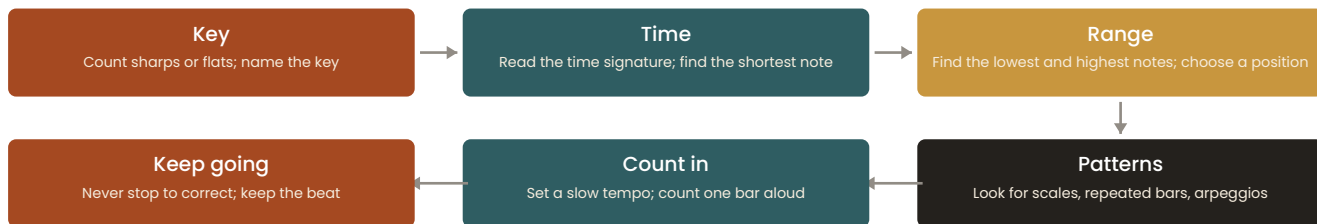
Classical and fingerstyle scores add small signs to the notes to show how the editor wants them played. This book prints fingerings above the TAB; in published classical music they are printed beside the notes.

Mark	Meaning
1 2 3 4 beside a note	Fretting-hand finger: index, middle, ring, little
0	Open string
A dash before the number (-2)	Guide finger: slide the same finger along the string to the new note
p i m a	Picking-hand finger: thumb, index, middle, ring (Chapter 1)
A number in a circle (circled 2, for example)	The string to play the note on
Roman numeral (V, VII)	Position: the 1st finger is at that fret
C V, CV or BV	Barre at the 5th fret (C for <i>cejilla</i> , B for barre); ½C V is a half barre
A curved line between different notes	Slur: play the second note with a hammer-on or pull-off

A graded reading method

Reading improves with daily volume, not with occasional long sessions. Read a little every day, from music slightly easier than what you can play from memory. The method below adds one new element at each level; stay at a level until you can read its exercises once through without stopping.

Before you play a new piece



Level	New element	Exercises
1	1st and 2nd strings, quarter and half notes	1a–1c
2	3rd string, 3/4 time	2a–2b
3	Bass strings and ledger lines below the staff	3a–3b
4	Eighth notes, key signatures with sharps	4a–4b
5	Reading in a higher position	5
6	Melody and bass together	6

CAREFUL · Reading, not memorising

Read each exercise no more than three times in a row. After that you are playing from memory. Move on and come back another day. Cover the TAB with a strip of paper so your eyes stay on the notation.

▶ TRACK 222 Level 1a: the 1st string

0 — 1 — 3 — 1 — 0 — 3 — 1 — 3 — 1 — 0 — 0

T
A
B

▶ TRACK 223 Level 1b: the 2nd and 1st strings

0 — 1 — 3 — 0 — 1 — 0 — 3 — 1 — 3 — 0 — 1

T
A
B

▶ **TRACK 224** Level 1c: Ode to Joy (Beethoven), notation only

▶ **TRACK 229** Level 4a: Frère Jacques (traditional) in G

5

Traditional French round. A second player can start two bars later.

▶ **TRACK 230** Level 4b: E minor, notation only

Bar 1: the F# on the 4th string comes from the key signature. Bar 3: so does the F# on the 1st string, 2nd fret.

▶ **TRACK 231** Level 5: position V, notation only

All notes in position V: 1st string frets 5, 7 and 8; 2nd string frets 5, 6 and 8; one note on the 3rd string at fret 7.

▶ **TRACK 232** Level 6: Ode to Joy with a bass line

Melody with stems up, bass with stems down. Read the bass note first in each bar, then the melody above it.

TRY THIS · Read something new every day

Collect easy music for any instrument in treble clef: hymn books, folk tune collections, violin and flute methods. Read one short melody a day in first position. Ignore the octave: the notes are played in the same places as guitar music.

Chapter 26 review · use it, train it, test it

Using it in real music

Classical repertoire. Classical pieces are published in notation with fingering marks, position numerals and string numbers. Reading them tells you not only which notes, but which string, finger and position the editor intends.

Playing with other musicians. Lead sheets, choir parts and melodies written for violin, flute or voice are in notation without TAB. Reading them lets you join in without asking for a guitar arrangement.

Checking TAB. TAB without rhythm is common online. Comparing it with a notated melody, or writing the rhythm in yourself, fixes most wrong transcriptions (Chapter 30).

How to train it

Drill	How	Time	Goal
Note-name flash	Point to random notes on a staff (or use flashcards) and name them aloud	3 min	30 notes per minute in first position
String by string	Play every natural note on one string, naming each aloud; change string each day	3 min	All six strings without hesitation
Daily sight-reading	One new 8-bar melody per day, TAB covered, metronome at a slow tempo	5 min	Read once through without stopping
Key signature naming	Name the major and minor key for 10 random key signatures	2 min	Each answer within 3 seconds
Position reading	Read a first-position melody again in position V or VII	4 min	Same melody, no open strings

Check your understanding

Write your answers, then check them in Appendix J.

1. Name the notes on the lines of the treble staff, bottom to top.

2. Why is guitar music written an octave higher than it sounds?

3. Which note is the open 2nd string, and where is it written?

4. A bar contains F $\sharp$, then later an F with no sign, on the same line. Which note is the second F?

5. A piece has one sharp in the key signature, ends on E and uses D $\sharp$ as an accidental. Which key is it in?

6. In the key of F major, how is every B on the 3rd string played in first position?

7. What does a Roman numeral VII above the staff mean?

8. Where can you play the E written in the top space of the staff, apart from the open 1st string?

MY NOTES

CHAPTER 27

Reading Rhythm



TRACKS

27

Rhythm notation shows exactly when each note starts and how long it lasts. It is a small system of note values, rests, dots, ties and time signatures, and every rhythm you will meet is built from it.

Pitch tells you which note to play; rhythm tells you when. A wrong note in the right place often passes unnoticed, but a right note in the wrong place never does. This chapter teaches you to read rhythm: the symbols, how to count them aloud, and a graded set of reading exercises on a single open string, so that nothing distracts from the timing.

Chapter 26 covers pitch. Chapter 37 builds timing and groove on top of what you learn here.

Beat, bar and tempo

The **beat** is the steady pulse you tap your foot to. **Tempo** is the speed of the beat, in beats per minute (bpm). Beats are grouped into **bars** (measures) by barlines, and the **time signature** at the start says how many beats each bar holds and which note value gets one beat.

Note values

Each note value is half the length of the one above it. In 4/4 the quarter note is one beat.

Note (US)	British name	Beats in 4/4	Rest	Count in 4/4
Whole note	Semibreve	4	Whole rest (hangs from the 4th line)	1 2 3 4
Half note	Minim	2	Half rest (sits on the 3rd line)	1 2
Quarter note	Crotchet	1	Quarter rest	1
Eighth note	Quaver	½	Eighth rest	1 &
Sixteenth note	Semiquaver	¼	Sixteenth rest	1 e & a

Eighth and sixteenth notes have flags; when several of them fall in the same beat they are joined by **beams** instead. Beams show the beats at a glance: one beamed group is usually one beat (or half a bar of eighths in 4/4).

▶ TRACK 233 Note values on the open 3rd string

Whole, half, quarter, eighth and sixteenth notes. Keep the beat in your foot steady while the notes get faster.

Rests

A rest is a measured silence with the same value as the note of the same name. On the guitar a rest has to be played: lift the fretting finger or touch the string with the picking hand so the note actually stops.

▶ **TRACK 234** Rests

Tablature for Track 234:

Staff	Bar 1	Bar 2	Bar 3	Bar 4
T	0	0	0	0
A	0	0	0	0
B				

Stop the string exactly on each rest. Bar 3 alternates eighth notes and eighth rests, so the notes are short and detached.

Dots and ties

A **dot** after a note adds half its value. A dotted half note lasts three beats; a dotted quarter lasts one and a half. A second dot adds half of the first dot.

A **tie** is a curved line joining two notes of the same pitch. The second note is not played again: its value is added to the first. Ties are used for notes that cross a barline, and to show where the beat falls inside a long note.

Written	Value in beats (4/4)	Same as
Dotted half	3	half tied to quarter
Dotted quarter	1½	quarter tied to eighth
Dotted eighth	¾	eighth tied to sixteenth
Half tied over a barline to a quarter	3	(spread over two bars)

▶ **TRACK 235** Dots and ties

Tablature for Track 235:

Staff	Bar 1	Bar 2	Bar 3	Bar 4	Bar 5
T	0	0	0	0	0
A	0	0	0	0	0
B					

Bar 3 writes a dotted quarter as a quarter tied to an eighth. Bar 4 ties the last note over the barline: do not pick again on beat 1 of bar 5.

Time signatures

The upper number of a time signature gives the number of beats, or in compound time the number of divided units. The lower number gives the note value: 4 is a quarter note, 8 an eighth, 2 a half.

Simple time divides each beat into two. **Compound time** divides each beat into three, so the beat is a dotted note. 6/8 has six eighth notes in a bar but only two beats, each a dotted quarter.

Time signature	Type	Beats per bar	One beat is	Count
2/4	Simple duple	2	quarter	1 & 2 &
3/4	Simple triple	3	quarter	1 & 2 & 3 &
4/4 (also written C)	Simple quadruple	4	quarter	1 & 2 & 3 & 4 &
2/2 (cut time)	Simple duple	2	half	1 & 2 &
6/8	Compound duple	2	dotted quarter	1 & a 2 & a
9/8	Compound triple	3	dotted quarter	1 & a 2 & a 3 & a
12/8	Compound quadruple	4	dotted quarter	1 & a 2 & a 3 & a 4 & a

▶ TRACK 236 Six eighths in 3/4: three beats

Beamed in pairs: 1 & 2 & 3 &. The bass note marks beat 1, the 4th string beats 2 and 3.

▶ TRACK 237 Six eighths in 6/8: two beats

Beamed in threes: 1 & a 2 & a. Same number of notes as the 3/4 example, but the accents fall on the 1st and 4th eighth notes.

▶ TRACK 238 3/4 against 6/8 on the same six eighths

Switching between the two groupings is called hemiola. It is common in Baroque dances, Latin American music and flamenco.

Counting aloud

Counting aloud is the most reliable way to read rhythm. Say every subdivision, even when no note is played on it, and play on the syllables where notes begin.

Subdivision	Spoken count (one beat)	Takadimi syllables
Quarter notes	1	ta
Eighth notes	1 &	ta di
Sixteenth notes	1 e & a	ta ka di mi
Eighth-note triplets	1 trip let	ta ki da
Compound beat (6/8)	1 & a	ta ki da

Counting sixteenth-note figures

The five common sixteenth-note figures. Count 1 e & a throughout and play only where a block begins.

TRACK 239 The five sixteenth figures

Tablature for Track 239:

Measure	T	A	B
1	0	0	0
2	0	0	0
3	0	0	0
4	0	0	0
5	0	0	0

Beat by beat: four sixteenths, eighth and two sixteenths, two sixteenths and eighth, dotted eighth and sixteenth, then sixteenth, eighth, sixteenth.

Syncopation

Syncopation places notes or accents on the weak parts of the beat (the "&" or "e" and "a") and holds them through the next strong beat. In notation this appears as an eighth followed by a quarter, or as a tie across a beat or a barline.

The most common syncopation in popular music is the **push** (anticipation): a chord arrives an eighth note early, on the "&" of 4, and is tied over the barline.

Syncopation patterns in 4/4

Straight eighths
Off-beats only
1 & (2) & 3 & (4) &
Push into next bar

1 e & a 2 e & a 3 e & a 4 e & a

TRACK 240 Syncopation and the push

Em

C

Tablature for Track 240:

Measure	T	A	B
1	0	0	2
2	0	0	2
3	0	0	2
4	0	0	2

Bar 1: 1 & (2) & 3. Bar 2: the note on the "&" of 2 is held through beat 3. Bar 3: the C chord arrives on the "&" of 4 and is tied into bar 4.

Triplets and swing

A **triplet** fits three notes into the time of two of the same value. Eighth-note triplets put three notes in one beat, marked with a 3 over the beam. Quarter-note triplets put three notes in two beats.

Swing (or shuffle) is a way of playing pairs of eighth notes: the first of each pair is long and the second short, in a ratio close to 2:1. Exact swing is the eighth-note triplet with the first two notes tied. Jazz and blues are usually written in straight eighths with the word *Swing* or a note at the top of the music, and you play them swung.

Straight eighths, triplets and swing

Straight eighths
Triplets
Swing (long-short)

1 2 3 4

The swing row is the triplet row with the middle note of each group tied to the first.

 TRACK 241 Triplets against straight notes

The second system of the musical score for 'The Bird Song' consists of two staves. The top staff is a treble clef with a key signature of one sharp (F#) and a 4/4 time signature. It contains four measures of music, each featuring a triplet of eighth notes. The first three measures are grouped by a bracket with a '3' above them. The fourth measure is a single eighth note. The bottom staff is a three-staff system labeled T, A, and B. It contains two measures of music, each with a single eighth note. The first measure is grouped by a bracket with a '3' above it. The second measure is a single eighth note.

Bar 1: quarters and triplets. Bar 2: two eighths against three triplets. Bar 3: written-out swing, the first two triplet notes tied.

TIP • Feeling triplets

Say "1 trip let 2 trip let" or the word "choc-o-late" once per beat. For swing, keep saying all three syllables but play only on the first and the last: "1 (trip) let 2 (trip) let".

Reading drills

The exercises below are on the open 3rd string, so that you can watch only the rhythm. Set a metronome at 60 bpm. Count one bar aloud before you start, keep counting while you play, and do not stop for mistakes. When an exercise is clean twice in a row, raise the tempo by 8 bpm or move on.

Exercise	New element
1	Quarters, halves, dotted halves, quarter rests
2	Eighth notes
3	Eighth rests and off-beats
4	Dotted quarter and eighth
5	Syncopation and ties
6	Sixteenth-note figures
7	3/4
8	6/8
9	Triplets
10	12/8

 TRACK 242 Exercise 1: quarters, halves and rests

The first staff of music is written in treble clef with a 4/4 time signature. It contains the following sequence of notes and rests: a quarter note G4, a quarter note A4, a quarter note B4, a quarter rest, a quarter note C5, a dotted half note D5, a quarter note E5, a quarter rest, a quarter note F5, and a quarter rest. The staff ends with a double bar line.

 TRACK 243 Exercise 2: eighth notes

The first staff of music is written in treble clef with a 4/4 time signature. It contains a sequence of notes: a quarter note G4, an eighth note A4, a quarter note B4, an eighth note C5, a quarter note D5, an eighth note E5, a quarter note F5, and an eighth note G5. This is followed by a quarter rest, then a quarter note G4, an eighth note F4, a quarter note E4, an eighth note D4, a quarter note C4, and finally a double bar line.

▶ TRACK 244 Exercise 3: off-beats



▶ TRACK 245 Exercise 4: dotted quarters



 TRACK 246 Exercise 5: syncopation and ties



 TRACK 247 Exercise 6: sixteenths



 TRACK 248 Exercise 7: 3/4



 TRACK 249 Exercise 8: 6/8



 TRACK 250 Exercise 9: triplets



 TRACK 251 Exercise 10: 12/8



Four dotted-quarter beats per bar. The long-short figure (quarter + eighth) in bar 1 is the shuffle feel of a slow blues.

TRY THIS · Rhythm from the page, then from the ear

Pick any exercise above, play it twice from the page, then close the book and play it from memory. Next, clap a rhythm from a song you know and write it down using the same note values. Writing rhythms is the fastest way to read them (Chapter 30).

Chapter 27 review · use it, train it, test it

Using it in real music

Strumming and riffs. Strum patterns, funk rhythms and rock riffs are written with exactly these symbols, often with x noteheads for muted strums. Reading them saves you from guessing a rhythm from a recording.

Classical and fingerstyle pieces. Two voices with different rhythms (a melody in eighths over a bass in half notes) are common. Count the shortest value and place both voices on that grid.

Playing with others. Bands and ensembles communicate in bars and beats: "the push into bar 9", "the triplet fill at the end of the bridge". Counting aloud gives you the same map.

How to train it

Drill	How	Time	Goal
Value ladder	Play the note-values example while naming each value aloud (whole, half, quarter, eighth, sixteenth)	2 min	Every value named and played without a pause
Count and clap	Clap one reading exercise while counting every subdivision aloud	2 min	No hesitation on rests or ties
Daily rhythm reading	One new four-bar rhythm per day on an open string, metronome on	5 min	Once through without stopping
Swing switch	Four bars straight eighths, four bars swung, alternating	3 min	A clear difference you can hear on a recording
Write what you hear	Write down the rhythm of a short phrase from a recording	5 min	Matches the recording when played back

Check your understanding

Write your answers, then check them in Appendix J.

1. How many sixteenth notes fit in a dotted quarter note?

2. What is the difference between a dot and a tie?

3. Is 3/4 simple or compound, and how many beats does it have?

4. How many beats are there in a bar of 6/8, and what note value is one beat?

5. Count the figure "dotted eighth + sixteenth" in one beat.

6. What is a push (anticipation)?

7. How is swing related to triplets?

8. What does a 3 over three beamed eighth notes mean?

MY NOTES

CHAPTER 28

Ear Training



Ear training connects what you hear to what you know: the interval between two notes, the quality of a chord, the function of a chord in a key. Ten minutes a day of focused listening makes learning songs, improvising and playing with others far easier.

Ear training is not a talent you have or lack. It is a set of recognition skills, each trained the same way: hear an example, name it, check the answer, repeat. This chapter trains the skills guitarists use most: intervals, chord qualities, chord functions in a key, common progressions and short melodies. Every example has an audio track; listen before you read the music.

The theory behind each skill is in earlier chapters: intervals in Chapter 3, triads and seventh chords in Chapter 10 and Chapter 11, and chords in a key in Chapter 13. Chapter 30 puts all of these skills to work on real recordings.

How to practise

- **Short and daily.** Ten minutes every day trains the ear far better than an hour once a week.
- **Sing.** Singing a note forces you to hear it precisely before you produce it. Your voice does not need to be good, only roughly in tune.
- **Name, then check.** Always commit to an answer before you look. A wrong answer you notice teaches more than a right guess.
- **Start with few choices.** Compare two things (major or minor, 4th or 5th) until you are always right, then add a third.
- **Use context.** Real music has a key. Hearing a note as a degree of the key (the 3rd, the 5th) is more useful than hearing an isolated interval.

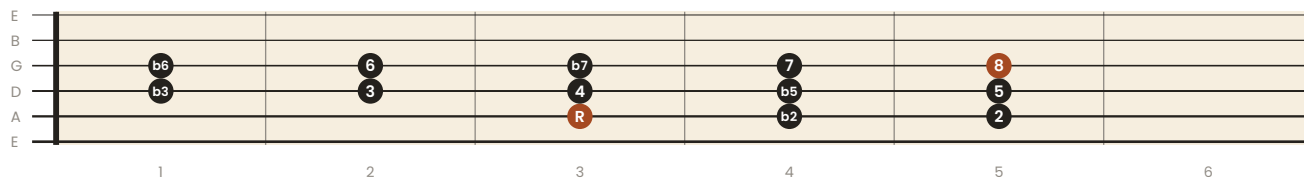
Intervals

An **interval** is the distance between two notes. A **melodic** interval has the notes one after the other; a **harmonic** interval has them together. Learn each ascending interval with a reference: the opening of a tune you know well that starts with that interval. The table names songs only; sing the opening to yourself and compare.

Interval	Half steps	Ascending reference	Descending reference
Minor 2nd	1	Jaws theme	Für Elise (Beethoven)
Major 2nd	2	Happy Birthday	Mary Had a Little Lamb
Minor 3rd	3	Greensleeves	Hey Jude (The Beatles)
Major 3rd	4	When the Saints Go Marching In	Swing Low, Sweet Chariot
Perfect 4th	5	Here Comes the Bride; Amazing Grace	Eine kleine Nachtmusik (Mozart)
Tritone	6	Maria (West Side Story); The Simpsons theme (first to third note)	
Perfect 5th	7	Twinkle, Twinkle, Little Star	The Flintstones theme
Minor 6th	8	The Entertainer (Joplin), the leap after the pick-up	
Major 6th	9	My Bonnie Lies over the Ocean	Nobody Knows the Trouble I've Seen
Minor 7th	10	Somewhere (West Side Story)	
Major 7th	11	Take On Me (a-ha), chorus	

Interval	Half steps	Ascending reference	Descending reference
Octave	12	Somewhere over the Rainbow	

Intervals above C



Every interval within the octave above C (5th string, 3rd fret), in one position. Play the root, then each interval, and sing both notes.

▶ **TRACK 252** Intervals 1: minor 2nd to tritone

m2 M2 m3 M3 P4 TT

T 0 0 0 0 0

A 5 5 0 0 0

B 3 3 2 2 3 3 3 3 3 3

Each interval is played melodically (lower note first), then harmonically. m2 B-C, M2 C-D, m3 E-G, M3 C-E, P4 C-F, tritone F-B.

▶ **TRACK 253** Intervals 2: perfect 5th to octave

P5 m6 M6 m7 M7 P8

T 0 0 1 1 2 2 1 1 0 0 1 1

A 0 0 2 2 2 2 0 0 3 3 1 1

B 3 3 2 2 3 3 0 0 3 3 3 3

P5 C-G, m6 E-C, M6 C-A, m7 D-C, M7 C-B, octave C-C. All notes are natural, so no accidentals are needed.

TIP · What harmonic intervals sound like

Octaves, 5ths and 4ths sound open and hollow. 3rds and 6ths sound sweet and full. 2nds and 7ths clash: the minor 2nd and major 7th most strongly. The tritone sounds restless and wants to resolve. Sort harmonic intervals into these three families first, then tell the members of each family apart.

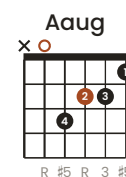
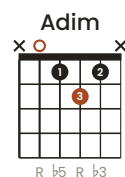
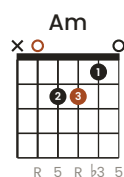
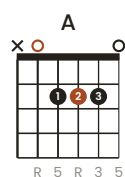
Chord qualities

A chord's **quality** comes from the intervals above its root. Learn to hear the four triads first, then the five common seventh chords. The examples below keep the root A, so only the quality changes.

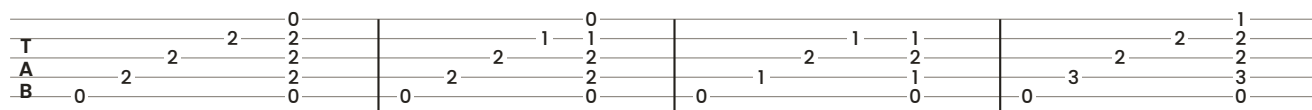
Quality	Formula	Sound
Major	1 3 5	Bright, stable
Minor	1 b3 5	Darker, still stable
Diminished	1 b3 b5	Tense, unstable, small
Augmented	1 3 #5	Unresolved, floating, "dreamlike"
Dominant 7th	1 3 5 b7	Bluesy; wants to move
Major 7th	1 3 5 7	Soft, smooth, jazzy
Minor 7th	1 b3 5 b7	Mellow, relaxed

Quality	Formula	Sound
Half-diminished (m7b5)	1 b3 b5 b7	Dark, leads to a dominant chord
Diminished 7th	1 b3 b5 $\flat$ 7	Very tense, symmetrical

The four triads on A

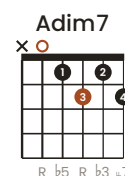
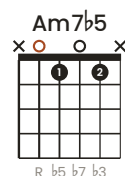
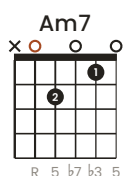
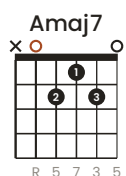
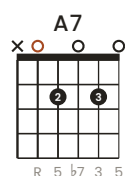


TRACK 254 Triads on A: major, minor, diminished, augmented

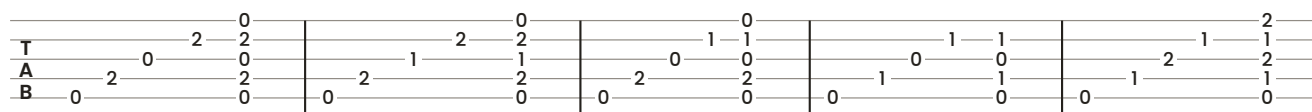


Each chord is arpeggiated, then played together. Listen to the 3rd (major or minor) and then to the 5th (perfect, diminished or augmented).

Five seventh chords on A



TRACK 255 Seventh chords on A



Dominant 7th, major 7th, minor 7th, half-diminished and diminished 7th. The difference between A7 and Amaj7 is one fret on the 3rd string.

TRY THIS · Major or minor

Ask someone to play A or Am, or use a random-chord app, twenty times. Name each before you look. When you score 20 out of 20 twice in a row, add the dominant 7th, then the major 7th, then the minor 7th.

Chord function in a key

Inside a key, each chord has a role. You do not need to recognise chords in isolation if you can hear where they lead.

Function	Chords (major key)	Effect
Tonic	I (also vi, iii)	Home, rest, resolution
Subdominant (predominant)	IV, ii	Moving away from home
Dominant	V, V7, vii°	Tension that pulls back to I

Hear the bass line first: it moves by clear steps and leaps (I up to IV is a 4th, IV to V is a step, V to I is a 4th up or a 5th down). Then decide whether each chord is major or minor. In a major key I, IV and V are major; ii, iii and vi are minor; vii° is diminished.

TRACK 256 Tonic, subdominant, dominant, tonic in C

C F G7 C

	0	1	1	0
T	1	1	0	1
A	0	2	0	0
B	2	3	0	2
	3		2	3
			3	

I – IV – V7 – I. Listen for the pull from G7 back to C: the B in G7 rises a half step to C, and the F falls to E.

TRACK 257 Hearing scale degrees after a cadence

C F G C

	0	1	3	0	0
T	1	1	0	1	
A	0	2	0	0	
B	2	3	0	2	
	3		2	3	
			3		

C F G C

	0	1	3	0	2
T	1	1	0	1	
A	0	2	0	0	
B	2	3	0	2	
	3		2	3	
			3		

C F G C

	0	1	3	0	0
T	1	1	0	1	
A	0	2	0	0	
B	2	3	0	2	
	3		2	3	
			3		

A cadence sets the key, then one note follows. Name its scale degree before reading on. The answers are 3 (E), 6 (A) and 7 (B). The 7th sounds as if it must rise to C.

Recognising progressions

Most songs use a handful of progressions. Learn their sound as a whole, the way you recognise a word without spelling it. For each, sing the bass line (the roots) along with the recording.

Four common progressions in C

C <i>I</i>	F <i>IV</i>	G <i>V</i>	C <i>I</i>
C <i>I</i>	G <i>V</i>	Am <i>vi</i>	F <i>IV</i>
Am <i>vi</i>	F <i>IV</i>	C <i>I</i>	G <i>V</i>
Dm <i>ii</i>	G7 <i>V7</i>	C <i>I</i>	C <i>I</i>

Rows: *I* – *IV* – *V* – *I*; *I* – *V* – *vi* – *IV*; *vi* – *IV* – *I* – *V*; *ii* – *V7* – *I*.

▶ TRACK 258 *I* – *V* – *vi* – *IV*

C G Am F C G Am F

0 3 0 1 0 3 0 1
T 1 0 1 1 1 0 1 1
A 0 0 2 2 0 0 2 2
B 2 0 2 3 2 0 2 3
3 2 0 3 2 0 0 3

The bass falls from C to G, rises to A, falls to F. This loop underlies a very large number of pop songs.

▶ TRACK 259 *ii* – *V7* – *I*

Dm G7 C Dm G7 C

1 1 0 1 0 1 0
T 3 0 1 3 0 1 0
A 2 0 0 2 0 0 0
B 0 0 2 0 0 2 3
2 2 3 2 2 3 3

The bass moves up a 4th twice (D to G, G to C). This is the central progression of jazz (Chapter 48).

▶ TRACK 260 Mystery progression A

0 0 1 3 0 0 1 3
T 1 1 1 0 1 1 1 0
A 0 2 2 0 0 2 2 0
B 2 2 3 0 2 2 3 0
3 0 2 3 3 0 2 3

Listen first, then name the four chords with Roman numerals. The answer is in the quiz answers (question 7).

▶ TRACK 261 Mystery progression B

0 1 0 3 0 1 0 3
T 1 1 1 0 1 1 1 0
A 2 2 0 0 2 2 0 0
B 2 3 2 0 2 3 2 0
0 3 3 2 0 3 3 2

Listen first. Which chord feels like home, and is it major or minor? The answer is in the quiz answers (question 8).

Melodic dictation

Dictation means writing down a melody you hear. It is the core skill of transcription, trained here on short phrases. Work in this order:

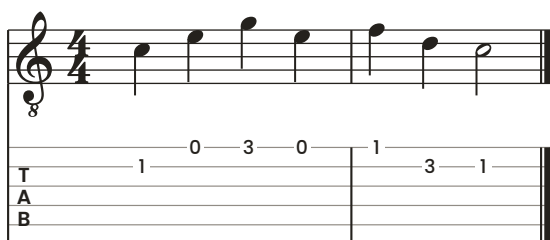
1. **Find the tonic.** Sing the note that sounds like home. Find it on the guitar.
2. **Rhythm.** Tap along and write the rhythm alone, using the counting from Chapter 27.
3. **Contour.** Draw the shape: up, down, step, leap.
4. **Pitches.** Name each note as a scale degree, using the intervals you know. Then find it on the guitar.
5. **Check.** Play your answer along with the recording. Correct the first wrong note and play again.

▶ TRACK 262 Dictation 1



Two bars in C, all steps. Cover the music, listen three times, write it down, then compare.

▶ TRACK 263 Dictation 2



Leaps through the C major chord, then steps down to the tonic.

▶ TRACK 264 Dictation 3



Starts on the 5th. Bar 1 contains a dotted quarter and an eighth.

▶ TRACK 265 Dictation 4



A minor. The fifth note, G#, is the raised 7th of harmonic minor: hear how it leads up to A.

Singing and playing

Singing connects the ear to the hands. These exercises take a few minutes and need nothing but the guitar.

- **Match the pitch.** Play a note in your vocal range, sing it, and hold it until it blends with the string. Then sing first and check with the guitar.
- **Sing the scale degrees.** Play a C major chord, then sing up the scale with numbers (1 2 3 4 5 6 7 8) or solfège (do re mi fa sol la ti do). Then sing 1 3 5 8 5 3 1.
- **Sing an interval, then check.** Play C, sing a perfect 5th above it, then play G to check. Change the starting note each time.
- **Sing the bass line.** While strumming a progression, sing the root of each chord. This trains you to hear the bass in recordings.
- **Play what you sing.** Sing a short phrase you make up, then find it on the guitar without searching note by note: start on the tonic and think in degrees.

CAREFUL · Range

Keep the singing range comfortable: roughly the octave from the open 5th string (A) to the A on the 3rd string, 2nd fret, for lower voices, or an octave above for higher voices. Strained singing teaches the wrong pitch.

A daily 10-minute routine

Time	Exercise	How
0–2	Pitch matching	Play five random notes in your range; sing each, then sing it first and check
2–4	Intervals	Name ten random intervals (app or partner); sing three from a given note
4–6	Chord quality	Name ten random chords: major and minor first, then sevenths
6–8	Function and progressions	Play a cadence in a random key, then name the degree of a note or the next chord
8–10	Dictation	Write down two bars from a recording or the dictation tracks, then check

Keep a short log: date, exercise, score. Progress in ear training is steady but slow, and a log shows it.

Chapter 28 review · use it, train it, test it

Using it in real music

Learning songs quickly. Recognising I – V – vi – IV or a ii – V – I by ear means you can play along with a song after hearing it once, and only check the details.

Improvising. Hearing chord quality and function tells you which notes will sound stable over each chord (Chapter 35).

Playing with others. In a jam, the bass line and the chord quality are often all you need. Singing the roots of a progression trains exactly that.

How to train it

Drill	How	Time	Goal
Interval pairs	Random harmonic intervals from two families (for example 4ths and 5ths), then add more	3 min	18 of 20 correct
Chord quality	Major, minor, dominant 7th, major 7th and minor 7th, played at random	3 min	18 of 20 correct
Bass line singing	Sing the roots of a song's progression while it plays	3 min	Every change on time
Degree after cadence	Cadence in C, then one random note in the scale; name its degree	2 min	9 of 10 correct
Two-bar dictation	One phrase per day from a recording or melody book	5 min	Correct pitches and rhythm after at most five hearings

Check your understanding

Write your answers, then check them in Appendix J.

1. How many half steps are in a perfect 5th, and which reference song starts with one?

2. Which interval opens Here Comes the Bride?

3. Which single note distinguishes a major triad from a minor triad?

4. Name the formula of a dominant 7th chord and describe its sound.

5. Which chords in a major key have a dominant function?

6. In a melodic dictation, what should you identify before writing any pitches?

7. Name the chords of Mystery progression A in Roman numerals.

8. Name the chords of Mystery progression B. Which chord is home?

MY NOTES

CHAPTER 29

Ear-Training Course



A 12-week graded course built on the skills from the previous chapter. Every week has quiz tracks that you answer on paper and mark against the answer key, so you always know where you stand.

Chapter 28 explained what to listen for: intervals, chord qualities, chord functions, progressions and dictation. This chapter turns those skills into a course. It runs for twelve weeks, one stage per week, ten minutes a day. Each week adds a little to what you already hear, and every week ends with a measurable result.

How the course works

Ten minutes a day, one week per stage. Work through the weeks in order. A typical week looks like this:

Day	What to do
1	Read the week's section. Play the reference example and do the singing exercise
2–3	Quiz A each day, then the singing exercise
4–5	Quiz B each day, then quiz A once more if you scored below 9
6–7	Quiz C (or the hardest quiz of the week). This is the test for the week

Using the quiz tracks. Each quiz has an audio track and an answer sheet with ten numbered lines. A click starts each question; then you hear the example and a short silence to write your answer. Play the track once from start to finish without pausing, write every answer on paper (not in the book, so you can reuse the sheet), then check your answers in Appendix J. Count your score out of 10 and write it in your log with the date.

The pass mark is 9 out of 10, twice in a row. Pass the hardest quiz of the week twice, on two different days, before you move on. If you score 9 once and 7 the next day, the skill is not yet reliable: keep going. If a week takes two or three weeks, that is normal. The early weeks go quickly for most players; weeks 5, 9 and 10 usually take longer.

Sing before you answer. After each example, sing what you heard before you write: both notes of an interval, the root and 3rd of a chord, the bass line of a progression, the melody of a dictation. Singing makes you hold the sound long enough to compare it with your references. A guess you have not sung is still a guess.

CAREFUL · Fixed tracks

Each quiz track is the same every time you play it, so after a few hearings you may remember the answers by their order. When a quiz starts to feel familiar, move to the next quiz of the week, or practise the same set with a partner or an ear-training app (Chapter 28 describes how). Use the tracks for the test at the end of the week.

The twelve weeks

Week	Topic	Quiz kinds
1	Perfect 5th, octave, perfect 4th	Intervals, ascending
2	Major and minor 3rd	Intervals, ascending
3	Major, minor, diminished and augmented triads	Chords
4	Every interval in the octave	Intervals, ascending
5	Descending and harmonic intervals	Intervals
6	Seventh chords	Chords
7	Scale degrees in major	Melody dictation
8	Progressions with I, IV and V	Progressions
9	Progressions with vi and ii	Progressions
10	Minor and minor pentatonic melodies	Melody dictation
11	Colour chords and scale types	Chords, scales
12	Mixed review and final exam	All kinds

Week 1: perfect 5th, octave and 4th

All three belong to the open, hollow family described in Chapter 28. Tell them apart with the reference songs from its table: Somewhere over the Rainbow for the octave, Twinkle, Twinkle for the 5th, Here Comes the Bride for the 4th. **Sing:** play the open low E, sing it, then sing the 5th (B) and the octave (E) above it. Check each note on the guitar.

▶ TRACK 266 Week 1 reference: 5th, octave and 4th over an E drone

P5 P8 P4

T 0 0 0 0 2 2 2 2 2 2 2 2

A 2 2 2 2 2 2 2 2 2 2 2 2

B 0 0 0 0 2 2 2 2 2 2 2 2

Each interval starts on E (4th string, 2nd fret) over the open low E. The low E keeps the key in your ear.

▶ TRACK 267 Week 1, quiz A: perfect 5th or octave

Answer with: perfect 5th, octave. 10 questions; a click starts each one.

- 1 _____ 2 _____ 3 _____ 4 _____ 5 _____
 6 _____ 7 _____ 8 _____ 9 _____ 10 _____

▶ TRACK 268 Week 1, quiz B: perfect 4th or perfect 5th

Answer with: perfect 4th, perfect 5th. 10 questions; a click starts each one.

- 1 _____ 2 _____ 3 _____ 4 _____ 5 _____
 6 _____ 7 _____ 8 _____ 9 _____ 10 _____

▶ TRACK 269 Week 1, quiz C: perfect 4th, perfect 5th or octave

Answer with: perfect 4th, perfect 5th, octave. 10 questions; a click starts each one.

- 1 _____ 2 _____ 3 _____ 4 _____ 5 _____
 6 _____ 7 _____ 8 _____ 9 _____ 10 _____

Week 2: major and minor 3rd

The 3rd decides whether music sounds major or minor, so this is the most useful interval to know. The major 3rd (When the Saints) sounds bright; the minor 3rd (Greensleeves) sounds darker and smaller. In the second quiz the 3rds are mixed with last week's 4th and 5th: first decide whether the interval is a 3rd at all (sweet and full) or a 4th or 5th (open and hollow). **Sing:** play E, sing up E G# B (1 3 5) and then E G B (1 b3 5). Feel which note changes.

▶ **TRACK 270** Week 2 reference: 3rds against the 4th and 5th

T		1	1		0	0		2	2		0	0
A	2		2		2		2		2		2	
B	0				0		0				0	

E to G# (major 3rd), E to G (minor 3rd), E to A (4th), E to B (5th), all over the open low E.

▶ **TRACK 271** Week 2, quiz A: major or minor 3rd

Answer with: minor 3rd, major 3rd. 10 questions; a click starts each one.

1 _____ 2 _____ 3 _____ 4 _____ 5 _____
 6 _____ 7 _____ 8 _____ 9 _____ 10 _____

▶ **TRACK 272** Week 2, quiz B: 3rds, 4th and 5th

Answer with: minor 3rd, major 3rd, perfect 4th, perfect 5th. 10 questions; a click starts each one.

1 _____ 2 _____ 3 _____ 4 _____ 5 _____
 6 _____ 7 _____ 8 _____ 9 _____ 10 _____

▶ **TRACK 273** Week 2, quiz C: 3rds, 4th, 5th and octave

Answer with: minor 3rd, major 3rd, perfect 4th, perfect 5th, octave. 10 questions; a click starts each one.

1 _____ 2 _____ 3 _____ 4 _____ 5 _____
 6 _____ 7 _____ 8 _____ 9 _____ 10 _____

Week 3: triads

The quiz plays each chord together, then as an arpeggio from the root, then together again. Listen to the 3rd first (major or minor), then to the 5th: diminished sounds tight and tense, augmented sounds stretched and unresolved (Chapter 28 has the sound of each). **Sing:** during the arpeggio, sing along 1 3 5; then sing the 3rd alone and decide whether it is the major or the minor 3rd from week 2.

▶ **TRACK 274** Week 3, quiz A: major or minor

Answer with: major, minor. 10 questions; a click starts each one.

1 _____ 2 _____ 3 _____ 4 _____ 5 _____
 6 _____ 7 _____ 8 _____ 9 _____ 10 _____

▶ **TRACK 275** Week 3, quiz B: major, minor or diminished

Answer with: major, minor, diminished. 10 questions; a click starts each one.

1 _____ 2 _____ 3 _____ 4 _____ 5 _____
 6 _____ 7 _____ 8 _____ 9 _____ 10 _____

▶ TRACK 276 Week 3, quiz C: all four triads

Answer with: major, minor, diminished, augmented. 10 questions; a click starts each one.

1 _____ 2 _____ 3 _____ 4 _____ 5 _____
 6 _____ 7 _____ 8 _____ 9 _____ 10 _____

Week 4: every interval in the octave

This week fills the gaps between the intervals you know, one family at a time: 2nds and 3rds, then 4th, tritone and 5th, then 6ths, 7ths and octave. Within each family, compare the two sizes: the minor 2nd is the tight half step of the Jaws theme, the major 2nd the step from the second to the third note of Happy Birthday; the tritone sits between the 4th and the 5th and sounds unstable. Use the reference table in Chapter 28 for the 6ths and 7ths. **Sing:** play C, then sing each interval of the quiz you are working on before you play it to check.

▶ TRACK 277 Week 4, quiz A: 2nds and 3rds

Answer with: minor 2nd, major 2nd, minor 3rd, major 3rd. 10 questions; a click starts each one.

1 _____ 2 _____ 3 _____ 4 _____ 5 _____
 6 _____ 7 _____ 8 _____ 9 _____ 10 _____

▶ TRACK 278 Week 4, quiz B: 4th, tritone and 5th

Answer with: perfect 4th, tritone, perfect 5th. 10 questions; a click starts each one.

1 _____ 2 _____ 3 _____ 4 _____ 5 _____
 6 _____ 7 _____ 8 _____ 9 _____ 10 _____

▶ TRACK 279 Week 4, quiz C: 6ths, 7ths and octave

Answer with: minor 6th, major 6th, minor 7th, major 7th, octave. 10 questions; a click starts each one.

1 _____ 2 _____ 3 _____ 4 _____ 5 _____
 6 _____ 7 _____ 8 _____ 9 _____ 10 _____

Week 5: descending and harmonic intervals

A descending interval is the same distance heard from the top. Either learn the descending references (Für Elise for the minor 2nd, Hey Jude for the minor 3rd), or sing the interval back upwards from the lower note and name it as an ascending one. For harmonic intervals, sort first into the three families from Chapter 28 (open, sweet, clashing), then name the member. **Sing:** after each harmonic interval, sing the lower note, then the upper one, then name the ascending interval you sang.

▶ TRACK 280 Week 5 reference: the same intervals three ways

m3

P5

Up, down, then together. The minor 3rd E–G and the perfect 5th E–B, over the open low E.

▶ TRACK 281 Week 5, quiz A: descending 3rds, 4th, 5th and octave

Answer with: minor 3rd, major 3rd, perfect 4th, perfect 5th, octave. 10 questions; a click starts each one.

1 _____ 2 _____ 3 _____ 4 _____ 5 _____
 6 _____ 7 _____ 8 _____ 9 _____ 10 _____

▶ **TRACK 282** Week 5, quiz B: harmonic intervals

Answer with: minor 3rd, major 3rd, perfect 4th, perfect 5th, major 6th, octave. 10 questions; a click starts each one.

1 _____ 2 _____ 3 _____ 4 _____ 5 _____
6 _____ 7 _____ 8 _____ 9 _____ 10 _____

▶ **TRACK 283** Week 5, quiz C: every interval, up or down

Answer with: minor 2nd, major 2nd, minor 3rd, major 3rd, perfect 4th, tritone, perfect 5th, minor 6th, major 6th, minor 7th, major 7th, octave. 10 questions; a click starts each one.

1 _____ 2 _____ 3 _____ 4 _____ 5 _____
6 _____ 7 _____ 8 _____ 9 _____ 10 _____

Week 6: seventh chords

Each seventh chord is a triad plus a 7th, so hear it in two steps: major, minor or diminished triad, then major or minor 7th on top. The sound of each chord is described in the chord-quality table of Chapter 28; play the seventh-chord track there before the quizzes. **Sing:** during the arpeggio, sing 1 3 5 7, then sing only the 7th and decide whether it is a half step or a whole step below the octave.

▶ **TRACK 284** Week 6, quiz A: maj7, 7 or m7

Answer with: major 7th, dominant 7th, minor 7th. 10 questions; a click starts each one.

1 _____ 2 _____ 3 _____ 4 _____ 5 _____
6 _____ 7 _____ 8 _____ 9 _____ 10 _____

▶ **TRACK 285** Week 6, quiz B: add the half-diminished chord

Answer with: major 7th, dominant 7th, minor 7th, half-diminished. 10 questions; a click starts each one.

1 _____ 2 _____ 3 _____ 4 _____ 5 _____
6 _____ 7 _____ 8 _____ 9 _____ 10 _____

▶ **TRACK 286** Week 6, quiz C: triads and seventh chords

Answer with: major, minor, major 7th, dominant 7th, minor 7th, half-diminished. 10 questions; a click starts each one.

1 _____ 2 _____ 3 _____ 4 _____ 5 _____
6 _____ 7 _____ 8 _____ 9 _____ 10 _____

Week 7: scale degrees in major

From now on you hear notes in a key, not as isolated intervals. Each question plays the tonic chord, then a short melody that starts on the tonic (1) and moves by steps and small leaps, never repeating a note straight away. It can dip below the tonic: write those notes as low 7 or low 5, and a note an octave up as high 1. Write the degrees, for example 1 2 3 or 1 low 7 1 2. Hear each note against the tonic: 3 and 5 belong to the chord and sound at rest, 2, 4 and 6 lean towards a chord tone, and 7 pulls up to 8. **Sing:** play the E major scale over the drone below, singing the degree numbers; then sing 1 3 5 3 1 and 1 2 3 2 1 from memory.

▶ **TRACK 287** Week 7 reference: E major scale degrees over a drone

The image shows a musical score for a reference track. The top staff is a treble clef with a key signature of one sharp (F#) and a 4/4 time signature. It contains the E major scale: E4, F#4, G#4, A4, B4, C#5, D#5, E5. The bottom staff has three lines labeled T, A, and B. The T line has notes E4, F#4, G#4, A4, B4, C#5, D#5, E5. The A line has notes E4, F#4, G#4, A4, B4, C#5, D#5, E5. The B line has notes E4, F#4, G#4, A4, B4, C#5, D#5, E5. Fingerings are indicated below the notes: 0, 2, 4, 1, 2, 0, 2, 4, 0.

Degrees 1 2 3 4 5 6 7 8 (E F# G# A B C# D# E). Sing each number with the note.

▶ TRACK 288 Week 7, quiz A: three-note melodies in major

Answer with: scale degrees of the major scale; 'low' marks notes below the tonic (the tonic chord sounds first). 10 questions; a click starts each one.

1 _____	2 _____
3 _____	4 _____
5 _____	6 _____
7 _____	8 _____
9 _____	10 _____

▶ TRACK 289 Week 7, quiz B: four-note melodies in major

Answer with: scale degrees of the major scale; 'low' marks notes below the tonic (the tonic chord sounds first). 10 questions; a click starts each one.

1 _____	2 _____
3 _____	4 _____
5 _____	6 _____
7 _____	8 _____
9 _____	10 _____

Week 8: progressions with I, IV and V

Each question plays four chords, each with its root in the bass. The first chord is always I, so it tells you the key. Follow the bass and the chord functions as described in Chapter 28: IV steps away from home, V pulls back to it. **Sing:** while the quiz plays, sing the roots of the chords with degree numbers (1, 4, 5); the bass line usually gives you the answer.

▶ TRACK 290 Week 8, quiz A: two progressions

Answer with: I IV V I; I V IV I. 10 questions; a click starts each one.

1 _____	2 _____
3 _____	4 _____
5 _____	6 _____
7 _____	8 _____
9 _____	10 _____

▶ TRACK 291 Week 8, quiz B: four progressions

Answer with: I IV V I; I V IV I; I IV I V; I I IV V. 10 questions; a click starts each one.

1 _____	2 _____
3 _____	4 _____
5 _____	6 _____
7 _____	8 _____
9 _____	10 _____

▶ TRACK 292 Week 8, quiz C: with the dominant 7th

Answer with: I IV V7 I; I V7 IV I; I IV I V7; I V7 I IV; I IV IV V7. 10 questions; a click starts each one.

1 _____	2 _____
3 _____	4 _____
5 _____	6 _____
7 _____	8 _____
9 _____	10 _____

Week 9: progressions with vi and ii

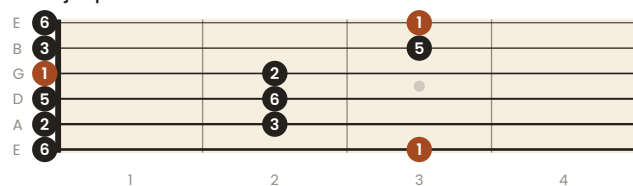
Two minor chords join the three major ones. vi shares two notes with I, so it sounds like a darker version of home; ii shares two notes with IV and usually leads to V. Hear each chord as major or minor, then place its root with the bass line. The common loops (I – V – vi – IV, I – vi – IV – V, ii – V – I) quickly become recognisable as whole shapes. **Sing:** play the loop below and sing the roots (1 6 4 5), then improvise a simple melody over it using only 1, 3 and 5 of each chord.

TRACK 293 Week 9 reference: I – vi – IV – V in G · LOOP · pop · 88 bpm

G <i>I</i>	Em <i>vi</i>	C <i>IV</i>	D <i>V</i>
---------------	-----------------	----------------	---------------

IMPROVISE WITH

G major pentatonic



Sing the roots G, E, C, D as 1, 6, 4, 5. Then sing the 3rd of each chord: B, G, E, F#.

TRACK 294 Week 9, quiz A: the four-chord loops

Answer with: I vi IV V; I V vi IV; vi IV I V; I IV V I. 10 questions; a click starts each one.

- | | |
|---------|----------|
| 1 _____ | 2 _____ |
| 3 _____ | 4 _____ |
| 5 _____ | 6 _____ |
| 7 _____ | 8 _____ |
| 9 _____ | 10 _____ |

TRACK 295 Week 9, quiz B: ii, V and I

Answer with: ii V I I; I vi ii V; I IV ii V; ii V7 I vi. 10 questions; a click starts each one.

- | | |
|---------|----------|
| 1 _____ | 2 _____ |
| 3 _____ | 4 _____ |
| 5 _____ | 6 _____ |
| 7 _____ | 8 _____ |
| 9 _____ | 10 _____ |

TRACK 296 Week 9, quiz C: all six chords

Answer with: I vi ii V; I iii IV V; I V vi IV; vi IV I V; ii V I vi; I IV vi V. 10 questions; a click starts each one.

- | | |
|---------|----------|
| 1 _____ | 2 _____ |
| 3 _____ | 4 _____ |
| 5 _____ | 6 _____ |
| 7 _____ | 8 _____ |
| 9 _____ | 10 _____ |

Week 10: minor melodies

In minor the tonic chord is minor and the 3rd, 6th and 7th are lowered (b3, b6, b7). The b3 gives the dark colour; the b6 sounds heavy and leans down to 5; the b7 is a whole step below the tonic and does not pull up as strongly as the 7 in major. The minor pentatonic leaves out 2 and b6, so its melodies move in larger steps. **Sing:** over the drone below, sing the minor scale up from 1 to 5 and back, then the minor pentatonic 1 – b3 – 4 – 5 – b7 – 8.

▶ TRACK 297 Week 10 reference: E minor pentatonic over a drone

8

T 0 2 0 3 0

A 2 0 2 0 0

B 0 0 0 0 0

Degrees 1 $\flat$ 3 4 5 $\flat$ 7 8 (E G A B D E).

▶ TRACK 298 Week 10, quiz A: four-note melodies in minor

Answer with: scale degrees of the minor scale; 'low' marks notes below the tonic (the tonic chord sounds first). 10 questions; a click starts each one.

- | | |
|---------|----------|
| 1 _____ | 2 _____ |
| 3 _____ | 4 _____ |
| 5 _____ | 6 _____ |
| 7 _____ | 8 _____ |
| 9 _____ | 10 _____ |

▶ TRACK 299 Week 10, quiz B: minor pentatonic melodies

Answer with: scale degrees of the minor pentatonic scale; 'low' marks notes below the tonic (the tonic chord sounds first). 10 questions; a click starts each one.

- | | |
|---------|----------|
| 1 _____ | 2 _____ |
| 3 _____ | 4 _____ |
| 5 _____ | 6 _____ |
| 7 _____ | 8 _____ |
| 9 _____ | 10 _____ |

▶ TRACK 300 Week 10, quiz C: five-note melodies in minor

Answer with: scale degrees of the minor scale; 'low' marks notes below the tonic (the tonic chord sounds first). 10 questions; a click starts each one.

- | | |
|---------|----------|
| 1 _____ | 2 _____ |
| 3 _____ | 4 _____ |
| 5 _____ | 6 _____ |
| 7 _____ | 8 _____ |
| 9 _____ | 10 _____ |

Week 11: colour chords and scale types

Suspended chords replace the 3rd with the 2nd (sus2) or the 4th (sus4), so they sound neither major nor minor: open and waiting. The augmented triad from week 3 returns here for contrast. The 6 chord is a major triad with an added 6th and sounds sweet and settled; the dominant 9th is a dominant 7th with a bright 9th on top. Last, four scales played up from the root: major, natural minor, Dorian (minor with a raised 6th) and Mixolydian (major with a lowered 7th). **Sing:** play D, Dsus4, D, Dsus2 and sing the note that moves on the 1st string.

▶ TRACK 301 Week 11 reference: D, Dsus4, D, Dsus2

D Dsus4 D Dsus2 D

The 3rd (F#) moves up to the 4th (G), back, then down to the 2nd (E). The other strings stay the same.

▶ TRACK 302 Week 11, quiz A: major, sus2, sus4 or augmented

Answer with: major, sus2, sus4, augmented. 10 questions; a click starts each one.

- | | | | | |
|---------|---------|---------|---------|----------|
| 1 _____ | 2 _____ | 3 _____ | 4 _____ | 5 _____ |
| 6 _____ | 7 _____ | 8 _____ | 9 _____ | 10 _____ |

▶ TRACK 303 Week 11, quiz B: 6, 9, maj7 or 7

Answer with: major 6th, dominant 9th, major 7th, dominant 7th. 10 questions; a click starts each one.

- | | | | | |
|---------|---------|---------|---------|----------|
| 1 _____ | 2 _____ | 3 _____ | 4 _____ | 5 _____ |
| 6 _____ | 7 _____ | 8 _____ | 9 _____ | 10 _____ |

▶ TRACK 304 Week 11, quiz C: which scale

Answer with: major, minor, dorian, mixolydian. 10 questions; a click starts each one.

- | | |
|---------|----------|
| 1 _____ | 2 _____ |
| 3 _____ | 4 _____ |
| 5 _____ | 6 _____ |
| 7 _____ | 8 _____ |
| 9 _____ | 10 _____ |

Week 12: review and final exam

The first three quizzes review the course across all kinds. Take them on days 1–3 and go back to the matching week for any quiz below 9. Take the final exam on one day, the three quizzes in a row, without replaying any question. A score of 27 out of 30 or more means you have finished the course. **Sing:** before each answer in the exam, sing what you heard once, as you have done all course.

▶ TRACK 305 Week 12, review A: every interval, up or down

Answer with: minor 2nd, major 2nd, minor 3rd, major 3rd, perfect 4th, tritone, perfect 5th, minor 6th, major 6th, minor 7th, major 7th, octave. 10 questions; a click starts each one.

- | | | | | |
|---------|---------|---------|---------|----------|
| 1 _____ | 2 _____ | 3 _____ | 4 _____ | 5 _____ |
| 6 _____ | 7 _____ | 8 _____ | 9 _____ | 10 _____ |

▶ TRACK 306 Week 12, review B: five-note melodies in major

Answer with: scale degrees of the major scale; 'low' marks notes below the tonic (the tonic chord sounds first). 10 questions; a click starts each one.

1 _____	2 _____
3 _____	4 _____
5 _____	6 _____
7 _____	8 _____
9 _____	10 _____

▶ TRACK 307 Week 12, review C: which scale

Answer with: major, minor, dorian, mixolydian. 10 questions; a click starts each one.

1 _____	2 _____
3 _____	4 _____
5 _____	6 _____
7 _____	8 _____
9 _____	10 _____

▶ TRACK 308 Final exam 1: harmonic intervals

Answer with: minor 2nd, major 2nd, minor 3rd, major 3rd, perfect 4th, tritone, perfect 5th, minor 6th, major 6th, minor 7th, major 7th, octave. 10 questions; a click starts each one.

1 _____	2 _____	3 _____	4 _____	5 _____
6 _____	7 _____	8 _____	9 _____	10 _____

▶ TRACK 309 Final exam 2: chord qualities

Answer with: major, minor, diminished, augmented, sus4, major 7th, dominant 7th, minor 7th, half-diminished, major 6th. 10 questions; a click starts each one.

1 _____	2 _____	3 _____	4 _____	5 _____
6 _____	7 _____	8 _____	9 _____	10 _____

▶ TRACK 310 Final exam 3: progressions

Answer with: I IV V I; I V vi IV; I vi IV V; ii V I vi; vi IV I V; I iii IV V; I IV vi V7. 10 questions; a click starts each one.

1 _____	2 _____
3 _____	4 _____
5 _____	6 _____
7 _____	8 _____
9 _____	10 _____

After the course

Ear training does not stop at week 12; it moves into real music. The next step is transcription: working out bass lines, chords and melodies from recordings (Chapter 30). Start with songs that use the progressions from weeks 8 and 9, and write down the bass line first. Keep five minutes of the daily routine for the hardest quiz set you have, and run the whole course again after six months: the second time goes much faster, and it shows how far your ear has come.

Songs. Weeks 8 and 9 cover the chords of most pop, folk and rock songs. Name the progression of every song you hear on the radio for a week.

Solos. Weeks 7 and 10 train scale degrees. When you hear a phrase you like, sing it and name its degrees before you look for it on the fretboard.

Drill	How	Time	Goal
Quiz of the week	The week's hardest quiz track, answers on paper, marked against the key	4 min	9 of 10, twice in a row
Sing the answer	Before writing each answer, sing what you heard	Always	Every answer sung first
Radio progressions	Name the chords of a song's chorus in Roman numerals	3 min	One song a day
Degree phrases	Sing a three-note phrase from a song, name the degrees, then play it	3 min	First try correct

Write your answers, then check them in Appendix J.

1. What is the pass mark for moving on to the next week?

2. What should you do before writing each answer?

3. In the melody quizzes, which note does every melody start on, and what plays first?

4. How do you name a descending interval if you only know ascending references?

5. Which degrees tell Mixolydian and Dorian apart from the major scale?

CHAPTER 30

Transcription



Transcribing means learning music directly from a recording and writing it down. It combines ear training, theory and fretboard knowledge, and it teaches you a style more thoroughly than any book.

A transcription can be a complete written score, a chord chart, or a solo you have learned by ear and never written down. The process is the same: listen closely, find the key, work out the bass, chords and melody, and check your result against the recording until they match. This chapter gives a step-by-step method and a worked example.

You will use every listening skill from Chapter 28, the reading skills from Chapter 26 and Chapter 27, and the fretboard knowledge from Chapter 4.

Choosing material

Choose recordings in which the part you want is clearly audible, at a moderate tempo, in a style you already know a little. Success on easy material builds the skills that make hard material possible.

Level	Good first material	Avoid at first
1	Folk or pop songs with three or four chords; simple vocal melodies	Dense mixes, heavy distortion
2	Bass lines; riffs; slow blues solos with few notes	Fast solos, unusual tunings
3	Fingerstyle pieces with melody and bass; jazz standards' melodies	Several guitars playing similar parts
4	Complete solos; chord voicings in jazz and bossa nova; complete arrangements	

TIP · Know it before you transcribe it

Listen to the whole song several times away from the guitar before you start. Sing the melody and the bass line. You cannot transcribe what you cannot sing back.

Tools

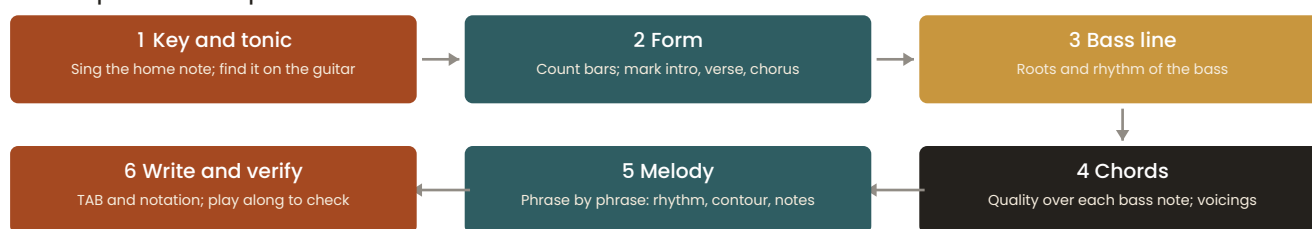
- **Slow-down software.** Plays the recording at 50–75% speed without changing the pitch. Many free and paid programs and most digital audio workstations can do this.
- **Looping.** Set a loop around one phrase, a few seconds long, and repeat it until you can sing it.
- **Equaliser.** Boost the low frequencies to hear the bass, or cut them to hear the guitar and voice.
- **Pitch shift.** If the recording is slightly sharp or flat (old recordings, or guitars tuned down a half step), shift the recording or retune the guitar until the open strings match.
- **Paper or notation software.** Write as you go. Memory is unreliable after ten minutes of looping.

CAREFUL · When notes fall between frets

If nothing on the guitar matches the recording, the track may be tuned to a different reference or the band may be tuned down. Many rock and blues recordings use half-step-down tuning (E♭ A♭ D♭ G♭ B♭ E♭). Check the lowest note of the song before you start.

The method

Transcription in six steps

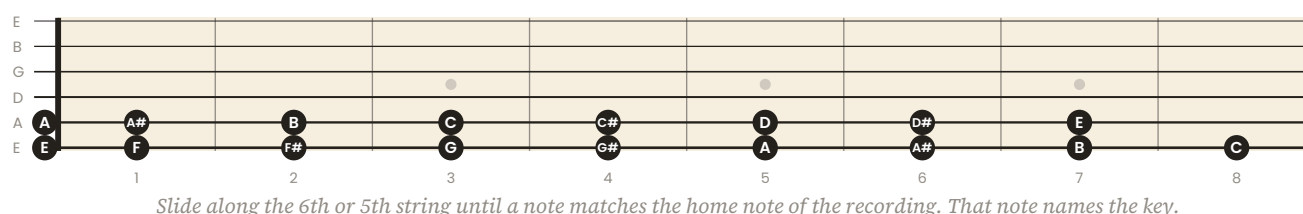


1. Find the key and the tonic

The **tonic** is the note that sounds like home. It is usually the last bass note of the song or section and often the first chord.

1. Hum the note that the song keeps returning to.
2. Find it on the 6th or 5th string by sliding up from the open string until it matches.
3. Decide major or minor: play the major and the minor triad on that root and hear which fits the song.
4. Play the scale of that key (or its pentatonic) along with the recording. Notes that clash tell you to look again.

Tonic search on the 6th and 5th strings



2. Map the form

Count the bars in each section and write a skeleton before you write any notes: "Intro 4 bars, verse 8 bars, chorus 8 bars, verse, chorus, solo 16 bars, chorus". Most popular songs use sections of 4, 8, 12 or 16 bars, so a count that comes out at 7 or 9 usually means a mistake or a deliberate extra bar worth noting.

3. Bass line first

The bass usually plays the root of each chord on the strong beats. Once you have the bass line, you have most of the progression. Listen with the low frequencies boosted, sing the bass part, then find it on the 6th to 4th strings. Write the bass notes and the bar in which each one changes.

4. Chords

For each bass note, decide the quality of the chord above it: major, minor, or a seventh chord (Chapter 28). Use the key to predict: in a major key, chords on degrees 1, 4 and 5 are usually major, on 2, 3 and 6 minor. Then find the **voicing**: open strings ringing, a barre high on the neck, a capo. The sound of the open strings and the register of the top note decide the position.

LISTEN FOR · Clues to the voicing

Open strings ring longer and brighter than fretted notes. A low note held under a moving melody suggests an open bass string. A chord whose top note moves while the rest stays still suggests a shape with one finger moving, such as the sus2 and sus4 changes around open D and A chords.

5. Melody and solos

Work one phrase at a time, usually two bars.

1. Loop the phrase and sing it until you can sing it without the recording.
2. Write the rhythm first, counting against the beat (Chapter 27).
3. Find the first note relative to the tonic, then each following note by interval or by scale degree.

4. Choose a position. The same notes exist in several places; pick the one that matches the sound (open or fretted, thick or thin string) and the fingering the player probably used.
5. Listen for the technique: slides, hammer-ons, bends and vibrato are part of the melody (Chapter 36).

TRACK 311 The same phrase in two positions

Bars 1–2 use open strings, bars 3–4 play exactly the same pitches fretted, mostly around the 2nd to 5th frets. The open version rings into the next note; the fretted version is drier. Listen to the recording to decide which the player used.

6. Write it down and verify

Write the transcription in the form you will use: a chord chart for accompaniment, TAB with rhythm for riffs and solos, notation and TAB for fingerstyle pieces. Always include:

- key, time signature and tempo;
- the form, with bar numbers or rehearsal letters;
- rhythm, not only fret numbers (TAB without rhythm cannot be played without the recording);
- techniques: slides, bends, hammer-ons, pull-offs and muting.

TRACK 312 Writing techniques into a transcription

An original A minor pentatonic phrase with a hammer-on, a pull-off, a slide and a full bend. Each technique is marked in the notation and in the TAB: write them in as soon as you hear them.

Then verify. Play along with the recording at full speed. Record yourself and listen back alongside the original. Mark anything that does not match, fix the first error, and play through again.

TRY THIS · Transcribe the bass line only

Choose a song you know well with three or four chords. Transcribe only the bass line for the first verse, writing one note per chord change with its bar number. Then add the chord quality above each bass note. This alone gives you a playable chord chart.

Worked example

The track below is an original eight-bar melody with a bass line, standing in for a recording. Listen to it several times before reading the steps, then follow the method and compare your results with each step.

TRACK 313 The recording

System 1:

T	3	2	0	3	0	3	3	2	0	3	2	0
A	2											
B	0		0		3			0		0		0

System 2:

T	2	3	0	0	3	0	2	3	0	2	0	2	3
A			4			0							0
B	2				3				0		0		

Shown here in TAB only. Try to transcribe it from the sound before looking at the steps.

Step 1: key and tonic. The melody and the bass both end on D, and bar 1 starts on a D bass note with an A and an F# above it. The home note is D. The D major triad fits; D minor clashes with the F# in the melody. The key is D major.

Step 2: form. Eight bars in two four-bar phrases. Bar 4 ends on E, the 2nd degree, which sounds unfinished; bar 8 ends on D, which sounds final. This question-and-answer shape is common in folk melodies.

Step 3: bass line. With the melody ignored, the bass plays two notes per bar: the root on beat 1 and the 5th on beat 3.

TRACK 314 Step 3: the bass line

System 1:

T							
A							
B	0		0	3	0	0	0

System 2:

T							
A							
B	2	4	3	0	0	0	0

Roots on beat 1: D, G, D, A, B, G, A, D.

Step 4: chords. Over each root, test major and minor. In D major, D, G and A are major and B is minor. The bass in bar 5 moves from B to F#, the 5th of B minor, and the melody outlines B, D and F#: the chord is Bm.

Step 4: the chord chart

D <i>I</i>	G <i>IV</i>	D <i>I</i>	A <i>V</i>
Bm <i>vi</i>	G <i>IV</i>	A <i>V</i>	D <i>I</i>

Step 5: melody. Sing the first phrase and find its first note: A, the 5th of D. Write the rhythm before the pitches: bar 1 is three quarter notes and two eighths, bar 2 a dotted quarter, an eighth and a half note. Then fill in the notes by step and leap.

Step 5: rhythm of bars 1 and 2

Bar 1

Bar 2

1 e & a 2 e & a 3 e & a 4 e & a 5 e & a 6 e & a 7 e & a 8 e & a

Write the rhythm on a grid of sixteenths before you name a single note.

D major in first position: where the melody lies

Every melody note of the example lies in this window, on strings 1 to 3.

 TRACK 315 Step 5: the melody alone

[illegible]

Most notes are chord tones on the beat. The C# in bar 7 is the 3rd of A, leading to D in the last bar.

Step 6: the complete transcription. Combine melody, bass and chord symbols. Choose the position: everything lies in first position, with the open D and A strings as bass notes.

▶ **TRACK 316** Step 6: the finished transcription

5

Melody with stems up, bass with stems down. Verify by playing along with the recording track at full speed.

NYLON · STEEL

Nylon: classical guitarists transcribe mainly to arrange: a melody and bass from a song or piano piece become a solo piece. Write the result in notation and TAB with fingerings, and check that the melody stays above the bass on separate strings.

Steel and electric: most transcription is of riffs, chord parts and solos. TAB with rhythm and technique markings is usually enough; add a chord chart for the form.

Chapter 30 review · use it, train it, test it

Using it in real music

Learning repertoire. Many songs have no reliable published version. A transcription you have checked against the recording is more accurate than most TAB found online.

Learning a style. Transcribing two or three solos by a player teaches their vocabulary, phrasing and tone in a way that reading a lesson does not. Play the transcribed phrases in new keys and over other songs.

Arranging. A transcribed melody, bass line and chord chart is the starting point for a solo guitar arrangement (Chapter 20).

How to train it

Drill	How	Time	Goal
Find the tonic	Play five random songs; find the home note on the 6th or 5th string within 30 seconds	3 min	Correct key for each song
Bass line	Transcribe the bass line of one verse	10 min	Every chord change on the right bar
Chord chart	Add chord qualities above a transcribed bass line and play along	10 min	Playable chart matching the recording

Drill	How	Time	Goal
Two-bar phrase	Transcribe two bars of melody or solo; sing first, then play	10 min	Pitches and rhythm match at full speed
Verify	Record yourself playing a transcription with the original	5 min	No audible differences

Check your understanding

Write your answers, then check them in Appendix J.

- ### 1. Why is the bass line usually transcribed before the chords?

- ## 2. How do you find the tonic of a song?

3. What does slow-down software do that simply playing a tape slower does not?

4. Every note in a recording seems to fall between two frets. What is the likely cause?

- ### 5. Why should you write the rhythm of a phrase before its pitches?

[illegible]

6. The same melody can be played in several positions. How do you choose?

7. In the worked example, which chord is in bar 5, and how do the bass and melody show it?

8. Name three things a finished transcription must include besides the notes.

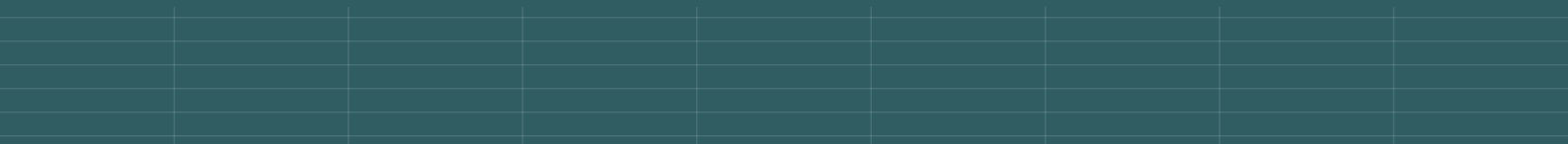
MY NOTES

PART EIGHT

Deeper Theory

Minor-key harmony, secondary dominants, borrowed chords, modulation, voice leading and improvising over changes.

- 31 Minor-Key Harmony
- 32 Secondary Dominants and Borrowed Chords
- 33 Modulation
- 34 Voice Leading and Reharmonisation
- 35 Improvising over Changes



CHAPTER 31

Minor-Key Harmony



A minor key is not one scale but three. The natural, harmonic and melodic forms share a tonic and a key signature, and minor-key music moves freely between them to get a leading note, a major V chord and smooth melodic lines.

Chapter 6 introduced the three forms of the minor scale, and Chapter 13 built triads on the natural minor scale and noted that the v chord is usually made major. This chapter follows that idea through: it harmonises the harmonic and melodic minor scales, shows how real minor-key progressions combine all three forms, and names the modes of the two scales, which supply most of the dominant colours used in jazz, flamenco and metal.

All examples are in A minor so the three scales can be compared note for note. Transpose them with the shapes at the end of the chapter.

What the raised 7th does to the chords

Chapter 6 shows why the natural minor's whole step between the 7th and the tonic (G to A in A minor) weakens the cadence, and how harmonic and melodic minor raise notes to fix it. In harmony, raising the 7th from G to G $\sharp$ has three consequences:

- The v chord becomes **E major** (E G $\sharp$ B) and the V7 becomes **E7** (E G $\sharp$ B D). Its 3rd, G $\sharp$, resolves up a half step to A.
- The vii chord becomes **G $\sharp$ $^\circ$** and the seventh chord **G $\sharp$ $^\circ$ 7** (G $\sharp$ B D F), a second dominant-function chord with two tritones.
- The raised 7th creates a gap of three half steps between F and G $\sharp$, an **augmented 2nd**. This interval gives harmonic minor its Middle Eastern and Eastern European sound, and it is awkward in a smooth melody.

Melodic minor also raises the 6th (F to F $\sharp$), which removes the augmented 2nd. In classical practice the 6th and 7th are raised going up and restored going down. Jazz and most modern theory use the raised form in both directions, and this chapter does the same when it says "melodic minor": its chords and modes below all use F $\sharp$ and G $\sharp$.

TRACK 317 The three forms of A minor

3

A reminder of the three forms, one bar each, as they are harmonised below. Listen for the leap from F to G $\sharp$ in bar 2 and how F $\sharp$ smooths it in bar 3.

Chords of harmonic minor

Stacking thirds from each degree of A harmonic minor gives the chords below. Compared with natural minor, every chord that contains G changes: III becomes augmented, v becomes major V, and the $\flat$ VII chord (G) is replaced by G $\sharp$ $^\circ$.

Degree	1	2	3	4	5	6	7
Triad	i	ii $^\circ$	$\flat$ III+	iv	V	$\flat$ VI	vii $^\circ$
In A	Am	B $^\circ$	C+	Dm	E	F	G $\sharp$ $^\circ$
Seventh	i(maj7)	ii $\emptyset$ 7	$\flat$ III(maj7 $\sharp$ 5)	iv7	V7	$\flat$ VI(maj7)	vii $^\circ$ 7
In A	AmMaj7	Bm7 $\flat$ 5	Cmaj7 $\sharp$ 5	Dm7	E7	Fmaj7	G $\sharp$ $^\circ$ 7

Diatonic seventh chords of A harmonic minor

R 5 7 $\flat$ 3 5 R $\flat$ 5 $\flat$ 7 $\flat$ 3 R 3 $\sharp$ 5 3 R 5 $\flat$ 7 $\flat$ 3 R 5 $\flat$ 7 3 5 R R 3 5 7 R $\flat$ 7 $\flat$ 3 $\flat$ 5

V7 and vii $^\circ$ 7 are the chords in everyday use. The i(maj7) and $\flat$ III(maj7 $\sharp$ 5) chords appear mainly in jazz and film music.

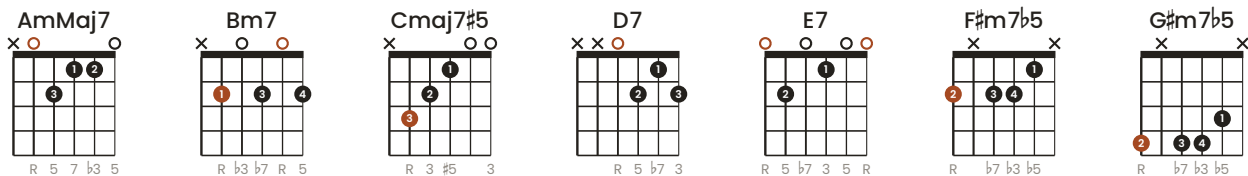
The two chords that matter most are E7 and G $\sharp$ $^\circ$ 7. Both contain G $\sharp$ and D, the tritone that resolves inward to A and C, the root and 3rd of Am. G $\sharp$ $^\circ$ 7 is a symmetrical chord: moved up three frets it is the same four notes (B $^\circ$ 7, D $^\circ$ 7, F $^\circ$ 7), so one shape covers four resolutions. Chapter 32 uses that property.

Chords of melodic minor

Raising the 6th as well changes every chord that contains F. Melodic minor has two major triads a whole step apart (IV and V), two diminished triads, and two dominant seventh chords.

Degree	1	2	3	4	5	6	7
Triad	i	ii	$\flat$ III+	IV	V	vi $^\circ$	vii $^\circ$
In A	Am	Bm	C+	D	E	F $\sharp$ $^\circ$	G $\sharp$ $^\circ$
Seventh	i(maj7)	ii7	$\flat$ III(maj7 $\sharp$ 5)	IV7	V7	vi $\emptyset$ 7	vii $\emptyset$ 7
In A	AmMaj7	Bm7	Cmaj7 $\sharp$ 5	D7	E7	F $\sharp$ m7 $\flat$ 5	G $\sharp$ m7 $\flat$ 5

Diatonic seventh chords of A melodic minor



The IV7 chord (D7 in A minor) is the Dorian sound of a minor key: the bright major 6th, F#, under a minor tonic. Am – D7 or Am7 – D9 (minor tonic, Dorian IV) is common in soul, funk and jazz. The tonic itself is often played as **Am6** (A C E F#) or **AmMaj7**, both from melodic minor.

The working minor key

Real minor-key music does not choose one form. It takes each chord from whichever form gives the best sound in context. The table lists the usual options for each degree. The first column comes from natural minor, the others from harmonic and melodic minor.

Step	Natural minor	Harmonic / melodic minor	Typical use
1	Am, Am7	AmMaj7, Am6	Tonic; m6 and mMaj7 for jazz and bossa
2	B°, Bm7b5	Bm, Bm7 (melodic)	iiø7 in the minor ii – V – i
b3	C, Cmaj7	C+, Cmaj7#5	Relative major; augmented as a passing chord
4	Dm, Dm7	D, D7 (melodic)	iv is standard; IV gives a Dorian colour
5	Em, Em7	E, E7, E7b9	V7 for cadences; v for modal, folk and rock
b6	F, Fmaj7	F#, F#m7b5 (melodic)	bVI is standard
7	G, G7	G#, G#°7	bVII in pop and rock; vii°7 in classical and jazz

The practical rule: **use natural minor for everything except the dominant**, and raise the 7th whenever the music leads back to the tonic. Pop and rock often keep the minor v or skip it entirely (i – bVI – bVII), which is why many minor-key rock songs sound modal rather than classical.

Typical minor-key progressions

Am <i>i</i>	Dm <i>iv</i>	E7 <i>V7</i>	Am <i>i</i>
Am <i>i</i>	F <i>bVI</i>	Bm7b5 <i>iiø7</i> E7 <i>V7</i>	Am <i>i</i>
Am <i>i</i> G <i>bVII</i>	F <i>bVI</i>	E <i>V</i>	E <i>V</i>

Row 1: i – iv – V7 – i, the basic cadence. Row 2: with bVI and the minor ii – V. Row 3: the Andalusian cadence, i – bVII – bVI – V.

▶ TRACK 318 Natural minor v against harmonic minor V7

Am Dm Em Am Am Dm E7 Am

T 1 3 1 1 3 0 1
A 2 2 0 2 2 1 2
B 2 0 2 2 0 0 2
0 0 0 0 0 0 0

The same cadence twice. In bar 2 the minor v (G on the 3rd string) leads nowhere; in bar 4 the G# of E7 rises to A.

The Andalusian cadence

The **Andalusian cadence**, Am – G – F – E (i – $\flat$ VII – $\flat$ VI – V), is explained in Chapter 14. In minor-key terms it takes three chords from natural minor and only the last, the major V, from harmonic minor. Over that E chord the scale is E Phrygian dominant, the sound Chapter 53 builds on.

TRACK 319 Andalusian cadence with a Phrygian dominant run

The musical notation shows the Andalusian cadence in 4/4 time. The first system contains measures 1, 2, and 3, with chords Am, G, and F respectively. The second system contains measures 4, 5, and 6, with chords E, E7, and Am respectively. A guitar tablature is provided below the staff notation, showing fingerings for the guitar.

Bar 5 runs down E Phrygian dominant (E F G $\sharp$ A B C D) and resolves G $\sharp$ to A.

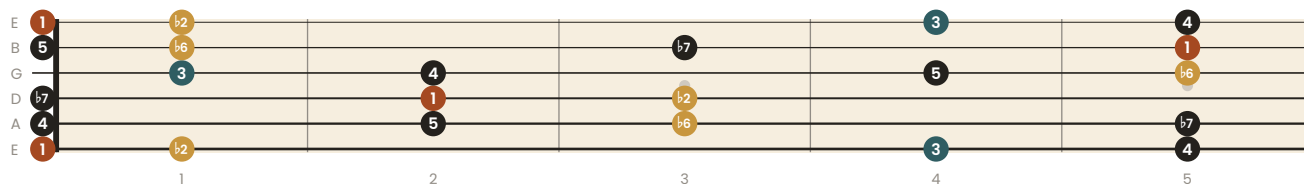
The modes of harmonic minor

Every seven-note scale has seven modes, one starting on each degree. Chapter 9 did this for the major scale. Harmonic minor and melodic minor have their own sets. The names below are the usual ones; several have alternatives.

Mode	Formula	From A harmonic minor	Seventh chord	Main use
1 Harmonic minor	1 2 $\flat$ 3 4 5 $\flat$ 6 7	A to A	mMaj7	Minor tonic, classical and metal lines
2 Locrian $\sharp$ 6	1 $\flat$ 2 $\flat$ 3 4 $\flat$ 5 6 $\flat$ 7	B to B	m7 $\flat$ 5	Over ii $\flat$ 7 in a minor ii – V
3 Ionian $\sharp$ 5	1 2 3 4 $\sharp$ 5 6 7	C to C	maj7 $\sharp$ 5	Rare
4 Dorian $\sharp$ 4	1 2 $\flat$ 3 $\sharp$ 4 5 6 $\flat$ 7	D to D	m7	Klezmer and Eastern European music ("Ukrainian Dorian")
5 Phrygian dominant	1 $\flat$ 2 3 4 5 $\flat$ 6 $\flat$ 7	E to E	7 (7 $\flat$ 9 $\flat$ 13)	V7 in minor keys; flamenco, klezmer, Middle Eastern music
6 Lydian $\sharp$ 2	1 $\sharp$ 2 3 $\sharp$ 4 5 6 7	F to F	maj7	Rare
7 Superlocrian $\sharp$ 7	1 $\flat$ 2 $\flat$ 3 $\flat$ 4 $\flat$ 5 $\flat$ 6 $\sharp$ 7	G $\sharp$ to G $\sharp$	$\circ$ 7	Over vii $\circ$ 7

Only **Phrygian dominant** is in everyday use. It is Phrygian with a major 3rd, or, more usefully, the A harmonic minor scale heard over its own V chord. Its $\flat$ 2 and $\flat$ 6 (F and C over E) are the $\flat$ 9 and $\flat$ 13 of E7: F falls to E, the 5th of Am, and C is held into the next chord as the 3rd of Am. Play it over every V7 that resolves to a minor chord.

E Phrygian dominant in open position



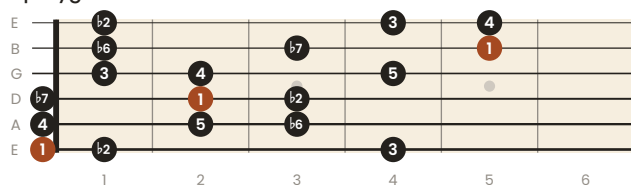
G# (teal) is the major 3rd; F and C (gold) are the b9 and b13 of E7. Compare with E Phrygian, which has G.

▶ **TRACK 320** Phrygian dominant vamp on E: E – F · LOOP · rock · 116 bpm

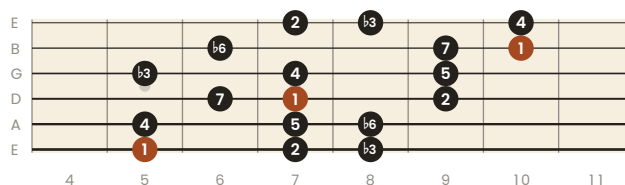


IMPROVISE WITH

E phrygian dominant



A harmonic minor



This is the V chord of A minor held as a home, the sound of flamenco and surf guitar. Over E aim for G# and let it rise to A; over F land on F and let it fall a half step to E when the E chord returns.

UNDER THE HOOD · Hungarian minor and the double harmonic scale

Raising the 4th of harmonic minor gives the **Hungarian minor** scale (1 2 b3 #4 5 b6 7), with two augmented 2nds. Its fifth mode, 1 b2 3 4 5 b6 7, is the **double harmonic** or **Byzantine scale**. Both are colours rather than key systems; they are used for melody over a static tonic.

The modes of melodic minor

Melodic minor's modes are central to jazz. Four of them sit over dominant or half-diminished chords and supply altered tensions that the major scale cannot.

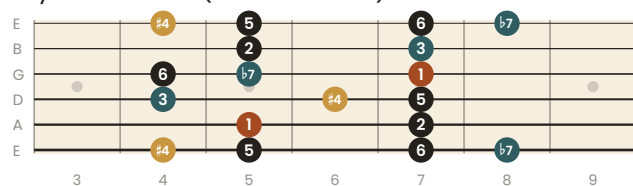
Mode	Formula	From A melodic minor	Seventh chord	Main use
1 Melodic minor	1 2 b3 4 5 6 7	A to A	mMaj7	Minor tonic: m6, mMaj7, m6/9
2 Dorian b2	1 b2 b3 4 5 6 b7	B to B	m7	Over sus4(b9) chords
3 Lydian augmented	1 2 3 #4 #5 6 7	C to C	maj7#5	Over maj7#5
4 Lydian dominant	1 2 3 #4 5 6 b7	D to D	7 (7#11)	Dominants that do not resolve down a 5th: IV7, bVII7, II7, tritone substitutes
5 Mixolydian b6	1 2 3 4 5 b6 b7	E to E	7 (7b13)	V7 to a minor chord, softer than Phrygian dominant
6 Locrian #2	1 2 b3 4 b5 b6 b7	F# to F#	m7b5	Over iiø7: Locrian with a natural 9
7 Altered	1 b2 #2 3 b5 #5 b7	G# to G#	7alt	V7 with b9, #9, b5 and #5

Find the parent scale with three short rules:

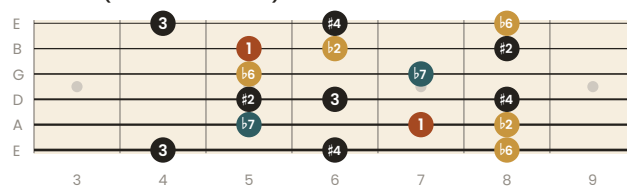
- **Lydian dominant** on X: melodic minor a perfect 5th above. D7#11 = A melodic minor.
- **Locrian #2** on X: melodic minor a minor 3rd above. Bm7b5 = D melodic minor.
- **Altered** on X: melodic minor a half step above. E7alt = F melodic minor (E F G G# Bb C D); G7alt = Ab melodic minor.

Two dominant modes of melodic minor

D Lydian dominant (A melodic minor)

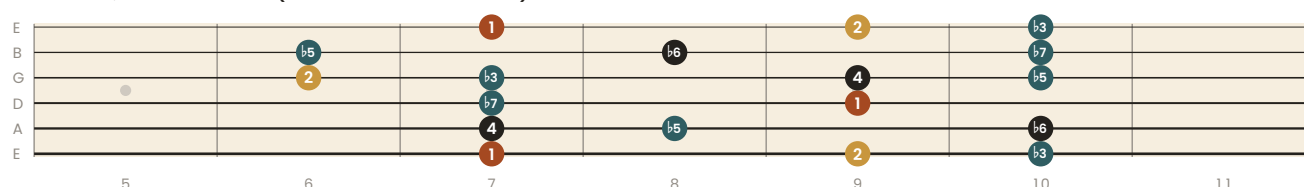


E altered (F melodic minor)



The #4 and the four alterations are gold, the 3rd and b7 teal. The engine labels the altered scale 1 b2 b3 b4 b5 b6 b7: read b3 as #9, b4 as the major 3rd and b6 as #5.

B Locrian #2 over Bm7b5 (from D melodic minor)

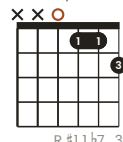


The natural 9th (C#, gold) is what separates Locrian #2 from Locrian.

The altered scale contains every alteration of a dominant chord and no natural 5th or 9th. Its notes spell a dominant chord on the 1, 3 (written b4) and b7, plus b9, #9, b5 (#11) and #5 (b13). The chord symbol 7alt asks for this sound.

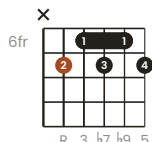
Dominant chords that call for melodic or harmonic minor modes

D7#11



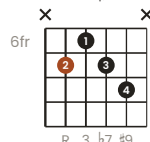
R #11 b7 3

E7b9



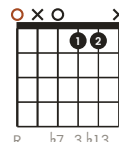
R 3 b7 b9 5

E7#9



R 3 b7 #9

E7b13



R b7 3 b13

D7#11 takes Lydian dominant. E7b9 takes Phrygian dominant or the half-whole diminished scale. E7#9 and E7b13 take the altered scale.

▶ TRACK 321 A melodic minor over AmMaj7
AmMaj7

Fifth position, one octave and a half up, then the AmMaj7 arpeggio down to A.

▶ TRACK 322 E altered resolving to Am
Am E7alt Am

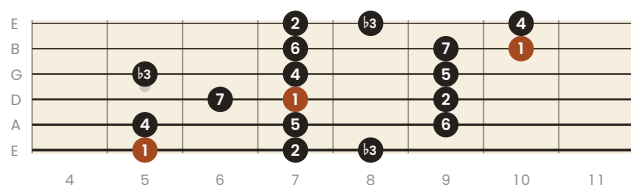
Bar 2 descends E altered: G# (3), G (#9), F (b9), E, D (b7), C (#5), Bb (b5, printed as A#), G#. The line ends by resolving G# to A.

TRACK 323 Melodic minor and altered loop: Am6 – E7alt · LOOP · bossa nova · 112 bpm

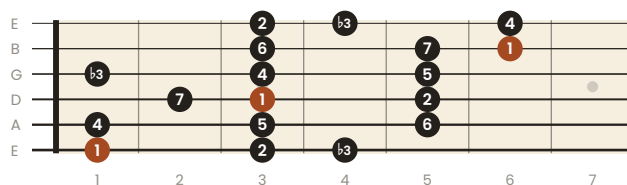
Am6 <i>i6</i>	Am6 <i>i6</i>	E7alt <i>V7alt</i>	E7alt <i>V7alt</i>
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IMPROVISE WITH

A melodic minor



F melodic minor



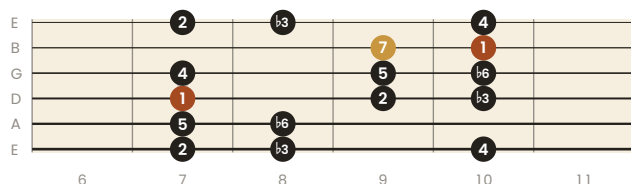
A melodic minor over Am6, F melodic minor (the E altered scale) over E7alt. Over E7alt aim for F (b9) and C (#5), then resolve F down to E and hold C into Am6, where it is the b3.

Fretboard shapes

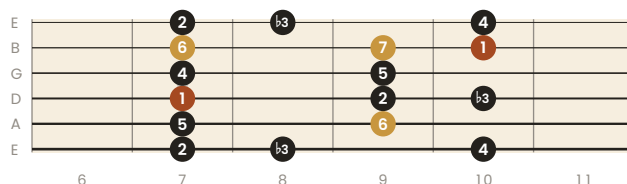
Chapter 6 shows all three forms in 5th position as edits of the natural minor shape: find every b7 and raise it one fret for harmonic minor; raise every b6 as well for melodic minor. The same edit works in any position. Here it is applied to the shape in 7th position, around the root on the 4th string.

Harmonic and melodic minor in 7th position

A harmonic minor



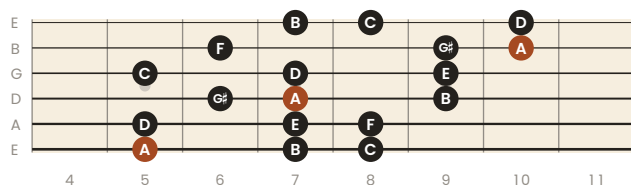
A melodic minor



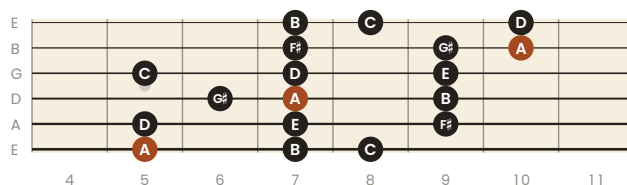
The changed notes are gold. Everything else is the natural minor shape in the same position.

Three notes per string from the root

A harmonic minor



A melodic minor



Because modes share notes with their parent, these shapes also give the modes. The A harmonic minor shape played over E7 is E Phrygian dominant; the A melodic minor shape played over D7 is D Lydian dominant.

TIP · Two altered sounds from one shape

Learn melodic minor in two positions. Then any altered dominant is one fret below a shape you know: for G7alt play A^b melodic minor, for E7alt play F melodic minor.

Where each sound is used

Sound	Where you hear it	Examples
Harmonic minor melody	Baroque and classical minor-key music, neoclassical metal	Bach's minor-key preludes; Yngwie Malmsteen
Phrygian dominant	Flamenco, surf, klezmer, Middle Eastern pop, metal	"Misirlou" (Dick Dale), "Hava Nagila" (traditional)
V7 to i (harmonic minor cadence)	Almost all minor-key jazz standards and ballads	"Summertime", "Autumn Leaves", "Sunny"
Melodic minor tonic (m6, mMaj7)	Jazz, bossa nova, film noir and spy themes	"Blue Bossa"; the final m(maj9) chord of the James Bond theme
Lydian dominant	II7 and IV7 chords, tritone substitutes, Brazilian music	The II7 in "Take the A Train"
Altered scale	Dominant chords in bebop and modern jazz	ii – V – i solos in any standard
Andalusian cadence	Flamenco, pop, rock, surf	"Hit the Road Jack", "Smooth", "Runaway"

Chapter 31 review · use it, train it, test it

Using it in real music

Reading a minor-key chart. When a chart in a minor key shows a major or dominant chord on the 5th degree (E or E7 in A minor), the melody and any solo over it use G#, not G. Everywhere else, default to natural minor unless the chart shows D or D7 (melodic minor colour) or an m6 or mMaj7 tonic.

Soloing over V7 in minor. Play Phrygian dominant first: it is the harmonic minor shape you already use for the key. When you want more tension, switch to the altered scale a half step above the root. Resolve G# to A or the b9 (F) to E on the tonic chord.

Writing. To make a natural minor progression sound classical, change v to V7 before the tonic. To make it sound modal and rock-like, do the reverse.

How to train it

Drill	How	Time	Goal
Three minors	Play A natural, harmonic and melodic minor in 5th position back to back, then in D and E	5 min	Switch forms without stopping at 80 bpm in eighths
Harmonise	Say and play the seventh chords of D harmonic minor and D melodic minor	5 min	All 14 chords named and played without a reference
Cadence loop	Loop i – iv – V7 – i; over V7 play only Phrygian dominant, resolving to the 3rd or root of i	5 min	Every resolution lands on beat 1
Altered locator	Call out a random dominant (B7, F7, Db7...) and play the melodic minor a half step up	3 min	Under 3 seconds per chord
Mode naming	Pick a melodic minor shape; play it from each degree and name the mode and chord	5 min	All seven modes named correctly

Check your understanding

Write your answers, then check them in Appendix J.

1. Why is the 7th degree of the minor scale raised?

2. Spell the triads of A harmonic minor.

3. Which two chords of A harmonic minor contain the tritone G#–D, and where does it resolve?

4. What is the fifth mode of harmonic minor and over which chord is it used?

5. Which melodic minor scale gives the altered scale on B $\flat$?

6. Name the melodic minor mode for D7#11 and its parent scale.

7. What are the seventh chords on degrees 4 and 5 of C melodic minor?

8. In the Andalusian cadence in A minor, which chord uses the raised 7th?

MY NOTES

CHAPTER 32

Secondary Dominants and Borrowed Chords



Most chromatic chords in tonal music come from two sources. Secondary dominants point at a chord in the key as if it were a temporary tonic; borrowed chords take their colour from the parallel minor or major.

Chapter 13 showed the seven chords that belong to a key, and Chapter 14 showed how they move. Most songs and almost all jazz standards also contain chords that are not in the key. Nearly all of them can be explained in one of two ways:

- **Secondary dominants and secondary leading-tone chords** make a diatonic chord sound briefly like a tonic. This is called **tonicisation**.
- **Borrowed chords** (modal interchange) take a chord from the parallel key, usually the parallel minor, without changing the tonic.

Both keep the key intact. When a passage settles in a new key for longer, it is a modulation, the subject of Chapter 33.

Tonicisation and V7/x

Every major and minor triad in a key can act as a temporary tonic. To point at it, play its dominant seventh: the chord a perfect 5th above it. The 3rd of that dominant is a half step below the target root, so it adds one leading note that is not in the key.

The chord is written V7/x ("five-seven of x"). In C major, A7 is V7/ii because it resolves to Dm; its 3rd, C#, is the leading note of D. The diminished vii° chord cannot be tonicised because it is not a stable triad, and V7/I is just V7.

Secondary dominant	Target	In C	Chromatic notes	In G	Chromatic notes
V7/ii	ii	A7 → Dm	C#	E7 → Am	G#
V7/iii	iii	B7 → Em	D#, F#	F#7 → Bm	A#, C#
V7/IV	IV	C7 → F	Bb	G7 → C	F
V7/V	V	D7 → G	F#	A7 → D	C#
V7/vi	vi	E7 → Am	G#	B7 → Em	D#

V7/IV is simply I7: a dominant seventh on the tonic. Its new note is the $\flat 7$ of the key, not a leading note, and it is the reason the I chord in a blues is a dominant seventh. V7/V is often called the **II7** or "supertonic dominant". V7/vi is the most common secondary dominant in pop music.

The secondary dominants of C major

V7/ii V7/iii V7/IV V7/V V7/vi

A7 B7 C7 D7 E7

A E G C# E B D# A B# C E Bb C D A C F# E B D G#

Each new note (C#, D# and F#, Bb, F#, G#) is the 3rd, 5th or 7th of the chord.

Secondary dominants in a progression

C <i>I</i>	A7 <i>V7/ii</i>	Dm <i>ii</i>	G7 <i>V7</i>	C <i>I</i>
E7 <i>V7/vi</i>	Am <i>vi</i>	D7 <i>V7/V</i>	G7 <i>V7</i>	C <i>I</i>

Each secondary dominant resolves down a perfect 5th to the chord it is named after.

▶ **TRACK 324** Secondary dominant: one chromatic inner voice

C A7 Dm G7 C

	1	2	1	1	0
T	0	0	3	1	1
A	2	2	2	0	0
B	3	0	0	2	2
			3	3	3

The note on the 2nd string climbs C – C# – D. The C# is what makes A7 point at Dm.

TIP · Major triads count too

A secondary dominant does not need its 7th. A major triad on II, III or VI (D, E or A in C major) does the same job when it moves to the chord a 5th below: the 3rd alone is the leading note. The C – E – Am – F idea in many pop ballads is I – V/vi – vi – IV.

Chains of dominants

A secondary dominant can resolve to another secondary dominant. Moving backwards round the circle of fifths from the tonic gives **extended dominants**: each chord is the V7 of the next. In C: E7 – A7 – D7 – G7 – C. Ragtime, early jazz and country use this chain constantly ("Salty Dog Blues"; "Sweet Georgia Brown" begins on VI7 and spends several bars on each dominant). The bridge of the "rhythm changes" form is III7 – VI7 – II7 – V7, two bars each.

▶ **TRACK 325** Extended dominants: E7 – A7 – D7 – G7 – C

E7 A7 D7 G7 C

	0	0	2	2	2	2	1	1	0
T	1	1	0	0	1	1	1	1	1
A	0	0	2	2	2	2	0	0	0
B	2	2	0	0	0	0	2	2	2
	0	0					3	3	3

The 7th of each chord falls a half step to the 3rd of the next (D to C#, G to F#, C to B, F to E).

Secondary ii – V

Any secondary dominant can be preceded by its own ii chord, giving a ii – V that points at a diatonic chord. Use m7 before a dominant whose target is major, and m7b5 before one whose target is minor:

Target	ii – V – target in C
ii (Dm7)	Em7b5 – A7 – Dm7
iii (Em7)	F#m7b5 – B7 – Em7
IV (Fmaj7)	Gm7 – C7 – Fmaj7
V (G7)	Am7 – D7 – G7
vi (Am7)	Bm7b5 – E7 – Am7

Jazz standards are built largely from these. "Autumn Leaves" alternates a ii – V – I in the relative major with one in the relative minor. Chapter 35 shows what to play over them.

Secondary leading-tone chords

The diminished seventh chord built on the leading note of a target is the **secondary leading-tone chord**, vii°7/x. It is the V7/x with its root removed and a b9 added: A7b9 (A C# E G Bb) minus A gives C#°7 (C# E G Bb). It sits a half step below the target root.

vii°7/x	Target	In C	Spelling
vii°7/ii	Dm	C#°7	C# E G Bb
vii°7/iii	Em	D#°7	D# F# A C
vii°7/IV	F	E°7	E G Bb Db
vii°7/V	G	F#°7	F# A C Eb
vii°7/vi	Am	G#°7	G# B D F

Played between two diatonic chords a whole step apart, these are called **passing diminished chords**: C – C#°7 – Dm7, or Dm7 – D#°7 – Em7. The bass rises by half steps. Gospel, swing and Brazilian choro use them constantly, and the I – #I°7 – ii7 – V7 turnaround is one of the most common openings in jazz.

Passing diminished chords in C

Diagram 1: C (3fr) - Root on 5th string, notes C, G, C, E.

Diagram 2: C#dim7 (3fr) - Root on 5th string, notes C#, G, Bb, E.

Diagram 3: Dm7 (5fr) - Root on 5th string, notes D, A, C, F.

Diagram 4: D#dim7 (5fr) - Root on 5th string, notes D#, A, C, F#.

Diagram 5: Em7 (7fr) - Root on 5th string, notes E, B, D, G.

The root on the 5th string climbs by half steps. The same diminished shape serves both passing chords.

▶ TRACK 326 Passing diminished chord in C

C C#dim7 Dm7 G7 C

T 5 5 6 3 5

A 5 3 5 4 5

B 5 5 7 3 5

3 4 5 3 3

UNDER THE HOOD · Why one shape covers four chords

A diminished seventh chord divides the octave into four minor 3rds, so each of its notes can be the root. C#°7, E°7, G°7 and Bb°7 are the same four notes. In C major, C#°7 can therefore lead to Dm (vii°7/ii), F (vii°7/IV, as E°7),

UNDER THE HOOD · Why one shape covers four chords (cont.)

A $\flat$ (as G $^\circ$ 7) or B (as B $^\circ$ 7): one chord, four resolutions. Chapter 33 uses this to change key.

Borrowed chords

A **borrowed chord** comes from the parallel key: the key with the same tonic but the other mode. A song in C major can use chords from C minor while C remains the tonic. The process is also called **modal interchange** or **modal mixture**.

Borrowed chord	From C minor	Chromatic note(s)	Typical progression
iv	Fm	A $\flat$	I – IV – iv – I (C F Fm C)
$\flat$ VI	A $\flat$	A $\flat$, E $\flat$	I – $\flat$ VI – $\flat$ VII – I (C A $\flat$ B $\flat$ C)
$\flat$ VII	B $\flat$	B $\flat$	I – $\flat$ VII – IV – I (C B $\flat$ F C)
$\flat$ III	E $\flat$	E $\flat$, B $\flat$	I – $\flat$ III – IV (C E $\flat$ F)
ii $\circ$ 7	Dm7b5	A $\flat$	ii $\circ$ 7 – V7 – I (Dm7b5 G7 C)
v	Gm	B $\flat$	I – v – IV (C Gm F), Mixolydian colour
i	Cm	E $\flat$	Major and minor tonic in alternation

The $\flat$ 3, $\flat$ 6 and $\flat$ 7 of the parallel minor are the source of every one of these chords. Three of them dominate pop and rock:

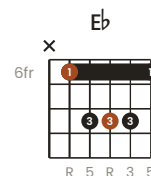
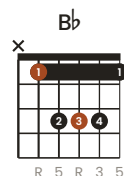
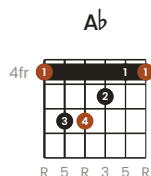
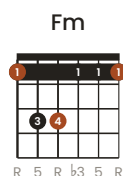
- iv in a major key is the minor subdominant. Its $\flat$ 6 (A $\flat$ in C) falls a half step to the 5th of I. Heard in "In My Life" (The Beatles) and in the I – III – IV – iv of "Creep" (Radiohead).
- $\flat$ VII gives a Mixolydian or rock sound. The coda of "Hey Jude" repeats I – $\flat$ VII – IV – I.
- $\flat$ VI – $\flat$ VII – I is a bright, heroic cadence: two major chords rising by whole steps to the tonic. It is the standard ending in film and game music.

Borrowed chords in C major

C <i>I</i>	F <i>IV</i>	Fm <i>iv</i>	C <i>I</i>
C <i>I</i>	A $\flat$ <i>$\flat$VI</i>	B $\flat$ <i>$\flat$VII</i>	C <i>I</i>
C <i>I</i>	E $\flat$ <i>$\flat$III</i>	F <i>IV</i>	C <i>I</i>

The numerals are computed against C major, so a flat sign marks every chord root taken from C minor.

The four common borrowed chords in C



Left to right, iv, $\flat$ VI, $\flat$ VII and $\flat$ III of C major.

▶ **TRACK 327** The minor iv chord in F

F B $\flat$ B $\flat$ m F

	1		3		2		1
T	1		3		2		1
A	2		3		3		2
B	3		3		3		3

The 2nd string falls D – D $\flat$ – C. The D $\flat$ is the b6 of F, borrowed from F minor.

▶ **TRACK 328** Rising cadence to the tonic in F

D $\flat$ E $\flat$ F

	6	6	8	8	10
T	6	6	8	8	10
A	6	6	8	8	10
B	4	4	6	6	8

bVI – bVII – I. One barre shape moves up two frets, then two more.

Borrowing from other modes is less common but follows the same logic. bII (D $\flat$ in C) comes from Phrygian; in first inversion it is the **Neapolitan sixth** of classical harmony, used before V in minor keys. In minor keys the borrowing often runs the other way: a major IV (D in A minor) comes from Dorian, and the major tonic at the end is the Picardy third.

CAREFUL · Borrowed or secondary

A major chord on bVII (B $\flat$ in C) is borrowed; a dominant seventh on bVII (B $\flat$ 7) is usually the **backdoor dominant**, iv7 – bVII7 – I (Fm7 – B $\flat$ 7 – C), which resolves to the tonic up a whole step rather than down a 5th. A major chord on III or VI (E or A in C) is not borrowed: it is a secondary dominant whether or not it resolves.

The Picardy third

A piece in a minor key that ends on a **major** tonic chord uses a **Picardy third** (*tierce de Picardie*). The raised 3rd turns the expected Am into A. It was the normal way to end a minor-key piece in the Renaissance and Baroque periods, and Bach's minor-key chorales and many of his preludes end this way. In modern songs it gives a lift at the very end.

▶ **TRACK 329** Picardy third in A minor

Am Dm E7 A

	1	3	0	0	0
T	1	3	0	0	2
A	2	2	1	1	2
B	2	0	0	0	2

The last chord has C $\sharp$ instead of C.

Recognising chromatic chords in charts

Read a chart in three passes: find the key, mark the diatonic chords, then classify everything else with the table below.

What you see	Analysis	Example in C
Dominant 7th that falls a 5th to the next chord	Secondary dominant V7/x	A7 → Dm
Major triad on II, III or VI	Secondary dominant (with or without the 7th)	E → Am
m7b5 or m7 immediately before such a dominant	Secondary ii – V	Em7b5 – A7 – Dm
°7 chord a half step below the next chord	Secondary leading-tone chord	C°7 → Dm7
°7 on the same root as the next chord (C°7 → C)	Common-tone diminished: decoration, not a dominant	C°7 → C
Minor chord on IV or V, major on bIII, bVI, bVII	Borrowed from the parallel minor	Fm, Ab, Bb
Dominant 7th a half step above the next chord	Tritone substitute (see Chapter 34)	D♭7 → C
Dominant 7th on bVII resolving up to I	Backdoor dominant	B♭7 → C
Many non-diatonic chords in a row, then a new cadence	Modulation (see Chapter 33)	... D7 → G and stays

Analyse this progression

C <i>I</i>	E7 <i>V7/vi</i>	Am <i>vi</i>	C7 <i>V7/IV</i>	F <i>IV</i>
Fm <i>iv</i>	C <i>I</i>	A7 <i>V7/ii</i>	Dm7 <i>ii7</i>	G7 <i>V7</i>
			C <i>I</i>	

Two secondary dominants, a borrowed iv, and a third secondary dominant leading into the final ii – V – I.

TRY THIS · Analyse three songs

Take three songs you play and write Roman numerals under every chord. Mark each non-diatonic chord as V/x, vii°7/x or borrowed. Most pop songs contain at least one; most jazz standards contain several in every chorus.

What to play over them

A secondary dominant temporarily changes the key, so the scale changes with it. Over V7 to a major target (V7/IV, V7/V), play Mixolydian on the chord root: the key's scale with the one new note. Over V7 to a minor target (V7/ii, V7/iii, V7/vi), play Phrygian dominant, the 5th mode of the target's harmonic minor. E7 → Am uses E Phrygian dominant (A harmonic minor), as in Chapter 31. Over a borrowed chord, use the parallel minor scale: C natural minor notes over Fm, Ab or Bb in C major.

Chapter 32 review · use it, train it, test it

Using it in real music

Playing by ear. When a song moves to a chord that sounds like it is "leaning" towards the next one, try the dominant seventh a 5th above the next chord. When the music suddenly darkens without losing its home, try iv or bVI.

Arranging. Turn a plain I – vi – ii – V into I – VI7 – II7 – V7 (C – A7 – D7 – G7) for a ragtime or swing feel. Put iv before the last I of a ballad for a softer ending.

Reading charts. Analyse before you play. Knowing that A7 in C is V7/ii tells you the C# is coming and where it resolves.

CHAPTER 33

Modulation



TRACKS

33

A modulation moves the tonic to a new note and confirms it with a cadence. The common methods differ only in how the old key is left, through a shared chord, a shared note, a sudden jump or a single dominant.

Chapter 32 showed how a chord outside the key can point briefly at another chord. If the music stays at the new chord, treats it as home and closes there with a cadence, the key has changed. That is a **modulation**. The difference from tonicisation is length and confirmation: a single V7/V followed by a return to I is tonicisation; a V7/V followed by a phrase that ends with a full cadence on V is a modulation to the dominant key.

A modulation is complete when three things are present:

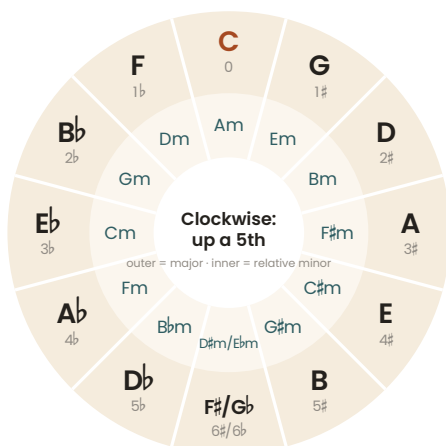
1. The new key's **leading note** (F# when moving from C to G).
2. A **dominant chord** of the new key resolving to its tonic.
3. Enough time, usually a phrase or more, for the new tonic to sound like home.

Closely related keys

Two keys are **closely related** when their key signatures differ by no more than one accidental. Every major key has five: its relative minor, the keys a 5th above and below (dominant and subdominant), and their relative minors. They are exactly the keys of the major and minor chords of the scale, ii to vi.

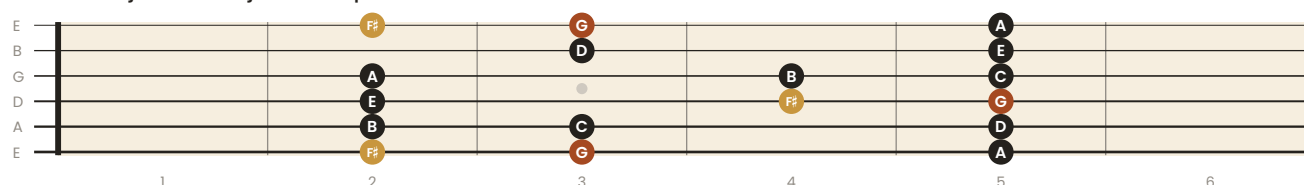
Home key	Dominant	Subdominant	Relative minor	Relative of dominant	Relative of subdominant
C major	G	F	Am	Em	Dm
G major	D	C	Em	Bm	Am
D major	A	G	Bm	F#m	Em
F major	C	Bb	Dm	Am	Gm
A minor	E minor	D minor	C	G	F

Moving to a closely related key changes one note of the scale or none, so it sounds smooth. Moving to a distant key (C to Eb, C to E, C to F#) changes several notes at once and sounds like a change of colour. The parallel key (C major to C minor) keeps the tonic but changes three notes; Chapter 32 treated borrowing from it, and a longer passage in it is a **change of mode**.



The closely related keys of C major are its neighbours on the circle of fifths, G and F, plus the three relative minors Am, Em and Dm.

From C major to G major in 2nd position



The G major notes in the position you know for C major. One note changes: F becomes F# (gold). Moving to F major would change B to Bb instead.

Pivot-chord modulation

A **pivot chord** belongs to both keys. The listener hears it as part of the old key, but it then behaves as a chord of the new key, usually as its ii or IV leading to V. Choose the pivot from the chords the two keys share.

C major chord	In C	In G	In F	In Am	In Em
C	I	IV	V	bIII	bVI
Dm	ii	not in key	vi	iv	not in key
Em	iii	vi	not in key	v	i
F	IV	not in key	I	bVI	not in key
G	V	I	not in key	bVII	bIII
Am	vi	ii	iii	i	iv

The strongest pivots are chords that become ii or IV in the new key, because they lead naturally to its V. Am is the classic pivot from C to G: vi in C, ii in G.

Pivot-chord modulation from C to G

C <i>I</i>	F <i>IV</i>	C <i>I</i>	Am <i>vi=ii</i>
D7 <i>V7</i>	G <i>I</i>	C <i>IV</i>	D7 <i>V7</i> G <i>I</i>

Am is vi in C and ii in G. From the D7 onwards every numeral refers to G.

▶ **TRACK 330** Pivot-chord modulation, C to G

C F C Am D7 G C D7 G G

	1	2	3	4	5	6	7	8	9	10
T	0	1	0	1	2	0	1	2	0	0
A	2	2	2	2	2	0	0	2	0	0
B	3	3	3	0	0	2	3	0	2	3

The F# in D7 is the new leading note. After it, C sounds like IV rather than home.

In minor keys the most common pivot modulation is to the relative major and back. In A minor, Dm is iv in A minor and ii in C, so Dm – G7 – C moves to C major in three chords; Bm7b5 – E7 – Am returns.

Direct modulation

A **direct** or **phrase modulation** has no pivot. One phrase ends in the old key and the next begins in the new one. It works best at a section boundary, between a verse and a chorus or before a repeat, and it allows any key, including distant ones. The new key is usually confirmed by its own cadence within a few bars.

TRACK 331 Direct modulation from C to A at a phrase boundary

Chord progression: C, Am, F, G, C, A

6

Chord progression: F#m, D, E, A

The second phrase repeats the first a minor 3rd lower, I – vi – IV V – I in A. No chord connects the two keys.

The second phrase repeats the first a minor 3rd lower, I – vi – IV V – I in A. No chord connects the two keys.

Truck-driver modulation

The most familiar direct modulation moves the whole song **up a half step or a whole step**, usually for the final chorus. The name comes from the sound of a lorry changing gear. It raises the energy without new material, and it is common in pop ballads, gospel and musical theatre: "Man in the Mirror" (Michael Jackson) goes up a half step, and "Love on Top" (Beyoncé) goes up a half step four times in its last minute.

The jump is often prepared by the V7 of the new key for one bar, which turns the direct modulation into a short dominant preparation: from G to A, the link chord is E7.

TRACK 332 Truck-driver modulation, G to A

Chord progression: G, Em, C, D, G, E7, A, F#m, D, E, A

Chord progression: G, Em, C, D, G, E7, A, F#m, D, E, A

I – vi – IV – V in G, the link chord E7, then the same progression in A.

TIP · On guitar it is two frets

With barre chords a whole-step modulation is the same shapes two frets higher, and a half-step modulation one fret higher. With open chords the new key needs new shapes, so for a live key change learn the progression as barre chords and the change becomes a slide.

Common-tone modulation

A **common-tone modulation** holds one note while the harmony underneath moves to a new key. It is the standard way to reach distant keys smoothly. The most useful are the **chromatic mediants**: major keys a major or minor 3rd away whose tonic triads share one note.

From C major	Common tone	Relationship
E major (E G $\sharp$ B)	E	Up a major 3rd
A $\flat$ major (A $\flat$ C E $\flat$)	C	Down a major 3rd
A major (A C $\sharp$ E)	E	Down a minor 3rd
E $\flat$ major (E $\flat$ G B $\flat$)	G	Up a minor 3rd

Film scores use these shifts constantly, and they are common in Romantic music and in late Beatles songs.

▶ TRACK 333 Common-tone modulation from C to E

C Am C E A B7 E

	C	Am	C	E	A	B7	E
T	0	0	0	0	0	2	0
3	1	1	1	0	2	0	0
5	0	2	0	1	2	2	1
7	2	2	2	2	2	1	2
	3	0	3	2	0	2	2
				0			0

The open 1st string, E, sounds in every chord from C to E. The cadence A – B7 – E confirms the new key.

Modulating with dominants and diminished chords

The quickest modulation is the **dominant of the new key**, or its ii – V. A secondary dominant that does not return becomes a modulation: in C, D7 – G followed by a cadence in G is a modulation to the dominant. Jazz standards change key this way every few bars. "All the Things You Are" passes through several keys by ii – V – I, and the bridge of "Have You Met Miss Jones" moves through keys a major 3rd apart. "Giant Steps" (John Coltrane) does the same at high speed.

A **diminished seventh chord** can be respelt to point at four different tonics, because each of its notes can act as the leading note. In C, B $^{\circ}$ 7 (B D F A $\flat$) normally resolves to C. The same four notes are D $^{\circ}$ 7 (to E $\flat$), F $^{\circ}$ 7 (to G $\flat$ or F $\sharp$) and G $\sharp$ 7 (to A or Am).

Spelling	Leading note	Resolves to
B $^{\circ}$ 7: B D F A $\flat$	B	C or Cm
D $^{\circ}$ 7: D F A $\flat$ C $\flat$	D	E $\flat$ or E $\flat$ m
F $^{\circ}$ 7: F A $\flat$ C $\flat$ E $\flat\flat$	F	G $\flat$ or G $\flat$ m
G $\sharp$ 7: G $\sharp$ B D F	G $\sharp$	A or Am

TRACK 334 One diminished chord, two resolutions

C Bdim7 C G#dim7 Am Dm E7 Am

T	1	3	1	3	1	3	0	1
A	0	1	0	1	2	2	1	2
B	2	3	2	3	2	0	0	2
	3	2	3	2	0		2	0

The same shape is played twice. First it is B°7 and resolves to C; then it is heard as G#°7 and leads to A minor, confirmed by iv – V7 – i.

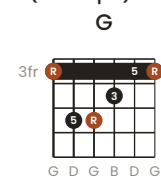
Modulating with shapes you know

On guitar a key change is a change of position. Three facts from Chapter 12 make it quick:

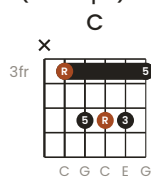
- **Up a 4th (to IV):** an E-shape barre with its root on the 6th string and an A-shape barre with its root on the 5th string at the same fret are a 4th apart. G at the 3rd fret (E-shape) to C at the 3rd fret (A-shape) is I to IV, so a modulation from G to C stays in position.
- **Up a 5th (to V):** the reverse. A-shape C at the 3rd fret to E-shape G at the 3rd fret.
- **Up a step:** move every shape up two frets (whole step) or one fret (half step).

Same fret, two keys

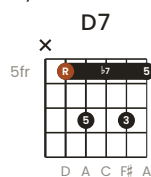
G (E-shape)



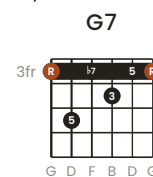
C (A-shape)



D7, V7 of G



G7, V7 of C



All four shapes sit between the 3rd and 7th frets. G and C are a 4th apart at the same fret, and each key's dominant seventh is close by.

For single-note playing the same idea applies to scales. A position you know in the old key contains all but one note of a closely related key. Before the modulation, find the note that changes (F to F# from C to G; B to Bb from C to F) and aim for it on the new V7.

Analysing modulations in charts

Work through the chart in this order:

1. **Find the cadences.** Mark every V – I and ii – V – I. Each one names a tonic, at least for that moment.
2. **Count how long each tonic lasts.** One or two bars that return to the home key are tonicisation. A phrase or section that ends with a cadence on a new tonic is a modulation.
3. **Look for the new accidental.** A modulation from C to G brings F#; from C to F, Bb. In written music a change of key signature settles it.
4. **Find the link.** Look back from the first chord that belongs only to the new key. The last chord that belongs to both keys is the pivot. If there is none, it is a direct modulation; if a single held note connects the chords, a common-tone modulation.
5. **Write numerals in both keys** under the pivot chord (vi = ii), then continue in the new key.

Analyse this progression

F <i>I</i>	Dm <i>vi</i>	Gm7 <i>ii7</i>	C7 <i>V7</i>
F <i>I</i>	Am <i>iii=v</i>	Em7 ^{b5} <i>iiø7</i>	A7 <i>V7</i>
Dm <i>i</i>	B ^b <i>bVI</i>	Gm7 <i>iv7</i>	A7 <i>V7</i>
Dm <i>i</i>	Gm7 <i>iv7=ii7</i>	C7 <i>V7</i>	F <i>I</i>

F major, then a pivot (Am, iii in F and v in D minor) into D minor with a full cadence, then back to F through Gm7, which is iv7 in D minor and ii7 in F.

TRY THIS · Find the key changes

Choose a jazz standard or a musical-theatre song you know and mark every cadence. Count the tonics it visits and decide, for each, whether it is tonicisation or a modulation. Most standards visit two to four keys in 32 bars.

Chapter 33 review · use it, train it, test it

Using it in real music

Learning a song. Mark the modulations first. A key change tells you when your scale position or your capo shapes have to change.

Arranging. A truck-driver modulation is the quickest way to lift a final chorus. For a smoother result, add the V7 of the new key in the last bar of the previous section.

Improvising. Over a modulation, play the new key's leading note on its V7 chord. A solo that hits F[#] on D7 tells the listener you have arrived in G.

How to train it

Drill	How	Time	Goal
Related keys	Name the five closely related keys of C, G, D, A, F and B ^b	3 min	30 keys in under two minutes
Pivot hunt	For C to G, C to F and C to Am, list every shared chord and its numeral in both keys	5 min	All pivots named with both numerals
Pivot loop	Play I – IV – vi=ii – V7 – I from C to G, then G to D, round the circle	5 min	Through six keys without stopping
Whole-step lift	Play a four-chord loop as barre chords, insert the new V7, repeat two frets higher	3 min	Clean change on the first beat of the new key
Scale switch	Over a looped C – Am – D7 – G, play C major and switch to G major on the D7	5 min	F [#] heard on every D7

Check your understanding

Write your answers, then check them in Appendix J.

1. What is the difference between tonicisation and modulation?

2. List the closely related keys of D major.

3. Which chord is the usual pivot from C major to G major, and what are its two numerals?

4. Which note changes when you move from C major to F major?

6. Name a key a major 3rd above C that can be reached by a common-tone modulation, and the common tone.

7. To which four keys can the diminished seventh chord B D F A^b lead?

8. On guitar, how do you go from I to IV at the same fret with barre chords?

MY NOTES

CHAPTER 34

Voice Leading and Reharmonisation



A chord progression is also a set of melodic lines, one for each note of the chords. Voice leading is the craft of moving those lines smoothly; reharmonisation changes the chords while keeping the melody, and relies on the same principles.

Every chord voicing is a stack of separate voices: a bass, a top note and one or two inner voices. When the chords change, each voice moves to a note of the next chord. If the voices move by small steps, the progression sounds connected and the hand moves little. If they all leap, the same chords sound blocky. Chapter 13 and Chapter 14 treated progressions as a sequence of chord symbols; this chapter treats them as lines.

Common tones and stepwise motion

Three principles cover most smooth voice leading:

1. **Keep common tones.** A note that belongs to both chords stays in the same voice.
2. **Move the other voices by step,** to the nearest note of the new chord.
3. **Let the bass leap if it must;** the upper voices should not.

How many common tones two triads share depends on the distance between their roots:

Root movement	Common tones	Example in C	Upper voices
By a 3rd (up or down)	2	C → Am (C, E)	One voice moves a step
By a 4th or 5th	1	C → F (C)	Two voices move a step
By a step	0	F → G	All voices move; move them against the bass

When the roots move by step, move the upper voices in **contrary motion** to the bass: if the bass goes up from F to G, let the upper voices fall from F A C to D G B or B D G. This avoids parallel 5ths and octaves, the main thing classical voice leading forbids, and it sounds better in any style.

TRACK 335 I – vi – IV – V7 – I on the top three strings

The top three strings never move more than two frets. C stays on the 3rd string for three chords, then falls to B and returns. The bass moves by leaps.

The same progression as smooth triads on strings 1–3

Each change moves one or two notes by a step.

TRY THIS • Voice-lead a song you know

Take a four-chord song and play it with triads on strings 1–3 only. For each change, find the voicing of the next chord that keeps the most common tones. Then do the same on strings 2–4.

Guide tones

In seventh chords the **3rd** and the **7th** decide the quality: major or minor 3rd, major or minor 7th. They are called **guide tones**. In a ii – V – I they form two smooth lines:

- The **7th** of each chord falls a half step to the **3rd** of the next.
- The **3rd** of each chord stays to become the **7th** of the next.

Chord	Dm7	G7	Cmaj7
Line A	C (♭7)	B (3)	B (7)
Line B	F (♭3)	F (♭7)	E (3)

This is why the ii – V – I sounds inevitable, and why jazz guitarists can outline it with two notes. The same rule holds round the whole circle of fifths: every chord whose root falls a 5th passes its 7th down by step to the next 3rd.

TRACK 336 Guide tones over Dm7 – G7 – Cmaj7

Dm7 G7 Cmaj7

	Dm7	G7	Cmaj7
T	6		5
A	5	4	4
B	5	3	3

Shell voicings: root plus the two guide tones. Line A goes C – B – B, line B goes F – F – E.

A **guide-tone line** is a melody made from these notes, one per chord, played in rhythm. It is the skeleton of a jazz solo over the changes. Chapter 35 builds solos from it.

Shell voicings for ii – V – I in C

Dm7 G7 Cmaj7

The guide tones are teal. Only one guide tone moves at each change.

Inner voices on the top four strings

Four-note voicings on strings 1–4 (drop-2 voicings; see Chapter 48) let all four voices move independently. A ii – V – I in drop-2 voicings moves only two voices at each change, each by a step, while the other two hold.

ii – V – I in drop-2 voicings, two positions

Dm7 G7 Cmaj7 Dm7 G7 Cmaj7

10fr 8fr 8fr 3fr 3fr 3fr

C F A D B F G D B E G C F C D A F B D G E B C G

Upper position: C F A D → B F G D → B E G C. Lower position: F C D A → F B D G → E B C G. In both, two voices move by step and two hold.

▶ **TRACK 337** ii – V – I with drop-2 voicings in two positions

Diagram illustrating the ii – V – I progression with drop-2 voicings in two positions. The chords shown are Dm7, G7, Cmaj7, Dm7, G7, and Cmaj7. The diagram shows the chord voicings on a staff and the corresponding fingerings for the Tenor (T), Alto (A), and Bass (B) voices.

	Dm7	G7	Cmaj7	Dm7	G7	Cmaj7
T	10	10	8	5	3	3
A	10	8	8	3	3	1
B	10	10	9	5	4	4
	10	9	9	3	3	2

Line clichés

A **line cliché** keeps a chord static and moves one voice chromatically; Chapter 14 explains it in full. In voice-leading terms it is the smallest possible motion: one voice moves a half step, every other voice holds. Inside the chord the minor form runs 1 – 7 – b7 – 6 (Am – AmMaj7 – Am7 – Am6); the major forms run 5 – #5 – 6 – b7 (C – C^{aug} – C6 – C7) or 1 – 7 – b7 – 6 (C – Cmaj7 – C7 – C6).

▶ **TRACK 338** Minor line cliché on Am, inner voice

Diagram illustrating the Minor line cliché on Am, inner voice. The chords shown are Am, AmMaj7, Am7, and Am6. The diagram shows the chord voicings on a staff and the corresponding fingerings for the Tenor (T), Alto (A), and Bass (B) voices.

	Am	AmMaj7	Am7	Am6
T	5	5	5	5
A	5	5	5	5
B	7	6	5	4
	0	0	0	0

The line moves to an inner voice on the 4th string, A – G# – G – F#, while the rest of the chord holds.

Reharmonisation

To **reharmonise** is to replace the chords under a melody with different ones that still fit it. Each technique below keeps the melody note inside the new chord (or as a tension of it) and usually improves the voice leading.

Diatonic substitution

Chords whose roots are a 3rd apart share two notes, so they can replace each other. Within a key they fall into three function groups:

Function	Chords in C	Shared notes
Tonic	C, Em, Am (I, iii, vi)	C and E between C and Am; E and G between C and Em
Subdominant	F, Dm (IV, ii)	F and A
Dominant	G, B ^o (V, vii ^o)	B and D

Replacing I with vi at the start of a repeat, or IV with ii before V, changes the colour without changing the function.

Tritone substitution

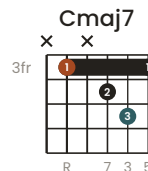
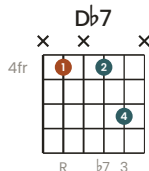
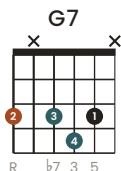
Any dominant seventh can be replaced by the dominant seventh whose root is a tritone (three whole steps) away. G7 (G B D F) and Db7 (Db F Ab Cb) share the tritone B–F (Cb is B), and the tritone is what makes a dominant resolve. The substitute's root lies a half step above the target, so the bass moves chromatically: Dm7 – Db7 – Cmaj7 instead of Dm7 – G7 – Cmaj7.

Dominant	Tritone substitute	Shared tritone	Resolves to
G7	Db7	B (Cb), F	C
C7	Gb7	E (Fb), Bb	F

Dominant	Tritone substitute	Shared tritone	Resolves to
D7	A $\flat$ 7	F $\sharp$ (G $\flat$), C	G
A7	E $\flat$ 7	C $\sharp$ (D $\flat$), G	D or Dm
E7	B $\flat$ 7	G $\sharp$ (A $\flat$), D	A or Am

The substitute's other two notes, D $\flat$ and A $\flat$, are the b5 and b9 of G. Over a G in the bass, D $\flat$ 7 therefore sounds like an altered G7 (G7b9b5), which is why tritone substitution and the altered scale (Chapter 31) are used together.

G7 and its tritone substitute D $\flat$ 7



The 3rd and 7th of G7 (B and F) become the 7th and 3rd of D $\flat$ 7. Both resolve to the same notes of C.

TRACK 339 ii – V – I and its tritone substitute in F

Gm7 C7 Fmaj7 Gm7 G $\flat$ 7 Fmaj7

	Gm7	C7	Fmaj7	Gm7	G $\flat$ 7	Fmaj7
T	3	5	1	3	2	1
A	3	3	2	3	3	2
B	3	3	2	3	2	2
	3	3	1	3	2	1

Bar 3 replaces C7 with G $\flat$ 7. The bass now falls G – G $\flat$ – F by half steps; the guide tones B $\flat$ and E are the same in both versions.

Chromatic approach chords

Any target chord can be approached from a half step above or below by a chord of the **same quality**: E $\flat$ m7 – Dm7, C $\sharp$ m7 – Dm7, D $\flat$ 7 – C7. The approach chord is a decoration, not a function; it works because every voice moves by a half step. Jazz arrangers use it before the downbeat of a new phrase, and funk and soul players slide into chords this way (see Chapter 45). Moving a single voicing up or down chromatically or diatonically in parallel is called **planing**; it is the basis of the sound of many fusion and film scores.

Inserting ii – V and secondary dominants

Any chord can be preceded by its own V7 or ii – V (Chapter 32). Combined with tritone substitution this gives most bebop reharmonisations.

A simple progression and a reharmonisation

F <i>I</i>	Dm <i>vi</i>	B $\flat$ <i>IV</i>	C <i>V</i>	F <i>I</i>
Fmaj7 <i>Imaj7</i>	D7 <i>V7/ii</i>	Gm7 <i>ii7</i>	G $\flat$ 7 <i>bII7</i>	Fmaj7 <i>Imaj7</i>

Second row: Dm becomes D7 (secondary dominant of Gm), B $\flat$ becomes Gm7 (diatonic substitute), C becomes G $\flat$ 7 (tritone substitute).

TRACK 340 Plain and reharmonised versions

F Dm B $\flat$ C F Fmaj7 D7 Gm7 G $\flat$ 7 Fmaj7

	1	1		0	1	1	7	3	2	1
T	1	3	3	1	1	2	5	3	3	2
A	2	2	3	0	2	2	7	3	2	2
B	3	0	3	2	3	2	5	3	2	1
			1	3		1		3		

CAREFUL · Keep the melody

A substitute chord must accept the melody note: it has to be a chord tone or a usable tension of the new chord. Over D $\flat$ 7 the melody notes F (3), A $\flat$ (5), B ($\flat$ 7) and G (#11) all work. A melody note of C, the major 7th of D $\flat$, clashes with the chord's $\flat$ 7 and rules the substitute out.

Chapter 34 review · use it, train it, test it

Using it in real music

Comping. Choose voicings by the shortest path, not by root position. In a ii – V – I, move from one drop-2 voicing to the nearest voicing of the next chord, and the top note becomes a melody of its own.

Arranging a song. Pick one change per chorus: a diatonic substitute in the verse, a tritone substitute at the turnaround, a line cliché under a long minor chord. Too many substitutions at once hide the song.

Reading jazz charts. A dominant seventh a half step above the next chord (D $\flat$ 7 $\rightarrow$ C) is almost always a tritone substitute. Play it with the same shell shape one fret above the target.

How to train it

Drill	How	Time	Goal
Common-tone triads	Play I – vi – IV – V in C, G and D on strings 1–3, keeping common tones	5 min	Hand moves no more than two frets per change
Guide-tone ii – V – I	Play only the 3rd and 7th of Dm7 – G7 – Cmaj7 round the circle of fifths	5 min	All 12 keys, half-step motion only
Drop-2 cycle	Play ii – V – I in drop-2 voicings on strings 1–4, two positions per key	5 min	Two voices move, two hold, in every key
Line cliché	Play the minor line cliché on Am, Dm and Em	3 min	Moving voice heard clearly above the rest
Tritone swap	Loop ii – V – I, alternating V7 and $\flat$ II7 every four bars	4 min	Bass moves by half steps without looking

Check your understanding

Write your answers, then check them in Appendix J.

1. How many common tones do C and Am share, and which are they?

2. When the bass moves by step, how should the upper voices move?

3. What are the guide tones of G7?

4. In Dm7 – G7, which guide tone moves and where does it go?

5. What is the tritone substitute for E7, and why does it work?

6. Spell the four chords of a minor line cliché on Dm.

7. Which chords can substitute for C in C major by diatonic substitution?

8. What is a chromatic approach chord?

MY NOTES

CHAPTER 35

Improvising over Changes



TRACKS

35

Playing over a progression that moves through several keys needs more than one scale. Chord-scale thinking chooses a scale for each chord; target-note thinking aims chord tones at the beats. Fluent players use both, and connect the targets with guide tones, enclosures and passing notes.

Over a single chord or a progression in one key, one scale covers everything (Chapter 7, Chapter 9). Jazz standards, many soul and pop songs and most music with secondary dominants change key every bar or two. Over such **changes** the soloist has to show each chord, or at least not contradict it. This chapter builds on the harmony of Chapter 31, Chapter 32 and Chapter 34 and turns it into a practice method.

Two ways of thinking

	Chord-scale thinking	Target-note thinking
Question asked	Which scale fits this chord?	Which chord tone do I land on, and when?
Main material	Seven-note scales and modes	Arpeggios, guide tones, approach notes
Strength	Every note "fits"; good for modal tunes and long chords	Outlines the harmony; lines sound like they move through the changes
Weakness	Lines can wander and miss the chord changes	Can sound like arpeggio exercises without scale connections
Typical style	Modal jazz, fusion, rock soloing	Bebop, swing, gypsy jazz, country

They are two views of the same thing. A chord scale is the chord's arpeggio (1 3 5 7) with the gaps filled by its tensions (9 11 13). Target-note thinking decides which of those notes go on the beats.

Chord-scale choices

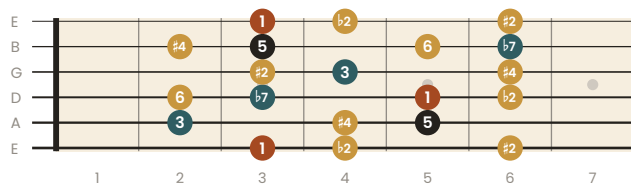
The table gives the usual first choice for each chord type, with alternatives. "Avoid" marks a scale note that clashes when held on a strong beat: it lies a half step above a chord tone.

Chord	Function	Scale	Avoid	Alternatives
maj7	I or IV	Ionian (I), Lydian (IV)	4 over I	Lydian over I for a modern sound
m7	ii	Dorian	none	Aeolian (vi), Phrygian (iii)
7	V to a major chord	Mixolydian	4	Bebop dominant, Lydian dominant
7	V to a minor chord	Phrygian dominant	4	Altered, half-whole diminished
7	Non-resolving (IV7, bVII7, II7, tritone subs)	Lydian dominant	none	Mixolydian
7alt, 7#9, 7b13	Altered V	Altered	none	Half-whole diminished (with 13)
7b9, 13b9	V	Half-whole diminished	none	Phrygian dominant
7#5, 9#5	V	Whole tone	none	Altered
m7b5	ii in minor	Locrian #2	none	Locrian
°7	Passing or vii°7	Whole-half diminished	none	Harmonic minor of the target
m6, mMaj7	i in minor	Melodic minor	none	Dorian (m6)

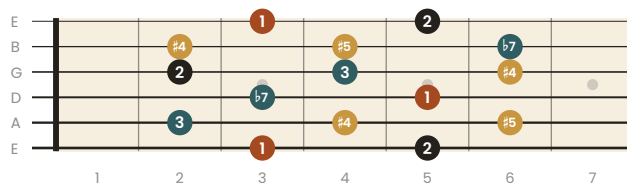
Learn this table one row at a time, always over a practice loop. The scales themselves are in Chapter 9 and Chapter 31.

Altered-dominant scales on G

G half-whole diminished



G whole tone



Half-whole diminished contains $\flat 9$, $\sharp 9$, $\sharp 11$ and 13 (gold) with a natural $5th$. Whole tone contains 9 , $\sharp 11$ and $\sharp 5$ and no $5th$. Both repeat every few frets: the diminished pattern every three frets, the whole-tone pattern every two.

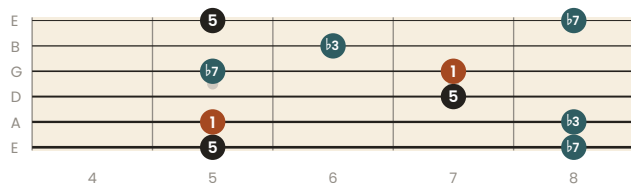
Target notes

A line sounds like it follows the changes when **chord tones fall on the strong beats** (1 and 3 in 4/4) and, above all, on the **first beat after each chord change**. The best targets at a change are the guide tones, the 3rd and 7th, because they define the new chord and are usually a half step from a note of the old one.

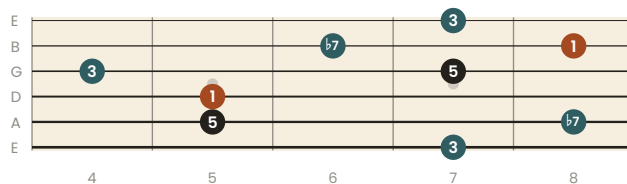
Learn the arpeggios of each chord in one position, so you can see them as dots within the scale shape you already use.

Arpeggios of Dm7, G7 and Cmaj7 in 5th position

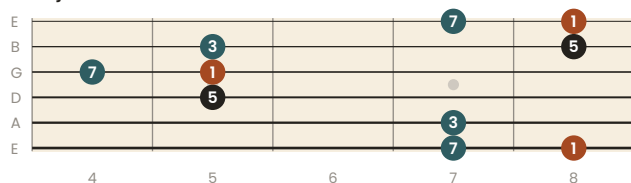
Dm7



G7



Cmaj7



The guide tones are teal. Notice that the 7th of each chord is a half step or a whole step above the 3rd of the next.

▶ TRACK 341 Two guide-tone lines through a cycle of fifths

Dm7 G7 Cmaj7 Fmaj7 Bm7 $\flat 5$ E7 Am7

	Dm7		G7		Cmaj7		Fmaj7		Bm7 $\flat 5$		E7		Am7	
T	6	5	6	4	5	4	5	2	3	2	3	1	0	1
A														
B														

Upper line: F F E E D D C C. Lower line: C B B A A G $\sharp$ G G. Each line alternates 3rd and 7th and never moves more than a half or whole step.

▶ **TRACK 342** A target-note line over ii – V – I

Dm7 **G7** **Cmaj7**

T 5 6 5 7 5 4 7 6 8 7 6 5 4 5

A 7 5 4 7 6 8 7 6 5 4 5

B 5 8 7 5 4 7 6 8 7 6 5 4 5

Bar 1 arpeggiates Dm7 and lands C (♭7) on the last eighth; bar 2 starts on B, the 3rd of G7. Bar 2 ends on F (♭7), which falls to E, the 3rd of Cmaj7.

TIP • Start with one note per chord

Before playing lines, play a single guide tone on beat 1 of every bar for a whole chorus. Then two notes per bar, then four. Increase the density only while every change still lands on a 3rd or 7th.

Enclosures and approach notes

An **enclosure** surrounds the target with notes above and below before landing on it. It delays the target by one or two eighths and makes the landing stronger. The common forms, aimed at E (the 3rd of Cmaj7):

Enclosure	Notes	Rule
Scale above, chromatic below	F – D♯ – E	The most common bebop enclosure
Scale below, scale above	D – F – E	Diatonic, softer
Double chromatic from above	G♭ – F – E	Two half steps down into the target
Double chromatic from below	D – D♯ – E	Two half steps up into the target
Chromatic above and below	A♭ – F♯ – G (target G, the 5th)	Works on any target; on E it equals the first row

A single **chromatic approach note** (a half step below or above the target on the off-beat before it) is the simplest version. Place the target on a beat, and the approach notes on the off-beats before it.

▶ **TRACK 343** Enclosures into the 3rds of G7 and Cmaj7

Dm7 **G7** **Cmaj7**

T 6 5 7 5 7 5 3 4 7 6 8 6 7 6 4

A 7 5 4 7 6 8 6 7 6 4

B 5 8 7 5 4 7 6 8 7 6 5 4 5

3

T 5 4 5 7 5 7 5

A 4 5 7 5 7 5

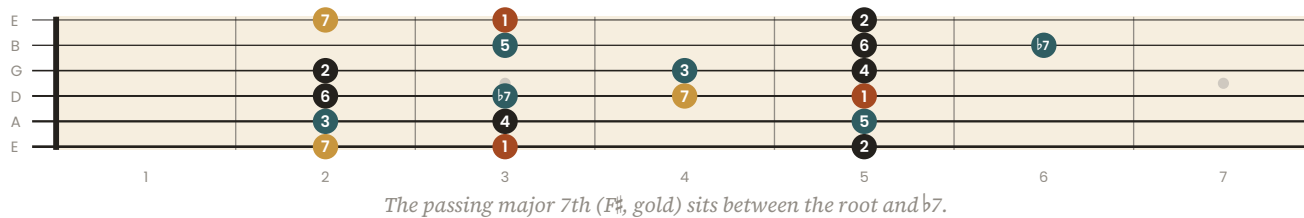
B 5 8 7 5 4 7 6 8 7 6 5 4 5

End of bar 1: C (above) and A♯ (below) enclose B, the 3rd of G7. End of bar 2: F and D♯ enclose E, the 3rd of Cmaj7.

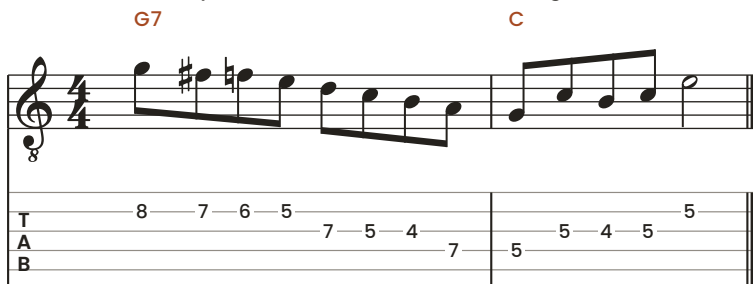
The bebop dominant scale

Mixolydian has seven notes, so a descending eighth-note run that starts on a chord tone on the beat soon puts non-chord tones on the beats. The **bebop dominant scale** adds the major 7th as a passing note between the root and the $\flat 7$. Starting from any chord tone on the beat and descending, the chord tones 1, 3, 5 and $\flat 7$ then fall on the beats.

G bebop dominant in 3rd position



▶ TRACK 344 Bebop dominant line over G7 resolving to C



Descending from G: the beats fall on G, F, D and B, all chord tones of G7. $F\sharp$ is the passing note.

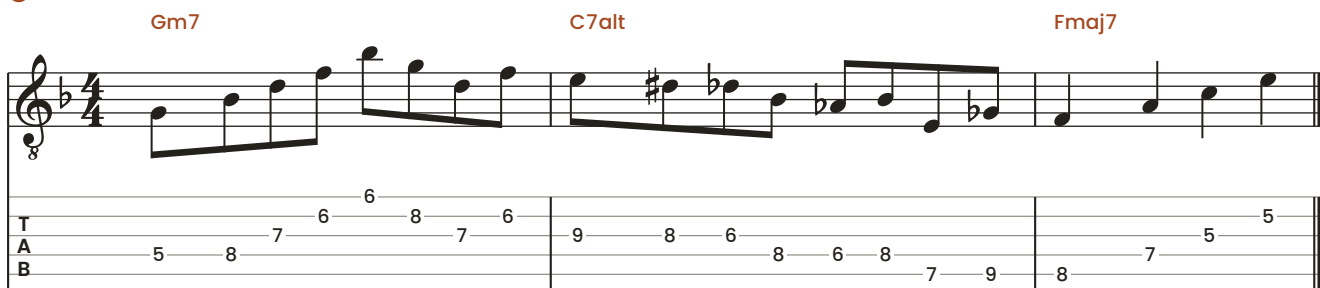
ii – V – I in major

The ii – V – I is two bars or four bars of changes leading to a tonic. Practise it in three layers, each over a practice loop:

1. **Arpeggios.** Play the four chord tones of each chord, up and down, in one position.
2. **Guide tones.** Land on the 3rd or 7th on beat 1 of each bar.
3. **Lines.** Connect the guide tones with scale notes, chromatic approach notes and enclosures.

Over the V7 in a major ii – V – I you can stay with Mixolydian or bebop dominant for an inside sound, or play the altered scale for tension that resolves strongly to the I chord.

▶ TRACK 345 Altered line in a ii – V – I in F



Bar 2 uses C altered ($D\flat$ melodic minor): E, $E\flat$, $D\flat$, $B\flat$, $A\flat$, then E and $G\flat$ enclose F, the root of the I chord.

TRACK 346 Practice loop: major ii – V – I in C · LOOP · jazz swing · 120 bpm

Dm7 <i>ii7</i>	G7 <i>V7</i>	Cmaj7 <i>Imaj7</i>	Cmaj7 <i>Imaj7</i>
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IMPROVISE WITH

C major

Grade	7	8	9	10	11	12	13
E	0	1	0	2	0	3	4
B	0	0	0	6	0	7	1
G	1	0	3	4	0	5	0
A	0	0	7	1	0	2	0
D	0	0	0	5	0	6	0
E	0	0	0	2	0	3	0

G altered

Work through the three layers above. For guide tones, play F (3rd of Dm7), F (7th of G7) and E (3rd of Cmaj7), then C, B and B; once those are easy, try the altered scale over G7 and resolve it to E or G.

ii – V – i in minor

In a minor ii – V – i every chord takes a different parent scale:

Chord in D minor	Scale	Parent scale
Em7b5	E Locrian (or Locrian #2)	F major (or G melodic minor)
A7 (A7b9, A7alt)	A Phrygian dominant or A altered	D harmonic minor, or Bb melodic minor
Dm6, DmMaj7	D melodic minor	D melodic minor
Dm7	D Dorian	C major

The important notes are the ones that change between chords: C#, the 3rd of A7 and the leading note of D, and the b9 and #5 of A7 (Bb and F). Bb falls to A, the 5th of Dm, and F is held as its 3rd. Target C# on the A7 and resolve it to D.

 TRACK 347 Minor ii – V – i line in D minor

Em7b5 A7alt Dm6

8

T
A
B

6 5 6 7 6 8 8 6 8 7 6 5 7 5

Bar 2: C# (3), Eb (b5), F (#5), G (b7), Bb (b9) from A altered. Eb falls a half step to D on the downbeat of bar 3, and the Dm6 arpeggio shows the B of D melodic minor.

CAREFUL · Altered sounds need a clear resolution

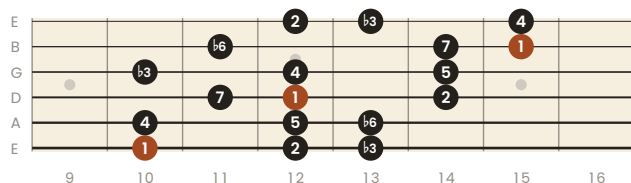
An altered note held over the V chord sounds like a mistake unless the next chord resolves it. End altered lines on a chord tone of the tonic, a half step or whole step away: $b9$ to 5, $\#9$ and $b5$ to 1, $b13$ to 3 in major (in minor it is the $b3$ and can simply be held).

TRACK 348 Practice loop: minor ii – V – i in D minor · LOOP · bossa nova · 120 bpm

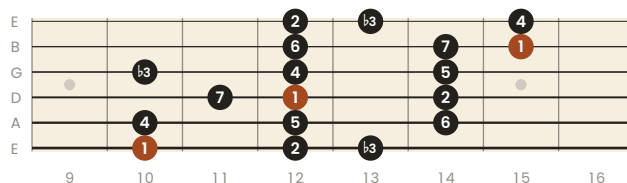
Em7b5 <i>iiø7</i>	A7b9 <i>V7b9</i>	Dm6 <i>i6</i>	Dm6 <i>i6</i>
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IMPROVISE WITH

D harmonic minor



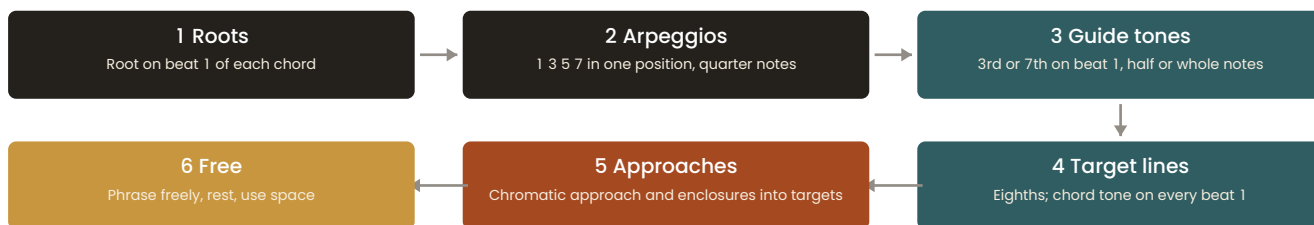
D melodic minor



D harmonic minor over Em7b5 and A7b9, D melodic minor over Dm6. Put C# on beat 1 of the A7b9 bar and resolve it to D; then try Bb falling to A.

Practising over a looping progression

A looper pedal, an app or a recorded practice loop is the most effective tool for this work, because the changes keep coming while you practise (see Chapter 42). Record two to four bars of the progression with a steady rhythm and practise in this order. Stay at each stage until it is easy at a slow tempo.



Six stages for any progression. Move to the next only when the previous one is easy at about 70 bpm.

Useful loops, from easy to hard:

Loop	Chords	Focus
Two-chord vamp	Dm7 – G7 (two bars each)	Dorian against Mixolydian; F to F, C to B
Major ii – V – I	Dm7 – G7 – Cmaj7 – Cmaj7	Resolving to the 3rd of I
Minor ii – V – i	Em7b5 – A7 – Dm6 – Dm6	C# on A7, altered tensions
Turnaround	Cmaj7 – A7 – Dm7 – G7	Chord changes every two beats; A7 is V7/ii
Cycle of fifths	Dm7 – G7 – Cmaj7 – Fmaj7 – Bm7b5 – E7 – Am7	Guide-tone lines through a whole cycle
12-bar jazz blues	F7 – Bb7 – F7 – Cm7 F7 – Bb7 – Bdim7 – F7 – Am7b5 D7 – Gm7 – C7 – F7 D7 – Gm7 C7	Secondary ii – V into Bb and Gm

TRY THIS · Same lick, every key

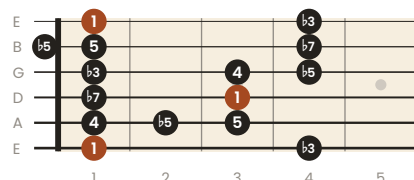
Take one two-bar ii – V – I line you like, from this chapter or a recording, and play it over the same progression in all twelve keys. Move around the circle of fourths (C, F, Bb, Eb ...). This forces you to know the line as degrees, not as frets.

TRACK 349 Practice loop: jazz blues in F · LOOP · jazz swing · 120 bpm

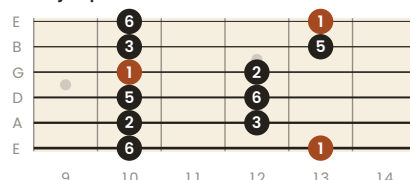
F7 <i>I7</i>	Bb7 <i>IV7</i>	F7 <i>I7</i>	Cm7 <i>v7</i>	F7 <i>I7</i>
Bb7 <i>IV7</i>	Bdim7 <i>#iv°7</i>	F7 <i>I7</i>	Am7b5 <i>iiiø7</i>	D7 <i>VI7</i>
Gm7 <i>ii7</i>	C7 <i>V7</i>	F7 <i>I7</i>	D7 <i>VI7</i>	Gm7 <i>ii7</i>
			C7 <i>V7</i>	

IMPROVISE WITH

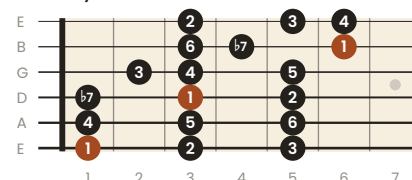
F blues



F major pentatonic



F mixolydian



Play F blues over the whole form first. Then aim for the 3rd of each new chord on beat 1: A over F7, D over Bb7, F# over D7 (leading into G in bar 9) and E over C7.

LISTEN FOR · Hearing the changes in a solo

Listen to a bebop soloist (Charlie Parker, Wes Montgomery, Joe Pass) and follow the chart while they play. Mark every place where you can hear the chord change in the melody alone. In most cases it is a 3rd or 7th on beat 1, reached by step or by an enclosure.

Chapter 35 review · use it, train it, test it

Using it in real music

Learning a standard. Before improvising, play the guide-tone lines of the whole tune. They show where the harmony moves and give you landing notes for every bar.

Blues and rock players. Keep your pentatonic vocabulary, but aim its phrases at the 3rd of each chord on beat 1. Over a 12-bar blues this one change makes a solo sound like it follows the chords.

Choosing between scale and arpeggio. Over a chord that lasts four bars or more, think in scales. Over chords that change every bar or faster, think in arpeggios and targets.

How to train it

Drill	How	Time	Goal
Arpeggio map	Play the arpeggios of Dm7, G7, Cmaj7 in 5th position, in time, changing on beat 1	5 min	No pause at the changes at 100 bpm in quarters
Guide-tone chorus	Over a looped ii – V – I, play only 3rds and 7ths, one per beat	5 min	Every beat a guide tone, half-step motion at changes
Enclosure targets	Enclose the 3rd of each chord of a looped ii – V – I	5 min	Target always on beat 1
Bebop run	Descend G bebop dominant from G, B, D and F on the beat	3 min	Chord tones on all four beats
Altered switch	Loop G7 – Cmaj7; alternate Mixolydian and altered lines over G7 every four bars	5 min	Each altered line resolves to a C chord tone
Minor ii – V – i	Loop Em7b5 – A7 – Dm6, targeting C# on A7	5 min	C# on beat 1 of every A7 bar

Check your understanding

Write your answers, then check them in Appendix J.

1. What is the difference between chord-scale and target-note thinking?
2. Which notes make the best targets at a chord change, and why?
3. Which scale is the first choice over G7 resolving to Cm?
4. In a minor ii – V – i in D minor, where do the $\flat 9$ and $\sharp 5$ of A7 go on the Dm chord?
5. Write an enclosure of B using a scale step above and a half step below.
6. What note does the bebop dominant scale add to Mixolydian, and why?
7. Over Bm7 $\flat 5$ in A minor, which scale adds a natural 9th to Locrian?
8. Name the six practice stages for a looping progression.

MY NOTES

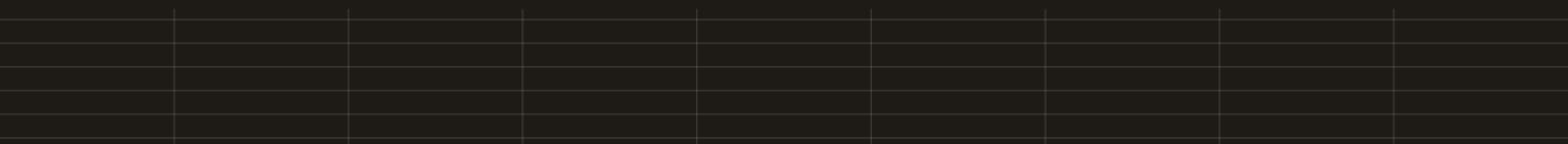
PART NINE

Technique and Time

Bends, vibrato, slides, harmonics and a complete rhythm training programme.

36 Expressive Techniques

37 Rhythm Mastery



CHAPTER 36

Expressive Techniques



Slurs, bends, vibrato, slides, harmonics and muting change how a note starts, moves and stops. They turn a correct line of notes into a phrase.

A note on the guitar has three phases: the attack, the sustain and the release. Picking or plucking decides the attack. The techniques in this chapter decide everything else: a slur replaces the attack with a softer one, a bend or slide changes the pitch while the note sounds, vibrato animates the sustain, and muting controls how and when the note stops.

All of them work on every guitar, but not to the same degree. Bends are central on electric, limited on steel-string and nearly absent on nylon. Slurs, slides, vibrato and harmonics are part of every style, from classical studies to blues.

Technique symbols in TAB

TAB uses letters and signs between or above the fret numbers. Printed TAB varies a little between publishers; these are the common forms.

Symbol	Meaning
5h7 / 7p5	Hammer-on from fret 5 to 7 / pull-off from fret 7 to 5
7b9	Bend fret 7 up to the pitch of fret 9
7b9r7	Bend, then release back to fret 7
7pb9	Pre-bend: bend silently, then pick and release
7~~	Vibrato
5/7 / 7\5	Slide up from fret 5 to 7 / slide down from fret 7 to 5
<12>	Natural harmonic at fret 12
x	Muted (dead) note: no pitch
P.M.	Palm mute; a dashed line shows how long it lasts
tr	Trill: rapid hammer-ons and pull-offs
(7)	Note held or tied from before; also used for a ghost note
full, ½	Size of a bend: whole step, half step

In this book's scores, slurs print as an arc with **H**, **P** or **sl.** above the TAB, bends as a curved arrow marked **full** or **½**, vibrato as a wavy line above the staff, and natural harmonics as diamond noteheads with the fret number in angle brackets.

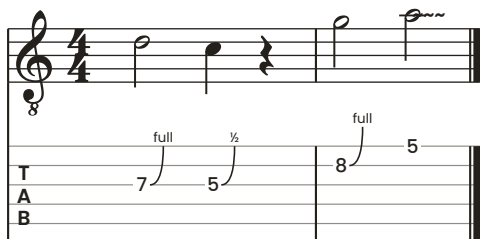
Hammer-ons, pull-offs and trills

A **hammer-on** sounds a higher note on the same string by bringing a fretting finger down hard, without picking again. A **pull-off** sounds a lower note: the fretting finger leaves the string with a slight sideways flick, plucking it as it goes. Lifting the finger straight up gives almost no sound. Together these are called **slurs**, or **ligados** in classical guitar. They make lines smoother and faster because the picking hand plays fewer notes.

- Hammer with the fingertip, close to the fret wire, from about 1 cm above the string. Speed of the finger matters more than force.
- For a pull-off, the lower note must already be fretted (or be the open string) before you pull.

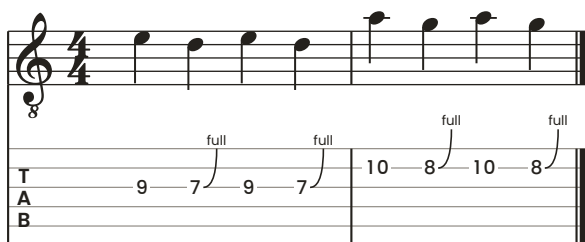
Bend to pitch. A bend has a target: the pitch of a fret higher up. Play the target first, then bend to match it. A bend that stops short sounds out of tune, and it is the most common fault in lead playing.

TRACK 352 Bends in A minor pentatonic



Full bend D to E (4 to 5) on the 3rd string, half bend C to C# (b3 to 3), then a full bend G to A (b7 to root) answered by A on the 1st string.

TRACK 353 Bend to pitch



Play the target note, then bend to it. The two notes should sound identical.

NYLON · STEEL

Nylon: bending barely changes pitch on nylon strings and the wide, flat fingerboard leaves little room. Use slides and vibrato instead; a small bend can give a blues inflection.

Steel: half-step and small whole-step bends on strings 1–3 work well with light-gauge strings (.011 or lighter).

Electric: with .009–.010 strings, whole-step and larger bends are easy, and bends on the wound 3rd and 4th strings are possible.

Vibrato

Vibrato is a regular wobble in pitch that makes a sustained note sing. Its speed and width are personal: experienced players can often be recognised by their vibrato alone.

Type	How	Where
Axial (classical)	Rock the fingertip along the length of the string, towards and away from the nut. The pitch moves slightly above and below the note	Nylon and classical; subtle and natural
Radial (bend) vibrato	Push the string slightly sideways and let it return, repeatedly. The pitch moves above the note only	Steel-string, electric, blues, rock
Wrist vibrato	Fingertip fixed; the hand rotates from the wrist with the thumb hooked over the neck	Wide blues and rock vibrato
Bent-note vibrato	Bend to pitch, then release slightly and bend back, repeatedly	Long sustained bends

- Practise vibrato to a metronome: two wobbles per click, then three, then four.
- Choose speed and width on purpose. Slow and wide sounds emotional; fast and narrow sounds intense.
- Let the note sound plain first, then add vibrato. This delayed vibrato is very vocal.
- Radial vibrato must return exactly to the note each time, or the note sounds sharp.

▶ **TRACK 354** Plain note, then the same note with vibrato

Listen for a steady speed and width. The vibrato should not change the note's centre pitch.

Slides

- **Legato slide (5/7):** pick once, slide to the new fret keeping firm pressure; the second note is not picked.
- **Shift slide:** slide, then pick the arrival note. It is a way of changing position audibly.
- **Slide in and out (/7 or 7\):** start from an undefined lower fret and glide into the note, or glide away from it at the end.
- **Glissando** (classical *gliss.*): an expressive slide between two melody notes, usually with the shifting finger.
- **Bottleneck slide:** a glass or metal tube on a finger, held directly over the fret wire without pressing the string down. It is usually played in Open G or Open D tuning (Chapter 25). Rest the slide lightly on the strings and mute behind it with a trailing finger.

Slides are also the easiest way to connect scale positions. The example moves from box 1 of A minor pentatonic into box 2 and back.

▶ **TRACK 355** Slides between pentatonic boxes 1 and 2

Legato slides up into box 2 on the 3rd and 2nd strings, a box 2 bend (D up to E on the 1st string), then slides back down on every string. Keep the finger pressing all the way through each slide, or the second note dies. Chapter 7 shows the single 3rd-string shift.

Where the slides connect box 1 and box 2

Box 1 occupies frets 5–8, box 2 frets 7–10. The shaded frets belong to both.

Putting slurs, bends, slides and vibrato together

In real playing these techniques appear in combination, often several in one beat. The lick below uses five of them in two bars. Learn it slowly with the TAB, then play it from memory and listen to whether each technique is audible: a bend that reaches its target, slurs as loud as the picked notes, a slide that keeps the note alive, and an even vibrato at the end.

TRACK 356 A short lick with every technique

Bend, hammer-on and pull-off, slide, bend and vibrato, all in box 1 of A minor pentatonic.

TRY THIS · Your own lick

Write a two-bar lick in A minor pentatonic box 1 that uses one bend, one slur and one slide, and ends on a long note with vibrato. Play it over the slow A minor loop below, or over a loop of your own (Chapter 42).

TRACK 357 Slow A minor loop for bends and vibrato · LOOP · 6/8 ballad · 64 bpm (dotted quarter)

Am <i>i</i>	Am <i>i</i>	F <i>bVI</i>	F <i>bVI</i>
Dm <i>iv</i>	Dm <i>iv</i>	E7 <i>V7</i>	E7 <i>V7</i>

IMPROVISE WITH

A minor pentatonic

Bend G up to A over Am and F (A is in both chords) and D up to E over E7, checking each target against the fretted note. Hold every long note plain for a beat, then add vibrato.

Harmonics

Touch a string lightly directly above a fret wire, without pressing it down, and pick it: you hear a bell-like **natural harmonic**. The touching finger divides the string into equal parts, and only the vibrations that have a still point (a node) under the finger sound. Remove the finger as soon as the note sounds.

Touch at fret	String divided into	Sounds above the open string
12	2 parts	One octave
7 (and 19)	3 parts	An octave and a perfect 5th
5 (and 24)	4 parts	Two octaves
about 3.9, just before the 4th fret wire (also 9 and 16)	5 parts	Two octaves and a major 3rd, slightly flat against the fretted note
about 3.2, just past the 3rd fret wire	6 parts	Two octaves and a perfect 5th

Natural harmonics and the notes they produce

Fret 12 (rust) and fret 5 (teal) give the open string's note, one and two octaves up. Fret 7 gives the 5th, fret 4 the major 3rd.

TRACK 358 Natural harmonics at frets 12, 7 and 5

String	Fret 12	Fret 7	Fret 5
T	<12>	<7>	<5>
A	<12>	<7>	<5>
B	<12>	<7>	<5>

Diamond notes show where to touch the string. Harmonics at frets 7 and 5 sound higher than written (see the table above).

- **Tuning check with harmonics.** The 5th-fret harmonic on one string matches the 7th-fret harmonic on the next higher string, for every pair except G and B. For the B string, compare the 7th-fret harmonic on the 6th string with the open B. Harmonics give pure intervals, which differ very slightly from the tempered tuning of the frets, so finish with a tuner or fretted octaves.
- **Artificial harmonics.** Fret a note, touch the string exactly 12 frets higher with the index finger, and pluck behind it with the thumb or ring finger. Any fretted note becomes a harmonic. Alternating these with ordinary notes gives **harp harmonics**. Both are covered in Chapter 21.
- **Pinch harmonics** (electric): pick so that the side of the picking thumb grazes the string just after the pick. Moving the picking point along the string changes which harmonic squeals out. They need distortion to be effective.

NYLON · STEEL

Nylon: natural harmonics ring clearly but softly; pluck firmly and close to the bridge. **Steel and electric:** harmonics are louder and sustain longer, and an electric's bridge pickup brings out harmonics at frets 5 and 4 that are hard to hear acoustically.

Muting and percussive sounds

Silence is part of the sound. Clean playing depends as much on stopping strings as on starting them.

Technique	How	Sound
Palm muting (P.M.)	Rest the edge of the picking hand lightly on the strings, right at the bridge saddle	Short, tight, chunky
Fret-hand muting	Relax the fretting fingers so they touch the strings without pressing	Percussive "chk" (funk, reggae)
Unused-string muting	Spare fingertips and the side of the fretting finger touch strings you are not playing	Clean single notes and chords
Picking-hand muting	Thumb or palm rests on lower strings; fingers stop strings after a note	Clean lines and staccato bass
Body taps and slaps	Heel, fingers or thumb strike the top or the strings (Chapter 21)	Percussive acoustic fingerstyle

Palm muting. Move the palm towards the neck and the mute gets heavier; move it onto the saddle and the note opens up. A common rock rhythm alternates muted and open strokes.

TRACK 359 Palm-muted chugs with open accents

Palm-mute every unaccented note on the open 6th string. Lift the palm slightly for the accented power chords. All downstrokes.

The muted chuck. Release the fretting pressure of a chord (or lay the picking hand on the strings) and strum: you get a pitchless click. On beats 2 and 4 it acts as a snare drum. Funk (Chapter 45) and reggae (Chapter 52) are built on it.

TRACK 360 Strumming with a chuck on 2 and 4

Play it on any chord, for example two bars each of Am and G. The x is a muted strum: release the fretting pressure but keep the fingers on the strings.

CAREFUL · Unwanted noise

Most noise in a beginner's sound comes from strings that should be silent. Check three places: open strings ringing under a fretted line (mute with the picking hand), the string below the one you fret (touch it with the fingertip), and strings above (touch them with the underside of the fretting finger).

More colours

- **Double stops:** two notes at once. 3rds and 6ths taken from the key sound sweet and are used in country, soul and classical music. Play the 6ths below with the pick and the middle finger, or with p and m, skipping the 2nd string.
- **Tremolo picking:** very fast, even down-up picking on one note (surf, mandolin style). The classical fingerstyle tremolo (p-a-m-i) is a different technique, covered in Chapter 21.
- **Rake:** drag the pick across muted strings into a target note, so the target gets a short percussive run-up.
- **Tapping:** a picking-hand finger hammers onto the fretboard and pulls off to a fretted note. Mostly an electric technique.
- **Ghost notes:** very soft, almost muted notes between the main notes. They add groove without adding pitch.

TRACK 361 Descending 6ths in C major on strings 3 and 1

Same fret on both strings gives a major 6th; the top note one fret lower gives a minor 6th. Slide between the shapes for a country or soul sound.

Chapter 36 review · use it, train it, test it

Using it in real music

Making a melody sing. A bend into the high note, vibrato on the long note and a slide between phrases turn a plain melody into a vocal line. Learn a vocal melody by ear (Chapter 30) and copy the singer's slides and vibrato on the guitar.

Classical expression. Axial vibrato, ligados and natural harmonics are written into a great deal of the classical repertoire, from Carcassi and Tárrega studies onwards.

Rhythm parts. Palm muting and the chuck turn strumming into a drum part. Rock riffs, funk and reggae depend on them more than on the chords.

How to train it

Drill	How	Time	Goal
Bend to pitch	Fret 9 on the 3rd string, then bend fret 7 to match; repeat on the 2nd string (10 and 8)	3 min	Every bend reaches its target
Vibrato to the click	Sustained note at 60 bpm: 2, then 3, then 4 wobbles per beat	3 min	Controlled speed and width
Slur strength	Hammer-on/pull-off pairs on every string, all finger pairs	3 min	Slurred notes as loud as picked ones
Harmonic map	Play the harmonics at frets 12, 7, 5 and 4 on all six strings and name each note	2 min	Clear harmonics; notes named without the chart
Palm-mute control	Eighth notes on the 6th string; move the palm from heavy to light mute and back over 4 bars	2 min	Smooth, deliberate change of mute

Check your understanding

Write your answers, then check them in Appendix J.

1. In TAB, what does "7b9r7" mean?

2. How do you check that a bend is in tune?

3. How does axial (classical) vibrato differ from radial (bend) vibrato?

4. Which note does the 7th-fret natural harmonic on the open A string produce?

5. Why should a pull-off move sideways rather than straight up?

6. Where does the picking hand rest for palm muting?

7. Why are bends rarely used on nylon-string guitars?

8. Which harmonic pair cannot be used to tune the G and B strings, and what can you use instead?

MY NOTES

CHAPTER 37

Rhythm Mastery



Timing is a skill like any other, and it responds to the same kind of deliberate practice as scales and chords. This chapter gives you the subdivisions, the patterns, the metronome methods and a ten-minute daily routine.

Listeners forgive a wrong note more easily than an unsteady beat. A song played with simple chords and solid time sounds finished; the same song with rushed changes sounds like practice. Chapter 27 teaches how rhythm is written. This chapter is about playing it: feeling the beat, dividing it, placing notes exactly, and choosing where in the beat you sit.

Four levels of rhythm skill

Level	What it means	You have it when
1. Pulse	Feeling a steady beat	You tap your foot evenly to any song
2. Subdivision (the key level)	Feeling the notes inside each beat	You switch between eighths, triplets and sixteenths without losing the beat
3. Placement	Putting notes exactly where you choose	Syncopations, rests and pushes land precisely
4. Feel	Sitting ahead of, on or behind the beat on purpose	Other musicians lock in with you easily

Each level depends on the one below. Most timing problems that look like level 3 or 4 problems (a strum pattern that wobbles, a solo that rushes) are really level 2 problems: the player is not feeling the subdivision.

Counting

Count out loud. It feels awkward at first and it is the fastest way to fix rhythm, because your voice keeps the subdivision going while your hands are busy. Use the counts from Chapter 27: "1 & 2 &" for eighths, "1 trip let" for triplets, "1 e & a" for sixteenths and "1 (trip) let" for swing.

The subdivision ladder

Every rhythm is built from subdivisions. Play one note or a muted chord, one bar of each subdivision, then go up and down the ladder without stopping. Keep your foot on the quarter-note beat throughout: the foot never speeds up, only the hand does.

TRACK 362 The subdivision ladder on the open 5th string

Quarters, eighths, triplets, sixteenths. Then come back down. The beat stays the same; only the number of notes inside it changes.

TRACK 363 Binary subdivisions on a sixteenth-note grid

TRACK 364 Triplets and swing on a triplet grid

Swing is the triplet grid with the middle note left out (Chapter 27). Practise moving between the triplet and swing rows without changing the beat: blues shuffles, jazz and much country live on that switch.

TRY THIS · The ladder with a click

Set the metronome to 60 bpm. Play two bars of each subdivision, up and down, counting aloud. When it is smooth, play only one bar of each, then change the order (eighths, sixteenths, triplets, quarters). Raise the tempo to 80 only when every change lands on beat 1.

Syncopation, ties and rests

Syncopation and the **push** (a chord that arrives on the "&" of 4 and is tied over the barline) are explained and notated in Chapter 27. Here the job is to play them so they make the music lean forward, without letting the beat itself move.

Rests are just as important. A short, deliberate silence gives a groove its shape, and in a band it leaves room for the drums and the voice.

TRACK 365 Five ways to place the same eighth-note bar

Accents are rust. In the push, the accented last eighth would be tied into the next bar.

TRACK 366 The push: chords that arrive early

Each new chord starts on the "&" of 4 and is tied over the barline. Do not play beat 1 of the next bar; feel it.

Grooves across styles

A handful of rhythmic cells appear in almost every style. Learn to clap and strum each one, and you will recognise them in songs from Havana to Nashville.

TRACK 367 Rhythmic cells in one bar of sixteenths

All cells are written here in sixteenths so that each fits one bar. They are usually written in eighth notes at twice this length: the tresillo and Charleston then fill one bar, the claves two bars (or one bar of cut time).

Cell	Where you hear it
Tresillo (3 + 3 + 2)	Latin music, New Orleans second line, rock and roll, countless pop choruses and dance tracks
Charleston	1920s jazz, swing comping, many pop rhythm parts
Son clave	Cuban son, salsa (Chapter 50)
Bossa clave	Bossa nova; the last stroke is one grid step later than in son clave (an eighth note in the usual two-bar notation)
Backbeat (2 and 4)	Rock, pop, soul, country, blues
Off-beat skank	Reggae (Chapter 52) and ska

TRACK 368 Tresillo strum

The tresillo at sixteenth-note level, twice per bar. Accent the first group; keep the hand moving down and up on every sixteenth, touching the strings only where a stroke is marked.

Training with the metronome

A metronome only helps if you listen to it. The stages below move the click further away so that your own sense of time does more of the work.

Click on	What it trains
1. Every beat	Basic steadiness
2. Beats 2 and 4 only (set the metronome to half tempo and treat each click as a backbeat)	Backbeat feel: how drummers think
3. Beat 1 only (one click per bar)	Your internal clock fills in beats 2–4
4. The "&"s (hear the click as off-beats)	Off-beat awareness; reggae and funk feel
5. Gap click: 2 bars on, 2 bars silent	Holding tempo alone; the click tells you if you drifted

Click on	What it trains
6. A drum loop instead of a click	Locking with a groove and the bass drum

SHORTCUT • The gap-click test

Many metronome apps can mute bars at random or on a cycle. If you finish the silent gap exactly with the click, your internal clock is solid. If you arrive early, you rush; late, you drag. Most players rush, especially in loud or exciting passages and on fills.

Tempo work. Raise the tempo with the tempo ladder (Chapter 38). For timing, also practise well below your comfortable tempo: very slow playing (50–60 bpm) exposes timing errors that speed hides, because each gap between notes is long enough to rush.

Feel: where you place the beat

Two players can hit every note at the same moment on paper and still feel different, because they place notes very slightly ahead of or behind the exact beat. This placement is called **feel** or **time feel**.

Placement	Effect	Typical styles
On top of the beat (slightly ahead)	Urgent, driving	Punk, bluegrass, some funk
In the middle	Neutral, tight	Pop, most rock
Behind the beat (laid back)	Relaxed, heavy	Reggae, soul, slow blues, neo-soul (Chapter 55)

- **Lock with the kick drum and the bass.** In a band, the guitar's strong strums should land with the kick drum, and short chords should leave space around the snare.
- **Practise both sides.** Play along with a drum loop and deliberately lean ahead for a minute, then behind for a minute. The difference is a few milliseconds, but you will hear it.
- **Consistency matters most.** A feel that is slightly behind but consistent sounds intentional. One that drifts sounds unsteady.

▶ TRACK 369 Locking with the drums

Kick
Snare
Guitar

1 e & a 2 e & a 3 e & a 4 e & a

The guitar's accented strokes coincide with the kick; it rests on the snare beats.

Odd metres

An odd metre has a number of beats that does not divide into equal twos or threes. Players feel it as a mix of groups of two and three. Dave Brubeck's "Take Five" is in 5/4; Pink Floyd's "Money" is mostly in 7/4; many Balkan dances are in 7/8 or 11/8.

▶ TRACK 370 5/4 grouped 3 + 2

Beats
Strum

1 & 2 & 3 & 4 & 5 &

▶ TRACK 371 Strumming in 5/4

1 & 2 & 3 & 4 & 5 &

▶ TRACK 372 7/8 grouped 2 + 2 + 3

2 + 2 + 3
3 + 2 + 2

1 2 3 4 5 6 7

▶ **TRACK 373** A riff in 7/8 (2 + 2 + 3)

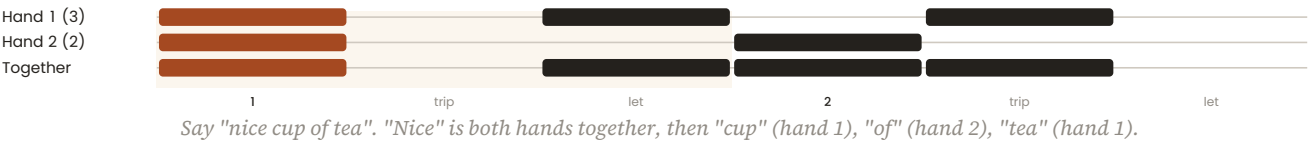
Count "1 2, 1 2, 1 2 3" or "quick, quick, slow". The accents mark the start of each group.

How to learn an odd metre. Count the groups out loud first ("1 2 1 2 1 2 3"), then clap the accents, then play. Do not try to feel the bar as one long unit: feel the groups.

Polyrhythm and cross-rhythm

A **polyrhythm** plays two different divisions of the same time span at once. The most common is 3 against 2. A **cross-rhythm** places a repeating group against the bar so that its accents shift, such as a three-note figure repeated in a 4/4 bar of sixteenths.

▶ **TRACK 374** 3 against 2



▶ **TRACK 375** 3 over 4: a three-sixteenth figure across a 4/4 bar



West African music layers several such patterns at once (Chapter 56), and flamenco's 12-beat cycles mix groups of 3 and 2 (Chapter 53).

TRY THIS · Tap it before you play it

Tap 3 against 2 on your knees, right hand on the threes, left on the twos, until you can keep it going for a minute while counting "1 2 3" out loud. Then swap hands. Then play the threes on muted strings while your foot taps the twos.

The daily 10-minute rhythm routine

This routine is the core rhythm practice of the book; other chapters refer back to it. Do it at the start of a session, after a short warm-up, with a metronome or drum loop. Change the tempo and the material each week.

Minutes	Exercise	Details
0–2	Subdivision ladder	Muted chord, counting aloud: quarters, eighths, triplets, sixteenths, and back
2–4	Backbeat strum	One strum pattern (Chapter 22) with the click on 2 and 4 only
4–6	Gap-click test	Hold the same pattern through 2 silent bars; note whether you rushed or dragged
6–8	Rhythm cell	Clap, then play, one pattern from this chapter: tresillo, clave, push, 3 against 2, or an odd metre
8–10	Play along	Play with a real recording and copy the guitarist's rhythm part exactly, including rests and accents

TIP · Record and look

Record yourself to a click in any recording software (Chapter 42) and look at the waveform against the grid. You will see exactly where you rush or drag. Then listen without looking: the ear is the final judge.

TIP · Move

Nod, tap your foot, sway. Rhythm lives in the body, and players who move with the beat keep better time than players who sit rigid.

NYLON · STEEL

Nylon: classical players usually play without a drummer, so the internal pulse carries everything. Count rests fully and practise pieces with a metronome before adding rubato (expressive tempo changes); rubato only works against a pulse the listener can feel. **Steel and electric:** players who accompany singers or play in bands should practise with drum loops as much as with a click, to develop groove and to lock with a kick drum.

Chapter 37 review · use it, train it, test it**Using it in real music**

Every song you play. Steady pulse and clear subdivision matter more than any chord or scale in this book. If a song sounds wrong and the notes are right, check the rhythm first.

Playing with others. Locking with drums and bass, and leaving space, are the skills bandmates value most.

Learning a new strum or riff. Clap and count it before you play it. If you cannot clap it, you cannot play it.

How to train it

Drill	How	Time	Goal
Subdivision ladder	Quarter → eighth → triplet → sixteenth on a muted chord, counting aloud	2 min	Every change lands on beat 1
Gap click	Click on for 2 bars, silent for 2; finish with the click	3 min	Within a sixteenth of the click after the gap
Click on 2 and 4	Metronome at half tempo, felt as the backbeat; play a strum pattern	3 min	Natural, relaxed groove
Push practice	Play a progression twice: once on beat 1, once pushed to the "&" of 4	3 min	Pushes land early and hold; beat 1 is felt, not played
Odd-metre count	Count and clap 5/4 (3 + 2) and 7/8 (2 + 2 + 3) for one minute each	2 min	Steady groups without stopping

Check your understanding

Write your answers, then check them in Appendix J.

1. Name the four levels of rhythm skill in this chapter.

2. When you play a push into the next bar, what do you do on beat 1 of that bar?

3. Which subdivision level are most strumming and soloing timing problems really about, and why?

4. Write the grouping of the tresillo.

5. How is 7/8 commonly grouped?

6. What does the gap-click test reveal?

7. In 3 against 2, which parts of the cycle do both rhythms share?

8. Why practise with the metronome on 2 and 4?

MY NOTES

PART TEN

Putting It to Work

Practice methods, common mistakes, and worked examples that combine everything in the book.

- 38 Practise Smarter
- 39 What to Avoid
- 40 How It All Fits Together

CHAPTER 38

Practise Smarter



Progress depends less on how long you practise than on how you practise. This chapter shows how to design a session, how to work on a hard passage, how to space and mix material so it sticks, and how to measure progress.

Two players can practise for the same number of hours and progress at very different rates. The difference is rarely talent. It is whether each minute has a goal, whether mistakes are caught and corrected, and whether material comes back often enough to stay in memory. These are habits, and you can learn them as you learn chords.

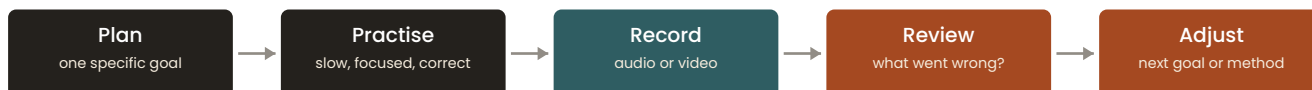
Designing a practice session

Every session, however short, answers four questions before you start.

Question	Example
Goal: what will be better at the end?	"The change from C to F in bar 6 at 70 bpm, three times in a row"
Material: what exactly will you play?	Bars 5–7 of the piece, plus the C and F chord shapes alone
Method: how will you practise it?	Slow, chunked, with a metronome; then in context
Measure: how will you know?	Metronome setting, number of clean runs, a recording

Write the goal at the top of a page or in a notes app. A goal you can measure ("three clean runs at 70 bpm") is far more useful than a vague one ("get better at barre chords").

The practice loop



Deliberate practice

Research on expert performance in music, sport and chess finds that the kind of practice that builds skill has four features. Playing pieces you already know is enjoyable and keeps them alive, but it is not this kind of practice.

1. **A specific goal just beyond your current level.** Not so easy that it is automatic, not so hard that you cannot get it right at any tempo.
2. **Full attention.** Twenty focused minutes achieve more than an hour with the television on. When your attention fades, stop or change the task.
3. **Immediate feedback.** A metronome, a tuner, a mirror, a recording or a teacher tells you what actually happened, not what you think happened.
4. **Repetition with correction.** Each repetition fixes something. Repeating a mistake practises the mistake.

Methods that work

Slow is fast. Play at a tempo at which you make no mistakes, even if that is half speed. The hands learn the movements you repeat, including the errors. Speed comes from clean, relaxed repetitions, not from forcing.

The tempo ladder. Find the tempo at which you can play a passage cleanly three times in a row. Raise the metronome by 4–8 bpm and repeat. When you fail twice, drop back 8–12 bpm, play cleanly, and climb again. Stop the ladder for the day after about ten minutes; the next day, start a little below yesterday's best.

Chunking. Break a passage into small pieces, sometimes only two or three notes. Master each piece, then join two pieces, then three. Mistakes usually happen at the joins (a position shift, a chord change, a string crossing), so practise each join as a chunk of its own: the last note of one chunk and the first note of the next.

TRACK 376 A passage for chunking

Am E Am

Chunk 1: bar 1, beats 1–2. Chunk 2: bar 1, beats 3–4. Then the join: the last two notes of bar 1 plus the first two of bar 2, where the chord changes. Practise the join most.

Backward chaining. Learn the last bar of a phrase first, then the last two bars, and so on. You always finish on the material you know best, and the ending, which most players neglect, gets the most practice.

Variation. Practise a hard passage in different ways: with dotted rhythms (long-short, then short-long), with accents in different places, louder and softer, with the eyes closed. Each variation forces the hands to know the notes rather than rely on one fixed motion.

TRACK 377 Rhythmic variations of bar 1

Straight, long-short, short-long. When the passage is clean in both dotted rhythms, the straight version usually feels easy.

Retrieval. After learning something, put the book away and play it from memory. Trying to recall it, even unsuccessfully, strengthens memory more than reading it again. The quizzes at the end of each chapter and the flashcards in the web companion (with spaced repetition that brings back the cards you miss) work the same way.

Mental practice. Away from the guitar, picture the fretboard and "play" a passage in your head, seeing each finger move and hearing each note. It is no substitute for playing, but it is a useful extra five minutes on a train.

TIP • One thing at a time

Correct one problem per repetition. If a passage has a wrong note, a rushed beat and a tense thumb, fix the note first, then the timing, then the tension. Trying to fix everything at once fixes nothing.

Spacing and mixing

Spaced repetition. Memory fades quickly after you first learn something, and each review slows the fading. Reviewing at growing intervals keeps material with the least total effort. A new scale, chord vocabulary or piece can follow this schedule.

Review	After first learning	Takes
1	Next day	5 min
2	2 days later	3 min
3	4 days later	3 min
4	1 week later	2 min
5	2 weeks later	2 min
6	1 month later	2 min

If a review goes badly, go back one step. If you use material in songs regularly, it reviews itself.

Short daily sessions. Motor skills continue to consolidate between sessions, particularly during sleep. Twenty minutes every day achieves more than two and a half hours once a week, and it is easier on the hands.

Interleaving. Mix two or three topics in a session instead of drilling one for an hour: ten minutes of a scale, ten of a chord change, ten of a piece, then back to the scale briefly. Interleaved practice feels harder and less fluent in the session, but retention a week later is better.

Use it in music. Every new scale, chord or arpeggio should appear in a song, a riff or an improvisation within a week. Material that is only ever drilled stays an exercise.

Session templates

Each template begins with a short warm-up (see Chapter 39 for a safe one) and ends with something musical. Swap the topics each week; keep the shape.

15 minutes (a busy day)

Time	Activity	Example
2 minutes	Warm-up	Slow chromatic 1-2-3-4 on each string; shake out the hands
8 minutes	One focus	The hardest two bars of your current piece, chunked, with the metronome
5 minutes	Play	Run the whole piece or song once, without stopping for mistakes

30 minutes (the standard session)

Time	Activity	Example
5 minutes	Warm-up and technique	Chromatic exercise; slow arpeggio pattern (Chapter 17) or alternate picking (Chapter 24)
10 minutes	Theory on the neck	One pentatonic box, one triad string set, or fretboard note drills (Chapter 4)
10 minutes	Repertoire	A piece or song; hardest two bars first, then in context
5 minutes	Create	Improvise over a loop, or write a four-chord progression

60 minutes (a long session)

Time	Activity	Example
5 minutes	Warm-up	Stretches, chromatic exercise, slow scale
10 minutes	Rhythm	The daily 10-minute rhythm routine (Chapter 37)
10 minutes	Technique	One technique at a time: bends, slurs, tremolo, barre chords
10 minutes	Theory on the neck	New material on the spaced-repetition schedule
15 minutes	Repertoire	New piece (chunked) and one older piece (played through)
5 minutes	Ear	Sing roots, name intervals, work out a short melody (Chapter 28)
5 minutes	Create and record	Improvise or play through something and record it

CAREFUL · Breaks

In a session longer than 30 minutes, stop for a minute every 20–25 minutes: put the guitar down, drop the shoulders, open and close the hands. Fatigue makes you squeeze, and squeezing is how most playing injuries start.

A 30-day plan

One focused task a day, about 20 minutes, on top of whatever you enjoy playing. If you miss a day, continue from where you stopped: consistency matters more than an unbroken chain. The plan suits a player who knows the open chords; skip days you already know well.

Day	Task (with the chapter to use)
Day 1	Name all open strings; tune by ear (Chapter 1)
Day 2	Notes on the 6th string, frets 0–12 (Chapter 4)
Day 3	Notes on the 5th string, frets 0–12 (Chapter 4)
Day 4	C, G, D, Em, Am: one-minute changes (Chapter 11)
Day 5	A strum pattern on G – D – Em – C (Chapter 22)
Day 6	Major scale formula; C major on one string (Chapter 5)
Day 7	G major scale in one position, two octaves (Chapter 5)
Day 8	Intervals: play and name 3rds and 5ths (Chapter 3)
Day 9	Circle of fifths: name every major key and its sharps or flats (Chapter 5)
Day 10	A fingerpicking pattern on C – Am – F – G (Chapter 18)
Day 11	A minor pentatonic, box 1 (Chapter 7)
Day 12	Improvise for two minutes with box 1 over a loop (Chapter 7)
Day 13	Box 2, and slides between boxes 1 and 2 (Chapter 36)
Day 14	Blues scale; 12-bar blues in A (Chapter 43)
Day 15	Triads on strings 3–2–1 in C major (Chapter 10)
Day 16	Triad inversions through C – Am – F – G (Chapter 10)
Day 17	E-shape barre chord at the 5th fret (Chapter 11)
Day 18	A-shape barre chord at the 3rd fret (Chapter 11)
Day 19	Chords in G and D with Roman numerals (Chapter 13)
Day 20	I – V – vi – IV in three keys (Chapter 14)
Day 21	Travis pattern on G – Em – C – D (Chapter 19)
Day 22	One chord in all five CAGED shapes (Chapter 12)
Day 23	A major and A minor arpeggio shapes (Chapter 16)
Day 24	Andalusian cadence in 3/4 or 6/8 (Chapter 31)
Day 25	Dorian against Aeolian over a drone (Chapter 9)

Day	Task (with the chapter to use)
Day 26	Drop D: play a riff (Chapter 25)
Day 27	Open G: I – IV – V with one finger (Chapter 25)
Day 28	Sixteenth-note funk strum on Em7 – A7 (Chapter 45)
Day 29	Analyse a favourite song in eight steps (Chapter 40)
Day 30	Record a complete song from start to finish

Tracking progress

Progress on the guitar is slow enough that you do not notice it from day to day. A written log makes it visible and shows you what to do next. After each session, write one line.

Session	Practised	Tempo, time	Result / next time
Session 1	Bars 5–8 of the study, chunked	60 bpm, 15 min	Bar 6 change clean; join 7–8 still slow
Session 2	Bars 5–8, then bars 1–8	63 bpm, 15 min	Bars 1–8 clean twice; try 66 next time
Session 3	A minor pentatonic box 2	80 bpm, 5 min	Spaced review 2; one slip on the 2nd string

Useful numbers to track: metronome tempo for a passage, the number of clean runs out of five, how many minutes a chord change takes to become smooth, and the number of songs you can play from memory.

Recording yourself

While playing, most of your attention goes to the hands, so you hear only part of what comes out. A recording hears everything. A phone is enough.

- **Record once a week**, the same piece or exercise, with the same phone position. Keep the files and compare with last month, not with yesterday.
- **Listen for one thing per playback**: first timing, then clean notes (buzzes, unwanted ringing), then dynamics and tone.
- **Use video for technique**. Watch for a bent fretting wrist, raised shoulders, a thumb squeezing the neck, fingers flying off the strings, or a strumming hand that stops between strokes.
- **Record performance takes**. Pressing record creates a little pressure. Learning to play through mistakes with the red light on is good preparation for playing for others.

TIP · Breaking a plateau

When progress stalls for weeks, the cause is usually one of three things: you always play at the same tempo (slow down, or use the tempo ladder), you always play the same material (change it), or you never check yourself (record and listen). Change one of them for two weeks.

Chapter 38 review · use it, train it, test it

Using it in real music

Learning a new piece. Read through it once slowly, mark the two hardest places, chunk those first, then chain the piece together. Put the new piece on the spaced-review schedule.

Planning your week. Use the 30-minute template as the default and keep the topics on a weekly rotation, so that theory, technique, rhythm and repertoire all get time.

How to train it

Drill	How	Time	Goal
Practice log	One line after each session: what, tempo, minutes, result	1 min	A visible record for a month
Tempo ladder	One hard passage: climb 4–8 bpm per three clean runs; record the top tempo	10 min	Top clean tempo rises week by week
Join practice	Find the hardest join in a piece; play only that join 10 times slowly	3 min	No hesitation at the join
Weekly recording	Record one piece or improvisation; compare with last month	5 min	Hear your improvement
Memory check	Play yesterday's new material without looking at the book	2 min	Retrieved, not re-read

Check your understanding

Write your answers, then check them in Appendix J.

1. When should you raise the metronome tempo, and by how much?

2. What are the four features of deliberate practice?

3. Where do most mistakes in a chunked passage occur, and what do you do about it?

4. Why are short daily sessions better than one long weekly session?

5. What is interleaving?

6. Name a spaced-repetition schedule for new material.

7. What is backward chaining?

8. Name three things to watch for when filming yourself.

MY NOTES

CHAPTER 39

What to Avoid



TRACKS

39

Most guitarists who stop improving are held back by a small number of habits in technique, practice and health. This chapter lists the common ones, how to recognise them and what to do instead.

Few players are held back by a lack of talent. They are held back by habits that slow them down without being noticed: too much tension, practice without a goal, theory learned as shapes without names, a guitar that is hard to play. Each habit is easy to fix once you see it. Read this chapter once now and again in a few months; different items will apply each time.

A quick diagnosis

Start from the symptom. The table points to the usual cause and the fix.

Symptom	Usual cause	Fix
Buzzing notes	Finger too far from the fret, too little pressure, or high/uneven frets	Fret just behind the wire; if the buzz stays, have the guitar checked
A string in a chord does not sound	A neighbouring finger is touching it	Arch the fingers; play the chord string by string to find the culprit
Hand tires after a few minutes	Squeezing with the thumb; action too high	Press lighter; check the setup (Chapter 41)
Chord changes stop the rhythm	Practising chords and rhythm separately	Keep strumming; let the change be late, then fix it slowly
Playing rushes when it gets exciting	No subdivision in your head	Count aloud; gap-click training (Chapter 37)
Solos all sound the same	Always in pentatonic box 1, always starting on beat 1	Use other boxes, target chord tones, start phrases off the beat
A piece never gets past a certain point	Always starting from the beginning	Practise the hardest bars first and in isolation (Chapter 38)
Pain in wrist, forearm or fingers	Tension, a bent wrist, too long without breaks	Stop. See the health section below

Fretting hand

Avoid	Do this instead
Squeezing the neck with the thumb (the "death grip")	Press just hard enough for a clean note. Use the weight of the arm pulling back, not the thumb pushing forward
Pressing in the middle of the gap between frets	Press just behind the fret wire: cleaner note, less effort
Fingers lifting high off the strings	Keep the fingers hovering close to the fretboard, ready for the next note
Collapsed fingertips that mute other strings	Arch the fingers and play on the very tips; trim the nails of the fretting hand
A sharply bent fretting wrist	Bring the neck up and the elbow in until the wrist is fairly straight
Lifting all fingers between chords	Leave common fingers down; move fingers that change as one shape

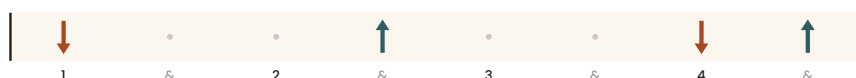
TRY THIS · The lightest pressure test

Pick a note at the 5th fret of the 3rd string repeatedly while releasing pressure until it buzzes, then add just enough to make it clean. That is all the pressure you need.

Picking hand and rhythm

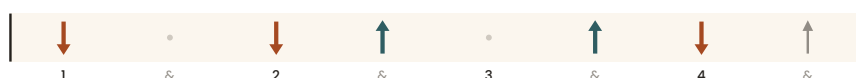
Avoid	Do this instead
Stopping the strumming hand between strums	Keep the hand moving down and up like a pendulum; miss the strings instead of stopping
Strumming from the elbow with a locked wrist	Let the wrist and forearm rotate; the motion should be loose
Gripping the pick hard	Hold it just firmly enough not to drop it; let it give a little on the strings
Big picking motions	Keep the pick close to the strings; small movements are faster and more accurate
Thumb drifting off the bass when fingerpicking	Practise the thumb part alone until it is automatic (Chapter 19)
Ignoring the metronome	Five minutes with a click or drum loop every day
Stopping at every mistake	In a run-through, keep going and fix the mistake afterwards

▶ TRACK 378 The pendulum: the hand never stops



Four strokes, eight motions. Down on every beat, up on every "&", whether or not the strings are struck. Sparse patterns like this are where a hand that stops loses the beat first.

▶ TRACK 379 Changing chords on the last upstroke

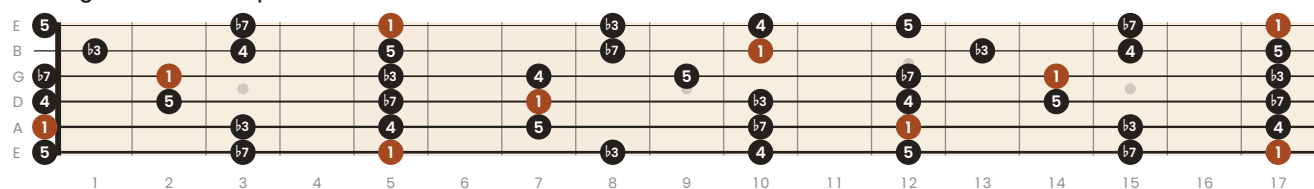


Old faithful (Chapter 22) with a light last upstroke (grey). Play it on the open strings while the fretting hand is already moving to the next chord, so the next downbeat stays on time.

Learning theory

Avoid	Do this instead
Memorising shapes without knowing the roots	Mark and name the root in every shape you learn
Soloing only in pentatonic box 1 ("box prison")	Learn boxes 2 and 5 next, and slide between them; play one solo per box
Learning scales you never use	Use every new scale in a song or improvisation within a week
Diving into modes too early	Master major, minor and pentatonic first; modes are variations of them (Chapter 9)
Reading only TAB	Learn chord symbols, Roman numerals and some standard notation (Chapter 26)
Learning theory only from books	Hear it: play every concept and listen to what it does

Leaving box 1: A minor pentatonic across the neck



Box 1 is shaded (frets 5–8). The same five notes continue in both directions. Learn the boxes on either side first, then connect them with slides (Chapter 7).

Practice habits

Avoid	Do this instead
Playing only what you already know	Spend at least a third of each session on something new or difficult
Practising fast and sloppy	Slow down until it is clean, then raise the tempo 4–8 bpm at a time
Starting a piece from the beginning every time	Start with the hardest bars and loop them in isolation
Noodling without a goal	Write one specific goal at the top of every session
Long sessions once a week	Short daily sessions: 20 minutes a day beats 3 hours on Sunday
Practising only alone	Play with recordings, friends or at an open mic; it builds real timing

Gear and setup

Avoid	Do this instead
Old, dull strings	Change them regularly; fresh strings play in tune and sound alive
A guitar with high action	Have it set up by a technician; it can greatly reduce the effort needed
Tuning only at the start of a session	Check the tuning between pieces, especially with new or nylon strings
The wrong capo	A flat capo on a flat classical fingerboard, a curved one on a radiused steel-string or electric fingerboard
A strap so long that the guitar hangs at the knees	Set the strap so the guitar sits at about the same height standing as sitting
Raising tuning far above standard (for example to Open E) on an acoustic	Tune down to Open D and play with capo 2; it avoids extra tension on the neck and top

NYLON · STEEL

Nylon: avoid resting the picking hand on the top, ragged or uneven nails, sitting without a footstool or support, and over-tightening new strings (they stretch for several days). Keep the guitar away from heaters and very dry air. **Steel and electric:** avoid anchoring the strumming hand so the wrist cannot move, pressing harder to make up for a high action, and leaving the guitar in a hot car. On electric, avoid practising only with heavy distortion: it hides unclear playing.

Health: tension, injury and warming up

Playing the guitar is a physical activity. Most playing-related problems come from too much tension held for too long, often combined with poor posture and too few breaks. The advice below is general. **If pain, numbness, tingling or loss of control persists or keeps returning, stop playing and see a doctor or a physiotherapist, ideally one who works with musicians.**

Signs of tension

- Raised shoulders, a clenched jaw, or holding your breath during hard passages.
- A thumb that leaves a mark on the back of the neck, or fingers that turn white at the tips.
- Fingers of the picking hand that stick out stiffly.
- A forearm that feels tight or warm after a few minutes.

Every 10 minutes, check shoulders, jaw, thumb and breathing, and let go of anything that is held.

Preventing injury

- **Breaks.** Stop for a minute every 20–25 minutes, and longer after an hour.
- **Build up slowly.** Increase practice time gradually, especially after a break of weeks, when changing to a heavier string gauge, or when learning a new stretch or barre.
- **Avoid long, repetitive overload.** Do not practise one hard movement (wide stretches, barre chords, fast tremolo) for long without variety.

- **Posture.** Keep the fretting wrist fairly straight, the shoulders level and the guitar brought to you (Chapter 1).
- **Warm hands.** Cold hands are stiff. Warm them with water or by moving before playing.
- **Pain is a signal, not a challenge.** Never "play through" pain. Some soreness in new fingertips is normal; sharp pain, aching joints or pain that lasts after playing are not.

CAREFUL · When to get help

See a professional promptly for pain that continues after you stop playing, pain at night, numbness or pins and needles in the fingers, swelling, or a finger that curls or loses control without your intending it. A rare condition called focal dystonia affects musicians' fine control and needs a specialist. Early advice usually means a shorter break.

A five-minute warm-up

1. **Loosen** (1 minute): roll the shoulders, shake out the hands, open and close the fists slowly. Move gently; do not force any stretch.
2. **Chromatic walk** (2 minutes): the exercise below, slowly, with relaxed fingers and the lightest pressure that gives a clean note.
3. **Finger independence** (1 minute): the second exercise, slowly.
4. **Easy music** (1 minute): a few chords or an easy scale before anything demanding.

▶ TRACK 380 Chromatic walk in 5th position

The exercise is a chromatic walk in 5th position, starting on the 5th fret. The first staff shows the treble clef and the first four strings (T, A, B, E). The second staff shows the bass clef and the last four strings (D, G, B, E). The notation includes fingerings (1-2-3-4 up, 4-3-2-1 down) and alternate picking (i and m).

One finger per fret (1-2-3-4 up, 4-3-2-1 down). Leave each finger down until the next string. Alternate picking, or alternate i and m.

▶ TRACK 381 Finger independence (fingers 1-3 and 2-4)

The exercise is finger independence (fingers 1-3 and 2-4) in 5th position, starting on the 5th fret. The first staff shows the treble clef and the first four strings (T, A, B, E). The second staff shows the bass clef and the last four strings (D, G, B, E). The notation includes fingerings (1-3 and 2-4) and alternate picking (i and m).

Two fingers move while the others stay close to the strings. Stop if the hand tightens.

Fingertips and calluses

New players' fingertips are sore for the first weeks, more on steel than on nylon. Short, frequent sessions (10–15 minutes, several times a day) build calluses faster and with less pain than long ones. Do not pick at peeling calluses, and avoid long playing straight after a hot bath, when the skin is soft.

Hearing

Loud amplifiers, drums and long headphone sessions can damage hearing permanently. Keep headphone volume moderate, take listening breaks, and use musicians' earplugs in loud rehearsals and concerts (Chapter 42).

Mindset

Avoid	Do this instead
Comparing yourself with players online	Compare yourself with your own recording from last month
Giving up on barre chords after a week	Most players need 6–8 weeks of a few minutes a day; use a capo and partial barres meanwhile
Waiting until you are "good enough" to play for others	Play for one friend, then a few; small performances build confidence
Treating mistakes as failure	Treat them as information: what, where, why. Then fix one

TIP · One fix a week

Choose one item from this chapter each week and work on it in every session. In two months you will have removed the habits that hold many players back for years.

Chapter 39 review · use it, train it, test it

Using it in real music

Diagnosing a problem. Stuck on a piece, a chord change or a tempo? Start with the diagnosis table; the cause is usually listed there.

Teaching or playing with friends. These are the first things good teachers fix. Spotting them in others helps you spot them in yourself.

How to train it

Drill	How	Time	Goal
Habit audit	Film yourself for 2 minutes; check fretting pressure, wrist angle, shoulders and strumming motion	5 min	One habit chosen to fix this week
Tension check	Every 10 minutes, pause and release shoulders, jaw, thumb and breath	30 sec	Playing relaxed by default
Lightest pressure	The pressure test on each finger, then on a barre chord	2 min	Clean notes with minimum force
Warm-up	The five-minute routine before each session	5 min	Warm, loose hands before hard material
One fix a week	Work on one item from the tables in every session	5 min	Habits replaced, not just noticed

Check your understanding

Write your answers, then check them in Appendix J.

1. What is the "death grip", and what should you do instead?

2. What should you do if you keep making the same mistake at full speed?

3. What is "box prison", and how do you escape it?

4. Why should the strumming hand keep moving when it is not striking the strings?

5. Which capo suits a classical guitar, and why?

6. Why is it better to tune an acoustic down to Open D and capo than

7. Name three warning signs that mean you should see a professional.

8. How do you build calluses with the least discomfort?

MY NOTES

CHAPTER 40

How It All Fits Together



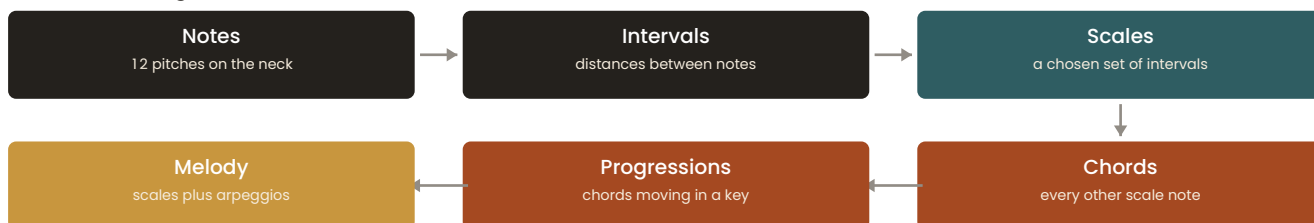
TRACKS

40

Worked examples that use the whole book at once. A progression is analysed, a solo is built over it, a minor-key cadence is arranged, a melody becomes a fingerstyle piece, and the main styles are summarised as recipes.

Every chapter so far has taken one subject at a time. Real music uses all of them at once: a song has a key, its chords have numbers, each chord has shapes all over the neck, and the melody or solo moves through a scale while landing on chord tones. This chapter works through complete examples to show the thinking, step by step.

How the building blocks connect



Arpeggios are the bridge between the last two boxes: they are chord tones played one at a time, so a melody built on them "speaks" the harmony. The fretboard is the map on which all six live at once.

Example 1: analysing a pop progression

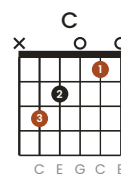
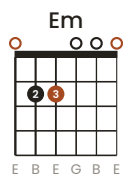
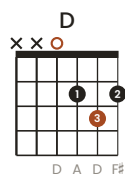
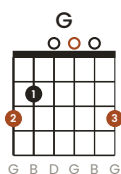
The progression G – D – Em – C, repeated, is one of the most common in pop music. Work through it in eight steps.

G – D – Em – C

G <i>I</i>	D <i>V</i>	Em <i>vi</i>	C <i>IV</i>	x4
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Step and question	Answer
1. Key: which major scale contains all the chords?	G major (G A B C D E F#). G sounds like home (Chapter 5)
2. Numerals: what are the Roman numerals?	I – V – vi – IV (Chapter 13)
3. Chords: how can I voice it?	Open chords; or barre chords: E-shape at the 3rd fret, A-shape at the 5th (Chapter 11)
4. Triads: what can a second guitar play?	Top-string triads with smooth voice leading around frets 2–5 (Chapter 10)
5. Scale: what do I solo with?	G major pentatonic: the same notes as E minor pentatonic (Chapter 7)
6. Position: where on the neck?	Open position, or the box at frets 12–15
7. Targets: which notes fit each chord?	G: G B D · D: D F# A · Em: E G B · C: C E G (Chapter 16)
8. Colour: how can I make it richer?	G – D/F# – Em7 – Cadd9, with a descending bass line

The progression in open chords

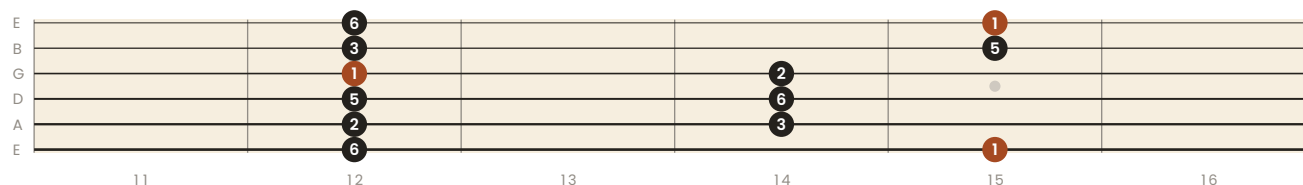


Step 8: the coloured version, with the bass falling G – F# – E – C



The 3rd and 4th fingers stay on fret 3 of strings 1 and 2 through G, Em7 and Cadd9, so the top of every chord rings the same notes. Chapter 47 builds a whole chord family on this anchor.

G major pentatonic at the 12th fret, labelled from G



The same five notes are E minor pentatonic box 1. They fit all four chords; what changes from chord to chord is which of them are chord tones.

HOW IT CONNECTS · Scale for moving, chord tones for landing

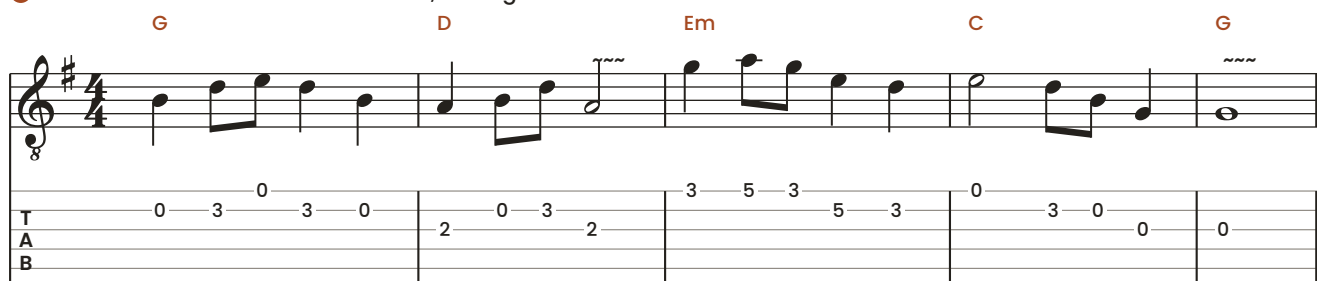
One set of five notes serves the whole progression, while the arpeggio notes change with each chord. Use the scale to move, and land on a chord tone when the chord changes. That single idea is the core of melodic improvising (Chapter 35).

Building a solo over the progression

A solo is a short composition. Build it in layers.

1. **Find the targets.** For each chord, choose one chord tone to land on at the start of the bar: B over G (the 3rd), A over D (the 5th), G over Em (the $\flat 3$), E over C (the 3rd). Thirds say the most about a chord.
2. **Connect them** with notes from G major pentatonic. Step-wise movement sounds like singing; leaps sound like an instrument.
3. **Make a motif.** A short rhythmic and melodic idea of three or four notes. Repeat it, then vary it: move it to another pitch, change one note, change its rhythm.
4. **Shape it.** Start low and simple, rise towards a high point near the end, and come back down to the root.
5. **Leave space.** Rests let a phrase breathe and give the listener time to hear it.
6. **Add expression.** Vibrato on long notes, a slide into a target, a hammer-on in a fast figure (Chapter 36).

▶ **TRACK 382** A solo over G – D – Em – C, ending on G



Every bar starts on a chord tone. Bar 3 repeats the shape of bar 1 higher up; the last two bars descend to the root.

TRY THIS · Your own solo

Use the loop below, or record your own (Chapter 42), and write your own four bars by the same method. First play only the four target notes, one per bar, as whole notes. Then add one connecting note before each target. Then add a motif. Record each version.

TRACK 383 Pop loop in G: I – V – vi – IV · LOOP · pop · 84 bpm

G <i>I</i>	D <i>V</i>	Em <i>vi</i>	C <i>IV</i>
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IMPROVISE WITH

G major pentatonic

G major

Play the four targets first (B, A, G and E on beat 1), then the written solo above, then your own. Repeat the motif from bar 1 over Em, as the written solo does, and end on G as the loop returns to the G chord.

Example 2: the Andalusian cadence in A minor

The progression Am – G – F – E descends step by step and sounds dramatic and Spanish. It is introduced in Chapter 14, explained in Chapter 31, is the backbone of flamenco (Chapter 53) and appears in many rock and pop songs.

The Andalusian cadence

Am <i>i</i>	G <i>bVII</i>	F <i>bVI</i>	E <i>V</i>
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- **Key:** A minor. Am is *i*, G is *bVII*, F is *bVI* and E is *V*.
- **The twist:** E major is not in A natural minor. It contains G#, the raised 7th borrowed from A harmonic minor (Chapter 6). The G# pulling up to A gives the cadence its strength when it returns to Am.
- **Scale choice:** A natural minor over Am, G and F; A harmonic minor over E (the same notes as E Phrygian dominant).
- **Arpeggio targets:** over E, aim for G# and B; resolve G# up to A when Am returns.
- **Bass line:** A – G – F – E is a falling scale. Play it with p on the 5th and 6th strings while *i*, *m* and *a* play the chords above: a classic nylon-string texture.

A harmonic minor in 5th position

The 7th (G#, gold) is the note that defines the E chord. Over Am, G and F, play G natural instead.

TRACK 384 The cadence arranged: falling bass, chords above

Am G F E Am

	0	0	3	3	1	1	0	0	0
T	1	1	0	0	1	1	0	0	1
A	2	2	0	0	2	2	1	1	2
B	0		3		1		0		0

The thumb plays the bass on beat 1 and lets it ring; *i*, *m* and *a* pluck the chords on beats 2 and 3. Listen for the G# in bar 4 rising to A in bar 5.

▶ **TRACK 385** A phrase over the cadence

Am G F E Am

5 6 8 5 | 8 6 5 7 | 5 7 5 6 | 5 5 4 6 | 7

T
A
B

Natural minor for three bars, then G# over E (bar 4), resolving up to A.

Arranging a melody for fingerstyle guitar

Turning a melody into a solo guitar piece follows the same three stages for almost any tune. The example uses the opening of the main theme from the last movement of Beethoven's Ninth Symphony ("Ode to Joy"), which is in the public domain. Chapter 20 develops the technique fully.

1. **Melody alone.** Find the melody in a position where it lies on the top two or three strings, so the bass strings are free.
2. **Add the bass.** Choose a chord for each bar or half bar and play its root on beat 1.
3. **Fill in.** Add an alternating bass or chord tones on the middle strings, keeping the melody the loudest voice.

▶ **TRACK 386** Stage 1: the melody alone

0 0 1 3 | 3 1 0 3 | 1 1 3 0 | 0 3 3

T
A
B

▶ **TRACK 387** Stage 2: melody and bass roots

C G C G

0 0 1 3 | 3 1 0 3 | 1 1 3 0 | 0 3 3

T
A
B

TRACK 388 Stage 3: melody over an alternating bass

0 0 1 3 | 3 1 0 3 | 1 1 3 0 | 0 3 3

T

A 2 2 | 0 0 | 2 2 | 0 0

B 3 3 | 3 3 | 3 3 | 3 3

The bass alternates root and a chord tone on the 4th string. Pluck the melody slightly louder than the bass, with a rest stroke if you play nylon strings.

TRY THIS · Arrange another tune

Choose a folk tune you know well, such as a traditional song or a nursery rhyme. Find its melody on strings 1–3, choose one chord per bar, and work through the three stages. Keep the arrangement simple enough to play at the tune's real tempo.

Style recipes

Every style combines the same building blocks in its own way. Use these recipes as starting points, then listen to players in each style and copy their favourite moves. The style workshops develop each one.

Style	Typical guitar, keys and tuning	Progression	Scales and sounds	Picking hand
Blues (Chapter 43)	Any; E or A; standard, Open G or Open D for slide	12-bar I7 – IV7 – V7	Minor pentatonic and blues scale; b3 bent towards 3	Shuffle strum, alternating-thumb picking
Folk and songwriting (Chapter 47)	Steel; G, D, C with capo; standard or double drop D	I – IV – V, I – V – vi – IV	Major pentatonic, drones	Strum patterns, Travis picking
Pop ballad	Any; C, G, D with capo	vi – IV – I – V	Major and minor pentatonic	Arpeggio patterns, 6/8 roll
Rock (Chapter 44)	Electric; E, A, D; standard or drop D	I – bVII – IV, power-chord riffs	Minor pentatonic, Mixolydian	Palm muting, downstroke riffs
Funk (Chapter 45)	Electric; E, A; standard	One-chord vamps on 7th and 9th chords	Dorian, Mixolydian, chord fragments	Sixteenth strum with chucks
Country (Chapter 46)	Steel or electric; G, C, E	I – IV – V with bass runs	Major pentatonic plus b3	Boom-chick, hybrid picking
Jazz (Chapter 48)	Archtop or nylon; F, Bb, C	ii7 – V7 – Imaj7	Arpeggios, chord scales	Comping on 2 and 4, Charleston
Bossa nova (Chapter 50)	Nylon; C, D minor	ii7 – V7 – Imaj7	Major scale, 7th and 9th chords	Thumb on 1 and 3, syncopated pinches
Flamenco (Chapter 53)	Nylon (flamenco); A minor / E Phrygian	Am – G – F – E	Phrygian, harmonic minor	Rasgueado, picado, p-a-m-i
Celtic (Chapter 54)	Any; D, G; DADGAD	I – bVII – IV over drones	Mixolydian, Dorian	Pinch patterns, light 6/8 strum
Reggae (Chapter 52)	Electric; minor keys	i – iv, I – IV – V	Minor pentatonic, triads	Off-beat skank, muted chops
Classical study (Chapter 58)	Nylon; E minor, A minor; standard or drop D	i – iv – V7 – i	Harmonic and melodic minor	p-i-m-a arpeggios, rest-stroke melody

A 12-week roadmap

For a player who knows the open chords and wants to connect the material in this book, this plan gives each topic two weeks in the order the chapters build on each other. For a complete year of study, use the plans in Appendix A.

Weeks and focus	Goal by the end
Weeks 1–2: open chords, rhythm, notes on strings 6 and 5	Change smoothly between C, G, D, Em and Am; find any root on strings 6 and 5
Weeks 3–4: major scale, keys, circle of fifths	Play G and C major in two octaves; name the chords in G and C
Weeks 5–6: pentatonic boxes 1 and 2, blues scale	Improvise over a 12-bar blues in A
Weeks 7–8: barre chords, triads on the top strings	Play I – V – vi – IV in any key two ways
Weeks 9–10: CAGED and arpeggios	Find a chord in all five shapes and arpeggiate it
Week 11: minor scales and modes	Hear the difference between Dorian, Aeolian and Mixolydian
Week 12: open tunings and review	Learn one song in Open G or DADGAD; revisit weak areas

Song analysis

SONG AND KEY

NUMERALS

VOICINGS

TRIADS

SCALE AND POSITION

ARPEGGIO TARGETS

COLOUR

PICKING HAND AND FEEL

Chapter 40 review · use it, train it, test it

Using it in real music

Analysing any song. Use the same eight steps for every new song: key, numerals, chords, triads, scale, position, arpeggio targets, colour. Ten minutes of analysis makes a song much faster to learn and remember. Appendix H has further examples.

Playing with others. When someone calls "ii – V – I in B $\flat$ " or "blues in G", you can work out every chord, scale and target note on the spot.

Drill	How	Time	Goal
Song analysis	Take one new song a week through the eight steps	10 min	A song analysed in under 10 minutes
Three ways to play	Play one progression as open chords, barre chords and triads, then solo over it	10 min	The whole book used on one song
Target notes	Over a looped progression, play only one chord tone per bar on beat 1, then add one approach note	5 min	Every bar lands on a chord tone
Three-stage arrangement	Take a short tune through melody, melody plus bass, melody plus alternating bass	10 min	One tune arranged per week

Write your answers, then check them in Appendix J.

1. The progression is Am – F – C – G. Which key is it in, and what are the numerals?

2. Which pentatonic scale fits the whole of Am – F – C – G?

3. Over the F chord in that progression, which notes are the chord tones?

4. The progression is E – A – B7. What are the numerals, and what could you solo with?

5. Dm7 – G7 – Cmaj7 is what kind of progression, in which key?

6. In Em – D – C – B, which scale fits best over the B chord, and why?

7. In the Andalusian cadence Am – G – F – E, which note does the E chord borrow, and where does it want to go?

8. What are the three stages of a simple fingerstyle arrangement?

PART ELEVEN

Gear and Sound

Choosing a guitar, amplification, effects, loopers and a minimal home recording setup.

- 41 Choosing a Guitar
- 42 Sound, Effects and Loopers

CHAPTER 41

Choosing a Guitar



Body size, scale length, neck shape, strings and setup decide how a guitar sounds and how it feels under your hands. Knowing what each does helps you buy wisely and get the best from the guitar you already own.

There is no best guitar, only a guitar that suits a player, a style and a pair of hands. A modest instrument that is well set up is easier to play and often sounds better than an expensive one that is not. This chapter explains the choices in order of how much they matter: the type of guitar, its size and neck, its setup and strings, and only then the woods and details.

Nylon, steel or electric

Chapter 1 compares the three types in detail. For choosing, the main question is what you want to play.

If you mainly want to play	Start with	Why
Classical pieces, flamenco, bossa nova	Nylon-string	The repertoire, technique and sound assume it; nylon is also kind to new fingertips
Fingerstyle folk, Celtic, open tunings	Steel-string, smaller body	Bright, ringing strings and long sustain suit drones and open tunings
Strummed songs with a voice	Steel-string, larger body	Volume and bass to accompany a singer
Rock, blues lead, funk, jazz, metal	Electric	Light strings for bends, sustain from the amplifier, easy access to high frets
A bit of everything, at home	Any; often a steel-string or an electric with headphones	Every shape and theory concept in this book works on all three

Many players own one of each type eventually. For a first guitar, choose the type that matches the music you listen to most: motivation matters more than any other factor.

Acoustic body shapes

Larger bodies, with wider lower bouts and deeper sides, generally give more volume and bass. Smaller bodies sound more focused and balanced and are more comfortable to hold, especially seated. Names vary between makers; the table lists common types from small to large.

Shape	Sound	Feel	Often used for
Travel or 3/4 size	Thinner, quieter	Short scale, light, easy to carry	Travel, children, practice anywhere
Parlour	Mid-focused, dry	Small and light	Blues, fingerstyle, smaller players
Classical	Warm, round, soft attack	Wide flat neck, low tension	Classical, flamenco, bossa nova
Concert / 00	Balanced, clear	Comfortable, smaller body	Fingerstyle, home playing
Auditorium / OM / 000	Balanced, articulate, even across the strings	Narrow waist, medium size	Fingerstyle, light strumming, recording
Grand auditorium	Versatile, good volume	Narrow waist, often a cutaway	All styles; live playing with a pickup
Dreadnought	Loud, strong bass, punchy	Large, square shoulders	Strumming, bluegrass, accompanying a voice
Jumbo	Large, rich bass and sustain	Very large; needs a long reach	Big strumming

A **cutaway** removes part of the upper bout so the hand can reach the high frets. It slightly reduces the body's air volume; the effect on sound is small.

Electric guitar types

On an electric guitar, the pickups, the amplifier and the player's hands shape the sound far more than the body.

Type	Sound	Typical styles
Solid body, single-coil pickups	Bright, glassy, clear; picks up mains hum	Blues, funk, country, indie, pop
Solid body, humbucking pickups	Thicker, warmer, louder output; hum-cancelling	Rock, hard rock, metal, blues
Solid body, P-90 (wide single-coil) pickups	Between the two: gritty and punchy	Rock, punk, blues
Semi-hollow body	Warm and airy, with a solid centre block that reduces feedback	Blues, jazz, rock and roll, indie
Hollow body (archtop)	Round, mellow, acoustic resonance; prone to feedback at high volume	Jazz, rockabilly

Many guitars combine pickup types, and a pickup selector switch (and coil-splitting on some humbuckers) gives several sounds from one instrument.

Dimensions that change the feel

Feature	What it changes	Typical values
Scale length (nut to saddle)	Longer = more string tension, brighter, slightly wider stretches; shorter = slinkier, easier stretches	Classical about 650 mm; steel-string about 629–648 mm; electric about 628 or 648 mm; short scale about 610 mm
Nut width	Space between the strings. Wider helps fingerstyle; narrower helps chords and bars	Steel 42–45 mm; classical 50–52 mm; crossover nylon 47–48 mm
Neck profile	The shape of the back of the neck: rounded C, flatter D, pointed V, thick U; thin or chunky	Personal; try several
Fingerboard radius	The curve across the fingerboard. Smaller radius = rounder, comfortable for chords; larger = flatter, easier for wide bends	Classical flat; electric about 184–400 mm (7.25–16 in)
Frets	Taller, wider frets make bends easier; low frets give more feel of the wood	Vintage, medium, jumbo
Body depth	Deeper = more bass and volume, harder to reach over	Thin electric to deep dreadnought

NYLON · STEEL

Nylon: classical guitars come in fractional sizes for younger and smaller players: 4/4 (650 mm scale) for adults and teenagers, 7/8 (about 615–630 mm) for smaller adults and teenagers, 3/4 (about 580 mm) for children around 8–11, and 1/2 (about 530 mm) for children around 5–8. **Crossover** nylon-string guitars have a narrower, slightly radiused neck, often with a cutaway and pickup, for steel-string players who want a nylon sound.

Woods and construction

Wood choices are among the most discussed and least clear-cut parts of guitar buying. Differences are real but usually smaller than construction, setup, strings and the individual instrument. Listen and compare rather than buying by specification.

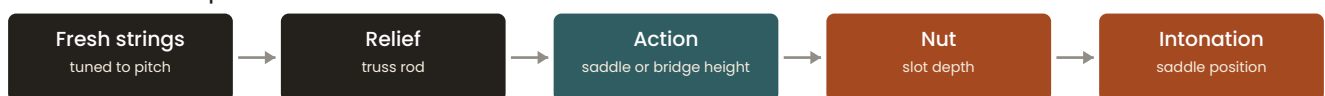
- **Solid versus laminate.** A solid top (one or two pieces of real wood) vibrates more freely than a laminate (layers glued together), and is the single most useful feature on an acoustic in a modest budget. Laminate backs and sides are tougher and less sensitive to humidity, which suits travel and beginners' guitars.

- **The top matters most** on an acoustic. Spruce is the most common: clear and articulate. Cedar is common on classical guitars and is usually described as warmer and quicker to respond at low volume. Mahogany tops sound drier and more compressed.
- **Back and sides** contribute less. Mahogany is often described as woody with strong mids, rosewood as fuller in bass and treble, maple as bright and focused. Many players cannot reliably tell these apart in blind tests; treat the descriptions as tendencies.
- **Electric guitars.** Body wood has a minor effect compared with pickups, amplifier and effects. Weight, comfort and the neck matter more.

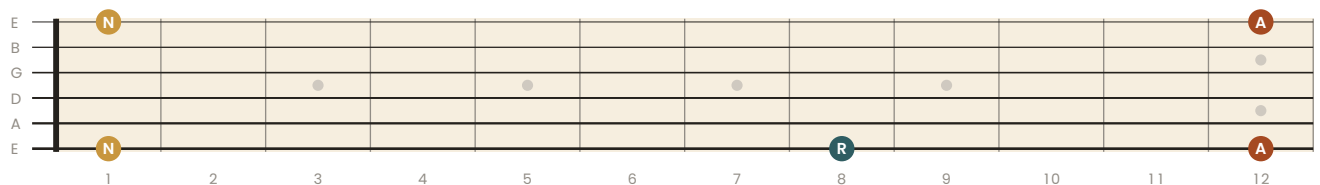
Setup basics

The **setup** is the adjustment of the guitar's playing parts: neck relief, action, nut height and intonation. A good setup makes a guitar easier to play in tune. Most music shops and guitar technicians offer it at modest cost, and many new guitars benefit from one.

The order of a setup



Where to check a setup



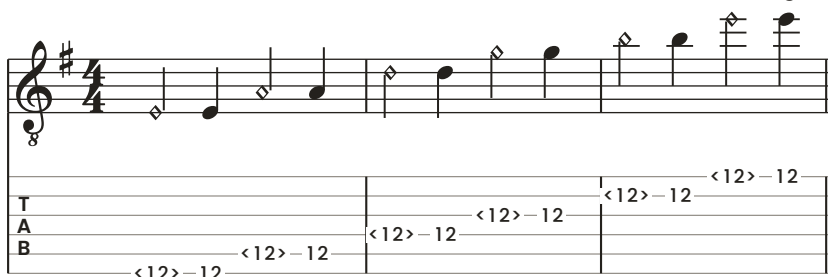
N: nut height, checked at the 1st fret. R: relief, checked around the 7th–8th fret. A: action, measured at the 12th fret on the 6th and 1st strings. Intonation is also checked at the 12th fret.

Check and method	Typical result
Relief: capo at the 1st fret, press the 6th string at the last fret, look at the gap at the 7th–8th fret	A very small gap, about the thickness of a business card (roughly 0.2–0.3 mm)
Action: measure from the top of the 12th fret to the bottom of the string	Steel: about 2.0–2.5 mm (6th string), 1.5–2.0 mm (1st). Electric: about 1.5–2.0 mm and 1.2–1.6 mm. Nylon: about 3.5–4 mm and 2.5–3 mm
Nut height: press each string at the 3rd fret and look at the gap over the 1st fret	A tiny gap; none at all causes buzzing, a large one makes first-position chords hard and sharp
Intonation: compare the 12th-fret harmonic with the fretted 12th-fret note on a tuner	The same reading. Fretted sharp: saddle moves away from the neck. Fretted flat: towards it

CAREFUL · Adjusting the truss rod

The truss rod sets relief, not action. Turn it only in small steps (an eighth to a quarter of a turn), give the neck time to settle, and stop if the nut is stiff. Classical guitars often have no adjustable rod. If you are unsure, take the guitar to a technician: a damaged truss rod is an expensive repair.

▶ TRACK 389 Intonation check: harmonic, then fretted note, on each string



Each pair should sound identical. A fretted note that is sharper than its harmonic is the most common intonation fault.

Strings

Type	Options	Notes
Steel acoustic	Extra light (.010), custom light (.011), light (.012), medium (.013): the gauge of the 1st string	Lighter is easier to press and bend; heavier is louder and fuller. Bronze and phosphor bronze are the common windings
Electric	.009, .010, .011 sets are most common	Nickel-wound; heavier for lower tunings
Nylon	Normal, high (hard) or extra-high tension; nylon or fluorocarbon trebles	Higher tension is louder and more direct, harder to press
Coated strings	Any type	Last longer before sounding dull; feel slightly smoother

- **When to change:** when strings sound dull, feel rough, or will not stay in tune or play in tune. With regular playing, often every 1–3 months for steel, 2–4 months for nylon.
- **Changing gauge** on a steel-string or electric can change the neck relief and the intonation. Check both afterwards.
- **New strings stretch.** Nylon strings in particular need several days to settle; retune often.

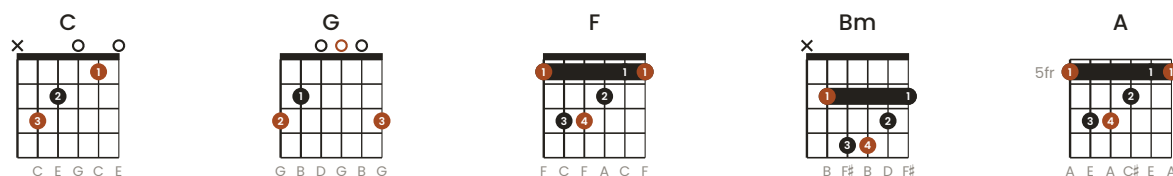
TIP · Caring for the instrument

Keep solid-wood guitars at about 45–55% relative humidity: very dry air (heated rooms in winter) can crack tops and leave fret ends sticking out; very damp air swells the wood and raises the action. Store the guitar in its case, away from radiators, windows and hot cars. Wipe the strings after playing and wash your hands before.

Buying checklist

1. **Play what you play.** Your usual chords, a barre chord, a scale up to the 12th fret, and a piece or riff you know well.
2. **Check comfort.** Neck width and profile, body size against your arm and chest, weight on a strap.
3. **Check the action and listen for buzzing** on every string at several frets.
4. **Check intonation:** the 12th-fret harmonic against the fretted note on each string.
5. **Look down the neck** from the headstock: it should be straight with only a slight forward dip, and not twisted.
6. **Check the tuners and electronics:** smooth tuning; on electrics and electro-acoustics, every control and switch without crackle.
7. **Compare blind if possible:** ask someone else to play two guitars while you listen from the front.
8. **Budget for a setup.** A well-adjusted modest guitar plays better than a badly adjusted expensive one.

A shop test in five chords



Open chords test the nut and first frets; the barre chords test action and comfort; A at the 5th fret checks tuning higher up the neck.

Used instruments

A used guitar can be good value, but check it more carefully.

Check	What to look for
Neck	Straight or slightly bowed, not twisted; truss rod turns and still has adjustment left
Frets	Grooves under the strings in the first five frets; buzzing or notes that choke when bent need fret work

Check	What to look for
Acoustic top and bridge	Bridge lifting at the back edge, a bulge behind the bridge, or a saddle sanded very low: signs that the neck angle needs resetting, which is a major repair
Cracks	Along the grain on the top, back and sides, especially near the soundhole and the lower edge
Electronics	Crackling pots, dead pickups, intermittent output jack
Hardware	Smooth tuners, no stripped screws, a bridge that holds its adjustment

If you are unsure, take a player or a technician with you, or ask the seller to let a technician look at it.

NYLON · STEEL

Nylon: check that the 12th-fret action is not excessive (above about 4.5 mm on the 6th string) and that the saddle still has height to lower. **Steel:** a dreadnought or jumbo with very high action may need a neck reset. **Electric:** most problems are adjustable or replaceable parts, which makes used electrics lower-risk.

Chapter 41 review · use it, train it, test it

Using it in real music

Buying your next guitar. Decide which type, body size and neck suit your style before you go shopping, then use the checklist and the five-chord test on every candidate.

Improving the guitar you own. Lighter strings, a lower action or a professional setup can change how a guitar plays far more than its price suggests.

How to train it

Drill	How	Time	Goal
Shop test routine	The same chords, barre chord and scale on every guitar you try	5 min	A fair comparison
Ear comparison	Record two guitars or two string sets playing the same phrase; listen without knowing which is which	5 min	Hear real differences in tone
Setup check	Measure action at the 12th fret, check relief and intonation	5 min	Know when your guitar needs a setup

Check your understanding

Write your answers, then check them in Appendix J.

1. Which acoustic body shape is known for strong bass and loud strumming?

2. What is the usual scale length of a full-size classical guitar?

3. Which part of an acoustic guitar affects its tone most: top, back or sides?

4. What does the truss rod adjust?

5. How do you check intonation, and which way do you move the saddle if the fretted note is sharp?

6. Single-coil or humbucker: which is quieter and thicker?

7. What humidity range suits solid-wood guitars?

8. Name two signs that a used acoustic may need a neck reset.

CHAPTER 42

Sound, Effects and Loopers



TRACKS

42

How to amplify an acoustic or electric guitar, which effects are useful and in which order, how to practise and perform with a looper, and how to record at home with a computer, an audio interface and headphones.

Sound does not stop at the strings. On an electric guitar the amplifier and effects are part of the instrument; an acoustic often needs a pickup or microphone to be heard in a room or recorded. A looper turns any exercise into music, and a small recording setup is the most honest practice tool there is. This chapter covers the essentials without assuming any particular brand of equipment.

The signal chain

Sound travels in a line from the strings to the listener's ears. Every link can shape it, and the order matters. These are the two most common chains.

Acoustic into a computer



Electric through an amplifier or modeller



An IR (impulse response) is a digital recording of how a speaker cabinet and microphone colour the sound. Modellers and amp simulation software use them in place of a real cabinet.

Amplifying an acoustic guitar

Method	Strengths	Weaknesses	Best for
Undersaddle piezo	Reliable, resists feedback well	Can sound hard or "quacky" on its own	Loud stages, strumming
Soundhole magnetic	Warm, often removable, no drilling	Steel strings only; less natural on fingerstyle	Steel-string, singer-songwriters
Contact transducer (on the top)	Woody, picks up percussive taps	Sensitive to placement	Fingerstyle, percussive playing
Internal microphone, blended with a pickup	Most natural sound from a built-in system	More prone to feedback	Quieter venues, fingerstyle
External microphone	The truest acoustic sound	Feedback at volume; you must stay still	Recording, classical guitar

TIP · Placing a microphone

For recording, a small-diaphragm condenser microphone about 15–30 cm from the guitar, pointed where the neck meets the body (around the 12th–14th fret), is a reliable starting point. Pointing at the soundhole gives more boom; moving towards the bridge gives more body and less detail. For a stereo sound, add a second microphone near the lower bout or farther back in the room.

CAREFUL · Feedback

Feedback (howling) happens when the guitar picks up its own amplified sound. To control it: turn down; stand beside or behind the speaker rather than in front of it; cut the low mids (about 100–250 Hz) where acoustic feedback usually starts; use the phase switch or notch filter on the preamp; or fit a soundhole cover.

Amplifying an electric guitar

Amplifier type	Character	Notes
Valve (tube) amp	Warm; distorts smoothly when pushed	Sounds best at higher volume; heavier, needs occasional valve replacement
Solid-state amp	Clean and reliable	Light and robust; clean sounds are a strength
Modelling amp or unit	Many amplifier and effect models in one box	Good at low volume and through headphones; can feed a mixing desk directly
Amp simulation software	Models running on a computer	Needs an audio interface; ideal for silent practice and recording

Gain and volume. On most amplifiers, *gain* (or *drive*) sets how hard the preamp is pushed, and therefore how much distortion you get; *master* or *volume* sets loudness. For a clean sound keep the gain low; for crunch, raise the gain and lower the master.

TIP · Practise clean

Distortion adds sustain and compression, which hides uneven picking, unwanted string noise and weak slurs. Practise technique with a clean or lightly overdriven sound, and add gain for the music.

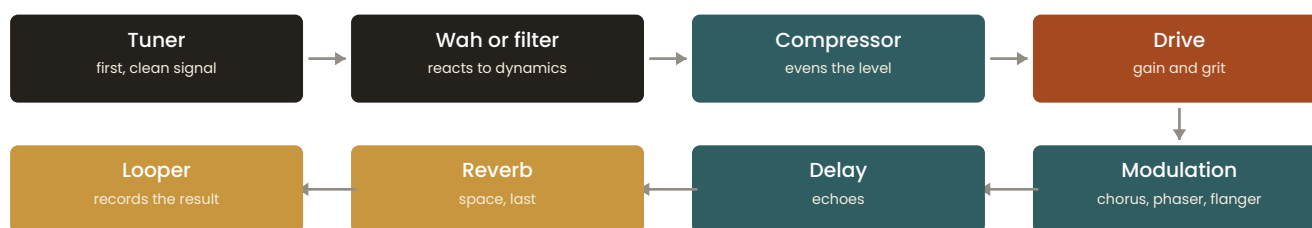
Effects

Effect	What it does	On acoustic	On electric
Tuner	Shows pitch; usually mutes the output while tuning	Essential	Essential
EQ	Boosts or cuts frequency ranges	The main tone tool; tames boom	Shapes the amp's tone
Compressor	Evens out loud and soft notes	Light: smoother fingerstyle	Funk, country, sustain
Overdrive, distortion, fuzz	Adds grit and sustain by clipping the signal	Rarely	The core of rock tone
Chorus	Adds a slightly detuned, moving copy	A touch for shimmer	Clean ballads, 1980s pop
Phaser, flanger	Sweeping, swirling tones	Rarely	Funk, psychedelic, rock
Wah, filter	A frequency peak swept by foot or by playing dynamics	No	Funk, rock leads
Octave, pitch shift	Adds a note an octave below or above	Bass lines when looping	Bass lines, synth sounds
Delay	Repeats what you play	Subtle repeats for solo pieces	Ambient textures, rhythmic echoes
Reverb	Simulates a room or hall	The most useful effect after EQ	Space and depth
Looper	Records and repeats a phrase	Accompany yourself	Practice and performance

Order of effects

When several effects are chained, each one processes everything that comes before it, so the order changes the sound. A compressor after a delay squashes the echoes; a drive after a reverb distorts the whole room. The order below is the usual starting point.

A conventional order



The logic is: dynamics effects first, then gain, then effects that move the sound, then time-based effects, and the looper last so that it records the finished sound. Reverb comes late because a room is the last thing sound passes through in real life. These are conventions, not laws: a delay before the drive, or the looper before the effects (so you can change effects over a fixed loop), both give useful sounds. Many amplifiers have an **effects loop**, which places time-based effects after the amp's own distortion.

Shaping the sound with EQ

Range	What you hear	Usual move
Below 80 Hz	Rumble, handling noise	Cut with a high-pass (low-cut) filter
80–250 Hz	Body and boom	Cut a little if boomy or feeding back
250–800 Hz	Boxiness, "mud"	Narrow cut if the sound is congested
800 Hz–2 kHz	Honk, nasal tone; also presence in a band mix	Small cut or boost to taste
2–5 kHz	Pick attack, clarity	Gentle boost for clarity
5–12 kHz	Air, sparkle, string noise	Small boost for sparkle (adds noise)

The open low E string has a fundamental of about 82 Hz and the open high E about 330 Hz; everything above that is overtones, which is where most of the guitar's tone lives. Cut before you boost: most problems come from too much of something.

NYLON · STEEL

Nylon: nylon strings have fewer high overtones, so avoid large treble boosts; a little more reverb than on steel suits the instrument. Magnetic soundhole pickups do not work with nylon strings. **Steel:** a gentle low cut and a narrow cut in the low mids usually make an amplified steel-string sound more natural. **Electric:** shape the sound mainly with the amp's own EQ and the pickup selector.

The looper

A looper records what you play and repeats it, so you can play over yourself. It is one of the most useful practice tools available: instant backing for scales, arpeggios, improvising and timing practice. Hardware loopers are pedals; software loopers run in recording programs and apps.

Action	What happens
Press once	Start recording
Press again	Stop recording and start playback: this sets the loop length
Press while looping	Overdub: add another layer
Hold (on most loopers)	Undo or redo the last layer
Stop (double-tap or a separate switch)	Stop playback; a long press usually clears the loop

Building a loop in layers



TRACK 390 Layers 1 and 2: chords and bass over C – Am – F – G

Diagram showing a 4-bar loop progression: C – Am – F – G. The top staff shows the chords in treble clef, 4/4 time. The bottom staff shows the bass line in bass clef, 4/4 time, with fret numbers for each string (T, A, B) across the four bars.

Chord progression: C – Am – F – G

Bass line (fret numbers):

Bar	T	A	B
1	0 0 0 0	1 1 1 1	0 0 2 3
2	0 0 0 0	2 2 2 2	0 2 0 0
3	1 1 1 1	2 2 2 2	3 3 3 3
4	3 3 3 3	0 0 0 0	0 0 0 0

Record the chords first (upper part), then overdub the bass (lower part) on the second pass. On an acoustic, an octave pedal on the bass layer gives a bass-guitar sound.

TRACK 391 Layer 3: a percussion pass

Diagram showing a 4-bar loop progression for percussion. The top staff shows the chords in treble clef, 4/4 time. The bottom staff shows the bass line in bass clef, 4/4 time, with fret numbers for each string (T, A, B) across the four bars.

Chord progression: C – Am – F – G

Bass line (fret numbers):

Bar	T	A	B
1	0 0 0 0	1 1 1 1	0 0 2 3
2	0 0 0 0	2 2 2 2	0 2 0 0
3	1 1 1 1	2 2 2 2	3 3 3 3
4	3 3 3 3	0 0 0 0	0 0 0 0

A bass note on 1 and 3 and a muted chuck on 2 and 4 make a simple drum part. Or tap the body: heel of the hand for the kick, fingertips for the snare.

TIP · Closing the loop

Count yourself in and press the switch exactly on beat 1 of the bar after your last bar, if anything slightly early: the foot is slower than you think. Practise closing a 4-bar loop to a metronome until the loop never stutters. Start with 2 or 4 bars. Many loopers and apps can quantise the loop to a set tempo.

Looper exercise	How	Builds
Progression and scale	Loop C – Am – F – G; improvise with C major pentatonic	Improvising (Chapter 7)
Chord-tone targets	Loop a progression; land on a chord tone on beat 1 of each bar	Arpeggios (Chapter 16)
Mode vamp	Loop Dm7 – G7 for two bars each; play D Dorian	Modes (Chapter 9)
Triad partner	Loop open chords; play top-string triads over them	Triads (Chapter 10)
Timing test	Loop a strum pattern; play along. Any drift is your timing, not the looper's	Rhythm (Chapter 37)
One-person band	Layer bass, chords and percussion; play the melody live	Arranging, performance

A minimal home recording setup

A complete practice and recording setup can be very small. Start at level 0 and add equipment only when you meet a real limitation.

Level	Equipment	What you can do
0: free	A computer or phone with its built-in microphone, recording software (a DAW), headphones	Record ideas, check timing against a click, use a tuner and metronome, play along with drum loops
1: minimal	Add a USB audio interface with an instrument (Hi-Z) input, closed-back headphones and a cable	Clean, low-latency recording of an electric or an acoustic with a pickup; amp and effect simulations
2: acoustic recording	Add a small-diaphragm condenser microphone (it needs 48 V phantom power from the interface) and a stand	Natural recordings of acoustic and classical guitar; microphone and pickup blended
3: performance	Add a looper pedal or footswitch and a small amplifier or powered speaker	Hands-free looping and effects; playing for others

A DAW (digital audio workstation) is recording software. Free and paid DAWs exist for every computer, and some come pre-installed. All of them offer multitrack recording, a metronome and basic effects;

many add drum loops, amp simulations and a speed control.

Setting it up

1. Connect the interface by USB. Plug the guitar into the instrument (Hi-Z or INST) input, or the microphone into a microphone input with phantom power switched on.
2. In the DAW's audio settings, select the interface as both input and output.
3. Set the sample rate to 48 kHz (44.1 kHz is also fine) and the buffer size to 64–128 samples while playing and recording.
4. Set the input gain so that your loudest playing peaks around –12 to –6 dBFS, never in the red.
5. Monitor: with a microphone or a clean pickup sound, use the interface's direct monitoring (no delay). With amp simulation, monitor through the software.
6. Use headphones when recording with a microphone, so the click does not leak into the recording.

Levels and latency

Levels. Digital recording has a hard ceiling at 0 dBFS; anything louder is clipped and cannot be repaired. Recording too quietly is less harmful, because modern interfaces have very low noise. Peaks around –12 dBFS leave room for the occasional louder strum.

Latency is the delay between playing a note and hearing it back through the computer. It comes mostly from the buffer: at 48 kHz, a 128-sample buffer adds about 2.7 ms on the way in and the same on the way out, plus a little for the converters. A total round trip under about 10 ms is rarely noticeable; for comparison, sound takes about 3 ms to travel one metre, so standing 3 m from an amplifier gives a similar delay. If you hear a delay, lower the buffer size or use direct monitoring. Raise the buffer (512 samples or more) for mixing large projects, when latency does not matter.

Use in practice	How
Metronome and count-in	Every exercise in this book
Drum loops or a virtual drummer	Groove practice; more musical than a click
Amp and effect simulation	Electric tones without pedals, at any volume
Multitrack recording	Weekly progress recordings; layer your own backing
Speed control without pitch change	Learn fast passages and transcribe solos (Chapter 30)
Looping	All the looper exercises above

CAREFUL · Protect your hearing

Hearing damage is permanent; the general advice on volume, breaks and earplugs is in Chapter 39. With this equipment in particular: set the headphone level before you start playing, not while a loop is running; turn down before plugging in or switching effects on, to avoid sudden loud pops; and remember that a high-gain sound feels quieter than it is.

Chapter 42 review · use it, train it, test it

Using it in real music

Playing live. Choose a pickup system, control feedback, and set EQ and reverb so that an acoustic sounds natural through a PA. On electric, set a clean and a driven sound you can switch between.

Practising alone. A looper or a DAW with drum loops turns every scale and arpeggio exercise into music.

Recording progress. A minimal interface and headphones let you record cleanly and hear yourself honestly (Chapter 38).

How to train it

Drill	How	Time	Goal
Loop closing	Record a 4-bar loop to a metronome until it repeats seamlessly	5 min	Perfectly timed loops
EQ sweep	Boost a narrow EQ band and sweep it through the range to find boomy and harsh spots, then cut them	5 min	Recognise problem frequencies by ear
Layered loop	Build chords, bass and percussion layers, then improvise for two minutes	10 min	A complete groove from one guitar
Weekly recording	Record one piece with the same settings every week	10 min	Honest, comparable progress

Check your understanding

Write your answers, then check them in Appendix J.

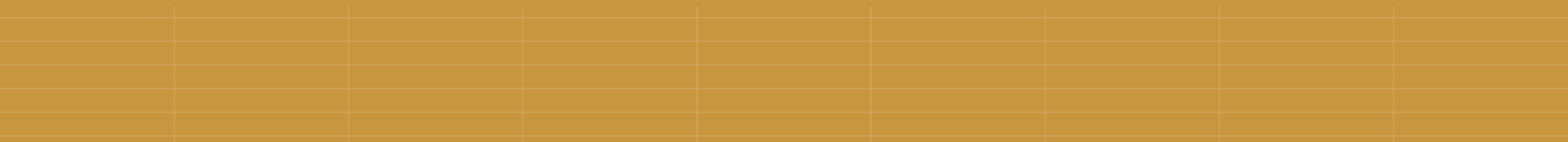
1. In a conventional pedal chain, where do delay and reverb go?
2. What is a good peak level when setting the interface input gain?
3. Which buffer size range gives low latency while playing and recording?
4. Where should a microphone point when recording an acoustic guitar?
5. Name two ways to reduce feedback from an amplified acoustic.
6. When should you press the looper switch to close a loop?
7. Which input on an audio interface is for a guitar plugged in directly?
8. Why does a soundhole magnetic pickup not work on a nylon-string guitar?

MY NOTES

Style Workshops

Fourteen musical traditions, each showing how theory and technique you already know become a style.

Blues
Rock and Riffs
Funk Rhythm
Country and Bluegrass
Folk and Songwriting
Jazz Harmony
Gypsy Jazz
Bossa Nova and Latin Rhythms
Choro and Brazilian Guitar
Reggae
Flamenco
Celtic Fingerstyle
Neo-Soul
West and Central African Guitar



CHAPTER 43

Blues



TRACKS

43

Three dominant chords, a twelve-bar form, a scale that sits between major and minor, and a rolling triplet feel. The blues is the foundation of rock, soul, jazz and much of country, and it has an acoustic tradition of its own.

A blues is defined by four things working together: a short repeating form, dominant 7th chords on every degree, melodies that mix the minor and major 3rd, and a triplet-based feel called the shuffle. None of these is difficult on its own. Playing them together, relaxed and in time, is the skill this workshop trains.

Ingredient	What it is	Background
Form	12-bar blues on I, IV and V; also 8-bar and minor forms	Chapter 14
Chords	Dominant 7th and 9th chords on every degree: A7, D7, E7 or A9, D9, E9	Chapter 11
Scales	Minor pentatonic and blues scale, mixed with the major 3rd and 6th	Chapter 7
Feel	Shuffle (swung eighths) or slow 12/8	Chapter 37
Technique	Bends, the ♭3 curl, vibrato, slides	Chapter 36

The 12-bar form

The standard 12-bar blues has three four-bar lines. The first stays on I, the second moves to IV and back, the third moves V – IV – I and ends on V to turn the form around. All three chords are dominant 7ths, even though only V7 is diatonic to the major key: the ♭7 on I and IV is part of the blues sound. The plain layout in A is charted in Chapter 14. In practice most players split bars 11 and 12 into a turnaround, I – IV then I – V, as below.

12-bar blues in A with a turnaround in bars 11–12

A7 I7	A7 I7	A7 I7	A7 I7
D7 IV7	D7 IV7	A7 I7	A7 I7
E7 V7	D7 IV7	A7 I7	D7 IV7
		A7 I7	E7 V7

Bars 11–12 move I – IV, I – V. The E7 on the last two beats sends the form back to bar 1.

Quick change, minor blues, 8-bar blues

Three variants cover most of what you meet at a jam. The **quick change** moves to IV in bar 2 and returns in bar 3. The **minor blues** uses minor 7th chords on i and iv and replaces the V – IV of bars 9–10 with ♭VI7 – V7 (in A minor: Am7, Dm7, F7 and E7; the full chart is in the minor-blues practice loop later in this chapter). The **8-bar blues** compresses the form; the version below is one of several common layouts.

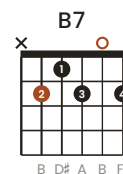
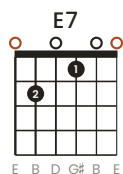
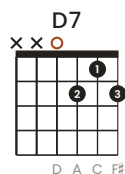
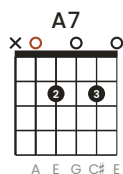
Quick change in E

E7 I7	A7 IV7	E7 I7	E7 I7
A7 IV7	A7 IV7	E7 I7	E7 I7
B7 V7	A7 IV7	E7 I7	B7 V7

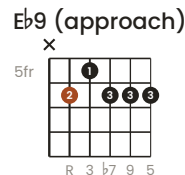
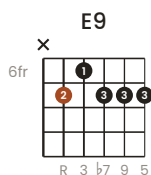
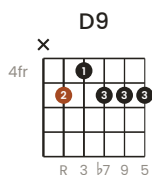
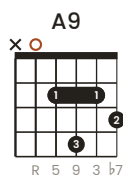
Dominant 7th and 9th chords

Open dominant 7th chords give the acoustic blues sound. For electric blues and anything in a horn-band style, the 9th chord replaces the 7th: it adds the 9th above the root, leaves out the low 5th and sits on the middle strings, out of the bass player's way.

Open dominant 7th chords



9th chords for a blues in A



D9 and E9 are the same shape two frets apart. Slide into any 9th chord from one fret above or below, on beat 4 of the previous bar or on the beat itself.

▶ TRACK 394 9th-chord stabs with a slide approach

D9 **E9**

	5	6	5	7	6	7
T	5	6	5	7	6	7
A	4	5	4	6	5	6
B	5	6	5	7	6	7

Short chords on the "let" of beats 2 and 4, horn-section style. Before beat 4's chord, slide in from a half step away: from above into D9, from below into E9.

Turnarounds

A turnaround fills bars 11 and 12 and leads back to the top of the form. The most common type moves a chromatic line against a fixed note and lands on V in bar 12. These two are standard vocabulary in E and A; learn them in both keys because most acoustic blues is played in one of these two.

▶ TRACK 395 Turnaround in E

E7 **B7**

	0	0	0	0	0	0	0	0	2
T	3	3	2	2	1	1	0	0	0
A									2
B									2

D - C# - C - B on the 2nd string under the open 1st string, then E and B7.

▶ **TRACK 396** Turnaround in A

A7 **E7**

T	5	5	5	5	5	5	5	5	0	0	0
A	5	5	4	4	3	3	2	2	0	1	1
B	0								0	0	0
									0	2	0

G – F $\sharp$ – F – E on the 4th string under E on the 2nd string. Fret 4.5 and 2.5 with the ring and index fingers, then let the middle finger take over.

Soloing

The core blues scale is the minor pentatonic with the $\flat 5$ added (Chapter 7). Over a dominant chord, though, its $\flat 3$ clashes with the chord's major 3rd, and that clash is the point. Blues players treat the 3rd as a flexible pitch: the $\flat 3$ bent slightly upward (a *curl*), the $\flat 3$ hammered or slid to the 3, or both played in the same phrase.

A blues scale, box 1, with the major 3rd added

Gold dots are the major 3rd (C $\sharp$). Reach them from the $\flat 3$ with a bend, a hammer-on or a slide.

Five working rules

- **Mix major and minor.** Over the I chord, curl or bend the $\flat 3$ toward the 3. Over IV, the $\flat 3$ of the key becomes the $\flat 7$ of the chord (C over D7 in A) and sounds settled.
- **Follow the chord change.** C over D7 moving to C $\sharp$ over A7 marks the change with a half step. One note per chord is enough.
- **Call and response.** Play a short phrase, then answer it, as a singer and a band do. A phrase of one or two bars followed by the same length of answer or rest fits the form.
- **Repeat and vary.** Play a phrase three times over the first, second and third lines, changing only the ending to suit the chord.
- **Leave space.** A bar of rest makes the next phrase stronger and gives the band room.

▶ **TRACK 397** Call and response over A7

A7

T	5	7	5	5	6	7	7	5	7	5	8	5	5	5	7	5	6	7	7
A																			
B																			

Bars 1 and 3 call high, bars 2 and 4 answer low. The hammer-on from C to C $\sharp$ moves from the $\flat 3$ of the scale to the 3rd of A7.

TRACK 398 Targeting the change: D7 to A7

D7 **A7**

T 8 7 5 7 5 6 7 5 7 7

The first bar ends on C, the $\flat 7$ of D7. The second starts on C $\sharp$, the 3rd of A7. The half step is the sound of the chord change.

TRACK 399 12-bar shuffle in A · LOOP · blues shuffle · 92 bpm

A7 <i>I7</i>	A7 <i>I7</i>	A7 <i>I7</i>	A7 <i>I7</i>
D7 <i>IV7</i>	D7 <i>IV7</i>	A7 <i>I7</i>	A7 <i>I7</i>
E7 <i>V7</i>	D7 <i>IV7</i>	A7 <i>I7</i>	E7 <i>V7</i>

IMPROVISE WITH

A minor pentatonic

A blues

A major pentatonic

Start with box 1 of the minor pentatonic and call and response. On later choruses, aim for C $\sharp$ over A7, C over D7 and G $\sharp$ over E7.

TRACK 400 Minor blues in A · LOOP · blues shuffle · 76 bpm

Am7 <i>i7</i>	Am7 <i>i7</i>	Am7 <i>i7</i>	Am7 <i>i7</i>
Dm7 <i>iv7</i>	Dm7 <i>iv7</i>	Am7 <i>i7</i>	Am7 <i>i7</i>
F7 <i>bVII7</i>	E7 <i>V7</i>	Am7 <i>i7</i>	E7 <i>V7</i>

IMPROVISE WITH

A minor pentatonic

A blues

No major 3rd here: stay with the minor pentatonic and the $\flat 5$. Over E7 in bar 10, land on G $\sharp$, its 3rd.

Acoustic blues traditions

Pre-war acoustic blues was played by solo singers accompanying themselves, so the guitar carries bass, rhythm and melody at once. Two regional approaches to the picking hand are worth learning in detail.

Style	Sound	Picking hand	Typical keys and tunings
Delta blues	Raw, driving, often with slide	Monotonic thumb: the root on every beat	E and A in standard tuning; Open G and Open D for slide

Style	Sound	Picking hand	Typical keys and tunings
Piedmont blues	Bouncy, ragtime-influenced	Alternating thumb between two bass strings, syncopated melody on top	C, G and D in standard tuning
Texas and country blues	Loose, vocal, free in form	Thumb and one finger, irregular bar lengths	E and A
Slow blues	Spacious, dramatic	Triplet strums or arpeggios in 12/8	Any

Delta: the monotonic thumb

The thumb plays the open low E on every beat, damped lightly with the heel of the hand. Above it, the fingers play short riffs and chord fragments on the top three strings. The thumb never stops, even when the melody pauses.

▶ TRACK 401 Delta-style figure in E

E7

Thumb on every dotted-quarter beat; the G to G $\sharp$ hammer-on is the minor-to-major 3rd move on the open strings.

Piedmont: the alternating thumb

Piedmont players (from the south-eastern United States) use the alternating bass of Travis picking (Chapter 19) under a syncopated melody, and favour ragtime progressions that move round the cycle of 5ths. The feel is usually straight or lightly swung.

▶ TRACK 402 Piedmont-style ragtime cycle in C

C A7 D7 G7

C – A7 – D7 – G7: each chord is the V7 of the next. Thumb on the beat, melody between the beats.

NYLON · STEEL

Nylon: the blues works well fingerstyle on nylon strings, especially the Piedmont style. Bends are small on nylon, so replace them with slides and wide vibrato, and use the rest stroke to bring out melody notes.

Steel and electric: bends of a half or whole step are standard. On acoustic steel-string, light-gauge strings make the $\flat 3$ curl and the half-step bends practical; on electric, the full and one-and-a-half-step bends of electric blues are normal.

Boogie in A

Original study · shuffle · 12/8 · 84 bpm (dotted quarter)

A complete 12-bar chorus that combines the boogie pattern, the walking line and the A turnaround. Play it with downstrokes and light palm muting throughout.

TRACK 403

A7

T																								
A	2	2	4	4	5	5	4	4	2	2	4	4	5	5	4	4	2	2	4	4	5	5	4	4
B	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

D7

T																								
A	0	4	2	4	5	4	2	4	2	2	4	4	5	5	4	4	0	4	2	4	5	4	2	4
B	0	4							0	0	0	0	0	0	0	0	0	4						

A7 **E7**

T																								
A	2	2	4	4	5	5	4	4	0	4	2	4	5	4	2	4	2	2	4	4	5	5	4	4
B	0	0	0	0	0	0	0	0	0	4							0	0	0	0	0	0	0	0

D7 **A7** **E7**

T																								
A	2	2	4	4	5	5	4	4	5	5	5	5	5	5	5	5	1	0						
B	0	0	0	0	0	0	0	0	0	5	4	4	3	3	2	2	2	2	4	4	2	0	0	0

PLAYING IT · Performance notes

Bars 4, 6 and 8 replace the dyads with the walking line. Keep the same long-short feel.

In bar 11 hold the E on the 2nd string with the little finger or ring finger and walk the 4th-string line down with the other fingers.

For a second chorus, play the quick change (D7 in bar 2) and end with A9 slid from B♭9.

LISTEN FOR • Listening

Delta and early acoustic: Charley Patton, Son House, Robert Johnson (*King of the Delta Blues Singers*), Skip James. **Piedmont:** Blind Blake, Blind Willie McTell, Rev. Gary Davis, Etta Baker. **Electric shuffle and boogie:** Jimmy Reed, John Lee Hooker, Muddy Waters (*Folk Singer* for his acoustic side). **Lead guitar:** T-Bone Walker, B.B. King (*Live at the Regal*), Freddie King, Albert King, Stevie Ray Vaughan. Listen for where the guitarist leaves space and how the shuffle feel changes between a medium and a slow blues.

Chapter 43 review • use it, train it, test it

Using it in real music

Jamming. The 12-bar blues is the common ground of rock, blues and jazz jams. Agree the key and the form (standard, quick change or minor), play the boogie pattern while others solo, and solo with call and response when your turn comes.

Rock, soul and pop. Boogie patterns, 9th-chord stabs and the $\flat 3$ -to-3 move turn up in rock riffs (Chapter 44), soul rhythm parts and funk (Chapter 45). The jazz blues in Chapter 48 is the same form with more chords.

How to train it

Drill	How	Time	Goal
Shuffle feel	Boogie pattern through the 12-bar in A at 70–100 bpm, metronome on 2 and 4	5 min	Even long-short feel, no rushing on the eighths
Form memory	Play the 12-bar with one strum per bar, eyes closed, in A, E and G	3 min	Never lose your place in the form
Turnarounds	Bars 11–12 in E and A, then straight back to bar 1	3 min	Land on beat 1 of the next chorus
Change targeting	Improvise over bars 4–7, landing on C over D7 and C $\sharp$ over A7	4 min	Hear the half-step move every time
Call and response	Two-bar phrase, then two bars of answer or silence, for a whole chorus	4 min	Phrases that start and end with the form

Check your understanding

Write your answers, then check them in Appendix J.

1. Write the chords of a standard 12-bar blues in E.

2. What is different in a quick-change blues?

3. In a minor blues in A, which chords usually sit in bars 9 and 10?

4. In the boogie pattern on A7, which notes alternate above the open A?

5. Why is C a strong note over D7 in a blues in A, and where does it want to move when the chord returns to A7?

6. What is a monotonic bass?

MY NOTES

CHAPTER 44

Rock and Riffs



Rock guitar is built on short repeated figures called riffs, played with power chords and single notes on the low strings, usually through a distorted amplifier. The style asks for tight muting, steady straight eighths and a strong sense of the backbeat.

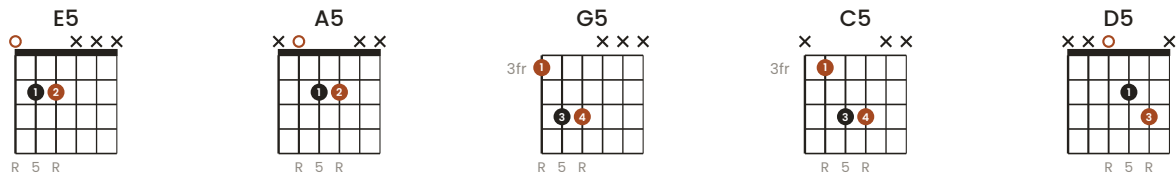
Rock grew out of the blues (Chapter 43) in the 1950s and kept its scales and its dominant-chord habits, but changed the feel and the sound. The eighth notes are mostly straight instead of swung, the drummer hits the snare hard on beats 2 and 4, and the guitar is amplified and distorted. Distortion adds sustain and weight, but it also turns full chords to mud, which is why rock relies on two-note and three-note shapes and on single-note riffs.

Element	Typical rock approach
Feel	Straight eighths, strong backbeat on 2 and 4; shuffle in blues-rock
Harmony	Power chords; i – bVI – bVII and I – bVII – IV progressions; riffs instead of chord changes
Scales	Minor pentatonic and blues scale, with chromatic passing notes
Technique	Downstroke eighths, palm muting, fretting-hand muting, bends, vibrato
Sound	Overdriven amplifier; bridge pickup for riffs, neck pickup for smooth leads

Power chords

Power chords (root and 5th, written E5, A5, G5) are introduced in Chapter 11. They are the rock chord because of distortion: a 3rd under heavy gain produces rough combination tones, while the plain 5th stays clean and solid. Two shapes cover the whole neck, root on the 6th string and root on the 5th, plus the open E5, A5 and D5 for riffs in open position.

Power chord shapes



Index on the root, ring and little finger on the 5th and the octave. The open E5, A5 and D5 use the open string as the root.

CAREFUL · Muting the unused strings

With distortion every unwanted string sounds. Let the underside of the index finger touch the strings above the power chord, and let the fingertip of the index finger touch the string below the root. Only the two or three fretted notes should ring when you strum all six strings.

Palm muting

Palm muting is explained in Chapter 36. With distortion it becomes the core of rock rhythm: most parts alternate palm-muted eighths on the low string with unmuted, accented power chords. Play the eighths with downstrokes only, which gives the even, heavy attack the style needs, and keep the mute the same weight from bar to bar so the chugs do not swell and fade.

▶ TRACK 404 Palm-muted eighths with accented chords

E5 G5 A5 E5 A5 G5 E5

P.M. -----| P.M. -----|

T
A
B

0 0 0 0 3 5 0 0 0 0 5 3 0

Palm-mute the open E eighths, lift the palm for the accented power chords as they move G5 – A5 and back. All downstrokes.

Rock rhythm patterns

Rock drumming puts the bass drum on beat 1 and the snare on beats 2 and 4; the guitar either doubles the eighth-note pulse or plays a syncopated figure against it. Four patterns cover a large part of the repertoire.

▶ TRACK 405 Four rock rhythm figures over a backbeat

Eighth drive Gallop 3+3+2 Push on 4& Snare

1 e & a 2 e & a 3 e & a 4 e & a

The gallop is an eighth and two sixteenths. The 3+3+2 figure is the tresillo (Chapter 37) twice per bar. A push lands a chord on the "&" of 4 and holds it over the barline.

TIP · Downstrokes or alternate picking

Eighth notes up to about 160 bpm sound heaviest with downstrokes only. Sixteenths and gallops need alternate picking (Chapter 24): down on the beat and the "&", up on the "e" and "a". Keep the palm mute in place while the pick moves.

Building riffs

A riff is a one- or two-bar figure that repeats and identifies a song. Most rock riffs are built in one of three ways, and many combine them.

Pentatonic riffs

The open-position E minor pentatonic (E G A B D) gives the low strings their own riff vocabulary. The open low E works as a home note that you return to between phrases.

▶ TRACK 406 Pentatonic riff in E

T
A
B

0 0 3 0 0 2 0 0 3 0 2 3

Bar 1 asks a question by ending on B; bar 2 answers by stepping down to G and back to E on the repeat.

Chromatic riffs

Chromatic passing notes fill the gaps in the pentatonic scale: G $\sharp$ between G and A, B $\flat$ between A and B. Place them on the off-beats so that the scale notes stay on the beats.

 TRACK 407 Chromatic riff in E

[illegible]

G# in bar 1 and Bb in bar 2 fall on the "and" of 2; the pentatonic notes land on the beats.

Pedal-tone riffs

A pedal-tone riff keeps returning to one low note, usually the open root, while a melody moves above it. The low note fills the gaps and keeps the riff driving even when the melody is slow.

 TRACK 408 Pedal-tone riff in E

8

P.M. --- P.M. --- P.M. --- P.M. --- P.M. --- P.M. --- P.M. ---

T
A
B

2 4 5 4 2 4 5 7 5

0 0 0 0 0 0 0 0 0 0 0 0 0 0

Two palm-muted sixteenths on the open E, then an unmuted melody note on the 4th string (E F# G A). Alternate picking throughout.

TRY THIS · Write your own

Take a two-bar riff and repeat bar 1 three times, changing only bar 2 on the fourth repeat: the shape AAAB. Keep the rhythm of bar 1 exactly the same each time; the listener recognises a riff by its rhythm first and its notes second.

Drop D

Lowering the 6th string a whole step to D gives a low D power chord on three open strings and turns every power chord on the bottom three strings into a one-finger barre. Tune the 6th string down until its 12th-fret harmonic matches the open 4th string. Drop D is common in hard rock, grunge and metal, and in acoustic music for a deeper bass (Chapter 25 covers other tunings).

Power chords in drop D

D5

D A D A D

F5

F C F C F

G5

G D G D G

A5

A F A F A

One finger flat across strings 6, 5 and 4. Moving a power chord is now a single movement of the hand.

Rock soloing basics

A musical staff diagram showing the notes of the A major scale across two octaves. The staff has five lines labeled E, B, G, D, A from top to bottom. The notes are: 1st octave (A, B, C, D, E, F, G, A) and 2nd octave (B, C, D, E, F, G, A, B). The notes are represented by black circles with numbers 1-7 and flats (b) for the second and sixth degrees. The notes are placed on the lines and spaces of the staff.

Three habits make pentatonic notes sound like rock lead guitar:

- **Repeat a short figure fast.** A three-note group in triplets, repeated four times, builds energy over one chord.
- **Bend and hold.** Bend the 4th of the scale up to the 5th (D to E in A) and hold it with vibrato. Check the pitch of the bend against the fretted target note.
- **End on a chord tone with vibrato.** A long note on the root or 5th, with steady vibrato, closes a phrase.

A5

The musical score for 'The Rose Tree' is presented in a standard musical notation format. The top staff is a treble clef with a 12/8 time signature. The melody consists of a series of eighth notes, each beamed together with a quarter note, creating a continuous eighth-note pattern. The notes are G4, A4, B4, C5, D5, E5, F5, G5, A5, B5, C6, D6, E6, F6, G6, A6, B6, C7, D7, E7, F7, G7, A7, B7, C8, D8, E8, F8, G8, A8, B8, C9, D9, E9, F9, G9, A9, B9, C10, D10, E10, F10, G10, A10, B10, C11, D11, E11, F11, G11, A11, B11, C12, D12, E12, F12, G12, A12, B12, C13, D13, E13, F13, G13, A13, B13, C14, D14, E14, F14, G14, A14, B14, C15, D15, E15, F15, G15, A15, B15, C16, D16, E16, F16, G16, A16, B16, C17, D17, E17, F17, G17, A17, B17, C18, D18, E18, F18, G18, A18, B18, C19, D19, E19, F19, G19, A19, B19, C20, D20, E20, F20, G20, A20, B20, C21, D21, E21, F21, G21, A21, B21, C22, D22, E22, F22, G22, A22, B22, C23, D23, E23, F23, G23, A23, B23, C24, D24, E24, F24, G24, A24, B24, C25, D25, E25, F25, G25, A25, B25, C26, D26, E26, F26, G26, A26, B26, C27, D27, E27, F27, G27, A27, B27, C28, D28, E28, F28, G28, A28, B28, C29, D29, E29, F29, G29, A29, B29, C30, D30, E30, F30, G30, A30, B30, C31, D31, E31, F31, G31, A31, B31, C32, D32, E32, F32, G32, A32, B32, C33, D33, E33, F33, G33, A33, B33, C34, D34, E34, F34, G34, A34, B34, C35, D35, E35, F35, G35, A35, B35, C36, D36, E36, F36, G36, A36, B36, C37, D37, E37, F37, G37, A37, B37, C38, D38, E38, F38, G38, A38, B38, C39, D39, E39, F39, G39, A39, B39, C40, D40, E40, F40, G40, A40, B40, C41, D41, E41, F41, G41, A41, B41, C42, D42, E42, F42, G42, A42, B42, C43, D43, E43, F43, G43, A43, B43, C44, D44, E44, F44, G44, A44, B44, C45, D45, E45, F45, G45, A45, B45, C46, D46, E46, F46, G46, A46, B46, C47, D47, E47, F47, G47, A47, B47, C48, D48, E48, F48, G48, A48, B48, C49, D49, E49, F49, G49, A49, B49, C50, D50, E50, F50, G50, A50, B50, C51, D51, E51, F51, G51, A51, B51, C52, D52, E52, F52, G52, A52, B52, C53, D53, E53, F53, G53, A53, B53, C54, D54, E54, F54, G54, A54, B54, C55, D55, E55, F55, G55, A55, B55, C56, D56, E56, F56, G56, A56, B56, C57, D57, E57, F57, G57, A57, B57, C58, D58, E58, F58, G58, A58, B58, C59, D59, E59, F59, G59, A59, B59, C60, D60, E60, F60, G60, A60, B60, C61, D61, E61, F61, G61, A61, B61, C62, D62, E62, F62, G62, A62, B62, C63, D63, E63, F63, G63, A63, B63, C64, D64, E64, F64, G64, A64, B64, C65, D65, E65, F65, G65, A65, B65, C66, D66, E66, F66, G66, A66, B66, C67, D67, E67, F67, G67, A67, B67, C68, D68, E68, F68, G68, A68, B68, C69, D69, E69, F69, G69, A69, B69, C70, D70, E70, F70, G70, A70, B70, C71, D71, E71, F71, G71, A71, B71, C72, D72, E72, F72, G72, A72, B72, C73, D73, E73, F73, G73, A73, B73, C74, D74, E74, F74, G74, A74, B74, C75, D75, E75, F75, G75, A75, B75, C76, D76, E76, F76, G76, A76, B76, C77, D77, E77, F77, G77, A77, B77, C78, D78, E78, F78, G78, A78, B78, C79, D79, E79, F79, G79, A79, B79, C80, D80, E80, F80, G80, A80, B80, C81, D81, E81, F81, G81, A81, B81, C82, D82, E82, F82, G82, A82, B82, C83, D83, E83, F83, G83, A83, B83, C84, D84, E84, F84, G84, A84, B84, C85, D85, E85, F85, G85, A85, B85, C86, D86, E86, F86, G86, A86, B86, C87, D87, E87, F87, G87, A87, B87, C88, D88, E88, F88, G88, A88, B88, C89, D89, E89, F89, G89, A89, B89, C90, D90, E90, F90, G90, A90, B90, C91, D91, E91, F91, G91, A91, B91, C92, D92, E92, F92, G92, A92, B92, C93, D93, E93, F93, G93, A93, B93, C94, D94, E94, F94, G94, A94, B94, C95, D95, E95, F95, G95, A95, B95, C96, D96, E96, F96, G96, A96, B96, C97, D97, E97, F97, G97, A97, B97, C98, D98, E98, F98, G98, A98, B98, C99, D99, E99, F99, G99, A99, B99, C100, D100, E100, F100, G100, A100, B100, C101, D101, E101, F101, G101, A101, B101, C102, D102, E102, F102, G102, A102, B102, C103, D103, E103, F103, G103, A103, B103, C104, D104, E104, F104, G104, A104, B104, C105, D105, E105, F105, G105, A105, B105, C106, D106, E106, F106, G106, A106, B106, C107, D107, E107, F107, G107, A107, B107, C108, D108, E108, F108, G108, A108, B108, C109, D109, E109, F109, G109, A109, B109, C110, D110, E110, F110, G110, A110, B110, C111, D111, E111, F111, G111, A111, B111, C112, D112, E112, F112, G112, A112, B112, C113, D113, E113, F113, G113, A113, B113, C114, D114, E114, F114, G114, A114, B114, C115, D115, E115, F115, G115, A115, B115, C116, D116, E116, F116, G116, A116, B116, C117, D117, E117, F117, G117, A117, B117, C118, D118, E118, F118, G118, A118, B118, C119, D119, E119, F119, G119, A119, B119, C120, D120, E120, F120, G120, A120, B120, C121, D121, E121, F121, G121, A121, B121, C122, D122, E122, F122, G122, A122, B122, C123, D123, E123, F123, G123, A123, B123, C124, D124, E124, F124, G124, A124, B124, C125, D125, E125, F125, G125, A125, B125, C126, D126, E126, F126, G126, A126, B126, C127, D127, E127, F127, G127, A127, B127, C128, D128, E128, F128, G128, A128, B128, C129, D129, E129, F129, G129, A129, B129, C130, D130, E130, F130, G130, A130, B130, C131, D131, E131, F131, G131, A131, B131, C132, D132, E132, F132, G132, A132, B132, C133, D133, E133, F133, G133, A133, B133, C134, D134, E134, F134, G134, A134, B134, C135, D135, E135, F135, G135, A135, B135, C136, D136, E136, F136, G136, A136, B136, C137, D137, E137, F137, G137, A137, B137, C138, D138, E138, F138, G138

Arena-style double stops

A double stop is two notes played together, usually on adjacent strings. Pentatonic double stops on the 2nd and 3rd strings, slid into place and repeated in triplets, are the stadium-rock version of the blues double stop. Resolve to the major 3rd (C# over A) for a bright ending, or to the b3 for a darker one.

▶ **TRACK 411** Double stops in A

A5

8

sl.

8 10 10 10 10 10 8 5

7 9 9 9 9 9 7 5

8 5 8 5 8 5 6

7 5 7 5 7 5 7

A 12/8 feel. Bar 1 slides D–G up to E–A. Bar 2 rocks between two pentatonic double stops in triplets and resolves to A and C#, the root and major 3rd.

NYLON · STEEL

Nylon and steel acoustic: without distortion, power chords sound thin. Use open-position riffs with ringing open strings, play full chords where the song allows, and replace palm-muted chugs with the muted strum (Chapter 22). Drop D works well on any acoustic guitar.

Electric: most riffs sit best on the bridge pickup with moderate gain; too much gain blurs fast riffs. Turn the guitar's volume control down for a cleaner sound without touching the amplifier. Chapter 42 covers amplifier and pedal settings.

Riff Study in E

Original study · straight eighths · 4/4 · 112 bpm

A complete 12-bar section with a riff (bars 1–4), a power-chord chorus (bars 5–8) and a pedal-tone bridge that returns to the riff (bars 9–12).

▶ **TRACK 412**

8

0 0 3 0 0 2

0 0 3 0 2 3

0 0 3 0 0 2

0 0 2 1 0 3 0

C5 D5 E5 G5 A5 C5 D5 E5

5

P.M. -----

5 5 5 5 5 5 5 5 7 7 7 7 7 7 7 7 2 2 2 2 5 7 5 7 2

3 3 3 3 3 3 3 3 5 5 5 5 5 5 5 5 2 2 2 2 5 7 3 5 2

0 0

9 A5 E5

P.M. -- P.M. -- P.M. -- P.M. -- P.M. -- P.M. -- P.M. --

T
A 2 4 5 4 2 4 5 7 5 7 2
B 0 0 0 0 0 0 0 0 0 0 3 0 0 2 0 0 3 4 5 0

PLAYING IT · Performance notes

Bars 5 and 6 are palm-muted throughout; bar 7 lifts the mute for the accented chords.

Bar 8 is the 3+3+2 figure from the rhythm patterns. Let each chord ring for its full length.

Bars 9–10 need alternate picking. Practise them alone at 80 bpm before playing the whole study.

TRACK 413 Rock loop in E minor · LOOP · rock · 116 bpm

E5 I5	E5 I5	C5 bVII5	D5 bVII5
-----------------	-----------------	--------------------	--------------------

IMPROVISE WITH**E minor pentatonic**

1 2 3 4

E blues

1 2 3 4

Use the open-position box for riffs between phrases and box 1 at the 12th fret for leads. End phrases on E or B.

LISTEN FOR · Listening

Riff builders: Chuck Berry, The Kinks, The Who, Cream, Led Zeppelin, Black Sabbath (Tony Iommi), Deep Purple, AC/DC (Malcolm and Angus Young). **Later styles:** Van Halen, Metallica (downstroke and gallop rhythms), Soundgarden and Rage Against the Machine (drop D), Nirvana, Queens of the Stone Age. Listen for how the guitar and bass double the riff, and where the riff stops to leave space for the vocal.

Chapter 44 review · use it, train it, test it**Using it in real music**

Playing in a band. In a rock band the guitar riff, the bass and the bass drum usually lock together. Learn a riff with the drummer's kick pattern in mind, and keep palm-muted parts quieter than the accented hits so that the dynamics come through.

Writing songs. A strong riff can carry a whole verse. Write two-bar riffs from the pentatonic scale, add one chromatic passing note, and try the result in drop D. The progressions in Chapter 14 give chorus changes that contrast with a riff-based verse.

How to train it

Drill	How	Time	Goal
Downstroke endurance	Palm-muted eighths on the open E at 120–160 bpm, downstrokes only	3 min	Even volume for a full minute without tension
Power-chord shifts	E5 – G5 – A5 – C5 – D5 on beat 1 of each bar, muting all other strings	3 min	No open-string noise between chords
Gallop	Gallop rhythm on the open E, then on E5 – G5 – A5	3 min	Even sixteenth, accent on the beat

Drill	How	Time	Goal
Riff variation	Play any riff from this chapter as AAAB, inventing a new bar B each time	5 min	Four different endings in a row
Double stops	Bar 2 of the double-stop example, looped, at 60–90 bpm (dotted quarter)	3 min	Both notes sound together, no stray strings

Check your understanding

Write your answers, then check them in Appendix J.

1. Which notes make up an E5 power chord, and why is it neither major nor minor?

2. Why are rock eighth-note riffs usually played with downstrokes only, and when do you switch to alternate picking?

3. What is the tuning of drop D, from the 6th string to the 1st?

4. In a chromatic riff, where do the chromatic passing notes usually fall?

5. In the pedal-tone riff in E, what does the open low E play between the melody notes, and why?

6. Describe the gallop rhythm.

7. Over an A chord, which double stop gives a bright major ending?

MY NOTES

CHAPTER 45

Funk Rhythm



TRACKS

45

Funk guitar is a rhythm instrument first. The picking hand moves in steady sixteenth notes, most of the strokes are muted, and short chords on the top strings fit between the bass line and the drums.

Funk developed in the 1960s from rhythm and blues and soul, above all in James Brown's bands. Its guitar parts are short, repeated and percussive. A funk guitarist rarely plays a full chord for a full bar: instead the hand keeps a sixteenth-note pulse going and lets only a few strokes sound as chords, with the rest muted. Harmony is usually static, one or two chords for a whole section, so the interest comes from rhythm and from small changes of voicing.

Element	Typical funk approach
Feel	Straight or slightly swung sixteenths; strong downbeat on beat 1
Harmony	Dominant 9th, 7#9 and 13th chords; minor 7th vamps; one-chord sections
Voicings	Three- and four-note shapes on the middle and top strings
Technique	Constant sixteenth strumming, muted scratches, fretting-hand damping
Sound	Clean amplifier, bright single-coil pickups, often light compression

The sixteenth-note engine

The strumming hand moves down and up in sixteenths all the time: down on the beat and the "&", up on the "e" and the "a". The rhythm comes from deciding which strokes hit the strings, which brush them muted and which miss. Because the motion never changes, the timing of every stroke is fixed by the hand itself. Use a loose wrist, a small movement, and strum mostly the top three or four strings.

▶ TRACK 414 The engine: constant sixteenths, accents on 2 and 4



Light grey strokes are muted scratches, dark strokes are chords. Count "1 e & a 2 e & a" aloud while you play.

Muted scratches

For a scratch (the muted chuck of Chapter 36, played on sixteenths), relax the fretting hand so that the fingers lie on the strings without pressing them down. The strum then produces a dry click with no pitch. Switching between pressing and relaxing, while the strumming hand keeps going, is the central skill of funk rhythm. The fretting hand decides what sounds; the picking hand only keeps time.

▶ TRACK 415 Chords and scratches



The direction of every stroke follows the constant motion: down on even slots, up on odd ones. x marks a muted scratch.

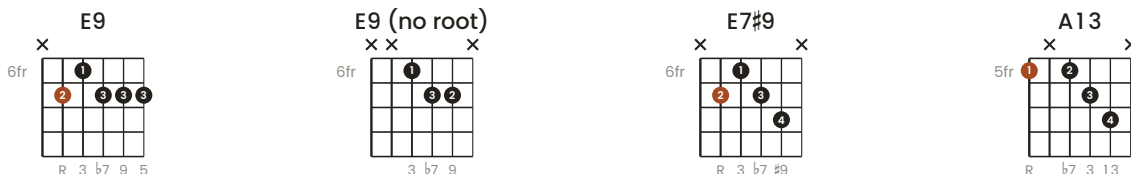
TRY THIS · Two-step practice

Set a metronome to 70 bpm. Play bar 1 of the pattern above with the fretting hand relaxed throughout, so every stroke is a scratch. Then press the chord only on the marked strokes. When it is even, raise the tempo by 4 bpm at a time up to 100.

Funk chords

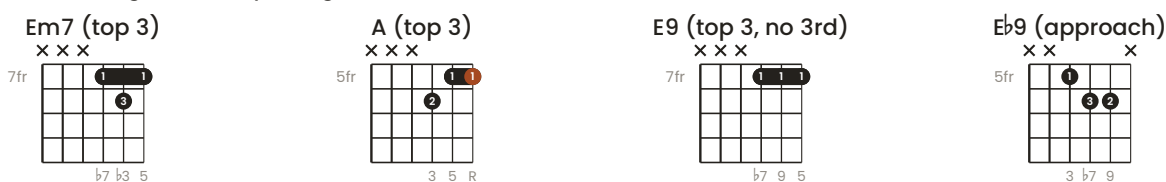
The dominant 9th is the standard funk chord. Funk players usually leave out the root (the bass player has it) and the 5th, and play the 3rd, $\flat 7$ and 9th on the middle strings. The 7#9 adds bite: its #9 is the same pitch as the minor 3rd, so the chord holds major and minor 3rds at once. Minor 7th chords played on the top three strings are the other common colour, often alternating with a major triad a 4th higher (a Dorian vamp, see Chapter 9).

Dominant funk voicings in E



The three-note E9 is the classic short chank. The A13 shape is the 13th version of a dominant chord with the root on the 6th string.

Small voicings on the top strings



Em7 to A is a $i\bar{7}$ – IV vamp in E Dorian. Each voice of Em7 moves down by step or a 3rd into A. The top-three E9 omits the 3rd, so it suits both E9 and Em9 grooves. Slide into any shape from a half step below.

Comping on the top strings

On the top three strings a chord takes up a narrow band of pitch, above the bass and the keyboard's left hand. That leaves room for a busy rhythm without clashing with other instruments. The example alternates Em7 and A twice per bar.

TRACK 416 Em7 – A vamp on the top strings

The notation shows a 4/4 bar with alternating Em7 and A chords. The guitar part is written for strings T, A, and B. The fret numbers are as follows:

String	Em7	A	Em7	A
T	7	7	7	7
A	8	8	8	8
B	7	7	7	7

Every rest is a muted scratch or a missed stroke; keep the hand moving in sixteenths throughout.

The one

James Brown asked his bands to play "on the one": the whole band lands hard on beat 1 of each bar, and everything after it is syncopated around that point. The bass player and the bass drum mark the one most strongly. The guitar either hits the one too or deliberately avoids it and answers just after it. Either way, everyone in the band must know where it is.

TRACK 417 How a funk groove locks together

The diagram shows the timing of the instruments over four measures. The measures are labeled 1, 2, 3, and 4. The instruments are: Hi-hat, Kick, Snare, Bass, and Guitar. The timing is as follows:

Measure	Hi-hat	Kick	Snare	Bass	Guitar
1	1	1	1	1	1
2	1	1	1	1	1
3	1	1	1	1	1
4	1	1	1	1	1

The bass doubles the kick on the one and the "e" of 2 and 3. The guitar avoids the one, hits the backbeat with the snare, and fills the sixteenth gaps the others leave open.

Single-note funk lines

Many funk parts are single-note lines on the 4th, 3rd and 2nd strings, played with the same constant sixteenth motion as the chords. The notes usually come from the Dorian or Mixolydian mode or the minor pentatonic with added 6th (Chapter 9). Mute every string you are not playing: the fretting-hand fingers lie across the strings, and the picking hand damps the bass strings.

▶ TRACK 418 Single-note line in E Dorian

Tablature for Track 418:

```

T
A
B
7-x-7-5-x-7-x-7-x-x-4-6-7-6-4-7-x-7-5-x-7-x-7-5-x-7-x-7
  
```

E, G, A, B, C#, D at the 7th position. The x noteheads are dead notes: the fretting finger touches the string without pressing it, and the pick strikes it for a percussive click.

CAREFUL · Damping is half the sound

If a single-note line rings, it stops sounding like funk. Lift the fretting finger slightly after each note so the string stops, and let the picking hand's palm touch the low strings. A short, dry note is the goal.

▶ TRACK 419 One-chord vamp on E9 · LOOP · funk · 100 bpm

E9 I9	E9 I9	E9 I9	E9 I9
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IMPROVISE WITH

E mixolydian

E minor pentatonic

Practise rhythm first: chord hits on E9 and scratches. Then play single-note lines from E Mixolydian; the G of the minor pentatonic is the bluesy #9.

▶ TRACK 420 Em7 – A Dorian vamp · LOOP · funk · 96 bpm

Em7 i7	A IV	Em7 i7	A IV
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IMPROVISE WITH

E dorian

E minor pentatonic

The C# of E Dorian is the 3rd of A; use it as the target note when the chord changes.

Locking with the drums

A funk guitar part works when it fits the drum pattern rather than competing with it. Four habits help:

- **Find the hi-hat.** The hi-hat usually plays the eighth or sixteenth grid; your hand should move with it exactly.
- **Accent with the snare.** A chord on beats 2 and 4 together with the snare gives the backbeat weight.

- **Fill the gaps.** Put your chords on sixteenths that the kick and bass leave empty.
- **Repeat exactly.** A funk part is repeated for minutes. Small variations are fine, but the basic pattern must stay the same each bar so that the band can lock to it.

NYLON · STEEL

Nylon: funk strumming works on nylon strings with the fingers (index for down and up strokes, or thumb down and index up), but chords sound softer and scratches less crisp. It suits Latin and Brazilian grooves better than hard funk (Chapter 50).

Steel and electric: the classic funk sound is an electric guitar with single-coil pickups on a clean amplifier, often the neck and middle pickups combined, with a light compressor. On steel-string acoustic, use a thin pick and strum close to the soundhole edge for a tighter attack.

Funk Study in E

Original study · sixteenth-note feel · 4/4 · 100 bpm

An 8-bar groove on E9 with a slide approach from E^b9, a single-note break and a closing 7#9 on the one.

▶ TRACK 421

E9

T	7	x	x	x	x	7	x	x	x	7	x	x	x	7	x	x	x	7	x
A	7	x	x	x	x	7	x	x	x	7	x	x	x	7	x	x	x	7	x
B	6	x	x	x	x	6	x	x	x	6	x	x	x	6	x	x	x	6	x

T	7	x	x	x	x	7	x	x	x	7	x	x	x	7	x	x	x	6	7
A	7	x	x	x	x	7	x	x	x	7	x	x	x	7	x	x	x	6	7
B	6	x	x	x	x	6	x	x	x	6	x	x	x	6	x	x	x	5	6

T																			
A																			
B	7	x	7			5	x	7	x			7			7	x	7		

E7#9

T	7	x	x	x	x	7	x	x	x	7	x	7	x	x	7	x		8	
A	7	x	x	x	x	7	x	x	x	7	x	7	x	x	7	x		7	
B	6	x	x	x	x	6	x	x	x	6	x	6	x	x	6	x		6	

PLAYING IT · Performance notes

The x chords are muted scratches. Learn the chord hits first with the scratches left out, then add them at a lower volume than the chords.

In bars 2 and 4 the Eb9 shape slides up one fret into E9 on the "&" of 3.

Bars 5–6 are the single-note line from this chapter. The G against E9 is the bluesy #9 that the closing E7#9 chord states as a chord.

LISTEN FOR · Listening

James Brown's bands: guitarists Jimmy Nolen, Alphonso "Country" Kellum and Catfish Collins (the late 1960s and early 1970s singles and live albums). **New Orleans:** The Meters (Leo Nocentelli). **Soul and disco:** Curtis Mayfield, Sly and the Family Stone, Earth, Wind & Fire (Al McKay), Chic (Nile Rodgers). **Later:** Prince, Tower of Power, Vulfpeck and Cory Wong. Listen to the guitar together with the hi-hat, and notice how little of each bar the guitar actually fills.

Chapter 45 review · use it, train it, test it**Using it in real music**

Soul, R&B and pop. The sixteenth engine and the small top-string voicings are standard in soul, disco, R&B and modern pop rhythm guitar. They also work in Chapter 55, which uses the same hand with richer chords.

Playing with a band. Ask the drummer which sixteenths the kick drum plays, and leave those for the bass. A part that is too busy is the most common problem: try removing one chord hit per bar and check whether the groove improves.

How to train it

Drill	How	Time	Goal
Constant motion	Muted sixteenths for one minute at 80–110 bpm, accent on 2 and 4	3 min	No gaps, no speeding up at the accents
Press and release	Bar 1 of the "Chords and scratches" pattern, chord pressed only on the marked strokes	4 min	Clean contrast between chord and scratch
Voicing switch	Em7 – A on the top strings, then E9 and E7#9 in the middle, one per bar	3 min	Changes without breaking the sixteenth motion
Single-note damping	The Dorian line at 80 bpm, each note short	3 min	No string rings past its sixteenth
One count	Play a groove and shout "one" on every beat 1 for a full minute	2 min	Never lose the downbeat

Check your understanding

Write your answers, then check them in Appendix J.

1. On which subdivisions does the strumming hand move down, and on which up?

2. How do you produce a muted scratch?

3. Which notes does the three-note E9 voicing on strings 4, 3 and 2 contain?

4. Spell E7#9 and explain why it sounds both major and minor.

5. What does "the one" mean in funk?

6. Why do funk guitarists often play chords on the top strings?

7. Which mode gives the i7 - IV vamp Em7 - A?

MY NOTES

CHAPTER 46

Country and Bluegrass



TRACKS

46

Country guitar is built on a bouncing bass and strum, bass runs that connect the chords, and bright major-key licks. Bluegrass adds speed, drive and a flatpicking tradition of fiddle tunes, crosspicking and the G-run.

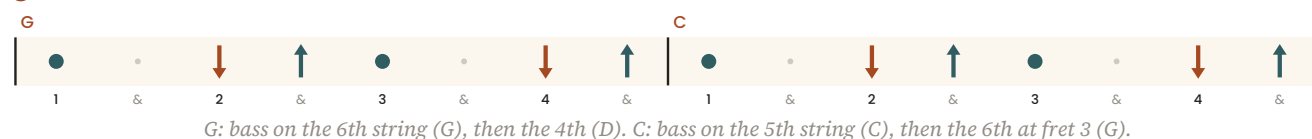
Country music comes from the Anglo-Celtic ballad and fiddle traditions of the American South, mixed with blues, gospel and popular song. Its guitar parts are mostly in major keys, mostly in open position or with a capo, and almost always tied to a strong two-beat feel: bass on 1 and 3, chord on 2 and 4. Bluegrass, which took shape in Bill Monroe's band in the 1940s, plays the same material faster, with acoustic instruments only, and gives the guitar both a driving rhythm role and flatpicked solos.

Element of the style	Country	Bluegrass
Feel	Two-beat "boom-chick", straight eighths, sometimes shuffle	Fast two-beat, straight eighths, pushed slightly ahead of the beat
Harmony	I – IV – V, I – vi – ii – V, secondary dominants	I – IV – V with occasional $\flat$ VII; II7 – V7 in fiddle tunes
Keys	G, C, D, E and A, often with a capo	G, C, D, A (G shapes with capo 2) and E
Technique	Bass runs, Carter style, hybrid picking, double stops, bends	Flatpicking, crosspicking, G-run, bass runs
Instrument	Steel-string acoustic; Telecaster-type electric	Steel-string dreadnought

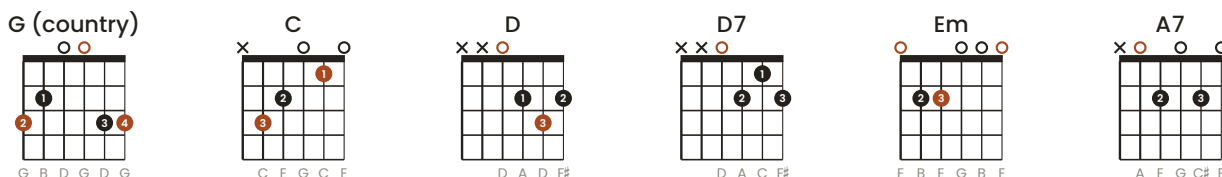
Boom-chick

Boom-chick (Chapter 22) is the country rhythm part: root on beat 1, alternate bass (usually the 5th) on beat 3, strums on 2 and 4. In country the strums cover the treble strings only, so that the bass stays clear, and the bass on beat 3 often drops below the root: on C it is the G on the 6th string. An upstroke on the "&" of 2 and 4 lightens the pattern.

▶ TRACK 422 Boom-chick in G and C



Country chord shapes in G



The G with the 2nd string at fret 3 (D on top of B) is the standard bluegrass and country G. It leaves the 1st and 2nd fingers free for bass runs and keeps the ring and little fingers anchored when changing to C and D.

Bass runs

Between chords, the bass walks through the scale to the root of the next chord. The run starts on beat 3 of the last bar of the old chord and lands on the new root on beat 1. Use the notes of the key's major scale; a chromatic note between two scale notes is also common.

TRACK 423 Bass runs in G

G **C**

4

G **D7** **G**

G to C walks up G – A – B – C. C to G walks down C – B – A – G. D7 to G walks down D – C – B – G.

TIP • The rule for any run

Count the scale steps between the two roots. Two steps fit on beats 3 and 4 as quarter notes; three steps need eighth notes on beat 4 or a start on beat 2. Always land on the new root on beat 1.

Carter-style picking

Maybelle Carter of the Carter Family played the melody on the bass strings with her thumb and filled the space between melody notes with down-up brush strokes on the treble strings. The result is melody and accompaniment from one guitar. Today the style is often played with a flatpick: pick the melody note, then brush down and up. Hammer-ons from open strings decorate the melody.

TRACK 424 Carter-style melody in C

C

3

G7 **C**

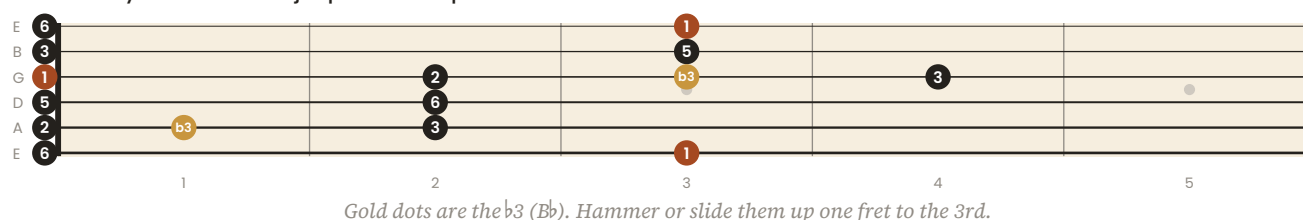
Melody notes (C, E, D–E, G, F, D, B, C) on the beat; brush strums down and up between them. Keep the chord shape held so the brushes ring.

Hybrid picking and country licks

Electric country players use hybrid picking (Chapter 24): the pick plus the middle and ring fingers. In country it serves two jobs: string skips and two-note pinches for 6ths and double stops, and the clucking "chicken picking" sound, made by snapping the finger notes (pull the string slightly away from the body before releasing it), often with the palm lightly muting the lower strings.

Country lead lines use the major blues scale (Chapter 7): the major pentatonic with the $\flat 3$ added, hammered or bent up to the 3, and with the 6th as a favourite resting note. Double stops in 3rds and 6ths, often slid into place, imitate the pedal steel guitar and the fiddle.

The country scale in G: major pentatonic plus $\flat 3$



▶ TRACK 425 Hybrid-picked lick in G

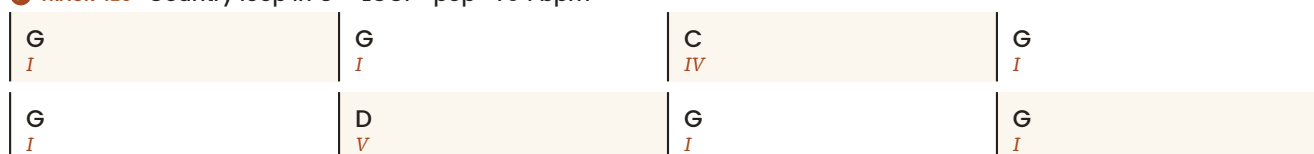


Bar 1 climbs the G major pentatonic with the $\flat B$ -B hammer-on. Bar 2 is a run of 6ths on strings 3 and 1; pick the 3rd string and pluck the 1st with the middle finger.

UNDER THE HOOD · Pedal-steel bends

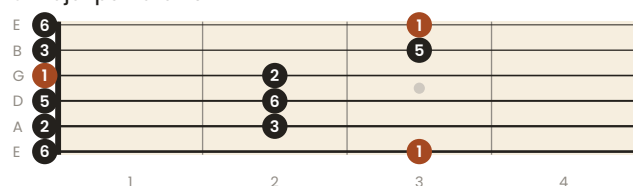
A pedal steel guitar raises single strings in pitch while other strings sustain. Guitarists imitate this by bending one string while holding a second note on a neighbouring string, for example bending the 3rd string from A to B at fret 2 while the 2nd string holds D at fret 3. The held note must stay in tune while the other moves; use the 3rd finger, supported by the 2nd, for the bend.

▶ TRACK 426 Country loop in G · LOOP · pop · 104 bpm

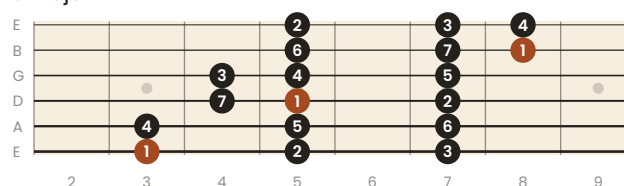


IMPROVISE WITH

G major pentatonic



G major

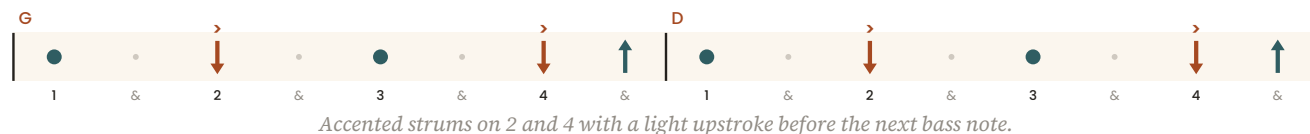


Play boom-chick with bass runs on one pass, then licks from the country scale. Add the $\flat B$ -B hammer-on and 6ths on strings 3 and 1.

Bluegrass rhythm

The bluegrass guitarist keeps the band's time. The pattern is the boom-chick, played louder, with a heavy pick and a strong strum on 2 and 4 that sits a fraction ahead of the beat. Bass runs connect the chords at the ends of phrases. Most bluegrass songs use three or four chords in eight-bar phrases.

▶ TRACK 427 Bluegrass rhythm in G and D



Typical bluegrass verse in G

G <i>I</i>	G <i>I</i>	C <i>IV</i>	G <i>I</i>
G <i>I</i>	G <i>I</i>	D <i>V</i>	D <i>V</i>
G <i>I</i>	G <i>I</i>	C <i>IV</i>	G <i>I</i>
G <i>I</i>	D <i>V</i>	G <i>I</i>	G <i>I</i>

Crosspicking

Crosspicking (Chapter 24) rolls a three-note pattern across three strings, so against a 4/4 bar of eight eighths it restarts every three notes and accents 3 + 3 + 2. Use strict alternate picking or a fixed down-down-up for each group, and let the strings ring like a banjo roll. In a band, the guitarist crosspicks behind a singer or as a contrast to single-note breaks.

▶ TRACK 428 The 3 + 3 + 2 accent pattern



▶ TRACK 429 Crosspicking on the middle strings

G

C

D

G

The roll moved down to strings 4, 3 and 2, for a darker, fuller sound under a singer or a fiddle. The accent falls on each restart of the three-note group.

The G-run

The G-run is the standard bluegrass ending phrase in G, associated with Lester Flatt. It descends from the 5th of the chord, passes through the $b3$ hammered to the 3, and lands on the low root. Many versions exist; the one below is typical. Players use it at the end of verses, after solos, and to signal the last line of a song.

TRACK 430 A typical G-run

G

T 3—0 2—0
A
B 1—2 3

D – B – A – G on strings 2 and 3, then B b hammered to B and the low G.

NYLON · STEEL

Steel: country and bluegrass belong to the steel-string acoustic. A dreadnought gives the loud bass that bluegrass rhythm needs; use a medium or heavy pick (0.8–1.2 mm).

Nylon: Carter style, boom-chick and bass runs sound soft and clear on nylon strings, played with thumb and fingers. Flatpicked fiddle tunes and hybrid-picked electric licks lose their attack.

Breakdown in G

Original study · flatpicking · 4/4 · 100 bpm

An 8-bar flatpicked melody in the style of a fiddle tune, ending with the G-run. It uses alternate picking (down on the beat, up on the "&") throughout.

TRACK 431

G C G

T 0 0 3—0 0 0 3—0 3—0 3—0 1—0 1—0 3—1 2—1 0 2—0 4—2—0 2—3
A
B 2—0

5 C D7 G

T 0 0 3—0 1—3—1—0 2—0—1— 2—0 3—0—2—0 3—1— 3—0 2—0
A
B 2—0 1—2—3

PLAYING IT · Performance notes

Strict alternate picking. The hammer-on in bar 8 is the only note not picked.

Practise at 70 bpm first. At full tempo, play a second guitar part with boom-chick and bass runs on the same chords.

Bar 4 descends from B to the low G across five strings; keep it even across the string changes.

LISTEN FOR • Listening

Foundations: the Carter Family (Maybelle Carter), Merle Travis, Chet Atkins. **Bluegrass:** Bill Monroe and his Blue Grass Boys, Flatt and Scruggs (Lester Flatt's rhythm and G-run), the Stanley Brothers with George Shuffler (an early crosspicker), Doc Watson, Clarence White, Tony Rice, Norman Blake, Bryan Sutton, Molly Tuttle. **Electric country:** James Burton, Roy Nichols, Albert Lee, Danny Gatton, Brent Mason, Brad Paisley. Listen to how the rhythm guitar's bass notes and the bass player's notes fit together.

Chapter 46 review • use it, train it, test it

Using it in real music

Accompanying singers. Boom-chick with bass runs makes a simple I – IV – V song sound complete on a solo guitar. Choose a capo position for the singer's key and keep the G, C, D and E shapes.

Jamming. Bluegrass and old-time jams use a small set of keys and forms. Strong, steady rhythm matters more than solos: when it is not your turn, play boom-chick with runs, and play quieter while others take a break.

How to train it

Drill	How	Time	Goal
Bass runs	G – C – G – D7 – G with a run into every chord, 80–110 bpm	4 min	Land on the new root on beat 1 every time
Carter style	Make up a melody on the bass strings over C and G7; brush between notes	4 min	Melody audible above the strums
Crosspicking	Strings 3–2–1 on G, C and D, accenting each group of three	3 min	Clear 3 + 3 + 2 accents with a metronome
Hybrid 6ths	Bar 2 of the hybrid lick, pick on string 3 and finger on string 1	3 min	Both notes exactly together
G-run	Play the G-run at the end of every 8-bar phrase of the bluegrass chart	3 min	Run starts on beat 1 of the last bar, without hesitation

Check your understanding

Write your answers, then check them in Appendix J.

1. In a boom-chick pattern on C, which bass notes do you play on beats 1 and 3?

2. Which notes walk the bass from G up to C?

3. What is Carter-style picking?

4. In a bar of three-string crosspicking, on which eighth notes do the accents fall?

5. How do you get the "chicken picking" sound?

6. Where does the G-run land, and where is it used?

MY NOTES

CHAPTER 47

Folk and Songwriting



TRACKS

47

In folk and singer-songwriter music the guitar accompanies a voice. Open-string chords, a capo, simple strumming and fingerpicking patterns, and a clear song form are the tools; the skill lies in choosing them to support the singer and the words.

Folk music in the broad sense covers traditional songs passed on by ear and the singer-songwriter tradition that grew from the folk revival of the 1950s and 1960s. In both, the guitar is an accompanying instrument: it gives the singer a key, a pulse and a harmony, and it shapes the song through dynamics and texture. Technique matters less than choices. A three-chord song with the right key, the right pattern and a well-placed fill is stronger than a complex part that covers the words.

Element	Typical approach
Instrument	Steel-string acoustic; nylon-string for softer ballads
Chords	Open chords with ringing strings: add9, sus2, sus4, 7th colours; slash chords with walking bass
Keys	Chosen for the voice, with a capo; shapes from G, C, D, E and A
Rhythm	Strumming in 4/4 and 3/4; fingerpicked arpeggios and Travis patterns
Form	Strophic ballads, verse-chorus songs, verse-chorus-bridge

Choosing a key: the capo

A capo shortens every string at once, so all the open-chord shapes move up together. Players keep the familiar G, C, D, E and A shapes and use the capo to put the song in the singer's key. To find the key, sing the highest note of the song (usually in the chorus) and move the capo until that note sits at the top of the comfortable range, not at its limit.

Capo	G shapes	C shapes	D shapes	E shapes	A shapes
0	G	C	D	E	A
1	A $\flat$	D $\flat$	E $\flat$	F	B $\flat$
2	A	D	E	F $\sharp$	B
3	B $\flat$	E $\flat$	F	G	C
4	B	E	F $\sharp$	G $\sharp$	C $\sharp$
5	C	F	G	A	D
7	D	G	A	B	E

The table also gives alternatives for the same key. A song in G can be played open with G shapes, with capo 5 and D shapes, or with capo 7 and C shapes. Two guitarists who choose different positions fill different registers instead of doubling each other.

NYLON · STEEL

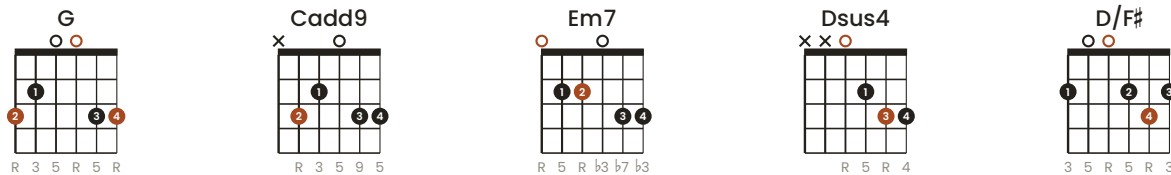
Nylon: classical guitars have a flat fingerboard and a wide neck, so they need a flat capo made for them; a curved steel-string capo leaves the middle strings buzzing.

Steel: use a capo that matches the curve of the fingerboard and clamp it just behind the fret, with the least pressure that stops buzzing. Too much pressure pulls the strings sharp; retune after fitting it.

Open-string voicings

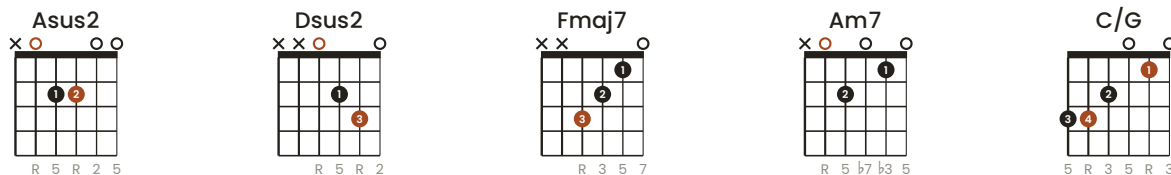
Folk guitar favours chords that let open strings ring. A useful family keeps the ring and little fingers on the 3rd fret of the 2nd and 1st strings while the other fingers move. The shared top notes (D and G) ring through every change, so the progression sounds connected even when the bass jumps.

The anchor-finger family in G



Ring and little fingers stay on fret 3 of strings 2 and 1 for the first four chords. D/F# puts the 3rd of D in the bass for walking lines from G to Em.

More open colours



Sus chords replace the 3rd with the 2nd or 4th and resolve back to the plain triad. Fmaj7 is the easy open substitute for the F barre chord.

Strumming for songs

One strumming pattern with dynamics is enough for most songs. Keep the hand moving down and up in eighths; let the chord sound on the strokes shown and miss on the others. Play verses quieter, with fewer strings or a lighter grip on the pick, and choruses louder and fuller.

▶ TRACK 432 Old faithful through the anchor-finger family

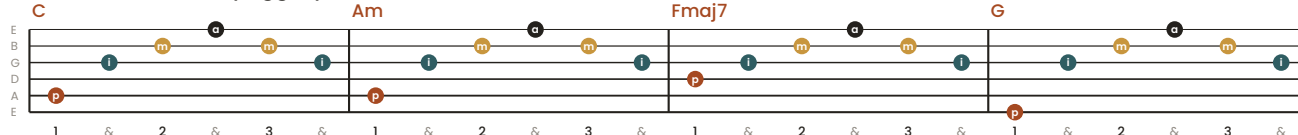


Old faithful (Chapter 22) on the four anchor-finger chords. Only the lower fingers move at each change, so the last upstroke of every bar can ring to the barline without being cut short.

Folk fingerpicking

Fingerpicked accompaniment gives a softer, more continuous texture than strumming. Two families cover most folk songs: arpeggio patterns, where the thumb plays the bass and the fingers roll through the treble strings (Chapter 18), and Travis picking, where the thumb alternates between two bass strings (Chapter 19). Waltzes and many ballads are in 3/4 and use a six-note arpeggio per bar.

▶ TRACK 433 A 3/4 arpeggio pattern



Thumb on the root, then i, m, a up the treble strings and back down. One pattern per bar.

TRACK 434 Travis-style accompaniment with anchor fingers

G **Cadd9**

Em7 **Dsus4** **D**

The thumb alternates between two bass strings on every beat; the fingers fill the off-beats. In bar 4 the little finger lifts from G to F# on the 1st string as Dsus4 resolves to D.

Writing progressions

Most songs use a handful of progressions, usually built from the chords of one key (Chapter 13). A progression of four chords, one or two bars each, repeated, is the usual starting point. What makes a song distinctive is the order of the chords, where the progression starts, and how the verse and chorus progressions differ.

Four common loops in G

G <i>I</i>	D <i>V</i>	Em <i>vi</i>	C <i>IV</i>
Em <i>vi</i>	C <i>IV</i>	G <i>I</i>	D <i>V</i>
G <i>I</i>	Em <i>vi</i>	C <i>IV</i>	D <i>V</i>
G <i>I</i>	F <i>bVII</i>	C <i>IV</i>	G <i>I</i>

I – V – vi – IV; vi – IV – I – V (the same chords started from the minor); I – vi – IV – V; I – bVII – IV – I (borrowed bVII, a folk-rock and Mixolydian sound).

Three techniques make simple progressions sound composed:

- **Descending bass.** Keep the upper chord shapes and walk the bass down the scale with slash chords. The line itself becomes a melody.
- **Pedal tones.** Keep one or two notes, often open strings, the same over changing bass notes. The anchor-finger family above does this.
- **Contrast between sections.** Give the verse a narrower range and fewer chords, and the chorus a new starting chord, usually IV or vi, so that the chorus arrives with a change of colour (Chapter 14).

Descending bass line in C

C <i>I</i>	G/B <i>V6</i>	Am <i>vi</i>	Am/G <i>vi/5</i>
F <i>IV</i>	C/E <i>I6</i>	Dm7 <i>ii7</i>	G <i>V</i>

The bass walks C – B – A – G – F – E – D, then lands on G (V) to return to C.

TRY THIS · Write a verse and a chorus

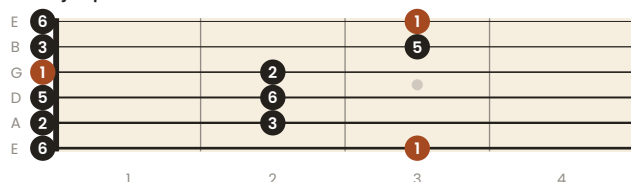
Choose a key with the capo. Write a four-chord loop for the verse using I, vi and IV. For the chorus, start on IV, use a different order and end on V. Sing a simple melody over each, then play verse – chorus – verse – chorus without stopping.

TRACK 435 Ballad loop in G · LOOP · ballad · 76 bpm

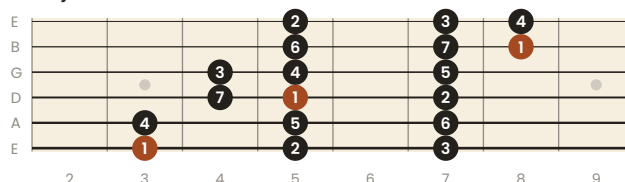
G <i>I</i>	Cadd9 <i>IVadd9</i>	Em7 <i>vi7</i>	Dsus4 <i>Vsus4</i>	D <i>V</i>
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IMPROVISE WITH

G major pentatonic



G major

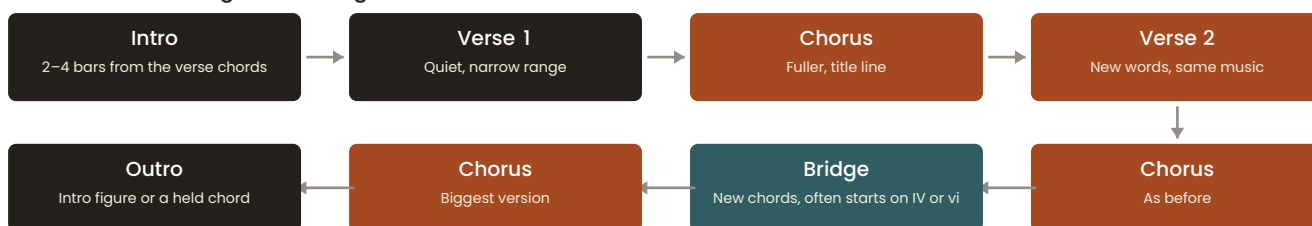


Use it to practise accompaniment patterns, or hum melodies over it to write a verse. For a melody on the guitar, start with the G major pentatonic.

Song forms

Form	Sections	Typical in
Strophic	A A A A (verse only, often with a refrain line)	Traditional ballads, early folk, hymns
Verse-chorus	Verse, chorus, verse, chorus, (bridge), chorus	Most pop, folk-pop and singer-songwriter songs
Verse-pre-chorus-chorus	Verse, pre-chorus (a build), chorus	Pop and rock
AABA (32-bar)	Two A sections, a contrasting B (bridge), A again	Tin Pan Alley songs, jazz standards, early pop
12-bar blues	Three four-bar lines	Blues and blues-based folk (Chapter 43)

A verse-chorus song with a bridge



Arranging the accompaniment around a voice

The voice carries the melody and the words; the guitar part should support both without competing.

- **Leave the vocal range clear.** Keep busy figures below or above the melody. A singer in the guitar's middle register is helped by bass-heavy picking or by high capo voicings.
- **Build with texture.** Fingerpick or play single strums in the first verse, strum in the chorus, add bass runs in the second verse, and play the final chorus fullest.
- **Fill the gaps, not the words.** Put fills, bass runs and hammer-ons in the space at the end of a vocal line, and return to plain chords when the singer comes back in.

- **Use stops and holds.** A held chord or a full bar of silence before a chorus or a key line focuses attention on the voice.
- **Make intros and outros from the song's own material.** Two or four bars of the verse pattern, or the chorus progression ending on I, are enough.

TRACK 436 A fill between vocal lines

Chord progression: G (bars 1-2), Em7 (bars 3-4)

The singer holds the last word over bar 1. The guitar fills bar 2 with a line that walks down to the root of Em7 (D – E – D – B – A – F# – E), and is back on a plain chord when the voice returns.

Song Accompaniment in G

Original study · fingerpicked verse and chorus · 4/4 · 88 bpm

A complete 12-bar accompaniment, verse (bars 1–8) and chorus (bars 9–12), using the anchor-finger chords. The verse keeps an even Travis-style pattern; the chorus switches to fuller pinched chords for contrast.

TRACK 437

Chord progression: G (bars 1-4), Cadd9 (bars 5-8), Em7 (bars 9-12)

Chord progression: Dsus4 (bars 1-4), D (bars 5-8), G (bars 9-12), Cadd9 (bars 13-16)

7

Em7 Dsus4 G Cadd9

10

G Dsus4 D G

PLAYING IT · Performance notes

Keep the ring and little fingers on fret 3 of strings 2 and 1 for the whole verse; only the bass fingers move.

In the chorus, pinch the bass and the first chord of each pair together with thumb and fingers, or play the chords as brush strokes with the index finger.

With a capo the same part fits any key; at capo 2 it sounds in A, at capo 4 in B.

LISTEN FOR · Listening

Folk revival: Woody Guthrie, Pete Seeger, Joan Baez, Bob Dylan's early acoustic albums. **British folk:** Bert Jansch, Davy Graham, Nick Drake, Sandy Denny. **Singer-songwriters:** Joni Mitchell (open tunings, see Chapter 25), Paul Simon, James Taylor, Leonard Cohen, Tracy Chapman. **Later:** Gillian Welch and David Rawlings, Laura Marling, Jason Isbell, Phoebe Bridgers. Listen to how little the guitar plays while the voice sings, and what it plays in the gaps.

Chapter 47 review · use it, train it, test it**Using it in real music**

Accompanying yourself. Pick the key for your voice first, then the capo position, then the pattern. Practise songs at performance volume: your strumming hand and voice must be able to work independently.

Accompanying others. Ask the singer for the key or the capo position before anything else. If two guitars play, one strums low while the other picks a capo-high part, or one plays a fingerpicked verse while the other waits for the chorus.

CHAPTER 48

Jazz Harmony



TRACKS

48

Jazz builds its harmony from seventh chords and the ii – V – I progression. On the guitar that means shell voicings and four-note drop voicings, smooth guide-tone movement from chord to chord, and comping rhythms that leave room for a soloist.

In jazz the basic chord is the seventh chord, not the triad, and colour tones (9ths, 11ths, 13ths and altered notes) are added freely. Most tunes are chains of ii – V – I progressions moving through several keys. The guitarist's main jobs are to state these chords clearly with compact voicings, to connect them smoothly, and to comp (accompany) with rhythms that support the soloist. This workshop covers the voicings and the comping; the theory of choosing notes over changes is in Chapter 35, and the principles of smooth voice movement are in Chapter 34.

The five chord qualities

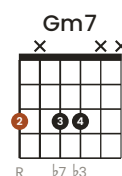
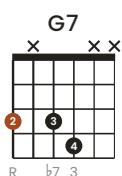
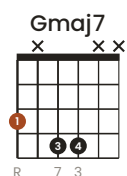
Chord	Formula	Typical extensions	Scale to play over it
maj7	1 3 5 7	9, #11, 13 (6)	Major (Ionian); Lydian for a brighter sound
m7	1 b3 5 b7	9, 11, 13	Dorian
7 (dominant)	1 3 5 b7	9, 13; altered: b9, #9, #11, b13	Mixolydian; altered scale or half-whole diminished for tension
m7b5	1 b3 b5 b7	11, b13 (9 in some styles)	Locrian (Locrian #2 when the 9 is used)
dim7	1 b3 b5 b7	Any note a whole step above a chord tone	Whole-half diminished

The 3rd and the 7th decide the quality of each chord: major 3rd with major 7th is maj7, minor 3rd with minor 7th is m7, major 3rd with minor 7th is dominant. The 5th adds little unless it is altered, so guitarists often leave it out.

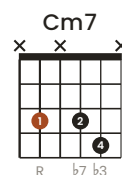
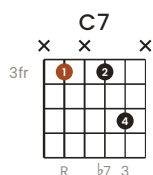
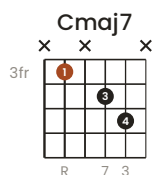
Shell voicings

A shell voicing plays only the root, 3rd and 7th. Shells are easy to play, sound clean, leave room for the bass player and the soloist, and show the harmony with nothing extra. Learn them with the root on the 6th string and on the 5th string, in three qualities: six shapes that work in every key.

Shells with the root on the 6th string (G)



Shells with the root on the 5th string (C)

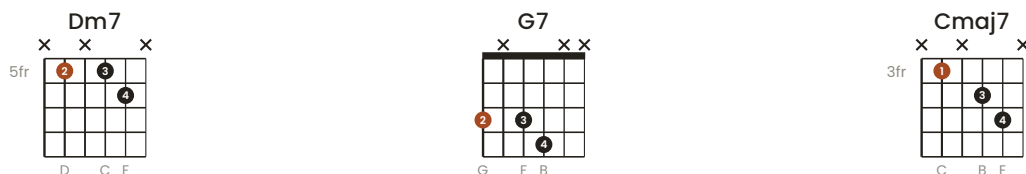


Mute the unused strings with the underside of the fretting fingers. Each shape moves to any root along its string.

The ii – V – I and guide tones

The ii – V – I is to jazz what I – IV – V is to folk. Alternate root-5 and root-6 shells through it and the hand hardly moves, because the 3rds and 7ths move by a half step or stay where they are. These two notes are the **guide tones**: the 7th of each chord falls a half step to the 3rd of the next, and the 3rd of each chord stays to become the 7th of the next.

ii – V – I in C with shells



D C F → G F B → C B E. The upper two notes of each shape are the guide tones.

	Dm7 (ii)	G7 (V)	Cmaj7 (I)
Upper guide tone	F (♭3)	F (♭7)	E (3)
Lower guide tone	C (♭7)	B (3)	B (7)

▶ TRACK 438 Guide tones over ii – V – I in C

Dm7 G7 Cmaj7

T	6	6	5	5
A	5	4	4	4
B	5	3	3	3

Only two notes per chord above the bass. C falls to B and F falls to E: the whole progression is two half-step resolutions.

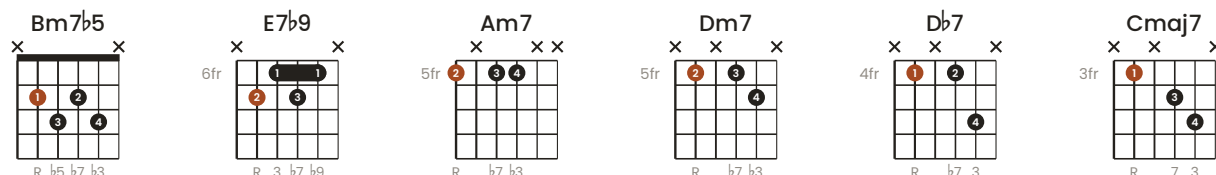
TRY THIS · Guide tones round the cycle

Play only the 3rd and 7th of each chord on the 3rd and 2nd strings (or 4th and 3rd) through ii – V – I in C, B♭ and A♭. Then do the same for the jazz blues later in this chapter. If you can play the guide tones, you can comp the tune.

Minor ii – V – i and tritone substitution

In a minor key the ii chord becomes m7♭5 and the V chord usually takes a ♭9, borrowed from the harmonic minor scale (Chapter 31). A second common move is the **tritone substitution**: replace a dominant 7th with the dominant 7th a tritone (six frets) away. Both chords share the same guide tones (G7 has B and F; D♭7 has F and C♭, which sounds as B), and the bass moves down by half steps. Chapter 32 and Chapter 35 explain both moves in detail.

Minor ii – V – i in A, and ii – ♭II7 – I in C



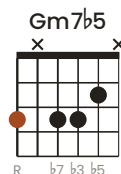
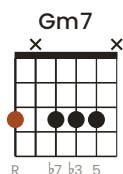
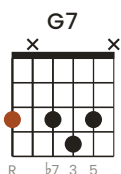
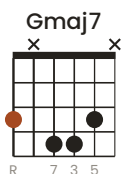
Left: Bm7♭5 – E7♭9 – Am7. Right: Dm7 – D♭7 – Cmaj7, with the bass falling D – D♭ – C.

Drop-2 and drop-3 voicings

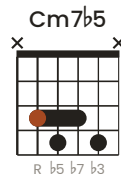
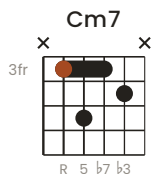
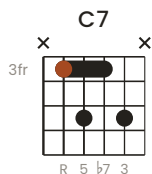
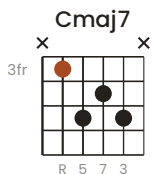
For a fuller sound, use four-note chords. Stacking a seventh chord in close position (all four notes within an octave) makes stretches that are often impossible on the guitar. Lowering one voice by an octave spreads the chord across the strings:

- **Drop-2:** take the second voice from the top of a close voicing down an octave. The result fits on four adjacent strings: strings 5–2 or 4–1.
- **Drop-3:** take the third voice from the top down an octave. The result has the root on the 6th string and skips the 5th string.

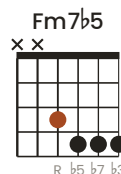
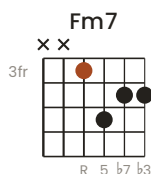
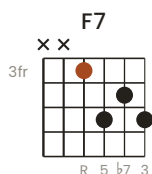
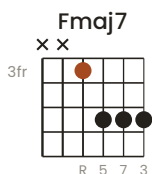
Drop-3, root on the 6th string (G)



Drop-2, root on the 5th string (C)



Drop-2, root on the 4th string (F)

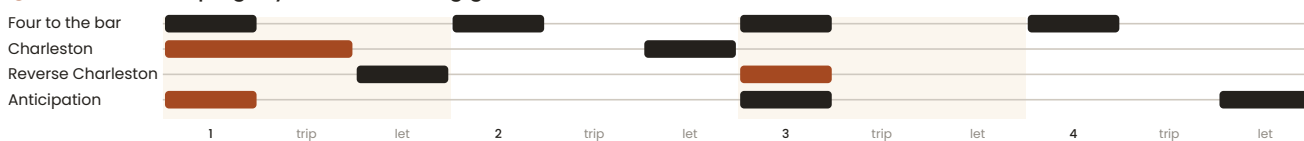


Each family changes quality by moving one or two notes by a half step. Learn the four qualities of one family before moving to the next.

Comping rhythms

To comp is to accompany with chords. Jazz eighth notes are swung (long-short, as in the blues shuffle), and comping chords are short and placed with care. Four rhythms cover most situations.

TRACK 439 Comping rhythms on a swing grid



The grid is in triplets, so the "S" falls on the third part of each beat. The anticipation lands on the "S" of 4 and belongs to the next bar's chord.

Four to the bar

In the big-band style associated with Freddie Green of the Count Basie Orchestra, the guitar plays a short chord on every beat with three-note shells on the 6th, 4th and 3rd strings. The chord is released just after it sounds, so the effect is closer to a drum than to a sustained chord.

▶ **TRACK 440** Four-to-the-bar shells on a blues in F

F7 Bb7 F7 Cm7 F7

T 2 2 2 2 7 7 7 7 2 2 2 2 4 4

A 1 1 1 1 6 6 6 6 1 1 1 1 3 3 2 2

B 1 1 1 1 6 6 6 6 1 1 1 1 3 3 1 1

Short quarter notes, all downstrokes, light on 1 and 3 and slightly heavier on 2 and 4. Release the pressure just after each chord sounds.

▶ **TRACK 441** ii – V – I in C · LOOP · jazz swing · 120 bpm

Dm7 <i>ii7</i>	G7 <i>V7</i>	Cmaj7 <i>Imaj7</i>	Cmaj7 <i>Imaj7</i>
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IMPROVISE WITH

D dorian

G mixolydian

C major

All three scales hold the same notes; what changes is the target. Land on F or C over Dm7, B or F over G7, and E or B over Cmaj7.

Rhythm changes and blues changes

Two forms make up a large part of jam-session repertoire. The **jazz blues** is the 12-bar blues (Chapter 43) with ii – V movements and a diminished passing chord. **Rhythm changes** is a 32-bar AABA form based on George Gershwin's song "I Got Rhythm" (1930): the A sections cycle through I – vi – ii – V, the bridge moves through dominant 7th chords around the cycle of 5ths.

▶ **TRACK 442** Jazz blues in F · LOOP · jazz swing · 132 bpm

F7 <i>I7</i>	Bb7 <i>IV7</i>	F7 <i>I7</i>	Cm7 <i>vi7</i>
Bb7 <i>IV7</i>	Bdim7 <i>#iv°7</i>	F7 <i>I7</i>	D7 <i>VI7</i>
Gm7 <i>ii7</i>	C7 <i>V7</i>	F7 <i>I7</i>	D7 <i>VI7</i>
Gm7 <i>ii7</i>	C7 <i>V7</i>	F7 <i>I7</i>	D7 <i>VI7</i>

IMPROVISE WITH

F minor pentatonic

F mixolydian

F major pentatonic

Comp along with shells, then solo. Blues phrases work throughout; for the jazz sound, outline the 3rd and 7th of each chord in bars 8–12.

Rhythm changes in Bb: A section

Bb <i>I</i>	G7 <i>VI7</i>	Cm7 <i>ii7</i>	F7 <i>V7</i>
Bb <i>I</i>	Bb7 <i>I7</i>	Eb <i>IV</i>	Edim7 <i>#iv°7</i>
Bb <i>I</i>	G7 <i>VI7</i>	Cm7 <i>ii7</i>	F7 <i>V7</i>
Bb <i>I</i>	G7 <i>VI7</i>	Cm7 <i>ii7</i>	F7 <i>V7</i>

The second A ends on Bb instead of the turnaround; the last A is the same as the second.

Rhythm changes in Bb: bridge

D7 III7	D7 III7	G7 VI7	G7 VI7
C7 II7	C7 II7	F7 V7	F7 V7

III7 – VI7 – II7 – V7, two bars each. Each chord is the V7 of the next.

Chord melody

In chord melody the guitarist plays the melody as the top note of each chord, with single melody notes between the chords. Drop-2 voicings on the top four strings are the usual starting point, because their highest note is on the 1st string. Find the melody note, pick a voicing of the right chord with that note on top, and fill the rest of the bar with the melody alone.

▶ TRACK 443 Chord-melody intro over ii – V – I

Dm7 G7 Cmaj7 C6

	5	3	1	1	0	1	0	3	0	0
T	3			3			1			1
A	5			4			4			2
B	3			3			2			3

Melody A – G – F, F – E – F, E – G – E, E. Each chord is a drop-2 voicing with the melody note on the 1st string; the last chord adds the root on the 5th string.

NYLON · STEEL

Nylon: jazz harmony sounds natural on nylon strings played with the fingers: the thumb takes the bass or root, the fingers pluck the upper notes together, as on a piano. This is the basis of bossa nova accompaniment (Chapter 50).

Steel and electric: the classic jazz guitar is a hollow-body archtop or semi-hollow electric with flatwound strings, played on the neck pickup with the tone control turned down. On a steel-string acoustic, play shells with a pick and keep the chords short.

Comping Study: Jazz Blues in F

Original study · swing eighths · 4/4 · 132 bpm

A complete chorus of the jazz blues in F, comped with shell voicings. It mixes the Charleston, the reverse Charleston and four-to-the-bar, and ends with a one-bar tag on F7.

▶ TRACK 444

F7 Bb7 F7 Cm7 F7 Bb7

	2	2	3	3	2	2	3	3	3	3
T	1	1	1	1	1	1	1	1	1	1
A										
B	1	1	1	1	1	1	1	1	1	1

6

Bdim7 F7 D7 Gm7 C7

11

F7 D7 Gm7 C7 F7

T 3 3 7 7 5 5 3 3 3 3 5 5 5 5

A 1 1 5 5 3 3 3 3 2

B 3 3 3 3 3 3 3 3 1

2 2 1 1 5 5 3 3 1

PLAYING IT · Performance notes

Swing the eighths: the Charleston's second chord falls on the last third of beat 2.

Bars 9–12 switch to four to the bar to drive the turnaround. Keep every chord short.

Bdim7 in bar 6 is a passing chord: its root B rises a half step from the B $\flat$ of bar 5 and leads on to C, the 5th of F7.

LISTEN FOR · Listening

Comping and rhythm: Freddie Green with the Count Basie Orchestra, Herb Ellis, Barney Kessel, Kenny Burrell. **Chord melody:** Joe Pass (*Virtuoso*), George Van Eps, Ted Greene, Ed Bickert. **Harmony and line:** Charlie Christian, Wes Montgomery, Jim Hall, Pat Martino, Pat Metheny, John Scofield, Julian Lage. Listen to a guitarist comping behind a soloist and notice how often the guitar rests.

Chapter 48 review · use it, train it, test it**Using it in real music**

Jazz standards. Most standards are chains of ii – V – I progressions in several keys. With shells in two string sets you can read and comp a lead sheet at sight; add drop-2 voicings when the band is small and the guitar has more space.

Richer pop and soul. Replace triads with maj7, m7 and 9th chords and move them by guide tones. The progression stays the same but sounds smoother and fuller (Chapter 55).

How to train it

Drill	How	Time	Goal
Shells in 12 keys	ii – V – I with shells around the cycle of 5ths (C, F, B $\flat$, E $\flat$...)	5 min	Any key without stopping, 80 bpm, one chord per bar
Guide-tone lines	Only the 3rd and 7th of each chord through the jazz blues in F	3 min	Hear every half-step resolution
Drop-2 qualities	One root, four qualities of one drop-2 family, then a new root	4 min	Change quality by moving only the changed notes
Comping rhythms	Jazz blues in F with the practice loop, alternating Charleston and four to the bar	5 min	Light, short, swinging chords
Chord melody	Harmonise the first four bars of a simple melody with drop-2 voicings	5 min	Melody note always on top

Check your understanding

Write your answers, then check them in Appendix J.

1. Which three notes make a shell voicing?

2. In the jazz blues in F, which guide tones does B $\flat$ 7 in bar 2 add against F7, and how far do they move?

3. Which scale fits the m7 chord of a ii – V – I?

4. In Dm7 – D $\flat$ 7 – Cmaj7, which bass line does the tritone substitution create?

5. How is a drop-2 voicing made?

6. What is the Charleston comping rhythm?

7. Which chord appears in bar 6 of a jazz blues in F, and what does it connect?

MY NOTES

CHAPTER 49

Gypsy Jazz



Gypsy jazz (jazz manouche) began in 1930s Paris with Django Reinhardt and Stéphane Grappelli. It combines swing tunes, minor-key harmony built on m6 chords, a percussive rhythm called la pompe, and fast arpeggio-based lines played with rest strokes.

Django Reinhardt, a Manouche (French Romani) guitarist, formed the Quintette du Hot Club de France with the violinist Stéphane Grappelli in 1934. The band had no drums: two rhythm guitars and a double bass kept time, and Reinhardt played lead. That line-up still defines the style. The rhythm guitars play la pompe, a short, clipped strum that works like a snare drum; the lead guitar plays loud, fast lines built from arpeggios and chromatic approach notes.

Element	Gypsy jazz approach
Feel	Swing at medium to very fast tempos; strong accent on 2 and 4
Harmony	m6 chords for minor, 6 and 6/9 chords for major, dominant 7ths with $b9$, diminished passing chords
Voicings	Three- and four-note shapes on the lower strings, root on the 6th or 5th string
Technique	La pompe; rest-stroke picking with a thick pick; arpeggios, chromatic runs, tremolo
Instrument	Selmer-Maccaferri-style steel-string guitar with a small oval or D-shaped soundhole

La pompe

Beats 1 and 3 get a short downstroke. Beats 2 and 4 get a very quick upstroke followed at once by an accented downstroke, and both are cut short by releasing the fretting-hand pressure the moment the chord sounds. The result is "boom, ch-chk, boom, ch-chk". The left hand does as much rhythmic work as the right: it squeezes the chord for the strum and relaxes it immediately afterwards.

▶ TRACK 445 La pompe



The upstroke falls on the last sixteenth before beats 2 and 4 and is barely audible. All four downstrokes are short; the ones on 2 and 4 are the loudest.

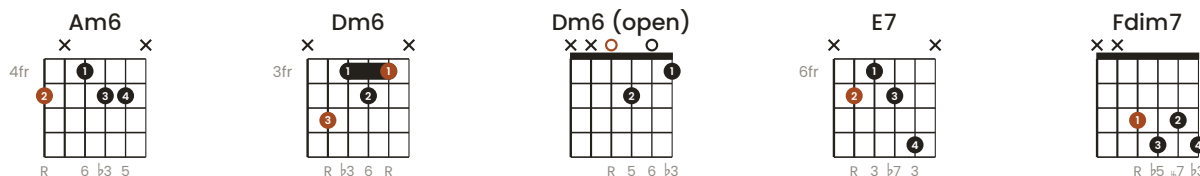
TIP • Keep it short and relaxed

Start at 100 bpm. If your forearm tightens, the strokes are too big: move from the wrist, and let the fretting hand do the muting. At 180 bpm and above, the upstroke often disappears and players simply accent 2 and 4.

Characteristic chords

Gypsy jazz replaces plain minor triads with minor 6th chords, and major triads with 6th and 6/9 chords. The m6 chord adds the major 6th to the minor triad: A C E F#. Dominant chords often carry a $b9$, and the diminished 7th chord a half step above the dominant's root stands in for it (Fdim7 over E7). The shapes are small and sit on the lower and middle strings, where la pompe sounds fullest.

Minor and dominant shapes



Am6 with the root on the 6th string, Dm6 with the root on the 5th, and E7 nearby, so the minor-swing changes need only small movements. Fdim7 is E7 $\flat$ 9 without its root.

Major colours



The 6th chord is the standard major tonic. The 6/9 is used for endings and in slower tunes.

The minor-swing progression

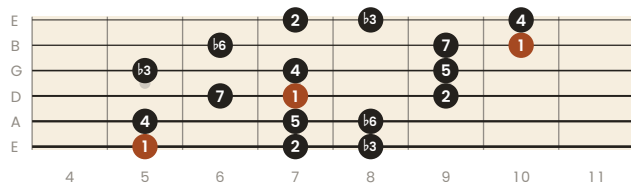
Many gypsy-jazz tunes in minor keys use a 16-bar form on i, iv and V, with the m6 chord on i and iv. The best known example is Reinhardt and Grappelli's "Minor Swing" (1937); the practice loop below uses a common version of that kind of form, not a transcription of the tune.

TRACK 446 Minor swing in A · LOOP · jazz swing · 160 bpm

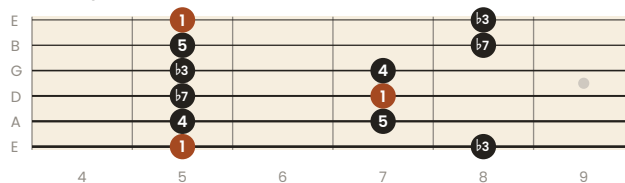
Am6 <i>i6</i>	Am6 <i>i6</i>	Dm6 <i>iv6</i>	Dm6 <i>iv6</i>
E7 <i>V7</i>	E7 <i>V7</i>	Am6 <i>i6</i>	Am6 <i>i6</i>
Dm6 <i>iv6</i>	Dm6 <i>iv6</i>	Am6 <i>i6</i>	Am6 <i>i6</i>
E7 <i>V7</i>	E7 <i>V7</i>	Am6 <i>i6</i>	Am6 <i>i6</i>

IMPROVISE WITH

A harmonic minor



A minor pentatonic



First play *la pompe* along with the loop. Then solo with arpeggios: Am6 and Dm6 on their chords, F diminished over E7. The harmonic minor contains all of them.

Rest-stroke picking

Gypsy-jazz picking is built for volume on an unamplified guitar. Use a thick pick (2–5 mm) and these rules:

- Moving to a higher-pitched string, play a downstroke that passes through the string and comes to rest on the next one (a rest stroke, as in classical *apoyando*).
- Each new string starts with a downstroke, even when that means two downstrokes in a row.
- Moving to a lower-pitched string, play an upstroke.
- On one string, alternate down and up as usual.

A line that climbs across the strings uses consecutive downstrokes, which gives the style its strong, even tone. Keep the wrist loose: the pick follows through rather than pushing.

Pompe and Arpeggios in A Minor

Original study · swing eighths · 4/4 · 132 bpm

A 16-bar chorus of the minor-swing progression. Bars 1–8 are rhythm guitar with la pompe on Am6, Dm6 and E7; bars 9–16 are a lead line built from the arpeggios of this chapter.

TRACK 449

Am6 Dm6

T	5	5	5	5	5	5	5	5	3	3	3	3	3	3	3	3
A	5	5	5	5	5	5	5	5	4	4	4	4	4	4	4	4
B	4	4	4	4	4	4	4	4	3	3	3	3	3	3	3	3
	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5

E7 Am6

T	9	9	9	9	9	9	9	9	5	5	5	5	5	5	5	5
A	7	7	7	7	7	7	7	7	5	5	5	5	5	5	5	5
B	6	6	6	6	6	6	6	6	4	4	4	4	4	4	4	4
	7	7	7	7	7	7	7	7	5	5	5	5	5	5	5	5

Dm6 Am6

T				5	6		7	5				5	8	5		7	5
A				4	7		6		7	4		4	5	7		5	
B	5	3	7													7	4
																	5

E7 Am6

T				4	7		7	4				5				5	
A				3	6		4	7		6		7	4		6	3	
B	5											7					7
																	5

PLAYING IT · Performance notes

In bars 1–8 play each chord as la pompe; the notation shows the downstrokes only.

Bar 11 approaches C from B a half step below. Bars 13–14 are F diminished over E7: up in bar 13, down in bar 14.

Follow the rest-stroke rules throughout the lead line, and swing the eighths.

Swing and gypsy-jazz jams. La pompe is the engine of every jam. Keep it steady and quiet enough for the soloist to be heard; a good pompe player is welcome in any session.

How to train it

Drill	How	Time	Goal
Pompe endurance	La pompe on Am6 for two minutes at 120 bpm, then 160 bpm	3 min	Even, relaxed, accents on 2 and 4
Minor-swing changes	The 16-bar progression with la pompe, eyes off the chart	4 min	Clean changes without losing the beat
Rest strokes	Am6 arpeggio up and down with the rest-stroke rules, 80–120 bpm	3 min	Even volume on every string
Arpeggio connection	Am6 – Dm6 – Fdim arpeggios in 5th position, one bar each	4 min	Move between arpeggios on the nearest note
Approach notes	Play each chord tone of Am6 preceded by the note a half step below	3 min	Approach note always on the off-beat

Write your answers, then check them in Appendix J.

6. Why did the Quintette du Hot Club de France need la pompe?

MY NOTES

CHAPTER 50

Bossa Nova and Latin Rhythms



Bossa nova sets jazz harmony over a quiet, syncopated samba pattern played by one guitarist's thumb and fingers. Afro-Cuban styles are organised around the clave, a two-bar rhythm that every part of the band fits around.

Bossa nova appeared in Rio de Janeiro in the late 1950s. João Gilberto reduced the samba percussion section to one guitar, Antônio Carlos Jobim wrote the songs, and the poet Vinicius de Moraes wrote many of the lyrics. The result is soft, intimate and harmonically rich. The same chapter covers the Afro-Cuban side of Latin music: son, cha-cha-chá and bolero, which share the guitar but are built on a different rhythmic system, the clave.

Three things define bossa nova on the guitar:

- **Feel.** The music is felt in two. The thumb plays the low surdo drum on beats 1 and 3 (in 4/4); the fingers play a syncopated figure taken from the samba percussion.
- **Harmony.** Seventh, ninth, thirteenth and altered chords, borrowed from jazz (Chapter 48), move by smooth voice leading. Root position triads are rare.
- **Touch.** Quiet, even and relaxed. Nothing is accented hard, and the voice or melody always sits on top.

The thumb and fingers pattern

Practise the thumb alone first. It plays half notes, steady as a heartbeat, usually the root on beat 1 and the 5th on beat 3. Then add the fingers. In the simplest pattern i, m and a pluck three chord notes together on 1, the & of 2 and 4, and the thumb joins them on 1.

▶ TRACK 450 One-bar pattern: fingers on 1, the and of 2, and 4

Diagram illustrating the one-bar pattern for Dm7, G7, and Cmaj7 chords. The diagram shows the fretboard with fingerings (i, m, a) and thumb (p) positions. The pattern is repeated for four measures, each corresponding to a different chord: Dm7, G7, Cmaj7, and Cmaj7. The thumb plays half notes on beats 1 and 3, while the fingers play a syncopated figure on beats 1, the & of 2, and 4.

Rust = thumb, teal = i, gold = m, ink = a. The thumb never stops; the fingers fall between its notes.

The full bossa nova pattern lasts two bars. The fingers follow the **bossa nova clave**: 1, the & of 2 and 4 in the first bar, then 2 and the & of 3 in the second. Against it the thumb keeps playing half notes. Count the eighth notes aloud and practise each hand alone before putting them together.

▶ TRACK 451 Two-bar bossa pattern on a ii – V – I in C

Diagram illustrating the two-bar bossa pattern on a ii – V – I in C. The diagram shows the fretboard with fingerings (i, m, a) and thumb (p) positions. The pattern is repeated for four measures, each corresponding to a different chord: Dm7, G7, Cmaj7, and Cmaj7. The thumb plays half notes on beats 1 and 3, while the fingers play a syncopated figure on beats 1, the & of 2, and 4.

The diagram includes a tablature section below the fretboard, showing the fingerings for each note in the pattern. The tablature is organized into four measures, each corresponding to a different chord: Dm7, G7, Cmaj7, and Cmaj7. The fingerings are as follows:

Measure	Chord	Thumb (T)	Index (i)	Middle (m)	Ring (a)
1	Dm7	6	6	6	6
2	G7	3	3	3	3
3	Cmaj7	5	5	5	5
4	Cmaj7	5	5	5	5

The thumb alternates root and 5th where the shape allows it (Dm7, Cmaj7) and repeats the root where it does not (G7).

TIP · Keep it quiet

Play the fingers with the flesh of the fingertip, close to the soundhole, and lift them just before the next pinch so the chords are slightly detached. The thumb is a little louder than the fingers, never the other way round.

Anticipation

Experienced players often change to the next chord one eighth note early, on the & of 4. The finger notes then anticipate the harmony while the thumb keeps the bass on the beat. Add this only when the basic pattern runs without thought.

Partido alto

Partido alto is a samba pattern, popular in instrumental Brazilian music from the 1970s onwards. It is written in 2/4 and counted in sixteenths. Over two bars there are seven finger strokes: off the beat at first, then four even eighth notes starting on beat 2, then off the beat again across the bar line. The thumb plays on every beat, with a slightly stronger stroke on beat 2.

Partido alto in two bars of 2/4

Thumb (surdo) Fingers

1 e & a 2 e & a 3 e & a 4 e & a

TRACK 452 Partido alto on Dm9 and G13

Dm9 **G13**

T	5	5	5	5	5	5	5	5	5	5	5	5	5	5
A	5	5	5	5	5	5	5	5	5	5	5	5	5	5
B	5	3	3	3	3	3	3	3	3	3	3	3	3	3

The last finger stroke of each two-bar cycle falls on the final sixteenth and leads into the next bar.

Bossa nova chord colours

Bossa voicings usually keep four notes on adjacent strings: a bass note on the 6th or 5th string and three upper notes the fingers can pluck together. The 5th is often left out to make room for a 9th, 11th or 13th.

Major, dominant and diminished colours

Cmaj7 **C6/9** **A7^b9** **Dm9** **G13** **C[#]dim7**

Minor-key colours

Am6 **Am7** **Bm7^b5** **E7^b9** **D9** **Fmaj7**

- **maj7** and **6/9** are the tonic chords. A 6/9 chord sounds more settled than maj7 and is the usual final chord.
- **m6** and **m9** replace plain minor chords. The m6 is typical of minor-key bossas.

- **7b9** is the dominant that leads strongly to a minor chord (A7b9 → Dm7, E7b9 → Am).
- **dim7** works as a passing chord between two chords a whole step apart: Cmaj7 – C#dim7 – Dm7.

Bossa progressions

Play each chart with the two-bar pattern, one chord per bar, using the voicings above.

ii – V – I with a secondary dominant

Cmaj7 <i>Imaj7</i>	A7b9 <i>VI7b9</i>	Dm9 <i>ii9</i>	G13 <i>V13</i>
Cmaj7 <i>Imaj7</i>	C#dim7 <i>#i°7</i>	Dm7 <i>ii7</i>	G7 <i>V7</i>

I – II7 – ii – V (a favourite opening move)

Cmaj7 <i>Imaj7</i>	Cmaj7 <i>Imaj7</i>	D9 <i>II9</i>	D9 <i>II9</i>
Dm7 <i>ii7</i>	G13 <i>V13</i>	C6/9 <i>I6/9</i>	C6/9 <i>I6/9</i>

Minor bossa

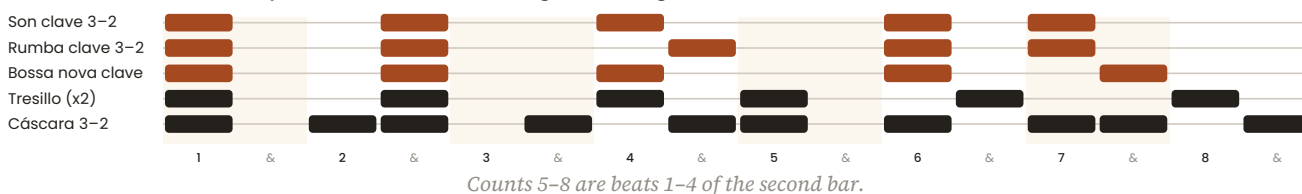
Am6 <i>i6</i>	Am6 <i>i6</i>	Bm7b5 <i>iiø7</i>	E7b9 <i>V7b9</i>
Am7 <i>i7</i>	D9 <i>IV9</i>	Dm9 <i>iv9</i>	E7b9 <i>V7b9</i>

The II7 chord (D9 in C) followed by ii (Dm7) is a step towards the dominant that appears in many bossa standards: only one note changes, F# falling to F. Listen for small moves like this; bossa harmony is mostly one or two voices moving by half step.

Clave and the Afro-Cuban rhythms

Son, salsa, rumba and much of Latin jazz are organised around the **clave**, a two-bar rhythm. One bar has three strokes (the *3 side*), the other two (the *2 side*). A song is said to be in 3–2 or 2–3 clave depending on which bar comes first, and the direction does not change within a section. Every part, including the guitar, places its accents so they agree with the clave.

▶ TRACK 453 Clave family over two bars of 4/4 (eighth-note grid)



- **Tresillo** is the 3 + 3 + 2 figure that forms the 3 side of the son clave. It underlies music across Latin America, from the habanera to the baião.
- **Rumba clave** moves the third stroke of the 3 side one eighth note later, which makes it more urgent.
- **Cáscara** ("shell") is the pattern the timbalero plays on the side of the timbales. It fills in the clave; practise it with your strumming hand on muted strings.

TRY THIS · Clap and step

Step quarter notes with your feet and clap son clave 3–2 at 80 bpm. When it is steady, say "three side" and "two side" at the start of each bar. Then switch to 2–3 by starting on the second bar.

Latin strumming and the guajeo

In a son or salsa group the **tres** (a Cuban guitar with three doubled courses) or the piano plays the **guajeo**, a repeating syncopated melody that outlines the chords and locks to the clave. On guitar, start with a strum that moves in steady eighth notes and accents the clave strokes. Then try a single-note guajeo on the I – IV – V – IV cycle typical of the son montuno.

TRACK 454 Eighth-note strum with clave accents (3–2)



Bar 1 on C (3 side), bar 2 on G7 (2 side). Light strokes keep the hand moving; only the clave strokes are accented.

TRACK 455 Guajeo-style ostinato on I – IV – V – IV (original)

The new chord arrives on the & of 2, ahead of beat 3. Played with 3–2 clave, the anticipation lines up with the 3 side's middle stroke.

Cha-cha-chá and bolero

Cha-cha-chá was developed in Cuba in the early 1950s by the violinist Enrique Jorrín. It is in 4/4 at a moderate tempo; the name imitates the dancers' triple step on 4 & 1. The guitar marks all four beats and doubles the & of 4.

Bolero, the Cuban and Mexican romantic song form, is slow and smooth. Guitar trios play a bass note on 1 and 3 with light brush strokes between them.

TRACK 456 Cha-cha-chá and bolero



Bars 1–2 cha-cha-chá on Am – E7 (about 120 bpm); bars 3–4 bolero on Am – E7 (about 70 bpm).

NYLON · STEEL

Nylon: bossa nova belongs to the nylon-string guitar. Play with the flesh of the fingers, or flesh and a little nail, for a round tone. The thumb strikes with the side, not the nail.

Steel: use bare fingers, not a pick, and play more quietly than feels natural. Short decay is part of the sound, so rest the side of the palm lightly near the bridge if the strings ring too long. Latin strumming patterns work well with a pick on either guitar.

Estudo em lá menor

Original study · bossa nova · 4/4 · 116 bpm

A minor-key bossa using the two-bar finger pattern and the voicings above. Bars 5–8 use the D9 → Dm9 half-step move from the bossa progressions.

TRACK 457

Am6 Bm7b5 E7b9

5 Am7 D9 Dm9 E7b9

9 Am6 Fmaj7 E7b9 Am6

rit.

T 2 2 2 2 1 1 1 1
A 1 1 1 1 3 3 3 3
B 2 2 2 2 2 2 2 1

T 0 0 0 5 5 5 5 1 1
A 0 0 0 5 5 5 5 3 3
B 0 0 5 4 5 4 3 5 3

T 2 2 2 0 0 1 1 1 2
A 1 1 1 1 1 3 3 3 1
B 2 2 2 3 2 1 1 1 2

T 0 0 0 3 2 3 2 0 0
A 0 0 0 3 2 1 1 0 0
B 0 0 0 3 2 1 1 0 0

PLAYING IT · Performance notes

Keep the three upper notes exactly together, as one soft pinch with i, m and a.

Bm7b5 (x2x231) and E7b9 (0xx131) share the index-finger partial barre at fret 1; move only what has to move.

Let the final Am6 ring. Bossa endings are quiet, often with a short rubato.

Practice loops

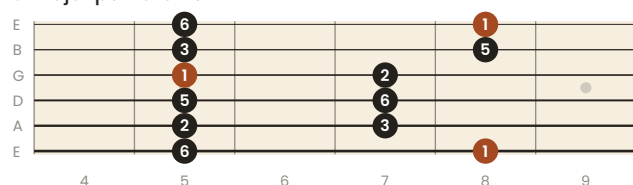
Play the bossa loop first with the two-bar thumb-and-fingers pattern, changing chords one eighth early once the pattern is steady. On the son montuno loop, start with the clave-accented strum, then switch to the guajeo, and keep every phrase in step with the 3–2 clave.

▶ **TRACK 458** Bossa nova loop in C • LOOP • bossa nova • 120 bpm

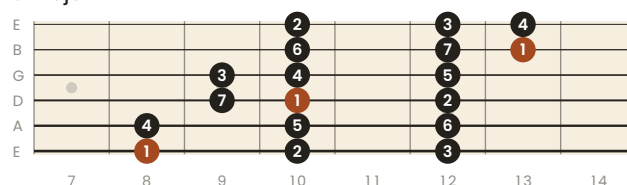
Cmaj7 <i>I maj7</i>	A7b9 <i>VI7b9</i>	Dm9 <i>ii9</i>	G13 <i>V13</i>
Cmaj7 <i>I maj7</i>	C#dim7 <i>#i°7</i>	Dm7 <i>ii7</i>	G7 <i>V7</i>

IMPROVISE WITH

C major pentatonic



C major



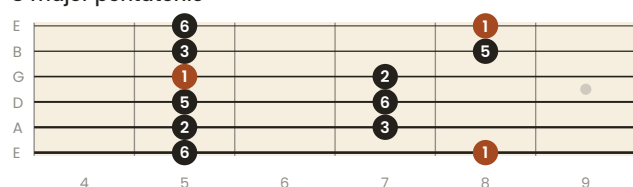
Improvise softly in long phrases. Over A7b9 aim for C# (its 3rd), which leads to D on the Dm9.

▶ **TRACK 459** Son montuno loop in C • LOOP • son montuno • 100 bpm

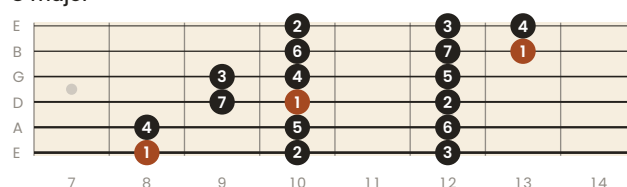
C <i>I</i>	F <i>IV</i>	G7 <i>V7</i>	F <i>IV</i>
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IMPROVISE WITH

C major pentatonic



C major



Play the guajeo from this chapter first, then vary it. Keep your phrases short and repetitive, like a tres player.

LISTEN FOR • Recordings to study

João Gilberto, *Chega de Saudade* (1959) and *João Gilberto* (1973, the "white album"), for the batida itself. Getz/Gilberto (1964, Stan Getz and João Gilberto with Jobim on piano). Antônio Carlos Jobim, *Wave* (1967). Luiz Bonfá and Baden Powell for solo guitar. For Cuban styles: Arsenio Rodríguez (tres and son montuno), Orquesta Aragón (cha-cha-chá), Trío Los Panchos (bolero), and the *Buena Vista Social Club* album (1997) with Compay Segundo.

Chapter 50 review • use it, train it, test it

Using it in real music

Accompanying a singer. The two-bar pattern under a voice is the core bossa skill. Keep the thumb steady even when the singer phrases freely, and let the fingers drop out on long held chords if the texture gets busy.

Playing with a band. In a Latin group, ask whether the tune is in 3-2 or 2-3 clave before you start. Your accents, anticipations and fills should agree with the clave, and a guitar part that crosses it sounds wrong to every player on the stand.

Reharmonising. Swap plain triads in any song for maj7, m9, 13 and 6/9 voicings, and put a 7b9 before each minor chord. With a bossa pattern underneath, almost any slow ballad can become a bossa.

Write your answers, then check them in Appendix J.

CHAPTER 51

Choro and Brazilian Guitar



TRACKS

Choro is Rio de Janeiro's instrumental tradition of fast, syncopated melodies and running bass lines. Around it sit samba, the rhythm behind bossa nova, and the baião of the Brazilian north-east.

Choro (literally "cry" or "lament") took shape in Rio de Janeiro in the 1870s, when local musicians played European dances such as the polka, waltz and schottische with Afro-Brazilian syncopation. The flautist Joaquim Callado led one of the first groups. Pixinguinha, Ernesto Nazareth, Jacob do Bandolim and Waldir Azevedo wrote much of the core repertoire in the following decades.

A choro group is called a **regional**. Its usual line-up:

Instrument	Role
Bandolim (mandolin), flute or clarinet	Melody, often virtuosic
Cavaquinho (small four-string guitar)	Chords in a tight, percussive sixteenth-note rhythm
Violão (six-string guitar)	Chords and bass in the middle register
Violão de sete cordas (seven-string guitar)	Bass lines and counter-melodies (the <i>baixaria</i>), played with a thumbpick
Pandeiro (frame drum with jingles)	The groove: continuous sixteenths with accents

Three things define the style on the guitar:

- **Feel.** 2/4 time, sixteenth-note subdivision and a light, forward-leaning swing. The basic syncopation is sixteenth–eighth–sixteenth inside one beat.
- **Harmony.** Functional and chromatic: secondary dominants, diminished passing chords and inversions that make the bass move by step (Chapter 32).
- **Counterpoint.** The guitar is not just accompaniment. Its bass line answers the melody in the gaps.

Choro form

Most choros are rondos in three sections, each usually 16 bars long and repeated: **A A B B A C C A**. A common key plan puts A in a major key, B in its relative minor (or the dominant) and C in the subdominant, so a choro in G major might move to E minor for B and C major for C. The return of A between the other sections is what makes the piece feel like a choro rather than a medley.

A section shape in G major (typical, original)

G <i>I</i>	G <i>I</i>	D7 <i>V7</i>	D7 <i>V7</i>
D7 <i>V7</i>	D7 <i>V7</i>	G <i>I</i>	G7 <i>I7</i>
C <i>IV</i>	Cm6 <i>iv6</i>	G/D <i>I64</i>	E7 <i>V17</i>
A7 <i>II7</i>	D7 <i>V7</i>	G <i>I</i>	D7 <i>V7</i>

Notice C to Cm6 (IV to minor iv, a borrowed chord), G/D (a second-inversion tonic over the 5th in the bass), and E7 – A7 – D7, a chain of dominants that falls by fifths back to G. All three are everyday choro moves.

The accompaniment pattern

The six-string guitarist plays the bass with the thumb on the beats and short chords with the fingers in the syncopated sixteenth–eighth–sixteenth figure (the Brazilian *síncope*). Keep the chords short and light, closer to the cavaquinho than to a strum.

Thumb and fingers in two bars of 2/4



▶ TRACK 460 Choro accompaniment on G – E7 – Am – D7

G E7 Am D7

T 0 0 0 0 | 0 0 0 0 | 1 1 1 1 | 2 2 2 2

A 0 0 0 0 | 1 1 1 1 | 2 2 2 2 | 1 1 1 1

B 0 0 0 0 | 0 0 0 0 | 0 0 0 0 | 0 0 0 0

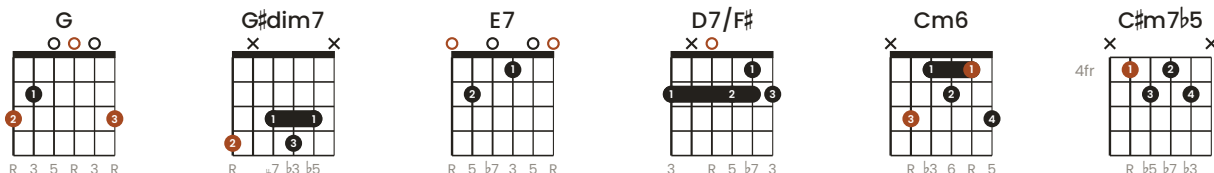
3 2 0 2 | 0 2 0 0 | 0 0 0 0 | 0 0 0 0

Bass on the beats (root, then 5th or 3rd); fingers on the e and a of each beat.

Choro chord vocabulary

Choro harmony is built from triads, dominant sevenths, diminished sevenths and minor sixths, often in inversion so that the bass line moves smoothly.

Common choro chords in G major



- **G#dim7** links G and Am7 by a chromatic bass step: G – G#dim7 – Am7 – D7. Any dim7 can be moved up or down three frets and keeps the same four notes.
- **D7/F#** puts the 3rd in the bass so the line can run F# – G into the tonic.
- **Cm6** is the borrowed minor iv. Its bass C falls naturally to B (G/B) or rises to C# (C#m7b5).
- **C#m7b5** replaces C in a bass line that climbs C – C# – D.

Baixaria: the running bass

The baixaria is the counter-melody of the seven-string guitar. Between the melody's phrases it fills the gaps with sixteenth-note runs that lead from one chord's bass note to the next by scale steps and chromatic approach notes. Three rules cover most of it:

1. Arrive on the next chord's bass note on the downbeat.
2. Approach it by step: from a half step below, a half step above, or by a scale run.
3. Play when the melody holds a long note or rests; stay on simple bass notes while the melody is busy.

▶ **TRACK 461** Baixaria on G – E7 – Am – D7 – G (original)

G E7 Am D7 G

Bar 1 falls chromatically G – F# – F to E; bar 2 descends through the E7 arpeggio to G#, the leading note of Am; bar 3 climbs chromatically C – C# to D; bar 4 falls by scale to F# and G.

NYLON · STEEL · Seven-string lines on six strings

Nylon: the six-string violão is a nylon-string instrument. Play baixarias with rest strokes of the thumb for weight. Seven-string players traditionally use steel strings and a thumbpick (*dedeira*) for more attack; a thumbpick also works on a nylon six-string.

Steel: a light thumbpick makes bass runs clearer. Keep the chords short so the bass can be heard.

Adapting: seven-string lines that go below low E (to D or C) can be played an octave higher, or tune the 6th string down to D for keys such as D major and D minor.

Samba batida

Samba is the music of Rio's carnival and its neighbourhoods, played by percussion ensembles of surdos, tamborins, caixas and pandeiros. On guitar the thumb imitates the surdo: a lighter note on beat 1 and a stronger one on beat 2. The fingers play a syncopated two-bar figure from the tamborim and cavaquinho. Bossa nova (Chapter 50) is a quieter, sparser version of the same idea.

▶ **TRACK 462** Samba pattern on Am7 and E7

Am7 E7

The accent on beat 2 is the surdo. In samba the third and fourth sixteenths of each beat are often played slightly early, which gives the music its lilt; learn it straight first.

Baião

Baião comes from the north-east of Brazil and was made famous nationally in the 1940s by the accordionist and singer Luiz Gonzaga with the lyricist Humberto Teixeira. It is in 2/4 and built on the zabumba drum's 3 + 3 + 2 figure, the same grouping as the Cuban tresillo. The trio of accordion, zabumba and triangle is the traditional line-up; the same rhythms feed forró and much of modern Brazilian jazz.

 TRACK 463 Baião bass and chords

[illegible]

The bass plays 3 + 3 + 2 sixteenths; the chords answer on the off-beats. A7 to G (I7 to $\flat$ VII) is a typical Mixolydian move.

Baião melodies favour two modes over a dominant seventh chord: **Mixolydian** (major scale with b7) and **Lydian dominant** (b7 and #4), which north-eastern musicians call the *escala nordestina*.

A Lydian dominant (escala nordestina) in 5th position

The #4 (D#) and b7 (G) in teal are the two notes that give the sound its colour.

Brazilian fingerstyle textures

Solo Brazilian guitar combines the parts of the regional in one pair of hands: bass line with the thumb, syncopated chords or melody with the fingers. Its main traditions are Garoto's harmonically adventurous pieces from the 1940s and 1950s, Heitor Villa-Lobos's concert works (the guitar *Choros No. 1* of 1920, the *Études* and *Preludes*), and Baden Powell's thumb-driven afro-sambas of the 1960s. Common textures:

- **Thumb ostinato with melody.** The thumb plays the samba surdo pattern on open bass strings; the melody sits on strings 1-3, often with open strings ringing against fretted notes.
- **Moving inner voice.** A chord is held while one inner note moves chromatically (for example C – C# – D – C# on the 3rd string), common in Garoto and in bossa nova.
- **Parallel shapes.** One fingering slides along the neck over an open bass string, giving changing harmonies against a pedal.

Chorinho em Sol

Original study · choro · 2/4 · 66 bpm

An eight-bar choro phrase for solo guitar, with melody over a stepping bass. It uses the síncope rhythm (bar 2), the chromatic F – F $\sharp$ approach (bars 3–4) and the borrowed Cm chord (bar 6).

TRACK 464

The musical score for 'Chorinho em Sol' is presented in two systems. Each system contains four bars of music. The first system features chords G, E7, Am, and D7. The second system features chords G, C, Cm, D7, and G. The melody is written in treble clef with a key signature of one sharp (F#). The bass line is written in a simplified notation with numbers 0-3 representing frets. The first system starts with a mezzo-forte (mf) dynamic marking.

PLAYING IT · Performance notes

Play the bass with rest strokes of the thumb and the melody with free strokes, so the melody stays light. In bar 2 the sixteenth–eighth–sixteenth figure is the síncope; do not let the eighth note become an accent. Practise at 50 bpm until the sixteenths are even, then bring it up. Choro is fast, but clarity comes first.

LISTEN FOR · Recordings to study

Pixinguinha's recordings with Benedito Lacerda (1940s). Jacob do Bandolim and the regional Época de Ouro, with Dino 7 Cordas on seven-string guitar: the model for baixaria. Waldir Azevedo (cavaquinho). Garoto (solo guitar). Raphael Rabello and Yamandu Costa (seven-string). For samba: Cartola, Paulinho da Viola. For baião: Luiz Gonzaga, Dominginhos. For solo Brazilian guitar: Baden Powell, *Os Afro-Sambas* (1966, with Vinicius de Moraes), and the guitar works of Heitor Villa-Lobos.

Practice loops

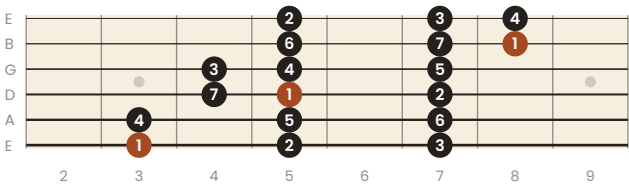
The first loop is the A-section chart from this chapter on a gentle samba-bossa groove: accompany it with the síncope pattern first, then replace the chords with baixaria runs between the roots. The second is a single A7 chord for hearing the two baião modes against each other.

▶ TRACK 465 Choro progression in G (samba-bossa feel) · LOOP · bossa nova · 100 bpm

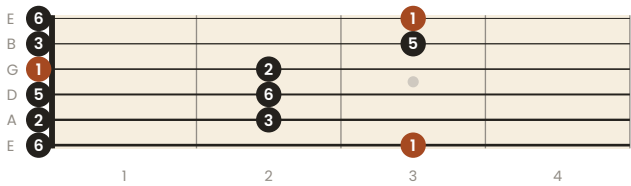
G I	G I	D7 V7	D7 V7
D7 V7	D7 V7	G I	G7 I7
C IV	Cm6 iv6	G/D I64	E7 VI7
A7 II7	D7 V7	G I	D7 V7

IMPROVISE WITH

G major



G major pentatonic



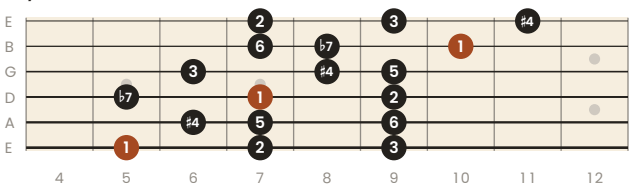
Practise baixarias over the loop: walk the bass from one chord's root to the next, landing on each downbeat.

▶ TRACK 466 Baião drone on A · LOOP · modal drone · 96 bpm

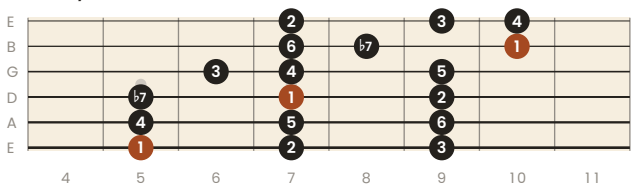
A7 I7

IMPROVISE WITH

A lydian dominant



A mixolydian



Alternate the two modes: D natural (Mixolydian) and D# (Lydian dominant) over the same A7.

Chapter 51 review · use it, train it, test it

Using it in real music

Rodas de choro. Choro is played in informal circles (*rodas*) where the guitarist reads chord charts and is expected to know the common progressions. Learn the A-section shape and the dim7 passing chords in G, D, C and A minor first.

Bass lines in any style. The baixaria rules (arrive on the downbeat, approach by step, fill the gaps) work in folk, country and jazz accompaniment as well. Try them over a ballad in 4/4.

Samba and baião in a band. Lock the thumb to the surdo or zabumba. The bass player and the guitarist must agree who plays the low notes; if there is a bass player, play the fingers' pattern only, higher up the neck.

How to train it

Drill	How	Time	Goal
Síncope	Sixteenth–eighth–sixteenth on one chord, with the thumb on beats, at 60 bpm	3 min	Chords short, thumb steady
Chart run	Accompany the A-section chart with the choro pattern, then with bass notes only	5 min	No stops at chord changes
Baixaria fills	Over G – E7 – Am – D7, improvise one run per two bars, landing on the next root	5 min	Every run lands on the downbeat

Write your answers, then check them in Appendix J.

CHAPTER 52

Reggae



TRACKS

52

Jamaican popular music developed from ska through rocksteady to reggae in less than a decade. In all three the rhythm guitar plays short, muted chords off the main beats and leaves the low end to the bass and drums.

Reggae is ensemble music. The bass carries the melody of the groove, the drums mark the beat in unusual places, and the guitar plays a chord so short it is almost a percussion sound. A good reggae guitar part is simple and exact: the same two chops per bar, in the same place, for the whole song.

Ska, rocksteady and reggae

Style and period	Tempo and feel	Guitar role
Ska, early 1960s	Fast (about 130–180 bpm), bright, driving	Up-chops on every off-beat ("and"), often doubled by piano and horns
Rocksteady, about 1966–1968	Slower and smoother; the bass becomes the lead voice	Chops on 2 and 4, more space, picked melodic lines doubling the bass
Reggae, from 1968	Slow to medium (about 60–90 bpm, felt in half time), heavy, laid back	The <i>skank</i> : short chops on 2 and 4; sometimes a double chop

The difference between ska up-chops and the reggae skank is partly a matter of counting. Count a reggae groove in fast quarter notes and the chops fall on every second eighth note, which sounds like ska slowed down. Count it slowly and they fall on 2 and 4. Musicians usually count reggae slowly, with the drummer's main accent on 3.

The band around you

Instrument	Role
Drums	One drop, rockers or steppers (see below). Hi-hat often plays swung or straight eighths
Bass	Deep, melodic, repetitive. The <i>riddim</i> (the instrumental part that identifies a song) lives in the bass line. Often rests on beat 1
Rhythm guitar	The skank on 2 and 4, short and bright
Second guitar	Picked single notes doubling the bass line an octave higher, muted with the palm
Organ	The <i>bubble</i> : a rocking pattern of short chords that fills the eighth notes between the skanks
Piano	Often doubles the guitar skank

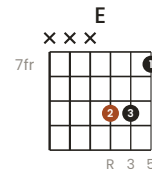
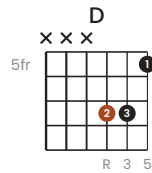
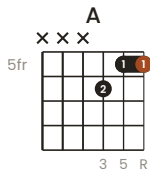
The chop

A chop is a chord cut off as soon as it sounds: the chord rings for an instant, then the fretting hand mutes it, like the muted chuck of Chapter 36 with a pitch in front.

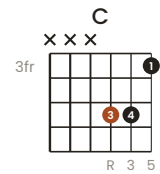
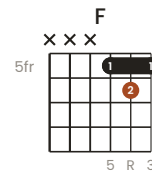
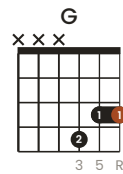
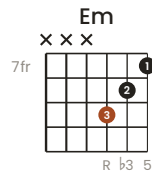
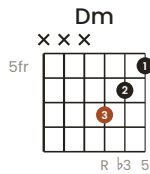
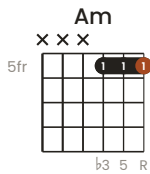
1. Fret a small three- or four-note chord on the top strings (the top-string triads of Chapter 10).
2. Strike the strings with a short, light downstroke.
3. Release the pressure of the fretting fingers immediately, keeping them on the strings. The chord should sound like "tchk", not ring.
4. Stay relaxed and play very slightly behind the beat. Reggae should feel patient.

Electric players usually use the bridge pickup with a bright, clean tone and sometimes a short spring reverb. On acoustic guitar, pick close to the bridge.

Top-string chords for I – IV – V in A



Minor-key chords around the 5th fret



All of these sit within a few frets of each other, so the fretting hand hardly moves. That is the point: small shapes are easy to mute and leave the lower register to the bass.

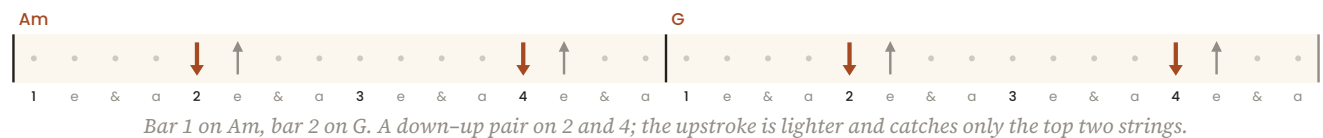
Rhythm patterns

Learn the basic skank first, then the double chop. The ska pattern, at a much faster tempo, shows where the skank came from. The off-beat skank in Chapter 22 (upstrokes on every "&") is the same part counted in fast quarter notes; many players use upstrokes when they feel it that way and downstrokes when they count slowly. Either works if the chop is short and exactly placed.

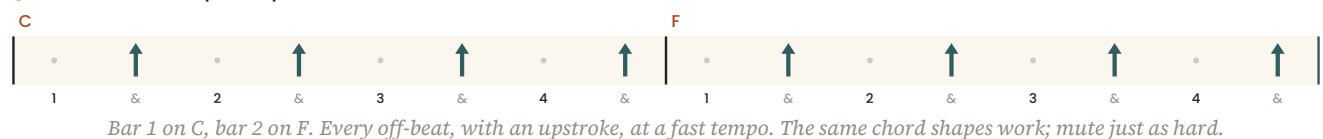
▶ TRACK 467 Reggae skank on A and D (count slowly)



▶ TRACK 468 Double chop (sixteenth notes)



▶ TRACK 469 Ska up-chops



TIP · Less is more

In a full band the guitar may play nothing but two chops per bar for the entire song. Resist the urge to fill. The space between the chops is part of the groove.

Drum feels: one drop, rockers and steppers

The guitar part barely changes from one drum feel to another, but the music feels completely different. Learn to recognise the three main feels so you can lock to them.

▶ TRACK 470 One drop (sixteenth-note grid, one bar)



▶ TRACK 471 Rockers and steppers



Rockers adds a kick on beat 1 to the one-drop kick on beat 3; steppers puts it on every beat. Sly Dunbar popularised both in the mid-1970s.

The bubble and bass-line doubling

The organ **bubble** fills the gaps between the skanks with short chords on the off-beat eighth notes and lower notes in between, so that together with the guitar every subdivision is covered without anyone playing all of them. A solo guitarist can suggest it with very light muted upstrokes between the chops, but in a band leave it to the keyboard.

The second guitar often plays the bass line an octave higher, picked with downstrokes and muted with the side of the palm at the bridge. This adds definition to the bass on small speakers and is a standard part of the reggae sound from rocksteady onwards. The study at the end of this chapter uses both parts.

Reggae progressions

Reggae harmony is simple: two to four chords, often a two-bar vamp for a whole song. Minor keys and the Aeolian $\flat VI$ and $\flat VII$ chords are especially common.

Two-chord vamps



I – IV in A, then i – iv in A minor.

Minor-key grooves in A minor



Major-key song form



I – V – vi – IV and I – vi – IV – V, the same pop progressions, given a reggae feel by the skank and the drums.

NYLON · STEEL

Nylon and steel: acoustic reggae works well for singing along. Chop near the bridge and mute with the fretting hand as on electric. Playing solo, slap the strings with the palm on beat 3 to imply the one drop, and use the thumb for a short bass note before it.

Electric: keep the tone clean and bright (bridge pickup, treble up, little bass). Too much gain or bass makes the chop sound blurred.

LISTEN FOR · Recordings to study

Ska: The Skatalites, Prince Buster, and the guitarist Ernest Ranglin. Rocksteady: Alton Ellis, The Paragons, and the guitarist Lynn Taitt. Reggae: Toots and the Maytals, *Funky Kingston* (1973); Bob Marley and the Wailers, *Catch a Fire* (1973) and *Exodus* (1977), for Carlton Barrett's one drop and Aston Barrett's bass; Burning Spear, *Marcus Garvey* (1975); Black Uhuru with Sly Dunbar and Robbie Shakespeare for rockers and steppers. Guitarist Earl "Chinna" Smith for rhythm and lead playing.

Riddim Study in A Minor

Original study · reggae · 4/4 · 72 bpm

Two guitar parts over the same four-bar riddim (i – iv – i – $\flat$ VII). Bars 1–4 are the rhythm-guitar skank with a double chop in bar 4; bars 5–8 double the bass line an octave higher.

TRACK 472

Am Dm Am G

mf

5 5 5 5 5 5 5 5 3 3 3 3

5 5 6 6 5 5 3 3 3 3

5 5 7 7 5 5 4 4 4 4

5

Am Dm Am G

7 7 5 7 5 7 7 6 7 5 7 5 5 5 4 5 7

PLAYING IT · Performance notes

Bars 1–4 use the top-string shapes from the chord row. Release every chop on the next sixteenth.

Bars 5–8 are picked with downstrokes and muted with the side of the picking hand on the saddle. The notes should be short and round.

Loop the eight bars, or record the skank and play the bass doubling over it.

Practice loop

The loop is the minor-key groove from this chapter over a one-drop beat. Skank on 2 and 4 for several passes, then play a bass-line doubling an octave up as in the study, and only then a melody.

TRACK 473 One-drop loop in A minor · LOOP · reggae one drop · 72 bpm

Am <i>i</i>	G $\flat$ VII	Am <i>i</i>	G $\flat$ VII
Am <i>i</i>	F $\flat$ VI	G $\flat$ VII	Am <i>i</i>

IMPROVISE WITH

A minor pentatonic

4 5 6 7 8 9

A aeolian

4 5 6 7 8 9 10 11

Skank the first time round, then improvise a melody. Leave space: reggae lead lines are short, vocal phrases.

Chapter 52 review · use it, train it, test it

Using it in real music

Acoustic songs. Off-beat chops give a pop or folk song a relaxed reggae lilt. Keep the bass notes out of the chord and let your voice carry the melody.

Playing in a band. Lock the skank to the snare or rim click and listen to the bass. If the drummer plays a one drop, there is nothing on beat 1: do not fill it.

Using a looper. Record the bass-line doubling part first, then the skank, then a melody or solo on top (Chapter 42).

How to train it

Drill	How	Time	Goal
Mute control	Chops on one chord, releasing pressure immediately after each strike	3 min	Short "tchk", no ringing strings
Skank with click	Click on 2 and 4 only at 70 bpm; play the skank exactly with it	5 min	Chops disappear into the click
Feel switching	Same skank over one drop, rockers and steppers loops, 8 bars each	5 min	Steady chops whatever the kick does
Bass doubling	Learn the bass line of a reggae recording by ear and play it an octave up, palm-muted	5 min	Matches the bass rhythm exactly
Ska tempo	Up-chops at 140, then 160, then 180 bpm	3 min	Relaxed wrist at full tempo

Check your understanding

Write your answers, then check them in Appendix J.

1. On which beats does the reggae skank fall when counted slowly?

2. What is a one-drop drum pattern?

3. How do you make a chop short?

4. How does a ska guitar part differ from a reggae skank?

5. What is the difference between rockers and steppers drumming?

6. What does the second guitar often play in reggae?

7. Which strings are normally used for skank chords, and why?

MY NOTES

CHAPTER 53

Flamenco



Flamenco is the song, dance and guitar tradition of Andalusia in southern Spain. Each of its forms has a fixed rhythmic cycle, most of its harmony is built on the Phrygian mode, and its right-hand techniques turn the guitar into a percussion instrument as well as a melodic one.

Flamenco grew among the Gitano (Roma) and Andalusian communities of southern Spain and became a professional art in the *café cantantes* of the later 19th century. Song (*cante*) is the centre of the tradition; dance (*baile*) and guitar (*toque*) grew around it. The guitarist accompanies singers and dancers, and since the 20th century has also played solo concert flamenco. The guitar's three jobs are to keep the rhythmic cycle, to supply the harmony, and to play melodic passages called *falsetas* between the sung verses.

Compás: the rhythmic cycle

Each flamenco form (*palo*) has its own **compás**: a cycle of beats with accents in fixed places. Many *palos* use a 12-beat cycle. Learn each compás by clapping (*palmas*) before you play it: loud claps on the accents, soft claps on the other beats, counting aloud.

▶ **TRACK 474** Soleá and alegrías: 12 beats, accents on 3, 6, 8, 10, 12

1 2 3 4 5 6 7 8 9 10 11 12

▶ **TRACK 475** Bulerías: the same accents, counted from 12

12 1 2 3 4 5 6 7 8 9 10 11

Seguiriyas: 2 + 2 + 3 + 3 + 2, accents on 1, 3, 5, 8, 11

1 2 3 4 5 6 7 8 9 10 11 12

The 12-beat cycle of soleá is felt as two bars of 3/4 followed by three bars of 2/4 (counts 1–6 in threes, 7–12 in twos), which is why the accents shift from every third beat to every second beat. Phrases and *falsetas* usually close on count 10, with 11 and 12 as a short tail. Bulerías uses the same accents at a much faster tempo and is usually counted from 12. Seguiriyas has five uneven beats, short-short-long-long-short, and is slow and intense.

Palo	Compás	Usual key	Character
Soleá	12 beats, slow	Por arriba or por medio	Serious, the foundation of the repertoire
Alegrías	12 beats, moderate	Major, typically E or A	Bright; from Cádiz
Bulerías	12 beats, fast, counted from 12	Por arriba or por medio	Playful, virtuosic, often the closing piece
Seguiriyas	12 counts, 2 + 2 + 3 + 3 + 2, slow	Por medio	Tragic, intense
Tangos	Binary (4/4)	Por medio	Earthy, steady groove
Rumba flamenca	Binary (4/4)	Andalusian cadence or major	Festive, popular dance music

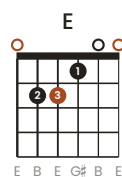
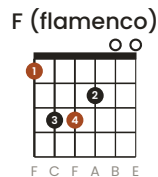
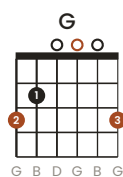
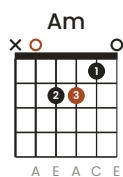
Flamenco harmony

Most flamenco harmony uses the **Phrygian mode** with a major chord on its tonic, sometimes called the *flamenco mode*. The **Andalusian cadence**, a descending four-chord progression, leads to that chord and is heard as resolving onto it (Chapter 31).

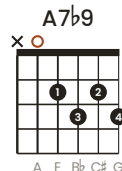
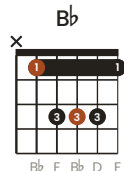
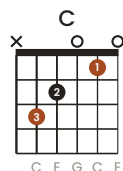
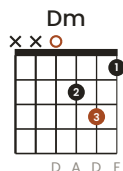
Guitarists name the key by the chord shapes they use, not by the sounding pitch, because the *cejilla* (capo) is moved to suit the singer:

- **Por arriba** ("from above"): E Phrygian, played with E-position chords. Cadence Am – G – F – E.
- **Por medio** ("from the middle"): A Phrygian, played with A-position chords. Cadence Dm – C – B $\flat$ – A.

Por arriba: Am – G – F – E



Por medio: Dm – C – B $\flat$ – A



The flamenco F keeps the open B and E strings ringing against F, A and C: technically Fmaj7#11. Dissonances like this, open strings rubbing against fretted notes, are central to the sound. The A7 $\flat$ 9 has B $\flat$, the b2 of A Phrygian, on top.

Andalusian cadence and the I – II move, por arriba

Am <i>iv</i>	G <i>bIII</i>	F <i>bII</i>	E <i>I</i>
F <i>bII</i>	E <i>I</i>	F <i>bII</i>	E <i>I</i>

Roman numerals relative to the Phrygian tonic E; heard from A minor, the same chords are the i – bVII – bVI – V of Chapter 14. Alternating bII and I (F – E) is the most common accompaniment move in soleá and bulerías.

Right-hand techniques

Technique	How it works	How to practise
Rasgueado	A strum in which the fingers flick out one after another. Curl the hand, then release little finger (<i>c</i> or <i>e</i>), ring (<i>a</i>), middle (<i>m</i>) and index (<i>i</i>) in quick succession across the strings	Start on muted strings, slowly and evenly. Then add an index upstroke: <i>c</i> – <i>a</i> – <i>m</i> – <i>i</i> – <i>i</i> $\uparrow$, and repeat it without gaps for a continuous roll
Golpe	A tap on the tap plate (<i>golpeador</i>) with the ring finger, usually together with a downstroke of <i>i</i> or <i>m</i> on an accent	Clap the soleá compás, then replace each accented clap with a strum plus golpe
Picado	Fast single-note scales with rest strokes, alternating <i>i</i> and <i>m</i>	Accent the first note of each group of four; raise the metronome 4 bpm at a time
Alzapúa	Thumb technique: a single-note rest stroke on a bass string, then a downstroke and an upstroke through the chord, usually in triplets	Slowly, bass – down – up, keeping the wrist loose and the thumb straight
Tremolo	Five-note tremolo: <i>p</i> – <i>i</i> – <i>a</i> – <i>m</i> – <i>i</i> (the classical tremolo has four: <i>p</i> – <i>a</i> – <i>m</i> – <i>i</i>)	Keep the four finger notes even; practise with a metronome on the thumb
Arpeggio and ligado	Fast <i>p</i> – <i>i</i> – <i>m</i> – <i>a</i> patterns and hammer-on / pull-off passages	Apply the patterns of Chapter 18 to the por arriba chords

▶ **TRACK 476** Soleá compás exercise por arriba

E **F** **E**

0 0 0 0 0 0 0 0 0 0 0 0

T 0 0 0 0 0 0 0 0 0 0 0 0

A 1 1 1 1 1 1 2 2 2 2 2 2

B 2 2 2 2 2 2 3 3 3 3 2 2

0 0 0 0 0 0 1 1 1 0 0 0

One 12-beat cycle written as four bars of 3/4. Light strokes with *i* on the unaccented counts; a strum with golpe, or a four-finger rasgueado ending on the beat, on 3, 6, 8, 10 and 12. The cycle moves to F on 7 and resolves to E on 10.

▶ **TRACK 477** Picado in E Phrygian

0 1 3 0 2 3 0 2 3 0 1 3 0 3 1 0 2 0 3 2 0 3 2 0 3 1 0

T 0 1 3 0 2 3 0 2 3 0 1 3 0 3 1 0 2 0 3 2 0 3 2 0 3 1 0

A 0 1 3 0 2 3 0 2 3 0 1 3 0 3 1 0 2 0 3 2 0 3 2 0 3 1 0

B 0 1 3 0 2 3 0 2 3 0 1 3 0 3 1 0 2 0 3 2 0 3 2 0 3 1 0

Rest strokes throughout, alternating *i* and *m*. Accent the first note of each group of four; keep the fingers close to the strings.

▶ **TRACK 478** Alzapúa on the Andalusian cadence por medio

Dm **C** **Bb** **A**

1 1 1 1 1 1 0 0 0 0 0 0

T 3 3 3 3 3 3 1 1 1 1 1 1

A 2 2 2 2 2 2 0 0 2 0 0 0

B 0 0 0 0 0 0 3 2 3 0 0 0

1 1 1 1 1 1 0 0 0 0 0 0

T 3 3 3 3 3 3 2 2 2 2 2 2

A 3 3 3 3 3 3 2 2 2 2 2 2

B 1 3 1 0 2 0

In each triplet the thumb plays the bass note as a rest stroke, then brushes down through the top strings and back up with the nail side of the thumb.

Rumba flamenca

Rumba flamenca is in 4/4 and uses the Andalusian cadence or simple major-key progressions. Its steady eighth-note strum is the easiest way into flamenco rhythm. The *x* is a muted chuck: the strumming hand slaps the strings (or plays a golpe) on beats 2 and 4.

TRACK 479 Rumba strum on Am – G – F – E

Am G F E

One bar each of Am, G, F and E. Play the F as the flamenco F (133200) with open B and E.

NYLON · STEEL · Flamenco and classical guitars

Nylon: flamenco guitars are nylon-strung like classical guitars but built lighter, traditionally with cypress back and sides (the *blanca*; rosewood instruments are called *negra*), with a lower action and a tap plate on each side of the soundhole. The sound is brighter, drier and more percussive. On a classical guitar every technique works, but fit a clear tap plate before practising golpes.

Steel: the harmony and the rumba strum work well on steel strings. Rasgueado on steel is hard on the nails and much louder; use it sparingly and never play golpes on an unprotected top.

Soleá Study por arriba

Original study · soleá · 3/4 (12-beat compás) · 120 bpm

Two compás cycles of 12 beats. The first is a short thumb falseta on the bass strings closing on count 10; the second strums the Andalusian cadence and lands on E on count 10. Count the accents aloud while you learn it.

TRACK 480

E F G E

Am G F E

5

0 1 3 1 0 3 1 0 1 1 3 3 3 0 0 1 0

T 0 0 0 3 3 3 0 0 0 0 0 0 0 0 0 0

A 1 1 1 0 0 0 0 0 0 0 0 0 0 0 0 0

B 2 2 2 0 0 0 0 0 0 0 0 0 0 0 0 0

0 0 0 2 2 2 3 3 3 2 2 2 1 1 1 0

PLAYING IT · Performance notes

Play the single notes with the thumb (rest strokes), and the accented chords with a downstroke of *i* plus a golpe, or a four-finger rasgueado that lands on the beat.

Bars 1–4 are counts 1–12 of the first cycle; bars 5–8 are the second cycle. The final E falls on count 10 and rings through 11 and 12.

When it is secure, replace the plain chords in bars 5–7 with continuous rasgueado that still lands on the accents.

CAREFUL · Nails, fingertips and the guitar top

- Rasgueado and golpe wear and crack nails. Build up a few minutes a day over several weeks, and stop if a nail splits or the fingertip hurts.
- Many players reinforce the nails with a thin layer of nail glue or a silk wrap. Learn the method from an experienced player and keep the glue off the skin.
- A golpe without a tap plate marks the finish and, over time, can damage the top. Use the golpeador, and tap with the flesh and nail of a, not with force from the arm.
- Rasgueado tension builds in the forearm. Keep the wrist loose, and shake the hand out between repetitions.

Practice loops

These loops are for harmony and melody, not for compás: the cadence loop runs in plain 6/8 bars, so practise the accents with palmas and the soleá exercise instead. Strum the cadence with rasgueados on the first passes, then switch to picado lines.

▶ **TRACK 481** Andalusian cadence in 6/8 · LOOP · 6/8 ballad · 80 bpm (dotted quarter)

Am <i>iv</i>	G <i>bIII</i>	F <i>bII</i>	E <i>I</i>
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IMPROVISE WITH

E phrygian dominant

E phrygian

Use G# over the E chord and G natural over Am and G. Try picado runs that land on E as the cadence arrives.

▶ **TRACK 482** E Phrygian drone · LOOP · modal drone · 80 bpm

E <i>I</i>

IMPROVISE WITH

E phrygian dominant

E phrygian

Explore the half step F – E above the drone, then the G# – A – G – F – E descent typical of flamenco melodies.

LISTEN FOR · Recordings to study

Ramón Montoya and Niño Ricardo (early solo flamenco guitar), Sabicas. Paco de Lucía, *Fuente y caudal* (1973) and *Almoraima* (1976), and his recordings accompanying the singer Camarón de la Isla. Manolo Sanlúcar, Tomatito, Gerardo Núñez and Vicente Amigo for later developments. For rumba: Peret (rumba catalana) and Paco de Lucía's rumba "Entre dos aguas" (1973).

Chapter 53 review · use it, train it, test it

Using it in real music

Accompanying dance and song. Flamenco accompaniment is about the compás first. A simple E – F alternation, played exactly in compás with clear accents, is more useful to a dancer than a fast falseta out of time.

Spanish colour in other styles. The Andalusian cadence, the flamenco F and a few bars of rumba strumming bring a Spanish flavour to pop, rock and film music. The Phrygian dominant mode (Chapter 9) is the scale behind many of these sounds.

Classical repertoire. Spanish concert music (Albéniz, Granados, Turina, Rodrigo) uses rasgueado, Phrygian harmony and picado scales; the techniques here prepare you for it.

How to train it

Drill	How	Time	Goal
Palmas	Clap the soleá compás with accents, counting aloud, then bulerías from 12	3 min	Accents in place without thinking
Rasgueado build-up	c – a – m – i on muted strings, then on E, at 60 bpm, one roll per beat	4 min	Even, rolling sound that lands on the beat
Picado	The picado exercise at 60, 68 and 76 bpm, rest strokes only	4 min	Even volume, both fingers sound alike
Alzapúa	Bass – down – up triplets on Dm – C – B♭ – A	3 min	Relaxed thumb, steady triplets
Rumba	Rumba strum on the Andalusian cadence with a clear chuck on 2 and 4	4 min	Groove holds at 104 bpm

Check your understanding

Write your answers, then check them in Appendix J.

1. Where are the accents in the soleá compás?

2. On which count is bulerías usually started?

3. How are the 12 counts of seguiriya grouped?

4. What do por arriba and por medio mean?

5. How do you turn the four-finger rasgueado into a continuous roll?

6. What is a golpe, and how do you protect the guitar?

7. How does the flamenco tremolo differ from the classical one?

MY NOTES

CHAPTER 54

Celtic Fingerstyle



TRACKS

54

Irish and Scottish dance music gives the guitar lilting compound rhythms, modal melodies and ornaments borrowed from the fiddle and pipes. DADGAD and drop D tuning let open strings drone and ring under the tune.

The traditional dance music of Ireland and Scotland is played by fiddles, flutes, whistles, pipes, accordions and concertinas, and learned by ear in informal sessions. The guitar entered the tradition in the 1960s and 1970s, first as an accompanying instrument and later as a solo voice. Two approaches are covered here: backing tunes in a session, and arranging melodies for solo fingerstyle guitar.

What defines the style:

- **Feel.** Dance rhythms with a characteristic lift: the stresses of the jig, reel and hornpipe, never a mechanical even pulse.
- **Harmony.** Modal melodies (Ionian, Mixolydian, Dorian, Aeolian; see Chapter 9) over drones and open fifths. Thirds are often left out of the chords so the tune can decide the mode.
- **Technique.** Ornaments (cuts, taps, rolls, slides, triplets) that decorate the rhythm, and fingerings that let open strings ring.

Tune types

Tune type	Metre	Feel
Jig (double jig)	6/8	Two groups of three eighths per bar: <i>DUM-da-da DUM-da-da</i>
Slip jig	9/8	Three groups of three; lilting and floating
Slide	12/8	Fast, four groups of three; from south-west Ireland
Reel	4/4 (often written 2/2)	Fast, even eighth notes, felt in two
Hornpipe	4/4	Slower than a reel, with dotted (swung) eighths and strong final beats
Polka	2/4	Bright and march-like; from south-west Ireland
Strathspey	4/4	Scottish; dotted rhythms and the "Scotch snap" (short note before a long one)
Air	Free	Song-like and rubato, often a song melody played without words

Most dance tunes have two parts, A and B, each 8 bars long and played twice: **AABB**, 32 bars in all. In a session a tune is usually played through three times, then the musicians change to another tune of the same type without stopping. A group of tunes played together is a *set*.

Rhythm for backing

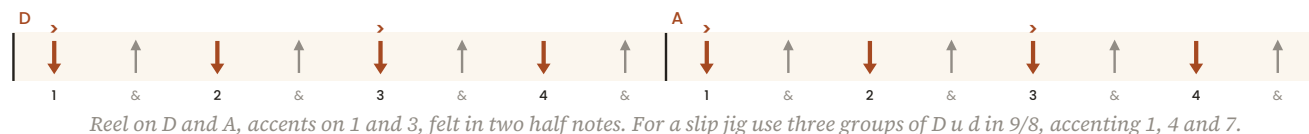
Session guitarists strum with a pick in a light, even motion and accent the main beats of the dance. Keep the volume below the melody instruments.

TRACK 483 Jig strum (6/8)



Bar 1 on D, bar 2 on G. Accents on 1 and 4. Each group of three is played down-up-down, so every group starts with a strong downstroke; at speed some players alternate strictly (D U D U D U) instead.

 TRACK 484 Reel strum



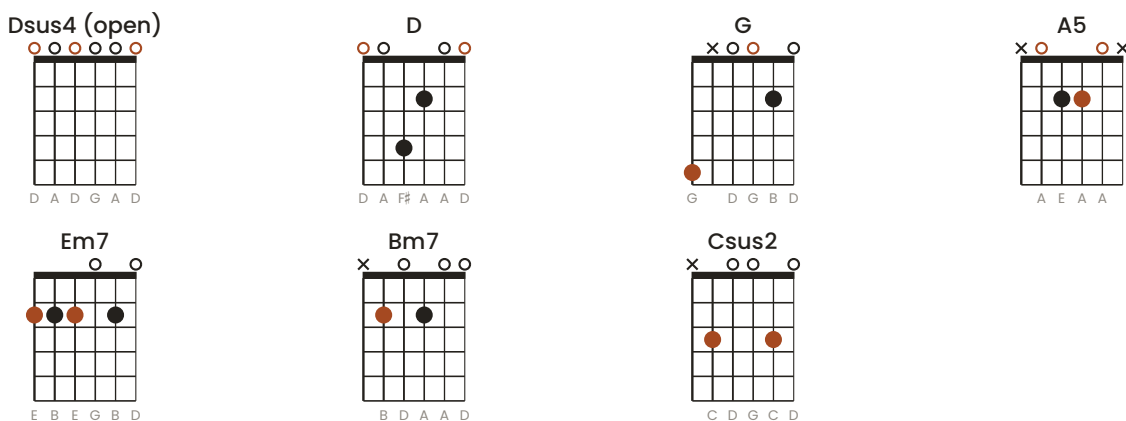
TIP • Following the tune

Many Irish and Scottish tunes are modal. A tune in D Mixolydian uses C, not C#, and wants D and C chords, not D and A. A tune in E Dorian wants Em and D. When you do not know the tune, play open fifths (D5, A5) or sus chords until you hear which third the melody uses.

DADGAD

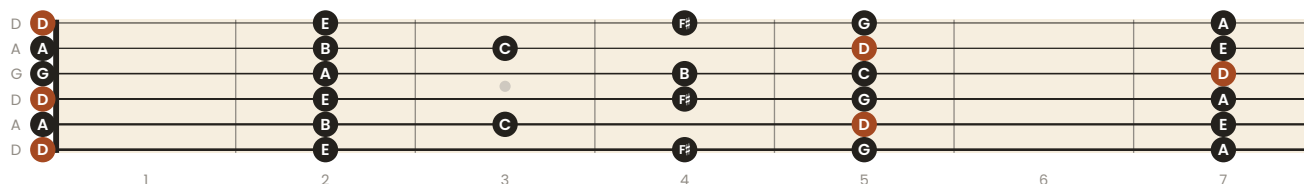
DADGAD tuning, D – A – D – G – A – D from the 6th string, was popularised by the British guitarist Davy Graham in the early 1960s and is now widely used by Irish and Scottish guitarists. The open strings form a Dsus4 chord; three of them are D or A, so the tuning drones on the tonic and fifth of D. It suits tunes in D major, D Mixolydian, D Dorian, G major, E Dorian and B minor. Chapter 25 covers how to retune safely.

Chords in DADGAD



Open strings ring through every shape. Thirds are added only where the tune needs them.

D Mixolydian in DADGAD, first position

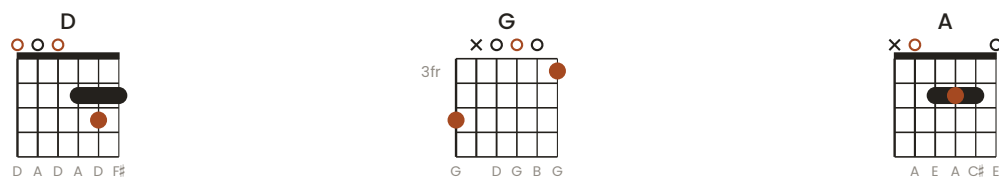


The D major scale in DADGAD is shown in Chapter 25; for Mixolydian tunes only C# becomes C. The open D and A strings fit both, which is why one tuning serves so many modal tunes.

Drop D

Drop D (D – A – D – G – B – E; Chapter 25) keeps standard tuning on the top five strings and adds a low D drone. It is a gentle first step for tunes in D and G, and for players who want standard-tuning chord shapes on the upper strings.

Drop D shapes for D, G and A



Ornaments

Ornaments in Irish music are rhythmic rather than melodic: they articulate and accent a note without adding a new melody. On the fiddle and flute they replace tonguing and bow changes; on the guitar they are played with slurs so that the picking hand keeps the rhythm.

Ornament	What it is	On guitar
Cut	A very short grace note above the main note, splitting a note or two repeated notes	Pull-off from a higher fret (or hammer and pull) so the grace note is barely heard
Tap (strike)	A very short grace note below the main note	Quick pull-off to a lower note and hammer back, or a hammer from below
Long roll	Note – cut – note – tap – note, filling a dotted quarter in a jig	One picked note, then all slurs
Short roll	Cut – note – tap – note, starting directly with the cut	As the long roll, without the first plain note
Slide	A glide into the main note from below	Slide one or two frets into the note
Triplet	Three quick notes in the time of one (jigs) or two (reels)	Picked triplet, or hammer-on and pull-off

TRACK 485 Ornaments on a jig phrase in DADGAD

Tuning: D A D G A D

Bar 1 plain. Bar 2: a cut (a flick of B) separates the two As. Bar 3: a roll on B: B, a cut from C#, B, a tap from open A, B. Bar 4: a triplet. Bar 5: a slide into F#. At full tempo the grace notes are quick clicks rather than notes.

Campanella

Campanella, fingering a scale so that consecutive notes ring into each other on different strings (Chapter 21), is the natural way to play tunes in DADGAD. The open A and D strings make it easy: a fretted G on the 3rd string followed by the open 2nd string, then B on the 3rd string while the A still rings. Harp-like runs of this kind are the signature of solo Celtic guitar.

TRACK 486 Campanella scale in DADGAD

Tuning: D A D G A D

Hold each fretted note as long as possible and let the open strings ring. The scale sounds legato even when played with free strokes.

Backing a session

- **Listen first.** Sit out the first time through a tune until you know its key, mode and chord changes.
- **Keep it simple.** One or two chords per bar is plenty. Open fifths and sus chords suit modal tunes better than full triads.
- **Follow, do not lead.** The melody players set the tempo. If the guitar is ahead or behind, it is the guitar that moves.
- **Bass lines.** Good session backing often moves the bass note while the upper strings drone, for example D – D/C – G/B – A.
- **Volume.** You should hear the melody instruments over your guitar at all times.

The Kitchen Door

Original study · jig, A part · DADGAD · 6/8 · 92 bpm (dotted quarter)

An eight-bar jig in D major with a drone bass on the open D strings. Play it twice (AA). It uses campanella in bars 4 and 7 and leaves room for ornaments once the notes are secure.

TRACK 487

Tuning: D A D G A D

5

D G A D

T
A
B

PLAYING IT · Performance notes

Thumb on the bass with free strokes; the melody with *i* and *m*, or with a pick (hybrid picking) if you prefer.

Add a cut on the second of two repeated notes, and a long roll on the held F# in bar 2 once you can play the tune at tempo.

Lilt the jig slightly, the first note of each group of three a fraction longer and stronger than the other two.

NYLON · STEEL

Nylon: a soft, intimate sound well suited to slow airs and harp tunes such as those of Turlough O'Carolan. Ornaments speak less clearly on nylon, so slur them firmly.

Steel: the usual choice for both backing and solo Celtic playing. The long sustain makes DADGAD drones and campanella shimmer. Retune by lowering strings, never by raising them (Chapter 25).

Practice loops

Both loops work in DADGAD or standard tuning. Back the jig loop with the jig strum until the lilt feels natural, then play phrases in groups of three over it. The drone is for modal ears: hold a melody note against it and hear what the third and seventh do.

▶ TRACK 488 Jig loop in D · LOOP · 6/8 ballad · 69 bpm (dotted quarter)

D <i>I</i>	G <i>IV</i>	D <i>I</i>	A <i>V</i>
D <i>I</i>	G <i>IV</i>	A <i>V</i>	D <i>I</i>

IMPROVISE WITH

D major pentatonic

D major

Improvise jig-like phrases in groups of three eighths; add cuts and rolls to repeated and long notes.

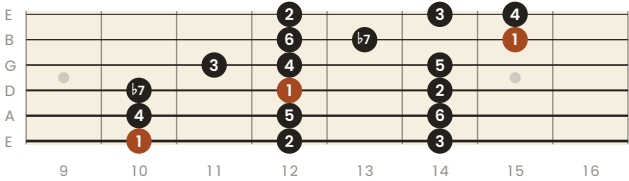
▶ **TRACK 489** Modal drone on D · LOOP · modal drone · 90 bpm

D5

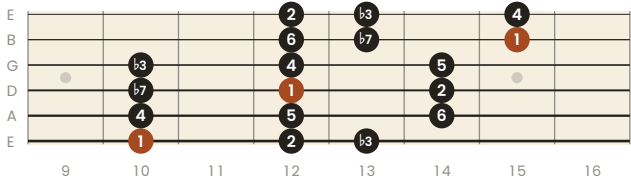
I5

IMPROVISE WITH

D mixolydian



D dorian



In DADGAD the open strings sound the drone with you. Switch between F# (Mixolydian) and F (Dorian) to hear the two modes.

LISTEN FOR · Recordings to study

Backing: Micheál Ó Domhnaill (The Bothy Band), Dáithí Sproule (Altan), Arty McGlynn, John Doyle. Solo Celtic fingerstyle: Pierre Bensusan, Tony McManus, El McMeen, Dick Gaughan. For the tunes themselves, listen to the melody players: the fiddlers Martin Hayes and Kevin Burke, the flute player Matt Molloy, the uilleann piper Liam O'Flynn. Traditional tunes to learn by ear: *The Kesh* and *Morrison's Jig* (jigs), *The Silver Spear* and *Drowsy Maggie* (reels), *The Kid on the Mountain* (slip jig), and O'Carolan's *Sí Bheag, Sí Mhór* (air).

Chapter 54 review · use it, train it, test it

Using it in real music

Solo sets. A slow air followed by a jig and a reel in DADGAD makes a complete solo set: one tuning, three moods.

Sessions. Backing jigs and reels with a steady, accented strum is the guitarist's main role. Learn the common tunes by ear so you can anticipate their changes.

Other styles. DADGAD drones and campanella runs work equally well in singer-songwriter accompaniment and in film-style instrumental pieces.

How to train it

Drill	How	Time	Goal
Jig lilt	Jig strum on D – G – A, accents on 1 and 4, at 100–120 bpm (dotted quarter)	3 min	Even lilt, no rushing on the downstrokes
Ornaments	Cut, tap and long roll on one note to a slow click, four bars each	4 min	Grace notes short and in time
Campanella	D major scale in DADGAD letting every note ring, up and down	3 min	No note stopped before the next sounds
Tune by ear	Learn one phrase of a traditional tune from a recording each day	5 min	A growing repertoire learned the traditional way
Modal backing	Back a D Mixolydian and an E Dorian tune with open fifths and sus chords	4 min	Chords never clash with the melody's third or seventh

Check your understanding

Write your answers, then check them in Appendix J.

1. What is the metre of a jig, and of a slip jig?
2. What is a cut?
3. What is a long roll made of?

4. Why does DADGAD suit tunes in D, and which modes besides D major does it serve?

5. In the DADGAD campanella scale, which string follows a fretted G on the 3rd string, and why?

6. What is the usual form of an Irish dance tune?

7. Why do session guitarists often leave the third out of chords?

MY NOTES

CHAPTER 55

Neo-Soul



TRACKS

55

Neo-soul combines soul and R&B song forms with jazz harmony and a relaxed, behind-the-beat hip-hop groove. The guitar plays extended chords, small embellishments between them, and very little else.

Neo-soul is a name given in the late 1990s to the music of artists such as D'Angelo, Erykah Badu, Maxwell, Lauryn Hill and Jill Scott. It looks back to 1970s soul (Curtis Mayfield, Marvin Gaye, Stevie Wonder), borrows chords from jazz, and takes its rhythmic feel from hip-hop production, in particular the loose, behind-the-beat swing of the producer J Dilla. The guitar style draws on Curtis Mayfield's and Jimi Hendrix's embellished ballad playing.

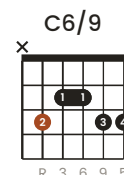
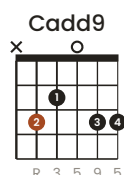
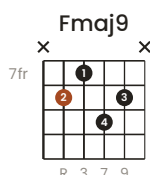
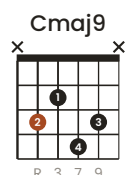
What defines the style:

- **Feel.** Slow to medium tempos (about 65–95 bpm), sixteenth-note swing, and parts placed slightly behind the beat. This is called playing *in the pocket*.
- **Harmony.** Ninth, eleventh and thirteenth chords, minor iv chords, chromatic slides and ii – V movement without the urgency of jazz.
- **Technique.** Small voicings on the middle and top strings, hammer-on and pull-off fills from the pentatonic scale between chord hits, double stops, and slides into chords from a half step away. A soft touch: fingers, or a pick held lightly, on the neck pickup.

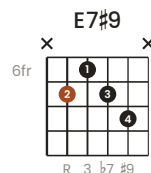
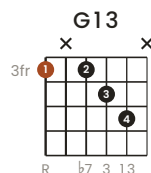
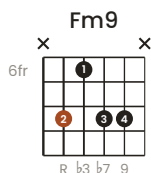
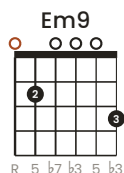
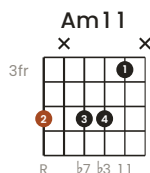
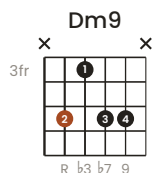
Extended voicings

Neo-soul voicings are four-note chords on adjacent strings that leave out the 5th (and often the root, when a bass player is present) to make room for 9ths, 11ths and 13ths. Chapter 48 explains how extensions are numbered.

Major colours



Minor and dominant colours



UNDER THE HOOD · Upper-structure triads

Many neo-soul sounds are a simple triad played over a different bass note. A minor triad a major 3rd above the root gives a maj7 chord (Em over C = Cmaj7). A major triad a 5th above the root gives a maj9 without its 3rd (G over C). A major triad a minor 3rd above the root gives m7 (C over A = Am7), and one a whole step below the root gives an 11 or 9sus4 chord (G over A = A11). On guitar this means familiar top-string triads become extended chords as soon as the bass plays a different root.

Embellished chord riffs

The core of the guitar style is a chord shape on the top strings that is decorated rather than strummed: hammer-ons from the 2nd or 4th into the 3rd, pull-offs to pentatonic notes, and short fills that return to the chord. The fretting hand holds the shape and lets one or two fingers move.

▶ TRACK 490 Embellished riff on Em9 Em9

The E minor triad at the 7th fret is held throughout. F# hammers onto G (9th to minor 3rd), and the fills use E minor pentatonic around the shape. Play the sixteenth notes with a light swing.

TRY THIS • Your own embellishments

Take any top-string triad from Chapter 10. Find the 2nd and 4th next to its 3rd, and the pentatonic notes within two frets of the shape. Make a one-bar riff that starts and ends on the full triad, with one hammer-on and one pull-off in between.

Double stops and slides

Double stops in 3rds and 6ths (Chapter 36) are a second source of fills. Neo-soul players favour 6ths on the 3rd and 1st strings, plucked with the thumb and middle finger (or pick and middle finger), and slide into the first one from a half step below so the fill starts with a soft, vocal scoop.

▶ TRACK 491 Sixths over Cmaj9 – Fmaj9 Cmaj9 Fmaj9

Diatonic sixths in C on strings 3 and 1. The slides approach the target from a half step below.

Whole chords can be slid the same way: play the voicing one fret below the target as a short sixteenth note and slide into it on the beat. Use it on one chord in a phrase, not every chord.

Pocket and dynamics

Neo-soul rhythm guitar leaves most of the bar empty. A typical comping part hits the chord on beat 1, then places one or two short stabs on off-beat sixteenths, and lets the drums and bass carry the rest.

- **Sixteenth-note swing.** The second and fourth sixteenths of each beat are played late, somewhere between straight and triplet feel. Copy the swing of the drummer, not a fixed ratio.
- **Behind the beat.** Place your chords a fraction late against the click. Rushing is the most common fault in this style.

- **Short and long.** Contrast long, ringing chords with very short stabs (release the fretting pressure), and play most of the part quietly.
- **Tone.** Neck pickup, clean amplifier, a little reverb and often chorus. On acoustic, play with the flesh of the fingers.

▶ **TRACK 492** Comping on Dm9 – G13 – Cmaj9

Dm9 **G13**

T	5	5	5	4	5	5	5	5
A	5	5	5	4	5	4	4	4
B	3	3	3	2	3	3	3	3
	5							

Cmaj9

T	3	3	2	3	3	3	3
A	4	4	3	4	4	4	4
B	2	2	1	2	2	2	2
	3						

The thumb (or pick) plays the root on beat 1; the fingers play short stabs on the upper three strings. Swing the sixteenths and sit behind the beat.

NYLON · STEEL

Steel and electric: the style is mostly played on electric guitar with a clean tone and the neck pickup. On a steel-string acoustic, light strings make the hammer-on fills much easier.

Nylon: the voicings and fills work well on nylon strings, where they sound closer to bossa nova (Chapter 50). Slides are shorter and quieter; let hammer-ons carry the embellishments instead.

Progressions

Most neo-soul songs loop two to four chords. The four progressions below cover the majority of them; play each with the comping pattern above.

ii – V – I with extensions

Dm9 <i>ii9</i>	G13 <i>V13</i>	Cmaj9 <i>Imaj9</i>	Cmaj9 <i>Imaj9</i>
--------------------------	--------------------------	------------------------------	------------------------------

Descending IV – iii – ii – I

Fmaj9 <i>IVmaj9</i>	Em7 <i>iii7</i>	Dm9 <i>ii9</i>	Cmaj9 <i>Imaj9</i>
-------------------------------	---------------------------	--------------------------	------------------------------

Minor iv to I (minor plagal cadence)

Fmaj9 <i>IVmaj9</i>	Fm9 <i>iv9</i>	Cmaj9 <i>Imaj9</i>	Cmaj9 <i>Imaj9</i>
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Minor vamp with a chromatic turnaround

Am11 <i>i11</i>	Am11 <i>i11</i>	Bbmaj7 <i>bIIImaj7</i>	Am11 <i>i11</i>
--------------------	--------------------	---------------------------	--------------------

Bbmaj7, the bII chord a half step above the tonic, works as a chromatic neighbour to Am11.

The minor iv (Fm9 in C) is borrowed from the parallel minor key; its Ab falls by a half step to G, which gives the cadence its soft, melancholy pull. Chromatic neighbour chords such as Bbmaj7 – Am11 come from the same habit of moving by a half step.

Late Set

Original study · neo-soul, swung sixteenths · 4/4 · 72 bpm

Eight bars combining the descending IV – iii – ii – I progression, pentatonic fills, sixths and the minor plagal cadence Fmaj9 – Fm9 – Cmaj9.

▶ TRACK 493

Fmaj9 **Em7**

T	8	8	8	10	8	10	8	9
A	9	9	9	9	9	9	9	9
B	7	7	7	7	7	7	7	7
	8							

Dm9 **Cmaj9**

T	5	5	5	8	5	6	5	5
A	5	5	5	5	5	5	5	5
B	3	3	3	3	3	3	3	3
	5							

Fmaj9 **Fm9**

T	8	9	10	8	7	8	8	8
A	9	9	10	9	7	8	8	8
B	7	7	7	7	7	6	6	6
	8					8		

Cmaj9

T	3	3	5	3	5	3	5
A	4	4	4	4	4	4	4
B	2	2	2	2	2	2	2
	3						

PLAYING IT · Performance notes

Hold each chord shape while the fills are played wherever possible, so the chord keeps ringing under them. Keep the stabs short by releasing the fretting pressure; keep the fills legato. Bars 5–6 (Fmaj9 – Fm9) move only one note, A to Ab. Bring it out.

Practice loops

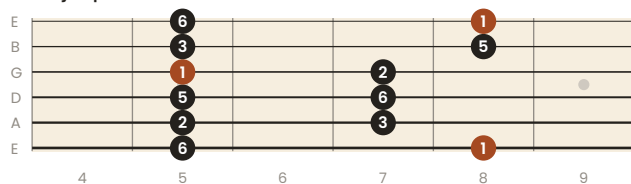
The ballad loop is the descending progression from the study: comp it with a chord on beat 1 and one or two short stabs, then fill the gaps with hammer-on figures. Record a pass and check that the stabs sit behind the beat. The minor groove is a Dorian vamp for single-note lines and sixths.

▶ TRACK 494 Neo-soul ballad in C · LOOP · ballad · 72 bpm

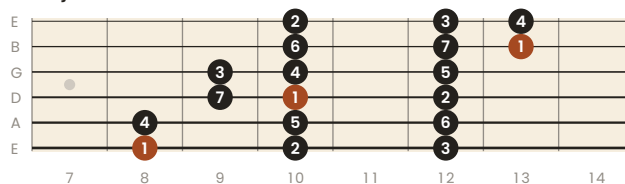
Fmaj9 <i>IVmaj9</i>	Em7 <i>iii7</i>	Dm9 <i>ii9</i>	Cmaj9 <i>I maj9</i>
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IMPROVISE WITH

C major pentatonic



C major



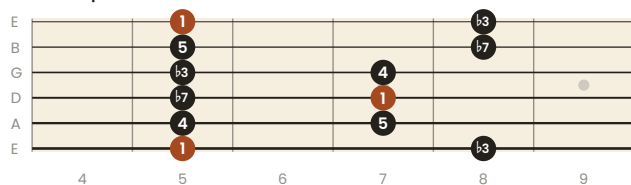
Alternate chord embellishments with short single-note fills. Play behind the beat.

▶ TRACK 495 Minor groove in A · LOOP · funk · 88 bpm

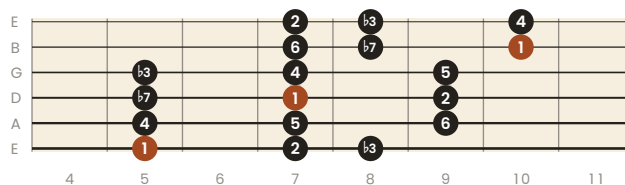
Am11 <i>i11</i>	Am11 <i>i11</i>	D9 <i>IV9</i>	D9 <i>IV9</i>
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IMPROVISE WITH

A minor pentatonic



A dorian



F#, the 6th of A Dorian, is the 3rd of D9: target it when the chord changes.

LISTEN FOR · Recordings to study

D'Angelo, *Voodoo* (2000). Erykah Badu, *Baduizm* (1997). Lauryn Hill, *The Miseducation of Lauryn Hill* (1998). Maxwell, *Urban Hang Suite* (1996). For the guitar style's roots: Curtis Mayfield's recordings with the Impressions and as a solo artist, and Jimi Hendrix's ballads such as "Little Wing". Later guitarists in the style include Isaiah Sharkey, Mark Lettieri, Tom Misch and Melanie Faye.

Chapter 55 review · use it, train it, test it

Using it in real music

Accompanying a singer. Neo-soul guitar is accompaniment first. Play the chord on beat 1 and fill only where the voice rests.

Reharmonising pop songs. Replace triads with maj9, m9 and 13 chords, add a minor iv before the tonic, and slow the groove down. A three-chord pop song becomes a neo-soul ballad.

Gospel and R&B. The same voicings, slides and fills are standard in modern R&B and gospel guitar.

How to train it

Drill	How	Time	Goal
Voicing changes	Dm9 – G13 – Cmaj9 – Fmaj9, one chord per beat at 60 bpm	3 min	Changes without gaps or buzzing
Hammer-on fills	Each top-string triad in C: 2nd or 4th hammered onto the 3rd, eight times	4 min	Clean, even slurs at full volume
Sixths	Diatonic sixths in C on strings 3 and 1, up and down with a slide into each	3 min	Both notes together, no 2nd string noise
Behind the beat	Comping example with a drum loop; record and check the stabs are late, not early	5 min	Relaxed, laid-back placement
Minor iv	Fmaj9 – Fm9 – Cmaj9 in three keys	3 min	Hear and play the falling half step

Check your understanding

Write your answers, then check them in Appendix J.

1. Spell Cmaj9.

2. Which note is usually left out of an extended voicing to make room for the 9th, 11th or 13th?

3. What is the minor plagal cadence in C major?

4. What chord do you get by playing an E minor triad over a C bass?

5. What does playing "in the pocket" mean in neo-soul?

6. How do you slide into a chord in this style?

MY NOTES

CHAPTER 56

West and Central African Guitar



Highlife from Ghana and Nigeria and Congolese rumba and soukous are built from short, repeating guitar parts that interlock with each other and with the drums. Each part is simple; the music lives in how they fit together.

The guitar reached the West African coast with sailors and traders in the late 19th and early 20th centuries, and by the 1920s it had become central to new urban popular music. This chapter covers two families: **highlife** (Ghana and Nigeria) with its acoustic ancestor **palm-wine** music, and **Congolese rumba** and its faster descendant **soukous** (Democratic Republic of the Congo and Republic of the Congo). Both have spread across the continent and influenced styles from East African benga to Afrobeat.

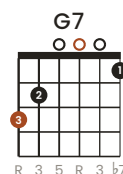
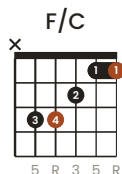
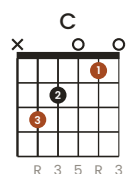
What they share:

- **Feel.** A steady dance groove, either in 4/4 with a strong off-beat lilt or in 12/8 with a triple subdivision, organised around a repeating bell or clave-like pattern.
- **Harmony.** Mostly major keys and the primary chords I, IV and V in repeating cycles. Interest comes from melody and rhythm rather than from harmonic surprise.
- **Technique.** Clean, bright tone. Short ostinatos (repeating figures) in the upper register, often in 3rds or 6ths, played with a pick on electric guitar or with thumb and index finger on acoustic.

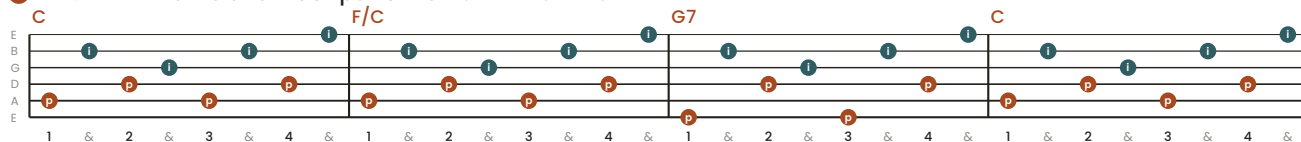
Palm-wine guitar

Palm-wine music, named after the drink served in the bars where it was played, is acoustic guitar music of the coast from Sierra Leone to Ghana and Nigeria. Its spread is often credited to Kru sailors from Liberia. The guitarist plays a bass line with the thumb and a syncopated melody with the index finger, usually in C or G with open chords, and sings over it. Ghanaian palm-wine players such as Kwaa Mensah and Koo Nimo, and S. E. Rogie in Sierra Leone, recorded the style from the 1950s onwards; it fed directly into guitar-band highlife.

Palm-wine chords in C



▶ TRACK 496 Thumb and index pattern on C – F – G7 – C



The thumb plays on every beat; the index finger answers on every off-beat. In palm-wine style the index often plays the melody notes, not just chord tones.

Highlife

Highlife developed in Ghana in the early 20th century and spread to Nigeria and Sierra Leone. Two strands grew side by side: **dance-band highlife** with horns, led by bandleaders such as E. T. Mensah, and **guitar-band highlife**, smaller groups of guitars, bass and percussion such as Nana Ampadu's African Brothers. In Nigeria, Victor Olaiya, Rex Lawson and Victor Uwaifo developed their own versions.

A highlife guitar section usually has two parts that interlock: a **lead** ostinato of single notes or double stops, often moving in eighth notes, and a **rhythm** part of short chord stabs on the off-beats. Neither part is complete on its own.

Interlocking highlife guitars (one bar of 4/4)

Lead guitar
Rhythm guitar
Bass (example)

1 e & a 2 e & a 3 e & a 4 e & a

The rhythm guitar avoids the main beats, where the lead and bass already play.

▶ TRACK 497 Highlife lead ostinato on I – IV – I – V in C

C F

8 5 5 5 8 8 5 5 | 8 5 6 5 8 6 5 6

3 C G

8 5 5 5 8 8 5 5 | 7 3 3 3 7 8 7 3

T
A
B

Single notes in 5th position on the top two strings, mostly chord tones with the 6th (A) as a colour note. Play with alternate picking and a light, even touch.

▶ TRACK 498 Highlife rhythm part (play against the lead)

C F

5 5 5 5 5 5 | 6 6 6 6 6 6

3 C G

5 5 5 5 5 5 | 3 3 3 3 3 3

T
A
B

Two-note chords (C and E, C and F, B and D) on strings 3 and 2, muted immediately after each stroke. Record one part with a looper and play the other over it.

Congolese rumba and soukous

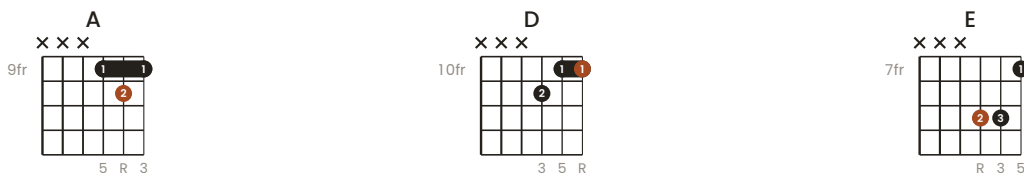
In the 1940s and 1950s, musicians in Léopoldville (now Kinshasa) and Brazzaville built a new music from Cuban son records, local rhythms and the electric guitar. Called **Congolese rumba**, it was led by bands such as Le Grand Kallé's African Jazz (with the guitarist Docteur Nico) and Franco's OK Jazz (later TPOK Jazz). From the late 1960s the fast dance section became more important, and in the 1980s Paris-based musicians such as Kanda Bongo Man and the guitarist Diblo Dibala made **soukous**, a faster style built largely from that section.

A Congolese band divides the guitars into three roles:

Role	What it plays
Solo (lead)	High, fast, repeating patterns and melodic phrases, especially in the <i>sebene</i>
Mi-solo (middle)	Arpeggios and counter-lines in the middle register, between lead and rhythm
Accompagnement (rhythm)	Steady chord patterns, often arpeggiated, that hold the cycle together

A song typically begins with a slower, sung section and then moves into the **sebene** (also *seben*), an instrumental dance section over a short repeating chord cycle such as I – IV – V – IV. The lead guitar plays cycling high-register patterns that change every few bars, while the singers and an *atalaku* (animateur) call out to the dancers.

Top-string triads for a *sebene* in A



Top-string triads from Chapter 10, chosen so that each keeps one note on the 2nd string as a pedal for the lead pattern.

▶ TRACK 499 Sebene lead pattern on A – E

A

E

The 2nd string acts as a pedal (A, then G#) while the melody moves on the 1st string. Pick alternately; keep the pedal notes slightly quieter than the melody.

TIP · Building a sebene part

Choose one shape per chord on the top two or three strings. Keep one string as a repeating pedal and move the other through chord tones. Repeat each pattern at least four times before varying it: the dancers need repetition, and the change only has an effect if the pattern was established first.

12/8 feel and bell patterns

Much West African music, and the traditional drumming behind it, is in 12/8: four main beats, each divided into three. The **standard bell pattern**, played on an iron bell, is the timeline for many Ewe, Yoruba and other West African rhythms. Its seven strokes fall in a 2 – 2 – 1 – 2 – 2 – 2 – 1 spacing across the twelve pulses.

▶ TRACK 500 12/8 timeline, pulse and guitar

Bell (standard pattern) 
 Dotted-quarter pulse 
 Guitar ostinato 

1 trip let 2 trip let 3 trip let 4 trip let

▶ TRACK 501 12/8 guitar ostinato on C and G

C G C G



Thumb on the main beats 1 and 3, fingers on the last pulse of beats 1 and 3 and on the first and third pulses of beats 2 and 4. Tap the dotted-quarter pulse with your foot and hear the bell pattern against it.

Practise switching between hearing the bar as four dotted quarters (12/8) and as six quarter notes (3/2 or 6/4): both are present in the bell pattern, and West African musicians move between them freely.

Progressions

Both styles repeat short cycles of primary chords. Play each chart with the parts above before moving on to the loops.

Highlife cycle

C I	F IV	C I	G V
C I	F IV	G V	C I

Sebene cycles

A I	D IV	E V	D IV
A I	E V	A I	E V

I – IV – V – IV, then a two-chord I – V cycle. A sebene can stay on one cycle for several minutes.

NYLON · STEEL

Steel and electric: highlife and soukous are electric-guitar styles: clean tone, bridge or middle pickup, a little reverb or chorus, and a pick. On a steel-string acoustic, the patterns work with a pick or with thumb and index.

Nylon: palm-wine guitar sounds good on nylon strings and suits the thumb-and-index technique well. The fast sebene patterns are harder on nylon because the strings are wider apart and the upper frets less accessible.

Palm-Wine Walk

Original study · palm-wine style · 4/4 · 104 bpm

An eight-bar piece in palm-wine style. The thumb plays on every beat; the index finger plays a syncopated melody from the chord shapes. Bars 1–4 start the melody off the beat; bars 5–8 start it on the beat.

TRACK 502

The musical score for 'Palm-Wine Walk' is presented in two systems, each containing four bars. The key signature is one flat (Bb), and the time signature is 4/4. The tempo is 104 bpm. The score is for guitar, with a melody line and a bass line. The melody is marked with a '7' (seventh) and 'mf' (mezzo-forte). The bass line is marked with fingerings (0, 1, 2, 3) and a '3' (triple). The chords are indicated above the staff: C, F/C, C, G7 in the first system, and C, F/C, G7, C in the second system.

PLAYING IT · Performance notes

Use only the thumb and index finger, as palm-wine players traditionally do. Rest the other fingers lightly on the top.

Keep the bass steady and slightly quieter than the melody; the syncopation should feel relaxed, not pushed.

Once you know it, sing or hum over it, or improvise new index-finger melodies from the same chord shapes.

Practice loops

Use the loops to practise interlocking, the core skill of these styles. Over the highlife loop, record the rhythm part with a looper and play the lead ostinato against it, then swap. Over the sebene loop, hold each lead pattern for at least four bars before you change it.

TRACK 503 Highlife loop in C · LOOP · pop · 112 bpm

C I	F IV	C I	G V
C I	F IV	G V	C I

IMPROVISE WITH

C major pentatonic

C major

Play the highlife lead ostinato, then invent your own: one short pattern per chord, repeated.

TRACK 504 Sebene cycle in A · LOOP · son montuno · 126 bpm

A I	D IV	E V	D IV
--------	---------	--------	---------

IMPROVISE WITH

A major pentatonic

A major

Build sebene patterns on the top two strings with a pedal note; change the pattern only every four or eight bars.

LISTEN FOR · Recordings to study

Palm-wine: Koo Nimo, Kwaa Mensah, S. E. Rogie. Highlife: E. T. Mensah and the Tempos, Nana Ampadu and the African Brothers, Victor Uwaifo, Rex Lawson, Prince Nico Mbarga ("Sweet Mother", 1976). Congolese rumba: Le Grand Kallé and African Jazz with Docteur Nico, Franco and TPOK Jazz, Tabu Ley Rochereau. Soukous: Kanda Bongo Man, Diblo Dibala, Pepe Kalle. For 12/8 drumming and bell patterns, recordings of Ewe and Yoruba percussion ensembles.

Chapter 56 review · use it, train it, test it

Using it in real music

Playing in a band. Agree who plays which role before you start. Two guitarists playing the same part cancel each other out; two interlocking parts make the groove.

Solo acoustic. Palm-wine technique gives you a complete accompaniment with two fingers. It works for calypso, folk and gentle pop songs as well.

Ostinato thinking. Short repeating parts that change slowly are a useful idea for any style built on a vamp, from funk (Chapter 45) to modal jazz.

How to train it

Drill	How	Time	Goal
Thumb and index	Palm-wine pattern on C – F/C – G7 – C at 90 bpm	4 min	Thumb never stops; index exactly between
Interlock	Loop the highlife rhythm part and play the lead over it, then swap	5 min	Both parts locked, no flaming
Sebene stamina	Sebene pattern at 100, 115, 128 bpm, 16 bars each	4 min	Even sixteenths, relaxed picking hand

Drill	How	Time	Goal
Bell pattern	Clap the 12/8 bell while stepping the dotted-quarter pulse	3 min	Steady for two minutes
12/8 ostinato	Play the 12/8 ostinato against the bell (recorded or clapped by a partner)	3 min	Hear both the 4-beat and 6-beat feel

Check your understanding

Write your answers, then check them in Appendix J.

- ### 1. What is a sebene?

2. Name the three guitar roles in a Congolese band.

- ### 3. Which fingers does palm-wine guitar traditionally use?

4. Write the 12/8 standard bell pattern.

- ### 5. Which chords make up most highlife and soukous progressions?

6. Why does the highlife rhythm guitar avoid the main beats?

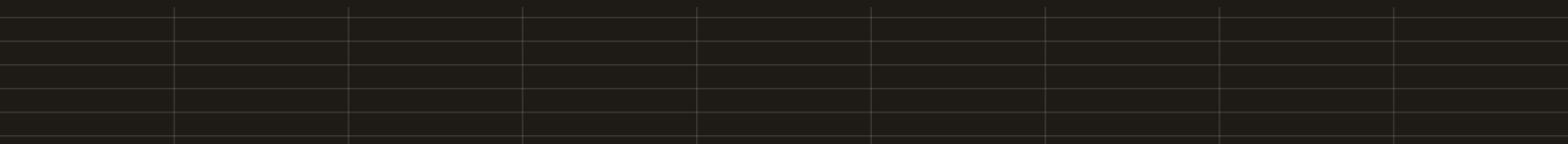
MY NOTES

PART THIRTEEN

Study Pieces

Complete pieces in notation and TAB, graded from first steps to concert studies, each applying ideas from the book.

- 57 Study Pieces: Elementary
- 58 Study Pieces: Intermediate
- 59 Study Pieces: Advanced



CHAPTER 57

Study Pieces: Elementary



TRACKS

Nine short pieces for roughly the first year of playing, from Grade 1 to a Grade 3 study that leads into the intermediate pieces. Each one is complete in notation and TAB and uses ideas from specific chapters, so you can learn a piece and see the theory working inside it.

The pieces in this chapter use first position, open strings and the simplest picking-hand patterns. They are ordered by difficulty, but each one also stands alone: if you have just worked through Chapter 17, go straight to the arpeggio studies; if you have just learned your first chords, start with the accompaniment study. The last piece, a Grade 3 study with a walking bass, is the step to Chapter 58.

Learn every piece from the notation and the TAB together. The TAB tells you where to play; the notation tells you how long each note lasts and which notes belong to the melody and which to the bass. Chapter 26 and Chapter 27 explain both in detail.

Piece	Grade and key	Techniques	Related chapters
Ode to Joy	Grade 1, C major	Melody on strings 1–3, alternating i and m	Chapter 26, Chapter 5
Thumb Walk	Grade 1, E minor	Thumb melody on the bass strings, rest stroke	Chapter 17, Chapter 6
Arpeggio Study in C	Grade 1, C major	p-i-m arpeggio, bass and treble on one staff	Chapter 18, Chapter 13
Boom-Chick Accompaniment in G	Grade 1, G major	Alternating bass with chord strokes	Chapter 22, Chapter 14
Two Voices in G	Grade 1–2, G major	Melody and bass together, sustained bass	Chapter 20, Chapter 10
Arpeggio Study in A minor	Grade 2, A minor	p-i-m-a arpeggio in 6/8, the F chord	Chapter 18, Chapter 31
Greensleeves	Grade 2, A minor	Melody over bass, 3/4 with pickup, raised 6th and 7th	Chapter 20, Chapter 6
Open-Position Blues in E	Grade 2, E	12-bar blues, pentatonic phrases, swing	Chapter 7, Chapter 43
Study in D with a Walking Bass	Grade 3, D major	Walking bass under a sustained melody, chromatic bass line, dynamics	Chapter 20, Chapter 14

How to learn a piece

1. Read through the score once without the guitar. Find the time signature, the key and any repeated bars.
2. Play the bass line alone, then the melody alone, at a slow tempo.
3. Put the two together two bars at a time. Keep the tempo slow enough that you never stop.
4. Raise the tempo in small steps only when the piece is clean at the current one. Chapter 38 describes this method in detail.

The pieces

Ode to Joy

Beethoven (1770–1827), arr. · Grade 1 · 4/4 · 84 bpm

The main theme of the finale of Beethoven's Symphony No. 9, simplified. It uses only the notes C to G on the first three strings, in first position. It practises reading quarter notes, the dotted quarter followed by an eighth, and a steady alternation of *i* and *m*. The chord symbols show the harmony in C major (Chapter 13); a teacher or friend can strum them while you play the tune.

TRACK 505

The musical score for 'Ode to Joy' is presented in four systems, each consisting of a treble staff and a three-string fretboard diagram. The key signature is C major, and the time signature is 4/4. The melody is written on the first three strings (Treble, A, B) in first position. Chord symbols (C, G) are placed above the melody. The fretboard diagrams show fingerings (0, 1, 3) and string numbers (T, A, B). The piece ends with a double bar line at measure 16.

PLAYING IT · Performance notes

Use one finger per fret: 1st finger for fret 1, 3rd finger for fret 3. Your hand stays in first position for the whole piece.

Alternate *i* and *m* strictly, even when a note repeats. Never pluck twice in a row with the same finger.

Count the dotted quarter in bars 4, 8 and 16 as "1 2 and": the eighth note falls on the "and" of beat 2.

In Beethoven's score the third line enters a beat early; this version keeps it on the beat to stay at grade 1.

Thumb Walk

Original study · Grade 1 · 4/4 · 72 bpm

A melody on the three bass strings, played entirely with the thumb. It builds the thumb control you need for every fingerstyle bass line (Chapter 17) and outlines a short progression in E minor, with the B7 chord taken from the harmonic minor scale (Chapter 6).

TRACK 506

Em Am G Em B7

5

Am Em B7 Em

PLAYING IT · Performance notes

Play every note with a rest stroke: the thumb pushes through the string and comes to rest on the next higher string.

Rest the fingers lightly on strings 1–3 for support. They also stop the treble strings from ringing.

The D# in bar 7 (string 4, fret 1) is the leading note of E minor. Let it rise to the final E.

Arpeggio Study in C

Original study · Grade 1 · fingerstyle · 3/4 · 92 bpm

A p-i-m arpeggio over open chords in C major. The thumb plays a bass note on beat 1 and lets it ring; i and m alternate on strings 2 and 1. The progression uses the I, ii, iii, IV, V7 and vi chords of C (Chapter 13) and the pattern is pattern 1 of Chapter 18.

TRACK 507

C Am Dm G7

p i m

T A B

5

C F G7 C

T 1 0 1 0 1

A

B 3

9

Am Em F C

T 1 0 1 0 1

A

B 0

13

F Dm G7 C

T 1 1 1 1 1

A 3

B 3

PLAYING IT · Performance notes

Hold each chord shape for the whole bar, so the bass note keeps sounding under the arpeggio.

For F, use the small four-string shape (xx3211): the 1st finger covers strings 1 and 2 at the 1st fret.

The thumb plays strings 4–6 and the fingers stay on strings 2 and 1. If the hand stays still, the right string is always under the right finger.

Boom-Chick Accompaniment in G

Original study · Grade 1 · pick or fingers · 4/4 · 100 bpm

The basic folk and country accompaniment: a bass note on beats 1 and 3 and a short chord on beats 2 and 4. The bass alternates between the root and another chord tone, as in Chapter 22. The progression is built from the I, IV, V and vi chords of G (Chapter 14). Use it to accompany any song with these chords.

TRACK 508

The musical score is presented in three systems, each with a treble clef, a key signature of one sharp (F#), and a 4/4 time signature. The notation includes a single bass note on the first and third beats of each measure, followed by a short chord stroke on the second and fourth beats. The chords are indicated by letters above the staff: G, C, G, D in the first system; G, C, G, D, G in the second system; and Em, C, D, G in the third system. Below the staff, the fretting hand positions are shown for the thumb (T), index (A), and bass (B) strings, with numbers indicating frets (0 for open, 1 for first, 2 for second, 3 for third). The bass string is consistently played on the first and third beats, while the other strings are played together on the second and fourth beats.

PLAYING IT · Performance notes

With a pick: play the bass note with a downstroke, then brush the top three strings with a short downstroke.
With fingers: thumb for the bass, then i, m and a together for the chord.

Keep the chord strokes short and light and the bass notes full. The contrast between low and high is what makes the pattern sound complete.

Fret the whole chord before you play the bass note. Only the picking hand moves within the bar.

Two Voices in G

Original study · Grade 1–2 · fingerstyle · 4/4 · 76 bpm

A first piece in two voices: a melody with stems up and a bass line in half notes with stems down. Every melody note is either a chord tone or a step between two chord tones, so the piece also shows how a tune sits on its harmony (Chapter 10, Chapter 20).

▶ TRACK 509

Sheet music for "Two Voices in G" in 4/4 time, featuring a melody with stems up and a bass line in half notes with stems down. The piece is in G major and consists of 12 measures.

Measures 1–4: Chords G, C, G, D. The melody starts on G4, moving up stepwise to B4. The bass line consists of half notes: G3, C3, G3, D3. Fingering: i, m, 3, 0, 3, 1, 0, 3, 0, 2, 2, 0.

Measures 5–8: Chords G, C, D, D7, G. The melody continues: C4, D4, E4, F#4, G4. The bass line: C3, F#2, D3, G2. Fingering: 0, 3, 3, 0, 3, 0, 1, 3, 0, 2, 1, 0.

Measures 9–12: Chords Em, C, D, G. The melody concludes: A4, B4, C5, B4. The bass line: C3, F#2, D3, G2. Fingering: 0, 2, 3, 0, 3, 0, 1, 2, 3, 2, 3, 3.

PLAYING IT · Performance notes

Let every bass note ring for its full two beats. Keep the fretting finger down until the next bass note is played. When a melody note and a bass note start together, pluck them at the same moment (a pinch of p and i or m). Play the melody slightly louder than the bass. Chapter 20 has exercises for this balance.

Arpeggio Study in A minor

Original study · Grade 2 · 6/8 · 52 bpm (dotted quarter)

A p-i-m-a-m-i arpeggio in 6/8, the ballad pattern of Chapter 18. The progression moves through the descending Am - G - F - E line, then uses the iv chord and the major V chord of the harmonic minor (Chapter 31). Each finger stays on its own string: i on 3, m on 2, a on 1.

TRACK 510

Arpeggio Study in A minor, 6/8 time, 52 bpm (dotted quarter). The piece features a descending chord progression: Am - G - F - E, followed by Dm, E7, and Am, then F, C, Dm, Am, Dm, E, E7, and finally Am.

The arpeggio pattern is p-i-m-a-m-i, with fingers i, m, a, m, i. The fretboard diagram shows the positions for each chord and the corresponding arpeggio notes.

Chord Progression:

- Am (1-4)
- G (2-3)
- F (1-2)
- E (1-2)
- Dm (2-3)
- E7 (1-2)
- Am (1-4)
- F (1-2)
- C (2-3)
- Dm (2-3)
- Am (1-4)
- Dm (2-3)
- E (1-2)
- E7 (1-2)
- Am (1-4)

Fretboard Diagram:

The diagram shows the fretboard for the guitar, with strings T (Treble), A (Alto), and B (Bass). The fret numbers are indicated below the strings.

Arpeggio Pattern:

The arpeggio pattern is p-i-m-a-m-i, with fingers i, m, a, m, i. The fretboard diagram shows the positions for each chord and the corresponding arpeggio notes.

PLAYING IT · Performance notes

Feel two beats in each bar (two groups of three eighths), not six. The bass note marks the first beat. Change chords during the last eighth of the bar. The open 3rd string on G and C gives you time. Play a slightly louder than *i* and *m* so the top line (E, G, F, E) sounds as a melody.

Greensleeves

Traditional, arr. · Grade 2 · 3/4 · 96 bpm

A simplified first half of Greensleeves, the traditional English tune, with one bass note per bar. Chapter 20 builds the complete arrangement with harmony notes and a moving bass; learn this version first. The melody uses the natural minor scale with the raised 7th (G $\sharp$) and, in the second ending, the raised 6th (F $\sharp$) of the melodic minor (Chapter 6). The piece starts with a one-beat pickup; count "3" before you play it.

▶ TRACK 511

The musical score for Greensleeves is presented in a simplified format. It begins with a treble clef and a 3/4 time signature. The melody is written on a single staff, with a bass line below it. The bass line consists of single notes, each lasting for one bar. The chords are indicated above the staff: Am, C, G, Em, Am, Dm, E, Am, E, Am. The score includes a first ending (1.) and a second ending (2.). Fingerings are indicated by numbers 1-4. Dynamics include *mp* (mezzo-piano). The piece starts with a one-beat pickup; count "3" before you play it.

PLAYING IT · Performance notes

Each bass note lasts the whole bar. Pluck it together with the first melody note of the bar, then let it ring. The dotted rhythm (dotted quarter, eighth, quarter) gives the tune its lilt. Count "1 2 and 3" and keep the eighth short and light.

Play the first eight bars, go back to the repeat sign, play bars 1–4 again and skip to the second ending (bar 9).

In the second ending the F $\sharp$ is on string 4, fret 4. Use the 4th finger and keep the 1st finger on the G $\sharp$ (string 3, fret 1).

The second half of the tune (the refrain) is in the full arrangement in Chapter 20. Try to work it out by ear before you look (Chapter 30).

Open-Position Blues in E

Original study · Grade 2 · pick or fingers · swing · 4/4 · 88 bpm

A 12-bar blues in E built from the E minor pentatonic scale in open position (Chapter 7). Each bar answers a bass note on the beat with a short pentatonic phrase. The form is the standard 12-bar blues of Chapter 43: four bars of I, two of IV, two of I, then V, IV, I and a final cadence.

▶ TRACK 512

E7 **E7**

A7

E7 **B7**

A7 **E7** **B7** **E7**

PLAYING IT · Performance notes

Swing the eighths: play each pair long-short, as the first and third notes of a triplet. The notation shows straight eighths, which is the usual convention for blues and jazz.

Stop each low note when the rest arrives. With a pick, lift the fretting finger or touch the string with the side of the picking hand.

Bar 11 runs down the whole scale in open position. Practise it alone until it is even, then join it to bar 12.

D

Bm

G

D/F#

Em

A7

D

13

T

A

B

2 3

2 3

3 0

2 3

3 2 0

2

2 3

2 0

0

3 2

2 0

0 2

0

PLAYING IT · Performance notes

Play every bass note with the thumb and give it its full beat. The bass is the moving part, so keep it even and slightly detached while the melody notes sustain above it.

Hold each melody note for its full length while the bass moves under it. In bar 1, keep the 1st finger on F# (string 1, fret 2) while the 4th finger plays C# (string 5, fret 4) under it.

Bars 11 and 12 climb B – C# – D – D# – E in the bass. Use the 4th finger for C# (string 5, fret 4) and the 1st finger for D# (string 4, fret 1).

Bars 13–16 repeat the opening with two-note chords in the melody. Play them forte, then slow down in bar 15 and let the final chord ring.

Chapter 57 review · use it, train it, test it

Using it in real music

Choosing the next piece. When a piece is clean at the written tempo, move to the next one in the table, even if the earlier one is not perfect. Return to it a week later; it will be easier.

Taking the pieces further. Play the accompaniment study under a friend singing or playing Ode to Joy or Greensleeves. Transpose Two Voices in G to C with the chords from Chapter 13. Improvise your own bar 11 for the blues with the same scale.

How to train it

Drill	How	Time	Goal
Hands separately	Play only the bass line of a two-voice piece, then only the melody	5 min	Both parts known from memory
Two-bar loops	Choose the hardest two bars; play them four times without stopping, then add the bar before	5 min	No hesitation at the bar line
Metronome ladder	Start 20 bpm below the written tempo; add 4 bpm after each clean run	5 min	Written tempo, clean, two days in a row
Bass sustain check	Play a two-voice piece and listen only to the bass: does each note last its full value?	3 min	No gaps in the bass

Check your understanding

Write your answers, then check them in Appendix J.

1. What do stems up and stems down mean in Greensleeves and Two Voices in G?
2. Greensleeves starts with a pickup of one beat in 3/4. Which beat of the count does the first note fall on?
3. In the Arpeggio Study in A minor, which finger plays string 1?

4. Which scale do the phrases in the Open-Position Blues come from?

5. What is the function of the B7 chord in Thumb Walk, and which scale does it come from?

6. In bar 1 of the Study in D with a Walking Bass, which notes does the bass play, and which chords do they imply?

MY NOTES

CHAPTER 58

Study Pieces: Intermediate



TRACKS

58

Seven pieces for grades 3–5. They combine independent bass and melody, barre chords, an alternate tuning, syncopation and position shifts, and each one draws on specific chapters of the book.

At this level a piece is no longer a single pattern repeated. The bass moves independently of the melody, chords change within the bar, and the picking hand has to switch between patterns. Each piece below concentrates on one of these skills. The table shows which chapter to review before you start.

Piece	Grade and key	Techniques	Related chapters
Travis Study in C	Grade 3, C major	Alternating thumb, melody on the off-beats, F barre	Chapter 19, Chapter 13
Waltz with a Moving Bass	Grade 3, G major	Thumb melody under sustained treble notes, 3/4	Chapter 20, Chapter 34
Étude in E minor	Grade 4, E minor	p-i-m-a-m-i in triplets, harmonic minor, half-diminished chord	Chapter 18, Chapter 31
Air in DADGAD	Grade 4, D	Open-string drones, modal melody (Mixolydian)	Chapter 25, Chapter 54
Silent Night	Grade 4, G major	Chord melody in 6/8, position shifts	Chapter 20, Chapter 12
Bossa Study in A minor	Grade 5, A minor	Thumb on 1 and 3, syncopated chords, extended voicings	Chapter 50, Chapter 11
Porch Rag	Grade 5, C major	Alternating bass, syncopated melody, circle of dominants	Chapter 19, Chapter 32

Working at this level

Learn the bass line of each piece first, on its own, until it runs without thought. In almost every piece here the thumb keeps time while the fingers play something less regular; if the thumb hesitates, the whole piece stops. Then add the melody one bar at a time.

Several pieces include notes that ring on after the next note is played (open strings in the DADGAD piece, bass notes in the waltz). The notation shows the written length; where a performance note says "let ring", keep the fretting finger down as long as the hand allows.

The pieces

Travis Study in C

Original study · Grade 3 · steel or nylon · 4/4 · 88 bpm

Travis picking in its plainest form (Chapter 19): the thumb alternates between the root and another chord tone on every beat, while *i* and *m* play on the off-beats. The melody note on beat 1 of each bar is pinched together with the bass. The progression uses the diatonic chords of C plus E7, the dominant of A minor (Chapter 32).

TRACK 514

Chord progression: C, C, Am, Am, Dm, G7, C, G7, C, E7, Am, F

8 *mf* *p* *i* *m* *i*

5

9

T
A
B

13

C Am Dm G7 C

0 1 3 1 0 1 0 1 1 3 1 0 1 3 0 1 0 3

3 2 2 0 2 0 2 0 0 0 0 3

3 3 0 0 0 3

PLAYING IT · Performance notes

On C, the 3rd finger moves between string 5 and string 6 (fret 3) while the other fingers stay down. Practise the bass alone until this move is automatic.

Mute the bass lightly with the side of the picking hand (palm muting, Chapter 19) so the thumb sounds even and the melody stands out.

Bar 12 needs a full F barre. If it is not yet secure, play the bass on string 4 only (4.3 throughout) with the small F shape.

Waltz with a Moving Bass

Original study · Grade 3 · nylon · 3/4 · 96 bpm

Here the thumb has the moving part. The bass walks in quarter notes, mostly by step, while the melody holds long notes above it. The two voices are written to avoid parallel octaves and fifths, the basic rule of Chapter 34, and the bass line creates inversions such as G/B and G/D (Chapter 10).

 TRACK 515

The image displays a musical score for the song "The Sound of Silence" by Simon & Garfunkel. It includes a guitar part and a bass part, both in G major and 3/4 time. The guitar part is written in treble clef with a key signature of one sharp (F#) and a 3/4 time signature. The bass part is written in bass clef. The score is divided into two systems, each with four measures. The first system is marked with a "5" at the beginning. The guitar part features a melodic line with dotted half notes and eighth notes, while the bass part provides a steady accompaniment with eighth and quarter notes. Chord diagrams are provided for each measure, showing the fret positions for the guitar. The chords are G, C, D7, G, Em, Am, D, and G. The guitar part is marked with a "5" at the beginning, indicating the fifth fret. The bass part is marked with a "5" at the beginning, indicating the fifth fret. The score is written in a standard musical notation style, with a key signature of one sharp (F#) and a 3/4 time signature. The guitar part is written in treble clef, and the bass part is written in bass clef. The score is divided into two systems, each with four measures. The first system is marked with a "5" at the beginning. The guitar part features a melodic line with dotted half notes and eighth notes, while the bass part provides a steady accompaniment with eighth and quarter notes. Chord diagrams are provided for each measure, showing the fret positions for the guitar. The chords are G, C, D7, G, Em, Am, D, and G. The guitar part is marked with a "5" at the beginning, indicating the fifth fret. The bass part is marked with a "5" at the beginning, indicating the fifth fret. The score is written in a standard musical notation style, with a key signature of one sharp (F#) and a 3/4 time signature. The guitar part is written in treble clef, and the bass part is written in bass clef. The score is divided into two systems, each with four measures. The first system is marked with a "5" at the beginning. The guitar part features a melodic line with dotted half notes and eighth notes, while the bass part provides a steady accompaniment with eighth and quarter notes. Chord diagrams are provided for each measure, showing the fret positions for the guitar. The chords are G, C, D7, G, Em, Am, D, and G. The guitar part is marked with a "5" at the beginning, indicating the fifth fret. The bass part is marked with a "5" at the beginning, indicating the fifth fret.

9

Em Am D7 G/B

13

C Am D7 G/D D7 G

PLAYING IT · Performance notes

Play every bass note with a rest stroke of the thumb so the line sounds legato and strong. The melody uses free strokes.

Keep each melody note sounding for its full length while the bass moves underneath. In bar 7 the 1st finger holds F# on string 1 while fingers 2, 3 and 4 walk up string 4.

Bars 15 and 16 end the piece with a cadence: C resolves to B and the bass F# rises to G. Slow down slightly in bar 15.

Étude in E minor

Original study · Grade 4 · nylon · 4/4 · 72 bpm

A continuous arpeggio in triplet eighths, p-i-m-a-m-i twice in each bar, in the style of the 19th-century guitar studies. The harmony is a minor-key progression from Chapter 31: i, iv, the major V7 from the harmonic minor, VI, and the half-diminished ii chord (F#m7b5) that leads to V. The thumb alternates between the root and a second bass note.

▶ TRACK 516

Em Am B7

p *i* *m* *a*

4

Em C Am

7

F#m7b5 B7 Em

10

C B7 Em

PLAYING IT · Performance notes

Feel one beat per group of three. The thumb notes fall on beats 1 and 3 and should be slightly stronger than the finger notes.

Bring out the top notes played by a (E, F#, G): together they form a slow melody above the arpeggio.

F#m7b5 in bar 7 is fingered 2x2210: 2nd finger on string 6, 3rd on string 4, 4th on string 3, 1st on string 2, with the open 1st string. Place it during the last triplet of bar 6.

Air in DADGAD

Original study · Grade 4 · DADGAD · 3/4 · 72 bpm

A slow air in DADGAD tuning (Chapter 25). The open D strings give a constant drone, and the melody, in D Mixolydian with its flat 7th (C natural, Chapter 9), moves between fretted and open strings so that notes overlap like a harp. This sound is central to the Celtic fingerstyle of Chapter 54.

TRACK 517

Tuning: D A D G A D

Sheet music for "Air in DADGAD" in DADGAD tuning (D A D G A D), 3/4 time, 72 bpm. The music is written for guitar and includes fingerings (p, m, i) and fret numbers (0, 2, 4, 5, 7).

The music is divided into three systems, each with a treble clef and a key signature of one sharp (F#). The time signature is 3/4. The first system is marked with a *p* (piano) dynamic. The second system is marked with a *m* (mezzo) dynamic. The third system is marked with a *i* (illegible) dynamic.

The first system (measures 1-4) is labeled with **D** and **G**. The second system (measures 5-8) is labeled with **D**, **G**, and **D**. The third system (measures 9-12) is labeled with **G**, **D**, **Csus2**, and **Dadd9**.

The guitar part is written in DADGAD tuning. The fret numbers for the strings (T, A, B) are as follows:

Measure	T	A	B
1	0	0	0
2	0	0	0
3	0	0	0
4	2	0	0
5	0	0	0
6	0	0	0
7	0	0	0
8	0	0	0
9	0	0	0
10	0	0	0
11	0	0	0
12	0	0	0

13

G D A D

7 5 4 2 0 2 4 0 0 0 0 0

T 7 5 4 2 0 2 4 0 0 0 0 0

A 0 0 0 0 0 2 0 0 0 0 0 0

B 5 0 0 0 0 0 0 0 0 0 0 0

PLAYING IT · Performance notes

Retune strings 6, 2 and 1 down a whole step (E to D, B to A, E to D). TAB numbers refer to the new tuning; the notation shows the pitches that sound.

Let every open string ring for as long as possible. In bar 3 the open A (string 2) should still sound under the next notes.

The G chord in bars 4, 7, 9 and 13 needs only one fretted bass note (string 6, fret 5); the open strings 4 and 3 complete the chord.

Silent Night

F. X. Gruber, arr. · Grade 4 · 6/8 · 50 bpm (dotted quarter)

Gruber's carol of 1818, arranged as a chord melody: the melody is always the highest note, chords sit below it on the strong beats, and the bass alternates between two chord tones. The harmony uses only I, IV and V7 in G. The D7 bars move to 5th position, using the A shape of Chapter 12, so the melody can stay on top (Chapter 20).

▶ **TRACK 518**

G

3 0 3 0 3 0 3 0

T 3 0 3 0 3 0 3 0

A 0 0 0 0 0 0 0 0

B 3 3 3 3 3 3 3 3

D7 G

5

5 5 3 3 3 3 3 3

T 5 5 3 3 3 3 3 3

A 5 5 0 0 0 0 0 0

B 0 0 0 0 0 0 0 2

9

C G

T 0 1 0 3 2 0 3 0 3 0 0 0

A 0 2 3 2 0 0 0 0

B 3 2 3 3 3 3

13

D7 G G/D

T 5 7 5 8 7 3 3 7

A 5 5 0 5 7 0 0 8

B 0 0 0 0 0 0 3 0

17

G D7 G

T 3 3 0 3 1 2 0 0 (0)

A 0 0 0 2 2 0 0 0

B 3 0 0 0 0 3 3 2

PLAYING IT • Performance notes

Count two beats per bar (two dotted quarters). The melody rhythm dotted quarter, eighth, quarter returns in almost every phrase.

Play the melody note of each chord with a, and the other chord notes more softly. Chapter 20 has exercises for this balance.

In bars 5, 6, 13 and 14 the 1st finger lies across strings 1–3 at the 5th fret. Shift to it during the last eighth note of the previous bar.

Bar 16 jumps to 7th position for the high B. Use the open D string in the bass to give yourself time to move.

Bossa Study in A minor

Original study · Grade 5 · nylon · 4/4 · 132 bpm

A bossa nova accompaniment with four-note voicings (Chapter 50). The thumb plays the bass on beats 1 and 3; the fingers play three-note chords in a syncopated two-bar pattern. The top note, E, stays on string 2, fret 5 through the first five chords: a common tone that ties together Am7, Am6, Dm9, G13 and Cmaj7 (Chapter 11). The harmony is a minor-key cycle ending with the ii – V – i of Chapter 31.

 TRACK 519

[illegible]

PLAYING IT · Performance notes

Pluck the three chord notes with i, m and a together, short and light. The thumb notes are longer and softer, like a bass drum.

Odd bars start with the chord on beat 1; even bars start with a rest and push the chord onto the off-beats. Count eighth notes aloud at first: "1 and 2 and 3 and 4 and".

Fmaj7 (bar 6) is a stretch: 1st finger on string 6, 2nd on string 3, 3rd on string 4, 4th on string 2 at the 5th fret. If it is too wide, leave out the bass F.

13

D7 G7 C G7 C

T 2 3 2 (2) 1 1 3 1 (1) 0 1 0 3 0 1 3 0 3 0

A 0 0 0 0 0 0 0 0 3 2 0 3 1 0

B 0 0 0 0 3 3 3 3 3 3 3 3 3 3

PLAYING IT · Performance notes

The tied notes are the point of the piece: pluck them on the off-beat and let them sound through the next bass note. Do not re-pluck the tied note.

Learn the bass alone first, then add the melody notes that fall on the beat (the pinches), and only then the off-beat notes.

Use a light palm mute on the bass strings and play the melody without muting, so the two parts have different colours (Chapter 19).

The chords are open shapes: A7 is x02020, D7 xx0212, G7 320001 and E7 020100. Fret the whole shape even where only some strings are played.

Chapter 58 review · use it, train it, test it**Using it in real music**

From study to repertoire. Each piece here stands for a whole repertoire. The Travis study and the rag lead to the alternating-bass styles of country blues and ragtime guitar; the waltz to the 19th-century classical studies of Carulli and Carcassi; the bossa study to the songbook of Chapter 50; the DADGAD air to Celtic slow airs. Appendix G lists pieces in each style.

Arranging your own. Silent Night shows the method for any simple tune: put the melody on the top strings, find a chord shape below each melody note, and add a bass note on the strong beats. Try it with a folk song you know well.

How to train it

Drill	How	Time	Goal
Thumb alone	Play only the lower voice of the Travis study or the rag, with a metronome on 2 and 4	5 min	Even bass at full tempo while counting aloud
Add one note	Add the melody notes one at a time: first those on the beat, then those between beats	5 min	No change in the bass when melody notes are added
Chord switch	In the étude and the bossa study, play only the first and last beat of each bar to practise the chord changes	5 min	Each new chord ready on the beat
Slow full play-through	Play the whole piece at 60% of the written tempo without stopping, even after mistakes	5 min	No stops; mark the bars that went wrong

Check your understanding

Write your answers, then check them in Appendix J.

1. In the Travis Study, what does the thumb play on each beat?

2. What does the C natural in the Air in DADGAD tell you about the mode?

3. Which chord in the Étude in E minor comes from the harmonic minor scale, and which note shows it?

4. In Porch Rag, why does each 7th chord lead naturally to the next one?

5. In a chord-melody arrangement such as Silent Night, where is the melody note in each chord?

MY NOTES

CHAPTER 59

Study Pieces: Advanced



TRACKS

59

Six concert studies for grade 6 and above. Each one concentrates on a technique that defines a style or a tradition, from barre arpeggios and two-part counterpoint to tremolo, flamenco picado, open-tuning grooves and jazz chord melody.

These pieces assume secure barre chords, fluent reading in all positions up to the 10th fret and a picking hand that can keep a pattern going while the fretting hand moves. They are longer than the earlier studies and some move quickly between positions. Practise them in sections, as described in Chapter 38, and learn each section to the written tempo before joining it to the next.

Piece	Grade and key	Techniques	Related chapters
Prelude in B minor	Grade 6, B minor	Sixteenth-note arpeggios, barre chords in several positions	Chapter 16, Chapter 12
Two-Part Study in A minor	Grade 6, A minor	Imitative counterpoint, independent voices	Chapter 34, Chapter 6
Open D Groove	Grade 6, D	Open D tuning, hammer-ons, pull-offs, slides, harmonics, percussion	Chapter 25, Chapter 36
Study in E Phrygian	Grade 6–7, E Phrygian	Picado, p-i-m-a arpeggios, thumb melody, Andalusian cadence	Chapter 53, Chapter 9
Tremolo Study in A minor	Grade 7, A minor	Classical tremolo p-a-m-i, melody on string 1 in high positions	Chapter 21, Chapter 31
Chord-Melody Study in F	Grade 7, F major	Ninth, thirteenth and altered chords, tritone substitution	Chapter 48, Chapter 35

Practising at this level

Separate the hands. Before you play any piece here, play through the left-hand shapes without the picking hand, moving from one chord or position to the next on a slow count. Most problems at this level are late position changes, not picking.

Practise the transitions. Mark every point where the hand changes position. Loop the last beat of the old position and the first beat of the new one until the shift is smooth, then widen the loop to a full bar on each side.

Record yourself. Long pieces hide uneven tempo. A recording played back at the end of a session tells you which bars rush and which drag. Chapter 38 describes this routine.

The pieces

Prelude in B minor

Original study · Grade 6 · nylon · 4/4 · 66 bpm

A sixteenth-note arpeggio (p-i-m-a-m-i-m-a) over a circle-of-fifths progression in B minor: i – iv – VII – III – VI – ii° – V7 – i (Chapter 31). Every chord is a movable barre shape from Chapter 12, and each one sits in a different position, so the piece trains fast, accurate barre changes. The two bass notes in each bar are the root and a second chord tone (Chapter 16).

TRACK 521

Bm **Em**

mf

p *i* *m* *a*

3

A **D**

5

G **C#m7b5**

7

F#7 **Bm**

9

G Em

11

C#m7b5 F#7 Bm

PLAYING IT · Performance notes

Each bar uses one barre shape: Bm and Em are A-shape minor barres (2nd and 7th fret), A is an E-shape barre at the 5th fret, D an A-shape barre at the 5th fret, G an E-shape barre at the 3rd fret and F#7 an E-shape 7th at the 2nd fret.

C#m7b5 is played with open strings 1–4 under a single fretted bass note, which gives you a full bar to prepare the F#7 barre.

Accent the thumb notes slightly. They form a bass line (B, E, A, D, G, C#, F#) that falls by fifths through the whole piece.

Change shape on the last sixteenth of each bar: release the old barre on the last note, then land the new one with the bass note.

Two-Part Study in A minor

Original study · Grade 6 · nylon · 4/4 · 72 bpm

Two independent voices in the manner of a Baroque two-part invention. The upper voice states a one-bar subject (a rising scale fragment and a falling fifth), and the lower voice imitates it an octave lower in the next bar. The subject then moves through D minor, G, C and F. The voices move mostly in contrary and oblique motion, with 3rds, 6ths and 10ths on the beats, which are the principles of Chapter 34. The cadences use the harmonic minor (Chapter 6).

TRACK 522

Am Dm A

mf *p*

i m i m

5

G C F Dm E7

T 0 2 0 1 3 0 0 1 0 1 5 3 1 0 1 3

A 0 2 0 1 3 0 0 1 0 1 3 0 1 0 1 3 0

B 2 3 3 0 2 3 5 3 3 0 1 2 0 1 3 0

9

Am C F

T 0 1 3 0 0 1 0 1 1 3 0 1 3 1 0 1

A 2 0 1 3 2 0 1 0 1 3 0 1 2 1 0 1

B 0 3 0 2 3 0 2 0 3 0 2 0 1 3 0 2 3 1

13

Dm E Am Dm E7 Am

T 1 0 3 1 0 0 1 1 3 0 3 1 0 1 0 1

A 0 0 0 0 0 0 2 0 0 3 1 0 1 2 2

B 0 4 0 0 0 0 0 0 2 0 0 0 0 0 0

PLAYING IT · Performance notes

Play the subject the same way every time it appears, in either voice: the four eighths legato, the quarter after them slightly detached. Imitation is only audible if each entry is phrased alike.

Hold every lower-voice note for its full length. In bars 3 and 7 the half notes on string 4 must keep sounding while the upper voice moves on strings 1 and 2.

In bar 4 the lower voice crosses onto string 3 (G and A) while the upper voice stays on string 2. Fret the lower notes with the 1st and 3rd fingers so the 4th finger stays free for the upper D.

Practise each voice alone, singing the other one. This is the most reliable way to hear two parts at once (Chapter 28).

Open D Groove

Original study · Grade 6 · Open D · steel · 4/4 · 96 bpm

A riff-based piece in Open D tuning (Chapter 25). A one-bar riff over a low D drone alternates open strings with hammer-ons and pull-offs, moves to a barre G at the 5th fret and an A at the 7th, and a middle section adds slides and 12th-fret harmonics (Chapter 36). The piece is built for a percussive picking hand: the rests in the bass are where the thumb slap falls (see the performance notes).

 TRACK 523

Tuning: D A D F# A D

The image displays a musical score for the song "The Sound of Silence" by Simon & Garfunkel. The score is written for guitar, vocal, and bass. The key signature is D major (two sharps) and the time signature is 4/4. The guitar part is in the treble clef, and the vocal and bass parts are in the bass clef. The score is divided into three systems, each with a measure number (1, 5, 9) at the beginning. The guitar part includes chords (D, G/D, A/D) and a melodic line with eighth and sixteenth notes. The vocal part includes lyrics and a melodic line with eighth and sixteenth notes. The bass part includes a melodic line with eighth and sixteenth notes. The score is written in a standard musical notation style, with a key signature of two sharps (F# and C#) and a time signature of 4/4. The guitar part is in the treble clef, and the vocal and bass parts are in the bass clef. The score is divided into three systems, each with a measure number (1, 5, 9) at the beginning. The guitar part includes chords (D, G/D, A/D) and a melodic line with eighth and sixteenth notes. The vocal part includes lyrics and a melodic line with eighth and sixteenth notes. The bass part includes a melodic line with eighth and sixteenth notes. The score is written in a standard musical notation style, with a key signature of two sharps (F# and C#) and a time signature of 4/4.

D G/D A/D D

PLAYING IT · Performance notes

Tune strings 6, 3, 2 and 1 down to D, F $\sharp$, A and D (6th and 1st a whole step, 3rd a half step, 2nd a whole step). The TAB refers to the new tuning; the notation shows the sounding pitches.

Percussion, which the notation cannot show: on beats 2 and 4, where the bass rests, slap the side of the thumb against strings 6 and 5 so they hit the frets with a short snare-like crack. On the off-beat rests in the upper voice, the fingertips of i and m can tap the top of the guitar near the soundhole. Learn the pitches first, then add the slap, then the taps.

Bar 9 is four natural harmonics at the 12th fret: touch the string directly over the fret wire without pressing, pluck, then lift the finger. Bar 12 plays three at once with the 1st finger lying flat across the strings.

Keep the hammer-ons and pull-offs as loud as the plucked notes. The pull-off is a small downward flick of the fretting finger, not a lift.

Study in E Phrygian

Original study · Grade 6–7 · nylon · 3/4 · 88 bpm

A flamenco-style study in E Phrygian, the mode on which most flamenco harmony rests (Chapter 9, Chapter 53). It opens with the Andalusian cadence Am – G – F – E as p-i-m-a arpeggios, continues with picado scale runs across the whole first position, then a thumb melody on the bass strings, and closes on the characteristic Fmaj7 $\sharp$ 11 – E cadence with open strings 1 and 2 ringing over both chords. There is no rasgueado: the piece trains the single-note techniques of flamenco.

TRACK 524

Am G F E

5 **E** **Fmaj7#11**

9 **E** **F** **G**

13 **F** **Fmaj7#11** **E**

PLAYING IT · Performance notes

Play the picado runs (bars 5–8) with alternating *i* and *m* rest strokes, strictly alternating even across strings. Start at half tempo and keep every note equally loud.

The thumb melody in bars 10–13 is played with the thumb alone, with rest strokes and a firm, slightly percussive attack. Rest the fingers on the treble strings.

Fmaj7#11 is the flamenco **F** chord: an **F** barre shape with strings 1 and 2 left open. The open **B** and **E** sound against the **F**, which is why the move to **E** sounds so strongly final.

Accented chords (bars 6, 8 and 16) are struck with the thumb, downwards across all strings, quickly enough to sound almost together.

Am E7 Am

13

8

8-8-8 7-7-7 5-5-5 4-4-4 4-4-4 7-7-7 10 10 10 8-8-8 7-7-7 5

T 7 5 6 7 6 7 5

A 0 7 6 7 6 7 5

B 0 7 6 7 6 7 5

rit.

PLAYING IT · Performance notes

Practise the right hand on the open 1st string first: p on string 5, then a, m, i on string 1, at a slow, strictly even tempo. Speed comes only after the four notes are equally spaced and equally loud.

The thumb notes are quarter notes in the lower voice; let them ring under the tremolo. The middle notes on strings 3 and 2 are also played by the thumb.

Each bar sits in one position: 5th position for A minor and D minor, 4th–7th for E7, 9th–12th for bars 9 and 10, 5th–8th for F. Shift during the thumb note at the start of the beat, never during the tremolo.

Aim for a melody that sounds sustained. If the repeated notes sound like separate attacks, play a, m and i lighter and closer to the string.

Chord-Melody Study in F

Original study · Grade 7 · jazz · nylon or electric · 4/4 · 80 bpm

A slow jazz chord-melody study with four- and five-note voicings: major 9th, minor 9th, 13th, altered dominants (7b9, 7b13) and a minor 6th chord (Chapter 48). The melody is always the top note of the chord or a single note between chords. The progression is a chain of ii – V movements in F with two substitutions from Chapter 35: Ab13 replaces D7 (tritone substitution) on the way to Gm9, and Bbm6 (iv minor, borrowed from F minor) leads back towards I.

TRACK 526

Fmaj9 Dm9 Gm9 C13 C7b9

8

mp

5 8 10 11 10 8

T 8 5 10 10 10 8

A 9 5 10 9 9 6

B 7 3 8 8 8 8

8 5 10 8 8

5

Am7 Ab13 Gm9 C7b9 C7b13

T	5	8	6	10	8	9
A	5		5	10	6	9
B	5		4	8	8	8
	5		4	10	8	8

9

Fmaj7 Bb13 Bbm6 Am7

T	10	8	8	6	6	5	8
A	9		7		6	5	
B	7		6		5	5	
	8		6		6	5	

13

Ab13 Gm9 C13 C7b9 C7b13 F6/9

T	6	4	10	10	8	9	8
A	5		10	9	6	9	8
B	4		8	8	8	8	7
	4		10	8	8	8	8

PLAYING IT · Performance notes

Play each chord with p on the bass note and i, m, a on the upper notes, all at once, with the top note slightly louder. On electric guitar, pluck with the fingers or use a pick for the bass and fingers for the rest (hybrid picking).

Most voicings skip a string: Fmaj9, Dm9 and Gm9 have their root on string 5 with string 6 muted; C13, Am7, Ab13, Bb13 and Bbm6 have their root on string 6 with string 5 muted by the underside of the bass-note finger. Bars 4, 8 and 15 change only one or two notes between chords (C13 to C7b9: A to G and E to Db). Keep the common notes pressed down and move only the fingers that change.

Let each chord ring until the next one. Swing is optional at this tempo; the piece works as a straight-eighth ballad.

Chapter 59 review · use it, train it, test it

Using it in real music

Toward the concert repertoire. The tremolo study prepares the classical tremolo pieces of Tárrega and Barrios; the two-part study prepares Bach's lute suites and the two-voice studies of Sor; the Phrygian study leads into the falsetas of soleá and farruca. Appendix G lists pieces in each of these styles at this level.

Make the techniques your own. Take the progression of the Prelude in B minor and play it with the tremolo pattern. Write a new riff in Open D on the same drone. Re-harmonise a standard melody with the voicings of the chord-melody study. Each technique is only learned once you can use it on material you did not get from the page.

How to train it

Drill	How	Time	Goal
Shape map	Play only the first beat of every bar as a block chord, in time, through the whole piece	5 min	Every position change lands on the beat
Tremolo on one string	p on string 5, a-m-i on string 1, metronome on the quarter note, raise by 4 bpm when even	5 min	Four even notes per beat at 80 bpm
Picado bursts	Four sixteenths then a quarter-note rest, on one string, then across strings; i-m rest strokes	5 min	Bursts clean at 120 bpm
Voice alone	Play one voice of the two-part study while singing the other	5 min	Both voices in time, from memory
Recorded run-through	Record the whole piece once at the end of the session and mark the uneven bars	5 min	A list of three bars to fix next time

Check your understanding

Write your answers, then check them in Appendix J.

1. In the Tremolo Study, when do you shift position, and why then?
2. What makes the F chord in the Study in E Phrygian an Fmaj7#11 rather than a plain F major chord?
3. In the Chord-Melody Study, which chord replaces D7 and why does it work?
4. How does the lower voice of the Two-Part Study relate to the upper voice in bars 1 and 2?
5. In the Open D Groove, why is the chord at the 7th fret written A/D?

MY NOTES

APPENDIX A

A 52-Week Plan



The learning tracks from the introduction, laid out week by week. One plan takes a new player from the first chord into learning track 3 in a year; a shorter plan takes a fluent player from learning track 3 into learning track 4 in six months.

This appendix turns the book into a course. Each week names the chapters to work from, the drills, practice loops and pieces to play, one ear-training task and a milestone to check at the end of the week. The order follows the learning tracks in *How to Use This Book* at the front of the book: Learning track 1 (Foundations), learning track 2 (Building), learning track 3 (Expanding) and learning track 4 (Advanced).

There are two plans. Choose the one that matches where you are now:

- **Plan A: weeks 1–52**, for players new to the guitar. It covers learning tracks 1 and 2 and starts learning track 3.
- **Plan B: weeks 1–26**, for fluent players who know the open and barre chords, the pentatonic boxes, the CAGED shapes and some fingerstyle. It covers learning track 3 and moves into learning track 4.

The daily session

Practise for 30–45 minutes a day, five or six days a week. A short daily session works better than one long session at the weekend. Every session has the same six blocks; the plan tells you what goes into each block this week. The shape follows the long template in Chapter 38 and starts with the daily 10-minute rhythm routine from Chapter 37.

Block	15 min	30 min	45 min	90 min	What goes in it
Warm-up	2	3	5	8	Slow chromatic exercise (Chapter 39) and this week's <i>Review</i> item
Rhythm	3	7	10	15	The daily rhythm routine; in short sessions, its first steps only
Technique	0	5	8	15	This week's drills: chord changes, picking hand, scale patterns
Theory or repertoire	7	8	12	30	New material from the Focus column, then this week's piece
Ear	0	4	5	10	The task in the Ear column
Play for fun	3	3	5	12	Songs you like, a song from Chapter 23, or improvising over a practice loop

If you have 15 minutes, keep the rhythm and repertoire blocks and do the ear task away from the guitar: sing intervals on a walk, or take a quiz on your phone. A week of 15-minute days counts as half a week, so spread the week's work over two.

If you have 90 minutes, take a break every 20–25 minutes (see the *Breaks* box in Chapter 38). Spend the extra time on repertoire and on the practice loops, not on new theory: each week already holds as much new material as most players can absorb.

How to use the plan

Read, then play. On the first day of a week, read the sections named in the Focus column (about 20 minutes, without the guitar if you like). On the other days, play.

Use the milestone as a test. At the end of the week, try the milestone once, at the stated tempo, without warming up on it first. If you pass, move on. If you nearly pass, move on and put the item in next week's warm-up. If you are far from it, repeat the week. Never repeat a week more than once: carry the weak item forward as a review task instead. The next quarter brings most material back anyway.

Spaced review is built in. Items marked *Review* bring back material from earlier weeks, at a higher tempo or in a new key. Do them in the warm-up or technique block; they take two or three minutes.

Log every session. Write one line after each session: what you practised, the metronome tempo and the result (Chapter 38 shows an example). At the end of each week, copy the result into the progress log at the end of this appendix. The log makes slow progress visible and shows you which week to repeat.

Record yourself. Record one take a week on your phone: the same piece or exercise, from the same position. The last week of every quarter is a showcase week. You record a complete study piece from start to finish and compare it with the previous showcase.

TIP · What the columns mean

Focus names the topic and its chapter. **Do this week** names drills, practice loops, strum grids and pieces by their titles in the chapters. **Ear** gives the week's ear task: *Course week 4* means week 4 of Chapter 29 (take its quizzes A to C and pass the last one with 9 of 10 twice, on different days), and *10A* means quiz A of course week 10. Look up any chord, scale or key you meet in Appendix C, Appendix D and Appendix E.

Songs from the songbook

Plan A brings in the songs of Chapter 23 once you know their chords and strum: the first ones in Quarter 1, the capo songs after week 20, the shuffle in the blues weeks and the 6/8 and reggae songs later. Songs marked *Play for fun* are not named in a week; add them to the *Play for fun* block from the week shown. In Plan B, any song fits the *Play for fun* block from week 1.

Song	Feel and level	From week	In the plan
Knockin' on Heaven's Door	4/4, beginner	5	Week 5
Brown Eyed Girl	4/4, beginner	8	Week 8
Have You Ever Seen the Rain	4/4 with chucks, intermediate	12	<i>Play for fun</i>
Let It Be	4/4, intermediate	19	Week 19
Stand By Me	4/4, beginner, capo 2	20	Week 20
Wagon Wheel	4/4 swung, beginner, capo 2	21	Week 21
I'm Yours	4/4 swung, beginner, capo 4	21	<i>Play for fun</i>
Before You Accuse Me	Shuffle, beginner to intermediate	23	Week 23
Wonderwall	4/4 sixteenths, intermediate, capo 2	25	<i>Play for fun</i>
The House of the Rising Sun	6/8, beginner	36	Week 36
Perfect	6/8, beginner, capo 1	38	Week 38
Piano Man	3/4, intermediate	38	<i>Play for fun</i>
Riptide	4/4 syncopated, beginner, capo 1	50	Week 50 (folk)
Zombie	4/4 sixteenths, beginner	50	Week 50 (rock)
Three Little Birds	Reggae, beginner	50	Week 50 (reggae)

Plan A: new to guitar (weeks 1–52)

Quarter 1, weeks 1–13: Learning track 1

Goal: hold and tune the guitar, play the open chords in time with the first strum patterns, use the basic fingerstyle strokes, find the natural notes on the neck and play the G major scale. Ear work uses Chapter 3: pitch matching, half steps and the first intervals. Showcase piece: **Boom-Chick Accompaniment in G** (Chapter 57).

Week	Focus	Do this week	Ear	Milestone
1	Posture, tuning (Chapter 1)	Posture check; Free and rest strokes; Alternate picking at 60 bpm; rhythm routine, step 1	Match each open string to a tuner note by ear	Tune up in under 3 minutes
2	Diagrams; half steps (Chapter 2, Chapter 3)	Read one example in every format; Chromatic recital; Em and Am; strum 1, All downs	Find the half steps by ear	Em – Am: 30 changes a minute
3	Note values; E, A and D (Chapter 3, Chapter 11)	Count, then strum; Clap and count; strum 2, Steady eighths; full rhythm routine	Sing back notes on the open strings	A – D in steady eighths at 70 bpm
4	G and C (Chapter 11, Chapter 22)	Open-chord changes: G – Em – C – D; strum 3, Old faithful; Silent pendulum	Major or minor: name chords a friend plays	G – C: 40 changes a minute
5	Changing in time (Chapter 22)	Last-upstroke changes; strums 4 and 5; Knockin' on Heaven's Door (Chapter 23)	Clap a song's rhythm before strumming it	Knockin' on Heaven's Door without stopping
6	Fingerstyle strokes (Chapter 17)	i m alternation; Thumb pendulum; the daily fingerstyle warm-up; <i>Review</i> strums 1–4	Interval songs: octave and 5th	Even i m eighths at 70 bpm
7	Planting; 3/4 time (Chapter 17, Chapter 22)	Planting cycle; Pinch alignment; strums 7 and 8; Waltz in 3/4: I – IV – V7 – I in D	Hear 3/4 or 4/4 in recordings	The waltz with the track, no stops
8	Open 7ths; strings 6 and 5 (Chapter 11, Chapter 4)	A7, D7, E7, G7, B7; fretboard plan, week 1; Note finder; Brown Eyed Girl at 110 bpm	Interval songs: major and minor 3rd	Any note on strings 6 and 5 in 2 seconds
9	Octave shapes; intervals (Chapter 4, Chapter 3)	Fretboard plan, week 2; One-string intervals; strum 9; Ballad in 6/8: vi – IV – I – V in C	Name 3rds and 5ths on one string	Natural notes on strings 6–3
10	Major scale; the F chord (Chapter 5, Chapter 11)	The C major scale on one string; Spell a key; fretboard plan, week 3; small F; start Boom-Chick	Sing do-re-mi over an open C	C major spelled and played
11	Keys and the circle (Chapter 5)	Move the shapes; Circle walk; G drone: hear the seven degrees; fretboard plan, week 4	Sing each degree over the drone	G major, two octaves, at 60 bpm
12	Practising well (Chapter 38)	Practice log; Tempo ladder on Boom-Chick; G major pop loop: I – vi – IV – V; <i>Review</i> F	Pick out a melody you know, by ear	Boom-Chick complete at 80 bpm
13	Showcase (Chapter 39)	Record Boom-Chick Accompaniment in G at 100 bpm; Habit audit	Retest weeks 1–12; note the weakest	Learning track 1 complete

Quarter 2, weeks 14–26: Learning track 2, first half

Goal: improvise with pentatonic boxes 1 and 2, play the first fingerpicking patterns and barre chords, name the chords in a key and the common progressions, and play a 12-bar blues. Chapter 29 starts this quarter, one course week per plan week. If you miss the pass mark, stay on that course week and let the Ear column fall behind; Quarter 3 leaves room to catch up. Showcase piece: **Open-Position Blues in E** (Chapter 57).

Week	Focus	Do this week	Ear	Milestone
14	Pentatonic box 1 (Chapter 7)	Box 1 up and down; Three notes, then five; A minor blues loop	Course week 1, pass quiz C	Box 1 from memory, eighths at 80 bpm
15	Box 2; power chords (Chapter 7, Chapter 11)	Sliding from box 1 into box 2; Box linking; Power-chord riff in E; <i>Review</i> Note finder	Course week 2, pass quiz C	Boxes 1 and 2 linked without stopping
16	Patterns 1–4 (Chapter 18)	Patterns 1–4 on C, Am, G and Em; Thumb alone; start Arpeggio Study in C	Course week 3, pass quiz C	Pattern 1 through C – Am – G – Em at 70 bpm
17	Patterns 5–8; 3/4 (Chapter 18)	Patterns 5–8; A waltz in E minor: i – iv – V7 – i; Arpeggio Study in C; <i>Review</i> strums 1–5	Course week 4, pass quiz C	Arpeggio Study in C complete, slowly
18	Triads, inversions (Chapter 10)	The four qualities on C; Inversion ladder: C major up strings 3–2–1; Quality switch	Course week 5, pass quiz C	C, F and G in three inversions on strings 3–2–1
19	Chords in a key (Chapter 13)	The diatonic triads of C major; Key charts; Numbers aloud; C and G in Appendix E; Let It Be; name its numerals	Course week 6, pass quiz C	The seven chords of C, G and D, named
20	Function; the capo (Chapter 13)	Same functions, different chords; Transpose a song with a capo; Stand By Me (capo 2); triads on strings 4–3–2	Course week 7, pass quiz B	One song played in two keys
21	Progressions, cadences (Chapter 14)	I – V – vi – IV in C with open chords; The four cadences in C; Four keys; Wagon Wheel (capo 2)	Course week 8, pass quiz C	I – V – vi – IV in four keys from memory
22	E-shape barre; minor loops (Chapter 11, Chapter 14)	Barre build-up; Barre progression in F; Andalusian cadence; Aeolian loop with open chords, then the Aeolian loop in A minor practice loop	Course week 9, pass quiz C	F barre with every string sounding
23	12-bar blues (Chapter 43)	Boogie pattern on A7, D7 and E7; Shuffle feel; Form memory; Before You Accuse Me play-along	Course week 10, pass quiz C	Before You Accuse Me: three choruses, form intact
24	Turnarounds, soloing (Chapter 43)	Turnaround in A; A blues phrase through the blue note; Call and response over A7; 12-bar shuffle in A loop; start Open-Position Blues in E	Course week 11, pass quiz C	One improvised chorus over the loop
25	A-shape barre; voice leading (Chapter 11, Chapter 10)	A-shape barres; Voice-led C – Am – F – G; the blues piece; <i>Review</i> Box linking, patterns 1–4	Course week 12, reviews A–C	Open-Position Blues in E at 70 bpm
26	Showcase	Record Open-Position Blues in E at 88 bpm; record a chorus over the shuffle loop	Final exam 1–3: 27 of 30 to pass	Recordings compared with week 13

Quarter 3, weeks 27–39: Learning track 2, second half

Goal: play the three minor scales, read simple notation in first position, find a chord in all five CAGED shapes, keep a Travis pattern going under changing chords and learn three more elementary pieces. Ear work comes from the learning track 2 chapters, plus one course quiz retaken each week, chosen to match the week's topic (10A means quiz A of course week 10; the pass mark is still 9 of 10). Showcase piece: Greensleeves (Chapter 57).

Week	Focus	Do this week	Ear	Milestone
27	Natural minor (Chapter 6)	A natural minor, one octave; Relative pairs; Natural minor loop: i – bVI – bVII	Drone practice; 10A	The relative minor of every key on the circle
28	Harmonic, melodic minor (Chapter 6)	A minor and E major: the raised 7th; Harmonic minor loop: i – iv – V7; start Arpeggio Study in A minor	Natural against harmonic; 10C	Three minor scales on A
29	Reading in first position (Chapter 26, Chapter 27)	Every natural note in first position; Levels 1a–1c; rhythm Exercises 1–3	Write what you hear; 7A	Level 1c at 60 bpm, no TAB
30	Accidentals; dots, ties (Chapter 26, Chapter 27)	Levels 2a–2b; Note-name flash; rhythm Exercises 4–5; Ode to Joy from the notation	Clap and write a heard rhythm; 7B	Arpeggio Study in A minor complete
31	CAGED in C (Chapter 12)	C major up the neck through the five shapes; CAGED ladder; Levels 3a–3b; Review Note finder	Chord quality by ear; 3C	C in all five shapes, in order
32	CAGED in any key (Chapter 12, Chapter 7)	One key, five shapes; Chord and box; A major pentatonic phrase; Country-pop loop in A	Sing the loop's roots; 9A	G and A in all five shapes
33	The Travis thumb (Chapter 19)	The thumb alone: C, G7 and the three-string bass; Four steps to the Travis pattern on C	Sing the bass of I – IV – V; 8C	Steady thumb on C – G7 at 70 bpm
34	Travis variations (Chapter 19)	The basic pattern through C – Am – F – G7; Change variation every bar; start Two Voices in G	Hear which bass note moves; 9B	Basic pattern through the changes at 70 bpm
35	Melody on top (Chapter 19)	Thumb-style study in C; G – C – D – G; Practice loop: ragtime changes in C; Two Voices in G	Cadence ear test; 9C	Two Voices in G complete
36	Greensleeves; 6/8 (Chapter 57, Chapter 27)	Start Greensleeves; rhythm Exercises 7–8; The House of the Rising Sun; Review Patterns 5–8	Sing the melody first; 2C	Greensleeves, bars 1–8
37	Position V (Chapter 26, Chapter 14)	Greensleeves with both endings; Level 5; Line cliché in A minor; Review CAGED in a random key	Hear the raised 6th and 7th; 11C	Greensleeves at 72 bpm
38	Review week (Chapter 38)	Every piece so far, once each; Perfect in 6/8; one fix from Breaking a plateau; Review inversions, barres	Weakest quiz; 5C and 6C	Five pieces from memory
39	Showcase	Record Greensleeves at 96 bpm; record the Travis pattern over the ragtime loop	Pick out a folk melody by ear	Learning track 2 complete

Quarter 4, weeks 40–52: into learning track 3

Goal: start the learning track 3 chapters (modes, open tunings, melody over bass, ear training, secondary dominants and one style workshop) and learn a Grade 3 study with a walking bass and a first intermediate piece. The Ear column works through Chapter 28, from intervals to dictation (the track names are its examples), with a course quiz where it matches the week. Showcase piece: **Travis Study in C** (Chapter 58).

Week	Focus	Do this week	Ear	Milestone
40	Slurs, slides (Chapter 36)	Start Travis Study in C; Hammer-ons and pull-offs through A minor pentatonic; Slides between pentatonic boxes 1 and 2	How to practise; Intervals 1	Travis Study in C, bars 1–8
41	What modes are (Chapter 9)	The drone switch; D drone for the drone switch; Parent lookup; Travis Study in C	Intervals 2	The parent scale of any mode on G
42	Modes on the neck (Chapter 9)	Dorian vamp: Am7 – D9; Mixolydian vamp: A – G; Characteristic-note phrases; Review CAGED ladder	Triads on A; 11C	Dorian and Mixolydian told apart by ear
43	Open G (Chapter 25)	Tuning of the week: Open G; I – IV – V in Open G; Fast retune; Travis Study in C	Seventh chords on A	Open G and back in 2 minutes
44	DADGAD, drop D (Chapter 25)	Common DADGAD chords; DADGAD air over a drone; Drop D riff; Slide on three strings	Tonic, subdominant, dominant, tonic in C	One short tune in DADGAD
45	Two voices (Chapter 20)	Reading two voices; Voice by voice; Counting two voices; start Study in D with a Walking Bass	Hearing scale degrees after a cadence	Travis Study in C complete, slowly
46	Arranging (Chapter 20)	Worked example: Greensleeves; Arranging checklist; Arrange another tune	I – V – vi – IV; ii – V7 – I	Eight bars of your own arrangement
47	Secondary dominants (Chapter 32)	Tonicisation and V7/x; V7/x in every key; Dominants round the circle (Chapter 14); the V7/x chords in bars 11–12 of Study in D	Mystery progression A; 9C	V7/ii, V7/IV and V7/V in C and G
48	Chains, borrowed chords (Chapter 32)	Extended dominants; Borrowed iv; Analyse three songs with a worksheet from Appendix H	Mystery progression B	Three songs analysed
49	A style workshop	Choose Chapter 47, Chapter 44 or Chapter 52; read it, play two of its drills	Dictation 1	The style's main rhythm at tempo
50	The style's loop	Ballad loop in G, Rock loop in E minor or One-drop loop in A minor; then Riptide, Zombie or Three Little Birds	Dictation 2	Two minutes over the loop, no stops
51	Review (Chapter 40)	Three ways to play one progression; Dorian vamp; Study in D at 72 bpm; Travis Study in C at tempo	Dictation 3	Travis Study in C at 80 bpm
52	Showcase	Record Travis Study in C at 88 bpm; improvise over the Dorian vamp; play all four showcases in order	Dictation 4	Four showcase recordings

Plan B: fluent player (weeks 1–26)

This plan assumes you have finished learning track 2 or have the same skills. Week 1 starts with a baseline recording of your best piece, so you can measure the six months. The Ear column runs Chapter 29 in the first quarter, one course week per plan week, and Chapter 28, Chapter 30 and course retakes in the second.

Quarter 1, weeks 1–13: Learning track 3

Goal: hear and play the modes, play in three open tunings, arrange a melody over a bass, use secondary dominants and jazz voicings, and work through one style workshop. Showcase piece: **Porch Rag** (Chapter 58).

Week	Focus	Do this week	Ear	Milestone
1	Modes, characteristic notes (Chapter 9)	Baseline recording; The drone switch; Brightness ladder	Course week 1, pass quiz C	Seven modes on one root, brightest to darkest
2	Modes on the neck (Chapter 9)	Dorian, Mixolydian, Lydian and Phrygian vamps; Characteristic-note phrases; start Étude in E minor	Course week 2, pass quiz C	One chorus per vamp, each mode audible
3	Open G, Open D (Chapter 25)	Tuning of the week; Slide phrase in Open G; Slide accuracy; Étude in E minor	Course week 3, pass quiz C	One I – IV – V song in Open G
4	DADGAD, drop D, Open C (Chapter 25)	Minor barres; Write in the tuning; start Air in DADGAD	Course week 4, pass quiz C	Fast retune between three tunings
5	Two voices (Chapter 20)	Voice by voice; Melody balance; Sixths; Étude in E minor complete	Course week 5, pass quiz C	Étude in E minor at 60 bpm
6	Arranging (Chapter 20)	Worked example: Greensleeves; Arrange another tune; Air in DADGAD; Review drone switch	Course week 6, pass quiz C	A 16-bar arrangement of your own
7	Secondary dominants (Chapter 32)	V7/x in every key; Extended dominants; start Porch Rag	Course week 7, pass quiz B	V7/x of each diatonic chord in C, F, G
8	Borrowed chords (Chapter 32)	Passing diminished; Borrowed iv; Chart analysis; Porch Rag	Course week 8, pass quiz C	Three charts analysed
9	Shells, ii – V – I (Chapter 48)	Shells in 12 keys; Guide tones round the cycle; Air in DADGAD complete	Course week 9, pass quiz C	Shell ii – V – I in six keys
10	Drop-2, comping (Chapter 48)	Drop-2 qualities; Comping rhythms; ii – V – I in C loop; Porch Rag	Course week 10, pass quiz C	Two minutes of comping over the loop
11	Jazz blues, chord melody (Chapter 48)	Jazz blues in F loop; Chord melody; Review Open G, modal vamps	Course week 11, pass quiz C	Three choruses of the F blues
12	Bossa nova (Chapter 50)	Thumb alone; Hands together; Clave clap; Bossa nova loop in C; start Bossa Study in A minor	Course week 12, reviews A–C	Bossa pattern steady at 120 bpm
13	Showcase	Record Porch Rag at 96 bpm; record Air in DADGAD; compare with the baseline	Final exam 1–3: 27 of 30 to pass	Learning track 3 core complete

Quarter 2, weeks 14–26: Learning track 4

Goal: use the full harmony of minor keys, lead voices smoothly, improvise through ii – V – I progressions with target notes, follow modulations and transcribe from recordings. Showcase piece: **Prelude in B minor** (Chapter 59); on a steel-string guitar, **Open D Groove**.

Week	Focus	Do this week	Ear	Milestone
14	Harmonic minor chords (Chapter 31)	Three minors; Harmonise; Bossa Study in A minor	Dictation 1 and 2	A harmonic minor in 7th chords
15	The working minor key (Chapter 31)	Cadence loop; Phrygian dominant vamp on E: E – F; start Prelude in B minor	Dictation 3 and 4	Bossa Study in A minor at 100 bpm
16	Melodic minor modes (Chapter 31)	Altered locator; Mode naming; Melodic minor and altered loop: Am6 – E7alt; Prelude	Mystery progressions A and B	Altered scale on any dominant
17	Common and guide tones (Chapter 34)	Common-tone triads; Guide-tone ii – V – I; Voice-lead a song you know; Prelude	Sing the bass of a song you know	One song re-voiced with common tones
18	Inner voices, reharmonising (Chapter 34)	Drop-2 cycle; Line cliché; Tritone swap; <i>Review</i> shells in 12 keys	6B and 6C: seventh-chord qualities	A 16-bar tune reharmonised
19	Target notes (Chapter 35)	Arpeggio map; Guide-tone chorus; Major ii – V – I in C loop; Prelude	7B and 12B: scale degrees	Every bar of the loop lands on a chord tone
20	Enclosures (Chapter 35)	Enclosure targets; Bebop run; Same lick, every key	5C and 12A: every interval	One lick in all 12 keys
21	Minor ii – V – i (Chapter 35)	Minor ii – V – i in D minor loop; Altered switch; Jazz blues in F: the last row as a loop	9C and 10C: progressions, minor melodies	Three choruses over the F blues
22	Pivot chords (Chapter 33)	Related keys; Pivot hunt; Pivot loop; Prelude complete, slowly	Name the new key in a recording	Pivot chords found in three songs
23	Direct modulation (Chapter 33)	Find the key changes; Whole-step lift; Scale switch; <i>Review</i> Phrygian dominant vamp	Sing the new tonic after each change	Prelude in B minor at 56 bpm
24	The method (Chapter 30)	Choosing material; Find the tonic; Transcribe the bass line only	Bass line of one chorus	Bass line written and checked
25	Worked example (Chapter 30)	Chord chart; Two-bar phrase; Verify; Prelude at tempo	A 16-bar melody or solo	16 bars played with the record
26	Showcase	Record Prelude in B minor at 66 bpm (or Open D Groove at 96); play your transcription with the record	One new tune learned by ear	Recordings compared with the baseline

Progress log

Copy this page once per quarter. At the end of each week, write one line: the dates, the number of days you practised, the milestone (met, nearly, or repeat) and the tempo you reached. Before the showcase, fill in the three summary lines.

Progress log for one quarter

WEEK ____ · DATES · DAYS PRACTISED · MILESTONE · TEMPO

WEEK ____ · DATES · DAYS PRACTISED · MILESTONE · TEMPO

WEEK ____ · DATES · DAYS PRACTISED · MILESTONE · TEMPO

WEEK ____ · DATES · DAYS PRACTISED · MILESTONE · TEMPO

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WEEK ____ · DATES · DAYS PRACTISED · MILESTONE · TEMPO

WEEK ____ · DATES · DAYS PRACTISED · MILESTONE · TEMPO

SHOWCASE PIECE · DATE RECORDED · TEMPO · FILE NAME

STRONGEST AREA THIS QUARTER · WEAKEST AREA

WEEKS TO REPEAT OR REVIEW NEXT QUARTER

When you finish

After Plan A, you are in learning track 3. Go on with Plan B from week 1. Its first weeks revisit modes and open tunings in more depth, then move to the chapters Plan A left out, such as Chapter 48 and the rest of Chapter 58. If your quiz scores in the last quarter were low, run Chapter 29 a second time alongside it.

After Plan B, stop following a fixed plan and build your own quarters on the same pattern: one theory chapter, one technique focus, one style workshop and one study piece per quarter, a showcase in week 13, and the progress log.

- **Styles.** Work through one style workshop a month, with its practice loops. Choose styles that build on what you have: Chapter 49 and Chapter 51 after jazz harmony, Chapter 53 after the Phrygian modes, Chapter 54 after DADGAD, Chapter 55 and Chapter 45 after voice leading.
- **Repertoire.** Take the next pieces from Chapter 59 and Appendix G. Keep two pieces in rotation: one new, one old.
- **Ear.** Do the 10-minute routine from Chapter 28 every day and transcribe one short passage a month (Chapter 30).
- **Rhythm.** Keep the daily rhythm routine. Change the rhythm cell each week and add the odd metres and polyrhythms from Chapter 37.
- **Review.** Once a month, play through every piece you have recorded. Any piece you can no longer play goes back into the warm-up for two weeks.

Playing with other people tests everything at once. When you can play a song from Chapter 23 or a practice-loop progression without looking at the page, take it to a friend, a jam session or an open mic.

APPENDIX B

Cheat Sheets

B

The formulas, maps and charts you will look up most often, collected in one place. Every table spells its examples from C or lists every key, so you can read off any scale or chord directly.

Scale formulas

Formulas count from the root (1). A flat or sharp before a number lowers or raises that degree compared with the major scale. Every scale is spelled from C.

Scale	Formula	From C	Chapter
Major (Ionian)	1 2 3 4 5 6 7	C D E F G A B	Chapter 5
Natural minor (Aeolian)	1 2 $\flat$ 3 4 5 $\flat$ 6 $\flat$ 7	C D $\flat$ E $\flat$ F G $\flat$ A $\flat$ B	Chapter 6
Harmonic minor	1 2 $\flat$ 3 4 5 $\flat$ 6 7	C D $\flat$ E $\flat$ F G $\flat$ A B	Chapter 6
Melodic minor (jazz, ascending)	1 2 $\flat$ 3 4 5 6 7	C D $\flat$ E $\flat$ F G A B	Chapter 6
Dorian	1 2 $\flat$ 3 4 5 6 7	C D $\flat$ E $\flat$ F G A $\flat$ B	Chapter 9
Phrygian	1 $\flat$ 2 $\flat$ 3 4 5 6 $\flat$ 7	C $\flat$ D $\flat$ E $\flat$ F G $\flat$ A $\flat$ B	Chapter 9
Lydian	1 2 3 $\sharp$ 4 5 6 7	C D E $\sharp$ F G A B	Chapter 9
Mixolydian	1 2 3 4 5 6 $\flat$ 7	C D E F G A $\flat$ B	Chapter 9
Locrian	1 $\flat$ 2 $\flat$ 3 4 $\flat$ 5 $\flat$ 6 $\flat$ 7	C $\flat$ D $\flat$ E $\flat$ F G $\flat$ A $\flat$ B	Chapter 9
Major pentatonic	1 2 3 5 6	C D E G A	Chapter 7
Minor pentatonic	1 $\flat$ 3 4 5 $\flat$ 7	C $\flat$ E $\flat$ F G $\flat$ B	Chapter 7
Blues	1 $\flat$ 3 4 $\flat$ 5 5 $\flat$ 7	C $\flat$ E $\flat$ F G $\flat$ G $\flat$ B	Chapter 7
Whole tone	1 2 3 $\sharp$ 4 $\sharp$ 5 $\flat$ 7	C D E $\sharp$ F $\sharp$ G $\flat$ B	Chapter 48
Diminished, half-whole	1 $\flat$ 2 $\sharp$ 2 3 $\sharp$ 4 5 6 $\flat$ 7	C $\flat$ D $\sharp$ E $\sharp$ F G A $\flat$ B	Chapter 48
Diminished, whole-half	1 2 $\flat$ 3 4 $\flat$ 5 $\sharp$ 5 6 7	C D $\flat$ E $\flat$ F G $\flat$ G $\sharp$ A B	Chapter 48
Bebop dominant	1 2 3 4 5 6 $\flat$ 7 7	C D E F G A $\flat$ B B	Chapter 48

Modes of harmonic and melodic minor

These are the minor-key scales of Chapter 31. The ones in everyday use are marked in bold.

Scale	Parent scale and degree	Formula	From C	Typical chord
Locrian $\sharp$ 6	Harmonic minor, 2nd	1 $\flat$ 2 $\flat$ 3 4 $\flat$ 5 6 $\flat$ 7	C $\flat$ D $\flat$ E $\flat$ F G $\flat$ A $\flat$ B	Cm7 $\flat$ 5
Dorian $\sharp$ 4	Harmonic minor, 4th	1 2 $\flat$ 3 $\sharp$ 4 5 6 $\flat$ 7	C D $\flat$ E $\sharp$ F G A $\flat$ B	Cm7
Phrygian dominant	Harmonic minor, 5th	1 $\flat$ 2 3 4 5 $\flat$ 6 $\flat$ 7	C $\flat$ D $\flat$ E F G $\flat$ A $\flat$ B	C7 $\flat$ 9 $\flat$ 13 (V7 in minor)
Hungarian minor	Harmonic minor with $\sharp$ 4	1 2 $\flat$ 3 $\sharp$ 4 5 $\flat$ 6 7	C D $\flat$ E $\sharp$ F G $\flat$ A B	CmMaj7
Dorian $\flat$ 2	Melodic minor, 2nd	1 $\flat$ 2 $\flat$ 3 4 5 6 $\flat$ 7	C $\flat$ D $\flat$ E $\flat$ F G A $\flat$ B	C7sus4($\flat$ 9)
Lydian augmented	Melodic minor, 3rd	1 2 3 $\sharp$ 4 $\sharp$ 5 6 7	C D E $\sharp$ F $\sharp$ G $\sharp$ A B	Cmaj7 $\sharp$ 5
Lydian dominant	Melodic minor, 4th	1 2 3 $\sharp$ 4 5 6 $\flat$ 7	C D E $\sharp$ F G A $\flat$ B	C7 $\sharp$ 11
Mixolydian $\flat$ 6	Melodic minor, 5th	1 2 3 4 5 $\flat$ 6 $\flat$ 7	C D E F G $\flat$ A $\flat$ B	C7 $\flat$ 13
Locrian $\sharp$2	Melodic minor, 6th	1 2 $\flat$ 3 4 $\flat$ 5 $\flat$ 6 $\flat$ 7	C D $\flat$ E $\flat$ F G $\flat$ A $\flat$ B	Cm7 $\flat$ 5
Altered	Melodic minor, 7th	1 $\flat$ 2 $\sharp$ 2 3 $\flat$ 5 $\sharp$ 5 $\flat$ 7	C $\flat$ D $\sharp$ E $\flat$ G $\sharp$ G $\flat$ B	C7alt

To find the parent melodic minor quickly: Lydian dominant on X uses the melodic minor a perfect 5th above X; Locrian #2 on X uses the melodic minor a minor 3rd above X; the altered scale on X uses the melodic minor a half step above X.

Chord formulas

Chord	Formula	From C	Chord	Formula	From C
C	1 3 5	C E G	C7	1 3 5 b7	C E G Bb
Cm	1 b3 5	C Eb G	Cmaj7	1 3 5 7	C E G B
Cdim	1 b3 b5	C Eb Gb	Cm7	1 b3 5 b7	C Eb G Bb
Caug	1 3 #5	C E G#	Cm7b5	1 b3 b5 b7	C Eb Gb Bb
Csus2	1 2 5	C D G	Cdim7	1 b3 b5 b7 (sounds as 6)	C Eb Gb A
Csus4	1 4 5	C F G	CmMaj7	1 b3 5 7	C Eb G B
C5	1 5	C G	C7sus4	1 4 5 b7	C F G Bb
C6	1 3 5 6	C E G A	C9	1 3 5 b7 9	C E G Bb D
Cm6	1 b3 5 6	C Eb G A	Cmaj9	1 3 5 7 9	C E G B D
Cadd9	1 3 5 9	C E G D	Cm9	1 b3 5 b7 9	C Eb G Bb D
C6/9	1 3 5 6 9	C E G A D	C11	1 (3) 5 b7 9 11	C G Bb D F
C7b9	1 3 5 b7 b9	C E G Bb Db	Cm11	1 b3 5 b7 9 11	C Eb G Bb D F
C7#9	1 3 5 b7 #9	C E G Bb D#	C13	1 3 5 b7 9 13	C E G Bb D A
C7#11	1 3 5 b7 #11	C E G Bb F#	Cmaj7#11	1 3 5 7 #11	C E G B F#
C7#5	1 3 #5 b7	C E G# Bb	C7alt	1 3 b7 + b9, #9, b5 or #5	C E Bb with Db, D#, Gb, G#

Other symbols for the same chords: Cmin or C- (Cm), C° (Cdim), C+ (Caug), CΔ7 (Cmaj7), C-7 (Cm7), Cø7 (Cm7b5), C°7 (Cdim7).

On guitar the 5th, and often the root, can be left out of extended chords: the 3rd and 7th (the guide tones) define the chord, and the tensions give it colour. Chapter 11 and Chapter 48 show the voicings.

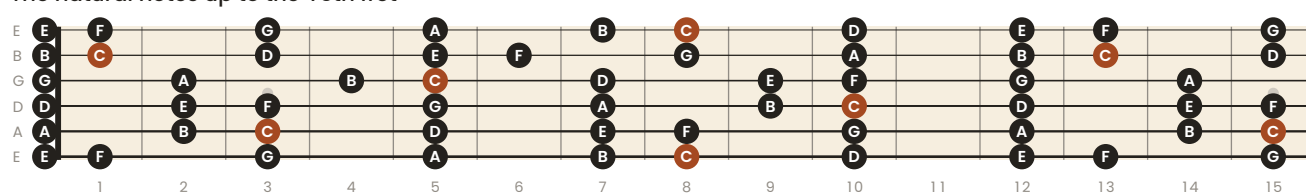
Intervals on the fretboard

Interval	Half steps	Same string	Next string up	Two strings up
Minor 2nd	1	1 fret up	4 frets back	
Major 2nd	2	2 frets up	3 frets back	
Minor 3rd	3	3 frets up	2 frets back	
Major 3rd	4	4 frets up	1 fret back	
Perfect 4th	5	5 frets up	Same fret	
Tritone	6	6 frets up	1 fret up	4 frets back
Perfect 5th	7	7 frets up	2 frets up	3 frets back
Minor 6th	8	8 frets up	3 frets up	2 frets back
Major 6th	9	9 frets up	4 frets up	1 fret back
Minor 7th	10	10 frets up		Same fret
Major 7th	11	11 frets up		1 fret up
Octave	12	12 frets up		2 frets up

"Up" means towards the higher-pitched string. Whenever the move crosses from the 3rd string (G) to the 2nd string (B), add one fret, because that pair of strings is tuned a major 3rd apart instead of a perfect 4th. Chapter 3 explains this B-string shift.

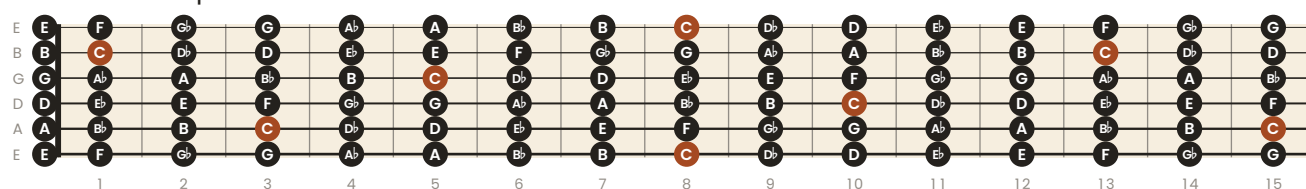
Every note on the neck

The natural notes up to the 15th fret



C is shown in rust. Natural notes are a whole step apart except B–C and E–F, which are a half step apart.

All twelve notes up to the 15th fret



Altered notes are named as flats; each one is also the sharp of the natural note below it (D $\flat$ = C $\sharp$, G $\flat$ = F $\sharp$). The pattern repeats from the 12th fret.

Diatonic chords in every major key

The pattern is the same in every key: I, IV and V are major; ii, iii and vi are minor; vii $^\circ$ is diminished. As seventh chords: Imaj7, ii7, iii7, IVmaj7, V7, vi7, vii $^\circ$ 7. Chapter 13 explains how the pattern arises.

Key	I	ii	iii	IV	V	vi	vii $^\circ$
C	C	Dm	Em	F	G	Am	Bdim
G	G	Am	Bm	C	D	Em	F $\sharp$ dim
D	D	Em	F $\sharp$ m	G	A	Bm	C $\sharp$ dim
A	A	Bm	C $\sharp$ m	D	E	F $\sharp$ m	G $\sharp$ dim
E	E	F $\sharp$ m	G $\sharp$ m	A	B	C $\sharp$ m	D $\sharp$ dim
B	B	C $\sharp$ m	D $\sharp$ m	E	F $\sharp$	G $\sharp$ m	A $\sharp$ dim
F $\sharp$	F $\sharp$	G $\sharp$ m	A $\sharp$ m	B	C $\sharp$	D $\sharp$ m	E $\sharp$ dim
F	F	Gm	Am	B $\flat$	C	Dm	Edim
B $\flat$	B $\flat$	Cm	Dm	E $\flat$	F	Gm	Adim
E $\flat$	E $\flat$	Fm	Gm	A $\flat$	B $\flat$	Cm	Ddim
A $\flat$	A $\flat$	B $\flat$ m	Cm	D $\flat$	E $\flat$	Fm	Gdim
D $\flat$	D $\flat$	E $\flat$ m	Fm	G $\flat$	A $\flat$	B $\flat$ m	Cdim

Diatonic chords in every minor key

The chords of the natural minor scale, plus the major V chord borrowed from the harmonic minor, which most minor-key music uses for its cadences (Chapter 31). Seventh chords: i7, ii $^\circ$ 7, $\flat$ IIIImaj7, iv7, v7 (or V7), $\flat$ VImaj7, $\flat$ VII7.

Key	i	ii $^\circ$	$\flat$ III	iv	v	$\flat$ VI	$\flat$ VII	V (harmonic)
Am	Am	Bdim	C	Dm	Em	F	G	E
Em	Em	F $\sharp$ dim	G	Am	Bm	C	D	B
Bm	Bm	C $\sharp$ dim	D	Em	F $\sharp$ m	G	A	F $\sharp$
F $\sharp$ m	F $\sharp$ m	G $\sharp$ dim	A	Bm	C $\sharp$ m	D	E	C $\sharp$
C $\sharp$ m	C $\sharp$ m	D $\sharp$ dim	E	F $\sharp$ m	G $\sharp$ m	A	B	G $\sharp$
G $\sharp$ m	G $\sharp$ m	A $\sharp$ dim	B	C $\sharp$ m	D $\sharp$ m	E	F $\sharp$	D $\sharp$
D $\sharp$ m	D $\sharp$ m	E $\sharp$ dim	F $\sharp$	G $\sharp$ m	A $\sharp$ m	B	C $\sharp$	A $\sharp$

Key	i	ii°	♭III	iv	v	♭VI	♭VII	V (harmonic)
Dm	Dm	Edim	F	Gm	Am	B♭	C	A
Gm	Gm	Adim	B♭	Cm	Dm	E♭	F	D
Cm	Cm	Ddim	E♭	Fm	Gm	A♭	B♭	G
Fm	Fm	Gdim	A♭	B♭m	Cm	D♭	E♭	C
B♭m	B♭m	Cdim	D♭	E♭m	Fm	G♭	A♭	F

Each minor key shares its notes with the major key in the same row of the previous table (Am with C, Em with G and so on): the relative major.

Capo chart

Find the chord shape you play along the top and the capo fret down the side; the cell gives the chord that sounds. To play in a given key with familiar shapes, find the key in the table and read off the shape and the fret.

Capo	C shape	A shape	G shape	E shape	D shape	Am shape	Em shape	Dm shape
1	C♯ / D♭	B♭	A♭	F	E♭	B♭m	Fm	E♭m
2	D	B	A	F♯	E	Bm	F♯m	Em
3	E♭	C	B♭	G	F	Cm	Gm	Fm
4	E	C♯ / D♭	B	A♭	F♯	C♯m	G♯m	F♯m
5	F	D	C	A	G	Dm	Am	Gm
6	F♯ / G♭	E♭	C♯ / D♭	B♭	A♭	E♭m	B♭m	G♯m
7	G	E	D	B	A	Em	Bm	Am

Example: a song in B♭ can be played with G shapes at capo 3, or with A shapes at capo 1. A capo does not change the shapes or the Roman numerals, only the sounding key (Chapter 13).

Common tunings

Tuning	Strings 6 to 1	Open-string chord	Chapter
Standard	E A D G B E	(Em11)	Chapter 1
Drop D	D A D G B E	(D5 on the bottom three strings)	Chapter 25
Double drop D	D A D G B D	(Gadd9/D)	Chapter 25
DADGAD	D A D G A D	Dsus4	Chapter 25, Chapter 54
Open D	D A D F♯ A D	D major	Chapter 25
Open G	D G D G B D	G major	Chapter 25
Open C	C G C G C E	C major	Chapter 25
Open E	E B E G♯ B E	E major	Chapter 25
Half step down	E♭ A♭ D♭ G♭ B♭ E♭	(standard, a half step lower)	Chapter 44

APPENDIX C

Chord Encyclopedia

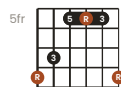
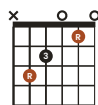


Every chord type in all twelve keys, in up to four voicings across the neck. The shapes were found by a search over the whole fretboard and checked note by note: each one contains exactly the chord's notes with the root in the bass, fits within four frets and needs no more than four fingers (a barre counts as one). Standard open and barre shapes come first where they exist. Dots show the degree of each note; the root is rust.

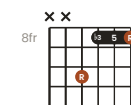
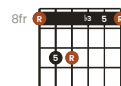
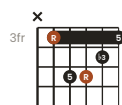
Where a chord has fewer than four boxes, the search found no other comfortable shape. Formulas and the theory behind each type are in Chapter 11 and Appendix B.

C

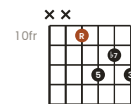
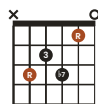
C
MAJOR
1 3 5
C E G



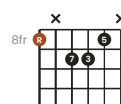
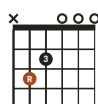
Cm
MINOR
1 b3 5
C E $\flat$ G



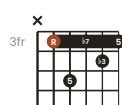
C7
DOMINANT 7TH
1 3 5 b7
C E G B $\flat$



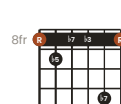
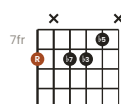
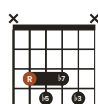
Cmaj7
MAJOR 7TH
1 3 5 7
C E G B



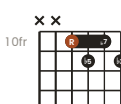
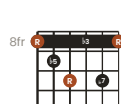
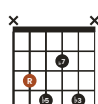
Cm7
MINOR 7TH
1 b3 5 b7
C E $\flat$ G B $\flat$



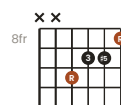
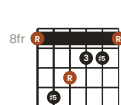
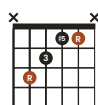
Cm7 $\flat$ 5
HALF-DIMINISHED
1 b3 b5 b7
C E $\flat$ G $\flat$ B $\flat$



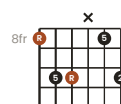
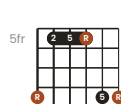
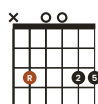
Cdim7
DIMINISHED 7TH
1 b3 b5 $\flat$ 7
C E $\flat$ G $\flat$ B $\flat$



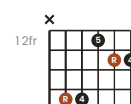
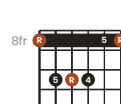
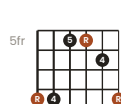
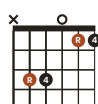
Caug
AUGMENTED
1 3 #5
C E G $\sharp$



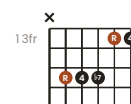
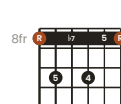
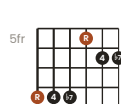
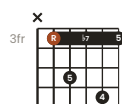
Csus2
SUSPENDED 2ND
1 2 5
C D G



Csus4
SUSPENDED 4TH
1 4 5
C F G

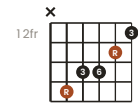
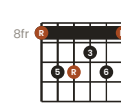
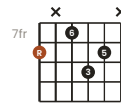
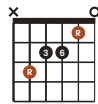


C7sus4
7TH SUSPENDED 4TH
1 4 5 b7
C F G B $\flat$

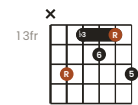
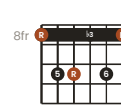
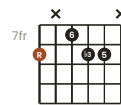
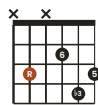


C6

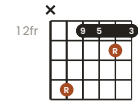
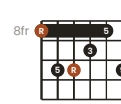
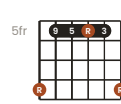
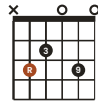
MAJOR 6TH
1 3 5 6
C E G A

**Cm6**

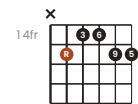
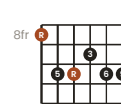
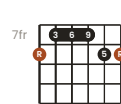
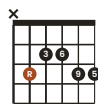
MINOR 6TH
1 b3 5 6
C E♭ G A

**Cadd9**

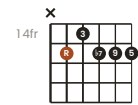
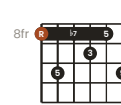
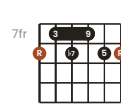
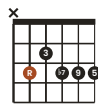
ADDED 9TH
1 3 5 9
C E G D

**C6/9**

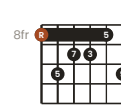
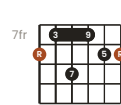
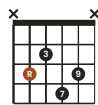
SIX-NINE
1 3 5 6 9
C E G A D

**C9**

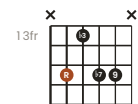
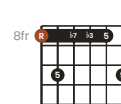
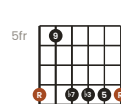
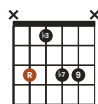
DOMINANT 9TH
1 3 5 b7 9
C E G B♭ D

**Cmaj9**

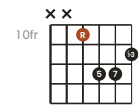
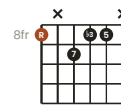
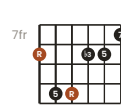
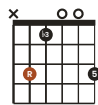
MAJOR 9TH
1 3 5 7 9
C E G B D

**Cm9**

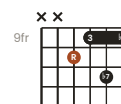
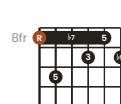
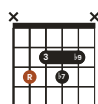
MINOR 9TH
1 b3 5 b7 9
C E♭ G B♭ D

**CmMaj7**

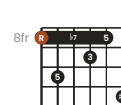
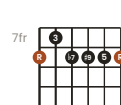
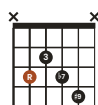
MINOR-MAJOR 7TH
1 b3 5 7
C E♭ G B

**C7b9**

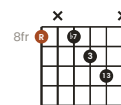
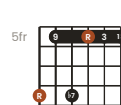
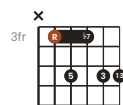
7TH FLAT 9
1 3 5 b7 b9
C E G B♭ D♭

**C7#9**

7TH SHARP 9
1 3 5 b7 #9
C E G B♭ D#

**C13**

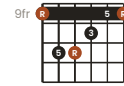
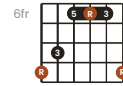
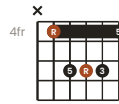
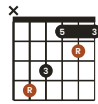
DOMINANT 13TH
1 3 5 b7 9 13
C E G B♭ D A



D^b / C[#]

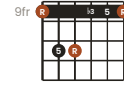
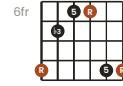
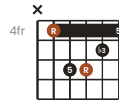
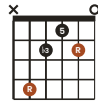
D^b

MAJOR
1 3 5
D^b F A^b



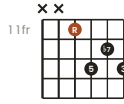
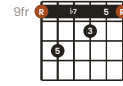
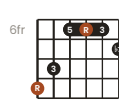
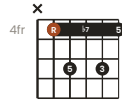
C[#]m

MINOR
1 b3 5
C[#] E G[#]



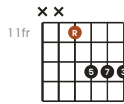
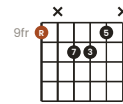
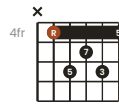
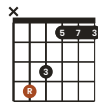
C[#]7

DOMINANT 7TH
1 3 5 b7
C[#] E[#] G[#] B



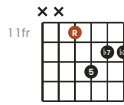
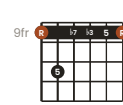
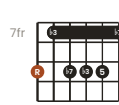
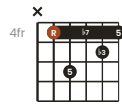
D^bmaj7

MAJOR 7TH
1 3 5 7
D^b F A^b C



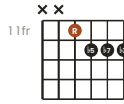
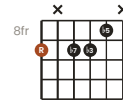
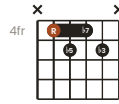
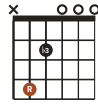
C[#]m7

MINOR 7TH
1 b3 5 b7
C[#] E G[#] B



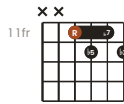
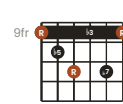
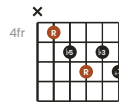
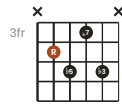
C[#]m7b5

HALF-DIMINISHED
1 b3 b5 b7
C[#] E G B



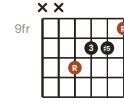
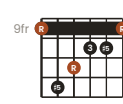
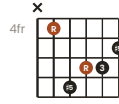
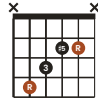
C[#]dim7

DIMINISHED 7TH
1 b3 b5 $\sharp$ 7
C[#] E G B^b



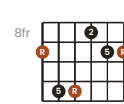
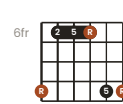
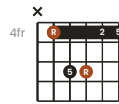
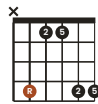
D^baug

AUGMENTED
1 3 $\sharp$ 5
D^b F A



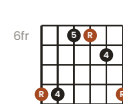
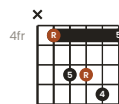
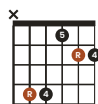
D^bsus2

SUSPENDED 2ND
1 2 5
D^b E^b A^b



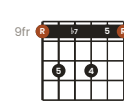
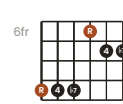
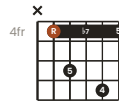
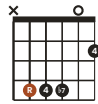
D^bsus4

SUSPENDED 4TH
1 4 5
D^b G^b A^b



C[#]7sus4

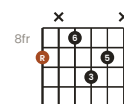
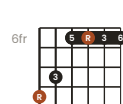
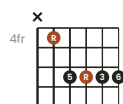
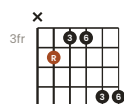
7TH SUSPENDED 4TH
1 4 5 b7
C[#] F[#] G[#] B



D $\flat$ 6

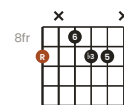
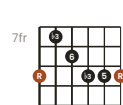
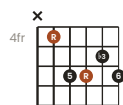
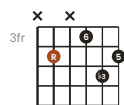
MAJOR 6TH

1 3 5 6

D $\flat$ F A $\flat$ B $\flat$ **C $\sharp$ m6**

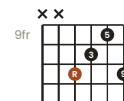
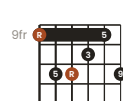
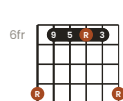
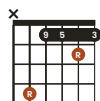
MINOR 6TH

1 b3 5 6

C $\sharp$ E G $\sharp$ A $\sharp$ **D $\flat$ add9**

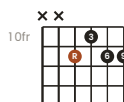
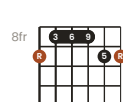
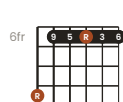
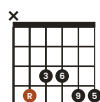
ADDED 9TH

1 3 5 9

D $\flat$ F A $\flat$ E $\flat$ **D $\flat$ 6/9**

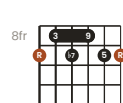
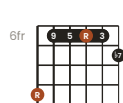
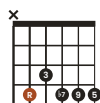
SIX-NINE

1 3 5 6 9

D $\flat$ F A $\flat$ B $\flat$ E $\flat$ **C $\sharp$ 9**

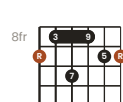
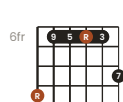
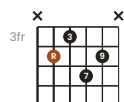
DOMINANT 9TH

1 3 5 b7 9

C $\sharp$ E $\sharp$ G $\sharp$ B D $\sharp$ **D $\flat$ maj9**

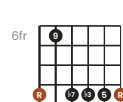
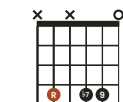
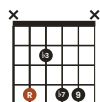
MAJOR 9TH

1 3 5 7 9

D $\flat$ F A $\flat$ C E $\flat$ **C $\sharp$ m9**

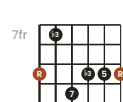
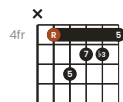
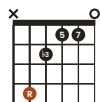
MINOR 9TH

1 b3 5 b7 9

C $\sharp$ E G $\sharp$ B D $\sharp$ **C $\sharp$ mMaj7**

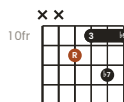
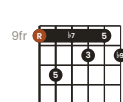
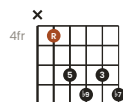
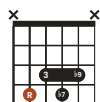
MINOR-MAJOR 7TH

1 b3 5 7

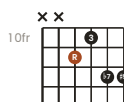
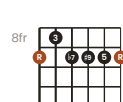
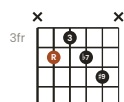
C $\sharp$ E G $\sharp$ B $\sharp$ **C $\sharp$ 7b9**

7TH FLAT 9

1 3 5 b7 b9

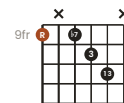
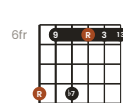
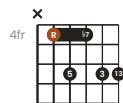
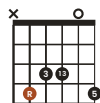
C $\sharp$ E $\sharp$ G $\sharp$ B D**D $\flat$ 7 $\sharp$ 9**

7TH SHARP 9

1 3 5 b7 $\sharp$ 9D $\flat$ F A $\flat$ C $\flat$ E**C $\sharp$ 13**

DOMINANT 13TH

1 3 5 b7 9 13

C $\sharp$ E $\sharp$ G $\sharp$ B D $\sharp$ A $\sharp$ 

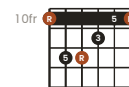
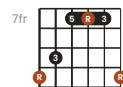
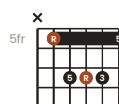
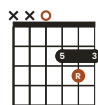
D

D

MAJOR

1 3 5

D F# A

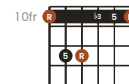
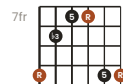
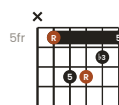
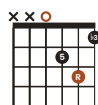


Dm

MINOR

1 b3 5

D F A

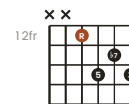
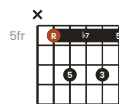
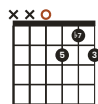


D7

DOMINANT 7TH

1 3 5 b7

D F# A C

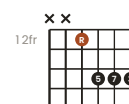
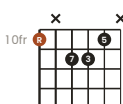
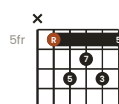
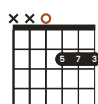


Dmaj7

MAJOR 7TH

1 3 5 7

D F# A C#

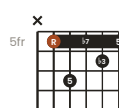
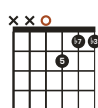


Dm7

MINOR 7TH

1 b3 5 b7

D F A C

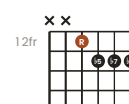
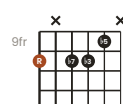
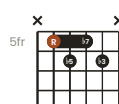
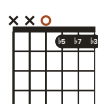


Dm7b5

HALF-DIMINISHED

1 b3 b5 b7

D F A b C

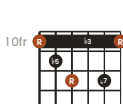
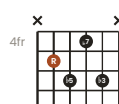
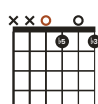


Ddim7

DIMINISHED 7TH

1 b3 b5 b7

D F A b C b

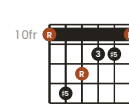
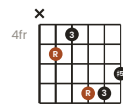
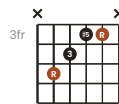
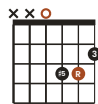


Daug

AUGMENTED

1 3 #5

D F# A#

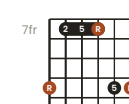
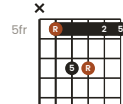
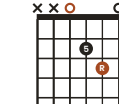
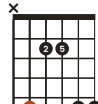


Dsus2

SUSPENDED 2ND

1 2 5

D E A

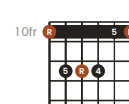
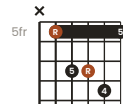
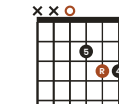
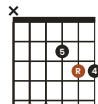


Dsus4

SUSPENDED 4TH

1 4 5

D G A

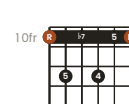
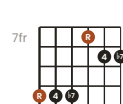
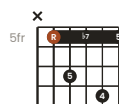
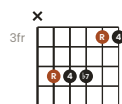


D7sus4

7TH SUSPENDED 4TH

1 4 5 b7

D G A C

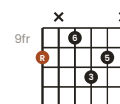
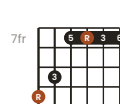
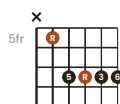
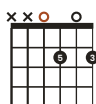


D6

MAJOR 6TH

1 3 5 6

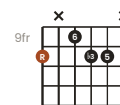
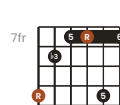
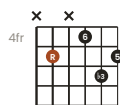
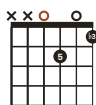
D F# A B

**Dm6**

MINOR 6TH

1 b3 5 6

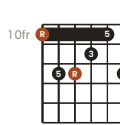
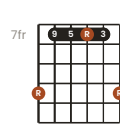
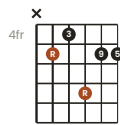
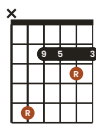
D F A B

**Dadd9**

ADDED 9TH

1 3 5 9

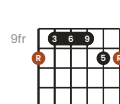
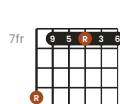
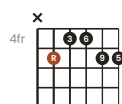
D F# A E

**D6/9**

SIX-NINE

1 3 5 6 9

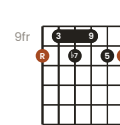
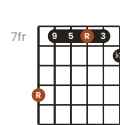
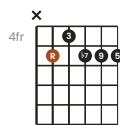
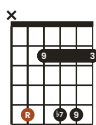
D F# A B E

**D9**

DOMINANT 9TH

1 3 5 b7 9

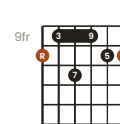
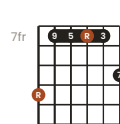
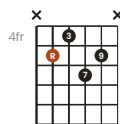
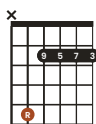
D F# A C E

**Dmaj9**

MAJOR 9TH

1 3 5 7 9

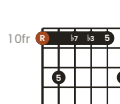
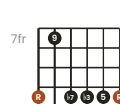
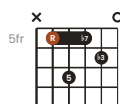
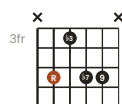
D F# A C# E

**Dm9**

MINOR 9TH

1 b3 5 b7 9

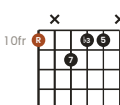
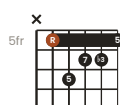
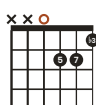
D F A C E

**DmMaj7**

MINOR-MAJOR 7TH

1 b3 5 7

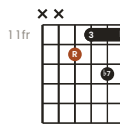
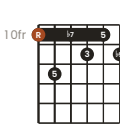
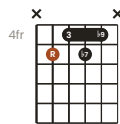
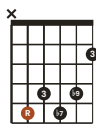
D F A C#

**D7b9**

7TH FLAT 9

1 3 5 b7 b9

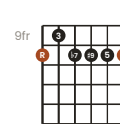
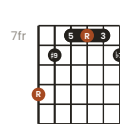
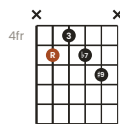
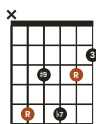
D F# A C E b

**D7#9**

7TH SHARP 9

1 3 5 b7 #9

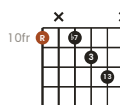
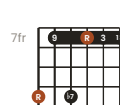
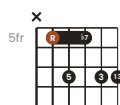
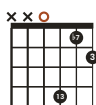
D F# A C E#

**D13**

DOMINANT 13TH

1 3 5 b7 9 13

D F# A C E B

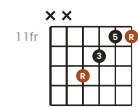
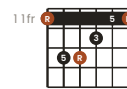
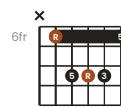
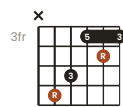


E $\flat$ / D $\sharp$

E $\flat$

MAJOR

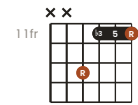
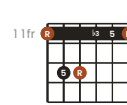
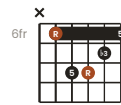
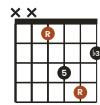
1 3 5

E $\flat$ G B $\flat$ 

E $\flat$ m

MINOR

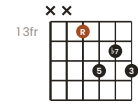
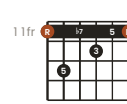
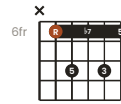
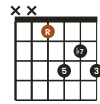
1 b3 5

E $\flat$ G $\flat$ B $\flat$ 

E $\flat$ 7

DOMINANT 7TH

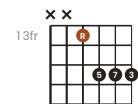
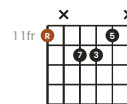
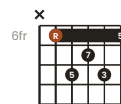
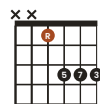
1 3 5 b7

E $\flat$ G B $\flat$ D $\flat$ 

E $\flat$ maj7

MAJOR 7TH

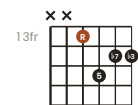
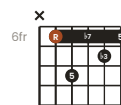
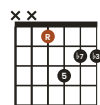
1 3 5 7

E $\flat$ G B $\flat$ D

E $\flat$ m7

MINOR 7TH

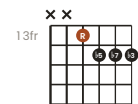
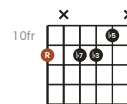
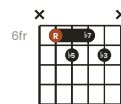
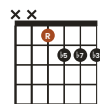
1 b3 5 b7

E $\flat$ G $\flat$ B $\flat$ D $\flat$ 

D $\sharp$ m7b5

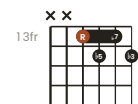
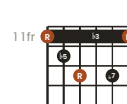
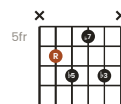
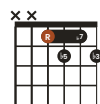
HALF-DIMINISHED

1 b3 b5 b7

D $\sharp$ F $\sharp$ A C $\sharp$ 

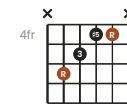
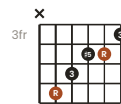
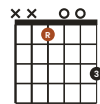
D $\sharp$ dim7

DIMINISHED 7TH

1 b3 b5 $\natural$ 7D $\sharp$ F $\sharp$ A C

E $\flat$ aug

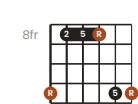
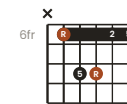
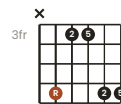
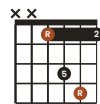
AUGMENTED

1 3 $\sharp$ 5E $\flat$ G B

E $\flat$ sus2

SUSPENDED 2ND

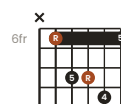
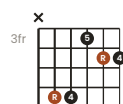
1 2 5

E $\flat$ F B $\flat$ 

E $\flat$ sus4

SUSPENDED 4TH

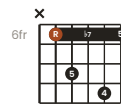
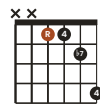
1 4 5

E $\flat$ A $\flat$ B $\flat$ 

E $\flat$ 7sus4

7TH SUSPENDED 4TH

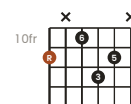
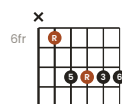
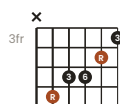
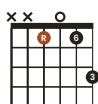
1 4 5 b7

E $\flat$ A $\flat$ B $\flat$ D $\flat$ 

E $\flat$ 6

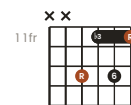
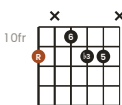
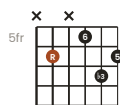
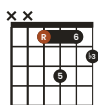
MAJOR 6TH

1 3 5 6

E $\flat$ G B $\flat$ C**E $\flat$ m6**

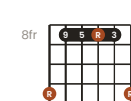
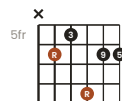
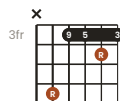
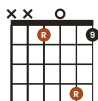
MINOR 6TH

1 b3 5 6

E $\flat$ G $\flat$ B $\flat$ C**E $\flat$ add9**

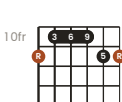
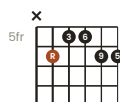
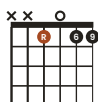
ADDED 9TH

1 3 5 9

E $\flat$ G B $\flat$ F**E $\flat$ 6/9**

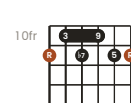
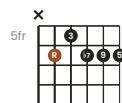
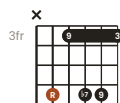
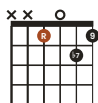
SIX-NINE

1 3 5 6 9

E $\flat$ G B $\flat$ C F**E $\flat$ 9**

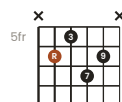
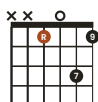
DOMINANT 9TH

1 3 5 b7 9

E $\flat$ G B $\flat$ D $\flat$ F**E $\flat$ maj9**

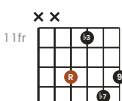
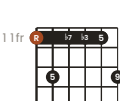
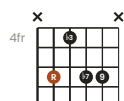
MAJOR 9TH

1 3 5 7 9

E $\flat$ G B $\flat$ D F**E $\flat$ m9**

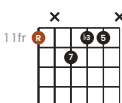
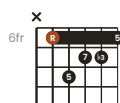
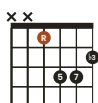
MINOR 9TH

1 b3 5 b7 9

E $\flat$ G $\flat$ B $\flat$ D $\flat$ F**E $\flat$ mMaj7**

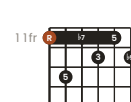
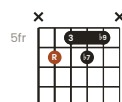
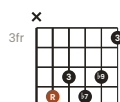
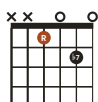
MINOR-MAJOR 7TH

1 b3 5 7

E $\flat$ G $\flat$ B $\flat$ D**E $\flat$ 7b9**

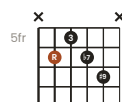
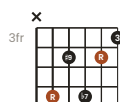
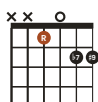
7TH FLAT 9

1 3 5 b7 b9

E $\flat$ G B $\flat$ D $\flat$ F $\flat$ **E $\flat$ 7#9**

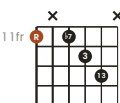
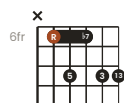
7TH SHARP 9

1 3 5 b7 #9

E $\flat$ G B $\flat$ D $\flat$ F#**E $\flat$ 13**

DOMINANT 13TH

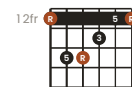
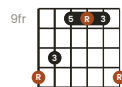
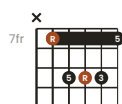
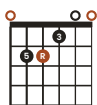
1 3 5 b7 9 13

E $\flat$ G B $\flat$ D $\flat$ F C

E

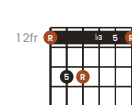
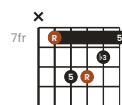
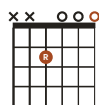
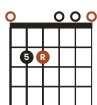
E

MAJOR
1 3 5
E G# B



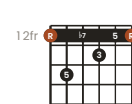
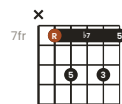
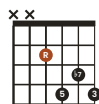
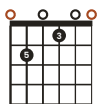
Em

MINOR
1 b3 5
E G B



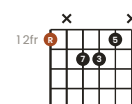
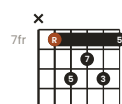
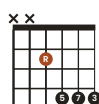
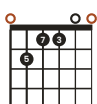
E7

DOMINANT 7TH
1 3 5 b7
E G# B D



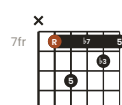
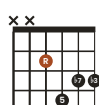
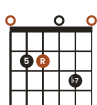
Emaj7

MAJOR 7TH
1 3 5 7
E G# B D#



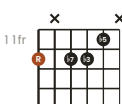
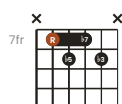
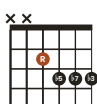
Em7

MINOR 7TH
1 b3 5 b7
E G B D



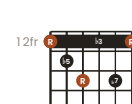
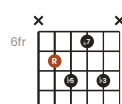
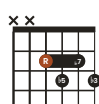
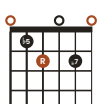
Em7b5

HALF-DIMINISHED
1 b3 b5 b7
E G Bb D



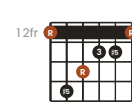
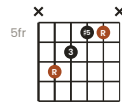
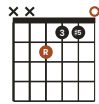
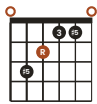
Edim7

DIMINISHED 7TH
1 b3 b5 #7
E G Bb Db



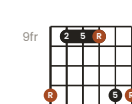
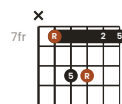
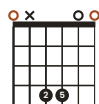
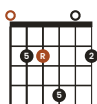
Eaug

AUGMENTED
1 3 #5
E G# B#



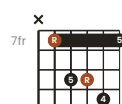
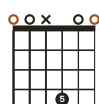
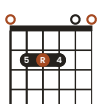
Esus2

SUSPENDED 2ND
1 2 5
E F# B



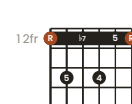
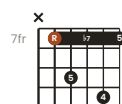
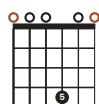
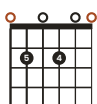
Esus4

SUSPENDED 4TH
1 4 5
E A B



E7sus4

7TH SUSPENDED 4TH
1 4 5 b7
E A B D

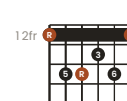
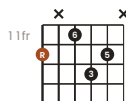
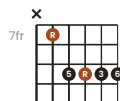
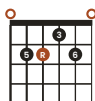


E6

MAJOR 6TH

1 3 5 6

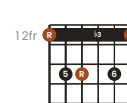
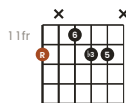
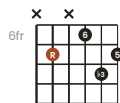
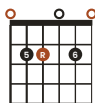
E G# B C#

**Em6**

MINOR 6TH

1 b3 5 6

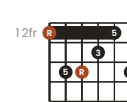
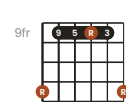
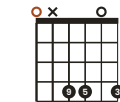
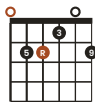
E G B C#

**Eadd9**

ADDED 9TH

1 3 5 9

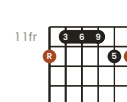
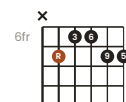
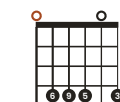
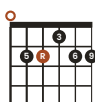
E G# B F#

**E6/9**

SIX-NINE

1 3 5 6 9

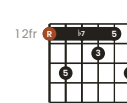
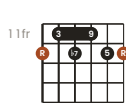
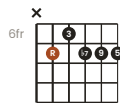
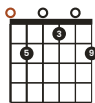
E G# B C# F#

**E9**

DOMINANT 9TH

1 3 5 b7 9

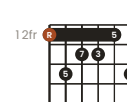
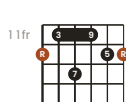
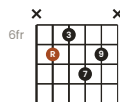
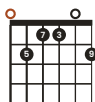
E G# B D F#

**Emaj9**

MAJOR 9TH

1 3 5 7 9

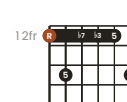
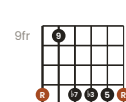
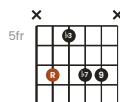
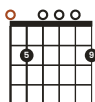
E G# B D# F#

**Em9**

MINOR 9TH

1 b3 5 b7 9

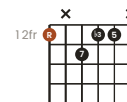
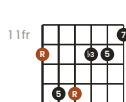
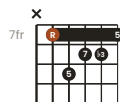
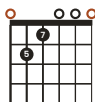
E G B D F#

**EmMaj7**

MINOR-MAJOR 7TH

1 b3 5 7

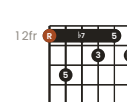
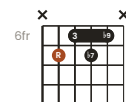
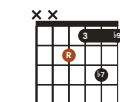
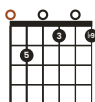
E G B D#

**E7b9**

7TH FLAT 9

1 3 5 b7 b9

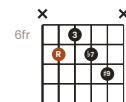
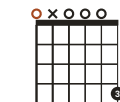
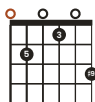
E G# B D F

**E7#9**

7TH SHARP 9

1 3 5 b7 #9

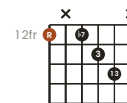
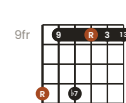
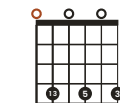
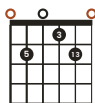
E G# B D F#

**E13**

DOMINANT 13TH

1 3 5 b7 9 13

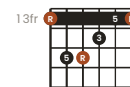
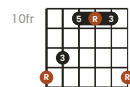
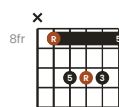
E G# B D F# C#



F

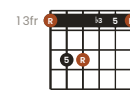
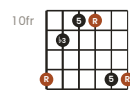
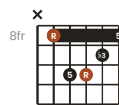
F

MAJOR
1 3 5
F A C



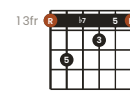
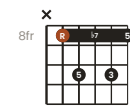
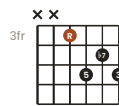
Fm

MINOR
1 b3 5
F A $\flat$ C



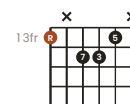
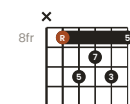
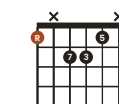
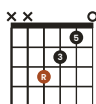
F7

DOMINANT 7TH
1 3 5 b7
F A C E $\flat$



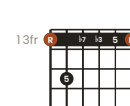
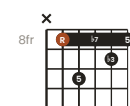
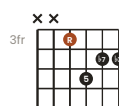
Fmaj7

MAJOR 7TH
1 3 5 7
F A C E

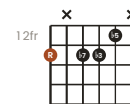
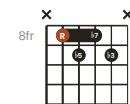
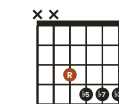
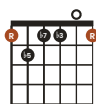


Fm7

MINOR 7TH
1 b3 5 b7
F A $\flat$ C E $\flat$

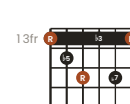
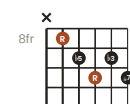
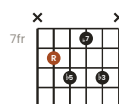
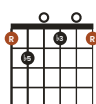
Fm7 $\flat$ 5

HALF-DIMINISHED
1 b3 b5 b7
F A $\flat$ C $\flat$ E $\flat$



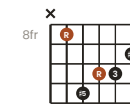
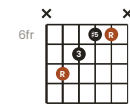
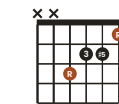
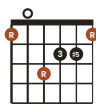
Fdim7

DIMINISHED 7TH
1 b3 b5 $\flat$ 7
F A $\flat$ C $\flat$ E $\flat$



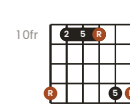
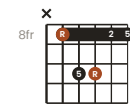
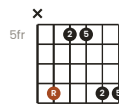
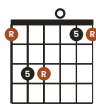
Faug

AUGMENTED
1 3 #5
F A C $\sharp$



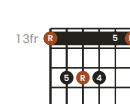
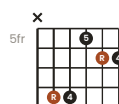
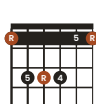
Fsus2

SUSPENDED 2ND
1 2 5
F G C



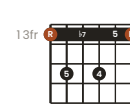
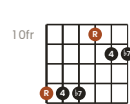
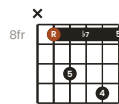
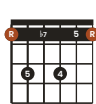
Fsus4

SUSPENDED 4TH
1 4 5
F E $\flat$ C



F7sus4

7TH SUSPENDED 4TH
1 4 5 b7
F E $\flat$ C E $\flat$

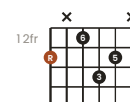
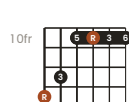
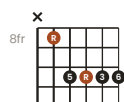
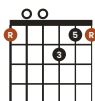


F6

MAJOR 6TH

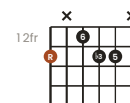
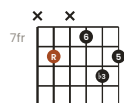
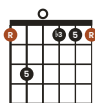
1 3 5 6

F A C D

**Fm6**

MINOR 6TH

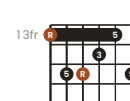
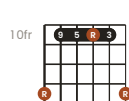
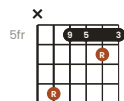
1 b3 5 6

F A $\flat$ C D**Fadd9**

ADDED 9TH

1 3 5 9

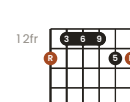
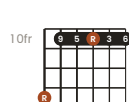
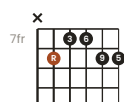
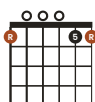
F A C G

**F6/9**

SIX-NINE

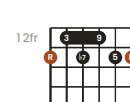
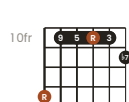
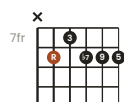
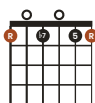
1 3 5 6 9

F A C D G

**F9**

DOMINANT 9TH

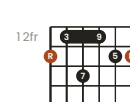
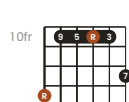
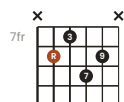
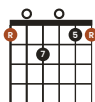
1 3 5 b7 9

F A C E $\flat$ G**Fmaj9**

MAJOR 9TH

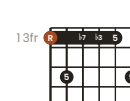
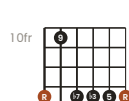
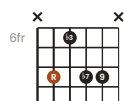
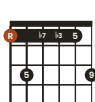
1 3 5 7 9

F A C E G

**Fm9**

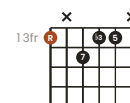
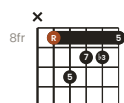
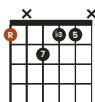
MINOR 9TH

1 b3 5 b7 9

F A $\flat$ C E $\flat$ G**FmMaj7**

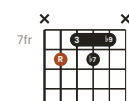
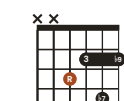
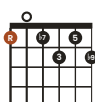
MINOR-MAJOR 7TH

1 b3 5 7

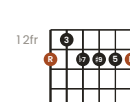
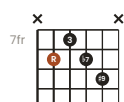
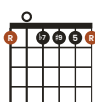
F A $\flat$ C E**F7 $\flat$ 9**

7TH FLAT 9

1 3 5 b7 b9

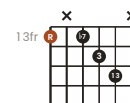
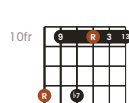
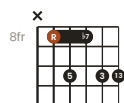
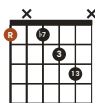
F A C E $\flat$ G $\flat$ **F7 $\sharp$ 9**

7TH SHARP 9

1 3 5 b7 $\sharp$ 9F A C E $\flat$ G $\sharp$ **F13**

DOMINANT 13TH

1 3 5 b7 9 13

F A C E $\flat$ G D

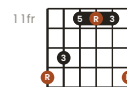
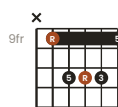
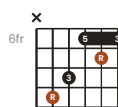
F# / Gb

F#

MAJOR

1 3 5

F# A# C#

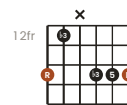
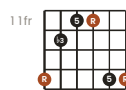
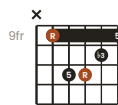


F#m

MINOR

1 b3 5

F# A C#

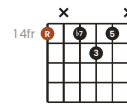
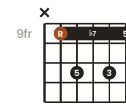
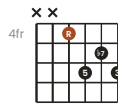


F#7

DOMINANT 7TH

1 3 5 b7

F# A# C# E

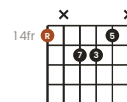
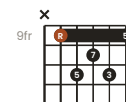
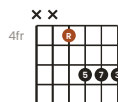
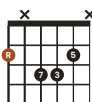


F#maj7

MAJOR 7TH

1 3 5 7

F# A# C# E#

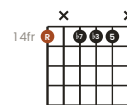
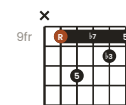
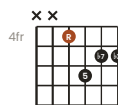


F#m7

MINOR 7TH

1 b3 5 b7

F# A C# E

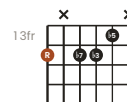
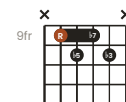
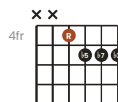
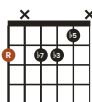


F#m7b5

HALF-DIMINISHED

1 b3 b5 b7

F# A C E

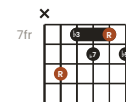
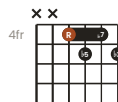
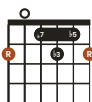


F#dim7

DIMINISHED 7TH

1 b3 b5 #7

F# A C Eb

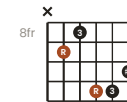
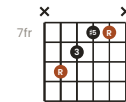
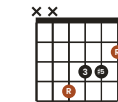
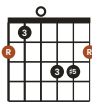


Gbaug

AUGMENTED

1 3 #5

Gb Bb D

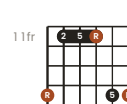
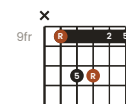
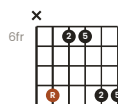


F#sus2

SUSPENDED 2ND

1 2 5

F# G# C#

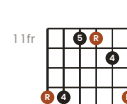


F#sus4

SUSPENDED 4TH

1 4 5

F# B C#

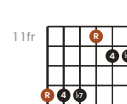
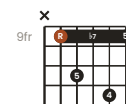
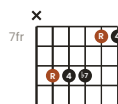


F#7sus4

7TH SUSPENDED 4TH

1 4 5 b7

F# B C# E

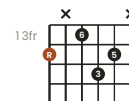
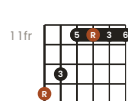
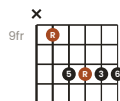
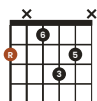


F#6

MAJOR 6TH

1 3 5 6

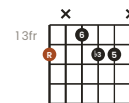
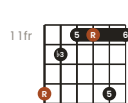
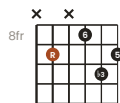
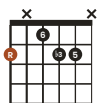
F# A# C# D#

**F#m6**

MINOR 6TH

1 b3 5 6

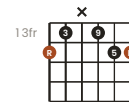
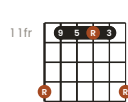
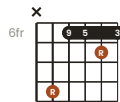
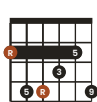
F# A C# D#

**F#add9**

ADDED 9TH

1 3 5 9

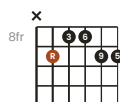
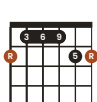
F# A# C# G#

**F#6/9**

SIX-NINE

1 3 5 6 9

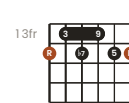
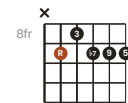
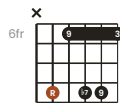
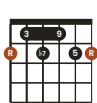
F# A# C# D# G#

**F#9**

DOMINANT 9TH

1 3 5 b7 9

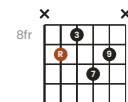
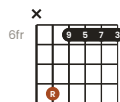
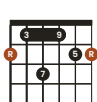
F# A# C# E G#

**F#maj9**

MAJOR 9TH

1 3 5 7 9

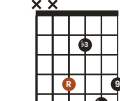
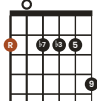
F# A# C# E# G#

**F#m9**

MINOR 9TH

1 b3 5 b7 9

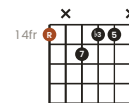
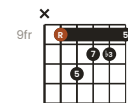
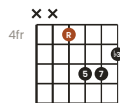
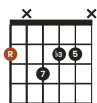
F# A C# E G#

**F#mMaj7**

MINOR-MAJOR 7TH

1 b3 5 7

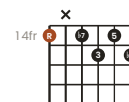
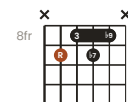
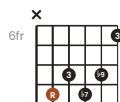
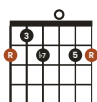
F# A C# E#

**F#7b9**

7TH FLAT 9

1 3 5 b7 b9

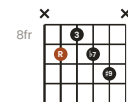
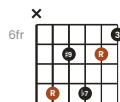
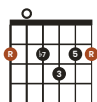
F# A# C# E G

**F#7#9**

7TH SHARP 9

1 3 5 b7 #9

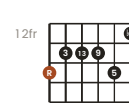
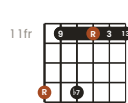
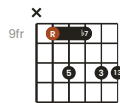
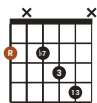
F# A# C# E G#

**F#13**

DOMINANT 13TH

1 3 5 b7 9 13

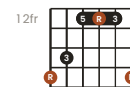
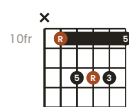
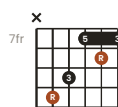
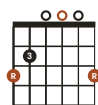
F# A# C# E G# D#



G

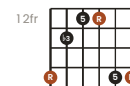
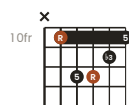
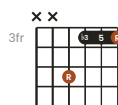
G

MAJOR
1 3 5
G B D



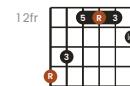
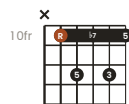
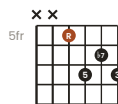
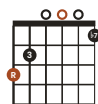
Gm

MINOR
1 b3 5
G B♭ D



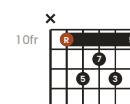
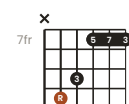
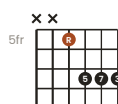
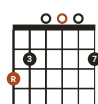
G7

DOMINANT 7TH
1 3 5 b7
G B D F



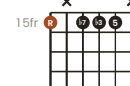
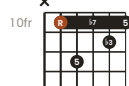
Gmaj7

MAJOR 7TH
1 3 5 7
G B D F♯



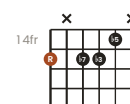
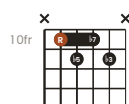
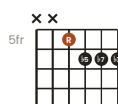
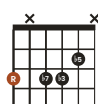
Gm7

MINOR 7TH
1 b3 5 b7
G B♭ D F



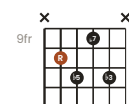
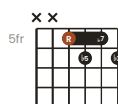
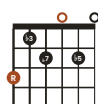
Gm7♭5

HALF-DIMINISHED
1 b3 b5 b7
G B♭ D♭ F



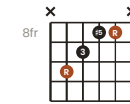
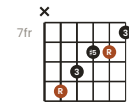
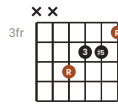
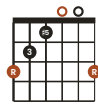
Gdim7

DIMINISHED 7TH
1 b3 b5 ♯7
G B♭ D♭ F♭



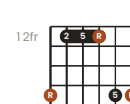
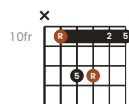
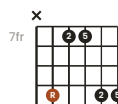
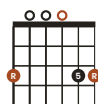
Gaug

AUGMENTED
1 3 #5
G B D♯



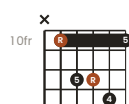
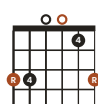
Gsus2

SUSPENDED 2ND
1 2 5
G A D



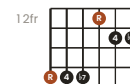
Gsus4

SUSPENDED 4TH
1 4 5
G C D



G7sus4

7TH SUSPENDED 4TH
1 4 5 b7
G C D F

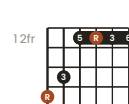
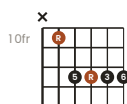
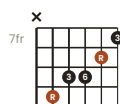
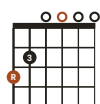


G6

MAJOR 6TH

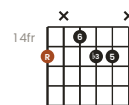
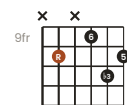
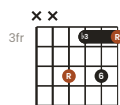
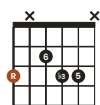
1 3 5 6

G B D E

**Gm6**

MINOR 6TH

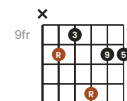
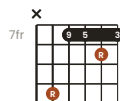
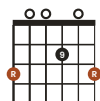
1 b3 5 6

G B $\flat$ D E**Gadd9**

ADDED 9TH

1 3 5 9

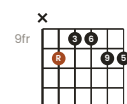
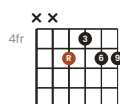
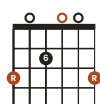
G B D A

**G6/9**

SIX-NINE

1 3 5 6 9

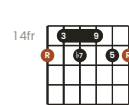
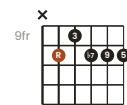
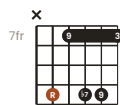
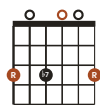
G B D E A

**G9**

DOMINANT 9TH

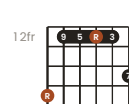
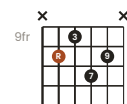
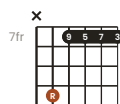
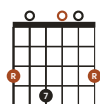
1 3 5 b7 9

G B D F A

**Gmaj9**

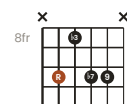
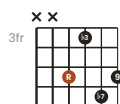
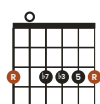
MAJOR 9TH

1 3 5 7 9

G B D F $\sharp$ A**Gm9**

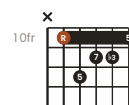
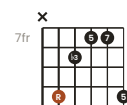
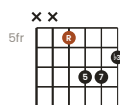
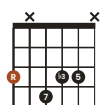
MINOR 9TH

1 b3 5 b7 9

G B $\flat$ D F A**GmMaj7**

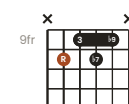
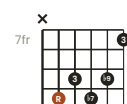
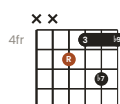
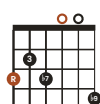
MINOR-MAJOR 7TH

1 b3 5 7

G B $\flat$ D F $\sharp$ **G7b9**

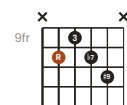
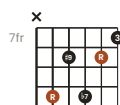
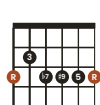
7TH FLAT 9

1 3 5 b7 b9

G B D F A $\flat$ **G7#9**

7TH SHARP 9

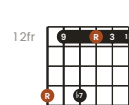
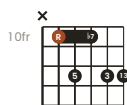
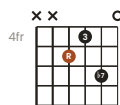
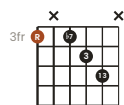
1 3 5 b7 #9

G B D F A $\sharp$ **G13**

DOMINANT 13TH

1 3 5 b7 9 13

G B D F A E



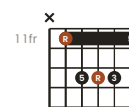
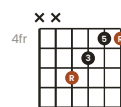
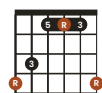
Ab / G#

Ab

MAJOR

1 3 5

Ab C Eb

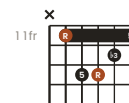
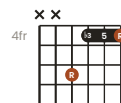
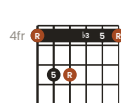
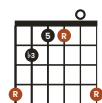


G#m

MINOR

1 b3 5

G# B D#

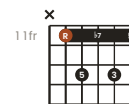
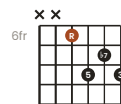
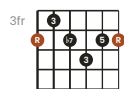


Ab7

DOMINANT 7TH

1 3 5 b7

Ab C Eb Gb

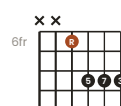
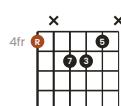


Abmaj7

MAJOR 7TH

1 3 5 7

Ab C Eb G

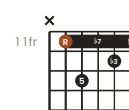
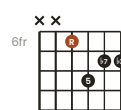
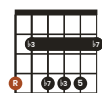


G#m7

MINOR 7TH

1 b3 5 b7

G# B D# F#

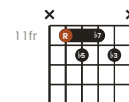
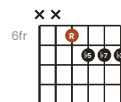
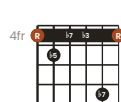
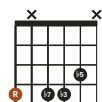


G#m7b5

HALF-DIMINISHED

1 b3 b5 b7

G# B D F#

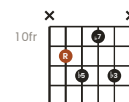
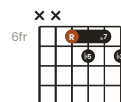
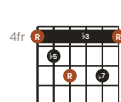
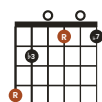


G#dim7

DIMINISHED 7TH

1 b3 b5 #7

G# B D F

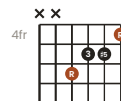
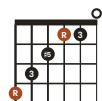


Abaug

AUGMENTED

1 3 #5

Ab C E

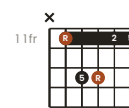
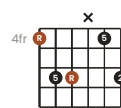
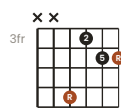
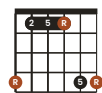


Ab7sus2

SUSPENDED 2ND

1 2 5

Ab Eb Eb

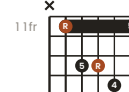
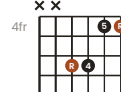
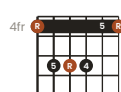
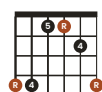


Ab7sus4

SUSPENDED 4TH

1 4 5

Ab Db Eb

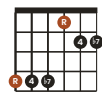


Ab7sus4

7TH SUSPENDED 4TH

1 4 5 b7

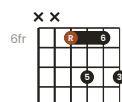
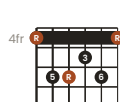
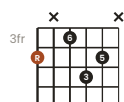
Ab Db Eb Gb



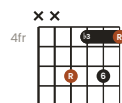
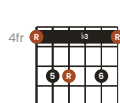
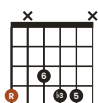
A $\flat$ 6

MAJOR 6TH

1 3 5 6

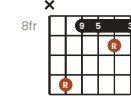
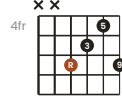
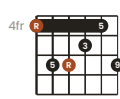
A $\flat$ C E $\flat$ F**G $\sharp$ m6**

MINOR 6TH

1 $\flat$ 3 5 6G $\sharp$ B D $\sharp$ E $\sharp$ **A $\flat$ add9**

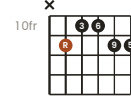
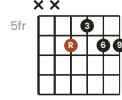
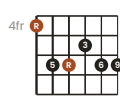
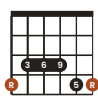
ADDED 9TH

1 3 5 9

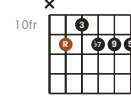
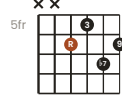
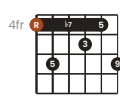
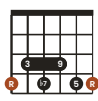
A $\flat$ C E $\flat$ B $\flat$ **A $\flat$ 6/9**

SIX-NINE

1 3 5 6 9

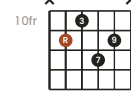
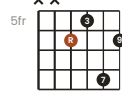
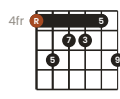
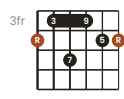
A $\flat$ C E $\flat$ F B $\flat$ **A $\flat$ 9**

DOMINANT 9TH

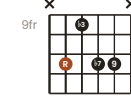
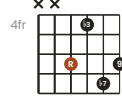
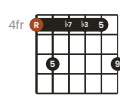
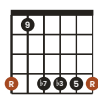
1 3 5 $\flat$ 7 9A $\flat$ C E $\flat$ G $\flat$ B $\flat$ **A $\flat$ maj9**

MAJOR 9TH

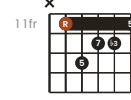
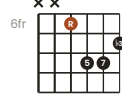
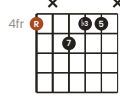
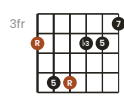
1 3 5 7 9

A $\flat$ C E $\flat$ G B $\flat$ **G $\sharp$ m9**

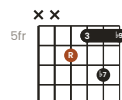
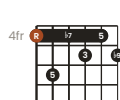
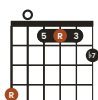
MINOR 9TH

1 $\flat$ 3 5 $\flat$ 7 9G $\sharp$ B D $\sharp$ F $\sharp$ A $\sharp$ **A $\flat$ mMaj7**

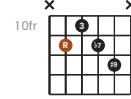
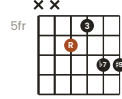
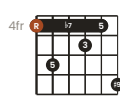
MINOR-MAJOR 7TH

1 $\flat$ 3 5 7A $\flat$ C $\flat$ E $\flat$ G**G $\sharp$ 7 $\flat$ 9**

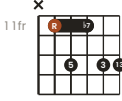
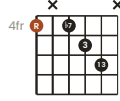
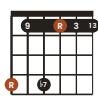
7TH FLAT 9

1 3 5 $\flat$ 7 $\flat$ 9G $\sharp$ B $\sharp$ D $\sharp$ F $\sharp$ A**A $\flat$ 7 $\sharp$ 9**

7TH SHARP 9

1 3 5 $\flat$ 7 $\sharp$ 9A $\flat$ C E $\flat$ G $\flat$ B**A $\flat$ 13**

DOMINANT 13TH

1 3 5 $\flat$ 7 9 13A $\flat$ C E $\flat$ G $\flat$ B $\flat$ F

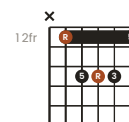
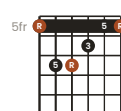
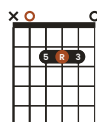
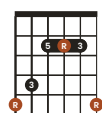
A

A

MAJOR

1 3 5

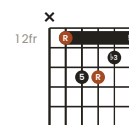
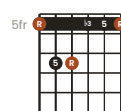
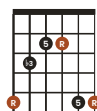
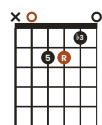
A C# E

**Am**

MINOR

1 b3 5

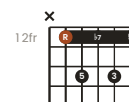
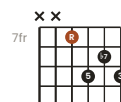
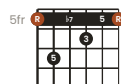
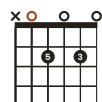
A C E

**A7**

DOMINANT 7TH

1 3 5 b7

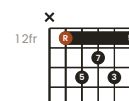
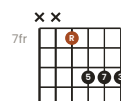
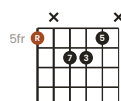
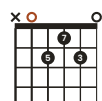
A C# E G

**Amaj7**

MAJOR 7TH

1 3 5 7

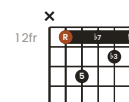
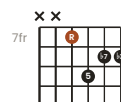
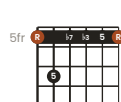
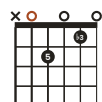
A C# E G#

**Am7**

MINOR 7TH

1 b3 5 b7

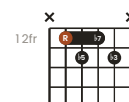
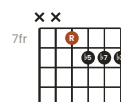
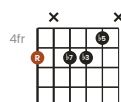
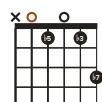
A C E G

**Am7b5**

HALF-DIMINISHED

1 b3 b5 b7

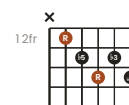
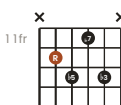
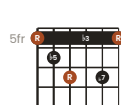
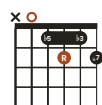
A C Eb G

**Adim7**

DIMINISHED 7TH

1 b3 b5 b7

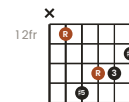
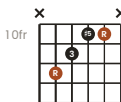
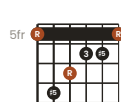
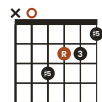
A C Eb Gb

**Aaug**

AUGMENTED

1 3 #5

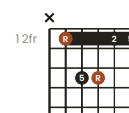
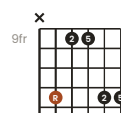
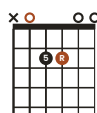
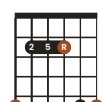
A C# E#

**Asus2**

SUSPENDED 2ND

1 2 5

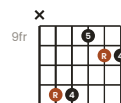
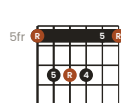
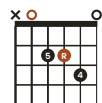
A B E

**Asus4**

SUSPENDED 4TH

1 4 5

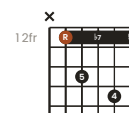
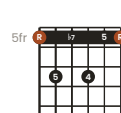
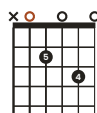
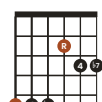
A D E

**A7sus4**

7TH SUSPENDED 4TH

1 4 5 b7

A D E G

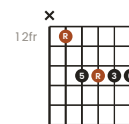
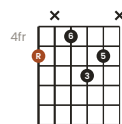
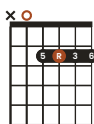
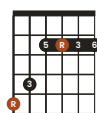


A6

MAJOR 6TH

1 3 5 6

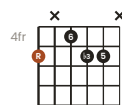
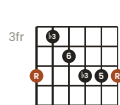
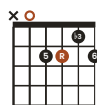
A C# E F#

**Am6**

MINOR 6TH

1 b3 5 6

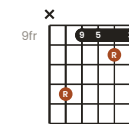
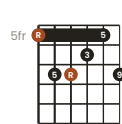
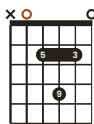
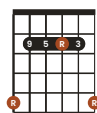
A C E F#

**Aadd9**

ADDED 9TH

1 3 5 9

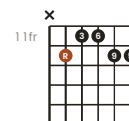
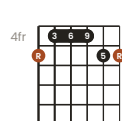
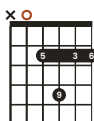
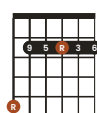
A C# E B

**A6/9**

SIX-NINE

1 3 5 6 9

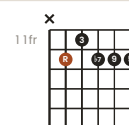
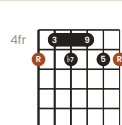
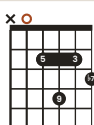
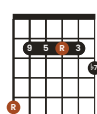
A C# E F# B

**A9**

DOMINANT 9TH

1 3 5 b7 9

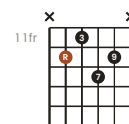
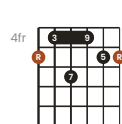
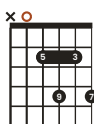
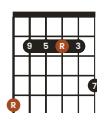
A C# E G B

**Amaj9**

MAJOR 9TH

1 3 5 7 9

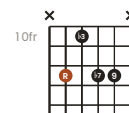
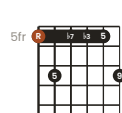
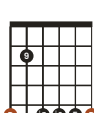
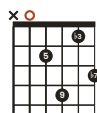
A C# E G# B

**Am9**

MINOR 9TH

1 b3 5 b7 9

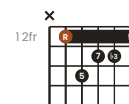
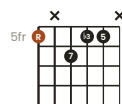
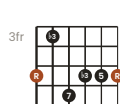
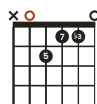
A C E G B

**AmMaj7**

MINOR-MAJOR 7TH

1 b3 5 7

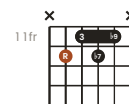
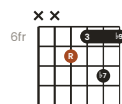
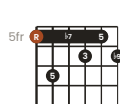
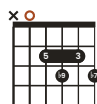
A C E G#

**A7b9**

7TH FLAT 9

1 3 5 b7 b9

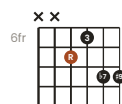
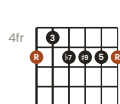
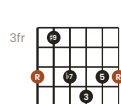
A C# E G Bb

**A7#9**

7TH SHARP 9

1 3 5 b7 #9

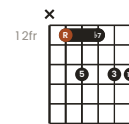
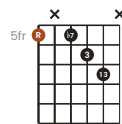
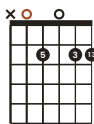
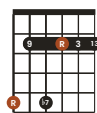
A C# E G B#

**A13**

DOMINANT 13TH

1 3 5 b7 9 13

A C# E G B F#

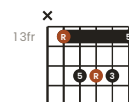
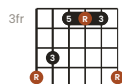
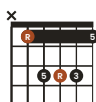


B $\flat$ / A $\sharp$

B $\flat$

MAJOR

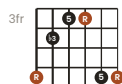
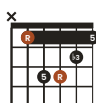
1 3 5

B $\flat$ D F

B $\flat$ m

MINOR

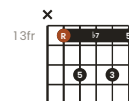
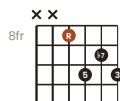
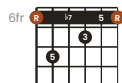
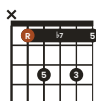
1 b3 5

B $\flat$ D $\flat$ F

B $\flat$ 7

DOMINANT 7TH

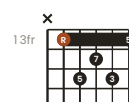
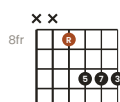
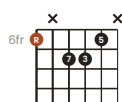
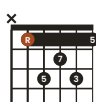
1 3 5 b7

B $\flat$ D F A $\flat$ 

B $\flat$ maj7

MAJOR 7TH

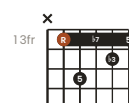
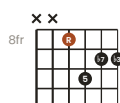
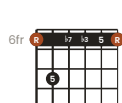
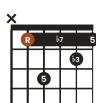
1 3 5 7

B $\flat$ D F A

B $\flat$ m7

MINOR 7TH

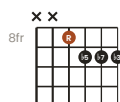
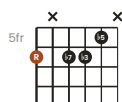
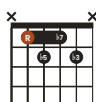
1 b3 5 b7

B $\flat$ D $\flat$ F A $\flat$ 

A $\sharp$ m7b5

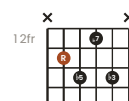
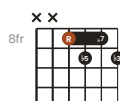
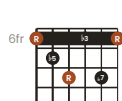
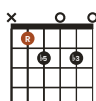
HALF-DIMINISHED

1 b3 b5 b7

A $\sharp$ C $\sharp$ E G $\sharp$ 

A $\sharp$ dim7

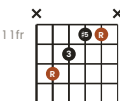
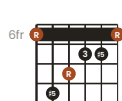
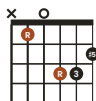
DIMINISHED 7TH

1 b3 b5 $\natural$ 7A $\sharp$ C $\sharp$ E G

B $\flat$ aug

AUGMENTED

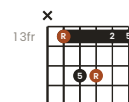
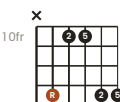
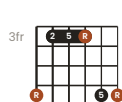
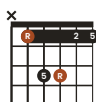
1 3 #5

B $\flat$ D F $\sharp$ 

B $\flat$ sus2

SUSPENDED 2ND

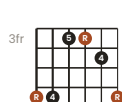
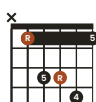
1 2 5

B $\flat$ C F

B $\flat$ sus4

SUSPENDED 4TH

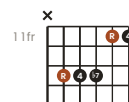
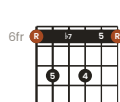
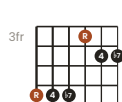
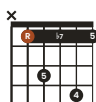
1 4 5

B $\flat$ E $\flat$ F

B $\flat$ 7sus4

7TH SUSPENDED 4TH

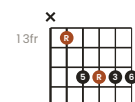
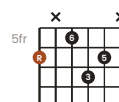
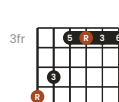
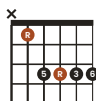
1 4 5 b7

B $\flat$ E $\flat$ F A $\flat$ 

B $\flat$ 6

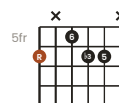
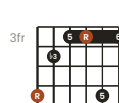
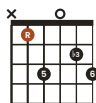
MAJOR 6TH

1 3 5 6

B $\flat$ D F G**B $\flat$ m6**

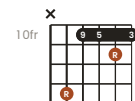
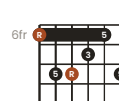
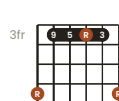
MINOR 6TH

1 b3 5 6

B $\flat$ D $\flat$ F G**B $\flat$ add9**

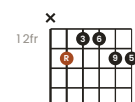
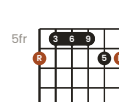
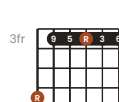
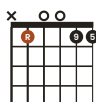
ADDED 9TH

1 3 5 9

B $\flat$ D F C**B $\flat$ 6/9**

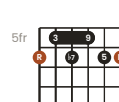
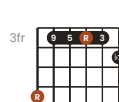
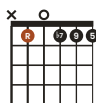
SIX-NINE

1 3 5 6 9

B $\flat$ D F G C**B $\flat$ 9**

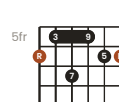
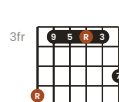
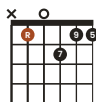
DOMINANT 9TH

1 3 5 b7 9

B $\flat$ D F A $\flat$ C**B $\flat$ maj9**

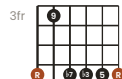
MAJOR 9TH

1 3 5 7 9

B $\flat$ D F A C**B $\flat$ m9**

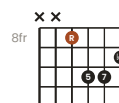
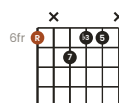
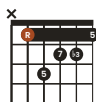
MINOR 9TH

1 b3 5 b7 9

B $\flat$ D $\flat$ F A $\flat$ C**B $\flat$ mMaj7**

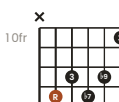
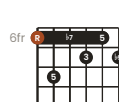
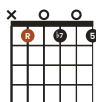
MINOR-MAJOR 7TH

1 b3 5 7

B $\flat$ D $\flat$ F A**B $\flat$ 7b9**

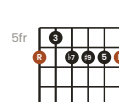
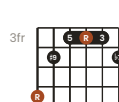
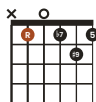
7TH FLAT 9

1 3 5 b7 b9

B $\flat$ D F A $\flat$ C $\flat$ **B $\flat$ 7#9**

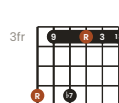
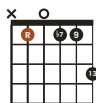
7TH SHARP 9

1 3 5 b7 #9

B $\flat$ D F A $\flat$ C#**B $\flat$ 13**

DOMINANT 13TH

1 3 5 b7 9 13

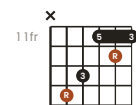
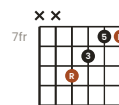
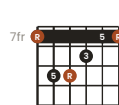
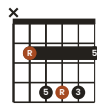
B $\flat$ D F A $\flat$ C G

B

B

MAJOR

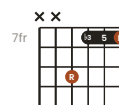
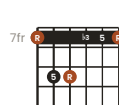
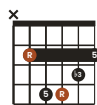
1 3 5

B D $\sharp$ F $\sharp$ 

Bm

MINOR

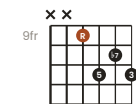
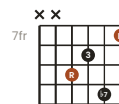
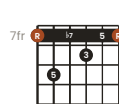
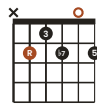
1 b3 5

B D F $\sharp$ 

B7

DOMINANT 7TH

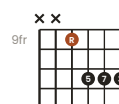
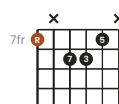
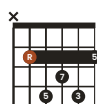
1 3 5 b7

B D $\sharp$ F $\sharp$ A

Bmaj7

MAJOR 7TH

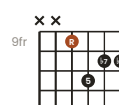
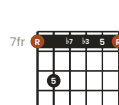
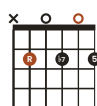
1 3 5 7

B D $\sharp$ F $\sharp$ A $\sharp$ 

Bm7

MINOR 7TH

1 b3 5 b7

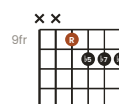
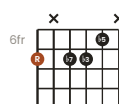
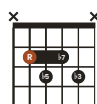
B D F $\sharp$ A

Bm7b5

HALF-DIMINISHED

1 b3 b5 b7

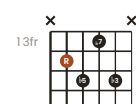
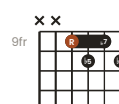
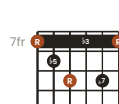
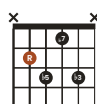
B D F A



Bdim7

DIMINISHED 7TH

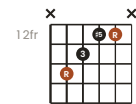
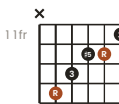
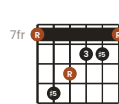
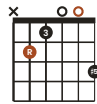
1 b3 b5 b7

B D F A $\flat$ 

Baug

AUGMENTED

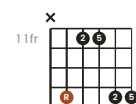
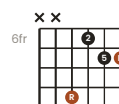
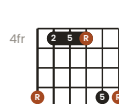
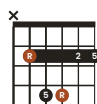
1 3 #5

B D $\sharp$ F $\sharp$ 

Bsus2

SUSPENDED 2ND

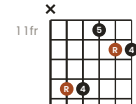
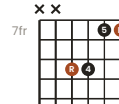
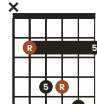
1 2 5

B C $\sharp$ F $\sharp$ 

Bsus4

SUSPENDED 4TH

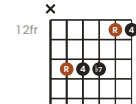
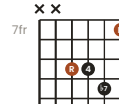
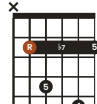
1 4 5

B E F $\sharp$ 

B7sus4

7TH SUSPENDED 4TH

1 4 5 b7

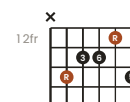
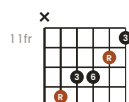
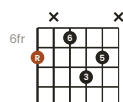
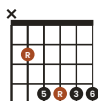
B E F $\sharp$ A

B6

MAJOR 6TH

1 3 5 6

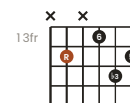
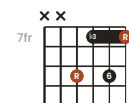
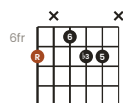
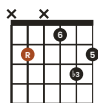
B D# F# G#

**Bm6**

MINOR 6TH

1 b3 5 6

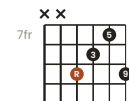
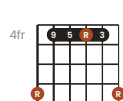
B D F# G#

**Badd9**

ADDED 9TH

1 3 5 9

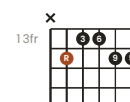
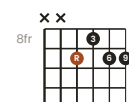
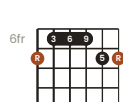
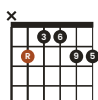
B D# F# C#

**B6/9**

SIX-NINE

1 3 5 6 9

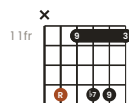
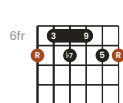
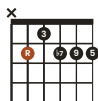
B D# F# G# C#

**B9**

DOMINANT 9TH

1 3 5 b7 9

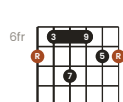
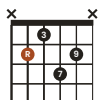
B D# F# A C#

**Bmaj9**

MAJOR 9TH

1 3 5 7 9

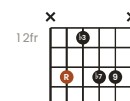
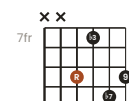
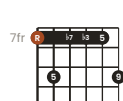
B D# F# A C#

**Bm9**

MINOR 9TH

1 b3 5 b7 9

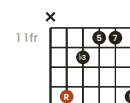
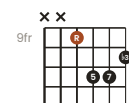
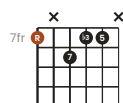
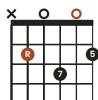
B D F# A C#

**BmMaj7**

MINOR-MAJOR 7TH

1 b3 5 7

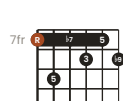
B D F# A#

**B7b9**

7TH FLAT 9

1 3 5 b7 b9

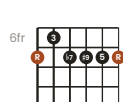
B D# F# A C

**B7#9**

7TH SHARP 9

1 3 5 b7 #9

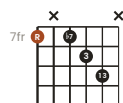
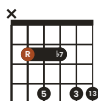
B D# F# A C#

**B13**

DOMINANT 13TH

1 3 5 b7 9 13

B D# F# A C# G#



APPENDIX D

Scale Atlas

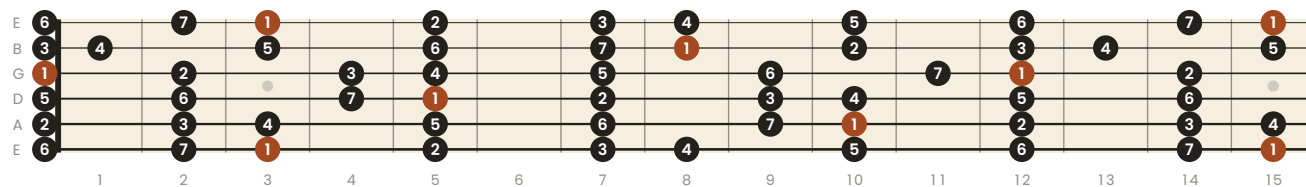


Twenty-five scales and modes, each with its formula, a map of the whole neck, a playable position, where it is used and a practice loop. Degrees are shown in the dots; the root is rust. Families follow Chapter 9, Chapter 7 and Chapter 31.

Ionian (major scale)

Formula	Notes in G	Family
1 2 3 4 5 6 7	G A B C D E F $\sharp$	Modes of the major scale

Sound. Bright and settled; the major 7th (F $\sharp$ in G) pulls up to the root, and the natural 4th (C) clashes with the major 3rd when held over the I chord. **Use.** Melodies and solos over major-key progressions in pop, rock, folk and country; maj7 and 6 chords on the I. **Chords.** G, Gmaj7, G6; the diatonic chords G, Am, Bm, C, D, Em **Heard in.** Do-Re-Mi (The Sound of Music); Joy to the World

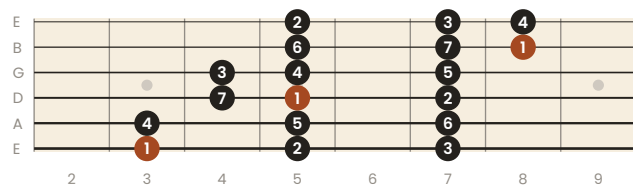


TRACK 527 Ionian (major scale) practice loop · LOOP · pop · 100 bpm

G <i>I</i>	Em <i>vi</i>	C <i>IV</i>	D <i>V</i>
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IMPROVISE WITH

G Ionian

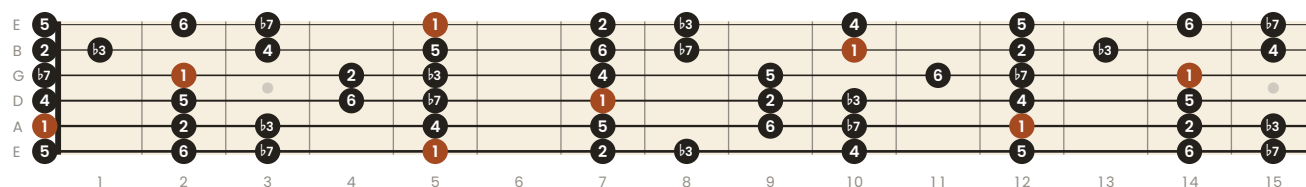


Play F $\sharp$ and resolve it up to G on the downbeat of each new bar; then hold C over the G chord and hear it ask to fall to B.

Dorian

Formula	Notes in A	Family
1 2 $\flat$ 3 4 5 6 $\flat$ 7	A B C D E F $\sharp$ G	Modes of the major scale

Sound. Minor but bright; the natural 6th (F $\sharp$ in A) lifts it above the darker natural minor. **Use.** Minor vamps in funk, soul, Latin rock and modal jazz; m7, m6 and m9 chords, and the ii chord in a ii – V. **Chords.** Am7, Am6, Am9; the vamp Am7 to D7 or D9 **Heard in.** Oye Como Va (Santana); So What (Miles Davis); Scarborough Fair (traditional, Simon & Garfunkel)

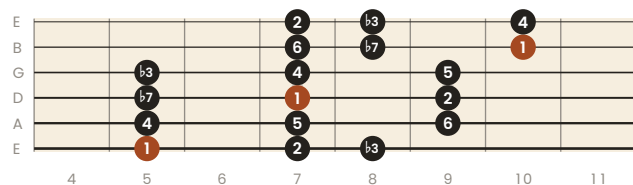


TRACK 528 Dorian practice loop · LOOP · funk · 96 bpm

Am7 <i>i7</i>	D9 <i>IV9</i>	Am7 <i>i7</i>	D9 <i>IV9</i>
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IMPROVISE WITH

A dorian

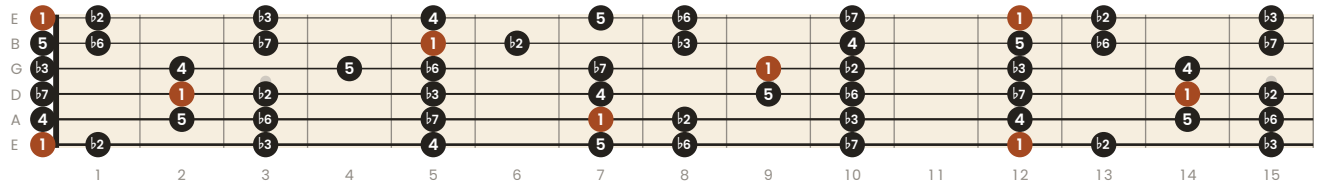


Target F $\sharp$ over Am7: it is the 3rd of the D9, so landing on it as the chord changes makes the Dorian colour obvious.

Phrygian

Formula	Notes in E	Family
1 $\flat 2$ $\flat 3$ 4 5 $\flat 6$ $\flat 7$	E F G A B C D	Modes of the major scale

Sound. Dark and tense; the $\flat 2$ (F in E) a half step above the root gives its Spanish, heavy-metal edge. **Use.** Metal riffs, film music and minor vamps where the $\flat \text{II}$ chord moves to i. **Chords.** Em, Em7, E5; the vamp Em to F/E or F5 to E5 **Heard in.** Symphony of Destruction (Megadeth)

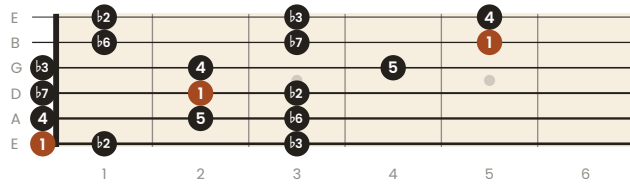


▶ **TRACK 529** Phrygian practice loop · LOOP · rock · 100 bpm

Em <i>i</i>	F/E <i>$\flat \text{II}/1$</i>	Em <i>i</i>	F/E <i>$\flat \text{II}/1$</i>
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IMPROVISE WITH

E phrygian

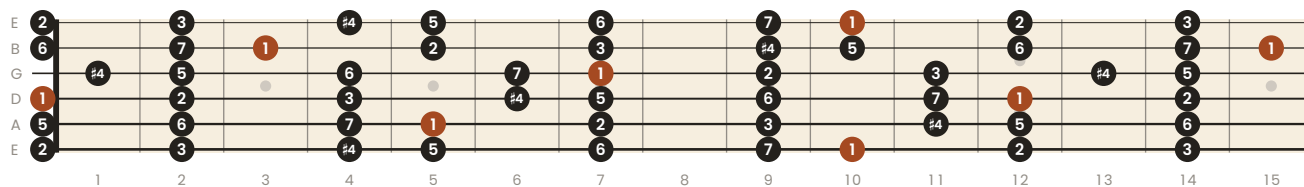


Trill between F and E on string 1 or 6, then rest on E; the half step down into the root is the Phrygian sound.

Lydian

Formula	Notes in D	Family
1 2 3 #4 5 6 7	D E F# G# A B C#	Modes of the major scale

Sound. Major and floating; the #4 (G# in D) replaces the 4th, so nothing in the scale pulls away from the major chord. **Use.** Dreamy major vamps in rock instrumentals and film music; maj7, maj7#11 and add9 chords, and the IV chord in a major key. **Chords.** D, Dmaj7, Dmaj7#11; the vamp D to E/D **Heard in.** Flying in a Blue Dream (Joe Satriani)

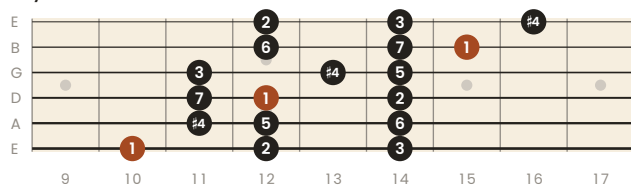


▶ **TRACK 530** Lydian practice loop · LOOP · pop · 92 bpm

D <i>I</i>	E/D <i>II/1</i>	D <i>I</i>	E/D <i>II/1</i>
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IMPROVISE WITH

D Lydian

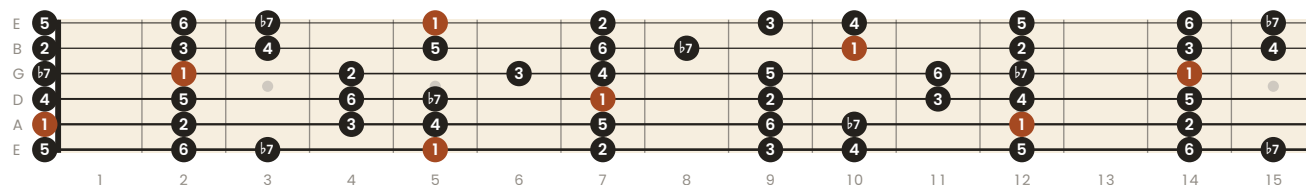


Land on G# over the E/D bar (it is the 3rd of E), then hold it over D: it sounds stable, not wrong.

Mixolydian

Formula	Notes in A	Family
1 2 3 4 5 6 $\flat 7$	A B C $\sharp$ D E F $\sharp$ G	Modes of the major scale

Sound. Major with a bluesy, unresolved edge; the $\flat 7$ (G in A) replaces the leading tone. **Use.** Rock, blues, folk and funk over dominant 7th chords and $\flat VII - IV - I$ progressions. **Chords.** A, A7, A9, A13; the progression A, G, D **Heard in.** Norwegian Wood (The Beatles); Sweet Home Alabama (Lynyrd Skynyrd)

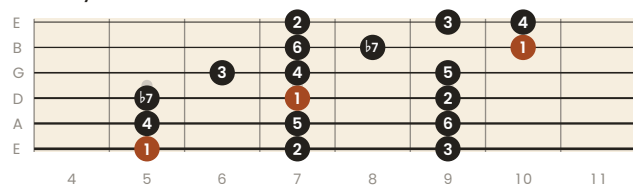


TRACK 531 Mixolydian practice loop · LOOP · rock · 104 bpm

A <i>I</i>	G <i>$\flat VII$</i>	D <i>IV</i>	A <i>I</i>
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IMPROVISE WITH

A mixolydian

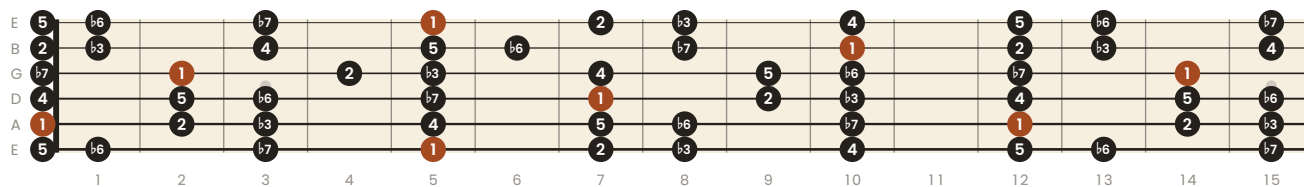


Target G as the chord changes to G, then bend or slide from G to A over the A chord to hear the $\flat 7$ resolve home.

Aeolian (natural minor)

Formula	Notes in A	Family
1 2 $\flat$ 3 4 5 $\flat$ 6 $\flat$ 7	A B C D E F G	Modes of the major scale

Sound. Sad and serious; the $\flat$ 6 (F in A) gives it the darker colour that Dorian lacks. **Use.** Minor-key rock, pop and ballads; the i - $\flat$ VII - $\flat$ VI movement and m7 chords on the i. **Chords.** Am, Am7; the progressions Am, G, F and Am, F, C, G **Heard in.** All Along the Watchtower (Bob Dylan, Jimi Hendrix); Stairway to Heaven, the solo (Led Zeppelin)

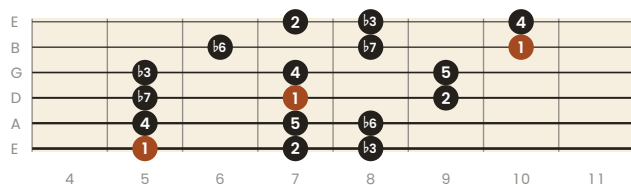


TRACK 532 Aeolian (natural minor) practice loop · LOOP · rock · 92 bpm

Am i	G $\flat$ VII	F $\flat$ VI	G $\flat$ VII
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IMPROVISE WITH

A aeolian

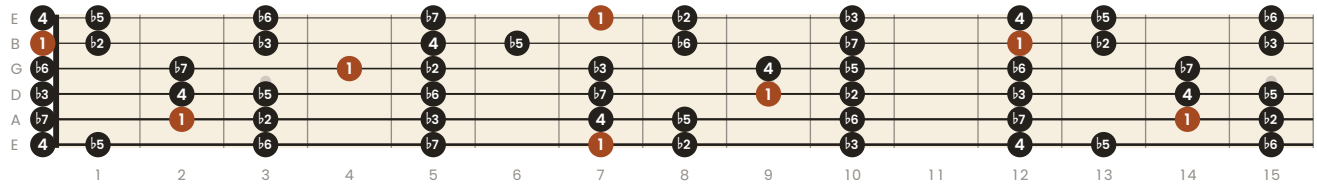


Over the F chord, lean on F and let it fall a half step to E as the progression returns to Am.

Locrian

Formula	Notes in B	Family
1 $\flat 2$ $\flat 3$ 4 $\flat 5$ $\flat 6$ $\flat 7$	B C D E F G A	Modes of the major scale

Sound. Unstable and dark; the $\flat 5$ (F in B) means the tonic chord is diminished, and the $\flat 2$ (C) adds tension. **Use.** Over $m7\flat 5$ chords (the vii of a major key, the ii of a minor key) and in some metal riffs; rarely a home key. **Chords.** $Bm7\flat 5$; the vamp $Bm7\flat 5$ to C/B Heard in. Army of Me (Björk)

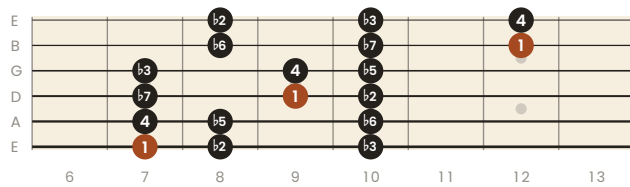


TRACK 533 Locrian practice loop · LOOP · rock · 90 bpm

Bm7$\flat 5$ <i>i$\flat$7</i>	C/B <i>$\flat$II/1</i>	Bm7$\flat 5$ <i>i$\flat$7</i>	C/B <i>$\flat$II/1</i>
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IMPROVISE WITH

B locrian

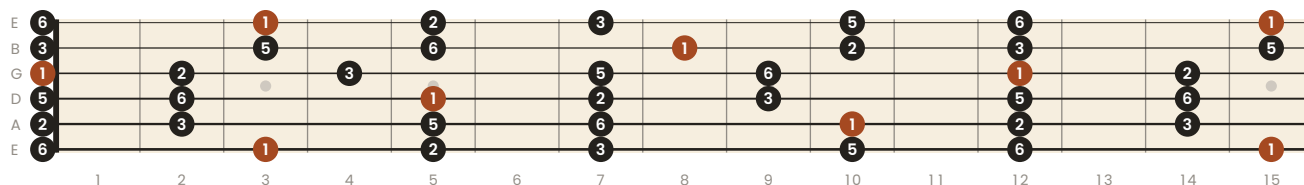


Play B then F in the low register so the tritone is clear, then step from C down to B to hear the $\flat 2$.

Major pentatonic

Formula	Notes in G	Family
1 2 3 5 6	G A B D E	Pentatonic and blues

Sound. Open and sweet; with no 4th or 7th it has no half steps, so every note sounds safe over the major chord. **Use.** Country, folk, rock and soul melodies and solos over major chords and major-key progressions. **Chords.** G, G6, Gadd9; the progression G, C, D **Heard in.** My Girl, the guitar riff (The Temptations); Amazing Grace

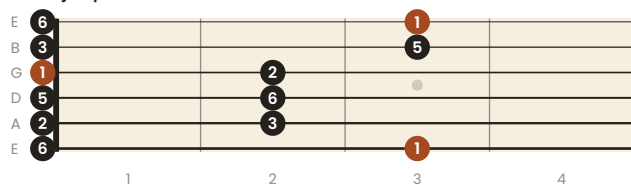


▶ **TRACK 534** Major pentatonic practice loop · LOOP · folk · 100 bpm



IMPROVISE WITH

G major pentatonic

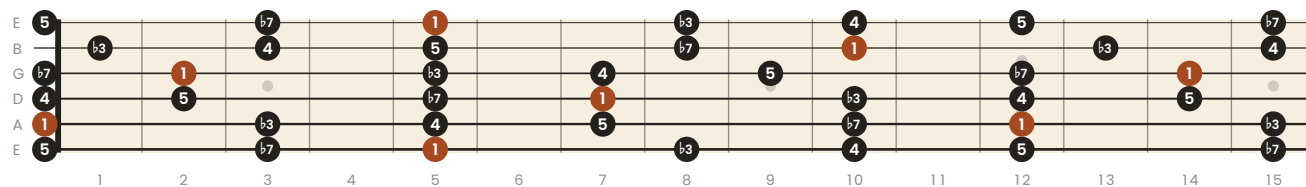


Make the 6th (E) your home note in licks over G; it gives the sweet country sound that the minor pentatonic cannot.

Minor pentatonic

Formula	Notes in A	Family
1 $\flat 3$ 4 5 $\flat 7$	A C D E G	Pentatonic and blues

Sound. Strong and bluesy; five notes with no half steps, built around the $\flat 3$ (C in A) and $\flat 7$ (G). **Use.** The first soloing scale in rock and blues: over minor chords, minor-key progressions and dominant blues. **Chords.** Am, Am7, A5; also A7 in a blues **Heard in.** Stairway to Heaven, the solo (Led Zeppelin); Albert King; Eric Clapton

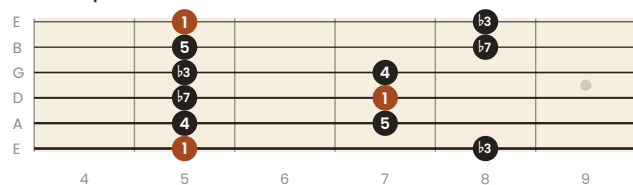


TRACK 535 Minor pentatonic practice loop · LOOP · rock · 96 bpm

Am7 <i>i7</i>	Dm7 <i>iv7</i>	Am7 <i>i7</i>	Em7 <i>v7</i>
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IMPROVISE WITH

A minor pentatonic



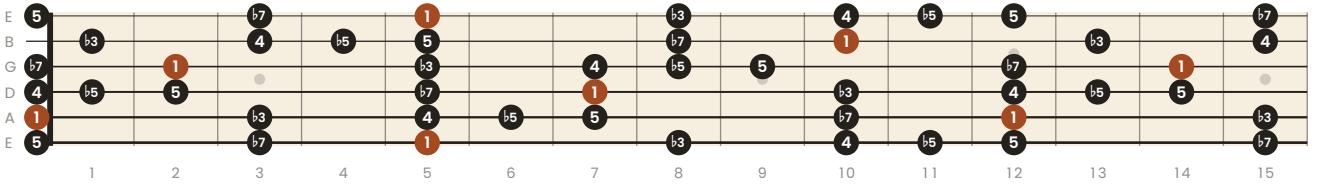
Bend the 4th (D) up a whole step to E on string 3 and match the pitch by ear; then resolve to A on the downbeat.

PENT

Blues scale

Formula	Notes in A	Family
1 $\flat 3$ 4 $\flat 5$ 5 $\flat 7$	A C D $E\flat$ E G	Pentatonic and blues

Sound. The minor pentatonic plus the $\flat 5$ ($E\flat$ in A), a passing blue note that adds grit. **Use.** Blues, rock and jazz blues soloing over dominant 7th chords and minor vamps. **Chords.** A7, D7, E7 in a 12-bar blues; Am7 **Heard in.** Smoke on the Water, the riff (Deep Purple); Sunshine of Your Love, the riff (Cream)

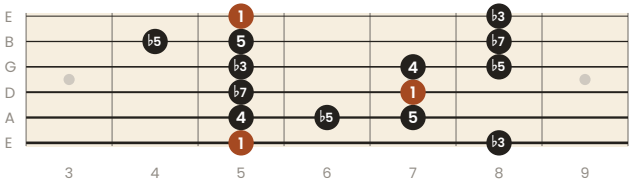


TRACK 536 Blues scale practice loop · LOOP · blues shuffle · 96 bpm

A7 <i>I7</i>	D7 <i>IV7</i>	A7 <i>I7</i>	A7 <i>I7</i>
D7 <i>IV7</i>	D7 <i>IV7</i>	A7 <i>I7</i>	A7 <i>I7</i>
E7 <i>V7</i>	D7 <i>IV7</i>	A7 <i>I7</i>	E7 <i>V7</i>

IMPROVISE WITH

A blues



Use $E\flat$ only in passing: slide from D through $E\flat$ to E, or hammer from $E\flat$ to E; do not rest on it.

Major blues

Formula	Notes in E	Family
1 2 $\flat 3$ 3 5 6	E F $\sharp$ G G $\sharp$ B C $\sharp$	Pentatonic and blues

Sound. The major pentatonic plus the $\flat 3$ (G in E), which slides up into the major 3rd (G $\sharp$). **Use.** Major-sounding blues, country, Southern rock and jump blues over dominant 7th chords. **Chords.** E7, A7, B7 in a 12-bar blues; E6 **Heard in.** B.B. King; Stevie Ray Vaughan

TRACK 537 Major blues practice loop · LOOP · blues shuffle · 112 bpm

E7 <i>I7</i>	A7 <i>IV7</i>	E7 <i>I7</i>	E7 <i>I7</i>
A7 <i>IV7</i>	A7 <i>IV7</i>	E7 <i>I7</i>	E7 <i>I7</i>
B7 <i>V7</i>	A7 <i>IV7</i>	E7 <i>I7</i>	B7 <i>V7</i>

IMPROVISE WITH

E major blues

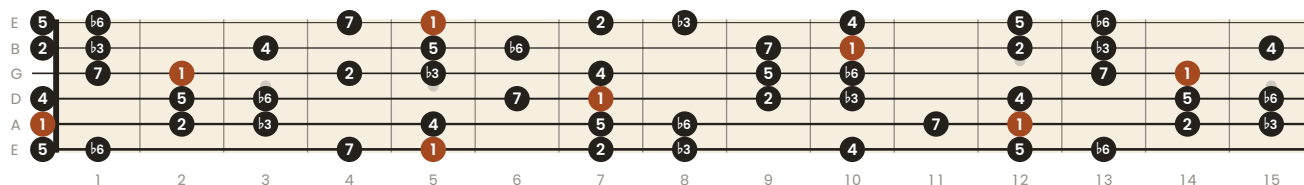
Hammer from G to G $\sharp$ on the I chord; the $\flat 3$ to 3 move is the sound. Over A7 drop G $\sharp$ and use G, the $\flat 7$ of A7.

Harmonic minor

Formula	Notes in A	Family
1 2 $\flat$ 3 4 5 $\flat$ 6 7	A B C D E F G $\sharp$	Harmonic minor family

Sound. Minor with a raised 7th (G $\sharp$ in A), which leaves an exotic augmented 2nd between F and G $\sharp$. Use. Minor-key cadences over V7 to i, classical and neoclassical metal, gypsy jazz and flamenco-tinged pop.

Chords. AmMaj7, Am; the cadence Am, Dm, E7, Am **Heard in.** Yngwie Malmsteen; Hotel California, the solos over F $\sharp$ 7 (Eagles)

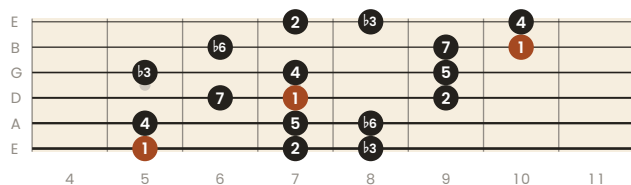


TRACK 538 Harmonic minor practice loop · LOOP · ballad · 76 bpm

Am <i>i</i>	Dm <i>iv</i>	E7 <i>V7</i>	Am <i>i</i>
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IMPROVISE WITH

A harmonic minor

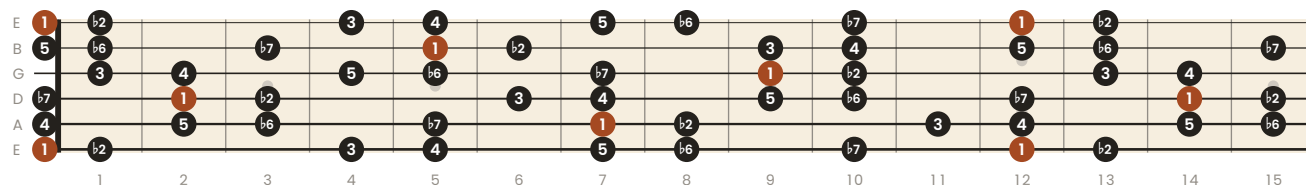


Over E7 play G $\sharp$ and resolve it up to A as Am arrives; then play F to G $\sharp$ slowly to hear the augmented 2nd.

Phrygian dominant

Formula	Notes in E	Family
1 $\flat 2$ 3 4 5 $\flat 6$ $\flat 7$	E F G $\sharp$ A B C D	Harmonic minor family

Sound. Phrygian with a major 3rd: the $\flat 2$ (F in E) and the 3rd (G $\sharp$) form an augmented 2nd with a Middle Eastern and flamenco colour. **Use.** Over the V7 chord in a minor key (E7 in A minor), flamenco, Middle Eastern, Jewish and Balkan music, and metal. **Chords.** E, E7, E7 $\flat 9$; the vamp E to F **Heard in.** Hava Nagila; White Rabbit (Jefferson Airplane); flamenco

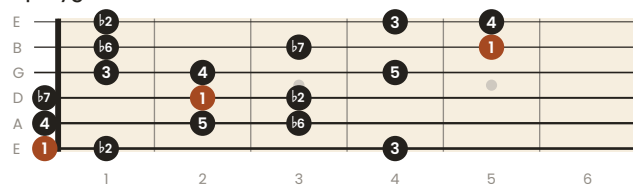


TRACK 539 Phrygian dominant practice loop · LOOP · rock · 110 bpm



IMPROVISE WITH

E phrygian dominant

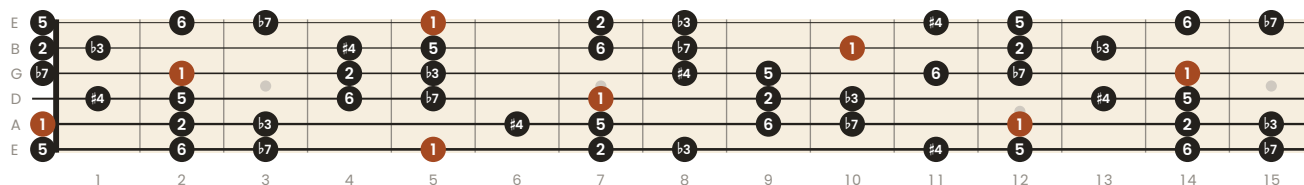


Play F then G $\sharp$ then F, then resolve F down to E; over the F chord let F sound as the root, over E as the $\flat 9$.

Dorian #4 (Ukrainian Dorian)

Formula	Notes in A	Family
1 2 $\flat 3$ #4 5 6 $\flat 7$	A B C D# E F# G	Harmonic minor family

Sound. Dorian with a raised 4th (D# in A), giving an augmented 2nd between C and D# and a crying, folk-like colour. **Use.** Klezmer, Eastern European and Balkan folk music; over m6 and m7 chords where a sharpened 4th is wanted. **Chords.** Am6, Am7; B7 over an A pedal **Heard in.** Klezmer (the Mi sheberakh mode); Romanian and Ukrainian folk music

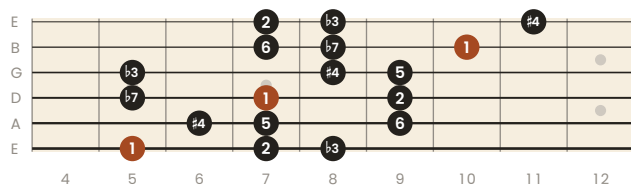


TRACK 540 Dorian #4 (Ukrainian Dorian) practice loop · LOOP · folk · 92 bpm

Am6 <i>i6</i>	B7/A <i>II42</i>	Am6 <i>i6</i>	B7/A <i>II42</i>
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IMPROVISE WITH

A dorian #4

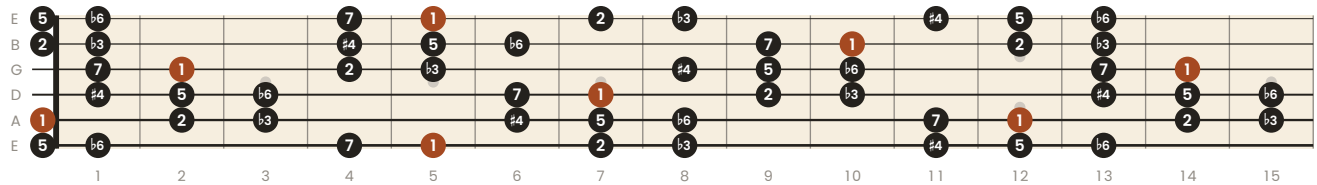


Play C, D#, E slowly and land on E; over the B7/A bar target D# as the 3rd of B7.

Hungarian minor

Formula	Notes in A	Family
1 2 $\flat$ 3 $\sharp$ 4 5 $\flat$ 6 7	A B C D $\sharp$ E F G $\sharp$	Harmonic minor family

Sound. Harmonic minor with a raised 4th (D $\sharp$ in A): two augmented 2nds (C to D $\sharp$, F to G $\sharp$) give a dramatic, exotic sound. **Use.** Neoclassical and metal soloing, film music and Romantic piano music in the Hungarian style; over minor chords and mMaj7 chords. **Chords.** AmMaj7, Am; E over an A minor tonic; Fmaj7 $\sharp$ 11 **Heard in.** Franz Liszt, Hungarian Rhapsodies

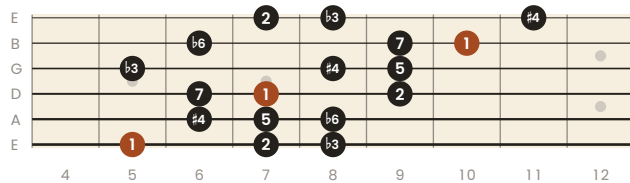


TRACK 541 Hungarian minor practice loop · LOOP · 6/8 ballad · 80 bpm (dotted quarter)

AmMaj7 <i>iMaj7</i>	E <i>V</i>	AmMaj7 <i>iMaj7</i>	Fmaj7 $\sharp$ 11 <i>bVIImaj7$\sharp$11</i>
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IMPROVISE WITH

A hungarian minor

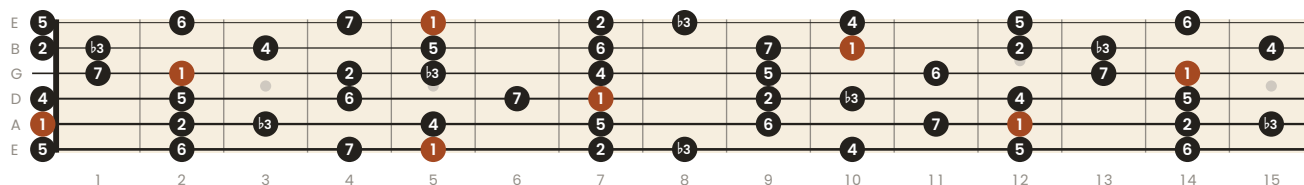


Play the run C, D $\sharp$, E, F, G $\sharp$, A in one position and hear both augmented 2nds; resolve D $\sharp$ up to E over the E chord.

Melodic minor (jazz minor)

Formula	Notes in A	Family
1 2 $\flat 3$ 4 5 6 7	A B C D E F $\sharp$ G $\sharp$	Melodic minor family

Sound. Minor 3rd with a major 6th and 7th (F $\sharp$ and G $\sharp$ in A): minor at the bottom, major at the top, smooth and modern. **Use.** Jazz over m6 and mMaj7 tonic chords; also the parent of the altered, Lydian dominant and Locrian $\sharp 2$ scales. **Chords.** AmMaj7, Am6, Am9 **Heard in.** The closing mMaj9 chord of the James Bond theme; ascending lines in Bach's minor-key music

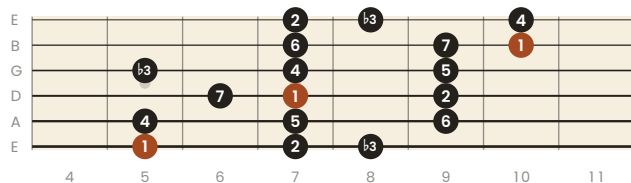


TRACK 542 Melodic minor (jazz minor) practice loop · LOOP · bossa nova · 120 bpm

AmMaj7 <i>iMaj7</i>	Am6 <i>i6</i>	AmMaj7 <i>iMaj7</i>	Am6 <i>i6</i>
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IMPROVISE WITH

A melodic minor

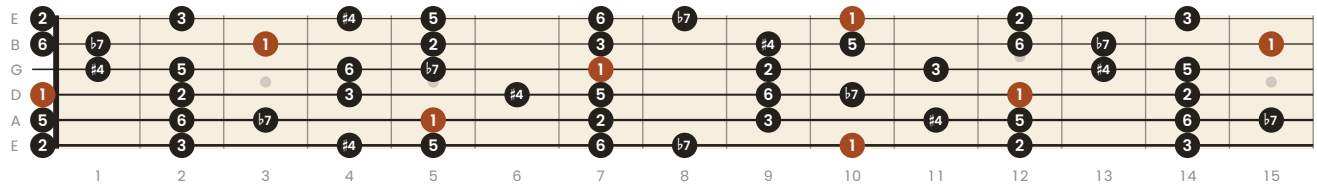


Over AmMaj7 target G $\sharp$ (the major 7th) and C (the $\flat 3$) in one phrase; the pair defines the sound.

Lydian dominant

Formula	Notes in D	Family
1 2 3 #4 5 6 b7	D E F# G# A B C	Melodic minor family

Sound. Mixolydian with a raised 4th (G# in D): a bright dominant sound with a #11 instead of the clashing natural 4th. **Use.** Over 7#11 and 9#11 chords, non-resolving dominants, the bVII7 chord and tritone substitutes in jazz, funk and fusion. **Chords.** D7#11, D9; the vamp D9 to E/D Heard in. The Simpsons theme (Danny Elfman)

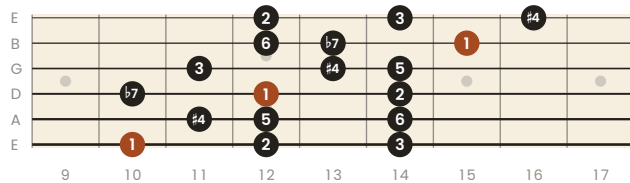


▶ **TRACK 543** Lydian dominant practice loop · LOOP · funk · 96 bpm

D9 I9	E/D II/1	D9 I9	E/D II/1
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IMPROVISE WITH

D lydian dominant

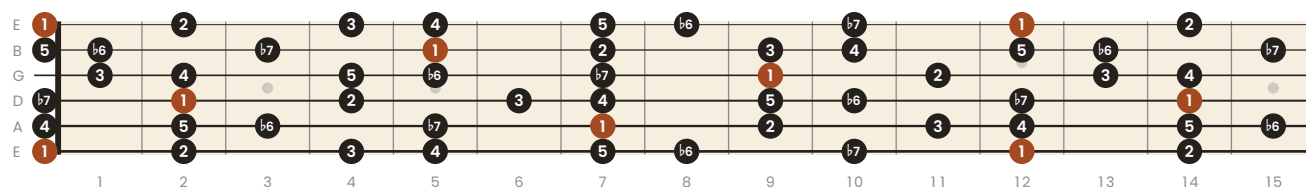


Hold G# against the D bass and compare it with G: G# floats, G wants to resolve. Target it as the 3rd of E/D.

Mixolydian $\flat 6$ (Aeolian dominant)

Formula	Notes in E	Family
1 2 3 4 5 $\flat 6$ $\flat 7$	E F $\sharp$ G $\sharp$ A B C D	Melodic minor family

Sound. A dominant scale with a minor 6th (C in E): the major 3rd is bright, the $\flat 6$ is sad, so it sounds bittersweet. **Use.** Over a V7 chord that resolves to a minor chord (E7 to Am or AmMaj7), and in film and rock ballad writing. **Chords.** E7, E7 $\flat 13$, Eaug; the vamp E to Am/E **Heard in.** Film scores and minor-key dominant chords; rarely the basis of a whole song

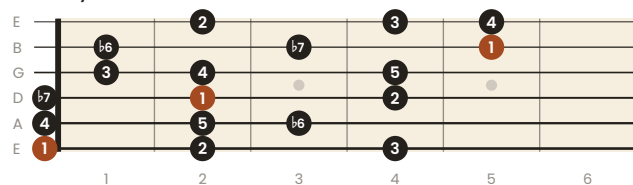


TRACK 544 Mixolydian $\flat 6$ (Aeolian dominant) practice loop · LOOP · ballad · 80 bpm

E <i>I</i>	Am/E <i>iv64</i>	E <i>I</i>	Am/E <i>iv64</i>
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IMPROVISE WITH

E mixolydian $\flat 6$

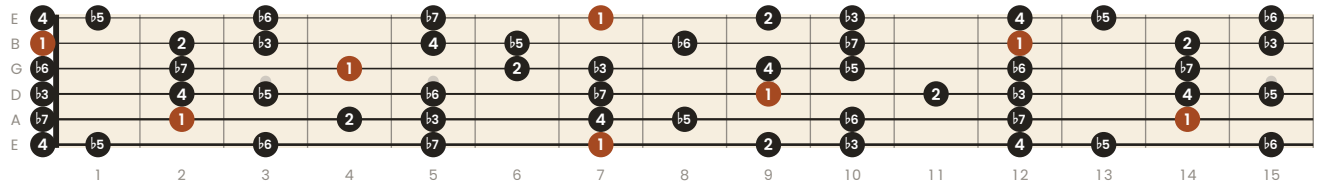


Over E play C and let it fall a half step to B; in the Am/E bar the same C becomes the 3rd of Am.

Locrian #2 (half-diminished)

Formula	Notes in B	Family
1 2 b3 4 b5 b6 b7	B C# D E F G A	Melodic minor family

Sound. Locrian with a natural 2nd (C# in B), so the m7b5 chord gets a smooth natural 9 instead of the harsh b9. **Use.** Jazz over m7b5 chords, especially the ii in a minor ii – V; it sounds more modern than plain Locrian. **Chords.** Bm7b5 (Bm9b5) **Heard in.** Jazz minor ii – V lines over half-diminished chords

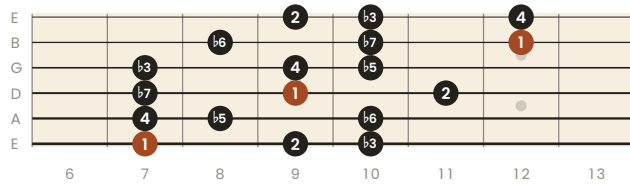


▶ **TRACK 545** Locrian #2 (half-diminished) practice loop · LOOP · bossa nova · 110 bpm

Bm7b5 <i>iø7</i>	Bm7b5 <i>iø7</i>	Bm7b5 <i>iø7</i>	Bm7b5 <i>iø7</i>
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IMPROVISE WITH

B locrian #2

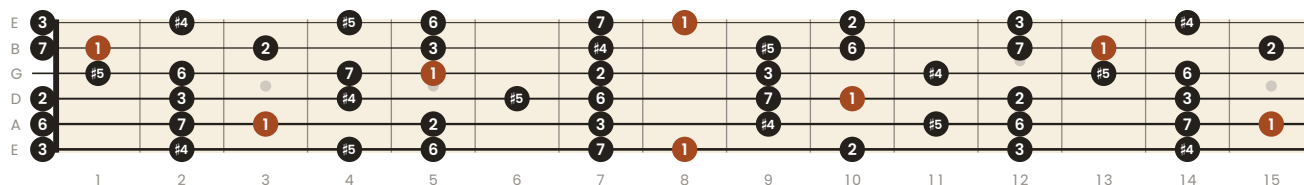


Hold C# over Bm7b5 and then play C: C# is calm and modern, C is the darker Locrian sound.

Lydian augmented

Formula	Notes in C	Family
1 2 3 #4 #5 6 7	C D E F# G# A B	Melodic minor family

Sound. Lydian with a raised 5th (G# in C): the #4 and #5 together make a bright, suspended, unsettled major sound. **Use.** Jazz and film music over maj7#5 chords, such as the bIII chord of a melodic minor key (Cmaj7#5 in A minor). **Chords.** Cmaj7#5; the vamp Cmaj7#5 to D/C **Heard in.** Modern jazz harmony over maj7#5 chords

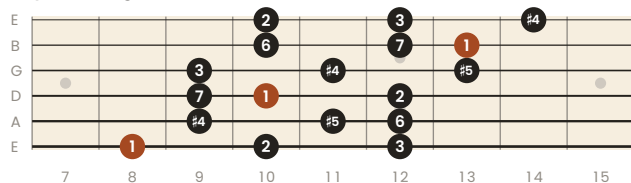


▶ **TRACK 546** Lydian augmented practice loop · LOOP · ballad · 72 bpm

Cmaj7#5 <i>Imaj7#5</i>	D/C <i>II/1</i>	Cmaj7#5 <i>Imaj7#5</i>	D/C <i>II/1</i>
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IMPROVISE WITH

C lydian augmented

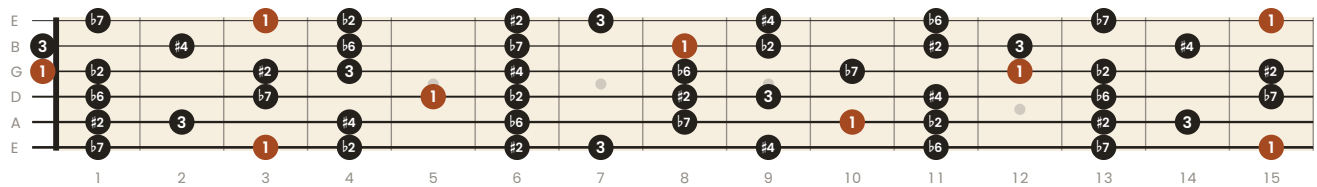


Target G# over Cmaj7#5 and compare it with G: the raised 5th is the note that makes the chord shimmer.

Altered (Superlocrian)

Formula	Notes in G	Family
1 $\flat 2$ $\sharp 3$ $\sharp 4$ $\flat 6$ $\flat 7$	G $\flat A$ A $\sharp$ B C $\sharp$ E $\flat$ F	Melodic minor family

Sound. Root, 3rd and $\flat 7$ plus every altered tension: $\flat 9$, $\sharp 9$, $\flat 5$ ($\sharp 11$) and $\sharp 5$ ($\flat 13$). Maximum tension that resolves strongly. **Use.** Jazz and fusion over altered dominant 7th chords (7alt, 7 $\sharp 9$, 7 $\flat 9$ $\flat 13$) resolving to the I or i chord. **Chords.** G7alt, G7 $\sharp 9$, G7 $\flat 13$; Dm7, G7alt, Cmaj7 **Heard in.** Bebop and post-bop jazz soloing over the V7 of a ii - V - I

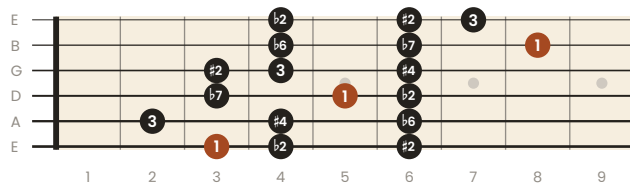


▶ **TRACK 547** Altered (Superlocrian) practice loop · LOOP · jazz swing · 120 bpm

Dm7 <i>v7</i>	G7alt <i>I7alt</i>	Cmaj7 <i>IVmaj7</i>	Cmaj7 <i>IVmaj7</i>
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IMPROVISE WITH

G altered

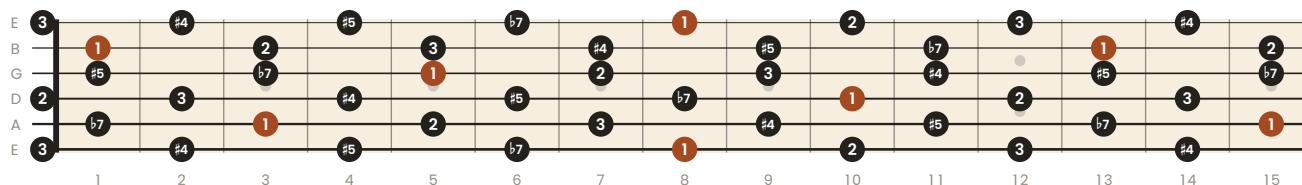


Over G7alt play A $\flat$ ($\flat 9$) and resolve it down to G, the 5th of Cmaj7; or E $\flat$ ($\flat 13$) down to E, the 3rd.

Whole tone

Formula	Notes in C	Family
1 2 3 #4 #5 b7	C D E F# G# Bb	Symmetric and bebop

Sound. Six notes a whole step apart, with no half steps: dreamy and floating, with the #4 and #5 (F# and G# in C). **Use.** Over augmented triads and 7#5 or 9#5 dominant chords; impressionist and film music. **Chords.** C7#5, C7b5, Caug **Heard in.** You Are the Sunshine of My Life, the intro (Stevie Wonder); Voiles (Claude Debussy)

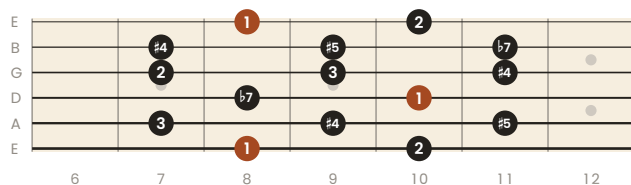


▶ **TRACK 548** Whole tone practice loop · LOOP · ballad · 76 bpm

C7#5 <i>I7#5</i>	C7b5 <i>I7b5</i>	C7#5 <i>I7#5</i>	C7b5 <i>I7b5</i>
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IMPROVISE WITH

C whole tone

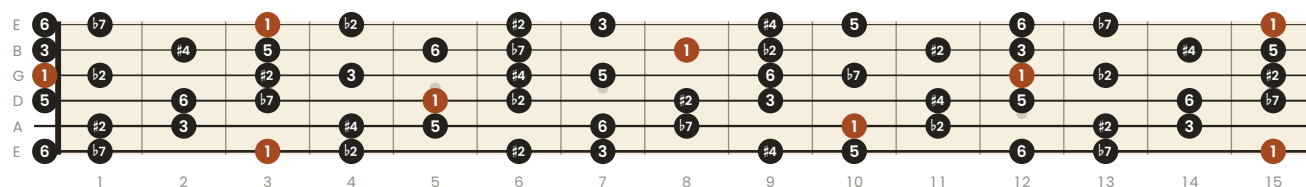


Play the scale as a run up two octaves, then target G# over C7#5 and F# over C7b5.

Half-whole diminished (dominant diminished)

Formula	Notes in G	Family
1 $\flat 2$ $\sharp 2$ 3 $\sharp 4$ 5 6 $\flat 7$	G $\text{A}\flat$ $\text{A}\sharp$ B C $\sharp$ D E F	Symmetric and bebop

Sound. Alternating half and whole steps from the root: $\flat 9$, $\sharp 9$, $\sharp 11$ and a natural 13 ($\text{A}\flat$, $\text{A}\sharp$, C $\sharp$, E in G) over a dominant 7th. **Use.** Jazz, fusion and blues over 7 $\flat 9$, 7 $\sharp 9$ and 13 $\flat 9$ chords; any minor-3rd shift of a line keeps it in the scale. **Chords.** G7 $\flat 9$, G7 $\sharp 9$, G13 $\flat 9$ **Heard in.** Jazz soloing over 13 $\flat 9$ chords; Petrushka (Igor Stravinsky) uses the same eight-note collection

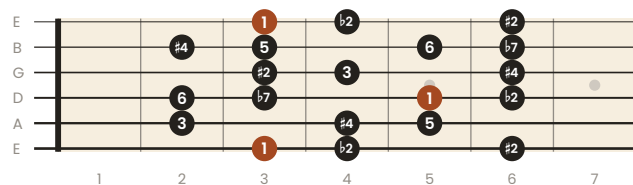


TRACK 549 Half-whole diminished (dominant diminished) practice loop · LOOP · jazz swing · 120 bpm

G7 $\flat 9$ <i>I7$\flat 9$</i>	G7 $\sharp 9$ <i>I7$\sharp 9$</i>	G7 $\flat 9$ <i>I7$\flat 9$</i>	G7 $\sharp 9$ <i>I7$\sharp 9$</i>
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IMPROVISE WITH

G half-whole diminished

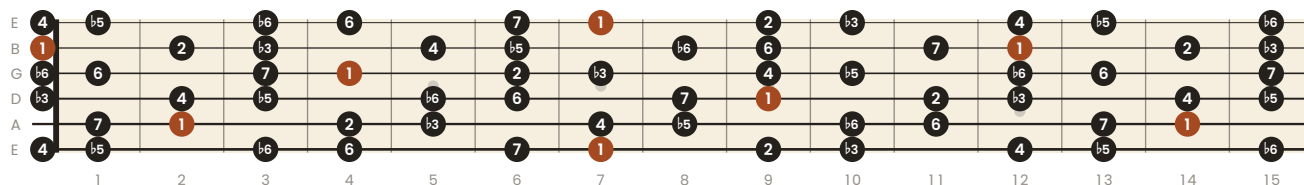


Play $\text{A}\flat$, $\text{A}\sharp$, B over G7 so the $\flat 9$ and $\sharp 9$ resolve up to the 3rd; then move the same lick up three frets.

Whole-half diminished

Formula	Notes in B	Family
1 2 $\flat 3$ 4 $\flat 5$ $\flat 6$ 7	B C $\sharp$ D E F G G $\sharp$ A $\sharp$	Symmetric and bebop

Sound. Alternating whole and half steps from the root: the diminished 7th chord plus a note a whole step above each chord tone. **Use.** Jazz over diminished 7th chords, including passing dim7 chords. **Chords.** Bdim7 (and the same chord as Ddim7, Fdim7, G $\sharp$ dim7); C $\sharp$ dim7 **Heard in.** Jazz soloing over diminished 7th chords

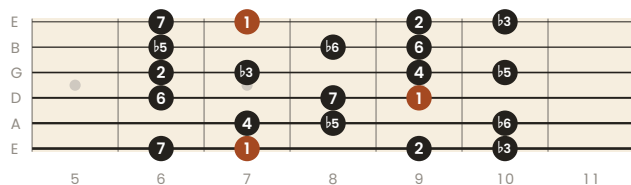


TRACK 550 Whole-half diminished practice loop · LOOP · ballad · 72 bpm

Bdim7 <i>i°7</i>	C $\sharp$ dim7 <i>ii°7</i>	Bdim7 <i>i°7</i>	C $\sharp$ dim7 <i>ii°7</i>
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IMPROVISE WITH

B whole-half diminished

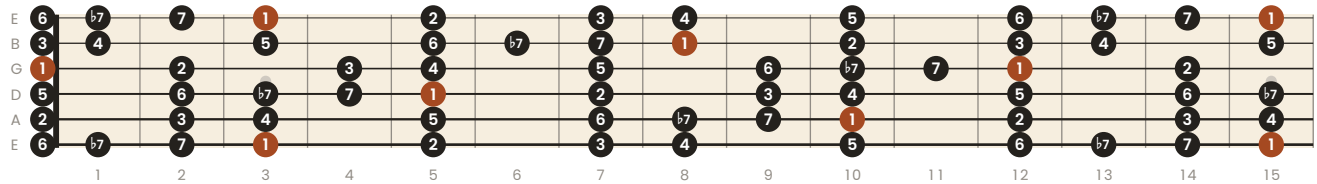


Over Bdim7 play each chord tone with its upper neighbour (C $\sharp$ to B, E to D, G to F, A $\sharp$ to G $\sharp$) and hear the chord sound through.

Bebop dominant

Formula	Notes in G	Family
1 2 3 4 5 6 $\flat$ 7 7	G A B C D E F F $\sharp$	Symmetric and bebop

Sound. Mixolydian plus a passing major 7th (F $\sharp$ in G), so a descending eighth-note line starting on a chord tone puts chord tones on the beats. **Use.** Bebop and swing lines over dominant 7th chords and the V7 (and its ii) in a ii – V. **Chords.** G7, G9, G13; the vamp Dm7 to G7 **Heard in.** Charlie Parker; Dizzy Gillespie

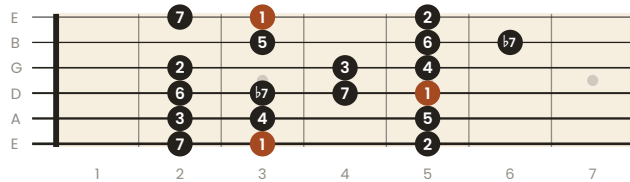


TRACK 551 Bebop dominant practice loop · LOOP · jazz swing · 140 bpm

Dm7 v7	G7 I7	Dm7 v7	G7 I7
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IMPROVISE WITH

G bebop dominant



Start on G on beat 1 and run down the scale in eighth notes: G, F $\sharp$, F, E, D, C, B, A; G, F, D and B fall on the beats.

APPENDIX E

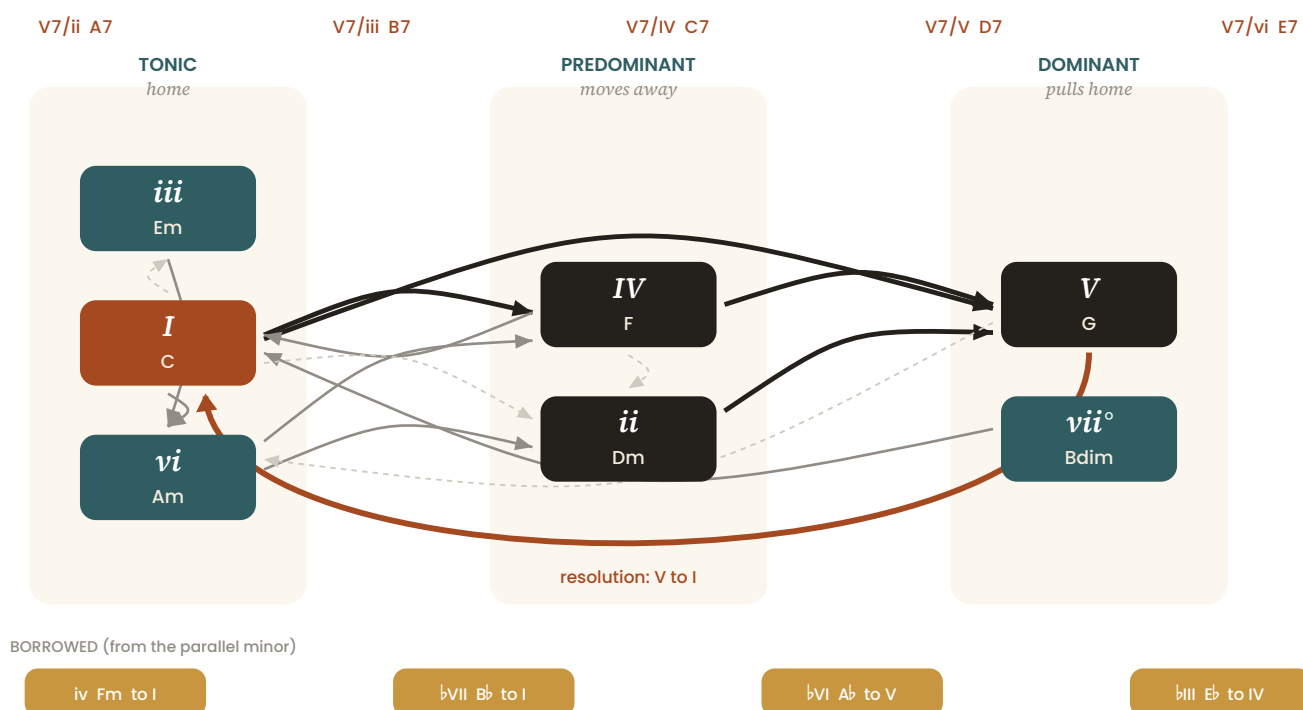
Progression Maps

E

Each map shows the chords of one key arranged by what they do. Tonic chords feel at home, predominant chords move away, dominant chords pull back home. Thick arrows are the main road (I – IV – V – I, ii – V – I); thin arrows are common moves; dotted arrows add colour. Above each map are the secondary dominants that lead into each chord (Chapter 32); below it are chords borrowed from the parallel key. Read a progression you know as a route on the map, then invent your own routes.

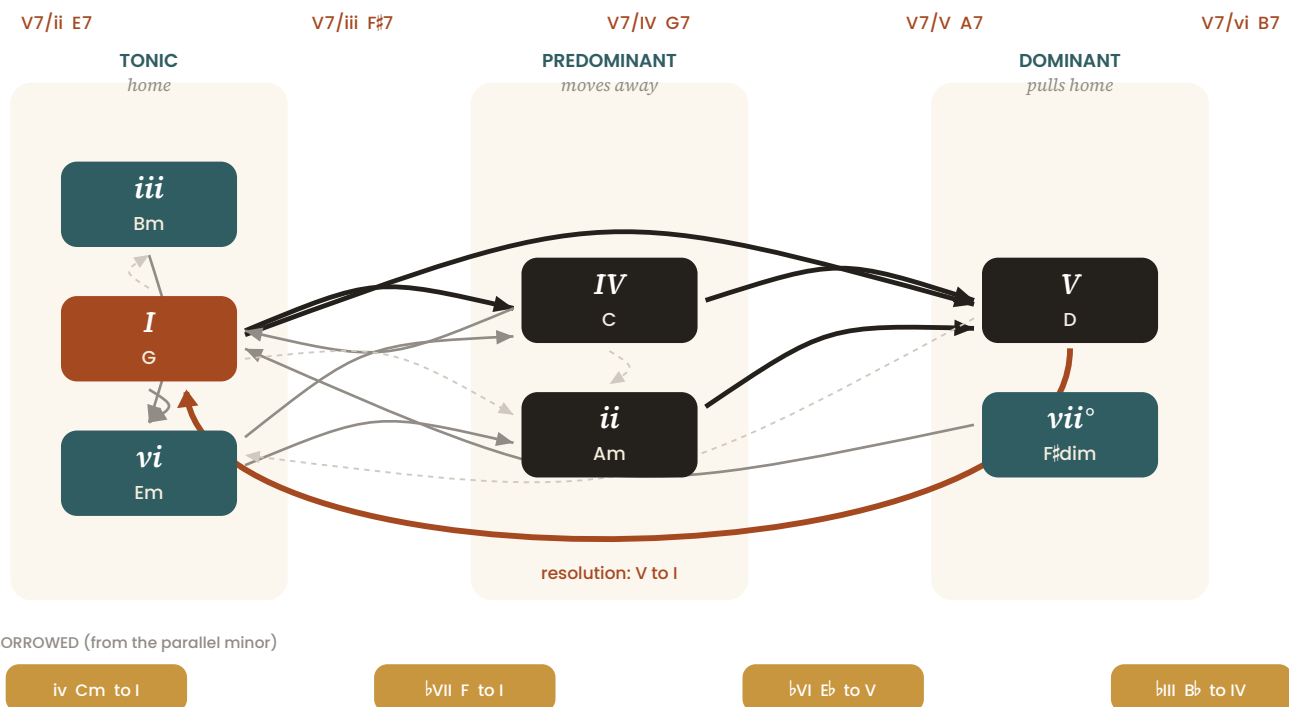
C major

SECONDARY DOMINANTS: each one leads to the chord it points at



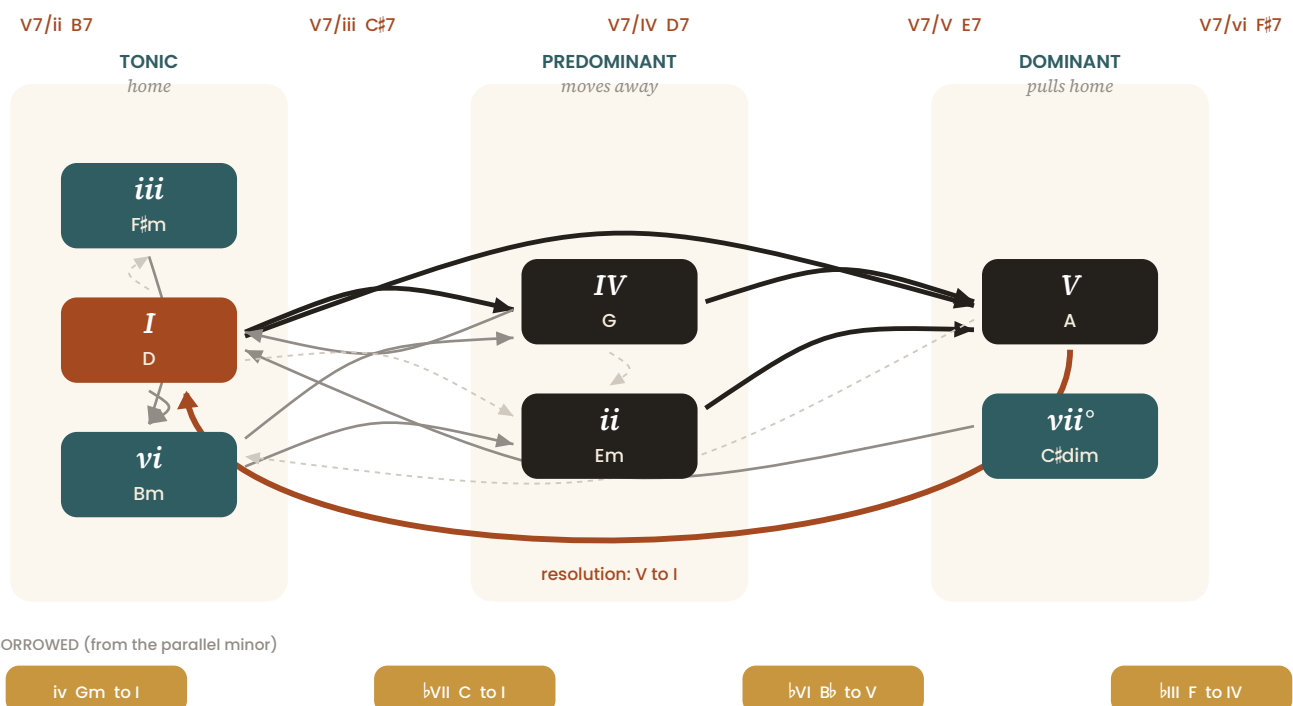
G major

SECONDARY DOMINANTS: each one leads to the chord it points at



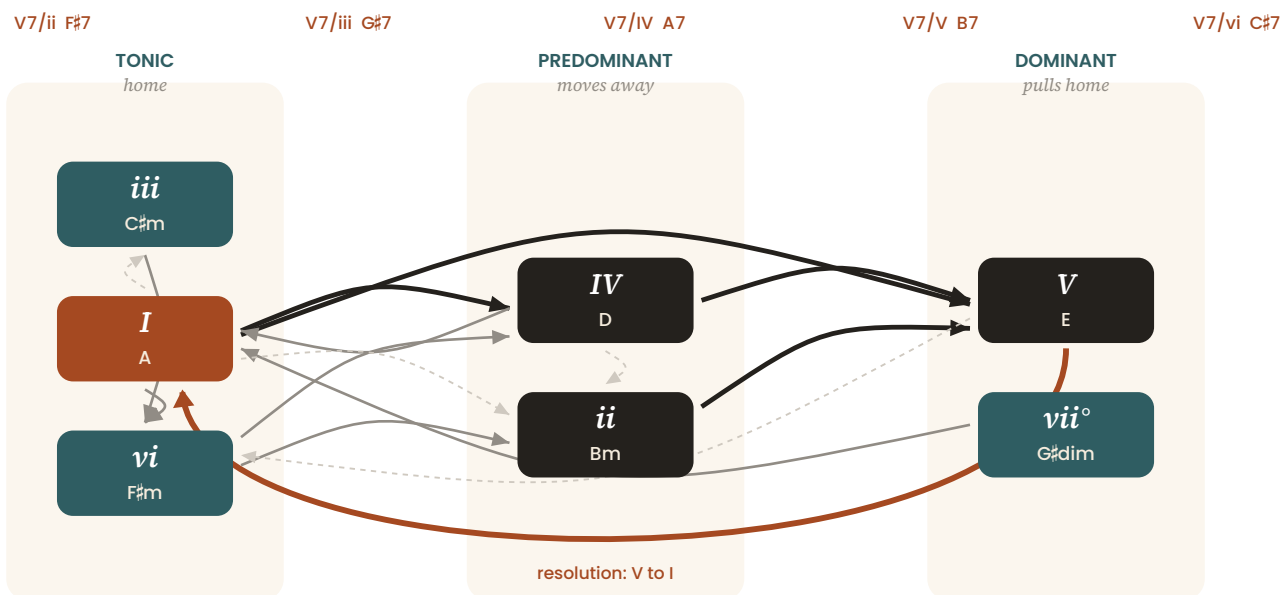
D major

SECONDARY DOMINANTS: each one leads to the chord it points at



A major

SECONDARY DOMINANTS: each one leads to the chord it points at



BORROWED (from the parallel minor)

iv Dm to I

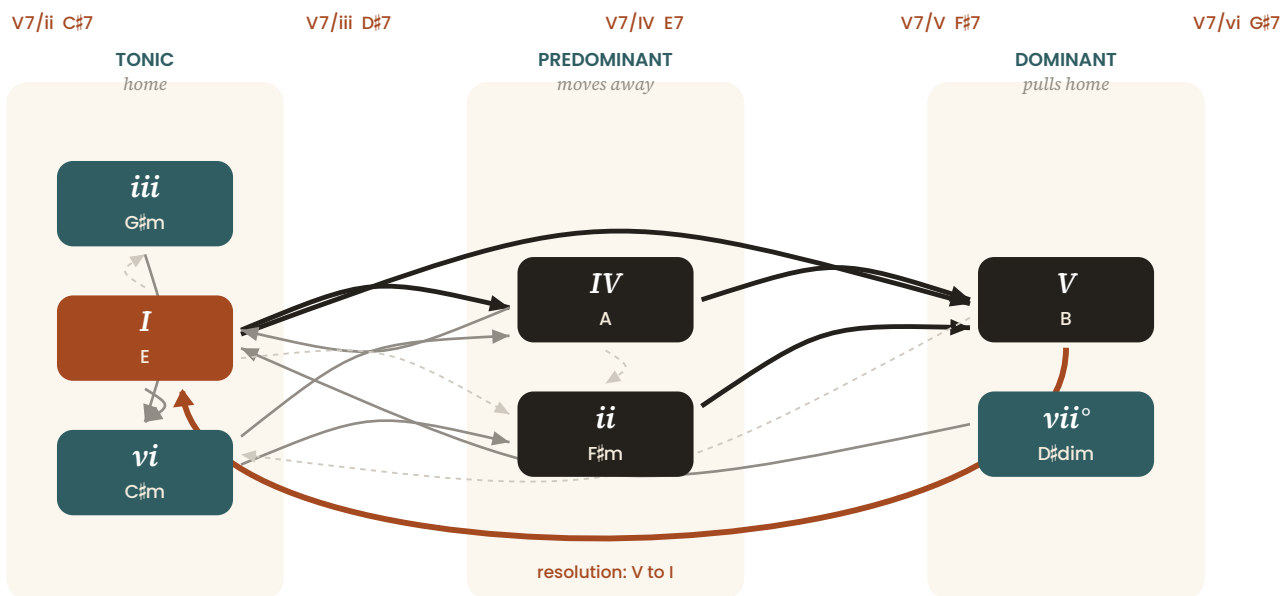
bVII G to I

bVI F to V

bIII C to IV

E major

SECONDARY DOMINANTS: each one leads to the chord it points at



BORROWED (from the parallel minor)

iv Am to I

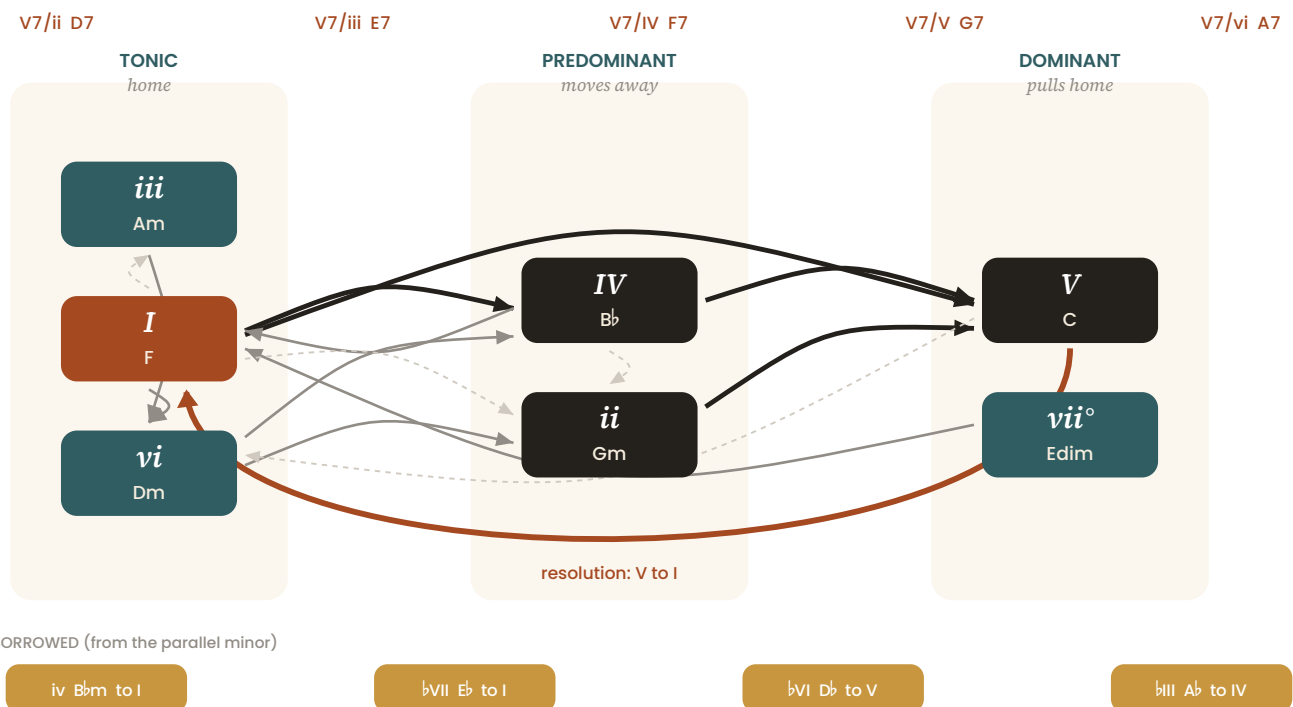
bVII D to I

bVI C to V

bIII G to IV

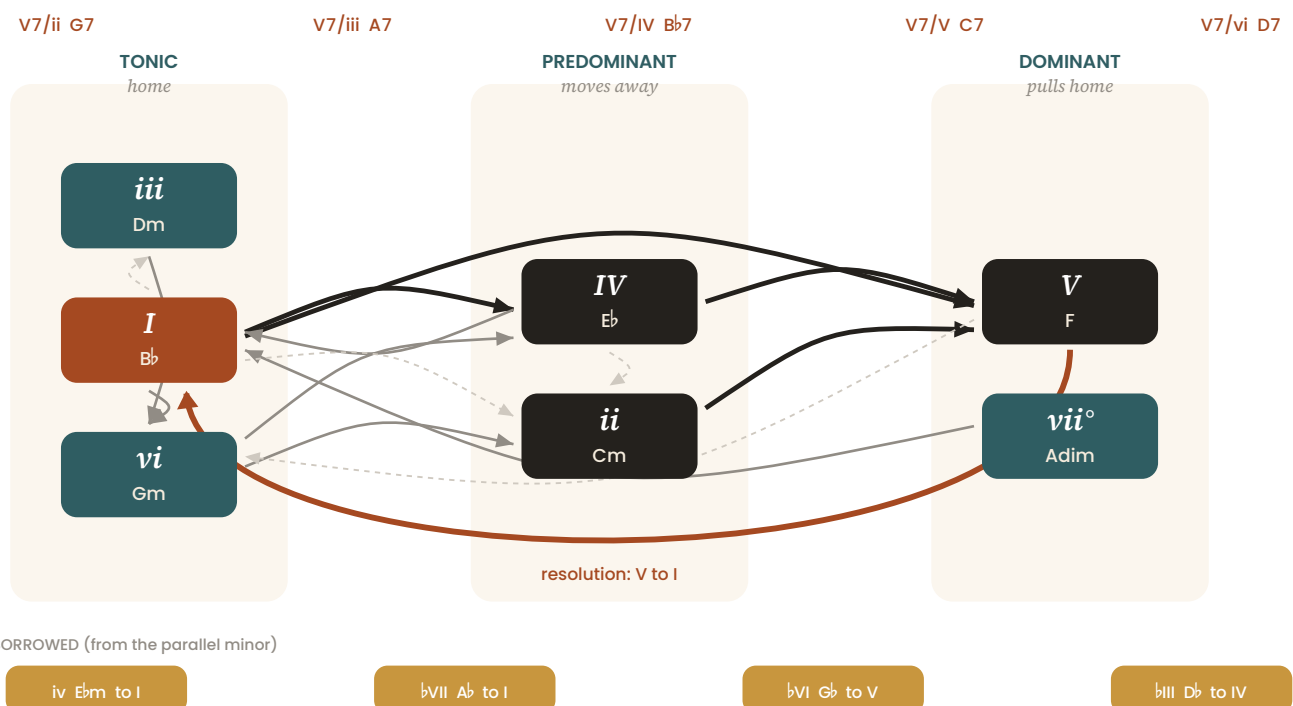
F major

SECONDARY DOMINANTS: each one leads to the chord it points at



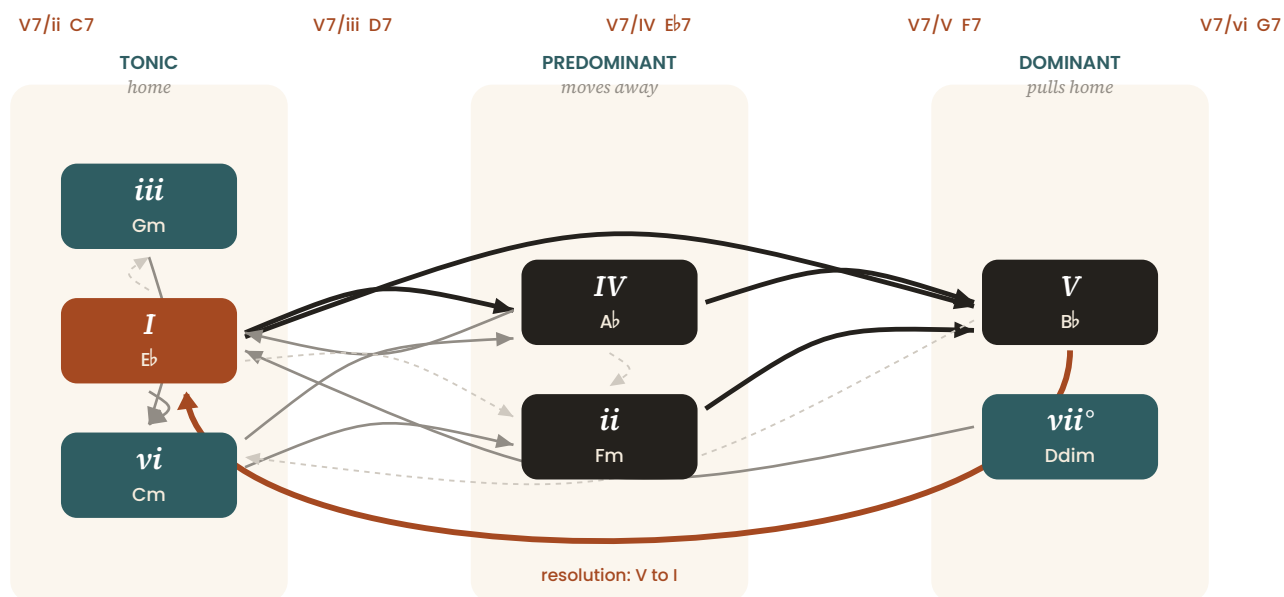
B $\flat$ major

SECONDARY DOMINANTS: each one leads to the chord it points at



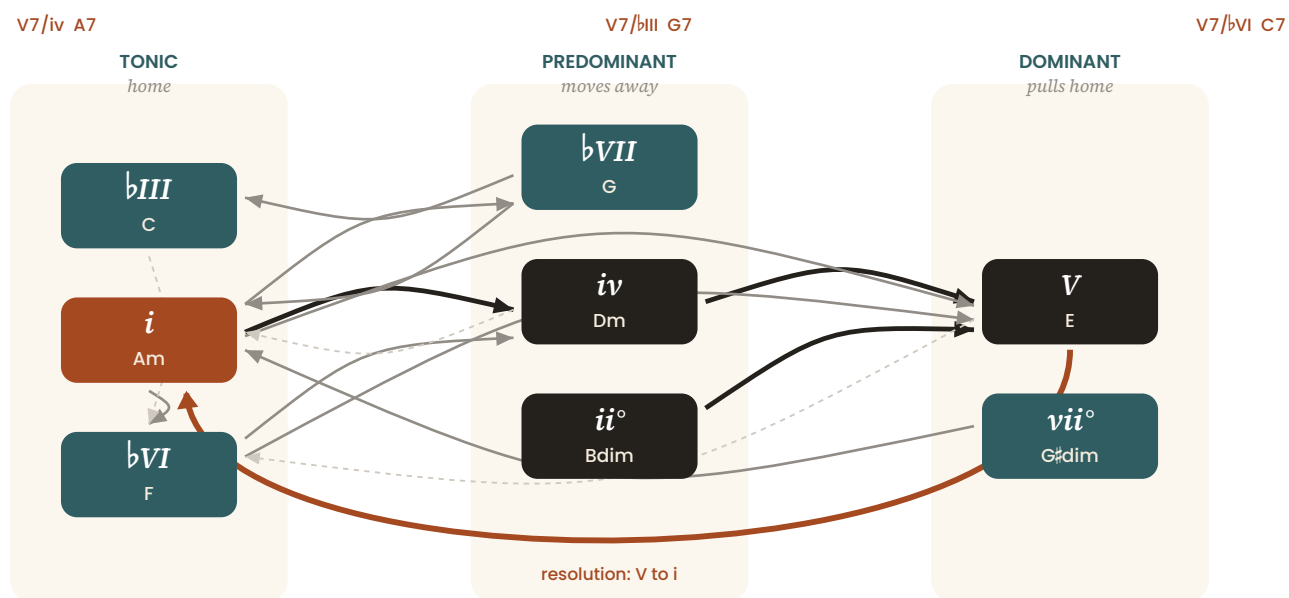
E♭ major

SECONDARY DOMINANTS: each one leads to the chord it points at



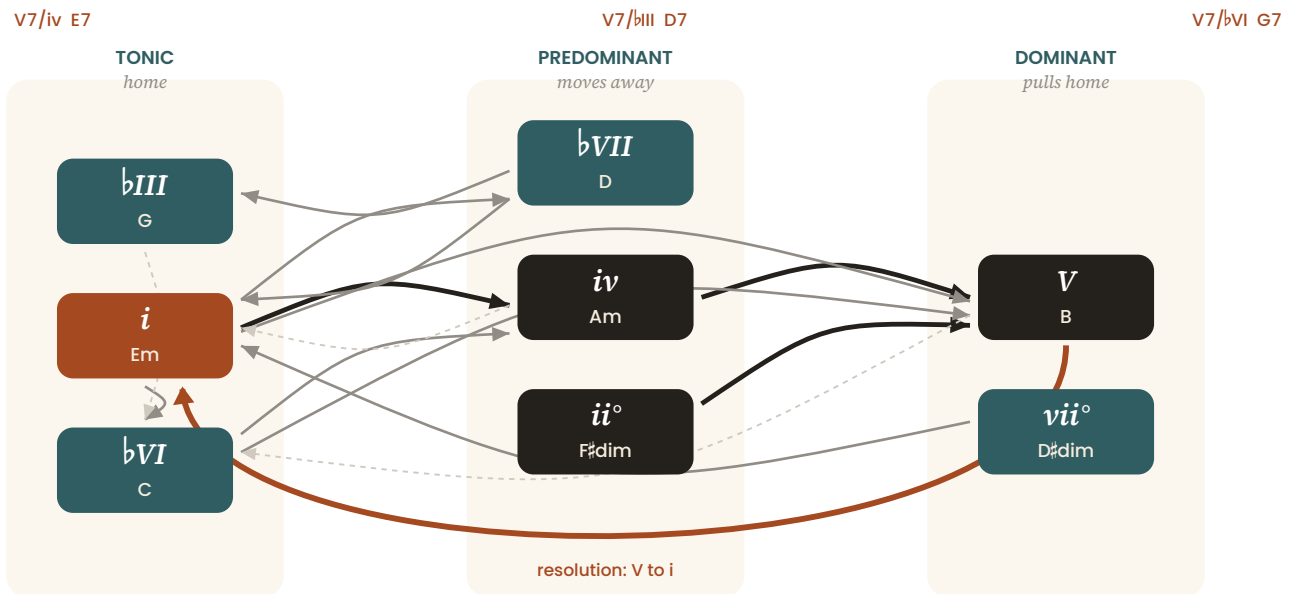
A minor

SECONDARY DOMINANTS: each one leads to the chord it points at



E minor

SECONDARY DOMINANTS: each one leads to the chord it points at



COLOUR CHORDS: Dorian IV, Neapolitan bII, Picardy I (as V of iv)

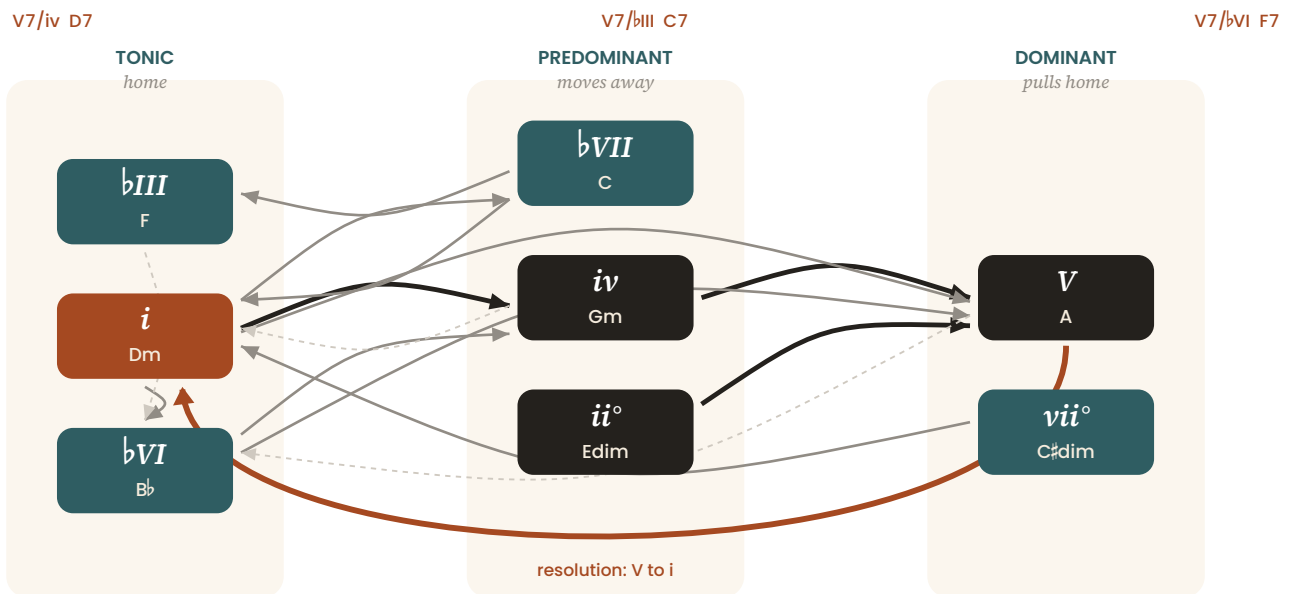
IV A to i

bII F to V

I E to iv

D minor

SECONDARY DOMINANTS: each one leads to the chord it points at



COLOUR CHORDS: Dorian IV, Neapolitan bII, Picardy I (as V of iv)

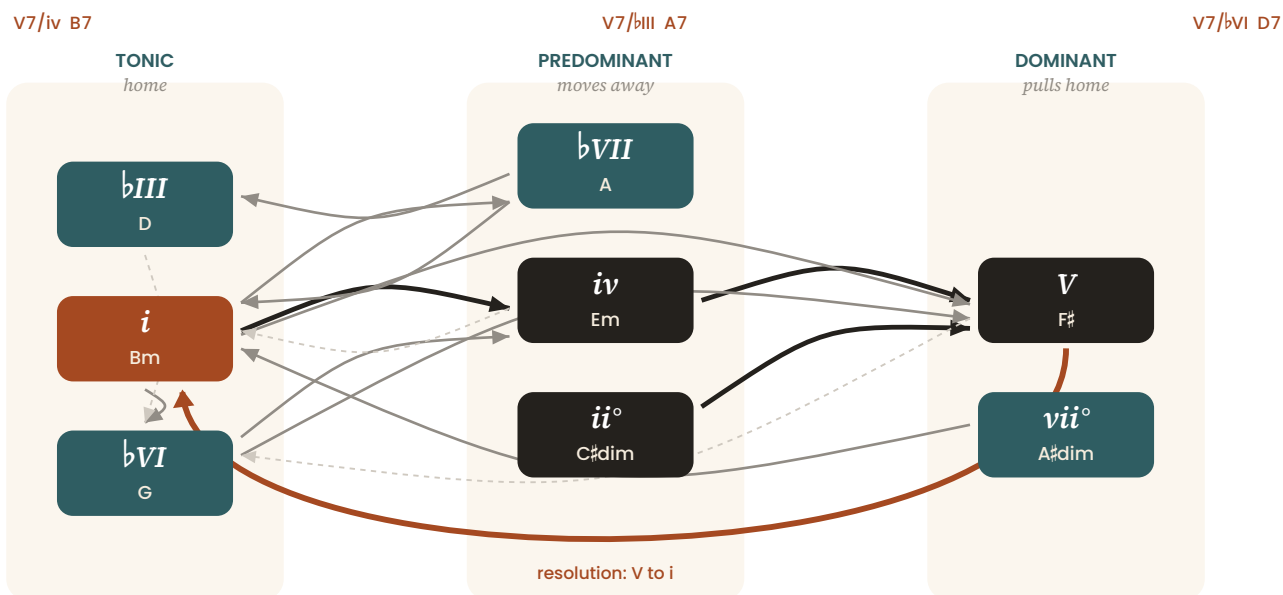
IV G to i

bII Eb to V

I D to iv

B minor

SECONDARY DOMINANTS: each one leads to the chord it points at



COLOUR CHORDS: Dorian IV, Neapolitan bII, Picardy I (as V of iv)

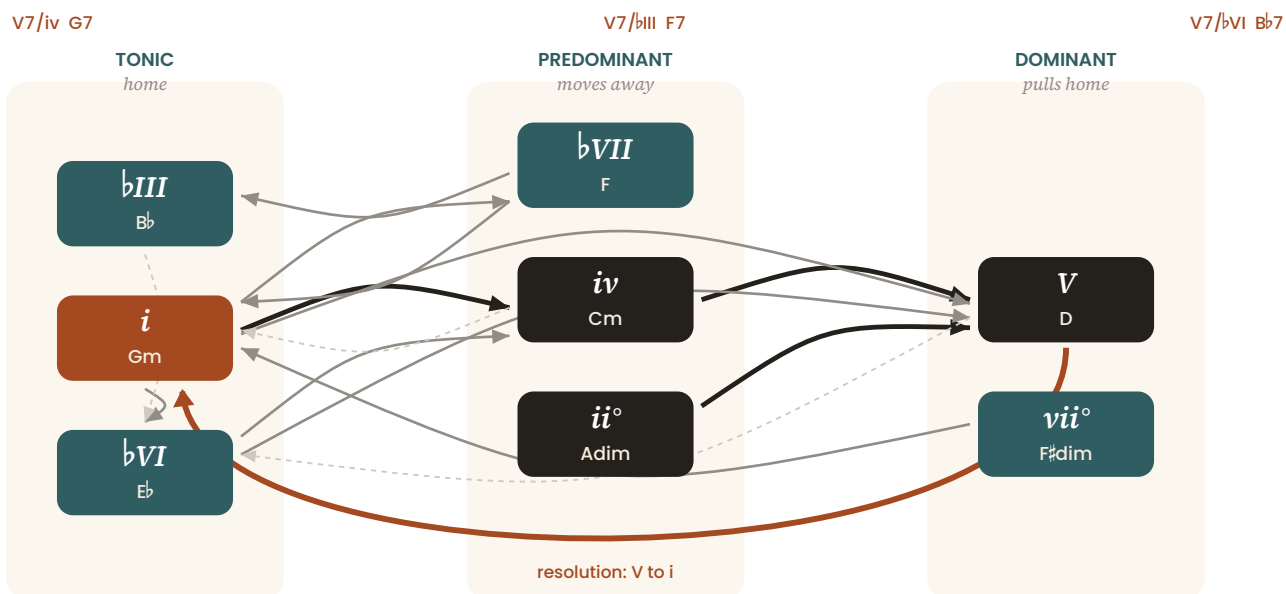
IV E to i

bII C to V

I B to iv

G minor

SECONDARY DOMINANTS: each one leads to the chord it points at



COLOUR CHORDS: Dorian IV, Neapolitan bII, Picardy I (as V of iv)

IV C to i

bII Ab to V

I G to iv

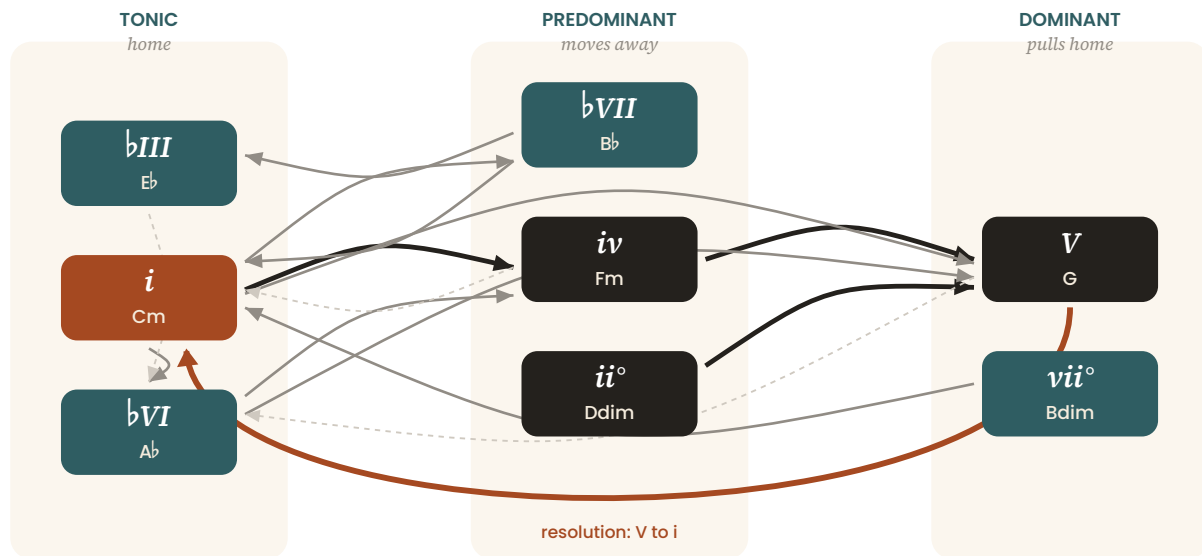
C minor

SECONDARY DOMINANTS: each one leads to the chord it points at

$V7/iv$ $C7$

$V7/bIII$ $Bb7$

$V7/bVI$ $Eb7$



COLOUR CHORDS: Dorian IV, Neapolitan bII , Picardy I (as V of iv)

IV F to i

bII Db to V

I C to iv

APPENDIX F

Glossary



Short definitions of the terms used in this book, in alphabetical order. The chapter named after a definition explains the idea in full.

Term	Meaning
Accidental	A sharp, flat or natural sign that raises, lowers or restores a note for the rest of the bar (Chapter 26).
Action	The height of the strings above the frets. A low action is easier to play but can buzz.
Aeolian	The natural minor scale: the mode starting on the 6th degree of the major scale (Chapter 9).
Alternate picking	Strict down-up picking, with downstrokes on the beat and upstrokes between beats (Chapter 24).
Alternating bass	A thumb pattern that alternates between the root and another chord tone on each beat (Chapter 19).
Alzapúa	Flamenco thumb technique that combines a single bass note with down- and upstrokes across several strings, in a fast repeating cycle (Chapter 53).
Anacrusis	One or more notes before the first full bar; also called a pickup or upbeat.
Anchor notes	Fretted notes that stay in place while the other fingers change chords, such as D and G at the 3rd fret of the top two strings under G, Em7, Cadd9 and Dsus4 (Chapter 15).
Andalusian cadence	The descending progression i – bVII – bVI – V, for example Am – G – F – E (Chapter 14).
Apoyando	Rest stroke: the finger plucks through the string and comes to rest on the next lower-pitched string.
Approach chord	A chord a half step above or below a target chord, usually of the same shape, played just before it (Chapter 15, Chapter 34).
Arpeggio	The notes of a chord played one after another instead of together (Chapter 16).
Articulation	How a note is attacked and released: legato, staccato, accented, muted.
Augmented	A triad with a major 3rd and a raised 5th (1 3 #5), or an interval one half step larger than perfect or major.
Backbeat	Accents on beats 2 and 4 of a 4/4 bar, the basis of most rock, pop and soul (Chapter 37).
Backdoor ii – V	The progression iv7 – bVII7 – I (Fm7 – Bb7 – C in C), a softer route to the tonic than ii – V – I (Chapter 14).
Baixaria	In choro, the counter-melody played in the bass register, traditionally on the seven-string guitar (Chapter 51).
Barre	One finger pressing several strings at the same fret.
Batida	The bossa nova guitar rhythm: thumb bass on the beats with syncopated finger chords (Chapter 50).
Bend	Pushing or pulling a string across the fretboard to raise its pitch (Chapter 36).
Blue note	A note bent or sung between major and minor: the b3, b5 and b7 heard against a major chord in blues. The b5 is the blue note added to the minor pentatonic to make the blues scale (Chapter 7).
Borrowed chord	A chord taken from the parallel key, such as iv or bVI in a major key; also called modal interchange (Chapter 32).
Box	A hand-sized scale pattern, usually one of the five pentatonic positions (Chapter 7).
Bulerías	A fast flamenco form in a 12-beat compás, usually counted from beat 12.
Cadence	A chord progression that closes a phrase: authentic (V – I), plagal (IV – I), half (ending on V) or deceptive (V – vi).

Term	Meaning
CAGED	A system that links five open chord shapes (C, A, G, E, D) to cover the whole neck (Chapter 12).
Call and response	A phrase answered by a second phrase, often on another instrument or in another register.
Campanella	A classical technique in which successive scale notes are played on different strings so that they ring into each other like bells.
Capo	A clamp across the strings that raises the pitch of all open strings by the same number of frets.
Chank	A short, muted chord stab in funk, often on the off-beat (Chapter 45).
Characteristic note	The note that sets a mode apart from the major or natural minor scale on the same root, such as the #4 of Lydian (Chapter 9).
Chord box	A diagram of one chord shape with the neck upright, low E on the left (Chapter 2).
Chord melody	A solo arrangement with the melody as the top note of each chord (Chapter 48).
Chord tone	A note that belongs to the current chord: root, 3rd, 5th or 7th.
Chromatic	Moving by half steps, or using notes outside the key.
Circle of fifths	The twelve keys arranged so that each is a perfect 5th from the next; neighbouring keys differ by one sharp or flat (Chapter 5).
Clave	The two-bar rhythmic key pattern of Afro-Cuban music, in 3-2 or 2-3 form (Chapter 50).
Close position	A chord voicing whose notes all fit within one octave (Chapter 10).
Common tone	A note shared by two chords, often held over the chord change (Chapter 34).
Compás	The rhythmic cycle of a flamenco form, often 12 beats with a fixed accent pattern (Chapter 53).
Comping	Accompanying a soloist or singer with chords, especially in jazz.
Compound interval	An interval larger than an octave, such as the 9th, 11th or 13th (Chapter 3).
Contrary motion	Two voices moving in opposite directions (Chapter 34).
Counterpoint	Two or more independent melodic lines played together.
Cut	In Celtic music, a very short grace note above the main note, used to separate repeated notes (Chapter 54).
DADGAD	The tuning D A D G A D, low to high, central to Celtic guitar (Chapter 25).
Dead note	A string muted by the fretting hand and struck for a percussive click; written x in TAB.
Diagonal pattern	A scale shape with three notes on every string, which climbs the neck as it crosses the strings; for the pentatonic, about one box per string (Chapter 8).
Diatonic	Using only the notes of the key.
Diminished	A triad with a minor 3rd and a flat 5th (1 b3 b5), or an interval one half step smaller than perfect or minor.
Dominant	The 5th degree of a key, or the chord built on it (V or V7), which leads strongly back to the tonic.
Dorian	The minor mode with a major 6th: 1 2 b3 4 5 6 b7 (Chapter 9).
Double stop	Two notes played together, usually on adjacent strings.
Drone	A sustained or repeated note under changing harmony or melody.
Drop-2 voicing	A four-note chord made by dropping the second-highest note of a close-position chord by an octave (Chapter 48).
Drop D	Standard tuning with the 6th string lowered to D.
Dynamics	The volume of the music, marked from pp (very soft) to ff (very loud).
Economy picking	Picking that uses a single sweep when crossing to the next string in the same direction (Chapter 24).
Enharmonic	Two names for the same pitch, such as F# and Gb.
Extension	A chord tone above the 7th: the 9th, 11th or 13th; also called a tension.
Falseta	A melodic instrumental passage in flamenco, played between sung verses.

Term	Meaning
Fill	A short phrase that fills a gap between vocal or melodic phrases.
Free stroke	Tirando: the finger plucks and moves up into the palm, clear of the next string.
Fretboard diagram	A diagram of the neck lying flat, high E at the top, showing many notes at once (Chapter 2).
Function	The role of a chord in a key: tonic, subdominant or dominant (Chapter 13).
Ghost note	A very soft, often muted note that adds rhythm more than pitch (Chapter 45).
Golpe	A tap of the ring finger on the guitar top, part of the rhythm in flamenco.
Groove	The repeating rhythmic feel of a piece, created by the interplay of the parts.
Guide tones	The 3rd and 7th of a chord; they define its quality and move smoothly to the next chord (Chapter 35).
Half-diminished	The m7b5 chord (1 b3 b5 b7), also written ø7; the ii chord of a minor key (Chapter 11).
Half step	The smallest interval in Western music: one fret on the guitar.
Hammer-on	Sounding a note by bringing a fretting finger down hard on the string, without picking it.
Harmonic	A bell-like tone produced by touching a string lightly over a node, such as the 12th, 7th or 5th fret (Chapter 36).
Harmonic minor	The natural minor scale with a raised 7th: 1 2 b3 4 5 b6 7 (Chapter 6).
Hemiola	A rhythm that groups the beats in threes against twos, or twos against threes.
Highlife	West African popular music, originally from Ghana, with interlocking guitar lines over a steady pulse (Chapter 56).
Hybrid picking	Using a pick and the free fingers of the picking hand together (Chapter 46).
Interval	The distance between two notes, named by number (3rd, 5th) and quality (major, minor, perfect).
Inversion	A chord with a note other than the root in the bass, such as C/E (Chapter 10).
Ionian	The major scale considered as a mode.
Key	The scale and tonic a piece is based on.
Key signature	The sharps or flats at the start of each staff, which apply to every note of that name.
La pompe	The gypsy-jazz rhythm guitar pattern: a short chord stroke on every beat, accented on 2 and 4, often with a quick upstroke before the beat (Chapter 49).
Leading note	The 7th degree of the major or harmonic minor scale, a half step below the tonic.
Ledger line	A short line above or below the staff for notes outside its range.
Legato	Smoothly connected notes; on guitar, often played with hammer-ons, pull-offs and slides.
Line cliché	One voice moving by half steps over a held chord, such as Am – AmMaj7/G# – Am7/G – D/F# (Chapter 14).
Locrian	The mode starting on the 7th degree of the major scale: 1 b2 b3 4 b5 b6 b7.
Lydian	The major mode with a raised 4th: 1 2 3 #4 5 6 7.
Lydian dominant	The 4th mode of melodic minor: 1 2 3 #4 5 6 b7, used over 7#11 chords.
Melodic minor	The minor scale with a raised 6th and 7th: 1 2 b3 4 5 6 7. In classical use it reverts to natural minor when descending.
Metronome	A device that clicks at a set tempo, in beats per minute (bpm).
Mixolydian	The major mode with a flat 7th: 1 2 3 4 5 6 b7.
Mode	A scale formed by starting a parent scale on a different degree (Chapter 9).
Modulation	A change of key within a piece (Chapter 33).
Motif	A short musical idea that is repeated and developed.
Natural minor	The minor scale formed from the 6th degree of the major scale: 1 2 b3 4 5 b6 b7.
Natural note	One of the seven notes A B C D E F G, without a sharp or flat (Chapter 3).
Neo-soul	A style of R&B whose guitar parts embellish chords with hammer-ons, double stops and small melodic fills (Chapter 55).

Term	Meaning
Octave	The interval of 12 half steps; the same note name at double or half the frequency.
One drop	A reggae drum pattern with the bass drum and snare on beat 3 and nothing on beat 1 (Chapter 52).
Open tuning	A tuning in which the open strings form a chord, such as Open G or Open D (Chapter 25).
Ostinato	A short pattern repeated throughout a passage.
Palm muting	Resting the side of the picking hand on the strings near the bridge to shorten and darken the notes.
Parallel key	The major and minor keys with the same tonic, such as C major and C minor.
Passing chord	A chord that connects two others, usually by moving the bass by step, such as F#dim7 between F and G (Chapter 14).
Passing note	A non-chord note that connects two chord tones by step.
Pedal point	A note held or repeated in one voice while the harmony changes above or below it.
Pentatonic	A five-note scale; the major and minor pentatonic are the most common (Chapter 7).
Phrygian	The minor mode with a flat 2nd: 1 b2 b3 4 5 b6 b7.
Phrygian dominant	The 5th mode of harmonic minor: 1 b2 3 4 5 b6 b7, used over V7 in minor keys and in flamenco.
Picado	Flamenco scale playing with alternating i and m rest strokes.
Picking map	A grid that shows which picking-hand finger plucks which string on each slot of a bar (Chapter 2).
Pivot chord	A chord that belongs to both keys in a modulation and connects them (Chapter 33).
Position	The fret at which the 1st finger sits; a region of the neck.
Position shift	Moving the whole fretting hand to a new position: by a slide, a stretch, an open string or a guide finger (Chapter 8).
Power chord	The root and 5th, often with the octave, written C5 (Chapter 11).
Practice loop	A recorded backing of drums, bass and chords that repeats without a gap, for comping and improvising; marked with a track number in this book (Chapter 2).
Progression	A sequence of chords (Chapter 14).
Pull-off	Sounding a lower note by pulling the fretting finger off the string with a slight sideways flick.
Push	A chord played an eighth note early, on the "&" of 4, and tied over the barline; also called an anticipation (Chapter 37, Chapter 15).
Quick change	In a 12-bar blues, moving to IV in bar 2 and back to I in bar 3 (Chapter 43).
Rasgueado	Flamenco strumming with the fingers flicked out one after another (Chapter 53).
Relative key	The major and minor keys that share the same notes, such as C major and A minor.
Resolution	The movement from a dissonant or unstable note or chord to a stable one.
Rest stroke	See apoyando.
Riff	A short repeated instrumental phrase that defines a song (Chapter 44).
Roman numerals	Symbols that name chords by their scale degree (I, ii, V7), independent of key (Chapter 13).
Root	The note a chord or scale is named after.
Scale degree	The position of a note within a scale, numbered from 1 (the tonic).
Secondary dominant	A dominant chord that resolves to a chord other than the tonic, such as A7 to Dm in C, written V7/ii (Chapter 32).
Shell voicing	A jazz chord of root, 3rd and 7th only.
Shuffle	A rhythm of long-short pairs on the triplet grid, typical of blues.
Skank	The short, off-beat chord chop of reggae guitar on beats 2 and 4 (Chapter 52).
Slash chord	A chord with a specified bass note, such as D/F#.
Slide	Moving a fretting finger along the string to a new fret while the note sounds.
Slur	A curved line joining notes played legato; on guitar, usually a hammer-on or pull-off.

Term	Meaning
Soukous	Congolese dance music with fast, high, interlocking guitar lines (Chapter 56).
Spice	A small change to a progression that keeps it recognisable but adds colour, motion or tension, such as a sus chord, a passing chord or a push (Chapter 15).
Staccato	Short, detached notes, marked with a dot above or below the note head.
Standard notation	Music written on the five-line staff, showing pitch and rhythm exactly (Chapter 26).
Strum grid	A diagram of a strumming pattern: down- and up-strums, misses, accents and chucks on a beat grid (Chapter 2).
Subdominant	The 4th degree of a key, or the chord built on it (IV); with ii, the chords that lead away from the tonic (Chapter 13).
Sus chord	A chord in which the 3rd is replaced by the 2nd (sus2) or the 4th (sus4).
Sweep picking	Playing one note per string with one continuous pick stroke across the strings, usually for arpeggios (Chapter 16).
Swing	Eighth notes played long-short, as the first and last notes of a triplet.
Syncopation	Accenting weak beats or off-beats, or placing notes between the beats.
TAB	Tablature: six lines for the six strings, with numbers showing the frets.
Tempo	The speed of the music, in beats per minute (bpm).
Three-notes-per-string pattern	A scale fingering with exactly three notes on every string; a major scale has seven of them (Chapter 5).
Thumb slap	A percussive strike of the thumb against the bass strings so that they hit the frets.
Tie	A curved line joining two notes of the same pitch into one longer note.
Time signature	The two numbers at the start of a piece: beats per bar and the note value of one beat (Chapter 27).
Tirando	Free stroke: the finger plucks and moves clear of the next string.
Tonic	The home note of a key, its 1st degree, and the chord built on it (I or i).
Transcription	Working out music from a recording by ear and writing it down (Chapter 30).
Transpose	To move music to a different key while keeping its intervals.
Travis picking	A fingerstyle technique in which the thumb alternates bass notes on every beat while the fingers play melody (Chapter 19).
Tremolo (classical)	A fingerstyle technique (p a m i) in which one melody note is repeated rapidly above a bass line.
Tremolo picking	Very fast repeated picking of one note with a pick.
Tresillo	The 3 + 3 + 2 rhythm in eighth notes, found in Latin, African and popular music.
Triad	A three-note chord built in 3rds: root, 3rd and 5th (Chapter 10).
Triplet	Three notes in the time of two, marked with a 3.
Tritone	The interval of six half steps (augmented 4th or diminished 5th); the unstable interval inside every dominant 7th.
Tritone substitution	Replacing a dominant 7th chord with the dominant 7th a tritone away, such as D $\flat$ 7 for G7 (Chapter 48).
Turnaround	A short progression at the end of a section that leads back to the start, such as I – vi – ii – V.
Twelve-bar blues	The standard blues form: four bars of I, two of IV, two of I, then V, IV, I, V (Chapter 43).
Two-string box	A pentatonic box cut down to two neighbouring strings: two notes on each, repeated up the neck in box order (Chapter 8).
Vamp	One or two chords repeated, used to set up a mode or groove for improvising (Chapter 9).
Vibrato	A regular wobble in pitch that makes a note sing (Chapter 36).
Voice leading	Moving each note of a chord smoothly to a note of the next chord (Chapter 34).
Voicing	The specific arrangement and spacing of a chord's notes on the guitar.
Walking bass	A bass line that moves in steady quarter notes, mostly by step, connecting the chords.

Term	Meaning
Whole step	Two half steps: two frets on the guitar.

APPENDIX G

Graded Repertoire



A list of real pieces and songs to learn or listen to, grouped by level and style. Each entry names the skills it uses and the chapters that explain them. No music is printed here; traditional and classical pieces are easy to find in free editions, and songs in published songbooks.

Learn one piece at each level for every few chapters you study. When a piece is comfortable, analyse it with a worksheet from Appendix H before moving on: finding the theory inside music you can already play is the fastest way to make it your own.

Levels use the same grades as the study pieces: Level 1 (grades 1–2) matches Chapter 57, Level 2 (grades 3–4) the first pieces of Chapter 58, Level 3 (grades 5–6) the grade 5 pieces of Chapter 58 and the grade 6 pieces of Chapter 59, and Level 4 (grade 7 and above) the last pieces of Chapter 59. Pieces marked *trad.* are traditional; songs by modern writers are named only as places to hear or learn an idea.

Level 1: first pieces (grades 1–2)

Open chords, simple strumming and arpeggios, melodies in first position.

Piece	Style	Skills	Chapters
Ode to Joy (Beethoven), melody	Classical	First-position notes, reading quarter notes	Chapter 26, Chapter 57
When the Saints Go Marching In (trad.)	Folk, jazz	I – IV – V in G or C, melody reading	Chapter 14, Chapter 27
Amazing Grace (trad. melody "New Britain")	Folk, gospel	3/4 strum or p-i-m pattern, I – IV – V	Chapter 22, Chapter 18
Drunken Sailor (trad.)	Folk	Dorian minor: Dm – C	Chapter 9
Carulli: short pieces from the Méthode, Op. 241	Classical	Free strokes, p-i-m arpeggios in C and A minor	Chapter 17, Chapter 18
Sor: lessons from Op. 60	Classical	Two voices in first position	Chapter 20
Knockin' on Heaven's Door (Bob Dylan)	Folk rock	G – D – Am and G – D – C, simple strumming	Chapter 11, Chapter 22
Three Little Birds (Bob Marley)	Reggae	A – D – E with the off-beat skank	Chapter 52
A twelve-bar shuffle in E or A	Blues	Root–5th–6th shuffle riff, 12-bar form	Chapter 43, Chapter 7

Level 2: building blocks (grades 3–4)

Fingerpicking patterns, minor keys, the first barre chords, alternating bass.

Piece	Style	Skills	Chapters
House of the Rising Sun (trad.)	Folk	6/8 arpeggio, Am – C – D – F and the major V (E)	Chapter 18, Chapter 31
Scarborough Fair (trad.)	Folk	Dorian melody, 3/4 arpeggios	Chapter 9
Greensleeves (trad.), complete	Folk	Melody over bass, harmonic and melodic minor	Chapter 6, Chapter 20
The Water Is Wide (trad.)	Folk	Travis picking, slash chords	Chapter 19, Chapter 10
Freight Train (Elizabeth Cotten)	Folk blues	Alternating thumb in C, melody on top	Chapter 19
Giuliani: right-hand studies, Op. 1 (first studies)	Classical	p-i-m-a patterns on C and G7	Chapter 18
Carcassi: Étude Op. 60 No. 1	Classical	Scales in C major and position shifts	Chapter 5, Chapter 4

Piece	Style	Skills	Chapters
Smoke on the Water riff (Deep Purple)	Rock	Parallel 4ths or power chords, palm muting	Chapter 44
Sweet Home Chicago (Robert Johnson)	Blues	Shuffle in E, turnaround	Chapter 43
Minor Swing (Django Reinhardt, Stéphane Grappelli)	Gypsy jazz	Am – Dm – E7 with la pompe	Chapter 49

Level 3: intermediate (grades 5–6)

Barre chords in all positions, independent bass, open tunings, syncopation, extended chords.

Piece	Style	Skills	Chapters
Spanish Romance (anon.)	Classical	Triplet arpeggios, barres, E minor to E major	Chapter 31, Chapter 12
Tárrega: Lágrima	Classical	Melody over bass, E major and E minor	Chapter 20, Chapter 33
Tárrega: Adelita	Classical	Expressive slurs, E minor and E major	Chapter 36
Bach: Bourrée in E minor, BWV 996	Classical	Two-part counterpoint	Chapter 34
Wildwood Flower (Carter Family style)	Country	Thumb melody with brushed chords	Chapter 46
Windy and Warm (John D. Loudermilk)	Country fingerstyle	Travis picking with a syncopated melody	Chapter 19
She Moved Through the Fair (trad.)	Celtic	DADGAD drones, modal melody	Chapter 25, Chapter 54
Carolan: Sí Bheag, Sí Mhór	Celtic	Melody in 3/4 with a bass line, often played in DADGAD	Chapter 54
St. James Infirmary (trad.)	Blues, jazz	Minor blues, the major V chord	Chapter 31
The Girl from Ipanema (Antônio Carlos Jobim)	Bossa nova	Batida, maj7 and 7 chords, a bridge that changes key	Chapter 50, Chapter 33
Corcovado (Antônio Carlos Jobim)	Bossa nova	Chromatic chord movement, extended voicings	Chapter 50, Chapter 34
Tico-Tico no Fubá (Zequinha de Abreu)	Choro	Fast single-note lines in minor and major	Chapter 51
Little Wing (Jimi Hendrix)	Rock, neo-soul	Chord embellishments with hammer-ons and double stops	Chapter 55
Papa's Got a Brand New Bag (James Brown; guitar Jimmy Nolen)	Funk	Sixteenth-note scratching on one chord	Chapter 45
Sweet Mother (Prince Nico Mbarga)	Highlife	Interlocking single-note patterns	Chapter 56

Level 4: advanced (grades 7 and above)

Tremolo, full-neck positions, complex harmony, concert repertoire.

Piece	Style	Skills	Chapters
Sor: Study in B minor, Op. 35 No. 22	Classical	Arpeggios with melody, minor-key harmony	Chapter 31, Chapter 16
Bach: Prelude in C minor, BWV 999 (D minor on guitar)	Classical	Continuous arpeggios, voice leading	Chapter 34, Chapter 16
Tárrega: Recuerdos de la Alhambra	Classical	Tremolo, A minor and A major	Chapter 21
Barrios: Una limosna por el amor de Dios	Classical	Tremolo with a moving bass	Chapter 21
Albéniz: Asturias (Leyenda), guitar arr.	Classical	Pedal tone, Phrygian colour, p-i alternation	Chapter 9, Chapter 53
Villa-Lobos: Prelude No. 1	Classical, Brazilian	Melody in the bass, parallel chord shapes	Chapter 20, Chapter 51

Piece	Style	Skills	Chapters
Joplin: The Entertainer, guitar arr.	Ragtime	Alternating bass, chromatic passing notes	Chapter 32, Chapter 19
Autumn Leaves (Joseph Kosma)	Jazz	ii – V – I in major and minor, chord melody	Chapter 48, Chapter 35
All the Things You Are (Jerome Kern)	Jazz	Modulation through several keys	Chapter 33, Chapter 35
Nuages (Django Reinhardt)	Gypsy jazz	Chromatic dominant chords, arpeggio soloing	Chapter 49
Entre dos aguas (Paco de Lucía)	Flamenco rumba	Picado, Phrygian harmony, rumba strum	Chapter 53
A soleá falseta from a flamenco method	Flamenco	12-beat compás, alzapúa, rasgueado	Chapter 53

Listening by style

Listen before you play. These recordings are reference points for the style workshops; each one shows the techniques of its chapter at a high level.

Style	Players to listen to	Listen for	Related chapter
Blues	Robert Johnson, B.B. King, Mississippi John Hurt	Shuffle feel, bends, call and response	Chapter 43
Rock	Jimmy Page, Angus Young, Keith Richards	Riffs, power chords, Open G tuning (Richards)	Chapter 44
Funk	Jimmy Nolen, Nile Rodgers	Sixteenth-note strumming, ghost notes	Chapter 45
Country and bluegrass	Merle Travis, Chet Atkins, Doc Watson	Alternating bass, crosspicking	Chapter 46
Folk and songwriting	Nick Drake, Joni Mitchell, Bert Jansch	Open tunings, fingerpicked accompaniment	Chapter 47
Jazz	Wes Montgomery, Joe Pass, Jim Hall	Chord melody, octaves, comping	Chapter 48
Gypsy jazz	Django Reinhardt	La pompe, arpeggio lines	Chapter 49
Bossa nova	João Gilberto, Baden Powell	Batida, extended voicings	Chapter 50
Choro	Dino 7 Cordas, Yamandu Costa	Seven-string bass lines, fast melodies	Chapter 51
Reggae	Bob Marley and the Wailers	Off-beat skank, one-drop groove	Chapter 52
Flamenco	Paco de Lucía, Sabicas, Tomatito	Picado, rasgueado, compás	Chapter 53
Celtic fingerstyle	Pierre Bensusan, Tony McManus	DADGAD, ornaments	Chapter 54
Neo-soul	Jimi Hendrix (ballads), Curtis Mayfield	Chord embellishments, double stops	Chapter 55
West and Central African	Koo Nimo, Franco (TPOK Jazz), Diblo Dibala	Palm-wine picking, interlocking ostinatos, sebene patterns	Chapter 56
Classical	Andrés Segovia, Julian Bream, John Williams	Tone, voicing, rest stroke melody	Chapter 17

Repertoire log

Write one piece on each line, with the dates you started it, could play it through and first played it for someone else.

Repertoire log

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APPENDIX H

Song Analysis



A seven-step method for analysing any piece you learn, a worked example, and blank worksheets to copy. Ten minutes of analysis tells you why a piece sounds as it does and makes it faster to learn and easier to remember.

Analyse a piece once you can play through it slowly, not before: the analysis is easier when the music is already under your fingers, and it then shows you what to practise. Work through the steps in order, writing the answers on a worksheet. The method extends the eight steps of Chapter 40 from a progression to a whole piece: it adds the form and a practice plan, and folds voicings, triads and colour into steps 2 and 6.

The seven steps

Step	What to find	How	Where to look
1 Form	The sections and where they repeat	Listen through once and mark each new section with a letter (A, B, C) and a name (intro, verse, chorus, bridge, outro). Count the bars in each	Chapter 30
2 Chords	Every chord, in order	Write a bar-by-bar chord chart. Use a slash for each beat if a bar holds more than one chord	Chapter 11, Chapter 2
3 Key	The tonic and the scale	Find the chord that feels like home (usually the first and last). Check which major or minor scale contains most of the chords	Chapter 13, Chapter 31
4 Numerals and outsiders	The function of each chord	Write a Roman numeral under every chord. Mark chords outside the key: secondary dominants, borrowed chords, the major V in minor, a modulation	Chapter 32, Chapter 33
5 Melody and solo	The scales and target notes	Name the scale of the melody. Mark which melody notes are chord tones on the strong beats. Choose a pentatonic or mode for improvising and the arpeggio for each chord	Chapter 7, Chapter 16, Chapter 35
6 Right hand and positions	How it is played	Name the strum or picking pattern and the bass string of each chord. Choose positions and CAGED shapes for melody and fills	Chapter 18, Chapter 22, Chapter 12
7 Plan	What to practise	List the two or three hardest spots and a drill for each. Set a target tempo	Chapter 38

Worked example: House of the Rising Sun

A traditional American song, best known in the 1964 arrangement in A minor with an arpeggiated 6/8 accompaniment. The analysis below uses that key and chord sequence.

Item	Answer
Title, source	House of the Rising Sun; traditional
Key, time, tempo	A minor; 6/8, about 60 dotted-quarter beats per minute
Tuning, capo	Standard tuning, no capo
Form	Instrumental intro, then verses of 16 bars; the same 16 bars repeat for every verse and the instrumental breaks

One verse (16 bars)

Am <i>i</i>	C <i>bIII</i>	D <i>IV</i>	F <i>bVI</i>
Am <i>i</i>	C <i>bIII</i>	E <i>V</i>	E <i>V</i>
Am <i>i</i>	C <i>bIII</i>	D <i>IV</i>	F <i>bVI</i>
Am <i>i</i>	E <i>V</i>	Am <i>i</i>	E <i>V</i>

Item	Answer
Chords in the key	Am (i), C (bIII) and F (bVI) belong to A natural minor
Chords from outside the key	D major (IV): the raised 6th, F#, from A Dorian or melodic minor. E major (V): the raised 7th, G#, from A harmonic minor
Cadences	Bars 7–8 are a half cadence on E (V); bars 13–16 alternate i and V, and the verse ends on V, which leads straight into the next verse
Scales for melody and solo	A minor pentatonic over the whole verse; A Dorian over D; A harmonic minor over E
Arpeggio targets	Am: A C E · C: C E G · D: D F# A · F: F A C · E: E G# B. Target the 3rds (C, E, F#, A, G#): they change with every chord
Picking pattern	p i m a m i in eighth notes (the ballad pattern of Chapter 18)
Bass strings	Am and C on string 5; D on string 4; F on string 4 (small shape) or 6 (barre); E on string 6
Tricky spots	The full F barre; the change from E back to Am at the start of each new line
Drills	Use Fmaj7 (xx3210) until the barre is secure; one-minute changes E to Am, then D to F
Plan	Pattern on Am alone at 50 bpm; then the 16 bars at 50; raise by 4 bpm per day to 60

The analysis explains the sound of the song: two chords from outside the natural minor scale (D and E) give it both the Dorian brightness of the major IV and the strong pull of the major V back to Am. These are the two most common alterations in minor-key music (Chapter 31).

TIP · Use the cheat sheets

The tables of diatonic chords in every major and minor key in Appendix B answer steps 3 and 4 for most songs in under a minute.

Blank worksheets

Copy or print the worksheets below. Each one covers a single piece; fill in what you can, and leave a field empty rather than guess.

Song analysis worksheet 1

TITLE · COMPOSER OR ARTIST · GUITAR

KEY · TIME SIGNATURE · TEMPO · TUNING AND CAPO

1 FORM: SECTIONS AND BAR COUNTS

2 CHORD CHART, LINE 1

2 CHORD CHART, LINE 2

2 CHORD CHART, LINE 3

3 KEY AND SCALE

4 ROMAN NUMERALS UNDER THE CHORDS

4 CHORDS FROM OUTSIDE THE KEY, AND WHERE THEY COME FROM

4 CADENCES AND MODULATIONS

5 SCALE OF THE MELODY · SCALES FOR SOLOING

5 ARPEGGIO AND TARGET NOTES

6 PICKING OR STRUM PATTERN · BASS STRING FOR EACH CHORD

6 POSITIONS AND CAGED SHAPES

7 TRICKY SPOTS

7 DRILLS FOR THEM

7 THIS WEEK · TARGET TEMPO

Song analysis worksheet 2

TITLE · COMPOSER OR ARTIST · GUITAR

KEY · TIME SIGNATURE · TEMPO · TUNING AND CAPO

1 FORM: SECTIONS AND BAR COUNTS

2 CHORD CHART, LINE 1

2 CHORD CHART, LINE 2

2 CHORD CHART, LINE 3

3 KEY AND SCALE

4 ROMAN NUMERALS UNDER THE CHORDS

4 CHORDS FROM OUTSIDE THE KEY, AND WHERE THEY COME FROM

4 CADENCES AND MODULATIONS

5 SCALE OF THE MELODY · SCALES FOR SOLOING

5 ARPEGGIO AND TARGET NOTES

6 PICKING OR STRUM PATTERN · BASS STRING FOR EACH CHORD

6 POSITIONS AND CAGED SHAPES

7 TRICKY SPOTS

7 DRILLS FOR THEM

7 THIS WEEK · TARGET TEMPO

Song analysis worksheet 3

TITLE · COMPOSER OR ARTIST · GUITAR

KEY · TIME SIGNATURE · TEMPO · TUNING AND CAPO

1 FORM: SECTIONS AND BAR COUNTS

2 CHORD CHART, LINE 1

2 CHORD CHART, LINE 2

2 CHORD CHART, LINE 3

3 KEY AND SCALE

4 ROMAN NUMERALS UNDER THE CHORDS

4 CHORDS FROM OUTSIDE THE KEY, AND WHERE THEY COME FROM

4 CADENCES AND MODULATIONS

5 SCALE OF THE MELODY · SCALES FOR SOLOING

5 ARPEGGIO AND TARGET NOTES

6 PICKING OR STRUM PATTERN · BASS STRING FOR EACH CHORD

6 POSITIONS AND CAGED SHAPES

7 TRICKY SPOTS

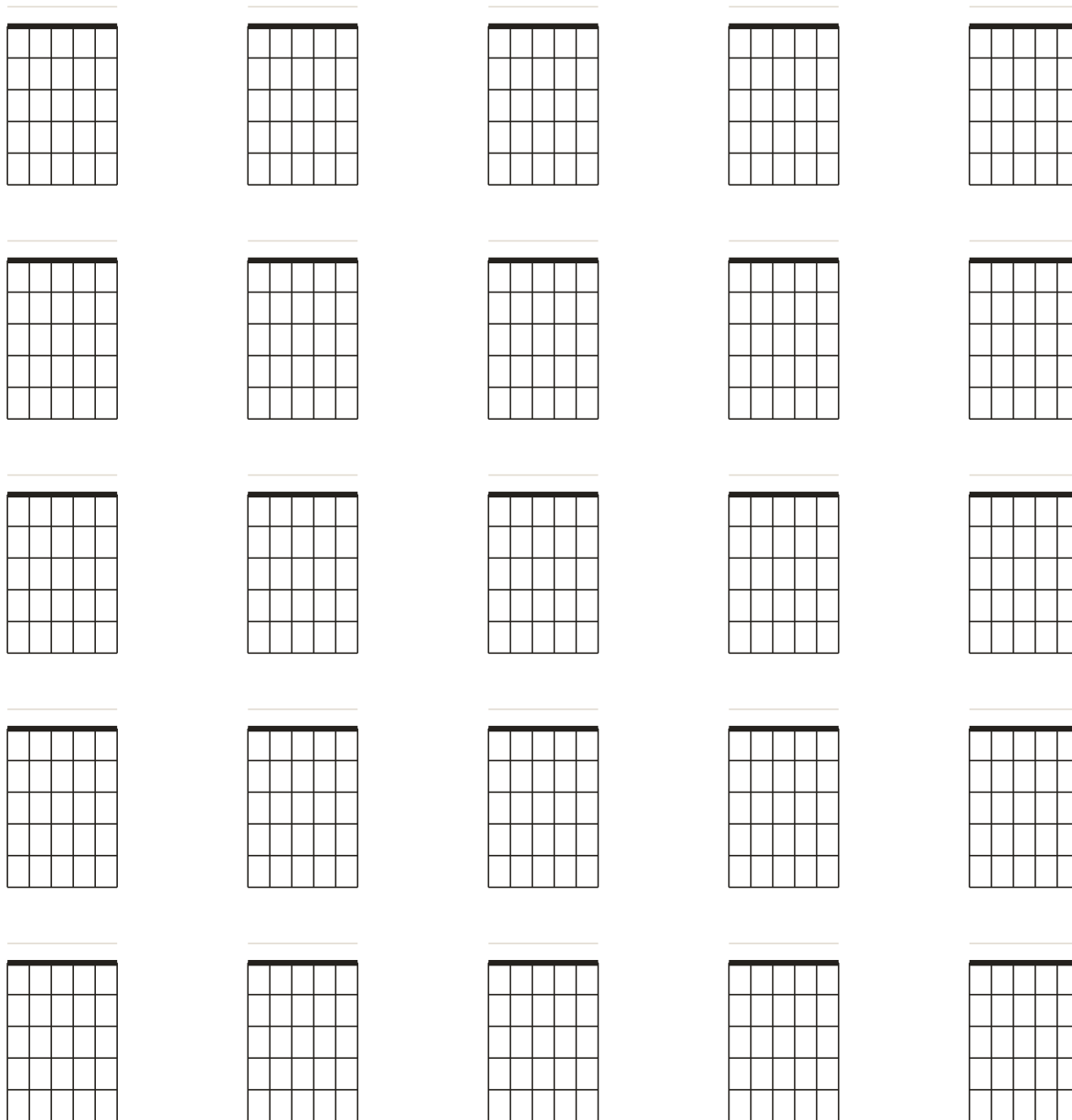
7 DRILLS FOR THEM

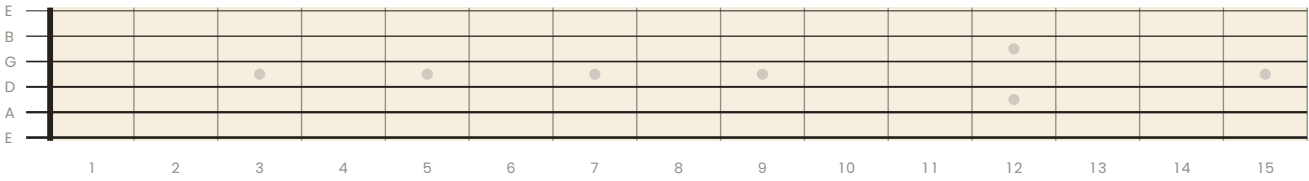
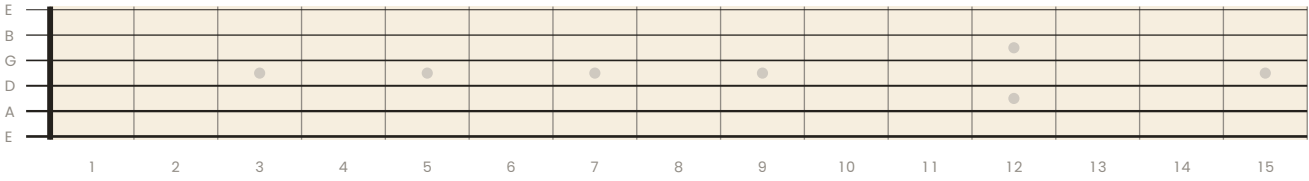
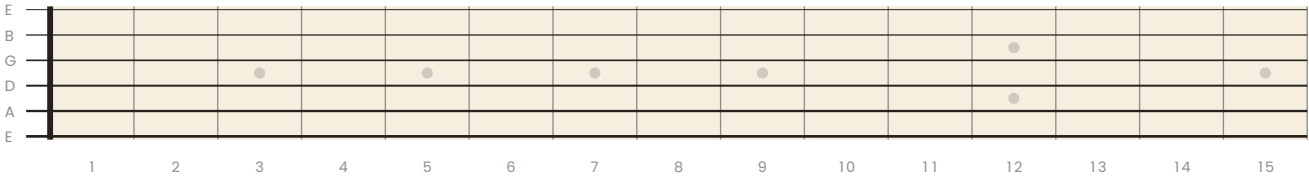
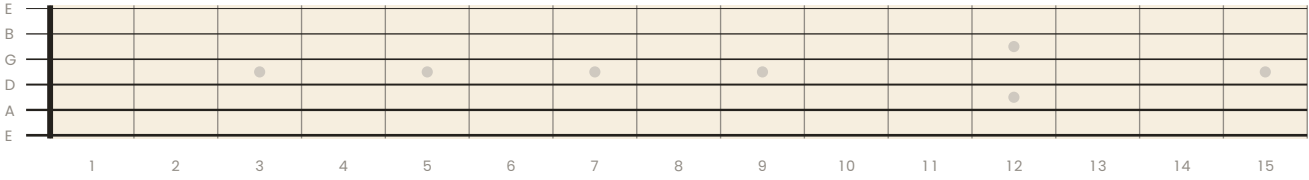
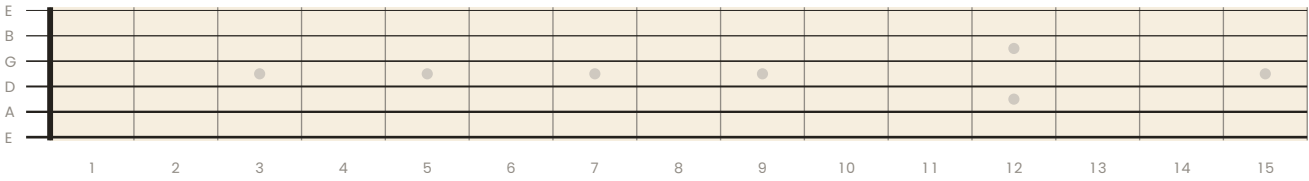
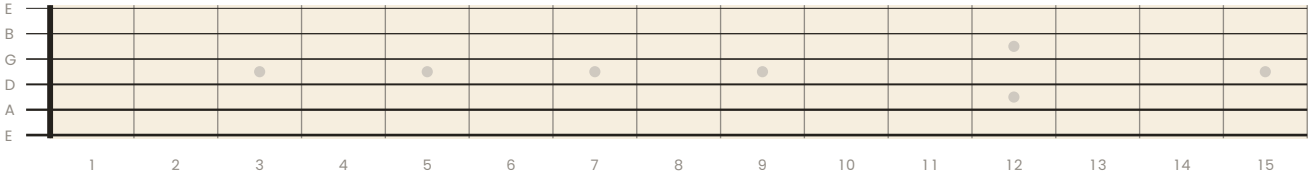
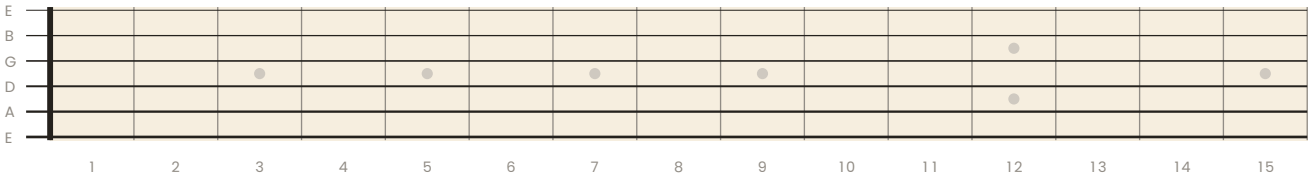
7 THIS WEEK · TARGET TEMPO

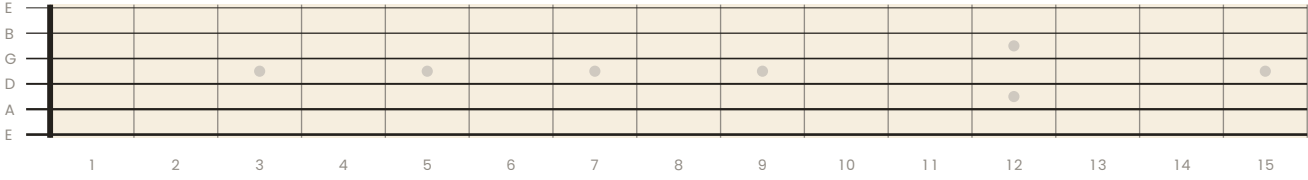
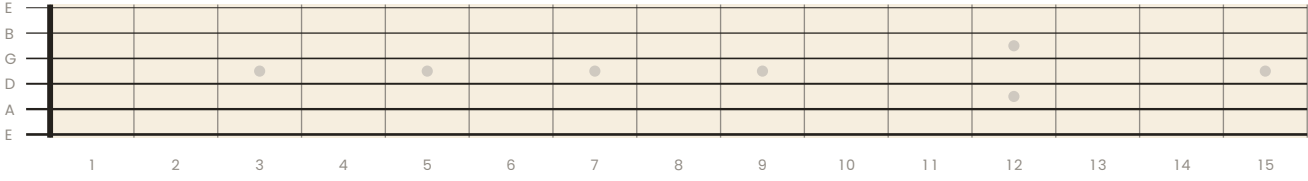
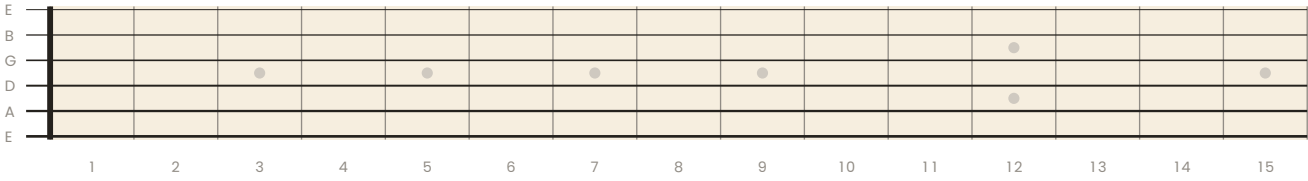
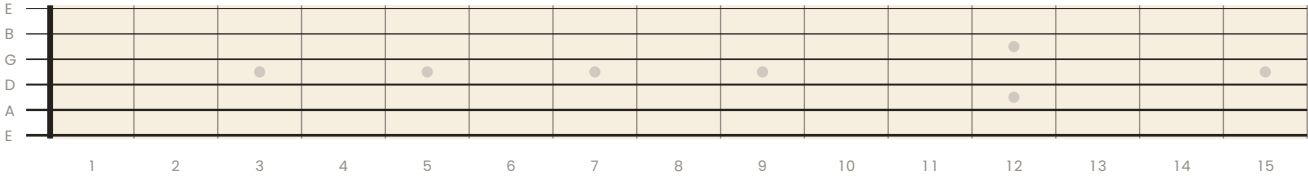
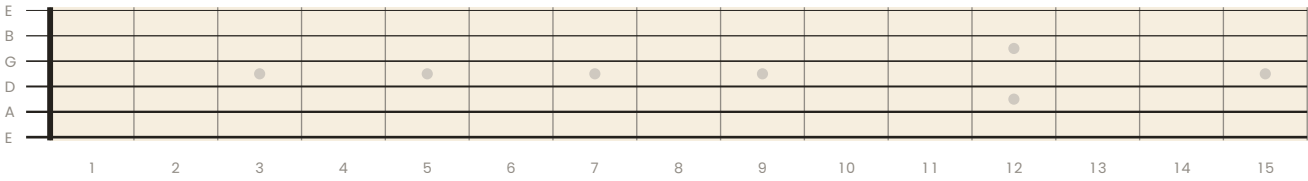
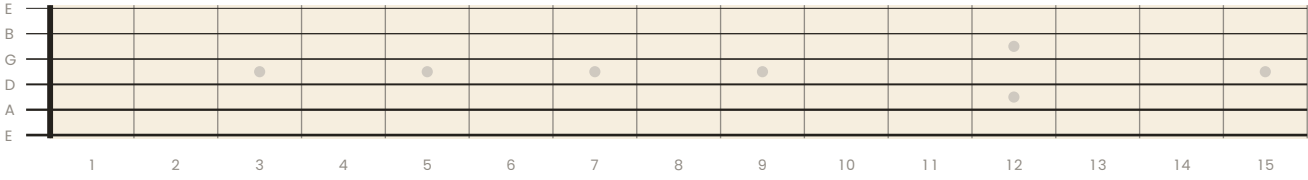
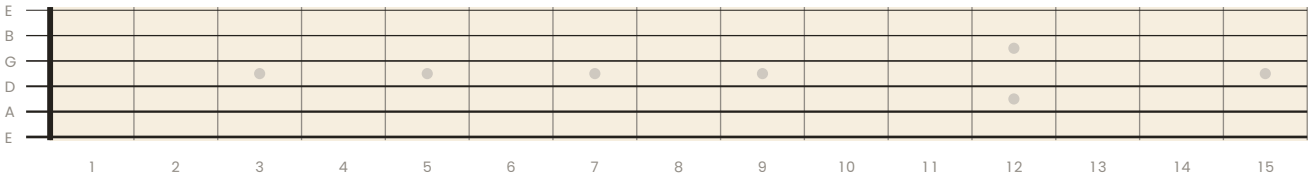
APPENDIX I

Blank Diagrams

Photocopy or print these pages to write out your own chords, scale shapes and ideas.







T
A
B

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B

T
A
B

APPENDIX J

Quiz Answers

Ear-training quizzes

Chapter 29 · Week 1, quiz A: perfect 5th or octave

1 perfect 5th; 2 octave; 3 perfect 5th; 4 octave; 5 perfect 5th; 6 octave; 7 octave; 8 perfect 5th; 9 octave; 10 octave

Chapter 29 · Week 1, quiz B: perfect 4th or perfect 5th

1 perfect 5th; 2 perfect 5th; 3 perfect 4th; 4 perfect 5th; 5 perfect 4th; 6 perfect 5th; 7 perfect 4th; 8 perfect 5th; 9 perfect 5th; 10 perfect 4th

Chapter 29 · Week 1, quiz C: perfect 4th, perfect 5th or octave

1 perfect 4th; 2 perfect 5th; 3 perfect 5th; 4 perfect 5th; 5 octave; 6 perfect 4th; 7 octave; 8 perfect 5th; 9 perfect 5th; 10 octave

Chapter 29 · Week 2, quiz A: major or minor 3rd

1 minor 3rd; 2 major 3rd; 3 minor 3rd; 4 major 3rd; 5 major 3rd; 6 minor 3rd; 7 major 3rd; 8 major 3rd; 9 minor 3rd; 10 major 3rd

Chapter 29 · Week 2, quiz B: 3rds, 4th and 5th

1 major 3rd; 2 minor 3rd; 3 major 3rd; 4 perfect 5th; 5 major 3rd; 6 perfect 4th; 7 minor 3rd; 8 perfect 5th; 9 perfect 5th; 10 minor 3rd

Chapter 29 · Week 2, quiz C: 3rds, 4th, 5th and octave

1 minor 3rd; 2 minor 3rd; 3 minor 3rd; 4 perfect 5th; 5 octave; 6 perfect 4th; 7 major 3rd; 8 perfect 4th; 9 perfect 4th; 10 minor 3rd

Chapter 29 · Week 3, quiz A: major or minor

1 minor; 2 major; 3 minor; 4 minor; 5 major; 6 minor; 7 major; 8 major; 9 major; 10 minor

Chapter 29 · Week 3, quiz B: major, minor or diminished

1 minor; 2 major; 3 minor; 4 diminished; 5 major; 6 minor; 7 diminished; 8 major; 9 diminished; 10 minor

Chapter 29 · Week 3, quiz C: all four triads

1 augmented; 2 diminished; 3 major; 4 major; 5 diminished; 6 minor; 7 augmented; 8 diminished; 9 minor; 10 augmented

Chapter 29 · Week 4, quiz A: 2nds and 3rds

1 minor 2nd; 2 major 2nd; 3 minor 2nd; 4 minor 2nd; 5 major 3rd; 6 minor 3rd; 7 minor 3rd; 8 major 2nd; 9 minor 3rd; 10 major 2nd

Chapter 29 · Week 4, quiz B: 4th, tritone and 5th

1 perfect 5th; 2 perfect 4th; 3 tritone; 4 tritone; 5 perfect 4th; 6 tritone; 7 perfect 5th; 8 perfect 4th; 9 perfect 4th; 10 tritone

Chapter 29 · Week 4, quiz C: 6ths, 7ths and octave

1 major 6th; 2 minor 6th; 3 major 6th; 4 minor 6th; 5 major 7th; 6 octave; 7 minor 6th; 8 major 7th; 9 minor 7th; 10 major 7th

Chapter 29 · Week 5, quiz A: descending 3rds, 4th, 5th and octave

1 perfect 5th; 2 perfect 4th; 3 perfect 5th; 4 octave; 5 perfect 5th; 6 perfect 5th; 7 octave; 8 major 3rd; 9 major 3rd; 10 minor 3rd

Chapter 29 · Week 5, quiz B: harmonic intervals

1 perfect 5th; 2 perfect 5th; 3 major 6th; 4 minor 3rd; 5 perfect 5th; 6 major 3rd; 7 perfect 4th; 8 major 6th; 9 perfect 4th; 10 major 3rd

Chapter 29 · Week 5, quiz C: every interval, up or down

1 minor 6th; 2 perfect 4th; 3 minor 7th; 4 perfect 5th; 5 major 6th; 6 minor 7th; 7 major 3rd; 8 minor 7th; 9 perfect 4th; 10 octave

Chapter 29 · Week 6, quiz A: maj7, 7 or m7

1 minor 7th; 2 major 7th; 3 minor 7th; 4 dominant 7th; 5 minor 7th; 6 dominant 7th; 7 major 7th; 8 minor 7th; 9 major 7th; 10 dominant 7th

Chapter 29 · Week 6, quiz B: add the half-diminished chord

1 major 7th; 2 dominant 7th; 3 minor 7th; 4 half-diminished; 5 dominant 7th; 6 half-diminished; 7 minor 7th; 8 major 7th; 9 major 7th; 10 minor 7th

Chapter 29 · Week 6, quiz C: triads and seventh chords

1 minor 7th; 2 dominant 7th; 3 minor; 4 major 7th; 5 minor; 6 minor; 7 major; 8 minor; 9 half-diminished; 10 minor 7th

Chapter 29 · Week 7, quiz A: three-note melodies in major

1 1 low 6 low 7; 2 1 2 1; 3 1 low 7 low 6; 4 1 low 7 3; 5 1 4 2; 6 1 4 5; 7 1 2 1; 8 1 3 5; 9 1 3 2; 10 1 3 4

Chapter 29 · Week 7, quiz B: four-note melodies in major

1 1 low 7 low 6 2; 2 1 2 3 4; 3 1 2 4 6; 4 1 low 7 1 low 6; 5 1 2 1 4; 6 1 low 7 low 6 low 5; 7 1 4 5 7; 8 1 3 4 5; 9 1 2 1 low 7; 10 1 low 6 2 3

Chapter 29 · Week 8, quiz A: two progressions

1 I IV V I; 2 I V IV I; 3 I IV V I; 4 I V IV I; 5 I IV V I; 6 I IV V I; 7 I V I V I; 8 I V IV I; 9 I V IV I; 10 I IV V I

Chapter 29 · Week 8, quiz B: four progressions

1 I I IV V; 2 I V IV I; 3 I IV I V; 4 I I IV V; 5 I IV V I; 6 I V IV I; 7 I IV V I; 8 I V IV I; 9 I IV V I; 10 I I IV V

Chapter 29 · Week 8, quiz C: with the dominant 7th

1 I IV V7 I; 2 I V7 IV I; 3 I IV IV V7; 4 I IV V7 I; 5 I IV IV V7; 6 I IV IV V7; 7 I V7 I IV; 8 I IV IV V7; 9 I V7 I IV; 10 I IV V7 I

Chapter 29 · Week 9, quiz A: the four-chord loops

1 I V vi IV; 2 I IV V I; 3 I vi IV V; 4 I IV V I; 5 vi IV I V; 6 I vi IV V; 7 vi IV I V; 8 vi IV I V; 9 I V vi IV; 10 vi IV I V

Chapter 29 · Week 9, quiz B: ii, V and I

1 I IV ii V; 2 ii V I I; 3 I IV ii V; 4 I IV ii V; 5 I vi ii V; 6 I vi ii V; 7 I vi ii V; 8 ii V7 I vi; 9 I IV ii V; 10 ii V7 I vi

Chapter 29 · Week 9, quiz C: all six chords

1 ii V I vi; 2 I iii IV V; 3 ii V I vi; 4 ii V I vi; 5 I V vi IV; 6 vi IV I V; 7 I iii IV V; 8 ii V I vi; 9 I IV vi V; 10 ii V I vi

Chapter 29 · Week 10, quiz A: four-note melodies in minor

1 1 2 low b7 b3; 2 1 4 b6 high 1; 3 1 2 b3 b6; 4 1 2 b3 1; 5 1 b3 2 low b7; 6 1 low b7 1 low b7; 7 1 b3 5 b3; 8 1 2 b3 b6; 9 1 low b6 2 b3; 10 1 2 b3 2

Chapter 29 · Week 10, quiz B: minor pentatonic melodies

1 1 b3 b7 5; 2 1 low 5 low b7 1; 3 1 b3 1 low b7; 4 1 4 5 high 1; 5 1 low b7 1 low b7; 6 1 4 b7 4; 7 1 b3 4 high 1; 8 1 low 5 low 4 1; 9 1 b3 1 low 5; 10 1 low b7 low 4 low b7

Chapter 29 · Week 10, quiz C: five-note melodies in minor

1 1 4 5 $\flat$ 6 5; 2 1 2 1 $\flat$ 3 1; 3 1 $\flat$ 3 1 $\flat$ 3 2; 4 1 2 4 $\flat$ 3 4; 5 1 2 1 2 4; 6 1 low $\flat$ 6
low $\flat$ 7 1 2; 7 1 low $\flat$ 6 2 $\flat$ 3 $\flat$ 6; 8 1 2 5 4 2; 9 1 $\flat$ 3 4 $\flat$ 7 $\flat$ 6; 10 1 4 $\flat$ 3 2 4

Chapter 29 · Week 11, quiz A: major, sus2, sus4 or augmented

1 sus4; 2 sus2; 3 sus4; 4 sus2; 5 augmented; 6 sus2; 7 sus2; 8 sus2;
9 sus4; 10 sus2

Chapter 29 · Week 11, quiz B: 6, 9, maj7 or 7

1 dominant 7th; 2 major 6th; 3 dominant 9th; 4 major 6th; 5
dominant 9th; 6 major 7th; 7 dominant 9th; 8 major 6th; 9 major
7th; 10 major 6th

Chapter 29 · Week 11, quiz C: which scale

1 minor; 2 dorian; 3 minor; 4 dorian; 5 mixolydian; 6 minor; 7
dorian; 8 minor; 9 mixolydian; 10 dorian

Chapter 29 · Week 12, review A: every interval, up or down

1 major 3rd; 2 minor 6th; 3 major 3rd; 4 minor 6th; 5 perfect 4th;
6 major 7th; 7 minor 6th; 8 tritone; 9 minor 7th; 10 minor 3rd

Chapter 29 · Week 12, review B: five-note melodies in major

1 1 low 7 low 6 low 7 low 6; 2 1 2 low 7 3 1; 3 1 low 6 low 7 2 3; 4 1
3 2 3 2; 5 1 low 7 low 6 2 low 7; 6 1 low 6 low 5 low 6 low 7; 7 1 low
7 low 6 low 5 low 7; 8 1 low 7 1 low 7 3; 9 1 low 7 2 3 5; 10 1 low 7
low 6 low 5 low 6

Chapter 29 · Week 12, review C: which scale

1 dorian; 2 mixolydian; 3 mixolydian; 4 major; 5 major; 6
dorian; 7 major; 8 dorian; 9 major; 10 mixolydian

Chapter 29 · Final exam 1: harmonic intervals

1 major 2nd; 2 minor 6th; 3 major 6th; 4 major 2nd; 5 major 7th;
6 minor 3rd; 7 tritone; 8 major 6th; 9 major 2nd; 10 perfect 5th

Chapter 29 · Final exam 2: chord qualities

1 augmented; 2 major 6th; 3 diminished; 4 major 7th; 5
diminished; 6 major 6th; 7 major; 8 sus4; 9 augmented; 10 sus4

Chapter 29 · Final exam 3: progressions

1 ii V I vi; 2 I IV vi V7; 3 I V vi IV; 4 I vi IV V; 5 vi IV I V; 6 I IV vi
V7; 7 I iii IV V; 8 vi IV I V; 9 I IV vi V7; 10 I V vi IV

Chapter quizzes

Chapter 1 · Know Your Instrument

1. E A D G B E
2. The 12th fret
3. Thumb, index, middle and ring finger of the picking hand
4. A rest stroke lands on the next lower-pitched string after plucking; a free stroke moves up into the palm without touching it
5. Yes. The tuning and fret spacing are the same, so the notes are the same
6. The ring finger
7. Tuning up keeps tension on the string so it does not slip flat afterwards

Chapter 2 · Reading the Diagrams

1. The 6th string (low E)
2. The top line
3. The top line of the box is the 5th fret, not the nut
4. Do not play that string
5. The 1st string (high E), as in TAB
6. One octave lower
7. Play the note on the 3rd string
8. A full barre at the 7th fret

Chapter 3 · The Building Blocks of Music

1. Two half steps, two frets
2. B–C and E–F
3. A $\flat$
4. A perfect 5th (7 half steps)
5. A minor 3rd (3 half steps)
6. Three
7. Three beats
8. The 3rd and 2nd strings (G and B)

Chapter 4 · Mapping the Fretboard

1. 1 = A, 2 = F, 3 = E, 4 = B
2. C
3. E
4. The 10th fret
5. At the 2nd fret (and the 14th)
6. At the 5th fret
7. The 4th fret
8. D

Chapter 5 · The Major Scale and Keys

1. W W H W W W H
2. D E F $\sharp$ G A B C $\sharp$ D
3. Three: F $\sharp$, C $\sharp$ and G $\sharp$
4. B $\flat$ major (B $\flat$ and E $\flat$)
5. A major
6. The leading note (leading tone)
7. B
8. E major

Chapter 6 · Minor Scales

1. E minor
2. F major
3. The 7th: G in natural minor, G $\sharp$ in harmonic minor
4. 1 2 $\flat$ 3 4 5 6 7
5. C minor
6. D $\sharp$
7. C minor, with three flats (B $\flat$, E $\flat$, A $\flat$)
8. An augmented 2nd (three half steps)

Chapter 7 · Pentatonic and Blues Scales

1. 1 $\flat$ 3 4 5 $\flat$ 7
2. E G A B D
3. E $\flat$, the $\flat$ 5
4. C major pentatonic
5. E minor pentatonic
6. The open position (fret 0), and again at the 12th fret
7. The 4th and the 7th
8. The 2nd and the $\flat$ 6

Chapter 8 · Playing Horizontally

1. Three frets, a minor 3rd
2. The 3rd and 2nd strings are a major 3rd apart instead of a perfect 4th (the B-string shift), so the notes on the 2nd string sit one fret higher
3. A, C and D at the 5th, 8th and 10th frets
4. The 3rd finger, from the second note on the string to the third
5. The open string sounds while the fretting hand is off the neck
6. On the A at the 10th fret of the 2nd string (skip a string, up three frets)

Chapter 9 · Modes

1. Mixolydian
2. The $\sharp$ 4
3. E F G A B C D
4. C major
5. D major
6. Aeolian

7. Lydian
8. F major

Chapter 10 • Triads

1. G B D; D F A
2. B D F; A^b C E
3. A minor 3rd with a major 3rd on top
4. 1st inversion; C/E
5. C major in 1st inversion (E G C)
6. The G and B strings are a major 3rd apart, not a 4th, so any shape crossing that pair shifts by one fret
7. The augmented triad splits the octave into three equal major 3rds, so each inversion has the same intervals
8. Keep common tones on the same string; move the other voices by the smallest possible step

Chapter 11 • Chords: Open, Barre and Beyond

1. G7: G B D F. Gmaj7: G B D F[#]
2. Cmaj7 has the major 7th (B); C7 has the minor 7th (B^b)
3. G major
4. E minor
5. C and G. It has no 3rd, so it is neither major nor minor
6. A D major chord with F[#] (its 3rd) as the lowest note
7. Csus4: C F G. Cadd9: C E G D
8. A dim7 chord is built from minor 3rds only, so it divides the octave equally and repeats every three frets

Chapter 12 • The CAGED System

1. The D shape
2. The E shape
3. The D shape, starting at the 5th fret (root on the 4th string, 5th fret)
4. The 4th string (also doubled on the 2nd string)
5. Lower every 3rd in the shape by one fret
6. Box 2
7. At the nut (open position): the A shape with its root on the open 5th string is the open A chord

Chapter 13 • Chords in a Key

1. D, Em, F[#]m, G, A, Bm, C[#]dim
2. E; Dm
3. Am7 – D7 – Gmaj7
4. Bdim (vii°); G7 (V7)
5. IV and ii are subdominant; V and vii° are dominant
6. Raising the 7th degree creates a leading note a half step below the tonic, which gives a much stronger pull back to i
7. B^b major
8. D shapes at capo 1, or C shapes at capo 3

Chapter 14 • Chord Progressions

1. D – A – Bm – G
2. E7, A7 and B7. V appears in bars 9 and 12
3. Em – D – C – B
4. Deceptive (interrupted) and plagal
5. E7 (V7/vi); it borrows G[#]
6. Cm
7. The upper notes stay the same while one voice (here the bass) moves down by half steps, so each chord is a small change from the last
8. The D major chord contains F[#], the raised 6th of A, which is the characteristic note of A Dorian

Chapter 15 • Spices: Making Progressions Your Own

1. The 1st finger leaves the 2nd string and the 4th finger frets the 3rd fret: C becomes D, the 9th

2. It keeps F, A and C and only replaces the F on the 1st string with open E, the major 7th, so the harmony and its function stay the same
3. C[#]dim7 (bass C – C[#] – D)
4. D and G; G, Em7, Cadd9, Dsus4 and A7sus4
5. Fm (iv) and B^b (bVII), both from C minor
6. It brings the chord in an eighth note early, on the "&" of 4, tied over the barline; nothing new is struck on beat 1

Chapter 16 • Arpeggios

1. Am7: A C E G. D major: D F[#] A
2. Cmaj7: C E G B. Bm7b5: B D F A
3. 1 b3 b5 b7
4. The 3rd and the 7th
5. G, B and D; B (the 3rd) is the most specific
6. C (the 7th of Dm7) falls a half step to B
7. Downwards (towards the floor), one continuous stroke across the strings

Chapter 17 • Fingerstyle Foundations

1. On the 3rd string (the next lower-pitched string)
2. The free stroke, because a rest stroke would land on and mute the next string, stopping the chord from ringing
3. On the 4th string (the next higher-pitched string)
4. Sequential planting, placing each finger as the previous one plays
5. The point of contact along the string, the angle of attack, and the direction of release (plus the surface: nail, flesh or pick)
6. The ring finger (a), because it shares tendons with the middle finger
7. So that it can play the bass strings without colliding with i
8. The thumb and one or more fingers plucking at the same moment

Chapter 18 • Fingerpicking Patterns

1. Pattern 2: p i m i
2. C: the 5th string; G: the 6th string; D: the 4th string
3. The open 1st string (E) belongs to all four chords, so it is a common tone
4. On the fourth eighth note, the second main beat of the bar
5. a, m, i, after the thumb (p a m i)
6. A note held or repeated, usually in the bass, while the harmony above it changes
7. A, G[#], G, F[#]
8. 2 + 2 + 3; the thumb plays the first note of each group

Chapter 19 • Travis Picking

1. Root on beats 1 and 3, alternate bass on beats 2 and 4
2. On the off-beats: i on the "&" of 1 and 3 (3rd string), m on the "&" of 2 and 4 (2nd string)
3. C (5th string, 3rd fret), E (4th string, 2nd fret), G (6th string, 3rd fret), E again
4. The open 5th string (A)
5. So that the alternating bass sounds short and percussive instead of blurring, while the treble strings ring
6. It starts on an off-beat and is held over the following beat, against the steady thumb
7. Each chord is the dominant (V7) of the next; the roots move by 4ths around the circle of fifths
8. A finger note. The thumb's pulse must never stop

Chapter 20 • Melody over Bass

1. Down; the melody stems point up
2. Yes. Each voice is counted separately, with rests shown where it is silent
3. The melody must be the highest note sounding at every moment

4. E, A, D and G (also C and the relative minors), because the roots of the main chords are open bass strings
5. Two voices moving in the same direction an octave plus a 3rd apart
6. A suspension (the 4th of E) that resolves to G $\sharp$, the 3rd of the chord
7. Bass alone, melody alone, then both together slowly, one bar at a time
8. Inner harmony notes; keep the melody and the bass

Chapter 21 · Fingerstyle Colours

1. The harmonics need no fretting, so the fretting hand is free to fret the bass notes
2. Fret a note, touch the string lightly with i 12 frets higher, and pluck with a or p behind the touching finger
3. Artificial harmonics alternated with ordinary plucked notes on other strings, blending like a harp
4. Playing consecutive scale or melody notes on different strings so that they ring into each other
5. c (little finger), a, m, i
6. Plucking close to the bridge; bright, thin and metallic
7. The thumb slap, because it strikes a string
8. A deeper push into the string before release

Chapter 22 · Strumming Patterns

1. On the beats: 1, 2, 3 and 4
2. D – D U – U D U (miss on the "&" of 1 and on beat 3)
3. A snare drum backbeat
4. A bass note on beats 1 and 3 (often alternating root and 5th) with strums on 2 and 4
5. On the last third of the beat (the third note of a triplet), after a miss on the middle one
6. They sound lighter and keep the bass from sounding on off-beats
7. Each group starts with a downstroke; the three-group ends with a downstroke and the next bar starts with one
8. On the last upstroke of the bar (the "&" of 4)

Chapter 23 · Strumming Songbook

1. F $\sharp$ m
2. It is the common teaching version; recordings vary the strum and are not transcribed exactly
3. It comes after only two beats, so the fingers have half a bar to move
4. Only on the off-beats ("&"), with short upstrokes
5. C, B, A and G, arriving on F
6. Find the beat and the metre (count in fours, threes or groups of three)

Chapter 24 · Pick Technique

1. The pad of the thumb and the side of the index finger at the first joint
2. Its rhythmic position: downstrokes on the beats (and on "&" in sixteenths), upstrokes on the off-beats between
3. Inside picking crosses between the two strings being played; outside picking plays their outer sides, going around them
4. Down, up, down; the last downstroke continues into the first downstroke on the next string
5. Release each note as the next one sounds, so the arpeggio does not ring as a chord
6. Using the pick together with the m and a fingers, which pluck the higher strings
7. 3 + 3 + 2
8. The fastest tempo at which you can play a passage several times in a row without mistakes and without tension

Chapter 25 · Open and Alternate Tunings

1. D G D G B D

2. Dsus4 (D, G and A; no 3rd)
3. G major (the IV chord)
4. The 2nd string (B). Lower it by one fret.
5. Tune to Open D (by lowering strings) and put a capo at the 2nd fret
6. G
7. Directly over the fret wire, not behind it
8. The 6th, 2nd and 1st strings, each lowered a whole step

Chapter 26 · Reading Standard Notation

1. E G B D F
2. So that the guitar's range fits on the treble staff with few ledger lines (shown by the small 8 under the clef)
3. B, on the middle line of the staff
4. F $\sharp$: the accidental lasts until the end of the bar
5. E minor (the relative minor of G major)
6. As B $\flat$, at the 3rd fret, because the key signature flattens it
7. Position VII: the 1st finger plays at the 7th fret
8. 2nd string 5th fret, 3rd string 9th fret, 4th string 14th fret

Chapter 27 · Reading Rhythm

1. Six
2. A dot adds half the note's value to the same note; a tie joins two written notes of the same pitch into one longer sound, often across a barline or beat
3. Simple triple: three quarter-note beats
4. Two beats, each a dotted quarter
5. Play on "1" and "a" (1 e & a)
6. A note or chord played an eighth early, on the "&" of 4, and tied over the barline
7. Swing eighths are triplets with the first two notes tied: long-short
8. A triplet: three eighth notes in the time of two (one quarter-note beat)

Chapter 28 · Ear Training

1. Seven; for example Twinkle, Twinkle, Little Star (or the Star Wars main theme)
2. A perfect 4th (ascending)
3. The 3rd: a major 3rd (4 half steps) or a minor 3rd (3 half steps) above the root
4. 1 3 5 $\flat$ 7; bluesy and unresolved, it wants to move (usually to the chord a 5th below)
5. V (and V7) and vii $^\circ$
6. The tonic, then the rhythm and the contour
7. I – vi – IV – V (C – Am – F – G)
8. vi – IV – I – V (Am – F – C – G). It can be heard as C major or as A minor (i – VI – III – VII); the ear decides by which chord feels like rest

Chapter 29 · Ear-Training Course

1. 9 out of 10 on the week's hardest quiz, twice in a row on different days
2. Sing what you heard: both notes, the chord tones, the bass line or the melody
3. The tonic (1); the tonic chord plays first to set the key
4. Sing it back upwards from the lower note and name it as an ascending interval
5. Mixolydian lowers the 7th ($\flat$ 7); Dorian lowers the 3rd and the 7th ($\flat$ 3, $\flat$ 7) but keeps the major 6th

Chapter 30 · Transcription

1. The bass usually plays the chord roots, so it gives the progression and confirms the key
2. Hum the note the song returns to (often the last bass note), find it on the 6th or 5th string, then test major and minor triads on it
3. It lowers the speed without changing the pitch

4. The recording is not at standard pitch (A = 440 Hz) or the guitar on it is tuned differently (for example a half step down)
5. Rhythm gives the frame: each pitch then has its place in the bar, and TAB without rhythm cannot be played without the recording
6. By sound (open or fretted, string thickness and register), by the technique heard (slides, bends) and by the most likely fingering
7. B minor: the bass moves from B to F# and the melody outlines B, D and F#
8. Any three of: key, time signature, tempo, form or bar numbers, rhythm, techniques (slides, bends, hammer-ons, pull-offs)

Chapter 31 · Minor-Key Harmony

1. To create a leading note a half step below the tonic, which turns v into a major V (or V7) and gives a strong cadence to i
2. Am, B°, C+, Dm, E, F, G#°
3. E7 and G#°7; the tritone resolves inward to A and C, the root and 3rd of Am
4. Phrygian dominant (1 b2 3 4 5 b6 b7), over V7 resolving to a minor tonic
5. B melodic minor (a half step above the root)
6. D Lydian dominant, the 4th mode of A melodic minor
7. F7 and G7
8. The final E (V) chord, whose 3rd is G#

Chapter 32 · Secondary Dominants and Borrowed Chords

1. E7 (resolving to A)
2. C#, the 3rd of A7 and the leading note of D
3. The vii° chord is diminished and cannot act as a temporary tonic
4. D#°7: D# F# A C (resolving to Em)
5. Fm (iv), Ab (bVI), Bb (bVII), Eb (bIII)
6. Bb is usually a borrowed bVII; Bb7 is usually the backdoor dominant resolving up a whole step to C
7. A major tonic chord ending a minor-key piece
8. I - V7/vi - vi - IV - iv (borrowed) - I

Chapter 33 · Modulation

1. Tonicisation points at a chord briefly and returns; modulation establishes the new tonic with a cadence and stays for a phrase or more
2. A major, G major, B minor, F# minor, E minor
3. Am: vi in C and ii in G
4. B becomes Bb
5. A direct modulation up a half or whole step, usually for a final chorus
6. E major, through the note E
7. C (as B°7), Eb (as D°7), Gb (as F°7), A or Am (as G#°7)
8. Change from an E-shape with the root on the 6th string to an A-shape with the root on the 5th string at the same fret

Chapter 34 · Voice Leading and Reharmonisation

1. Two: C and E
2. In contrary motion, to avoid parallel 5ths and octaves
3. B (3rd) and F (b7)
4. The 7th of Dm7 (C) falls a half step to B, the 3rd of G7; F stays and becomes the 7th of G7
5. Bb7; both contain the tritone G# (Ab) and D
6. Dm, DmMaj7, Dm7, Dm6 (the moving voice D - C# - C - B)
7. Am (vi) and Em (iii)
8. A chord of the same quality a half step above or below the target, used as a decoration

Chapter 35 · Improvising over Changes

1. Chord-scale thinking chooses a scale for each chord; target-note thinking chooses which chord tones fall on the strong beats and at the changes
2. The 3rd and 7th (guide tones), because they define the chord and usually lie a half step from a note of the previous chord
3. G Phrygian dominant (5th mode of C harmonic minor)
4. The b9 (Bb) falls a half step to A, the 5th; the #5 (F) is held as the 3rd
5. C - A# - B
6. The major 7th as a passing note, so that descending lines from a chord tone on the beat keep chord tones on the beats
7. B Locrian #2 (from D melodic minor)
8. Roots, arpeggios, guide tones, target lines, approach notes and enclosures, free playing

Chapter 36 · Expressive Techniques

1. Pick fret 7, bend it up to the pitch of fret 9, then release it back to fret 7
2. Play the target fret first, then bend the lower note until it sounds the same
3. Axial vibrato rocks the finger along the string and moves the pitch both above and below the note; radial vibrato pushes the string sideways and only raises the pitch
4. E (an octave and a 5th above the open A)
5. The sideways flick plucks the string; lifting straight up makes almost no sound
6. Lightly on the strings right at the bridge saddle
7. Nylon strings and the wide, flat fingerboard make bends change the pitch very little; slides and vibrato are used instead
8. The 5th/7th-fret pair does not work between G and B; compare the 7th-fret harmonic on the 6th string with the open B string

Chapter 37 · Rhythm Mastery

1. Pulse, subdivision, placement and feel
2. Nothing: the chord is held from the "&" of 4, so beat 1 is felt, not played
3. Level 2, subdivision: a pattern wobbles or a solo rushes because the player is not feeling the notes inside the beat
4. 3 + 3 + 2
5. 2 + 2 + 3 (also 3 + 2 + 2 or 2 + 3 + 2)
6. Whether your internal clock rushes or drags when the click is silent
7. Only the first beat; the other notes alternate between the two rhythms
8. It makes the click act as a backbeat, the way drummers and bands feel time, and forces you to supply beats 1 and 3 yourself

Chapter 38 · Practise Smarter

1. After three clean runs in a row, by about 4–8 bpm
2. A specific goal just beyond your level, full attention, immediate feedback, and repetition with correction
3. At the joins between chunks; practise each join as a chunk of its own
4. Skills consolidate between sessions, especially during sleep, and frequent reviews slow forgetting; it is also easier on the hands
5. Alternating two or three topics within a session instead of drilling one topic in a long block
6. Review after about 1 day, 2 days, 4 days, 1 week, 2 weeks and 1 month
7. Learning a phrase from the end: the last bar first, then the last two bars, and so on
8. For example: a sharply bent fretting wrist, raised shoulders, a squeezing thumb, fingers flying off the strings, a strumming hand that stops

Chapter 39 · What to Avoid

1. Squeezing the neck with the thumb; press only hard enough for a clean note, just behind the fret, using the arm's weight
2. Slow down until it is clean, isolate the difficult spot, then raise the tempo gradually
3. Soloing only in pentatonic box 1; learn the neighbouring boxes and connect them with slides
4. The constant down-up motion keeps time; misses are motions that clear the strings, so the hand is always in the right place
5. A flat capo, because the classical fingerboard is flat
6. Tuning up raises string tension, which stresses the neck and top and can break strings
7. For example: pain that continues after you stop playing, numbness or tingling, swelling, pain at night, or loss of control in a finger
8. Short, frequent sessions rather than long ones

Chapter 40 · How It All Fits Together

1. C major (relative A minor): vi – IV – I – V
2. A minor pentatonic (the same notes as C major pentatonic)
3. F, A and C
4. I – IV – V7 in E major; E major pentatonic (or E minor pentatonic and the blues scale for a bluesy sound)
5. A ii – V – I in C major
6. E harmonic minor, because B major contains D $\sharp$, the raised 7th of E minor
7. G $\sharp$, from A harmonic minor; it rises to A when Am returns
8. Melody alone; melody with bass roots; melody with bass and chord tones or an alternating bass

Chapter 41 · Choosing a Guitar

1. The dreadnought (also the jumbo)
2. About 650 mm
3. The top (soundboard)
4. Neck relief (the slight forward bow of the neck), not the action directly
5. Compare the 12th-fret harmonic with the fretted 12th-fret note on a tuner; if the fretted note is sharp, move the saddle away from the neck
6. The humbucker
7. About 45–55% relative humidity
8. The bridge lifting, a very low saddle, a bulging top behind the bridge, or very high action that cannot be lowered

Chapter 42 · Sound, Effects and Loopers

1. Near the end, after drive and modulation (or in the amplifier's effects loop), with only the looper after them
2. Around –12 to –6 dBFS, never reaching 0 dBFS (the red)
3. About 64–128 samples
4. Roughly where the neck meets the body (around the 12th–14th fret), about 15–30 cm away
5. For example: turn down, stand beside or behind the speaker, cut the low mids, use the phase switch or a notch filter, fit a soundhole cover
6. Exactly on beat 1 of the bar after the loop ends, if anything slightly early
7. The instrument input, labelled Hi-Z or INST
8. Magnetic pickups sense the movement of steel strings; nylon is not magnetic

Chapter 43 · Blues

1. E7 E7 E7 E7 | A7 A7 E7 E7 | B7 A7 E7 B7
2. The IV chord appears in bar 2, and the music returns to I in bar 3
3. F7 and E7 (bVI7 and V7)
4. E and F $\sharp$ (the 5th and 6th), with G (the b7) in the extended pattern
5. C is the b7 of D7; it moves up a half step to C $\sharp$, the 3rd of A7

6. The thumb plays the same bass note, usually the root, on every beat

Chapter 44 · Rock and Riffs

1. E and B (root and 5th), often with the octave E; it has no 3rd
2. Downstrokes give an even, heavy attack; switch to alternate picking for sixteenths, gallops and eighths faster than about 160 bpm
3. D A D G B E
4. On the off-beats, between pentatonic notes on the beats
5. Two palm-muted sixteenths, which keep the riff driving while the melody moves slowly on the 4th string
6. An eighth note followed by two sixteenths, repeated on every beat
7. A and C $\sharp$ (root and major 3rd)

Chapter 45 · Funk Rhythm

1. Down on the beat and the "&", up on the "e" and the "a"
2. Relax the fretting hand so the fingers touch the strings without pressing them, then strum
3. G $\sharp$, D and F $\sharp$ (3rd, b7 and 9th); the root and 5th are left out
4. E G $\sharp$ B D G; the $\sharp$ 9 (G) is the same pitch as the minor 3rd, heard against the major 3rd G $\sharp$
5. Beat 1 of the bar, which the whole band, especially the bass and kick drum, plays strongly
6. They take up a narrow, high range, leaving room for the bass and keyboards and allowing a busy rhythm without clashes
7. E Dorian

Chapter 46 · Country and Bluegrass

1. C (5th string, fret 3) on beat 1 and G (6th string, fret 3) on beat 3
2. G, A, B, arriving on C
3. Melody played on the bass strings with thumb or pick, with brush strums on the treble strings between melody notes
4. On eighths 1, 4 and 7: beat 1, the "&" of 2, and beat 4
5. Hybrid picking with a snapped finger note: pull the string slightly away from the body before releasing it so it slaps the frets
6. On the low G root; at the ends of verses, solos and songs in bluegrass

Chapter 47 · Folk and Songwriting

1. The key is the same, but the chords sound in a higher register, so the two guitars fill different ranges instead of doubling each other
2. Open with G shapes, capo 5 with D shapes, capo 7 with C shapes
3. Ring and little fingers, on fret 3 of the 2nd and 1st strings
4. A form in which the same music is repeated for every verse
5. In the gaps between vocal lines, typically at the end of a phrase, not under the words
6. C B A G F E D G
7. The 4th replaces the 3rd and tends to resolve down a half step to the 3rd

Chapter 48 · Jazz Harmony

1. The root, 3rd and 7th
2. D and A $\flat$ (3rd and b7 of B $\flat$ 7); they replace A and E $\flat$ of F7, each a half step away
3. Dorian
4. A chromatic descent: D – D $\flat$ – C
5. From a close-position four-note chord, drop the second note from the top down an octave
6. A chord on beat 1 and another on the "&" of 2
7. Bdim7; it connects B $\flat$ 7 in bar 5 to F7 in bar 7

Chapter 49 • Gypsy Jazz

1. A short downstroke on beats 1 and 3, and on beats 2 and 4 a quick upstroke followed by an accented, very short downstroke; the fretting hand releases the chord after each stroke
2. Am6: A C E F $\sharp$; Dm6: D F A B
3. When moving to a higher-pitched string, the downstroke passes through the string and comes to rest on the next one; each new string starts with a downstroke
4. F diminished (F A $\flat$ B D); it contains the 3rd, 5th, b7 and b9 of E7
5. m6 chords for minor, 6 and 6/9 chords for major
6. The band had no drums; the rhythm guitars, with the bass, supplied the beat and the backbeat

Chapter 50 • Bossa Nova and Latin Rhythms

1. Half notes on beats 1 and 3 of each 4/4 bar, usually the root then the 5th, imitating the surdo drum
2. Bar 1: 1, & of 2, 4. Bar 2: 2, 3
3. The last stroke of the 2 side is one eighth note later: & of 3 instead of 3
4. C E G A D
5. A dominant 7b9, for example A7b9 $\rightarrow$ Dm7 or E7b9 $\rightarrow$ Am
6. The dancers' triple step on 4 & 1
7. A repeating syncopated melodic figure, played by the tres or piano in son and salsa, that outlines the chords and locks to the clave

Chapter 51 • Choro and Brazilian Guitar

1. Rio de Janeiro, in the 1870s
2. A three-part rondo, usually AABACCA, each section often 16 bars
3. The bass counter-line of the choro guitar (especially the seven-string), linking chords by scale and chromatic runs
4. C (sometimes B), below the low E
5. Beat 2 of the 2/4 bar
6. 3 + 3 + 2 sixteenths in 2/4 (dotted eighth, dotted eighth, eighth)
7. #4 and b7

Chapter 52 • Reggae

1. Beats 2 and 4
2. Kick drum and rim click together on beat 3, with nothing on beat 1
3. Release the pressure of the fretting fingers right after the strike, keeping them on the strings
4. Ska uses upstroke chops on every off-beat at a fast tempo; reggae uses chops on 2 and 4 at a slower tempo
5. Rockers puts the kick on beats 1 and 3; steppers puts it on every beat
6. The bass line doubled an octave higher, picked and palm-muted
7. The top three or four strings: small shapes mute easily and leave the low register to the bass

Chapter 53 • Flamenco

1. On 3, 6, 8, 10 and 12
2. On 12
3. 2 + 2 + 3 + 3 + 2, with accents on 1, 3, 5, 8 and 11
4. Flamenco key played with E-position chords (E Phrygian, cadence Am – G – F – E) and with A-position chords (A Phrygian, cadence Dm – C – B $\flat$ – A), wherever the capo is
5. Add an index upstroke after c – a – m – i (c – a – m – i – i $\uparrow$) and repeat it without gaps
6. A tap on the top with the ring finger, usually with a downstroke on an accent; play it on a tap plate (golpeador)
7. It has four finger notes after the thumb (p i a m i) instead of three (p a m i)

Chapter 54 • Celtic Fingerstyle

1. A jig is in 6/8; a slip jig is in 9/8
2. A very short grace note above the main note, used to separate or accent it
3. Note – cut – note – tap – note
4. Three open strings are D or A, so the tonic and 5th drone under the tune; it also suits D Mixolydian, D Dorian, G major, E Dorian and B minor
5. The open 2nd string (A), so the G keeps ringing under the A
6. Two 8-bar parts, each played twice (AABB), 32 bars in all
7. Many tunes are modal, and an open fifth or sus chord will not clash with the third or seventh the melody uses

Chapter 55 • Neo-Soul

1. C E G B D
2. The 5th
3. Fm (or Fm9) to C: the minor iv chord resolving to I
4. Cmaj7
5. Locking to the groove with swung sixteenths, placed slightly behind the beat, with soft dynamics
6. Play the same voicing one fret below as a short note and slide it into the target chord on the beat

Chapter 56 • West and Central African Guitar

1. The fast instrumental dance section of Congolese rumba and soukous, led by repeating lead-guitar patterns over a short chord cycle
2. Solo (lead), mi-solo (middle) and accompagnement (rhythm)
3. The thumb (bass) and the index finger (melody)
4. x.x.xx.x.x.x: seven strokes spaced 2 – 2 – 1 – 2 – 2 – 2 – 1
5. The primary chords I, IV and V in repeating cycles
6. The lead guitar and bass already play there; the off-beat stabs interlock with them

Chapter 57 • Study Pieces: Elementary

1. Stems up show the melody voice; stems down show the bass voice played by the thumb
2. Beat 3
3. The ring finger, a
4. E minor pentatonic (E G A B D)
5. It is the dominant (V) of E minor, taken from the E harmonic minor scale (it contains D $\sharp$)
6. D, C $\sharp$, B and A: D, D/C $\sharp$, Bm and Bm/A, a bass line walking down the D major scale

Chapter 58 • Study Pieces: Intermediate

1. It alternates between the root of the chord and a second bass note (usually the 3rd or 5th), one note per beat
2. The melody uses the flat 7th of D, so it is D Mixolydian rather than D major
3. B7, the major V7 chord; its 3rd, D $\sharp$, is the raised 7th of E minor
4. Each one is the dominant of the next: A7 is V of D, D7 is V of G, G7 is V of C (a chain of secondary dominants)
5. On top: it is always the highest note of the chord

Chapter 59 • Study Pieces: Advanced

1. During the thumb note at the start of a beat, never during the tremolo, so the repeated melody notes stay evenly spaced
2. Open strings 1 and 2 add E (the major 7th) and B (the #11) to the F major triad
3. A $\flat$ 13, the tritone substitution: A $\flat$ 7 and D7 share the same tritone (C and G $\flat$ /F $\sharp$), and its root moves down a half step to G
4. It imitates the upper voice's one-bar subject an octave lower, one bar later
5. The barre on strings 1–3 gives an A major triad, while the open low D keeps droning underneath

APPENDIX K

Audio Track List



Every example marked with a ► track number has a recording. Each one comes at full tempo and at a slow practice tempo with a count-in.

Track	Chapter	Example	Type
1	1	Open strings in notation and TAB	score
2	2	A C major chord played one note at a time, then together	score
3	2	Slurs, a slide, a bend and vibrato	score
4	2	C major up and down in first position	score
5	2	A simple p-i-m-a pattern on A minor	picking
6	2	Reading a strum grid: accents and a chuck	strum
7	2	Three layers of a basic groove	rhythm
8	3	Every interval above C	score
9	3	One note in every common value	score
10	3	Subdivisions of a 4/4 bar	rhythm
11	3	Waltz in 3/4: accent on beat 1	rhythm
12	4	Natural notes on the 6th string	score
13	4	One note everywhere: every C from low to high	score
14	4	C major on the 2nd string	score
15	4	Tuning reference: fretted note, then the next open string	score
16	5	G major, one octave, with its key signature	score
17	5	A short melody in D major	score
18	5	Pattern 1 in groups of three	score
19	5	G major pop loop: I – vi – IV – V	jam
20	5	G drone: hear the seven degrees	jam
21	6	A natural minor, one octave	score
22	6	Natural minor loop: i – ♭VI – ♭VII	jam
23	6	A harmonic minor, one octave	score
24	6	A melodic minor, up in the melodic form, down in the natural form	score
25	6	A minor and E major: the raised 7th in the harmony	score
26	6	Harmonic minor loop: i – iv – V7	jam
27	7	Box 1 up and down	score
28	7	Sliding from box 1 into box 2	score
29	7	A minor blues loop	jam
30	7	A major pentatonic phrase	score
31	7	Country-pop loop in A	jam
32	7	A blues phrase through the blue note	score
33	7	12-bar shuffle in A	jam
34	8	A phrase that travels along two strings	score
35	8	A minor pentatonic up the 1st string	score

Track	Chapter	Example	Type
36	8	A major scale on one string	score
37	8	One-string lick on the 2nd string	score
38	8	Two-string run through six boxes	score
39	8	Two-string lick on the 4th and 3rd strings	score
40	8	Two-octave diagonal run	score
41	8	Slide shift from box 1 to box 3	tab
42	8	Stretch shift from box 1 to box 2	score
43	8	Open-string shift to the 12th fret	tab
44	8	Octave-shape jump with one cell	score
45	8	Guide-finger shift on the 3rd string	score
46	8	The lick in five places	score
47	8	A minor rock loop for horizontal practice	jam
48	9	Ionian, Mixolydian and Dorian on A	score
49	9	Dorian vamp: Am to D	picking
50	9	Mixolydian vamp: A to G	picking
51	9	Lydian vamp: C to D/C	picking
52	9	Phrygian vamp: Em to F	picking
53	9	D drone for the drone switch	jam
54	9	Dorian vamp: Am7 – D9	jam
55	9	Mixolydian vamp: A – G	jam
56	9	Lydian vamp: Cmaj7 – D/C	jam
57	9	Phrygian vamp: Em – F	jam
58	9	Aeolian loop: Em – C – D	jam
59	10	The four qualities on C	score
60	10	Inversion ladder: C major up strings 3–2–1	score
61	10	Voice-led C – Am – F – G	score
62	10	Triads over an open D drone	score
63	11	Open-chord changes: G – Em – C – D	strum
64	11	Power-chord riff in E	score
65	11	Barre progression in F	strum
66	11	Suspensions around D: D – Dsus4 – D – Dsus2 – D	score
67	11	Descending bass with slash chords	score
68	12	C major up the neck through the five shapes	score
69	12	I – IV – V in G, low and high positions	score
70	12	Pentatonic phrase around the E-shape C chord	score
71	13	The diatonic triads of C major, root position on strings 5–4–3	score
72	13	Diatonic 7th chords in C	score
73	13	Same functions, different chords	score
74	13	G shapes at capo 2, sounding in A	score
75	14	I – V – vi – IV in C with open chords	score
76	14	Andalusian cadence	score
77	14	12-bar boogie pattern in A	score
78	14	The four cadences in C	score

Track	Chapter	Example	Type
79	14	Line cliché in A minor	score
80	14	I – iii – IV – iv – I in C	score
81	14	Front door, then back door	score
82	14	Mixolydian rock in D	score
83	14	Mixolydian rock loop in D	jam
84	14	Aeolian loop with open chords	score
85	14	Aeolian loop in A minor	jam
86	14	Rising bass in C	score
87	14	Descending bass in D	score
88	14	Two deceptive cadences, then the real ending	score
89	14	Chords over an open D	score
90	14	Dominants round the circle	score
91	14	Dorian vamp: Am7 – D	score
92	15	Plain, then spiced: G – C – D – G	score
93	15	Plain, then spiced: Am – F – E – Am	score
94	15	Plain, then spiced: D – G – A – D	score
95	15	Plain, then spiced: Em – C – G – D	score
96	15	Plain, then spiced: E – C#m – A – B	score
97	15	Plain, then spiced: C – Dm – G7 – C	score
98	15	Plain, then spiced: G – C with a walk-up	score
99	15	Plain, then spiced: C – F – G – C	score
100	15	Plain, then spiced: G – Em – C – Am – D – G	score
101	15	Plain, then spiced: Am – F – C – G	score
102	15	Spice practice loop in G	jam
103	15	Whole song, plain	score
104	15	Whole song, lightly spiced	score
105	15	Whole song, fully spiced	score
106	16	Outlining C – Am – F – G in open position	score
107	16	7th-chord arpeggios through ii – V – I in C	score
108	16	Three-string sweeps through C – Am – F – G	score
109	16	p – i – m – a over Am and E	picking
110	17	Free stroke: i and m alternating on open strings	score
111	17	Rest stroke: one octave of C major, alternating i and m	score
112	17	Thumb: rest strokes on the bass strings, then a walking line	score
113	17	Thumb independence: steady bass, off-beat fingers	score
114	17	Planting practice: ascending (full) and descending (sequential)	picking
115	17	Pinch independence on C and G7	picking
116	17	Warm-up study in E minor	score
117	18	Patterns 1–4 on C, Am, G and Em	picking
118	18	Patterns 5–8 on D, Dm, E and A	picking
119	18	A waltz in E minor: i – iv – V7 – i	picking
120	18	The Andalusian cadence as a 6/8 ballad	picking
121	18	Tremolo study on A minor and E	score

Track	Chapter	Example	Type
122	18	C – G/B – Am – Am7/G with the pinch pattern	picking
123	18	Line cliché in A minor	score
124	18	Am – Fmaj7 – C – G6 with a rise-and-fall pattern	picking
125	18	Sliding shapes over an open D pedal	score
126	18	Dsus4 – D – Dsus2 and Asus4 – A – Asus2 colours	score
127	18	Walking bass in E Dorian under open-string pinches	score
128	18	A three-note finger cycle against a steady thumb	picking
129	18	Am7 – Fmaj7 – Am7 – E7 in 5/4	picking
130	18	D – C – G – D in 7/8 (2 + 2 + 3)	picking
131	18	A Dorian vamp, melody on the 1st string	score
132	18	Practice loop: a 6/8 ballad in A minor	jam
133	19	The thumb alone: C, G7 and the three-string bass	score
134	19	Four steps to the Travis pattern on C	picking
135	19	The basic pattern through C – Am – F – G7	score
136	19	Syncopated, backbeat and double-stop variations	picking
137	19	Thumb-style study in C	score
138	19	G – C – D – G	picking
139	19	A – D – E7 – A	picking
140	19	Hammer-ons on A, D and E	score
141	19	Em – D – C – B7	picking
142	19	E7 – A7 – D7 – G	picking
143	19	C – A7 – D7 – G7: the same chain, ending in C	picking
144	19	Practice loop: ragtime changes in C	jam
145	19	Walk-ups and walk-downs in G	score
146	20	Two voices, two rhythms	score
147	20	A moving melody over held bass notes	score
148	20	Parallel tenths and contrary motion in C	score
149	20	Sixths on strings 3 and 1 over an open D and A bass	score
150	20	Step 2: the melody alone (first eight bars)	score
151	20	Step 3: melody and bass roots	score
152	20	Greensleeves	piece
153	21	Harmonics over a moving bass	score
154	21	Harp harmonics on C and G	score
155	21	A campanella scale in G	score
156	21	Tremolo study on the 2nd string	score
157	21	A percussive groove	rhythm
158	21	Thumb slap with a backbeat chuck	strum
159	21	Two rasgueado rhythms	rhythm
160	21	One pattern, three colours	picking
161	22	1. All downs	strum
162	22	2. Steady eighths	strum
163	22	3. Old faithful	strum
164	22	4. Folk drive	strum

Track	Chapter	Example	Type
165	22	5. Accented old faithful	strum
166	22	6. Backbeat chuck	strum
167	22	7. Boom-chick	strum
168	22	8. Waltz	strum
169	22	9. 6/8 lilt	strum
170	22	10. Off-beat skank	strum
171	22	11. Sixteenth funk	strum
172	22	12. Pop push	strum
173	22	Old faithful in notation	score
174	22	3 + 3 + 2 accents	strum
175	22	7/8 as 2 + 2 + 3	strum
176	22	Pop: I – V – vi – IV in G with old faithful	strum
177	22	Country: boom-chick with alternating bass in C	strum
178	22	Waltz in 3/4: I – IV – V7 – I in D	strum
179	22	Ballad in 6/8: vi – IV – I – V in C	strum
180	22	Funk: sixteenth groove on a two-chord vamp	strum
181	22	Practice loop: strum along in G	jam
182	22	The descending bass, strummed	strum
183	22	Shuffle strum on E7 and A7	strum
184	23	Stand By Me: verse and chorus	jam
185	23	Brown Eyed Girl: verse	jam
186	23	Knockin' on Heaven's Door: verse and chorus	jam
187	23	Wagon Wheel: verse and chorus	jam
188	23	Zombie: the whole song	jam
189	23	I'm Yours: verse and chorus	jam
190	23	Riptide: verse and chorus	jam
191	23	Let It Be: verse	jam
192	23	Have You Ever Seen the Rain: verse	jam
193	23	Wonderwall: verse	jam
194	23	The House of the Rising Sun: verse	jam
195	23	Perfect: verse	jam
196	23	Piano Man: first half of the verse	jam
197	23	Three Little Birds: chorus	jam
198	23	Before You Accuse Me: 12-bar blues	jam
199	24	Chromatic 1-2-3-4 with alternate picking	score
200	24	String crossing: inside and outside	score
201	24	Economy picking: three notes per string in G major	score
202	24	Sweep arpeggios on Am and E	score
203	24	A minor pentatonic in string pairs	score
204	24	Crosspicking on G – C – D – G	score
205	24	Hybrid picking: alternating bass with pick, m and a above	score
206	24	Hybrid double stops: pick and m together	score
207	24	Tremolo-picked melody in A minor	score

Track	Chapter	Example	Type
208	24	Four subdivisions at one tempo	rhythm
209	25	I – IV – V in Open G	score
210	25	DADGAD air over a drone	score
211	25	Drop D riff	score
212	25	Open C arpeggios	score
213	25	Slide phrase in Open G	score
214	26	Every natural note in first position	score
215	26	Bass strings: reading below the staff	score
216	26	The 1st string above the staff	score
217	26	Accidentals in C major	score
218	26	Key of D major: two sharps from the signature	score
219	26	Key of F major: one flat from the signature	score
220	26	Reading in position V	score
221	26	Reading in position VII	score
222	26	Level 1a: the 1st string	score
223	26	Level 1b: the 2nd and 1st strings	score
224	26	Level 1c: Ode to Joy (Beethoven), notation only	score
225	26	Level 2a	score
226	26	Level 2b: notation only	score
227	26	Level 3a: Twinkle, Twinkle (traditional)	score
228	26	Level 3b: notation only	score
229	26	Level 4a: Frère Jacques (traditional) in G	score
230	26	Level 4b: E minor, notation only	score
231	26	Level 5: position V, notation only	score
232	26	Level 6: Ode to Joy with a bass line	score
233	27	Note values on the open 3rd string	score
234	27	Rests	score
235	27	Dots and ties	score
236	27	Six eighths in 3/4: three beats	score
237	27	Six eighths in 6/8: two beats	score
238	27	3/4 against 6/8 on the same six eighths	rhythm
239	27	The five sixteenth figures	score
240	27	Syncopation and the push	score
241	27	Triplets against straight notes	score
242	27	Exercise 1: quarters, halves and rests	score
243	27	Exercise 2: eighth notes	score
244	27	Exercise 3: off-beats	score
245	27	Exercise 4: dotted quarters	score
246	27	Exercise 5: syncopation and ties	score
247	27	Exercise 6: sixteenths	score
248	27	Exercise 7: 3/4	score
249	27	Exercise 8: 6/8	score
250	27	Exercise 9: triplets	score

Track	Chapter	Example	Type
251	27	Exercise 10: 12/8	score
252	28	Intervals 1: minor 2nd to tritone	score
253	28	Intervals 2: perfect 5th to octave	score
254	28	Triads on A: major, minor, diminished, augmented	score
255	28	Seventh chords on A	score
256	28	Tonic, subdominant, dominant, tonic in C	score
257	28	Hearing scale degrees after a cadence	score
258	28	I – V – vi – IV	score
259	28	ii – V7 – I	score
260	28	Mystery progression A	score
261	28	Mystery progression B	score
262	28	Dictation 1	score
263	28	Dictation 2	score
264	28	Dictation 3	score
265	28	Dictation 4	score
266	29	Week 1 reference: 5th, octave and 4th over an E drone	score
267	29	Week 1, quiz A: perfect 5th or octave	earquiz
268	29	Week 1, quiz B: perfect 4th or perfect 5th	earquiz
269	29	Week 1, quiz C: perfect 4th, perfect 5th or octave	earquiz
270	29	Week 2 reference: 3rds against the 4th and 5th	score
271	29	Week 2, quiz A: major or minor 3rd	earquiz
272	29	Week 2, quiz B: 3rds, 4th and 5th	earquiz
273	29	Week 2, quiz C: 3rds, 4th, 5th and octave	earquiz
274	29	Week 3, quiz A: major or minor	earquiz
275	29	Week 3, quiz B: major, minor or diminished	earquiz
276	29	Week 3, quiz C: all four triads	earquiz
277	29	Week 4, quiz A: 2nds and 3rds	earquiz
278	29	Week 4, quiz B: 4th, tritone and 5th	earquiz
279	29	Week 4, quiz C: 6ths, 7ths and octave	earquiz
280	29	Week 5 reference: the same intervals three ways	score
281	29	Week 5, quiz A: descending 3rds, 4th, 5th and octave	earquiz
282	29	Week 5, quiz B: harmonic intervals	earquiz
283	29	Week 5, quiz C: every interval, up or down	earquiz
284	29	Week 6, quiz A: maj7, 7 or m7	earquiz
285	29	Week 6, quiz B: add the half-diminished chord	earquiz
286	29	Week 6, quiz C: triads and seventh chords	earquiz
287	29	Week 7 reference: E major scale degrees over a drone	score
288	29	Week 7, quiz A: three-note melodies in major	earquiz
289	29	Week 7, quiz B: four-note melodies in major	earquiz
290	29	Week 8, quiz A: two progressions	earquiz
291	29	Week 8, quiz B: four progressions	earquiz
292	29	Week 8, quiz C: with the dominant 7th	earquiz
293	29	Week 9 reference: I – vi – IV – V in G	jam

Track	Chapter	Example	Type
294	29	Week 9, quiz A: the four-chord loops	earquiz
295	29	Week 9, quiz B: ii, V and I	earquiz
296	29	Week 9, quiz C: all six chords	earquiz
297	29	Week 10 reference: E minor pentatonic over a drone	score
298	29	Week 10, quiz A: four-note melodies in minor	earquiz
299	29	Week 10, quiz B: minor pentatonic melodies	earquiz
300	29	Week 10, quiz C: five-note melodies in minor	earquiz
301	29	Week 11 reference: D, Dsus4, D, Dsus2	score
302	29	Week 11, quiz A: major, sus2, sus4 or augmented	earquiz
303	29	Week 11, quiz B: 6, 9, maj7 or 7	earquiz
304	29	Week 11, quiz C: which scale	earquiz
305	29	Week 12, review A: every interval, up or down	earquiz
306	29	Week 12, review B: five-note melodies in major	earquiz
307	29	Week 12, review C: which scale	earquiz
308	29	Final exam 1: harmonic intervals	earquiz
309	29	Final exam 2: chord qualities	earquiz
310	29	Final exam 3: progressions	earquiz
311	30	The same phrase in two positions	score
312	30	Writing techniques into a transcription	score
313	30	The recording	score
314	30	Step 3: the bass line	score
315	30	Step 5: the melody alone	score
316	30	Step 6: the finished transcription	score
317	31	The three forms of A minor	score
318	31	Natural minor v against harmonic minor V7	score
319	31	Andalusian cadence with a Phrygian dominant run	score
320	31	Phrygian dominant vamp on E: E – F	jam
321	31	A melodic minor over AmMaj7	score
322	31	E altered resolving to Am	score
323	31	Melodic minor and altered loop: Am6 – E7alt	jam
324	32	Secondary dominant: one chromatic inner voice	score
325	32	Extended dominants: E7 – A7 – D7 – G7 – C	score
326	32	Passing diminished chord in C	score
327	32	The minor iv chord in F	score
328	32	Rising cadence to the tonic in F	score
329	32	Picardy third in A minor	score
330	33	Pivot-chord modulation, C to G	score
331	33	Direct modulation from C to A at a phrase boundary	score
332	33	Truck-driver modulation, G to A	score
333	33	Common-tone modulation from C to E	score
334	33	One diminished chord, two resolutions	score
335	34	I – vi – IV – V7 – I on the top three strings	score
336	34	Guide tones over Dm7 – G7 – Cmaj7	score

Track	Chapter	Example	Type
337	34	ii – V – I with drop-2 voicings in two positions	score
338	34	Minor line cliché on Am, inner voice	score
339	34	ii – V – I and its tritone substitute in F	score
340	34	Plain and reharmonised versions	score
341	35	Two guide-tone lines through a cycle of fifths	score
342	35	A target-note line over ii – V – I	score
343	35	Enclosures into the 3rds of G7 and Cmaj7	score
344	35	Bebop dominant line over G7 resolving to C	score
345	35	Altered line in a ii – V – I in F	score
346	35	Practice loop: major ii – V – I in C	jam
347	35	Minor ii – V – i line in D minor	score
348	35	Practice loop: minor ii – V – i in D minor	jam
349	35	Practice loop: jazz blues in F	jam
350	36	Hammer-ons and pull-offs through A minor pentatonic	score
351	36	Trills on the 3rd and 2nd strings	score
352	36	Bends in A minor pentatonic	score
353	36	Bend to pitch	score
354	36	Plain note, then the same note with vibrato	score
355	36	Slides between pentatonic boxes 1 and 2	score
356	36	A short lick with every technique	score
357	36	Slow A minor loop for bends and vibrato	jam
358	36	Natural harmonics at frets 12, 7 and 5	score
359	36	Palm-muted chugs with open accents	score
360	36	Strumming with a chuck on 2 and 4	strum
361	36	Descending 6ths in C major on strings 3 and 1	score
362	37	The subdivision ladder on the open 5th string	score
363	37	Binary subdivisions on a sixteenth-note grid	rhythm
364	37	Triplets and swing on a triplet grid	rhythm
365	37	Five ways to place the same eighth-note bar	rhythm
366	37	The push: chords that arrive early	score
367	37	Rhythmic cells in one bar of sixteenths	rhythm
368	37	Tresillo strum	strum
369	37	Locking with the drums	rhythm
370	37	5/4 grouped 3 + 2	rhythm
371	37	Strumming in 5/4	strum
372	37	7/8 grouped 2 + 2 + 3	rhythm
373	37	A riff in 7/8 (2 + 2 + 3)	score
374	37	3 against 2	rhythm
375	37	3 over 4: a three-sixteenth figure across a 4/4 bar	rhythm
376	38	A passage for chunking	score
377	38	Rhythmic variations of bar 1	score
378	39	The pendulum: the hand never stops	strum
379	39	Changing chords on the last upstroke	strum

Track	Chapter	Example	Type
380	39	Chromatic walk in 5th position	score
381	39	Finger independence (fingers 1-3 and 2-4)	score
382	40	A solo over G – D – Em – C, ending on G	score
383	40	Pop loop in G: I – V – vi – IV	jam
384	40	The cadence arranged: falling bass, chords above	score
385	40	A phrase over the cadence	score
386	40	Stage 1: the melody alone	score
387	40	Stage 2: melody and bass roots	score
388	40	Stage 3: melody over an alternating bass	score
389	41	Intonation check: harmonic, then fretted note, on each string	score
390	42	Layers 1 and 2: chords and bass over C – Am – F – G	score
391	42	Layer 3: a percussion pass	strum
392	43	Triplets, shuffle and backbeat	rhythm
393	43	Boogie pattern on A7, D7 and E7	score
394	43	9th-chord stabs with a slide approach	score
395	43	Turnaround in E	score
396	43	Turnaround in A	score
397	43	Call and response over A7	score
398	43	Targeting the change: D7 to A7	score
399	43	12-bar shuffle in A	jam
400	43	Minor blues in A	jam
401	43	Delta-style figure in E	score
402	43	Piedmont-style ragtime cycle in C	score
403	43	Boogie in A	piece
404	44	Palm-muted eighths with accented chords	score
405	44	Four rock rhythm figures over a backbeat	rhythm
406	44	Pentatonic riff in E	score
407	44	Chromatic riff in E	score
408	44	Pedal-tone riff in E	score
409	44	Drop D riff	score
410	44	Repeated triplet lick and bend in A	score
411	44	Double stops in A	score
412	44	Riff Study in E	piece
413	44	Rock loop in E minor	jam
414	45	The engine: constant sixteenths, accents on 2 and 4	strum
415	45	Chords and scratches	strum
416	45	Em7 – A vamp on the top strings	score
417	45	How a funk groove locks together	rhythm
418	45	Single-note line in E Dorian	score
419	45	One-chord vamp on E9	jam
420	45	Em7 – A Dorian vamp	jam
421	45	Funk Study in E	piece
422	46	Boom-chick in G and C	strum

Track	Chapter	Example	Type
423	46	Bass runs in G	score
424	46	Carter-style melody in C	score
425	46	Hybrid-picked lick in G	score
426	46	Country loop in G	jam
427	46	Bluegrass rhythm in G and D	strum
428	46	The 3 + 3 + 2 accent pattern	rhythm
429	46	Crosspicking on the middle strings	score
430	46	A typical G-run	score
431	46	Breakdown in G	piece
432	47	Old faithful through the anchor-finger family	strum
433	47	A 3/4 arpeggio pattern	picking
434	47	Travis-style accompaniment with anchor fingers	score
435	47	Ballad loop in G	jam
436	47	A fill between vocal lines	score
437	47	Song Accompaniment in G	piece
438	48	Guide tones over ii – V – I in C	score
439	48	Comping rhythms on a swing grid	rhythm
440	48	Four-to-the-bar shells on a blues in F	score
441	48	ii – V – I in C	jam
442	48	Jazz blues in F	jam
443	48	Chord-melody intro over ii – V – I	score
444	48	Comping Study: Jazz Blues in F	piece
445	49	La pompe	strum
446	49	Minor swing in A	jam
447	49	Rest-stroke arpeggio on Am6	score
448	49	Arpeggio line over Am6 – E7 – Am6	score
449	49	Pompe and Arpeggios in A Minor	piece
450	50	One-bar pattern: fingers on 1, the and of 2, and 4	picking
451	50	Two-bar bossa pattern on a ii – V – I in C	score
452	50	Partido alto on Dm9 and G13	score
453	50	Clave family over two bars of 4/4 (eighth-note grid)	rhythm
454	50	Eighth-note strum with clave accents (3–2)	strum
455	50	Guajeo-style ostinato on I – IV – V – IV (original)	score
456	50	Cha-cha-chá and bolero	strum
457	50	Estudo em lá menor	piece
458	50	Bossa nova loop in C	jam
459	50	Son montuno loop in C	jam
460	51	Choro accompaniment on G – E7 – Am – D7	score
461	51	Baixaria on G – E7 – Am – D7 – G (original)	score
462	51	Samba pattern on Am7 and E7	score
463	51	Baião bass and chords	score
464	51	Chorinho em Sol	piece
465	51	Choro progression in G (samba-bossa feel)	jam

Track	Chapter	Example	Type
466	51	Baião drone on A	jam
467	52	Reggae skank on A and D (count slowly)	strum
468	52	Double chop (sixteenth notes)	strum
469	52	Ska up-chops	strum
470	52	One drop (sixteenth-note grid, one bar)	rhythm
471	52	Rockers and steppers	rhythm
472	52	Riddim Study in A Minor	piece
473	52	One-drop loop in A minor	jam
474	53	Soleá and alegrías: 12 beats, accents on 3, 6, 8, 10, 12	compas
475	53	Bulerías: the same accents, counted from 12	compas
476	53	Soleá compás exercise por arriba	score
477	53	Picado in E Phrygian	score
478	53	Alzapúa on the Andalusian cadence por medio	score
479	53	Rumba strum on Am – G – F – E	strum
480	53	Soleá Study por arriba	piece
481	53	Andalusian cadence in 6/8	jam
482	53	E Phrygian drone	jam
483	54	Jig strum (6/8)	strum
484	54	Reel strum	strum
485	54	Ornaments on a jig phrase in DADGAD	score
486	54	Campanella scale in DADGAD	score
487	54	The Kitchen Door	piece
488	54	Jig loop in D	jam
489	54	Modal drone on D	jam
490	55	Embellished riff on Em9	score
491	55	Sixths over Cmaj9 – Fmaj9	score
492	55	Comping on Dm9 – G13 – Cmaj9	score
493	55	Late Set	piece
494	55	Neo-soul ballad in C	jam
495	55	Minor groove in A	jam
496	56	Thumb and index pattern on C – F – G7 – C	picking
497	56	Highlife lead ostinato on I – IV – I – V in C	score
498	56	Highlife rhythm part (play against the lead)	score
499	56	Sebene lead pattern on A – E	score
500	56	12/8 timeline, pulse and guitar	rhythm
501	56	12/8 guitar ostinato on C and G	score
502	56	Palm-Wine Walk	piece
503	56	Highlife loop in C	jam
504	56	Sebene cycle in A	jam
505	57	Ode to Joy	piece
506	57	Thumb Walk	piece
507	57	Arpeggio Study in C	piece
508	57	Boom-Chick Accompaniment in G	piece

Track	Chapter	Example	Type
509	57	Two Voices in G	piece
510	57	Arpeggio Study in A minor	piece
511	57	Greensleeves	piece
512	57	Open-Position Blues in E	piece
513	57	Study in D with a Walking Bass	piece
514	58	Travis Study in C	piece
515	58	Waltz with a Moving Bass	piece
516	58	Étude in E minor	piece
517	58	Air in DADGAD	piece
518	58	Silent Night	piece
519	58	Bossa Study in A minor	piece
520	58	Porch Rag	piece
521	59	Prelude in B minor	piece
522	59	Two-Part Study in A minor	piece
523	59	Open D Groove	piece
524	59	Study in E Phrygian	piece
525	59	Tremolo Study in A minor	piece
526	59	Chord-Melody Study in F	piece
527	D	Ionian (major scale) practice loop	jam
528	D	Dorian practice loop	jam
529	D	Phrygian practice loop	jam
530	D	Lydian practice loop	jam
531	D	Mixolydian practice loop	jam
532	D	Aeolian (natural minor) practice loop	jam
533	D	Locrian practice loop	jam
534	D	Major pentatonic practice loop	jam
535	D	Minor pentatonic practice loop	jam
536	D	Blues scale practice loop	jam
537	D	Major blues practice loop	jam
538	D	Harmonic minor practice loop	jam
539	D	Phrygian dominant practice loop	jam
540	D	Dorian #4 (Ukrainian Dorian) practice loop	jam
541	D	Hungarian minor practice loop	jam
542	D	Melodic minor (jazz minor) practice loop	jam
543	D	Lydian dominant practice loop	jam
544	D	Mixolydian b6 (Aeolian dominant) practice loop	jam
545	D	Locrian #2 (half-diminished) practice loop	jam
546	D	Lydian augmented practice loop	jam
547	D	Altered (Superlocrian) practice loop	jam
548	D	Whole tone practice loop	jam
549	D	Half-whole diminished (dominant diminished) practice loop	jam
550	D	Whole-half diminished practice loop	jam
551	D	Bebop dominant practice loop	jam